Passage — Product Requirements Document

Status	Draft v0.2 — for review by Ben Russell and Zach Demanche
Date	2026-10-21
Owners	@BenjaminSRussell (product), @zdemanche (engineering/security)
Method	Moonaan Scribe requirements workflow (understand → research → draft → validate → publish). Jira/Confluence research was unavailable this session; research covered public AX/SAP documentation and both owners' GitHub accounts (no prior ERP work found).
Companion	roadmap/waves.json is the machine-readable source for the 50 waves; roadmap/README.md is generated from it.
Thesis	Emulate the top-tier manufacturing ERPs (S/4HANA, D365 F&SCM, Infor LN, Epicor Kinetic, IFS, Plex) feature-for-feature, then be **10× easier, faster, more accurate and more open** — see Part 0A.

Document map

This master document (Part 0) holds the problem, goals, scope, delivery model, phase-level requirements and NFRs. The detailed parts hold wave-level requirements and are assembled into `docs/PRD-full.md` by `scripts/build_prd.py`.

Part	File	Content
0	`docs/PRD.md` (this file)	Problem, goals, personas, principles, scope map, delivery model, phase-level FRs, NFRs, risks
0A	prd/part-00-thesis.md	The 10× thesis: pain-point ledger (usability, speed, accuracy, openness), interoperability contract, what we keep from AX and SAP
1	prd/phase-0-foundation.md	Waves 1–5: pipeline, identity/tenancy, metadata kernel, events/workflow, shell/audit/observability
2	prd/phase-1-master-data.md	Waves 6–10: enterprise structure, party model, products, engineering data, migration framework
3	prd/phase-2-financial-core.md	Waves 11–15: universal journal, currencies/ledgers, tax, close, reporting
4	prd/phase-3-procure-to-pay.md	Waves 16–20: inventory ledger, purchasing, receiving/matching, AP/payments, costing/supplier portal
5	prd/phase-4-manufacturing.md	Waves 21–25: production orders, MRP, scheduling, process/lean, shop floor/IoT
6	prd/phase-5-order-to-cash.md	Waves 26–30: pricing, orders/ATP, fulfillment, billing/rev rec, AR/collections — go-live candidate
7	prd/phase-6-logistics.md	Waves 31–35: WMS, mobile, quality, transportation, EDI/e-invoicing/integration hub
8	prd/phase-7-treasury-controlling.md	Waves 36–40: cash/bank, fixed assets, budgeting/cost accounting, product costing, intercompany/consolidation
9	prd/phase-8-projects-people-service.md	Waves 41–45: projects, HR, payroll/expenses, EAM, field service
10	prd/phase-9-intelligence-golive.md	Waves 46–50: data platform, AI agents, localization/compliance, migration factory, scale/go-live
11	prd/appendices.md	Glossary, cross-module data model, event/API conventions, security model, NFR detail, numbering, localization, testing standards, traceability, references, competitive benchmark
12	prd/coverage-ax.md	Completeness oracle: every AX 2012 R3 / D365 F&SCM module → forms, inquiries/reports, periodic jobs, setup forms, each mapped to a wave or flagged GAP; gap register
13	prd/coverage-sap.md	Completeness oracle: every SAP ECC / S/4 component → transactions, reports, periodic processing, IMG customizing, each mapped to a wave or flagged GAP; gap register

Each wave section follows the fixed template in prd/TEMPLATE.md: scope and pain points solved → AX/SAP parity map → functional requirements → data model → APIs/events → UX → business rules → acceptance criteria → test plan → deployment → migration → dependencies/risks/open questions → parameters and setup → inquiries and reports → printed/electronic documents → periodic jobs → duties and privileges → message catalog → performance budgets → localization hooks.

Completeness rule. A wave is not specified until every row of Parts 12 and 13 that maps to it is either covered by a numbered FR in that wave's section or explicitly marked SUPERSEDED/OUT-OF-SCOPE with a reason. The gap registers (12.99, 13.99) must be empty before Phase 0 starts.

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1. Problem statement

Microsoft Dynamics AX (Axapta → AX 2012 → D365 Finance & Supply Chain) and SAP (ECC → S/4HANA) are the systems of record for most mid-to-large manufacturers, distributors and services firms. Both are:

- **Closed and expensive to change** — customization lives in proprietary languages (X++, ABAP) and licensing/upgrade costs dominate TCO.
- **Architecturally dated** — monolithic application servers, synchronous batch frameworks, and vendor-locked cloud operation (LCS, RISE).
- **Hard to integrate and extend with AI** — bolted-on copilots that can't act safely across the system.

Passage is a ground-up rebuild of the complete feature surface of AX/SAP as an open-architecture, cloud-agnostic platform: Kubernetes-native, Postgres-per-domain, Kafka-eventing, metadata-driven, with AI agents as first-class, governed actors.

2. Goals and non-goals

Goals

1. **Feature parity**: every AX 2012 / D365 F&SCM and SAP ECC / S/4 module in scope (§5) has an equivalent in Passage, mapped wave-by-wave.

2. **Provable delivery**: every one of the 50 waves ends with an artifact that is (a) **provable** — automated tests and fixtures that demonstrate correctness, (b) **testable** — a stakeholder can exercise it in a running environment, and (c) **deployable** — it ships to a Kubernetes environment through the GitOps pipeline.
3. **Sequential, top-down**: waves are strictly ordered; each wave depends only on earlier waves (dependency graph is a chain with stated back-references). A manufacturer can go live at the end of wave 30; full AX/SAP parity at wave 50.
4. **Cloud-agnostic and self-hostable**: runs on any conformant Kubernetes (EKS/GKE/AKS/bare-metal/k3d) with no provider-specific managed services in the critical path.
5. **Extensible without forking**: customers extend entities, workflows, forms and reports through metadata and sanctioned extension points, never by overlayering code.
6. **Financially rigorous**: a single append-only universal journal is the only source of truth; every subledger reconciles to it by automated check.
7. **10× on four measured axes** (defined in Part 0A): easier (≤ 1/10 the clicks per task; new user productive in < 30 min), faster (MRP incremental < 2 min, continuous close, real-time statements), more accurate (zero unexplained subledger-to-GL breaks, < 0.5% duplicate master data), open (any system on the customer's network connects in < 1 day through REST/GraphQL/Kafka/webhooks/MCP/EDI/OPC-UA/SFTP with one identity and one policy engine).
8. **Best-of-breed manufacturing**: planning, scheduling, shop floor, quality, costing and warehouse are benchmarked against S/4HANA, D365 SCM, Infor LN, Epicor Kinetic, IFS Cloud and Plex (Appendix L), not only against AX 2012.

Non-goals (v1)

- Replacing CRM/marketing automation suites (we integrate with Salesforce/HubSpot; a native CRM is v2).
- Retail POS and e-commerce storefront (we integrate with Shopify/Medusa; native POS is v2).
- Country packs beyond the first set (US, UK, DE, FR, IN, MX, CA) — framework ships in wave 48; more packs are v1.x.
- Supporting SAP's or Microsoft's proprietary extension languages. Migration tooling moves **data**, not X++/ABAP code.

3. Users and personas

Persona	Primary jobs	Waves that serve them first
Platform engineer / SRE	Deploy, observe, upgrade, scale, secure	1–5, 48, 50
System administrator	Tenants, users, roles, extensions, number sequences, audit	2, 3, 5, 48
Master-data steward	Parties, products, BOMs, org structure, migration	6–10
Controller / accountant	Journal, close, statements, tax, FX, subledgers	11–15, 19, 30, 36–40
Buyer / AP clerk	Requisition to payment	17–20
Production planner / supervisor	Orders, MRP, scheduling, shop floor	21–25
Sales rep / order desk / AR clerk	Quote to cash	26–30
Warehouse manager / operator	WMS, mobile, shipping, quality, transport	28, 31–35
Project manager / HR / payroll admin	Projects, people, payroll, assets, service	41–45
Analyst / executive	Dashboards, KPIs, copilots	46–47
Implementation partner	Migration, localization, cutover	10, 48–50

4. Product principles

1. **Journal first.** Nothing is financially true until it is in the universal journal. Subledgers are projections; they reconcile automatically every day.
2. **Metadata over code.** Entities, forms, workflows, reports and extensions are data. Code generators produce schema, API, clients and UI from it.
3. **Events are the integration surface.** Every state change is an immutable CloudEvent on Kafka via transactional outbox — no dual writes, no polling.
4. **Everything is tenant-scoped and policy-authorized.** Postgres RLS + OPA on every request; agents inherit (never exceed) their user's permissions.
5. **Idempotent, replayable, auditable.** Posting APIs take idempotency keys; audit is derived from CDC, not from application code that can forget.
6. **Ship the thin vertical slice.** A wave is done when a stakeholder can run the business process on a deployed environment with reconciled numbers — not when the code is written.

5. Scope map — AX/SAP modules → Passage phases

AX / D365 module	SAP module	Passage phase (waves)
AOT, security, workflow, batch, LCS, number sequences, DB log	DDIC, PFCG, Business Workflow, SM36, STMS, SNRO, CDHDR	**0 Foundation (1–5)**
Org model, global address book, products, BOM/route, DIXF	Enterprise structure, Business Partner, material master, CS/CA, LSMW	**1 Master data (6–10)**
General ledger, dimensions, currencies, tax, period close, financial reporting	FI-GL/ACDOCA, ledgers, FTXP, OB52, FSV	**2 Financial core (11–15)**
Inventory management, procurement & sourcing, AP, inventory costing	MM-IM, MM-PUR, LIV, FI-AP, F110, material ledger	**3 Inventory & P2P (16–20)**
Production control, master planning, scheduling, process/lean, MES	PP, PP-PI, REM, KANBAN, PP/DS, MII	**4 Manufacturing & planning (21–25)**
Trade agreements, sales, shipping, invoicing, rev rec, AR, collections	SD pricing, SD orders, LE-SHP, billing, RAR, FI-AR, FSCM	**5 Order-to-cash (26–30)**
Warehouse management, mobile app, quality, TMS, AIF/ER/EDI	EWM, RF, QM, TM, IDoc/PI, DRC	**6 Warehouse, quality & logistics (31–35)**
Cash & bank, fixed assets, budgeting/cost accounting, costing sheets, intercompany/consolidation	Cash mgmt, FI-AA, CO, product costing, Group Reporting	**7 Treasury, assets & controlling (36–40)**

AX / D365 module	SAP module	Passage phase (waves)
Projects, HR, payroll, expense, asset management, service	PS, HCM/SF, PY, Concur, PM, CS/FSM	**8 Projects, people & service (41–45)**
Entity store/Power BI, Copilot, localizations/SoD, upgrade tooling, LCS prod	BW/SAC, Joule, country versions/GRC, conversion tools, Basis	**9 Intelligence, compliance & go-live (46–50)**

Full per-wave mapping with references is in roadmap/waves.json (`replaces` field).

6. Delivery model — what a "wave" is

A wave is a 4–8 week increment with four mandatory outputs, enforced by the wave issue template and the Definition of Done:

Output	Definition	Evidence required to close the wave
Deliverable	The capabilities listed in the wave's `deliverable` array, built on the agreed stack.	Merged PRs linked from the wave issue.
Proof	The automated tests, fixtures, property tests and performance targets in `proof`.	CI run link; fixture datasets committed; perf report.
Deploy	What is running where, as stated in `deploy`.	Argo CD app healthy in `stage`; demo URL; recorded walkthrough for phase-exit waves.
References	The AX/SAP documentation and open-source/standards we built against.	Linked in the wave issue; ADRs for any deviation.

Sequencing rules

- Wave *n* may start when wave *n–1* has passed Proof and Deploy; overlapping *build* is allowed, overlapping *close* is not.
- Every 5th wave is a **phase-exit wave** with a stakeholder sign-off (demo + reconciliation evidence).
- Wave 30 is the **first go-live candidate** (manufacturer buy → make → sell → collect). Wave 50 is **production go-live + v1.0**.

7. Functional requirements (phase level)

Phase-level requirements below are the contract; wave-level detail (deliverables, proofs, references) is normative in `roadmap/waves.json` and rendered in `roadmap/README.md`.

FR-0 Foundation (waves 1–5)

- **FR-0.1** The platform deploys to any conformant Kubernetes via GitOps with signed images, SBOMs and digest-pinned rollback.
- **FR-0.2** Identity is federated (OIDC/SAML/SCIM); authorization is AX-style roles/duties/privileges compiled to OPA policies; tenant isolation is enforced by Postgres RLS.
- **FR-0.3** Business entities are declared in metadata that generates schema, API, clients, validation and UI; customers extend entities without code.
- **FR-0.4** All state changes publish CloudEvents via transactional outbox; approvals and batch jobs run on a durable workflow engine with timers, retries and human tasks.
- **FR-0.5** A single web shell renders any entity as accessible (WCAG 2.1 AA) list/detail/grid; number sequences support gap-free mode; audit is field-level and immutable; every request is traceable end-to-end.

FR-1 Master data (waves 6–10)

- **FR-1.1** Legal entities, operating units, sites, warehouses and effective-dated multi-purpose hierarchies; fiscal calendars; ISO reference data; UoM conversions.
- **FR-1.2** One party per real-world organization/person with layered roles, duplicate detection and merge, tax-id validation, governance workflow.
- **FR-1.3** Products with product/storage/tracking dimension groups, released products per entity, attributes/classification, GS1 identifiers, configurator.
- **FR-1.4** Multi-level versioned BOMs and formulas, routings/operations/resources/calendars, engineering change management with where-used.
- **FR-1.5** A migration framework with AX and SAP connectors, declarative mapping, validation, reconciliation and idempotent re-runs.

FR-2 Financial core (waves 11–15)

- **FR-2.1** Append-only double-entry universal journal with dimensions, account structures, idempotent atomic posting, balance projections rebuilt from the journal.
- **FR-2.2** Three-currency amounts per line, rate types/providers, parallel ledgers, FX revaluation and translation.
- **FR-2.3** Tax determination/calculation/posting for sales & use, VAT/GST and withholding; pluggable external providers; settlement and statutory extracts (SAF-T).
- **FR-2.4** Period control per ledger × module, close checklists, allocations, accrual schemes, year-end close.
- **FR-2.5** Statement designer and standard statements with drill-down to source document; audit trail reporting.

FR-3 Inventory & P2P (waves 16–20)

- **FR-3.1** Immutable inventory transactions with physical/financial status and dimensions; journals; reservations; cycle counting; GL integration.
- **FR-3.2** Requisition → RFQ → agreement → PO with policy-driven approvals, versioned change management and vendor pricing.
- **FR-3.3** Receiving with GR/IR accrual; vendor invoice capture; 2/3-way matching with tolerances and exception workflow; ERS.
- **FR-3.4** AP open-item settlement, payment proposals, ISO 20022/NACHA/check payment media, positive pay, prepayments.
- **FR-3.5** FIFO/LIFO/weighted-average/standard costing with inventory close and GL reconciliation; supplier portal.

FR-4 Manufacturing & planning (waves 21–25)

- **FR-4.1** Production order lifecycle with picking, route consumption, RAF, WIP and variance settlement; subcontracting.
- **FR-4.2** MRP/MPS with coverage groups, planned orders, action/delay messages, pegging, statistical forecasting.
- **FR-4.3** Finite/infinite capacity scheduling with Gantt and optional CP-SAT optimization.
- **FR-4.4** Process manufacturing (formulas, potency, co/by-products, catch weight, batch attributes) and lean (production flows, kanban).
- **FR-4.5** Shop-floor kiosk with time registration, IoT ingestion (OPC-UA/MQTT), OEE, offline tolerance.

FR-5 Order-to-cash (waves 26–30)

- **FR-5.1** Deterministic pricing engine with condition tables/access sequences, discounts, promotions, rebates, explain.
- **FR-5.2** Quotations, sales orders, ATP/CTP, credit management, holds, multi-channel intake.
- **FR-5.3** Release, pick/pack/ship, packing slips with COGS, carrier integration, drop-ship.
- **FR-5.4** Invoicing (one-off, consolidated, subscription), ASC 606/IFRS 15 revenue recognition.
- **FR-5.5** Cash application, collections/dunning/disputes, RMA, rebate settlement, customer portal with payments. **Go-live candidate.**

FR-6 Warehouse, quality & logistics (waves 31–35)

- **FR-6.1** Location directives, work templates, waves, license plates, replenishment, load planning.
- **FR-6.2** Mobile scanner app with configurable flows, GS1 parsing, offline queue, label printing.
- **FR-6.3** Quality orders by rule, sampling, usage decisions, inventory blocking, NCR/CAPA, CoA, supplier scorecards.
- **FR-6.4** Load/route planning, rating, tendering, tracking, freight reconciliation, landed cost.
- **FR-6.5** EDI (X12/EDIFACT), Peppol e-invoicing, integration hub (APIs, webhooks, catalog, developer portal).

FR-7 Treasury, assets & controlling (waves 36–40)

- **FR-7.1** Bank accounts, statement import and auto-reconciliation, cash positioning/forecasting, bank connectivity with vaulted keys.
- **FR-7.2** Fixed assets with multi-book depreciation, full transaction set, CIP from projects, IFRS 16 leases.
- **FR-7.3** Budgeting with workflow and budget control on commitments; cost accounting ledger with allocations and ABC.
- **FR-7.4** Costing sheets, BOM cost roll-up, standard cost activation and revaluation, variance decomposition.
- **FR-7.5** Intercompany trade/accounting with mirror documents and reconciliation; group consolidation with eliminations and translation. **Finance parity.**

FR-8 Projects, people & service (waves 41–45)

- **FR-8.1** WBS, resources, hour/expense/item journals, T&M/fixed-price/milestone billing, WIP and POC accounting, EVM.
- **FR-8.2** Workers/positions/jobs with effective dating, absence, benefits, self-service, talent basics, PII controls.
- **FR-8.3** Country-pluggable gross-to-net payroll (US + one EU reference) and expense management with policies and card feeds.
- **FR-8.4** Functional locations/assets, preventive and condition-based plans, work orders with costs to asset/GL, IoT triggers.
- **FR-8.5** Installed base, agreements/SLAs, cases, field work orders, dispatch optimization, technician mobile app.

FR-9 Intelligence, compliance & go-live (waves 46–50)

- **FR-9.1** CDC lakehouse (Iceberg) + semantic layer with governed metrics, embedded dashboards, document designer, catalog/lineage.
- **FR-9.2** MCP-exposed APIs; copilot UI; governed agents with approval gates, evals in CI, cost budgets, model-agnostic provider layer.
- **FR-9.3** Localization framework and first country packs; electronic reporting designer; SoD engine; SOC 2 control evidence; privacy requests.
- **FR-9.4** Transactional migration packs for AX and SAP, parallel-run diff harness, cutover runbooks, coexistence adapters.
- **FR-9.5** Scale/perf targets met, multi-region DR, operations and documentation complete, first customer in production, v1.0 released.

8. Non-functional requirements

Area	Requirement
Availability	99.9% per tenant (v1), 99.95% target with multi-region (wave 50). RPO 5 min, RTO 1 h.
Performance	Journal posting 5k lines/s sustained; generated entity reads p95 < 50 ms; pricing p95 < 20 ms; order intake p99 < 500 ms; dashboard lag p95 < 60 s.
Scale	10k concurrent users; 1M journal lines/h; 100k order lines/h; 50k-item MRP in < 10 min.
Security	OIDC/SAML/SCIM; OPA authorization on every request; Postgres RLS; signed images/SBOM; secrets in Vault; SoD engine; pen-test clean at high/critical before wave 48 exit.
Compliance	SOC 2 TSC control mapping, GDPR/CCPA subject requests, SAF-T, Peppol, ISO 20022, GS1.
Accessibility	WCAG 2.1 AA for every generated form and every hand-built workspace (axe in CI).
Observability	OpenTelemetry traces/metrics/logs for 100% of services; SLO dashboards; alerting.
Portability	No provider-managed service in the critical path; Helm + Terraform/OpenTofu modules per provider; k3d local parity.
Extensibility	Entity/enum/field/workflow/form/report extensions as metadata; versioned public APIs with additive-change guarantees.
Data integrity	Every subledger reconciles to the universal journal daily by automated check; no negative inventory unless policy allows; gap-free fiscal sequences.
Auditability	Field-level immutable audit from CDC; every agent action logged with inputs/outputs/approver.

9. Technical architecture (summary — see ADR-0001)

- **Runtime**: Kubernetes (any conformant distribution), Argo CD GitOps, Envoy Gateway, cert-manager.

- **Services**: Go for ledgers/engines (journal, inventory, pricing, MRP, scheduling), TypeScript (Node) for workflow/integration/UI-adjacent services, Python for forecasting/ML/agent evals.
- **Data**: Postgres per domain (CloudNativePG), RLS, Atlas migrations generated from metadata; Apache Iceberg lakehouse with Trino/ClickHouse for analytics.
- **Eventing**: Kafka (Strimzi), Debezium CDC + transactional outbox, CloudEvents + AsyncAPI, schema registry.
- **Workflow**: Temporal for approvals, batch, agents.
- **Identity/authz**: Keycloak, OPA/Rego.
- **Frontend**: React + module federation shell; PWA for shop floor, warehouse and field mobile.
- **AI**: MCP server over Passage APIs; Claude as default model via a provider-agnostic layer; eval harness in CI.
- **Observability**: OpenTelemetry → Prometheus/Tempo/Loki/Grafana.

10. Success metrics

Milestone	Metric
Wave 5	Platform demo: deploy → login → extend entity → approve workflow → trace a request, all live.
Wave 15	A period closed in Passage with statements that tie to the trial balance to the cent.
Wave 30	Demo manufacturer runs one simulated month buy → make → sell → collect; all subledgers reconcile; stakeholder sign-off.
Wave 40	Group consolidation produced; finance parity sign-off by a practicing controller.
Wave 50	First customer month-end in production; load/DR targets met; v1.0 released.
Ongoing	Every wave: CI green, proofs committed, Argo app healthy in stage, references linked.

11. Risks and mitigations

Risk	Mitigation
Scope is enormous (50 waves)	Strict sequencing; each wave independently demoable; phase-exit sign-offs; `waves.json` is the single plan of record and is re-estimated each phase exit.
Accounting correctness	Fixture sets reviewed by a practicing accountant per finance wave; daily automated subledger-to-GL reconciliation from wave 16 onward.
Postgres-per-domain vs cross-domain transactions	Sagas over Temporal + outbox; the journal posting API is the only cross-domain financial truth; no distributed 2PC.
Cloud-agnostic ambition increases ops burden	Reference stack is entirely CNCF/open-source with Helm charts; provider-specific Terraform modules are thin and optional.
Payroll/tax/localization legal exposure	Ship provider adapters (Avalara/ADP-style) alongside internal engines; country packs gated by regulator-schema fixtures.
AI agent misuse	Agents inherit user permissions via OPA, approval gates, evals in CI, kill switch, cost budgets.
Two-person team	GitHub Projects + wave issues as the operating system; AI-assisted generation (Moonaan sdlc `generate-architecture` → `generate-backlog` → `generate-code`) per wave.

12. Open questions

1. Licensing model for v1.0 (open-core vs. source-available vs. commercial) — decide before wave 35 (integration hub/public APIs).
2. First real customer: identify by wave 20 to shape go-live scope at wave 30.
3. Which EU country is the payroll/localization reference (DE vs. UK)?
4. Hosting of the demo environments (which provider first, even though the stack is agnostic)?

13. Next steps in the Moonaan SDLC pipeline

1. **This PRD** is the requirements source.
2. Run `/sdlc:generate-architecture docs/PRD.md docs/architecture/` per phase to produce the architecture document and compiled fact sheets.
3. Run `/sdlc:generate-backlog` per wave to turn `waves.json` entries into epics/stories with dependency ordering (GitHub Issues are the backlog store for now; Jira when the Moonaan proxy is reachable).
4. Run `/sdlc:generate-code` per story.

Validation checklist (Scribe step 4)

- [x] Every functional requirement is traceable to a wave with a provable, testable, deployable output.
- [x] Every wave names the AX and SAP features it replaces (grounding).
- [x] Every wave links external references (standards, vendor docs, open-source implementations).
- [x] Non-functional requirements are measurable.
- [x] Sequencing and go-live milestones are explicit.
- [ ] Reviewed by @zdemanche.
- [ ] Accountant review of finance-wave fixtures (owner TBD).

Part 0A — The 10× Thesis: What Top-Tier Manufacturing ERPs Get Wrong and How Passage Fixes It

Passage is not a cheaper AX or a friendlier SAP. It emulates the full capability surface of the best manufacturing ERPs — SAP S/4HANA, Dynamics 365 Finance & Supply Chain, Infor LN / CloudSuite Industrial, Epicor Kinetic, IFS Cloud and Plex — and then removes the structural reasons those systems are slow, error-prone, expensive to use and closed. "10×" is measured, per wave, on four axes:

Axis	What we measure	Baseline (typical AX 2012 / D365 / S/4 deployment)	Passage target
Easier	Clicks and screens per task; training hours to competence; share of tasks completed without a super-user	Sales order with 5 lines: 40–60 clicks across 4 forms; AP clerk training 3–5 days	≤ 1/10 of clicks; a new user completes a guided task in < 30 min
Faster	Latency of the business answer (not the screen): MRP, inventory close, ATP, period close, report	MRP regeneration 2–8 h nightly; inventory close 1–6 h; month-end close 5–10 business days	MRP incremental < 2 min, full < 10 min; close continuous; statements real-time
More accurate	Reconciliation breaks per month; silent rounding/FX errors; duplicate master data; manual journal share	Subledger-to-GL breaks every close; 3–8% duplicate customers/vendors; 20–40% of journals are corrections	Zero unexplained breaks (checked daily by machine); < 0.5% duplicates; corrections < 5%
Open	Days to connect a new system; share of data reachable by API/event; integration projects per year	Weeks per interface (AIF/IDoc mapping projects); large share of logic reachable only from the UI	Any system on the network connects in < 1 day with no middleware; 100% of entities and events reachable

0A.1 The structural pain-point ledger

Each pain point below names the AX/SAP mechanism that causes it, the root cause, the Passage design that removes it, and the wave where it ships. Wave files repeat this at wave granularity under "Pain points solved (10× lens)".

A. Usability

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
U1	Dozens of forms to complete one business task; fields scattered across tabs and FastTabs	AX `SalesTable` form with 20+ tabs; SAP `VA01` header/item/schedule-line screens	Forms mirror tables, not tasks	Task-oriented screens generated from metadata with the *task* as the unit (order intake, receipt, close task); line grids with inline edit; the full document on one scrollable page	5, 27
U2	Lookups require knowing the code (vendor account, main account, item number)	`MainAccount` lookup in `LedgerJournalTrans`; SAP GL account search help	Keys are the primary identifier; search is secondary	Global typeahead on name/description/any attribute, ranked by recency and role; keys shown but never typed	5
U3	Personalization is per-user, fragile, lost on upgrade	AX usage data (`SysLastValue`); SAP GUI variants	Personalization stored outside metadata	Saved views are first-class, versioned, shareable, role-defaulted; survive upgrades because they target metadata ids	5
U4	Batch jobs hide status; users poll "Batch jobs" form to see if posting finished	`BatchJob`/`BatchJobHistory`; SAP `SM37`	Fire-and-forget batch framework with no UI contract	Every long-running action returns a tracked workflow with progress, ETA and a notification; the originating screen updates live	4, 5
U5	Error messages name internal objects ("Account structure does not allow this combination")	AX infolog; SAP message classes	Validation written for developers	Validation messages are metadata with user text, remediation link and the exact field; errors shown at the field as you type	3, 5
U6	Shop-floor and warehouse UIs are RF text menus or require a PC	AX `JmgRegistration` terminal; SAP ITSmobile/RF	UI frameworks predate touch and offline	Touch-first PWA kiosk and scanner apps with offline queues; one scan completes a step	25, 32
U7	Excel round-trips to do bulk edits	AX Excel add-in; SAP GUI copy/paste	No bulk-edit grid	Every list page supports multi-row inline edit, paste from spreadsheet with validation preview, and undo	5
U8	Approval requires logging in to the ERP	AX workflow work items; SAP Business Workplace (`SBWP`)	Workflow inbox is inside the client	Approvals in email/Slack/Teams/mobile with one-tap approve, full context and the same audit trail	4
U9	Copilot features are bolted on and read-only	D365 Copilot summaries; SAP Joule Q&A	AI layer has no safe action surface	Every API is an MCP tool under the user's permissions; agents draft, users approve; explain-this-record everywhere	47

B. Speed

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
S1	MRP runs overnight; planners work on stale data; a changed order waits until tomorrow	AX `ReqCalc` regeneration; SAP `MD01` total planning	Full regeneration over all items; pegging recomputed from scratch	Net-change planning engine keyed by events (order/receipt/BOM change) with incremental re-plan of affected low-level codes; full regen < 10 min for 50k items	22
S2	Inventory close/recalculation blocks posting for hours	AX `InventClosing`; SAP material ledger `CKMLCP`	Settlement of issues to receipts is a batch over all transactions	Running costing (continuous settlement on each issue for FIFO/avg) so close is a verification pass, not a computation; partitioned by item	20
S3	ATP is computed per line with locks; order entry slows under load	AX `InventOnHand`/CTP; SAP ATP with `VBBE` requirements	Row-level locking on on-hand and requirement tables	Cumulative ATP projection maintained from the inventory event stream; reads lock-free; promise computed in memory	27
S4	Month-end close takes 5–10 days of manual steps	AX period close checklist is a list; SAP closing cockpit sequences manual transactions	Postings are batch; reconciliations are manual	Continuous close: every subledger reconciles to the universal journal daily by machine; close checklist tasks auto-complete when their check passes	14, 16+
S5	Reports run against OLTP and time out, or against a cube that is a day old	AX SSRS on `GeneralJournalAccountEntry`; SAP BW loads	No streaming path from transaction to analytics	CDC → Iceberg lakehouse with p95 < 60 s lag; statements served from pre-aggregated balances rebuilt from the journal	11, 46

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
S6	Posting a 10k-line journal takes minutes and locks the period	`LedgerJournalCheckPost`; SAP `FB50` with `BSEG` inserts	Row-by-row posting through class hierarchies; synchronous number sequences	Append-only journal with batch insert, idempotency keys and asynchronous balance projection: 5,000 lines/s sustained	11
S7	Deploying a customization takes a release window	AX model store deploy with AOS restart; SAP transports with downtime	Code and metadata are compiled into the runtime	Metadata extensions apply live; code ships via GitOps with digest rollback in < 15 min	1, 3

C. Accuracy

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
A1	Subledgers drift from the GL (inventory value ≠ inventory account; AR aging ≠ AR control)	`InventSum` vs `GeneralJournalAccountEntry`; SAP `BSID` vs `SKB1` balances	Subledgers post to the GL through separate code paths with their own rounding	One posting API; subledgers are projections of the same event stream; daily machine reconciliation with zero unexplained breaks as a release gate	11, 16, 19, 30
A2	Rounding and FX differences appear as "penny" entries	AX `LedgerJournalTrans` rounding to accounting currency; SAP `BSEG` rounding differences	Rounding at line level then summing	Three-currency amounts stored per line; rounding applied once at document level with explicit rounding lines; ISO 4217 minor units enforced	12
A3	Duplicate customers/vendors/items	`CustTable` created freely; SAP `KNA1` duplicates across sales orgs	No dedup at creation; roles not layered on one party	One party per real-world entity with fuzzy-match blocking at creation (precision ≥ 0.95) and governed merge	7
A4	Wrong tax because tax group ↔ item tax group intersections are configured by hand	`TaxGroup`/`TaxItemGroup` intersection; SAP `FTXP` condition records	Determination is a configuration lattice with no test harness	Determination engine with a 500-scenario fixture pack reviewed by accountants; explain-this-tax on every line	13
A5	Negative inventory and phantom on-hand	`InventSum` allows negative physical when `InventModelGroup` permits; SAP `MMBE` vs `MSEG` timing	Projection updated by application code that can skip	On-hand is a projection of immutable inventory events; policy enforced at the event write; projection rebuilt from events and compared nightly	16
A6	Costing variances nobody can explain	`InventCostTrans` variances; SAP `KKS1`	Variance types computed after the fact with no linkage	Variance decomposition attached to each ended order with lot-size, quantity, price and substitution components that sum exactly	21, 39
A7	Master data changed without approval or trace	Direct edit of `VendTable` bank account; SAP `XK02`	Audit log optional (`SysDatabaseLog`) and slow	Field-level immutable audit from CDC for every entity; sensitive fields (bank accounts, credit limits) require workflow	5, 7

D. Openness and interoperability

#	Pain point today	Where it lives in AX / SAP	Root cause	Passage answer	Wave
O1	Each interface is a project (AIF service, IDoc mapping, custom data entity)	AX AIF/`DMF` entities; SAP `WE20` partner profiles, PI/PO	Integration surface is bespoke per object	Every entity and event is published automatically (OpenAPI, AsyncAPI, MCP) from metadata; new entity ⇒ new API with no work	3, 4, 35
O2	External systems poll for changes	`DMF` change tracking; SAP change pointers (`BDCP2`)	No event stream	Kafka topics per entity with CloudEvents, replay from any offset, webhooks for partners	4
O3	MES/PLM/SCADA need a separate middleware product	SAP MII, PI/PO; AX via BizTalk/Logic Apps	ERP speaks only its own protocols	Native OPC-UA/MQTT ingestion, EDI engine, Peppol, SFTP, MCP — eight integration surfaces with one auth model	25, 35
O4	Vendor cloud lock-in (LCS, RISE)	D365 only on Microsoft-managed Azure; S/4 Cloud on RISE	Platform tied to the vendor's operations	Any conformant Kubernetes; Helm charts; self-host or managed	1
O5	Customizations block upgrades	X++ overlayering; ABAP modifications	Extension model is code in the vendor's layer	Extensions are metadata and sanctioned hooks; additive API guarantees; upgrades are image promotions	3
O6	Analytics requires exporting to BYOD/BW	Entity store; BW extractors	No open analytical surface	Iceberg tables readable by any engine (Trino, Spark, DuckDB, Power BI, Tableau) with governed metrics	46

0A.2 Interoperability contract ("all systems on the network can use it")

Passage treats the ERP as a service on the customer's network, not a destination. Every capability is reachable through the same eight surfaces with one identity and one policy engine:

Surface	Used by	Auth	Guarantee
REST (OpenAPI 3.1, generated per entity)	Apps, scripts, partners	OIDC bearer token or API key, OPA-authorized	p95 < 50 ms simple reads; additive versioning
GraphQL (read model)	BI tools, portals	Same	Cross-entity reads without N+1
Kafka topics (CloudEvents, AsyncAPI)	MES, WMS, data platforms, other ERPs	mTLS + ACLs per tenant	At-least-once, ordered per key, 7-day replay
Webhooks	SaaS partners	Signed (HMAC) deliveries	Retries with backoff, dead-letter, replay
MCP	AI agents (Claude and others)	Delegated user token; agent ≤ user permissions	Every tool call audited
EDI (X12/EDIFACT via AS2/SFTP) and Peppol	Trading partners, tax authorities	Partner certificates	Acknowledgments, SLA monitor
OPC-UA / MQTT	Machines, PLCs, SCADA	Device certificates	10k events/min per plant, no loss on broker restart

Surface	Used by	Auth	Guarantee
File / SFTP	Banks, legacy systems	Keys in Vault	ISO 20022, BAI2, NACHA, CSV with schema validation

Rule: **if it is not reachable through these surfaces, it does not ship.** Each wave's API/event section is reviewed against this rule before close.

0A.3 How "deep knowledge of AX and SAP" is applied

We deliberately keep what those systems got right and name it in every wave's parity map:

- **From AX:** the role/duty/privilege security model; inventory dimensions (product/storage/tracking) as a first-class concept; `InventTrans` status semantics; trade-agreement search order; number sequences per legal entity; financial dimensions with account structures; coverage groups; lean production flows.
- **From SAP:** the universal journal (ACDOCA) as single source of truth; parallel ledgers and currency types; condition technique for pricing and tax; posting periods per module; EWM's location-directive rigor; QM inspection lots and usage decisions; results analysis for projects; Group Reporting eliminations.

And we remove the parts that exist only because of their era: batch-first processing, table-shaped forms, code-based extensions, closed integration layers, vendor-bound operations.

Part 0 — Phase 0: Foundation & Platform Kernel (waves 1–5)

0.0 Phase overview

Business outcome at phase exit. At the end of wave 5 nothing business-facing exists, but everything a business module needs does: a change merged to `main` reaches a Kubernetes environment with digest-pinned rollback in under 15 minutes; a user signs in once through a federated IdP and lands in a legal entity; every API call is authorized by OPA against AX-style roles/duties/privileges and fenced by Postgres row-level security; an entity declared in YAML becomes a table, an API, a client and a form without hand-written code; every state change is a CloudEvent on Kafka via a transactional outbox; approvals and batch jobs run on Temporal; number sequences support gap-free mode; every field change is in an immutable, hash-chained audit log; and a click in the shell can be followed through gateway, service, database and Kafka consumer in one trace. This layer is what AX calls the AOT, SysSecurity, batch and workflow frameworks, number sequences and SysDatabaseLog, and what SAP calls DDIC, PFCG, SM36/SM37, Business Workflow, SNRO and CDHDR/CDPOS — rebuilt as separately deployable services on the ADR-0001 stack.

Phase-exit sign-off criteria

- `make deploy-local` on a clean laptop produces a working k3d environment with all Phase 0 services in under 10 minutes; the same Helm/Argo manifests are healthy in the cloud `stage` cluster.
- Stakeholder demo executes, live, the wave-5 success metric from PRD §10: deploy a change → log in via a federated IdP → extend the Vendor-placeholder entity with a field and an enum value → run a 3-level approval → open the audit entry → open the trace in Tempo.
- Table-driven authorization suite (≥ 20 duties × allow/deny) and RLS cross-tenant suite are green; effective-permission report equals OPA decision for 100 sampled resources.
- Golden-generation suite for 10 reference entities is deterministic across 3 consecutive CI runs; additive v2 change passes, breaking change fails.
- Outbox chaos test shows zero lost and zero duplicated events over 10 kill cycles; 1M-row batch resumes after pod kill with 0 duplicates.
- Gap-free number sequence Jepsen-style test (10k concurrent requests) shows 0 gaps, 0 duplicates.
- 100% of generated forms pass axe WCAG 2.1 AA; audit entry visible within 2 s (p95); every service exports OTel traces, metrics and logs.
- SLO dashboards exist for every Phase 0 service with alert routes to the PagerDuty-compatible webhook.

Cross-wave data model summary

Entity	Owning service	Key relationships
Release, Environment	platform (wave 1, GitOps repo + registry DB)	Release → Environment (promotion history)
Tenant, LegalEntity	identity (wave 2; regenerated from metadata in wave 3)	Tenant 1→N LegalEntity; every row in every domain carries tenant_id
User, Role, Duty, Privilege, RoleAssignment	identity / authz (wave 2)	User N↔N Role via RoleAssignment scoped to LegalEntity; Role 1→N Duty 1→N Privilege
EntityDefinition, FieldDefinition, EnumDefinition, TypeDefinition, ExtensionPackage	metadata-registry (wave 3)	Entity 1→N Field; Field → Type or Enum; ExtensionPackage → Entity (additive)
OutboxEvent, TopicDefinition, SchemaVersion	every service / events (wave 4)	Outbox row → CloudEvent; TopicDefinition → SchemaVersion (from EntityDefinition)
WorkflowDefinition, WorkflowInstance, WorkItem, BatchJob, BatchRun, BatchTask	workflow (wave 4)	Instance → Definition version; WorkItem → Instance + assignee User; BatchRun 1→N BatchTask
NumberSequence, NumberSequenceAllocation	numseq (wave 5)	Sequence scoped to Tenant/LegalEntity; Allocation → Sequence (gap-free audit)
AuditEntry, AuditField	audit (wave 5)	Entry → any entity/record; hash chain per tenant
SavedView, Personalization, ImportJob	shell-bff / importexport (wave 5)	SavedView → EntityDefinition + User; ImportJob → EntityDefinition

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.identity.user.created.v1 / .deactivated.v1	identity	audit, workflow (assignee cache), shell	user_id, tenant_id, email, status
dev.passage.authz.role-assignment.granted.v1 / .revoked.v1	authz	audit, OPA bundle builder	user_id, role_id, legal_entity_ids, effective_from/to
dev.passage.authz.policy.compiled.v1	authz	OPA sidecars (bundle pull), audit	bundle_version, sha256, role_count
dev.passage.metadata.entity.published.v1	metadata-registry	generator, events (topic/schema), shell (form cache)	entity_name, version, diff summary
dev.passage.metadata.extension.applied.v1	metadata-registry	generator, audit, shell	package_id, entity_name, fields_added, enum_values_added
dev.passage.workflow.instance.submitted.v1 / .completed.v1	workflow	owning entity service, audit, shell inbox	instance_id, entity, record_id, outcome
dev.passage.workflow.workitem.assigned.v1 / .escalated.v1	workflow	shell inbox, notifications	workitem_id, assignee_id, due_at
dev.passage.batch.run.completed.v1 / .failed.v1	workflow (batch)	shell job monitor, alerting	job_id, run_id, rows_processed, error
dev.passage.audit.entry.recorded.v1	audit	shell audit viewer (live), analytics (wave 46)	entry_id, entity, record_id, operation, hash

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Metadata kernel under-specified before business waves need it	Medium	High	10 reference entities in wave 3 include the shapes of Party, Product, JournalLine; golden tests frozen at phase exit	Product
Two-person team cannot operate Kafka + Temporal + Keycloak + CNPG	Medium	High	All operators installed via Helm umbrella; runbooks per component; k3d parity so every incident reproduces locally	Engineering
OPA policy compile from roles produces unexpected denies	Medium	Medium	Effective-permission viewer and `decision` debug endpoint; deny-reason strings in every 403	Engineering
RLS misconfiguration leaks tenant data	Low	Critical	`FORCE ROW LEVEL SECURITY` on every table; app role NOBYPASSRLS; CI test queries every table cross-tenant	Security
Gap-free sequence becomes a throughput bottleneck	Medium	Medium	Continuous mode only where fiscal law requires it; non-continuous is the default with block pre-allocation	Engineering
Audit volume from CDC overwhelms Postgres	Medium	Medium	Monthly partitions; audit flag per field; PII masking at capture; 10-year retention tiered to object storage from wave 46	Engineering

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0.1 Wave 1: Delivery Pipeline & Cluster Baseline

0.1.1 Scope and business context

AX ships code as LCS deployable packages built on a build VM and applied with AxUpdateInstaller; model store exports (`AxUtil export`) and database restores are the rollback. SAP moves ABAP through STMS transport requests (SE09/SE10 workbench requests, DEV → QAS → PRD routes, `tp import`), with back-out by transport of copies. Both couple release to a vendor tool and give no software bill of materials, no cryptographic signature and no sub-hour rollback. Wave 1 replaces this with a GitOps pipeline on any conformant Kubernetes. At the end of the wave an engineer can merge a pull request and watch the change reach `dev`, promote the same image digest to `stage`, and roll back to the previous digest, each in under 15 minutes. Out of scope: multi-region, production-grade HA Postgres, secrets rotation policy, cost reporting, and any business service other than the `hello-ledger` placeholder.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Applying an LCS deployable package to a D365 sandbox takes 2–4 h of downtime and a service request for production; a failed apply means a database restore	Package = whole-system binary; environment managed by vendor tooling	Per-service OCI image + Helm chart, promoted by digest through Argo CD; rolling update with no downtime	Merge → running in env: 4 h → < 15 min; downtime per release: hours → 0
STMS transport import to PRD (STMS_IMPORT) is a manual queue with no automatic back-out; a bad transport is reversed by hand-building a transport of copies	Transports carry object deltas, not immutable artifacts	Immutable digests; rollback is a one-line overlay revert	Rollback: 1–3 h and a Basis ticket → < 5 min, self-service
No SBOM or signature for an AX model or SAP transport; security teams cannot answer "which CVE is in production"	Build tooling predates supply-chain attestation	Syft SBOM, Grype/Trivy gate, cosign signature verified at admission	CVE-to-affected-environment lookup: days of manual inventory → one query; unsigned artifacts in prod: 0
OneBox/VHD developer environments (AX) or NetWeaver dev editions (SAP) take days to provision and drift from production	Monolithic app server image per developer	`make deploy-local` on k3d with the same charts as cloud	New engineer to a running environment: 2–5 days → < 10 min
Customers cannot host on their own infrastructure without LCS (D365) or RISE (S/4)	Vendor-managed cloud is the only supported topology	One OpenTofu interface across k3d/EKS/GKE/AKS/bare-metal; nothing provider-specific in the critical path	Supported deployment targets: 1 → 5 with identical manifests

0.1.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Package a release	LCS deployable package, `AxUtil export` model store, AXModel files	STMS transport request (E070/E071), `tp export`	OCI image (digest) + Helm chart version in `deploy/`	improved
Build and validate	Build VM, `AxBuild.exe`, Best Practice checks	SE80 activation, ATC (code inspector SCI), ABAP unit	GitHub Actions: lint, unit, container build, SBOM, scan	improved
Environment tiers	LCS dev/test/UAT/prod topology	STMS transport route DEV→QAS→PRD, STMS_QA	Argo CD app-of-apps per environment: dev, stage, prod	full
Promote between tiers	LCS "apply package" to the next environment	STMS import queue, `tp import`, STMS_IMPORT	`passage promote --to stage --digest sha256:…` opens a PR against the env overlay	full
Rollback	Restore DB + model store, "AOT revert"	Transport of copies back-out, SNOTE de-implement	Argo CD rollback to prior digest, `passage rollback --env stage`	improved
Code signing / trust	Strong-name model signing (`AxUtil … /key`)	Digitally signed SAP Notes, TMS trusted systems	cosign keyless signatures, policy-controller admission denies unsigned images	improved
Hotfix path	KB hotfix via LCS, binary update	SNOTE note implementation, SPAU	Hotfix branch → cherry-pick → same pipeline, same gates	full
Deployment log / history	LCS environment history, AxUpdateInstaller logs	STMS import log, SLOG/ALOG	Argo CD history + GitHub Deployments API + `platform.release` table	improved
Local dev environment	OneBox VM / VHD from LCS	SAP NetWeaver developer edition, mini-basis	`make deploy-local` on k3d, same charts as cloud	improved

0.1.3 Functional requirements

- **FR-1.1 Monorepo layout** — Must. The system shall keep all Passage source in one repository with the layout `services/go/<svc>`, `services/ts/<svc>`, `services/py/<svc>`, `web/shell`, `web/remotes/<name>`, `deploy/charts/<svc>`, `deploy/envs/{dev,stage,prod}`, `infra/modules/<provider>`, `metadata/entities`, `docs/`.

    - Notes: each service owns a `Dockerfile`, `Makefile`, `CODEOWNERS` entry and a Helm chart; CI path filters build only changed services; a `go.work` and `pnpm` workspace root pin toolchain versions (Go 1.23, Node 22, Python 3.12).
- **FR-1.2 Cluster bootstrap module** — Must. The system shall provide an OpenTofu module `infra/modules/cluster` with one variable interface (`provider ∈ {k3d, eks, gke, aks, baremetal}`, `name`, `node_pools[]`, `kubernetes_version`) that outputs a kubeconfig and installs the base add-ons.
    - Notes: base add-ons are Argo CD, Envoy Gateway, cert-manager, External Secrets Operator, CloudNativePG operator, sigstore policy-controller, metrics-server; provider modules contain only networking, node pools and storage classes; k3d module runs with 3 agent nodes and 8 GB RAM.
- **FR-1.3 Argo CD app-of-apps** — Must. The system shall define a root Application per environment that reconciles child Applications from `deploy/envs/<env>/apps/*.yaml`, each pinning a chart version and an image digest.
    - Notes: sync policy automated for dev, manual-approve PR for stage and prod; prune and self-heal enabled; sync waves order operators (–2), databases (–1), services (0), gateway routes (1).
- **FR-1.4 Digest-only promotion** — Must. The system shall promote images by immutable digest; tags are informational only.
    - Notes: `passage promote` rewrites `image.digest` in the target overlay and opens a PR with the SBOM diff and scan summary; promotion to prod requires the digest to have run in stage for ≥ 60 minutes with no SLO burn alert.
- **FR-1.5 CI pipeline** — Must. The system shall run on every PR and on `main`: lint (golangci-lint, eslint, ruff), unit tests with coverage ≥ 70% per service, container build (BuildKit, reproducible), Syft SBOM (SPDX 2.3), Grype and Trivy scan, cosign sign and attest.
    - Notes: required checks block merge; `main` builds push to the registry under `ghcr.io/passage/<svc>@sha256:…`; median PR pipeline ≤ 8 minutes; cache keyed by lockfile hash.
- **FR-1.6 Vulnerability gate** — Must. The system shall fail the build on any CRITICAL finding, or any HIGH finding older than 14 days with a fix available, unless an exception file `.vuln-exceptions.yaml` lists the CVE with an expiry ≤ 30 days.
- **FR-1.7 Signature verification admission** — Must. The system shall reject any pod whose image lacks a valid cosign signature from the repository identity (`https://github.com/passage/*` OIDC issuer) in namespaces labelled `passage.dev/verified=true`.
    - Notes: SLSA provenance attestation level 2 at wave 1, level 3 target by wave 48; a negative test deploys an unsigned image and asserts the admission denial message.
- **FR-1.8 `hello-ledger` placeholder** — Must. The system shall deploy a Go service with one Postgres table that proves the path: migration on start, REST endpoints, health probes, TLS ingress, metrics endpoint.
- **FR-1.9 Local parity** — Must. `make deploy-local` shall stand up k3d, Argo CD and `hello-ledger` behind `https://app.local.passage.dev` with a locally trusted cert-manager CA in under 10 minutes on a 16 GB laptop.
- **FR-1.10 Rollback** — Must. The system shall roll back any environment to the previous release by digest in ≤ 5 minutes via `passage rollback --env <env>` (which reverts the overlay commit) or the Argo CD history UI.
    - Notes: database migrations in Phase 0 are additive-only so rollback never requires a down migration; a rollback records a `Release` row with `kind=rollback`.
- **FR-1.11 Secrets and config** — Must. The system shall source all secrets from Vault through External Secrets Operator; no secret value appears in Git, Helm values or CI logs.
    - Notes: config via ConfigMap generated from `deploy/envs/<env>/values.yaml`; secret names follow `passage/<env>/<svc>/<key>`.

0.1.4 Data model

Release (`platform.release`, registry DB, also materialized in Git history)

Field	Type	Required	Notes
id	uuid	yes	PK
service	text	yes	matches chart name
image_digest	text	yes	`sha256:` + 64 hex; unique with service
git_sha	char(40)	yes	commit that produced the build
chart_version	text	yes	semver
sbom_url	text	yes	SPDX artifact in registry
signed	bool	yes	cosign verification result
status	enum	yes	built, scanned, signed, deployed_dev, deployed_stage, deployed_prod, superseded, rolled_back
created_at	timestamptz	yes	

Environment (`platform.environment`)

Field	Type	Required	Notes
name	text	yes	PK; dev, stage, prod, local-
cluster_provider	enum	yes	k3d, eks, gke, aks, baremetal
argo_root_app	text	yes	app-of-apps name
promotion_policy	enum	yes	auto, pr_approval
soak_minutes	int	yes	default 0 dev, 60 stage→prod
base_url	text	yes	`https://app.<env>.passage.dev`

HelloLedgerEntry (`hello_ledger.entry`)

Field	Type	Required	Notes
id	uuid	yes	PK
memo	varchar(200)	yes	
amount_minor	bigint	yes	integer minor units, USD/EUR examples: 12345 = $123.45

Field	Type	Required	Notes
currency	char(3)	yes	ISO 4217
created_at	timestamptz	yes	default now()
created_by	text	yes	`anonymous` until wave 2

Invariants: one `Release` row per (service, image_digest); at most one `deployed_<env>` release per service per environment; `Environment.name` is immutable. Indexing: `release(service, created_at desc)`; `entry(created_at)`.

0.1.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/healthz, /readyz	Liveness / readiness	n/a	none
GET	/v1/version	git sha, digest, chart version	n/a	none
GET	/v1/hello-ledger/entries?limit=&cursor=	List entries	n/a	none (token required from wave 2)
POST	/v1/hello-ledger/entries	Create entry	`Idempotency-Key` header, 24 h window	Maintain hello ledger
GET	/v1/hello-ledger/entries/{id}	Fetch one	n/a	none
GET	/v1/platform/releases?service=&env=	Release history	n/a	View platform releases
POST	/v1/platform/releases/{id}/promote	Open promotion PR	release id + env	Maintain platform releases

CLI commands `passage promote`, `passage rollback`, `passage status` call these endpoints or edit the overlay directly.

Event	Trigger	Payload fields	Consumers
dev.passage.platform.release.built.v1	CI publishes signed image	release_id, service, image_digest, git_sha, sbom_url	registry DB writer
dev.passage.platform.release.promoted.v1	Argo sync Healthy in target env	release_id, environment, actor, previous_digest	audit (wave 5), shell notifications
dev.passage.platform.release.rolledback.v1	Rollback sync Healthy	release_id, environment, actor, reason	audit, alerting
dev.passage.hello.entry.created.v1	Entry insert	entry_id, amount_minor, currency	none (proves outbox path in wave 4)

In wave 1 events are written as CloudEvents JSON to stdout and to the registry DB; Kafka transport replaces the sink in wave 4 without changing types.

0.1.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Argo CD UI (per env)	Sync, view history, rollback	app, health, sync status, revision, digest	Platform engineer (Argo RBAC via GitHub team)
GitHub PR checks	View gate results, SBOM diff	lint, unit, build, scan, sign	repository write
hello-ledger page `/`	Create entry, list entries	memo, amount, currency, created_at	none

Interaction rules: promotion PRs are templated with a checklist (soak time met, scan clean, rollback target named); CI annotates failing lines inline; `make deploy-local` prints a single URL table at the end.

0.1.7 Business rules and edge cases

- **BR-1.1** Release status machine: built → scanned → signed → deployed_dev → deployed_stage → deployed_prod → superseded; any deployed_* may go to rolled_back; no transition skips `signed`.
- **BR-1.2** A tag may be rewritten; a digest may not. Any overlay referencing a tag fails the `deploy-lint` CI check.
- **BR-1.3** Promotion to prod requires soak ≥ `Environment.soak_minutes` in stage with zero `SLOBurnRateHigh` alerts; the check is computed in UTC with ISO 8601 timestamps.
- **BR-1.4** Two concurrent promotions of the same service to the same environment: the second PR is auto-closed with a comment pointing at the first.
- **BR-1.5** A vulnerability exception past its expiry date fails the build as if no exception existed.
- **BR-1.6** Rollback target is always the last `deployed_<env>` release with a different digest; if none exists the command refuses.
- **BR-1.7** Images are retained for 180 days or while referenced by any environment, whichever is longer; SBOMs retained 7 years.
- **BR-1.8** A hotfix branch must be cut from the prod digest's `git_sha`; CI refuses `hotfix/*` branches whose merge base is not a deployed_prod release.

0.1.8 Acceptance criteria

- **AC-1.1** Given a PR with an injected lint error, When CI runs, Then the `lint` check fails and merge is blocked. (FR-1.5)
- **AC-1.2** Given a PR with a failing unit test, When CI runs, Then `unit` fails and the failing assertion is annotated inline. (FR-1.5)
- **AC-1.3** Given a clean macOS or Linux laptop with Docker, When `make deploy-local` runs, Then Argo CD and hello-ledger are reachable over TLS within 10 minutes. (FR-1.9)
- **AC-1.4** Given a signed image in dev, When `passage promote --to stage` is merged, Then Argo shows Healthy with that digest within 15 minutes of merge. (FR-1.3, FR-1.4)
- **AC-1.5** Given stage running digest B after A, When `passage rollback --env stage` runs, Then Argo history shows A active within 5 minutes and a `rolled_back` Release row exists. (FR-1.10)
- **AC-1.6** Given an unsigned image applied to a verified namespace, When the pod is created, Then admission denies it with a message naming the missing signature. (FR-1.7)
- **AC-1.7** Given an image with one CRITICAL CVE, When CI runs, Then the scan gate fails; adding a valid exception with expiry in 10 days passes it. (FR-1.6)

- **AC-1.8** Given a secret in Vault, When hello-ledger starts, Then it reads `DATABASE_URL` via ESO and the value is absent from Git, values files and CI logs (grep check). (FR-1.11)

0.1.9 Test plan and fixtures

Fixtures: `fixtures/pipeline/negative-lint.go` (committed lint failure behind a build tag), `fixtures/pipeline/unsigned-image.yaml`, `fixtures/pipeline/cve-exception.yaml` (expired and valid), `fixtures/hello-ledger/entries-100.json` (100 entries in USD and EUR).

Test	Type	Target	Pass criterion
Lint/unit negative	CI contract	pipeline	both injected failures block merge
Local bootstrap	E2E	`make deploy-local`	< 10 min, all pods Ready, TLS valid
Promote and rollback	E2E	Argo CD	both < 15 min, history correct
Admission unsigned	Negative	policy-controller	pod denied
SBOM presence	Contract	registry	SPDX attached to every digest
hello-ledger API	Unit + contract	handlers	100% of OpenAPI paths covered
Build reproducibility	Property	BuildKit	same source → same digest twice

Reconciliation: `Release` rows with status `deployed_<env>` equal the digests in each env overlay and the running pods' image IDs (nightly check).

0.1.10 Deployment and operations

Charts: `deploy/charts/platform-base` (umbrella for operators), `deploy/charts/hello-ledger`. Config: `values.yaml` per env; env vars `PASSAGE_ENV`, `OTEL_EXPORTER_OTLP_ENDPOINT` (used from wave 5), `DATABASE_URL` (ESO). Migrations: Atlas, additive-only, run as an init job. Feature flags: none. Scheduled jobs: nightly release reconciliation. SLOs: pipeline p50 ≤ 8 min, promotion merge→Healthy p95 ≤ 15 min, hello-ledger availability 99.9%. Alerts: `ArgoAppDegraded` > 10 min, `PipelineFailureRate` > 20% over 1 h, `AdmissionDeniedSpike`. Runbook entries: Argo out of sync, failed migration job, expired local CA. Rollback: `passage rollback`, then verify `/v1/version`.

0.1.11 Migration from AX / SAP

Nothing is migrated. LCS packages, AX model store layers and SAP transport requests contain code (X++/ABAP), which PRD §2 excludes. The only carry-over is organizational: the STMS DEV→QAS→PRD route maps to dev→stage→prod and LCS environment names are recorded in `Environment.description` for operator familiarity.

0.1.12 Dependencies, risks, open questions

Depends on: nothing (first wave).

Risk	Likelihood	Impact	Mitigation
k3d cannot run the full stack in 16 GB by wave 5	Medium	Medium	Profiles: `make deploy-local PROFILE=min` omits observability
Keyless cosign requires outbound access to Fulcio/Rekor	Medium	Low	Private sigstore instance documented for air-gapped installs
GitHub Actions minutes cost	Low	Low	Self-hosted runner module in `infra/`

Open questions: (1) First cloud provider for `dev/stage` (PRD §12 Q4) — default EKS; decide by wave 2 start. (2) Private registry vs ghcr.io for customers — default ghcr.io with mirror instructions; decide by wave 5. (3) Mesh (none vs Linkerd) for mTLS — default none, gateway-to-service mTLS only; decide by wave 4.

0.2 Wave 2: Identity, Tenancy & Authorization

0.2.1 Scope and business context

AX 2012 authorizes through `SecurityRole` → `SecurityDuty` → `SecurityPrivilege` → `SecurityEntryPoint` with access levels (NoAccess, Read, Update, Create, Correct, Delete), table permissions in the AOT, extensible data security policies (XDS) and `UserInfo` rows tied to Active Directory; companies are isolated by `DataAreaId` on every per-company table. SAP uses `SU01` user masters (USR02), `PFCG` roles (AGR_DEFINE, AGR_1251 field values, AGR_USERS), authorization objects such as `S_TCODE`, `F_BKPF_BUK` (company code) and `M_MATE_WRK` (plant), `SU24` proposals, `SU53` failed-check traces, and `MANDT` for client isolation. Wave 2 gives Passage one identity provider (Keycloak) federated to Entra ID, Okta or Google, SCIM provisioning, an AX-shaped role model compiled to Rego, OPA as a sidecar on every service, Envoy Gateway JWT validation with per-tenant rate limits, and Postgres RLS on every table. At the end of the wave a user logs in once, lands in a legal entity, and every API call is allowed or denied by policy, not code. Out of scope: segregation-of-duties conflict analysis (wave 48), position hierarchies (wave 42), record-level XDS-style data policies beyond tenant and legal entity (wave 15 adds dimension security), customer-facing portals (waves 20, 30).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Diagnosing "no authorization" in SAP needs SU53 on the user's session, then SUIM, then PFCG edit, role generation and user comparison (PFCG → Authorizations → Generate → User tab → User comparison): 6 screens and often a transport	Authorization data lives in profiles generated from roles; failures are only visible in the user's own buffer	Every 403 carries `duty_missing`; the effective-permission viewer shows the decision trace for any user × entity × action; assignment takes effect in ≤ 30 s without a transport	Access-fix cycle: 1–3 days (ticket + transport window) → < 5 min; screens: 6 → 1

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX security development: a new entry point needs a privilege, duty and role in the AOT, synchronization and an AOS restart; D365 needs a build and package	Security objects are AOT metadata compiled into the model store	Role/duty/privilege are data compiled to a Rego bundle on save	Role change to live: 1–2 h (AX) / 4 h (D365 package) → ≤ 5 s compile + 10 s propagation
Company isolation relies on every X++ query honoring `DataAreaId` (and `crossCompany` misuse) or every ABAP select carrying MANDT; leaks are code bugs	Isolation enforced in application code, not the database	Postgres RLS with `FORCE ROW LEVEL SECURITY`; the app role cannot bypass it	Code paths that can leak a tenant: every query → 0; cross-tenant CI test on 100% of tables
Joiners/leavers: SU10 mass maintenance or D365 user import run manually; leavers keep access for days	No standard provisioning protocol in ECC; D365 ties to Entra only	SCIM 2.0 server with deprovision revoking sessions in ≤ 60 s	Deprovision latency: days → < 60 s; manual user admin steps per joiner: ~10 → 0
MFA and federation to Okta/Google are unsupported on AX 2012; S/4 needs SAP IAS/SNC projects	IdP coupling to AD/Entra or SAP's own stack	Keycloak brokering any OIDC/SAML IdP with TOTP/WebAuthn	Federation to a new IdP: a multi-week project → < 1 h of configuration

0.2.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
User master	`UserInfo`, SysUserInfoPage form, AD import wizard	SU01, USR02/USR21, SU10 mass maintenance	identity `User` entity, Admin UI Users, SCIM /Users	full
Role	`SecurityRole`, SysSecRoles form	PFCG single/composite/derived roles, AGR_DEFINE	authz `Role` (+ `derived_from` for org-level variants)	full
Duty / privilege	`SecurityDuty`, `SecurityPrivilege`, `SecurityRoleTaskGrant`, `SecurityTaskEntryPoint`	Authorization objects/fields (TOBJ, AGR_1251), SU24 defaults	`Duty`, `Privilege`, `Permission(entity, action, scope)`	full
Access levels	NoAccess/Read/Update/Create/Correct/Delete on entry points and tables	ACTVT 01/02/03/06 etc. per object	action enum: read, create, update, correct, delete, post, approve	full
Company isolation	`DataAreaId`, `CompanyInfo`, `changecompany()`	MANDT client, BUKRS org level in roles	tenant_id RLS + legal_entity_id scope on RoleAssignment	improved
Org-scoped roles	`OMUserRoleOrganization` (role valid for selected legal entities)	Derived roles with org-level values (BUKRS, WERKS)	`RoleAssignment.legal_entity_ids[]` or `all`	full
Effective permissions	Security development tool, "View permissions"	SUIM, SU56 user buffer, SU53 trace	`GET /v1/authz/effective-permissions`, decision debug endpoint	improved
Identity federation / MFA	Entra ID (AAD) only on D365; AD on AX 2012	SAML via Identity Authentication Service, SNC	Keycloak OIDC/SAML brokering, TOTP/WebAuthn MFA	improved
Provisioning	AD sync, D365 user import	SAP IPS SCIM, CUA	SCIM 2.0 server (RFC 7644)	full

0.2.3 Functional requirements

- **FR-2.1 Keycloak as IdP** — Must. The system shall deploy Keycloak with one realm per tenant, brokering OIDC and SAML 2.0 to external IdPs, issuing RS256 JWTs with claims `sub`, `tid` (tenant), `email`, `preferred_username`, `pv` (policy bundle version), `amr`.
  - Notes: access token TTL 10 minutes, refresh 8 hours, absolute session 12 hours; MFA (TOTP or WebAuthn) required for roles flagged `mfa_required`; realm export committed as Helm values for reproducibility.
- **FR-2.2 SCIM provisioning** — Must. The system shall expose `/scim/v2/Users` and `/scim/v2/Groups` per tenant with filter, PATCH and bulk (≤ 1,000 operations) per RFC 7644; group membership maps to role assignment via a configurable mapping table.
  - Notes: deprovision sets `status=deprovisioned`, revokes sessions within 60 s, never deletes the row (audit needs it).
- **FR-2.3 Tenant and legal entity model** — Must. The system shall model `Tenant` → `LegalEntity` → `Site` (shell only; detailed in wave 6) and stamp `tenant_id` on every table in every domain database.
  - Notes: `legal_entity_id` is stamped on per-company tables (AX `SaveDataPerCompany=Yes` equivalent, declared in metadata in wave 3); a user's current legal entity is a session attribute, not a token claim, so switching does not require re-login.
- **FR-2.4 Row-level security** — Must. Every table shall have `ENABLE ROW LEVEL SECURITY` and `FORCE ROW LEVEL SECURITY` with policy `tenant_id = current_setting('app.tenant_id')::uuid`; services connect as role `passage_app` with `NOBYPASSRLS` and set the variable with `SET LOCAL` inside every transaction.
  - Notes: a missing `app.tenant_id` yields zero rows, not an error; migrations run as `passage_migrate`, which is also NOBYPASSRLS; a CI check fails if any table lacks the policy.
- **FR-2.5 Role/duty/privilege model** — Must. The system shall store Role → Duty → Privilege → Permission where Permission = (entity, action, field_mask?, condition?), and compile the closure into one Rego bundle per tenant.
  - Notes: actions are read, create, update, correct, delete, post, approve, export; `correct` retains AX semantics (edit date-effective rows); 20 seeded duties ship (Maintain users, Maintain security roles, Maintain vendors, View vendors, Maintain entity metadata, Apply extensions, Maintain workflows, Approve workflow, Maintain batch jobs, View batch history, Maintain number sequences, View audit log, Import data, Export data, Maintain platform releases, View platform releases, Maintain tenants, Maintain legal entities, Maintain saved views, Maintain hello ledger).
- **FR-2.6 Policy compilation** — Must. The system shall compile a bundle within 5 s of any role/duty/privilege/assignment change and distribute it to OPA sidecars via bundle polling (interval 10 s) with the version in `pv`.
  - Notes: a token whose `pv` is older than the active bundle by more than 2 versions is still honored (bundles are backward compatible); deny reasons are returned as `{"code":"AUTHZ_DENIED","duty_missing":"Maintain vendors"}`.
- **FR-2.7 OPA decision point** — Must. Every service shall call its OPA sidecar with `input = {subject, resource, action, context}` and enforce the decision before any database call; p95 decision latency ≤ 2 ms, p99 ≤ 5 ms.
  - Notes: `resource` carries entity, id, tenant_id, legal_entity_id and the field list for field-level masks; the gateway performs coarse checks (token validity, tenant match), services perform fine checks.

- **FR-2.8 Gateway** — Must. Envoy Gateway shall validate JWT signature, issuer and audience against the tenant realm's JWKS (cached 5 min), reject tokens with `tid` not matching the host's tenant, and enforce per-tenant rate limits (default 1,000 rps sustained, 2,000 burst; 429 with `Retry-After`).
- **FR-2.9 Role assignment with scope and effective dates** — Must. The system shall assign roles to users with `legal_entity_ids` (or `all`) and `effective_from/effective_to` (ISO 8601 dates, tenant time zone, inclusive).
  - Notes: expired assignments are excluded at compile time; a nightly job recompiles bundles for tenants with assignments crossing midnight.
- **FR-2.10 Admin UI** — Must. The system shall provide screens for users, roles, duties, privileges, assignments and an effective-permission viewer that renders the OPA decision for a chosen user × entity × action × legal entity.
- **FR-2.11 Service accounts and agents** — Should. The system shall support non-human principals (`kind=service|agent`) with client-credentials tokens and role assignments; an agent principal may hold only roles that are a subset of its sponsoring user's roles.

0.2.4 Data model

User (`identity.user`)

Field	Type	Required	Notes
id	uuid	yes	PK; equals Keycloak `sub`
tenant_id	uuid	yes	FK tenant; RLS
username	varchar(100)	yes	unique per tenant, case-insensitive
email	varchar(254)	yes	unique per tenant; PII
display_name	varchar(200)	yes	PII
kind	enum	yes	human, service, agent
status	enum	yes	invited, active, locked, disabled, deprovisioned
default_legal_entity_id	uuid	no	FK legal_entity
external_id	varchar(200)	no	SCIM externalId
mfa_enrolled	bool	yes	default false
last_login_at	timestamptz	no	
created_at, updated_at, version	timestamptz, timestamptz, int	yes	optimistic lock

Role / Duty / Privilege / Permission (`authz.*`)

Field	Type	Required	Notes
id	uuid	yes	PK (each table)
tenant_id	uuid	yes	`sys` rows have the reserved system tenant and are read-only
code	varchar(64)	yes	unique per tenant; e.g. `VendMaintain`
name, description	varchar(120), text	yes/no	localized labels in wave 48
layer	enum	yes	sys, partner, customer
parent_id	uuid	no	Role→Duty, Duty→Privilege via link tables `role_duty`, `duty_privilege`, `privilege_permission`
entity, action	varchar(64), enum	yes (Permission)	action ∈ read, create, update, correct, delete, post, approve, export
field_mask	text[]	no (Permission)	fields hidden when present
condition	jsonb	no (Permission)	Rego-compatible expression, e.g. `{"legal_entity_in":"$subject.legal_entities"}`
mfa_required	bool	yes (Role)	

RoleAssignment (`authz.role_assignment`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
user_id	uuid	yes	FK user
role_id	uuid	yes	FK role
legal_entity_ids	uuid[]	yes	empty array means all
effective_from	date	yes	default today
effective_to	date	no	null = open-ended
granted_by	uuid	yes	FK user
reason	text	no	required when `role.break_glass=true`

Tenant / **LegalEntity**: id, code (varchar 10, unique), name, status (active, suspended), default_currency (char 3), time_zone (IANA), created_at; LegalEntity adds tenant_id and `country` (ISO 3166-1 alpha-2).

Invariants: a user has at most one active assignment per (role, legal entity set) overlapping in time; `sys` layer rows are immutable in customer tenants; every Permission entity name must exist in the metadata registry from wave 3. Indexes: `user(tenant_id, lower(email))` unique; `role_assignment(tenant_id, user_id)`; `role_assignment(tenant_id, effective_to)` for expiry sweeps. Keycloak keeps its own Postgres (CNPG) per environment.

0.2.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/POST	/v1/identity/users	List / create users	POST: Idempotency-Key	Maintain users
PATCH	/v1/identity/users/{id}	Update status, default legal entity	version check (If-Match)	Maintain users
GET/POST/PUT	/scim/v2/Users, /scim/v2/Groups	SCIM provisioning	SCIM semantics	SCIM client credential
GET/POST/PUT	/v1/authz/roles, /duties, /privileges	Maintain the model	PUT by code	Maintain security roles
GET/POST/DELETE	/v1/authz/assignments	Grant / revoke	POST: natural key	Maintain security roles
GET	/v1/authz/effective-permissions?user_id=&legal_entity_id=	Flattened permissions	n/a	Maintain security roles or self
POST	/v1/authz/decision	Dry-run an OPA decision (debug)	n/a	Maintain security roles
POST	/v1/authz/policies/compile	Force recompile	bundle version	Maintain security roles
GET/POST	/v1/tenants, /v1/legal-entities	Tenancy shells	POST: code	Maintain tenants / Maintain legal entities
POST	/v1/session/legal-entity	Switch current legal entity	n/a	any active user

Commands validate: email RFC 5322; role code `^[A-Za-z][A-Za-z0-9]{2,63}$`; assignment dates `effective_from ≤ effective_to`; a revoke of a `sys`-layer role from the last `Tenant owner` is rejected (409).

Event	Trigger	Payload fields	Consumers
dev.passage.identity.user.created.v1	User insert (UI, SCIM, federation first login)	user_id, tenant_id, email, kind, source	workflow, audit, shell
dev.passage.identity.user.status-changed.v1	status transition	user_id, from, to, actor, reason	audit, session revoker
dev.passage.authz.role-assignment.granted.v1 / revoked.v1	assignment insert/delete	assignment_id, user_id, role_id, legal_entity_ids, effective_from, effective_to	policy compiler, audit
dev.passage.authz.policy.compiled.v1	bundle built	tenant_id, bundle_version, sha256	OPA sidecars, audit
dev.passage.identity.login.succeeded.v1 / failed.v1	Keycloak event listener	user_id, ip, idp, amr	audit, anomaly alert

0.2.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Users list/detail	Invite, lock, disable, assign roles, set default legal entity	username, email, status, kind, last login, MFA	Maintain users
Roles	Create, add duties, mark MFA/break-glass	code, name, layer, duty count, user count	Maintain security roles
Duties / Privileges	Create, add privileges / permissions	code, entity, action, field mask	Maintain security roles
Role assignment	Grant with legal entity picker and dates	user, role, legal entities, from, to, reason	Maintain security roles
Effective permissions	Pick user × legal entity, inspect decision trace	entity, action, allow/deny, via role → duty → privilege	Maintain security roles
Legal entity switcher (shell header)	Switch current legal entity	code, name	any

Rules: all grids are keyboard navigable (wave 5 shell completes this); role edits are saved explicitly, not on blur; a decision trace shows the first allowing path and all denying reasons; the switcher remembers the last legal entity per user in `Personalization` (wave 5).

0.2.7 Business rules and edge cases

- **BR-2.1** User status machine: invited → active; active ↔ locked (5 failed logins in 15 min locks for 30 min, admin may unlock); active → disabled → active; any → deprovisioned (terminal).
- **BR-2.2** Deny by default: no matching Permission means deny; explicit deny permissions do not exist (AX semantics), so removing access means removing the grant.
- **BR-2.3** Access level ordering for merge: delete > correct > update > create > read; the highest level across all roles wins per entity (AX `SecurityEntryPoint` union behavior).
- **BR-2.4** Legal-entity scoping: a permission is effective only in legal entities listed on the assignment; `all` includes legal entities created later.
- **BR-2.5** Effective dates are evaluated in the tenant time zone at compile time and at 00:05 nightly; a grant with `effective_from` in the future compiles into the bundle inactive.
- **BR-2.6** Revoking a session: status change to locked/disabled/deprovisioned revokes refresh tokens; access tokens remain valid at most 10 minutes (TTL), which is the documented worst case.
- **BR-2.7** Cross-tenant request (token `tid` ≠ host tenant) returns 404 rather than 403 to avoid tenant enumeration.
- **BR-2.8** Rate limit is per tenant per gateway replica, aggregated through the gateway's shared rate-limit service; service accounts count against the same bucket.
- **BR-2.9** Concurrency: two admins editing the same role use optimistic locking (`version`); the loser gets 409 with the current version.
- **BR-2.10** Agent principals cannot be granted `Tenant owner`, `Maintain security roles` or break-glass roles.

0.2.8 Acceptance criteria

- **AC-2.1** Given a user with duty "Maintain vendors", When they `PUT /vendors/{id}` (stub entity), Then 200; Given a user without it, Then 403 with `duty_missing`. Table-driven across 20 duties. (FR-2.5, FR-2.7)
- **AC-2.2** Given tenant A's app session, When a query filters `tenant_id = <tenant B>`, Then zero rows are returned under `passage_app`. (FR-2.4)
- **AC-2.3** Given two Keycloak realms federated to a stand-in Keycloak IdP in CI, When a user logs in through each, Then both sessions carry the correct `tid` and land in the user's default legal entity. (FR-2.1, FR-2.3)
- **AC-2.4** Given 100 random (user, entity, action, legal entity) tuples, When effective-permission report and `POST /v1/authz/decision` are compared, Then they agree 100%. (FR-2.6, FR-2.10)
- **AC-2.5** Given a SCIM PATCH deactivating a user, When processed, Then `status=deprovisioned`, refresh tokens are revoked within 60 s and the row persists. (FR-2.2)

- **AC-2.6** Given an assignment with `effective_to` yesterday, When the nightly recompile runs, Then the user's requests for that role's permissions are denied. (FR-2.9)
- **AC-2.7** Given 1,500 rps from one tenant for 10 s, When routed through the gateway, Then requests above the limit receive 429 with `Retry-After`. (FR-2.8)
- **AC-2.8** Given a role flagged `mfa_required`, When a user without MFA logs in, Then Keycloak forces enrollment before issuing a token. (FR-2.1)
- **AC-2.9** Given an agent principal sponsored by user U, When an assignment outside U's roles is attempted, Then 422 with `AGENT_EXCEEDS_SPONSOR`. (FR-2.11)

0.2.9 Test plan and fixtures

Fixtures: `fixtures/authz/duties-20.yaml` (20 duties with allow/deny expectations, 240 rows), `fixtures/identity/users-50.json` (two tenants, three legal entities), `fixtures/identity/keycloak-realm-ci.json` (two realms + broker), `fixtures/authz/decision-sample-100.json`.

Test	Type	Target	Pass criterion
Duty matrix	Contract (table-driven)	gateway + OPA + stub service	240/240 expected codes
RLS cross-tenant	Integration	every table (generated list)	0 rows for foreign tenant; CI fails on any table without policy
Federation login	E2E (Playwright)	Keycloak broker	2 IdPs, correct claims
Effective vs decision	Property	compiler	100/100 agree
Bundle compile latency	Performance	compiler	≤ 5 s for 500 roles / 2,000 duties
OPA decision latency	Performance	sidecar	p95 ≤ 2 ms at 2,000 rps
SCIM conformance	Contract	SCIM server	RFC 7644 test suite subset (filter, PATCH, bulk)

Reconciliation: count of Keycloak users per realm equals `identity.user` rows with status ≠ deprovisioned; sum of permissions in the compiled bundle equals the flattened SQL closure.

0.2.10 Deployment and operations

Charts: `keycloak` (with CNPG database, 2 replicas), `identity`, `authz` (compiler + bundle server), `opa` as a sidecar template in `deploy/charts/_lib`, `envoy-gateway` policies (`SecurityPolicy` for JWT, `BackendTrafficPolicy` for rate limits). Config: `KEYCLOAK_ISSUER_TEMPLATE`, `OPA_BUNDLE_URL`, `AUTHZ_BUNDLE_POLL_SECONDS=10`. Migrations: additive Atlas. Feature flags: `authz.enforce` (log-only mode for the first week in dev). Scheduled jobs: nightly assignment expiry recompile 00:05 tenant time; JWKS refresh. SLOs: login success ≥ 99.9%, token issuance p95 ≤ 300 ms, OPA p95 ≤ 2 ms, bundle propagation ≤ 30 s. Alerts: `AuthzDenySpike` (> 5× baseline), `BundleStale` (> 5 min), `KeycloakDown`. Runbook: stale bundle, locked-out tenant owner (break-glass via Keycloak admin console), JWKS rotation. Rollback: digest rollback; bundles are versioned so a rollback of authz re-serves the prior bundle.

0.2.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `UserInfo` (id, name, networkAlias, enable)	User	`networkAlias` → username; `enable=0` → disabled
AX `SecurityUserRole` (user, securityRole, validFrom/validTo)	RoleAssignment	dates carried; `OMUserRoleOrganization` rows → `legal_entity_ids`
AX `SecurityRole` / `SecurityDuty` / `SecurityPrivilege` + `SecurityRoleTaskGrant`, `SecurityTaskEntryPoint`	Role / Duty / Privilege	custom (CUS/VAR layer) roles only; SYS roles are mapped by name to Passage seeds
SAP USR02 (BNAME, UFLAG, GLTGV/GLTGB)	User	UFLAG 64/128 → locked; GLTGV/GLTGB → status dates
SAP AGR_USERS (AGR_NAME, UNAME, FROM_DAT, TO_DAT)	RoleAssignment	dates carried
SAP AGR_DEFINE / AGR_1251 (OBJECT, FIELD, LOW/HIGH)	Role / Duty / Privilege	authorization objects map via a crosswalk file to Passage permissions; BUKRS values → legal_entity_ids

Transformations run through the wave 10 migration framework; this wave ships only the crosswalk format and a dry-run report. Not migrated: passwords (federated), SU24 proposals, AX entry-point grants on forms that have no Passage equivalent (reported as "unmapped" with counts).

0.2.12 Dependencies, risks, open questions

Depends on: wave 1 (cluster, Argo, Vault/ESO, gateway).

Risk	Likelihood	Impact	Mitigation
Keycloak realm-per-tenant does not scale past ~500 tenants	Low (v1)	Medium	Organizations feature evaluated by wave 48
Rego closure bundle size for large role models	Medium	Medium	Partial evaluation; bundle per tenant; limit 50k permissions
SAML IdPs with non-standard attribute names	Medium	Low	Mapper templates for Entra ID, Okta, ADFS

Open questions: (1) Field-level masks in P0 or deferred to wave 15 — default: masks compiled but enforced only for PII-flagged fields; decide by wave 3. (2) Session legal entity in a cookie vs header — default header `X-Passage-Legal-Entity`; decide by wave 5. (3) Tenant realm per tenant vs shared realm with groups — default per tenant; decide by wave 2 mid-point.

0.3 Wave 3: Metadata Kernel (Entities, Fields, Extensions)

0.3.1 Scope and business context

In AX every business object is an AOT artifact: a Table with properties (`TableGroup`, `SaveDataPerCompany`, `PrimaryIndex`, `ClusteredIndex`, `TitleField1/2`, `CacheLookup`), Fields typed by Extended Data Types and Base Enums, Relations with `DeleteActions`, Indexes with `AlternateKey`, and table methods (`validateField`, `validateWrite`, `validateDelete`, `initValue`, `insert`, `update`) in X++; D365 adds data entities for OData and the extension model that replaced overlayering (SYS/GLS/FPK/SLN/ISV/VAR/CUS/USR layers). In SAP the equivalents are DDIC objects: tables (DD02L), fields (DD03L), domains (DD01L), data elements (DD04L), foreign keys (DD05S/DD08L), search helps (DD30L), append structures and `CI_*` customer includes, delivery classes and `SE11` activation, with BAPIs and CDS/OData exposing them. Wave 3 replaces both with a YAML entity definition language and a generator: declare an entity once and get the Postgres migration, Go model and repository, REST handlers, OpenAPI 3.1, TypeScript client and JSON Schema. Customers extend any entity with fields, enum values and declarative rules as metadata applied to a running environment. At wave end `Tenant`, `LegalEntity` and `User` from wave 2 are regenerated from metadata and the hand-written code is deleted. Out of scope: generated forms (wave 5), AsyncAPI topics (wave 4), scripted customer hooks (open question), views/maps/table collections (wave 46 analytics covers views).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Adding one field to VendTable in D365: table extension, EDT, form extension, data entity extension, label, build, deployable package, LCS apply — a 2–5 day developer task with an environment outage	Schema, API, form and entity are separate artifacts kept in sync by hand	One `x_` field in an `ExtensionPackage`: migration, API, client, JSON Schema and (wave 5) form generated and applied with zero downtime	Field addition: 2–5 days + outage → < 60 s, no outage; artifacts touched by hand: 6 → 1
SAP append structure on LFA1 needs SE11, activation (table conversion risk on large tables), CMOD/BAdI for validation, SE51 screen exit, and a transport through DEV→QAS→PRD	DDIC activation rewrites tables; UI exits are separate from DDIC	Additive `ADD COLUMN` with fast default, `CREATE INDEX CONCURRENTLY`, declarative validation in the same package	Lock time on a 100M-row table: minutes → < 2 s; transport cycle: 1–2 weeks → one PR
Integration partners (MES, PLM, WMS) must reverse-engineer OData entities or BAPI parameter lists; no machine-readable contract versioning	APIs are hand-exposed per entity (data entities, SEGW projects)	Every entity publishes OpenAPI 3.1 and JSON Schema with additive-change guarantees enforced in CI	Time for an external team to consume a new entity: weeks → the day it is published; breaking changes reaching partners: 0
Silent type drift: AX `real` fields and SAP `CURR` with DECIMALS=2 round amounts differently in X++/ABAP arithmetic vs. the database	Type semantics defined in two places (EDT/domain and language runtime)	One type system (`decimal(p,s)`, `money` + currency, `half_even`) enforced at the API and column	Rounding discrepancies between API and storage: possible → 0 (property-tested)
Finding "where is this field used" requires cross-reference tools (AX cross-reference update takes hours; SAP SE11 where-used across packages)	Reflection is an offline build artifact	Registry `/v1/metadata/*` answers entity, field, relation and extension queries live	Where-used answer: hours (after a cross-reference build) → < 20 ms

0.3.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Table definition	AOT Tables node, `TableGroup`, `SaveDataPerCompany`, `RecId`, `RecVersion`, `DataAreaId`, `Partition`	DD02L, delivery class, MANDT, SE11	`EntityDefinition` YAML; system fields id, tenant_id, legal_entity_id, version, created_*, updated_*	full
Typed fields	Extended Data Types (StringSize, Label, HelpText), `SysDictField`	Domains (DD01L) + data elements (DD04L)	`TypeDefinition` (EDT) + `FieldDefinition`	full
Enumerations	Base Enums, `SysDictEnum`, extensible enums (D365)	Domain fixed values (DD07L), check tables	`EnumDefinition` / `EnumValue`, customer values ≥ 1000	full
Relations / FK	Relations node, `ForeignKey`, `DeleteActions` None/Cascade/Restricted	DD05S/DD08L foreign keys, check tables, cardinality	`RelationDefinition` with on_delete restrict/cascade/set_null	full
Indexes	Indexes node, `AllowDuplicates`, `AlternateKey`	DD12L secondary indexes, unique flag	`IndexDefinition` unique/non-unique, partial predicate	full
Validation hooks	`validateField`, `validateWrite`, `validateDelete`, `initValue`, `modifiedField`	Table maintenance events, BAdIs, check tables	lifecycle hooks: initValue, validateField, validateWrite, validateDelete, afterWrite, afterDelete	full (declarative + registered code)
API exposure	Data entities (IsPublic, PublicCollectionName), OData, custom services	BAPIs, RFC, OData via SEGW / RAP, CDS views	generated REST (OpenAPI 3.1) per entity; GraphQL deferred	partial (no OData $batch)
Extensions	Extensions (`.extension`), Chain of Command, extensible enums	Append structures, CI_* includes, Z/Y namespace	`ExtensionPackage` (customer layer), `x_` field prefix	improved (zero-downtime apply)
Versioning / diff	Model store versions, compare tool	Version management (SE09 versions), transport diffs	`MetadataVersion` + `/diff` endpoint, snapshot golden tests	improved

0.3.3 Functional requirements

- **FR-3.1 Entity definition language** — Must. The system shall accept entity definitions in YAML validated by a published JSON Schema (`metadata/schema/entity.v1.json`) with keys `name`, `domain`, `table_group` (main, group, transaction, worksheet_header, worksheet_line, parameter, reference), `per_legal_entity`, `title_fields[]`, `fields[]`, `enums[]`, `relations[]`, `indexes[]`, `hooks[]`, `audit`, `api`.
  - Notes: `name` is UpperCamelCase ≤ 48 chars; `domain` selects the owning Postgres database; one entity per file under `metadata/entities/<domain>/`.
- **FR-3.2 Type system** — Must. The system shall support field types string(n ≤ 4000), text, int32, int64, decimal(p ≤ 38, s ≤ 12), money (decimal(28,6) + currency reference), bool, date, datetime (timestamptz), time, uuid, enum, json, bytes, and named `TypeDefinition`s (EDT-equivalents) that bundle type, length, label, help, format and lookup entity.
  - Notes: `money` fields always pair with a `currency` char(3); `decimal` default scale 6 for quantities, 2 for displayed amounts, with rounding `half_even`.
- **FR-3.3 Field attributes** — Must. Each field shall declare `required`, `unique`, `default` (literal or `now()`, `current_user()`, `numseq(<code>)` from wave 5), `computed` (expression over sibling fields, evaluated server-side), `visibility` (always, advanced, hidden), `pii` (none, personal, sensitive), `audit` (bool, default inherits entity), `allow_edit`, `allow_edit_on_create`, `label`, `help`, `min`, `max`, `pattern`.
- **FR-3.4 System fields** — Must. Every generated table shall carry `id uuid`, `tenant_id uuid`, `legal_entity_id uuid` (when `per_legal_entity=true`), `version int` (RecVersion), `created_at`, `created_by`, `updated_at`, `updated_by`, and optional `deleted_at` for soft delete.

    - Notes: RLS policy generated per FR-2.4; `version` increments on every update and is required in `If-Match`.
- **FR-3.5 Migration generation** — Must. The generator shall emit Atlas migrations that are deterministic for a given definition version; additive changes (new nullable field, new enum value, new index) generate forward-only migrations; destructive changes require an explicit `breaking: true` flag and a new API major version.
    - Notes: `ADD COLUMN … DEFAULT` uses Postgres fast default; index creation uses `CONCURRENTLY`; migrations run as a pre-sync Argo hook job.
- **FR-3.6 Code generation** — Must. For each entity the generator shall emit a Go struct, repository (list with filter/sort/cursor pagination, get, create, update, patch, delete), HTTP handlers, OpenAPI 3.1 document, TypeScript client (fetch-based, typed), and JSON Schema for request validation.
    - Notes: list filter grammar `?filter=field op value` with ops eq, ne, gt, ge, lt, le, in, like, is_null; max page size 500, default 50; cursor is opaque base64.
- **FR-3.7 Lifecycle hooks** — Must. The system shall execute hooks in order initValue → validateField (per changed field) → validateWrite → write → afterWrite, and validateDelete → delete → afterDelete; declarative hooks are expressions returning `{ok, code, message, field}`; coded hooks are Go/TS functions registered by name in the owning service.
    - Notes: validation errors return 422 with `errors[]{field, code, message}`; a coded hook that panics rolls back the transaction and returns 500 with a trace id.
- **FR-3.8 Metadata registry** — Must. The system shall store every definition version, expose diff between versions, publish an immutable version on approval, and serve the current version to generators, the gateway and the shell with ETag caching.
    - Notes: registry DB is `metadata`; publishing emits `dev.passage.metadata.entity.published.v1`; registry read p95 ≤ 20 ms.
- **FR-3.9 Extension packages** — Must. Customers shall create an `ExtensionPackage` (layer customer or partner) that adds fields (prefixed `x_`), enum values (code ≥ 1000), indexes, declarative hooks and labels to any `sys` entity; applying a package generates and runs the migration, regenerates OpenAPI/JSON Schema and invalidates the shell form cache with zero downtime.
    - Notes: limits: 200 extension fields per entity, 50 enum values per enum per package; packages are versioned and reversible only while additive.
- **FR-3.10 Backward compatibility** — Must. CI shall run the previous published version's generated TypeScript client against the new server and fail on any breaking change (removed field, type change, required added, enum value removed); additive changes pass.
- **FR-3.11 Golden tests** — Must. Ten reference entities (Tenant, LegalEntity, User, Role, VendorStub, ProductStub, JournalLineStub, Address, Note, Attachment) shall have committed snapshots of migration, OpenAPI and client; CI fails on any byte-level drift.
- **FR-3.12 Performance** — Must. A generated entity API shall serve 1,000 rps of simple reads with p95 < 50 ms on the 3-node dev cluster and 200 rps of creates with p95 < 120 ms.

0.3.4 Data model

EntityDefinition (`metadata.entity`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	`sys` tenant for shipped entities
name	varchar(48)	yes	unique with domain and version
domain	varchar(32)	yes	Postgres database / service
version	int	yes	monotonically increasing
status	enum	yes	draft, review, published, deprecated
table_group	enum	yes	FR-3.1 values
per_legal_entity	bool	yes	default true for main/transaction
title_fields	text[]	no	≤ 2
audit	bool	yes	default true for main/transaction
api_major	int	yes	default 1
spec	jsonb	yes	canonical YAML as JSON
spec_sha256	char(64)	yes	drift detection
published_at, published_by	timestamptz, uuid	no	

FieldDefinition (`metadata.field`)

Field	Type	Required	Notes
id	uuid	yes	PK
entity_id	uuid	yes	FK entity version
name	varchar(48)	yes	snake_case; `x_` reserved for extensions
type	enum	yes	FR-3.2
type_def_id	uuid	no	FK TypeDefinition
enum_id	uuid	no	FK EnumDefinition when type=enum
length, precision, scale	int	no	per type
required, unique	bool	yes	
default_expr	text	no	
computed_expr	text	no	
visibility	enum	yes	always, advanced, hidden
pii	enum	yes	none, personal, sensitive
audit	bool	yes	
layer	enum	yes	sys, partner, customer

Field	Type	Required	Notes
extension_package_id	uuid	no	FK when layer ≠ sys
ordinal	int	yes	form/column order default

ExtensionPackage (`metadata.extension_package`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	
code	varchar(32)	yes	unique per tenant
layer	enum	yes	partner, customer
version	int	yes	
status	enum	yes	draft, validated, applied, failed, retired
targets	jsonb	yes	entities, fields, enum values, hooks, labels
applied_at, applied_by	timestamptz, uuid	no	
migration_id	text	no	Atlas migration name

EnumDefinition / EnumValue / TypeDefinition / RelationDefinition / IndexDefinition follow the same shape: id, entity or enum FK, name/code, layer, ordinal; EnumValue has `code int` (sys < 1000, extensions ≥ 1000) and `label`; RelationDefinition has `from_field`, `to_entity`, `to_field`, `on_delete`; IndexDefinition has `fields[]`, `unique`, `where`.

Invariants: a published version is immutable; field names unique per entity across layers; an extension cannot change a `sys` field's type or required flag; every `enum` field references an enum; `title_fields` must be string type. Indexing: `entity(tenant_id, domain, name, version)` unique; `field(entity_id, ordinal)`. Partitioning: none (metadata volume < 1M rows).

0.3.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/v1/metadata/entities?domain=&status=	List entity definitions	n/a	any authenticated
GET	/v1/metadata/entities/{name}	Current published version (ETag)	n/a	any
PUT	/v1/metadata/entities/{name}/versions/{v}	Create/replace a draft version	by version	Maintain entity metadata
POST	/v1/metadata/entities/{name}/versions/{v}/publish	Publish	version	Maintain entity metadata
GET	/v1/metadata/entities/{name}/diff?from=&to=	Structured diff	n/a	any
POST	/v1/metadata/compat-check	Classify a diff as additive/breaking	n/a	Maintain entity metadata
GET/POST	/v1/metadata/extensions	List / create packages	POST: code+version	Apply extensions
POST	/v1/metadata/extensions/{id}/validate, /apply, /retire	Lifecycle	package version	Apply extensions
GET	/v1/metadata/openapi/{entity}	Generated OpenAPI 3.1	n/a	any
GET	/v1/metadata/types, /v1/metadata/enums	EDT and enum catalogs	n/a	any
GET/POST/PATCH/DELETE	/v1///{id}	Generated entity API	POST: Idempotency-Key; PATCH: If-Match	per Permission(entity, action)

Event	Trigger	Payload fields	Consumers
dev.passage.metadata.entity.published.v1	publish	entity, domain, version, api_major, diff_summary{added, changed, removed}	generator pipeline, events (wave 4), shell
dev.passage.metadata.entity.deprecated.v1	deprecate	entity, version, sunset_date	shell, API docs
dev.passage.metadata.extension.applied.v1	apply success	package_id, code, version, entities[], fields_added, enum_values_added, migration_id	audit, shell, generator
dev.passage.metadata.extension.failed.v1	apply failure	package_id, stage, error	alerting
dev.passage.metadata.schema.migrated.v1	Atlas job completes	domain, migration_id, duration_ms	audit

0.3.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Entity catalog	Browse, open, view diff, publish	name, domain, version, status, field count, extensions	Maintain entity metadata
Entity designer (YAML + form view)	Add field/enum/relation/index/hook, validate, save draft	name, type, required, default, pii, visibility	Maintain entity metadata
Extension packages	Create, validate, apply, retire, view migration log	code, layer, version, status, targets	Apply extensions
Type and enum catalogs	Create EDT, add enum values	code, base type, length, label	Maintain entity metadata

Rules: the designer validates against the JSON Schema on every keystroke (debounced 300 ms) and against the registry on save; apply shows a dry-run migration SQL with an estimated lock time and refuses if > 2 s; keyboard: `Ctrl+S` save draft, `Ctrl+Shift+P` publish.

0.3.7 Business rules and edge cases

- **BR-3.1** Entity version status machine: draft → review → published → deprecated; published versions are immutable; a deprecated version serves for 180 days after `sunset_date` announcement.
- **BR-3.2** Breaking vs additive: removing a field, narrowing a type, adding `required` without default, removing an enum value, or renaming are breaking; everything else is additive. Breaking requires `api_major + 1` and a parallel `/v{n+1}` route for ≥ 1 release.
- **BR-3.3** Enum value codes are permanent; retiring a value sets `retired=true` and the value remains readable but not assignable.
- **BR-3.4** Extension apply is transactional per domain database: migration, registry update and cache invalidation succeed together or the package goes to `failed` with the SQL error.
- **BR-3.5** Concurrent publishes of the same entity: second publish gets 409 and must rebase on the new version.
- **BR-3.6** Decimal rounding at write uses the field's scale with `half_even`; values exceeding precision return 422 `VALUE_OUT_OF_RANGE`.
- **BR-3.7** Soft-deleted rows (`deleted_at` set) are excluded from list and get by default; `?include_deleted=true` requires the `delete` action.
- **BR-3.8** `on_delete: restrict` returns 409 `REFERENCED_BY` with the referencing entity and count; cascade is limited to 10,000 child rows per request.
- **BR-3.9** PII `sensitive` fields are masked (`***`) in responses unless the Permission grants the field explicitly; they are never included in list endpoints.

0.3.8 Acceptance criteria

- **AC-3.1** Given the 10 reference entities, When the generator runs 3 times, Then the migration, OpenAPI and client snapshots are byte-identical and match the committed golden files. (FR-3.5, FR-3.6, FR-3.11)
- **AC-3.2** Given a customer package adding `x_risk_rating` (enum) and enum value `PreferredPlus (1001)` to VendorStub, When applied in dev, Then the API accepts the field within 60 s, no request fails during apply, and the wave-5 form shows it. (FR-3.9)
- **AC-3.3** Given a published v1 client, When the server publishes an additive v2, Then the v1 contract suite passes; Given a breaking change without `breaking: true`, Then CI fails naming the change. (FR-3.10, BR-3.2)
- **AC-3.4** Given 1,000 rps of `GET /v1/identity/users/{id}` for 5 minutes on the dev cluster, When measured at the gateway, Then p95 < 50 ms and error rate < 0.1%. (FR-3.12)
- **AC-3.5** Given a `validateWrite` declarative hook `amount > 0`, When a client posts `amount = -1`, Then 422 with `field=amount`. (FR-3.7)
- **AC-3.6** Given Tenant, LegalEntity and User regenerated from metadata, When the wave-2 test suites run, Then they pass unchanged and the hand-written handlers are removed from the repo. (FR-3.6)
- **AC-3.7** Given a diff request between versions 3 and 5, When called, Then the response lists each added/changed/removed field with classification additive/breaking. (FR-3.8)
- **AC-3.8** Given a field with `pii=sensitive`, When listed by a user without the field permission, Then the field is absent; when fetched by id, Then masked. (FR-3.3, BR-3.9)

0.3.9 Test plan and fixtures

Fixtures: `metadata/entities/_reference/*.yaml` (10 entities), `fixtures/metadata/golden/` (snapshots), `fixtures/metadata/ext-vendor-risk.yaml`, `fixtures/metadata/breaking-change-cases.yaml` (12 breaking, 12 additive cases), `fixtures/load/users-100k.sql` (100k users for the read benchmark).

Test	Type	Target	Pass criterion
Golden generation	Snapshot	generator	zero drift
Schema validation	Unit	JSON Schema	100% of invalid fixtures rejected with path
Compat classifier	Table-driven	compat-check	24/24
Extension apply	E2E	registry + Atlas + API	applied, 0 failed requests during apply (k6 background at 100 rps)
Hook ordering	Property	repository	hooks fire in FR-3.7 order for random mutations
Read benchmark	Performance (k6)	generated API	p95 < 50 ms @ 1,000 rps
RLS on generated tables	Contract	every generated table	policy present and effective

Reconciliation: registry field count per published entity equals `information_schema.columns` for the generated table minus system fields; OpenAPI property count equals field count minus hidden fields.

0.3.10 Deployment and operations

Charts: `metadata-registry` (Go, CNPG `metadata` database), `passage-gen` runs in CI and as an in-cluster Job for extension apply. Config: `REGISTRY_URL`, `GEN_OUTPUT_DIR`, `ATLAS_DEV_URL` (scratch database for migration planning). Migrations: generated; applied by Argo pre-sync hook with `atlas migrate apply --tx-mode all`. Feature flags: `metadata.extensions.enabled` per tenant. Scheduled jobs: nightly drift check (registry vs live schema). SLOs: registry availability 99.9%, read p95 ≤ 20 ms, extension apply ≤ 60 s. Alerts: `SchemaDrift`, `ExtensionApplyFailed`, `MigrationJobFailed`. Runbook: failed migration (mark package failed, run `atlas migrate down` only in dev), registry restore from CNPG backup. Rollback: registry and generated services roll back together by digest; because migrations are additive the previous digest runs against the newer schema.

0.3.11 Migration from AX / SAP

Source	Passage entity	Notes
AX AOT Tables (via `SysDictTable` export or AxModel XML) with `TableGroup`, `SaveDataPerCompany`	EntityDefinition	only customer-layer (CUS/VAR/USR) tables and fields are candidates; SYS tables map to Passage entities by crosswalk
AX EDTs, Base Enums	TypeDefinition, EnumDefinition	enum values keep their integer codes where < 1000
AX table extensions (`.extension` XML)	ExtensionPackage	field additions become `x_` fields
SAP DD02L/DD03L for Z*/Y* tables, append structures, CI_* includes	EntityDefinition / ExtensionPackage	DD04L data elements → TypeDefinition; DD07L fixed values → EnumValue

Transformation is a definition import (`passage-gen import-ax`, `import-sap`) producing draft YAML for review; nothing is auto-published. Not migrated: X++ table methods and ABAP user exits (code), SAP delivery-class semantics, AX `CacheLookup` and `Partition`.

0.3.12 Dependencies, risks, open questions

Depends on: wave 2 (tenant_id/RLS, duties `Maintain entity metadata`, `Apply extensions`; the three entities to regenerate).

Risk	Likelihood	Impact	Mitigation
Generator becomes a bottleneck for every later wave	High	High	Generator is itself golden-tested; business waves add entities by YAML only; escape hatch for hand-written endpoints alongside generated ones
Expression language for hooks grows into a scripting language	Medium	Medium	CEL-style expression subset frozen at wave 3; scripted hooks decided in OQ 2
Large-table `ADD COLUMN` with non-constant default locks	Low	Medium	Dry-run lock estimate; non-constant defaults applied by backfill job

Open questions: (1) GraphQL alongside REST — default no until wave 35; decide by wave 10. (2) Customer scripted hooks (sandboxed TS or WASM) — default declarative only through wave 30; decide by wave 20. (3) Soft delete default for main tables — default on for master data groups, off for transactions; decide by wave 6.

0.4 Wave 4: Event Backbone, Workflow & Batch

0.4.1 Scope and business context

AX runs approvals through the workflow framework (`WorkflowTable`, `WorkflowVersionTable`, `WorkflowWorkItemTable`, `WorkflowTrackingStatusTable`, `WorkflowTrackingTable`; approval, task and conditional-decision elements; work item statuses Pending/Delegated/Completed/Canceled; escalation and due dates) and background work through the batch framework (`BatchJob`, `Batch`, `BatchJobHistory`, `BatchGroup`, `RunBaseBatch`, `SysOperationServiceController` with `SysOperationExecutionMode` Synchronous/Asynchronous/ReliableAsynchronous/ScheduledBatch, job statuses Waiting/Executing/Ended/Error/Hold/Canceling/Canceled); D365 adds business events. SAP has Business Workflow (SWDD, work items SWWWIHEAD, SWIA/SWI1, SBWP inbox, event linkage SWE2, deadline monitoring), background jobs (SM36 definition, SM37 monitor, TBTCO/TBTCP, statuses Scheduled/Released/Ready/Active/Finished/Canceled, job classes A/B/C, SM62 events) and IDoc status processing (EDIDC/EDIDS status 03/51/53/64, BD87). Wave 4 replaces these with Strimzi Kafka carrying CloudEvents, a transactional outbox in every service via Debezium, a schema registry fed by entity metadata, Temporal for approvals and batch, a configuration-driven approval chain, a partitioned checkpointing batch framework and a job monitor. Out of scope: notification channels beyond in-app and email (wave 46 adds Teams/Slack), position-hierarchy assignee resolution (wave 42), line-item workflows (wave 17 first consumer), IDoc/EDI payload semantics (wave 35).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Downstream systems (MES, WMS, BI) poll `ModifiedDateTime` on AX tables or wait for the nightly IDoc/BD21 change-pointer job; IDocs stuck in status 51/64 are reprocessed by hand in BD87	No reliable push of changes; outbound integration is batch	Every committed change is a CloudEvent on Kafka via Debezium outbox within 500 ms; consumers subscribe, never poll	Change-to-consumer latency: 15 min–24 h → < 2 s p95; manual reprocessing runs per week: dozens → 0 (DLQ with reason, replayable)
AX workflow approvals: the workflow batch job (`WorkflowMessageProcessingTask`) runs every minute and the work-item list is refreshed manually; escalation "due dates" silently fail when the batch server is down	Workflow runtime is a polling batch job on the AOS	Temporal workflows with durable timers; work item pushed to the inbox by event	Submit-to-inbox latency: 1–5 min → < 2 s; missed escalations due to batch outage: possible → 0 (timers survive restarts)
SAP Business Workflow (SWDD) changes need an ABAP workflow developer, a transport and SWU3 checks; amount bands for release strategies live in customizing tables (CL20N/CL24N)	Chains are code and classification objects, not configuration	Designer with amount bands, conditions and SLAs as versioned data, activated per legal entity	Approval chain change: 1–3 weeks → < 30 min by an administrator; developer involvement: required → none
A failed AX batch job over 1M rows (e.g. inventory recalculation) restarts from zero; SM37 jobs show "Canceled" with a dump in ST22 and no restart point	Batch frameworks have no checkpoints or partitions	Partitioned tasks with checkpoints every 10k rows; resume from last checkpoint; rerun failed partitions only	Rework after a failure at 90%: 90% of the run → < 1%; mean time to resume: hours → < 1 min
Exactly-once delivery requires custom dedupe tables in X++/ABAP and still double-posts under AOS/work-process crashes	Dual writes (DB + queue) in application code	Outbox in the same transaction; consumer library dedupes on CloudEvents `id`	Duplicate side effects under chaos: observed → 0 over 10 kill cycles

0.4.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Change notification	Business events (D365 `BusinessEventsBase`), AIF outbound ports, alerts (`EventInbox`)	Event type linkage SWE2, change pointers BD21, IDoc outbound	CloudEvents on Kafka via outbox; AsyncAPI per topic	improved
Reliable publish	AIF queue (`AifGatewayQueue`), dual-write risk in X++	tRFC/qRFC (SM58), IDoc status 03	Debezium outbox event router; exactly-once effect	improved
Workflow definition	Workflow editor, `WorkflowTypeTable`, `WorkflowDocument` class, approval/task elements	SWDD workflow builder, tasks TS*, agent rules PFAC	`WorkflowDefinition` JSON (steps, conditions, SLAs), designer UI	full
Work items	`WorkflowWorkItemTable`, Work list, Delegate/Approve/Reject/Request change	SWWWIHEAD, SBWP inbox, forward/resubmit	`WorkItem` with approve/reject/delegate/request_change/recall	full
Escalation and deadlines	Time limits, escalation path, `WorkflowEscalation*`	Deadline monitoring (latest end), SWWDHEX	Temporal timers; SLA + escalation action	full
Workflow history	`WorkflowTrackingTable`, tracking status form	SWIA, workflow log (SWI1/SWEL)	Temporal history + `WorkflowInstance` timeline API	improved

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Background jobs	`BatchJob`, `Batch`, `BatchGroup`, `SysOperation` data contracts, recurrence (`SysRecurrence`)	SM36/SM37, TBTCO/TBTCP, variants, job classes A/B/C	`BatchJob` definition, `BatchRun`, `BatchTask`, cron recurrence, priority	full
Job monitoring	Batch job history, `BatchJobHistory`, Infolog	SM37, job log (TBTC5), SM50	Job monitor UI, run log, Prometheus metrics	improved
Restart / reprocess	Rerun batch job, no checkpoints	BD87 IDoc reprocess, job copy in SM37	checkpoint per partition; resume from last checkpoint	improved

0.4.3 Functional requirements

- **FR-4.1 Kafka and topics** — Must. The system shall run Strimzi-managed Kafka (3 brokers in cloud, 1 in k3d, KRaft) with topics named `passage.<domain>.<aggregate>.v<n>`, partition key = aggregate id, 12 partitions default, retention 7 days (entity-state topics compacted), and CloudEvents 1.0 JSON envelopes with extensions `tenantid`, `legalentityid`, `traceparent`, `actor`.
  - Notes: no per-tenant topics; consumers filter by `tenantid` header; max message 1 MB, larger payloads use a claim-check reference (object storage URL).
- **FR-4.2 Transactional outbox** — Must. Every service shall write domain changes and an `outbox_event` row in the same transaction; Debezium (one connector per domain database, pgoutput) with the outbox event router publishes and never writes to Kafka from application code.
  - Notes: outbox columns id, aggregate_type, aggregate_id, event_type, payload (jsonb), headers (jsonb), created_at; rows deleted after 24 h by a sweep; publish lag p95 ≤ 500 ms.
- **FR-4.3 Schema registry** — Must. Event payload schemas shall be JSON Schema derived from the metadata registry (`dev.passage.<domain>.<entity>.<verb>.v<n>` ↔ entity version) registered with BACKWARD_TRANSITIVE compatibility; CI rejects an incompatible schema.
- **FR-4.4 AsyncAPI generation** — Must. The generator from wave 3 shall emit an AsyncAPI 3.0 document per domain listing topics, event types and schemas; published at `/v1/events/asyncapi/{domain}`.
- **FR-4.5 Consumer library** — Must. Go and TS libraries shall provide idempotent consumption keyed by CloudEvents `id` (dedupe table per consumer group, 7-day window), ordered processing per partition, retry with exponential backoff (1 s → 5 min, 10 attempts) and a per-topic dead-letter topic `…dlq`.
- **FR-4.6 Temporal cluster** — Must. The system shall run Temporal (CNPG persistence, 1 namespace per environment, `tenant_id` as a search attribute) with Go and TS SDK wrappers that inject tenant, actor and trace context into every workflow and activity.
  - Notes: workflow ids `wf-{tenant}-{entity}-{record_id}-{seq}`; history retention 90 days; activity timeouts default 30 s start-to-close, heartbeat for long activities.
- **FR-4.7 Approval workflow template** — Must. The system shall provide a configurable approval workflow with steps (assignee resolution: user, role, expression, `record.owner`, manager placeholder), conditions (expression over the document, e.g. `amount_minor > 1000000`), completion policy (single, all, majority, percentage n), SLA (ISO 8601 duration) and escalation action (delegate_to, auto_approve, auto_reject, notify_only), and actions submit, approve, reject, delegate, request_change, recall, cancel.
  - Notes: a step with zero resolved assignees escalates immediately to the workflow administrator; delegation may chain at most 3 levels; delegation records keep the original assignee for audit.
- **FR-4.8 Workflow designer** — Must. Administrators shall configure approval chains per entity with amount bands and conditions in a form (no code), version them, and activate one version per entity × legal entity.
- **FR-4.9 Batch job framework** — Must. The system shall define jobs (`BatchJob`: handler name, parameters JSON Schema, partition strategy rangeIhashIlist, partition count ≤ 256, concurrency ≤ 32, priority A/B/C, retry policy), schedule them by cron (tenant time zone) or on demand, execute each partition as a Temporal activity with checkpoints every N rows (default 10,000) and resume from the last checkpoint after failure.
  - Notes: statuses waiting, scheduled, executing, ended, error, hold, canceling, canceled; max run time default 6 h; a job may declare `exclusive=true` to prevent overlapping runs.
- **FR-4.10 Job monitor** — Must. The system shall show runs, tasks, progress (rows/sec, ETA), logs and errors; allow cancel, hold, resume and rerun-failed-partitions; retain history 90 days.
- **FR-4.11 Metadata-changed events** — Must. The metadata registry shall emit `dev.passage.metadata.entity.published.v1` through its own outbox (first production consumer of the backbone).
- **FR-4.12 Chaos and replay** — Should. The platform shall support topic replay from an offset or timestamp per consumer group and ship a chaos test that kills a service between commit and publish.

0.4.4 Data model

OutboxEvent (`<domain>.outbox_event`, every service)

Field	Type	Required	Notes
id	uuid	yes	PK; becomes CloudEvents `id`
tenant_id	uuid	yes	RLS; copied to header `tenantid`
aggregate_type	varchar(64)	yes	entity name; routes to topic
aggregate_id	uuid	yes	partition key
event_type	varchar(128)	yes	`dev.passage.<domain>.<entity>.<verb>.v<n>`
payload	jsonb	yes	CloudEvents `data`
headers	jsonb	yes	traceparent, actor, legalentityid, schema_version
created_at	timestamptz	yes	CloudEvents `time`

WorkflowInstance (`workflow.instance`)

Field	Type	Required	Notes
id	uuid	yes	PK; equals Temporal workflow id suffix
tenant_id, legal_entity_id	uuid	yes	
definition_id, definition_version	uuid, int	yes	FK WorkflowDefinition

Field	Type	Required	Notes
entity, record_id	varchar(48), uuid	yes	document under approval
status	enum	yes	submitted, in_review, change_requested, approved, rejected, recalled, canceled, failed
submitted_by, submitted_at	uuid, timestamptz	yes	
current_step	int	no	1-based
completed_at, outcome_reason	timestamptz, text	no	
temporal_run_id	varchar(64)	yes	

WorkItem (`workflow.work_item`)

Field	Type	Required	Notes
id	uuid	yes	PK
instance_id	uuid	yes	FK
step	int	yes	
assignee_id	uuid	yes	FK user
original_assignee_id	uuid	no	set when delegated
status	enum	yes	pending, claimed, approved, rejected, delegated, escalated, change_requested, canceled, expired
due_at	timestamptz	yes	SLA
acted_at, comment	timestamptz, text	no	

BatchJob / BatchRun / BatchTask (`workflow.batch_*`): BatchJob has id, tenant_id, code (unique), handler, params_schema jsonb, partition_strategy, partition_count, concurrency, priority, cron, time_zone, exclusive, retry_policy jsonb, status (active, hold); BatchRun has id, job_id, params jsonb, status (FR-4.9), scheduled_at, started_at, ended_at, rows_total, rows_done, error, triggered_by; BatchTask has id, run_id, partition_no, status, checkpoint jsonb, attempts, last_error, rows_done.

Invariants: exactly one active `WorkflowDefinition` version per (entity, legal_entity); a WorkItem is terminal once acted; `BatchRun` of an exclusive job cannot overlap another run of the same job; outbox rows are never updated. Indexing: `outbox_event(created_at)`; `work_item(tenant_id, assignee_id, status)`; `instance(tenant_id, entity, record_id)`; `batch_run(job_id, scheduled_at desc)`. Partitioning: `work_item` and `batch_task` by month from wave 30.

0.4.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/v1/events/topics, /v1/events/asyncapi/{domain}	Topic catalog, AsyncAPI	n/a	any
GET/POST	/v1/events/schemas/{type}	Read / register schema	by type+version	Maintain entity metadata
POST	/v1/events/replay	Replay topic for consumer group from offset/time	request id	Maintain batch jobs
GET/POST/PUT	/v1/workflows/definitions, /{id}/versions/{v}, /{id}/activate	Definitions	by version	Maintain workflows
POST	/v1/workflows/instances	Submit a document	entity+record_id+seq	per entity `post/update`
GET	/v1/workflows/instances/{id}, /{id}/timeline	Status and history	n/a	submitter, assignee, Maintain workflows
POST	/v1/workflows/instances/{id}/actions	recall, cancel	action+instance version	submitter / Maintain workflows
GET	/v1/workitems?assignee=me&status=pending	Inbox	n/a	any
POST	/v1/workitems/{id}/actions	approve, reject, delegate, request_change	workitem id + action	assignee (Approve workflow)
GET/POST/PUT	/v1/batch/jobs, /{id}	Define jobs	by code	Maintain batch jobs
POST	/v1/batch/jobs/{id}/runs	Start run	Idempotency-Key	Maintain batch jobs
GET/POST	/v1/batch/runs/{id}, /{id}/cancel, /{id}/resume, /{id}/rerun-failed	Monitor and control	run id + action	View batch history / Maintain batch jobs

Validation: approve/reject require a comment when the definition flags `comment_required`; delegate target must hold the step's role; a run request against a `hold` job returns 409.

Event	Trigger	Payload fields	Consumers
dev.passage.workflow.instance.submitted.v1	submit	instance_id, entity, record_id, definition_version, submitted_by	owning service (sets record status), audit
dev.passage.workflow.workitem.assigned.v1	step entered / delegated	workitem_id, instance_id, assignee_id, due_at, step	shell inbox, notifier
dev.passage.workflow.workitem.escalated.v1	SLA timer fired	workitem_id, from_assignee, to_assignee, action	notifier, audit
dev.passage.workflow.instance.completed.v1	terminal status	instance_id, outcome (approved/rejected/canceled/recalled), completed_at	owning service, audit, shell
dev.passage.batch.run.started.v1 / completed.v1 / failed.v1	run transitions	job_code, run_id, rows_total, rows_done, duration_ms, error	job monitor, alerting, audit
dev.passage.metadata.entity.published.v1	registry outbox	(see wave 3)	events service (topic/schema create)

0.4.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Work items (inbox)	Approve, reject, delegate, request change, bulk approve	document, step, submitted by, due, age	Approve workflow
Workflow instance timeline	View steps, comments, timers; recall	status, step, assignee, acted at, SLA	submitter / Maintain workflows
Workflow designer	Add step, condition, SLA, escalation; version; activate	entity, legal entity, bands, assignees	Maintain workflows
Batch jobs	Define, schedule, hold, run now	code, handler, cron, last run, next run	Maintain batch jobs
Job monitor	Watch progress, cancel, resume, rerun failed, view log	run, status, rows/sec, ETA, errors by partition	View batch history
Topic catalog	Browse topics, schemas, consumer lag	topic, partitions, schema version, lag	Maintain batch jobs

Rules: bulk approve is limited to 100 items per action and requires a confirmation listing document totals; the inbox refreshes from events within 2 s; designer validates expressions against the entity's current metadata; keyboard `a` approve, `r` reject, `d` delegate on a focused row.

0.4.7 Business rules and edge cases

- **BR-4.1** Instance status machine: submitted → in_review → (approved | rejected | change_requested → submitted on resubmit); submitted/in_review → recalled (by submitter, before any approval) or canceled (by administrator); any → failed on unrecoverable activity error (alert raised).
- **BR-4.2** Work item status machine: pending → claimed → approved | rejected | change_requested; pending/claimed → delegated (new item created) | escalated (on SLA) | expired (SLA with `auto_reject`) | canceled (instance terminal).
- **BR-4.3** SLA timers use the tenant time zone and the tenant work calendar when one exists (wave 6); in P0 they are wall-clock; the timer survives Temporal worker restarts.
- **BR-4.4** Completion policy `majority` rounds up (3 of 4); `percentage 60` on 5 assignees needs 3; a single reject under `all` rejects the step immediately.
- **BR-4.5** Exactly-once effect: a consumer that sees a CloudEvents `id` already in its dedupe table acknowledges without side effects; the dedupe table is keyed (consumer_group, event_id) and swept after 7 days.
- **BR-4.6** Ordering is guaranteed per aggregate id only; consumers must not assume cross-aggregate order.
- **BR-4.7** Batch checkpoint is committed in the same transaction as the processed rows; after resume, rows with ids ≤ checkpoint are skipped, so handlers must process in key order within a partition.
- **BR-4.8** Retry policy: activity failures retry up to 5 times with backoff 2 s × 2^n capped at 5 min; after that the task is `error` and the run continues other partitions; run ends `error` if any task is `error`.
- **BR-4.9** A workflow definition change never alters running instances; they finish on their version.
- **BR-4.10** Poison events after 10 delivery attempts go to the DLQ with the last error in header `dlqreason`; a DLQ depth > 0 for 10 min alerts.

0.4.8 Acceptance criteria

- **AC-4.1** Given a service killed by SIGKILL between DB commit and Debezium capture, 10 times in a loop, When it restarts, Then every committed outbox row appears on Kafka exactly once (consumer dedupe table shows 0 duplicates, 0 gaps). (FR-4.2, FR-4.5)
- **AC-4.2** Given a 3-level chain (role → user → expression) with 4-hour SLAs, When step 2's assignee delegates and step 3 times out with `delegate_to admin`, Then the instance completes approved after simulated clock advance and the timeline shows 5 work items. (FR-4.7)
- **AC-4.3** Given a batch job over 1,000,000 synthetic rows in 16 hash partitions, When the worker pod is killed at ~40%, Then the run resumes and ends with rows_done = 1,000,000 and no duplicate side effects. (FR-4.9)
- **AC-4.4** Given a schema change that removes a required field, When CI registers it, Then the registry rejects it with the incompatible path. (FR-4.3)
- **AC-4.5** Given an entity publish in the registry, When the outbox is captured, Then a topic `dev.passage.metadata.entity.v1` message with type `dev.passage.metadata.entity.published.v1` exists within 1 s. (FR-4.11)
- **AC-4.6** Given a submitter recalls after step 1 approved, When the action is sent, Then 409 `RECALL_NOT_ALLOWED`. (BR-4.1)
- **AC-4.7** Given a consumer group replay from `T-1h`, When executed, Then events are redelivered and the dedupe table suppresses side effects for already-processed ids. (FR-4.12)
- **AC-4.8** Given an exclusive job with a running instance, When a second run is requested, Then 409 `RUN_OVERLAP`. (FR-4.9)
- **AC-4.9** Given a step whose assignee expression resolves to nobody, When the step is entered, Then it escalates to the workflow administrator within 5 s and emits `workitem.escalated`. (FR-4.7)

0.4.9 Test plan and fixtures

Fixtures: `fixtures/events/chaos-outbox.yaml` (kill schedule), `fixtures/workflow/chain-3-level.json`, `fixtures/workflow/completion-policies.json` (12 cases), `fixtures/batch/rows-1m.sql` (synthetic), `fixtures/events/schema-breaking-cases/` (8 breaking, 8 compatible).

Test	Type	Target	Pass criterion
Outbox chaos	Chaos / integration	Debezium + consumer lib	0 dup, 0 loss over 10 kills
Approval chain	Temporal test server (time-skipping)	workflow	expected timeline, timers fire
Completion policy	Property	step evaluator	12/12 fixtures; random assignee sets obey rounding
Batch resume	Integration	batch framework	1M rows, resume, idempotent
Schema compatibility	Contract	registry	16/16 classified correctly
Publish lag	Performance	outbox → Kafka	p95 ≤ 500 ms at 2,000 events/s
DLQ routing	Negative	consumer lib	poison event lands in DLQ with reason

Reconciliation: count of outbox rows created in a window equals Kafka messages for that domain in the same window (nightly check, tolerance 0); sum of `BatchTask.rows_done` equals `BatchRun.rows_done`.

0.4.10 Deployment and operations

Charts: `strimzi-kafka` (cluster CR), `debezium-connect` (one connector per domain DB, secrets via ESO), `schema-registry`, `temporal` (frontend/history/matching/worker, CNPG persistence), `workflow` (TS workers + API), `events` (catalog API). Config: `KAFKA_BOOTSTRAP`, `SCHEMA_REGISTRY_URL`, `TEMPORAL_ADDRESS`, `TEMPORAL_NAMESPACE`, `OUTBOX_SWEEP_HOURS=24`. Migrations: outbox table added to every domain by generator; `logical_decoding` and `wal_level=logical` set in CNPG clusters. Feature flags: `workflow.enforce` per entity. Scheduled jobs: outbox sweep hourly, DLQ report daily, Temporal history archival. SLOs: publish lag p95 ≤ 500 ms, consumer lag ≤ 10 s p95, workflow action p95 ≤ 300 ms, batch scheduler drift ≤ 30 s. Alerts: `DebeziumConnectorFailed`, `ConsumerLagHigh`, `DLQNonEmpty`, `TemporalWorkflowFailedRate`, `BatchRunError`. Runbook: connector restart with slot reset, replay procedure, stuck workflow reset via Temporal CLI, DLQ reprocess. Rollback: digest rollback; schemas are backward compatible so older consumers keep working; Temporal workers are versioned with build ids.

0.4.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `WorkflowTable` / `WorkflowVersionTable` (approval elements, conditions)	WorkflowDefinition	exported via workflow export XML; steps mapped where assignee type is user/role; hierarchy-based assignees flagged for wave 42
AX `BatchJob` / `Batch` / `SysRecurrence`	BatchJob	only jobs whose handler has a Passage equivalent (listed per later wave); recurrence → cron
SAP TBTCO/TBTCP (job name, program, variant, period)	BatchJob	same rule; variants → params
SAP SWDD workflow templates (WS*), PFAC rules	WorkflowDefinition	manual rebuild from a generated inventory report
In-flight work items (`WorkflowWorkItemTable`, SWWWIHEAD)	not migrated	cut over by completing or canceling in the source system (wave 49 runbook)

Not migrated: workflow history (kept read-only in the source for audit), AIF/IDoc queues, batch job logs.

0.4.12 Dependencies, risks, open questions

Depends on: wave 3 (entity metadata for schemas and outbox generation, registry as first producer), wave 2 (actor identity, duties `Approve workflow`, `Maintain workflows`, `Maintain batch jobs`, `View batch history`).

Risk	Likelihood	Impact	Mitigation
Debezium replication slot bloat when connector is down	Medium	High	Slot lag alert at 1 GB; max WAL retention 20 GB; runbook
Temporal operational complexity on k3d	Medium	Medium	Single-binary dev mode for `PROFILE=min`
Expression language in conditions diverges from metadata hooks	Medium	Medium	Same expression engine as wave 3

Open questions: (1) JSON vs Avro/Protobuf for payloads — default JSON Schema (matches metadata kernel); decide by wave 4 start. (2) Email provider in P0 — default SMTP relay via ESO secret; decide by wave 5. (3) Temporal namespace per tenant vs shared with search attribute — default shared; decide by wave 48.

0.5 Wave 5: UX Shell, Number Sequences, Audit & Observability

0.5.1 Scope and business context

AX users work in list pages and details forms generated from the AOT (form datasources, `ListPageInteraction`, action panes, FactBoxes, `SysQueryForm` advanced filters, `SysLastValue` personalization, `SysTableLookup` lookups, Infolog) and D365 workspaces, saved views and the Excel add-in; SAP users work in SAP GUI dynpros and ALV grids with layout variants, F4 search helps, SE16N, and the Fiori launchpad with catalogs and groups. Document numbering comes from AX `NumberSequenceTable` / `NumberSequenceReference` / `NumberSequenceScope` (shared, company, legal entity, fiscal calendar period, operating unit) with continuous and non-continuous modes and `NumberSequenceList` pre-allocation, or SAP `SNRO` number range objects with `NRIV` intervals, buffering (`NRIV_LOKAL`) and gap-free settings. Change tracking is AX `SysDatabaseLog` / `SysDatabaseLogSetup` (Insert/Update/Delete/RenameKey) or SAP change documents (`CDHDR`/`CDPOS`, `SCDO` objects, `DBTABLOG` table logging). Monitoring is LCS environment monitoring or Solution Manager / ST03N / SM21 / ST22. Wave 5 replaces all of this with a React module-federation shell, a metadata-driven form renderer, a number sequence service with a Jepsen-tested gap-free mode, an audit service that derives field-level history from CDC into a hash-chained append-only store, OpenTelemetry across every service into Prometheus/Tempo/Loki/Grafana, and CSV/XLSX import-export against any entity. At wave end the admin workspace runs at `app.<env>.passage.dev` and Phase 0 is demonstrated to stakeholders. Out of scope: full-text search across documents (wave 46), report designer (wave 15), mobile PWA (wave 32), localization of labels (wave 48), fiscal-period scopes for sequences (wave 6 supplies calendars; scope type is reserved here).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Editing 50 vendor records in AX means 50 details-form round trips (open, edit, save) or an Excel add-in session that fails on the first validation error; SAP needs an LSMW recording or XK02 one by one	Forms are record-at-a-time; bulk edit is an external tool	Grid inline edit, bulk edit of a field across ≤ 1,000 selected rows, validated per row with per-row error reporting	Clicks for 50 edits: ~300 → ~6; failed bulk sessions due to one bad row: all-or-nothing → partial with error file
Continuous number sequences in AX serialize on `NumberSequenceTable` for the whole transaction, causing lock waits on sales-order entry; SAP gap-free NRIV without buffering shows as `NRIV` lock contention in SM12	Lock held for the caller's full transaction	Allocation row with reserved → used/released lifecycle; lock held only for the allocation statement	Gap-free allocation p99: 100s of ms under load → ≤ 30 ms; buffered p99 ≤ 2 ms; gaps and duplicates: 0 (Jepsen-tested)
AX `SysDatabaseLog` must be enabled per table/field in `SysDatabaseLogSetup` and slows every write via triggers in application code; SAP change documents exist only where the developer called `CHANGEDOCUMENT_*` function modules	Audit is opt-in application code that can be forgotten	Audit derived from CDC on every `audit=true` field; no code path can skip it; hash chain proves integrity	Fields with audit coverage: whatever developers remembered → 100% of flagged fields; write-path overhead: trigger cost → 0

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Tracing a slow save: AX trace parser / SQL profiler sessions, SAP ST05/SAT/STAD per work process; no end-to-end id across UI, server, DB and interfaces	No distributed trace context	W3C `traceparent` from the shell through gateway, service, Postgres, outbox, Kafka consumer and Temporal	Time to locate the slow span of one user action: hours of expert time → < 1 min from the audit entry's trace link
SAP GUI and AX client forms fail basic accessibility; keyboard navigation differs per form; ALV variants and `SysLastValue` are per-user and non-shareable	Forms hand-built per screen; no accessibility gate	Generated forms with ARIA grid semantics, one keyboard model, axe in CI; saved views shareable by role	WCAG 2.1 AA violations on generated forms: unknown → 0 in CI; shared views: 0 → any role

0.5.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Navigation / workspaces	Navigation pane, area pages, D365 workspaces and tiles	Fiori launchpad (FLP), catalogs/groups, SAP menu (SE43)	Shell navigation by workspace, tiles bound to saved views	full
List page	List page framework, `ListPageInteraction`, action pane, quick filter	ALV grid (SALV), SE16N, layout variants	Generated list page with grid, filters, saved views	full
Details form / line grids	Details form, `FormRun`, datasources, line grid	Dynpro header/item screens, table controls	Generated detail page with header + line grids	full
Lookups	`SysTableLookup`, EDT relation lookups	F4 search helps (DD30L), personal value list	Generated lookup from `RelationDefinition` with recent values	full
Advanced filter / query	`SysQueryForm`, `Query` ranges, saved queries	Dynamic selections, SE16N selection screen	Filter builder over the list filter grammar; saved as view	full
Personalization	`SysLastValue`, usage data, D365 personalization	Layout variants, user parameters (SU3)	`SavedView`, `Personalization` entities per user and shared	full
Number sequences	`NumberSequenceTable`, `NumberSequenceReference`, `NumberSequenceScope`, `NumberSeq::newGetNum`, continuous flag	SNRO objects, NRIV intervals, buffering NRIV_LOKAL, gap-free flag	numseq service: sequence, scope, continuous/non-continuous, block pre-allocation	full
Database log / change documents	`SysDatabaseLog`, `SysDatabaseLogSetup`, database log viewer	CDHDR/CDPOS, SCDO, DBTABLOG (SCU3)	audit service from CDC; `AuditEntry` / `AuditField`; viewer per record	improved (hash chain)
Excel integration	Excel add-in (OData data entities), Open in Excel	ALV export, Analysis for Office, LSMW upload	CSV/XLSX import-export per entity with staging and validation	partial (no live Excel binding)
Monitoring	LCS environment monitoring, Event Viewer, SQL insights	Solution Manager, ST03N, SM21, ST22, STAD	OpenTelemetry → Grafana/Tempo/Loki/Prometheus; SLO dashboards	improved

0.5.3 Functional requirements

- **FR-5.1 Shell** — Must. The system shall ship a React host (module federation, `web/shell`) that loads remotes by manifest, renders workspace navigation, a global search box (title fields of any entity the user may read, p95 ≤ 300 ms), the legal entity switcher, the work item inbox badge, and a notification tray; first contentful paint ≤ 1.5 s on a 10 Mbps connection.
  - Notes: remotes are versioned by digest and pinned per environment; the shell degrades to an error tile when a remote fails to load.
- **FR-5.2 Metadata-driven forms** — Must. The renderer shall generate, from `EntityDefinition`, a list page (grid, quick filter, filter builder, saved views, column chooser), a detail page (sections by `visibility`, tabs when > 24 fields, line grids for child relations, lookups from relations, attachments and notes panels) and dialogs for create; no entity needs hand-written UI to be usable.
  - Notes: field order from `ordinal`; `advanced` visibility collapses into a "More" section; `hidden` never renders; `pii=sensitive` renders masked with a reveal action gated by permission.
- **FR-5.3 Grid** — Must. The grid shall support virtual scrolling to 100,000 rows, inline edit with per-cell validation, multi-row select, bulk edit of one field across selected rows (≤ 1,000), column sort/resize/reorder/pin, group by one column, and export of the current view.
- **FR-5.4 Keyboard-first** — Must. Every action shall be reachable by keyboard: `/` focus search, `n` new, `Ctrl+S` save, `Ctrl+Enter` save and close, `Esc` cancel, `Ctrl+Shift+F` filter builder, arrows and `Tab` for cell navigation, `F2` edit cell, `Enter` open row, `Ctrl+Shift+V` saved views; focus order follows reading order; shortcuts are listed in a `?` overlay.
- **FR-5.5 Accessibility** — Must. All shell screens and generated forms shall pass axe-core WCAG 2.1 AA rules with zero violations in CI (Playwright + axe), support 200% zoom without horizontal scroll, and expose ARIA grid semantics.
- **FR-5.6 Validation timing** — Must. Field-level validation runs on blur (client JSON Schema, then `validateField` server call when the field has a hook); write validation runs on save; server `422 errors[]` map to fields and unresolved errors show in a summary list with focus jump.
- **FR-5.7 Saved views and personalization** — Must. Users shall save filter + sort + columns + grouping as a `SavedView` (personal or shared with a role), set one as default per entity, and have column widths/order and last legal entity persisted as `Personalization`.
- **FR-5.8 Number sequences** — Must. The numseq service shall define sequences with `code`, `scope_type` (shared, legal_entity, legal_entity_fiscal_period (reserved), operating_unit (reserved)), `format` segments (constant, number with fixed width 1–18, year `YYYY`/`YY`, legal entity code, scope value), `smallest`, `largest`, `next`, `continuous` flag, `manual` flag, `allow_to_lower`, `block_size` (non-continuous, default 50, max 1,000), `wrap` flag, and `in_use` flag; total formatted length ≤ 40.
  - Notes: references bind (entity, field, legal entity) to a sequence, mirroring `NumberSequenceReference`; `numseq(<code>)` is a valid field default from wave 3.
- **FR-5.9 Gap-free and buffered allocation** — Must. Continuous sequences shall allocate inside the caller's transaction with a `SELECT … FOR UPDATE` on the sequence/scope row and record an `Allocation` row (status reserved → used | released); a rollback releases the number to be reused by the next caller; p99 ≤ 30 ms at 500 rps per sequence. Non-continuous sequences shall pre-allocate a block per service instance and serve from memory, p99 ≤ 2 ms; unused numbers in a block are lost on restart (documented, like `NRIV_LOKAL`).
- **FR-5.10 Audit capture** — Must. The audit service shall consume Debezium change events for every table whose entity or field has `audit=true`, and write `AuditEntry` (operation insert/update/delete, actor from `updated_by`, `trace_id` from the transaction's outbox header or `pg_current_xact_id` correlation, `occurred_at`) and `AuditField` rows for changed fields only, with `pii=sensitive` values stored as SHA-256 digests plus masked display.

    - Notes: entry visible in the viewer within 2 s (p95) of commit; the audit database is `audit`, separate from domain databases; role `passage_app` has INSERT and SELECT only.
- **FR-5.11 Immutable hash chain** — Must. Each `AuditEntry` shall store `prev_hash` (previous entry for the same tenant) and `hash = SHA-256(prev_hash ‖ canonical_json(entry + fields))`; a daily job writes the chain head per tenant to an anchor table and to object storage; `GET …/verify` recomputes a range.
- **FR-5.12 Audit viewer** — Must. From any record the user shall open "History" showing entries with user, time, trace id, operation and before/after per field, filterable by field and date, exportable; a global audit workspace searches by entity, record, user, date range.
- **FR-5.13 OpenTelemetry** — Must. Every service shall emit traces (W3C `traceparent` propagated gateway → service → Postgres spans → outbox header → Kafka consumer → Temporal activities), RED metrics (`http_server_requests`, `db_query_duration`, `kafka_consume_lag`, `temporal_activity_duration`) and structured JSON logs with `trace_id`, `tenant_id`, `actor`; the shell emits browser spans for navigation and API calls.
    - Notes: sampling 100% in dev, tail-based 10% + 100% of errors and > 1 s in stage/prod; PII never in span attributes (lint rule).
- **FR-5.14 Dashboards and alerting** — Must. Grafana shall ship per-service SLO dashboards (availability, latency, error budget burn), platform dashboards (Kafka lag, Temporal, Postgres, Argo), and Alertmanager routes to a PagerDuty-compatible webhook with severities page/ticket/notify.
- **FR-5.15 Import/export** — Must. Users shall import CSV or XLSX into any entity through staging (header mapping saved as a template per entity, type coercion, validation using the entity's JSON Schema and hooks, error file with row and field) and export any list view to CSV/XLSX respecting RLS, permissions and PII masks; 100,000 rows import ≤ 5 min; runs as a wave-4 batch job.

0.5.4 Data model

NumberSequence (`numseq.sequence`) and **NumberSequenceAllocation** (`numseq.allocation`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
code	varchar(20)	yes	unique per tenant (`Vend_Acct`, `SO_Num`)
scope_type	enum	yes	shared, legal_entity, legal_entity_fiscal_period, operating_unit
scope_value	varchar(32)	no	legal entity code etc.; unique with code
format	jsonb	yes	segments; e.g. `[{"const":"SO-"},{"year":"YYYY"},{"const":"-"},{"number":6}]`
smallest, largest, next	bigint	yes	1 ≤ smallest ≤ next ≤ largest ≤ 10^18
continuous, manual, allow_to_lower, wrap, in_use	bool	yes	defaults false except in_use
block_size	int	yes	1 for continuous; 50 default
version	int	yes	optimistic lock
Allocation: id, sequence_id, number, status (reserved, used, released), tx_ref, requested_by, requested_at, used_at	—	—	only for continuous sequences

AuditEntry (`audit.entry`, partitioned monthly) and **AuditField** (`audit.field`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id, legal_entity_id	uuid	yes/no	
entity	varchar(48)	yes	
record_id	uuid	yes	
operation	enum	yes	insert, update, delete
actor_id	uuid	yes	from updated_by / created_by; `system` for jobs
occurred_at	timestamptz	yes	commit time
trace_id	char(32)	no	W3C trace id
tx_id	bigint	yes	Postgres xid for grouping
seq	bigint	yes	per-tenant monotonic
prev_hash, hash	char(64)	yes	SHA-256 hex
AuditField: entry_id, field, old_value jsonb, new_value jsonb, pii enum, digest char(64)	—	—	old/new null when masked

SavedView (`ui.saved_view`) and **Personalization** (`ui.personalization`)

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id, owner_id	uuid	yes	
entity	varchar(48)	yes	
name	varchar(60)	yes	unique per owner + entity
filter, sort, columns, group_by	jsonb	yes/no	list grammar from FR-3.6
shared_with_role_id	uuid	no	null = personal
is_default	bool	yes	one per owner + entity
version	int	yes	
Personalization: user_id, key (varchar 80), value jsonb, updated_at	—	—	e.g. `grid.Vendor.columns`, `session.legal_entity`

ImportJob (`io.import_job`): id, tenant_id, entity, template_id, file_ref, status (uploaded, validating, validated, loading, completed, failed, canceled), rows_total, rows_ok, rows_failed, error_file_ref, batch_run_id, created_by, created_at.

Invariants: `next ≤ largest` unless `wrap`; for continuous sequences the set of `used` allocation numbers has no gaps between `smallest` and `next-1`; `hash` of entry n equals SHA-256(prev_hash of n ‖ canonical(n)) and `prev_hash` of n+1 equals `hash` of n; one default SavedView per (owner, entity). Indexing: `sequence(tenant_id, code, scope_value)` unique; `allocation(sequence_id, number)` unique; `entry(tenant_id, entity, record_id, occurred_at desc)`; `entry(tenant_id, seq)` unique; `saved_view(tenant_id, entity, owner_id)`. Partitioning: `audit.entry` and `audit.field` by month; retention 10 years for `transaction` table-group entities, 3 years otherwise, enforced by partition detachment to object storage from wave 46.

0.5.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/POST/PUT	/v1/numseq/sequences, /{id}	Maintain sequences and references	PUT: If-Match	Maintain number sequences
POST	/v1/numseq/sequences/{code}/next?scope=	Allocate (continuous: inside caller tx via `tx_ref`)	`Idempotency-Key` returns the same number	internal / per entity create
POST	/v1/numseq/allocations/{id}/use, /release	Finalize or release a continuous allocation	allocation id	internal
GET	/v1/audit/entries?entity=&record_id=&actor=&from=&to=	Query history	n/a	View audit log (or owner of record)
GET	/v1/audit/entries/{id}	Entry with fields	n/a	View audit log
POST	/v1/audit/verify?tenant=&from_seq=&to_seq=	Recompute chain	n/a	View audit log
GET/POST/PUT/DELETE	/v1/ui/views, /{id}	Saved views	PUT: If-Match	Maintain saved views (own)
GET/PUT	/v1/ui/personalization/{key}	Personalization	last write wins	self
GET	/v1/ui/forms/{entity}	Form definition (ETag)	n/a	any
GET	/v1/ui/search?q=&limit=	Global search	n/a	any (results filtered by read permission)
POST/GET	/v1/io/imports, /{id}, /{id}/validate, /{id}/load, /{id}/errors	Import lifecycle	POST: file sha256	Import data
POST	/v1/io/exports	Export a view (async, returns file ref)	request id	Export data

Event	Trigger	Payload fields	Consumers
dev.passage.numseq.number.allocated.v1	continuous allocation `used` or `released`	sequence_code, scope_value, number, status, tx_ref, actor	audit
dev.passage.audit.entry.recorded.v1	entry committed	entry_id, tenant_id, entity, record_id, operation, hash, seq	shell live history, analytics (wave 46)
dev.passage.audit.chain.anchored.v1	daily anchor	tenant_id, seq, hash, anchor_ref	compliance evidence (wave 48)
dev.passage.io.import.completed.v1 / failed.v1	import terminal	import_id, entity, rows_ok, rows_failed, error_file_ref	shell inbox, audit
dev.passage.ui.view.shared.v1	saved view shared	view_id, entity, role_id, owner	shell cache

0.5.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Admin workspace (home)	Tiles: users, roles, entities, extensions, workflows, jobs, sequences, audit, imports	counts, failed runs, pending work items	per tile duty
Generated list page (any entity)	New, open, inline edit, bulk edit, filter, save view, export, history	columns from metadata, status, updated	entity read/update
Generated detail page	Save, save and close, delete, open related, history, attachments	sections, tabs, line grids, lookups	entity actions
Number sequences	Create, set format, test next, mark in use	code, scope, format preview, next, continuous	Maintain number sequences
Audit viewer (record and global)	Filter, verify chain, export	time, actor, operation, field, old, new, trace link	View audit log
Import/export	Upload, map, validate, load, download errors	entity, template, rows, status	Import data / Export data
Observability (Grafana, linked)	Open trace from any record history; SLO dashboards	trace id, service, latency, error budget	Platform engineer

Rules: inline edit commits per row on `Enter` or row blur with optimistic locking (409 shows a diff and lets the user reload or overwrite with permission); bulk edit previews affected rows and runs as a batch job above 200 rows; validation messages appear under the field and in a summary list; personalization never changes required-field visibility; the trace link from the audit viewer deep-links into Tempo with the `trace_id`.

0.5.7 Business rules and edge cases

- **BR-5.1** Continuous allocation lifecycle: reserved → used (on caller commit) | released (on caller rollback or 10-minute reservation timeout); a released number is the next number served, preserving gap-freedom; `next` only advances when no released numbers exist.
- **BR-5.2** Two concurrent continuous requests serialize on the row lock; the lock is held only for the allocation statement, not the caller's whole transaction, which is why `Allocation` rows exist.
- **BR-5.3** Non-continuous blocks are per pod; a pod restart loses at most `block_size − 1` numbers per sequence; `allow_to_lower=false` forbids editing `next` downward.
- **BR-5.4** Year segment is evaluated in the legal entity's time zone at allocation time; a sequence with a year segment and `continuous=true` resets `next` to `smallest` on 1 January only when `reset_on_year=true` (default false).

- **BR-5.5** `in_use=false` sequences reject allocation with 409 `SEQUENCE_NOT_IN_USE`; `manual=true` sequences require the caller to supply the number and enforce uniqueness per scope.
- **BR-5.6** Reaching `largest` with `wrap=false` returns 409 `SEQUENCE_EXHAUSTED` and raises an alert at 90% consumption.
- **BR-5.7** Audit entries are never updated or deleted by the application; a verification mismatch raises `AuditChainBroken` (page severity) and freezes further anchoring until acknowledged.
- **BR-5.8** Updates that change only `updated_at/version` produce no audit entry; updates to `audit=false` fields produce an entry with no `AuditField` rows only if another audited field changed.
- **BR-5.9** Deleting a record writes an entry with all audited fields as `old_value`; soft delete is an update with field `deleted_at`.
- **BR-5.10** Saved view sharing: a shared view is read-only to others; copying creates a personal view; deleting a shared default view reverts affected users to the system default.
- **BR-5.11** Import loads are all-or-nothing per file by default; `partial=true` loads valid rows and produces the error file for the rest; re-importing the same file sha256 is a no-op with 200 and the previous import id.

0.5.8 Acceptance criteria

- **AC-5.1** Given the 10 reference entities plus 5 extension cases, When Playwright + axe runs list, detail and create dialogs, Then 0 WCAG 2.1 AA violations and the keyboard suite completes every action without a mouse. (FR-5.2, FR-5.4, FR-5.5)
- **AC-5.2** Given a continuous sequence and 10,000 concurrent allocation requests with 10% random rollbacks (Jepsen-style with clock skew and partition injection), When the run ends, Then used numbers are contiguous, unique and `next − smallest = count(used)`. (FR-5.9, BR-5.1)
- **AC-5.3** Given a non-continuous sequence under 2,000 rps across 3 pods, When measured, Then p99 ≤ 2 ms and no duplicates. (FR-5.9)
- **AC-5.4** Given a field change on any audited entity, When committed, Then the audit viewer shows user, time, old/new and the trace id within 2 s (p95 over 1,000 changes). (FR-5.10, FR-5.12)
- **AC-5.5** Given a tampered `AuditField.new_value` (edited directly in the DB in test), When `verify` runs, Then it reports the first broken seq and `AuditChainBroken` fires. (FR-5.11, BR-5.7)
- **AC-5.6** Given a click "Save" on a vendor stub, When the request flows, Then one trace in Tempo contains spans for gateway, service, Postgres, outbox publish, Kafka consumer (audit) and the shell. (FR-5.13)
- **AC-5.7** Given a CSV of 100,000 vendor stubs with 1% invalid rows, When imported with `partial=true`, Then 99,000 rows load in ≤ 5 min and the error file lists 1,000 rows with field and code. (FR-5.15)
- **AC-5.8** Given a user saves a view with filter and columns and marks it default, When they reopen the list, Then the view applies; another user in the shared role sees it read-only. (FR-5.7, BR-5.10)
- **AC-5.9** Given the SLO dashboard for `identity`, When error rate is forced to 2% for 10 min in stage, Then `SLOBurnRateHigh` reaches the webhook with severity page. (FR-5.14)

0.5.9 Test plan and fixtures

Fixtures: `fixtures/numseq/jepsen-config.yaml` (10k clients, nemesis schedule), `fixtures/audit/tamper-cases.sql`, `fixtures/io/vendors-100k.csv` (1% invalid), `fixtures/ui/keyboard-script.json` (every shortcut), `fixtures/ui/entities-15.yaml` (10 reference + 5 extension variants), `fixtures/otel/trace-expectation.json` (span list for AC-5.6).

Test	Type	Target	Pass criterion
axe + keyboard	E2E (Playwright)	shell + renderer	0 violations, 100% shortcuts
Gap-free Jepsen	Property / chaos	numseq	0 gaps, 0 dups under partitions and skew
Buffered throughput	Performance	numseq	p99 ≤ 2 ms @ 2,000 rps
Audit latency	Integration	CDC → audit	p95 ≤ 2 s
Chain verify	Negative	audit	tamper detected at exact seq
Trace continuity	Integration	OTel pipeline	all expected spans share one trace id
Import 100k	Performance	importexport	≤ 5 min, error file correct
Form renderer snapshot	Snapshot	renderer	15 entities × 3 pages stable
Grid virtualization	Performance (browser)	grid	100k rows scroll at ≥ 50 fps on reference laptop

Reconciliation: for each continuous sequence, `count(used allocations) = next − smallest − count(released outstanding)`; audit entries per table per day equal Debezium change count for audited tables; import `rows_ok + rows_failed = rows_total`.

0.5.10 Deployment and operations

Charts: `shell-web` (static assets behind the gateway, remotes manifest in ConfigMap), `forms-renderer` (remote), `shell-bff` (TS: views, personalization, search, SSE), `numseq` (Go, CNPG `numseq`), `audit` (Go consumer + API, CNPG `audit`, monthly partition job), `importexport` (TS, object storage bucket via ESO credentials), `observability` umbrella (OTel Collector DaemonSet + gateway, Prometheus, Tempo, Loki, Grafana, Alertmanager). Config: `REMOTES_MANIFEST_URL`, `OTEL_TRACES_SAMPLER=parentbased_traceidratio`, `AUDIT_RETENTION_YEARS`, `NUMSEQ_RESERVATION_TIMEOUT_SECONDS=600`, `ALERT_WEBHOOK_URL` (ESO). Migrations: additive; `audit.entry` partitions created 3 months ahead by a scheduled job. Feature flags: `ui.inline_edit`, `ui.bulk_edit`, `audit.capture` per domain. Scheduled jobs: audit anchor 00:30 UTC, partition maintenance weekly, sequence exhaustion check hourly, import staging cleanup after 7 days. SLOs: shell availability 99.9%, form load p95 ≤ 800 ms, numseq continuous p99 ≤ 30 ms, audit visibility p95 ≤ 2 s, trace ingestion loss < 0.1%. Alerts: `SLOBurnRateHigh` (multi-window 1h/6h), `AuditLagHigh` (> 10 s), `AuditChainBroken`, `SequenceNearExhaustion`, `RemoteLoadFailure`, `OTelCollectorDropping`. Runbook: audit consumer rebuild from Kafka retention, sequence reservation stuck, remote rollback (manifest pin), Grafana datasource auth. Rollback: digest rollback per chart; shell remotes roll back by manifest pin without redeploying the host.

0.5.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `NumberSequenceTable` (NumberSequence, Txt, Format, Lowest, Highest, NextRec, Continuous, Manual, InUse, Wrap) + `NumberSequenceScope`	NumberSequence	`Format` `###-######` converted to segments; `NextRec` → `next` after cutover freeze; scope `DataAreaId` → legal entity code
AX `NumberSequenceReference` (DataTypeId → NumberSequenceId)	sequence reference	EDT name mapped to Passage entity/field by crosswalk
SAP NRIV (OBJECT, SUBOBJECT, NRRANGENR, FROMNUMBER, TONUMBER, NRLEVEL, EXTERNIND)	NumberSequence	SUBOBJECT (BUKRS) → scope value; EXTERNIND='X' → manual; buffering settings from TNRO → block_size
AX `SysDatabaseLog` (table, recId, logType, createdBy, createdDateTime, data container)	AuditEntry/AuditField (read-only import)	loaded as historical entries with `source=ax`, excluded from the hash chain (separate chain `legacy`)
SAP CDHDR/CDPOS (OBJECTCLAS, OBJECTID, CHANGENR, USERNAME, UDATE, UTIME, TABNAME, FNAME, VALUE_OLD, VALUE_NEW)	AuditEntry/AuditField	same treatment; OBJECTID resolved to Passage record ids via the migration run ledger (wave 10)
D365 saved views / AX `SysLastValue` usage data; SAP ALV layout variants (LTDX)	SavedView	optional; only named, shared variants are imported

Not migrated: AX Infolog history, SAP SM21/ST22 logs, SAP DBTABLOG (table logging) beyond customizing tables, personal (unnamed) usage data. Number sequence `next` values are taken at cutover (wave 49) and never earlier.

0.5.12 Dependencies, risks, open questions

Depends on: wave 4 (CDC stream for audit, batch jobs for import/export, events for inbox), wave 3 (form definitions from metadata, `numseq()` defaults), wave 2 (duties `Maintain number sequences`, `View audit log`, `Import data`, `Export data`, `Maintain saved views`), wave 1 (gateway, TLS, charts).

Risk	Likelihood	Impact	Mitigation
Generated forms are usable but ugly; stakeholders judge the platform by them	High	Medium	Design tokens and a 15-entity visual review before the demo; hand-built workspaces for admin home
Audit consumer falls behind on bulk loads	Medium	Medium	Partitioned consumer group (12), batching 500 rows per commit, `AuditLagHigh` alert
Gap-free sequences used where not required (throughput)	Medium	Medium	Default non-continuous; continuous requires a documented legal reason in the sequence description
Tail sampling hides a rare slow path	Low	Medium	100% sampling of > 1 s and errors; exemplar links from metrics

Open questions: (1) Grid implementation (in-house vs. open-source data grid library) — default open-source MIT grid with virtualization; decide by wave 5 start. (2) Audit anchor target (object storage only vs. external timestamping) — default object storage with WORM policy; decide by wave 48. (3) SSE vs WebSocket for the inbox — default SSE; decide by wave 5 mid-point. (4) Whether `legal_entity_fiscal_period` scope ships in wave 5 or wave 6 — default wave 6 with the enum reserved now; decide by wave 6 start.

Part 1 — Phase 1: Enterprise Structure, Master Data & Engineering Data (waves 6–10)

1.0 Phase overview

Business outcome at phase exit. A master-data steward can model the demo manufacturer end to end on Passage without touching AX or SAP: three legal entities, two plants, five warehouses and their hierarchies; one governed party record per customer, vendor and contact; a 500-item, 50-master product catalog released per legal entity; multi-level BOMs and routings under engineering change control; and a migration framework that has loaded all of it from a synthetic AX extract with 100% reconciliation. Every later phase (journal in wave 11, inventory in wave 16, orders in wave 26) filters by this enterprise structure, posts against these parties and products, and loads its own legacy data through this framework.

Phase-exit sign-off criteria

- `make demo-seed` loads the demo manufacturer (3 legal entities, 2 sites, 5 warehouses, 3 hierarchy purposes) in under 2 minutes and every list page re-scopes on legal-entity switch (property test over 20 entities, 0 leaks).
- Duplicate detection on the 10k-party fixture achieves precision ≥ 0.95 and recall ≥ 0.90; merge leaves zero dangling references (referential-integrity audit returns 0 rows).
- Variant explosion of a 3×10×10 master creates 1,000 variants in < 5 s; GS1 check-digit tests pass for all 40 published GS1 vectors.
- 7-level, 2,000-component BOM explosion completes in < 500 ms and equals the golden result byte for byte.
- Synthetic AX extract (50k customers, 20k vendors, 100k items, 10k BOMs) migrates in < 30 min; re-run produces 0 changes; reconciliation report shows 0 unexplained exceptions.
- Stakeholder walkthrough recorded: create party → release product → build BOM → approve ECO → run migration.

Cross-wave data model summary

Entity	Owning service	Key relationships
LegalEntity	org-service (Go)	tenant 1:N legal_entity; legal_entity 1:N fiscal_calendar_assignment
OperatingUnit	org-service	type ∈ {department, cost_center, business_unit, value_stream}; node of HierarchyVersion
Site / Warehouse / Location	org-service	legal_entity 1:N site 1:N warehouse 1:N location
HierarchyType / HierarchyVersion / HierarchyNode	org-service	purpose-bound, effective-dated tree over OrgUnit ids
PostalAddress / AddressPurpose	org-service	referenced by Party, Site, Warehouse, LegalEntity
FiscalCalendar / FiscalPeriod	org-service	consumed by ledger (wave 11) for period status
UnitOfMeasure / UomConversion	org-service	product-specific conversions live in product-service
Party (Person/Organization)	party-service (Go)	1:N PartyRole; 1:N PartyAddress; 1:N TaxRegistration
Customer / Vendor / Contact / Employee role	party-service	role rows keyed (party_id, legal_entity_id)
DuplicateCandidate / MergeCase	party-service	pairs of party_id with score; Temporal workflow id
Product / ProductMaster / ProductVariant	product-service (Go)	master 1:N variant; dimension_group FK
ReleasedProduct	product-service	(product_id, legal_entity_id) with site/warehouse defaults
ItemGroup / ItemModelGroup / Category	product-service	referenced from ReleasedProduct
ConfigurationModel	product-service (Python solver)	master 1:N model; constraints in JSON
BomHeader / BomVersion / BomLine	engineering-service (Go)	version → (product_id, site_id, effective range)
RouteHeader / RouteVersion / Operation / OperationRelation	engineering-service	operation N:1 resource requirement
Resource / ResourceGroup / Capability / WorkCalendar	engineering-service	calendar consumed by scheduling (wave 23)
EngineeringChangeRequest / EngineeringChangeOrder	engineering-service	ECO 1:N affected BomVersion/RouteVersion
MigrationProject / MigrationRun / MappingSpec / StagingRow / ReconciliationResult	migration-service (TS + Go loaders)	run N:1 project; row N:1 run

Cross-wave event catalog

Event name	Producer	Consumers	Payload summary
dev.passage.org.legalentity.created.v1	org-service	party, product, ledger (w11), shell	legal_entity_id, code, name, country, currency, fiscal_calendar_id
dev.passage.org.hierarchy.version.activated.v1	org-service	security (OPA bundle), reporting, procurement	hierarchy_type_id, version_id, effective_from, node_count
dev.passage.org.site.created.v1 / warehouse.created.v1	org-service	product (released defaults), inventory (w16)	id, legal_entity_id, parent ids, address_id
dev.passage.org.fiscalcalendar.generated.v1	org-service	ledger (w11), period close (w14)	calendar_id, pattern, period_count, year
dev.passage.party.created.v1 / updated.v1	party-service	search index, migration, audit	party_id, type, name, primary_address_id
dev.passage.party.customer.created.v1 / vendor.created.v1	party-service	AR (w30), AP (w19), sales (w26), purchasing (w17)	party_id, legal_entity_id, account_number, group, terms
dev.passage.party.merged.v1	party-service	every service holding party_id FKs	survivor_id, merged_ids[], reference_rewrite_count
dev.passage.party.change.approved.v1	party-service (Temporal)	audit, notification	party_id, field, old, new, approver
dev.passage.product.created.v1 / variant.created.v1	product-service	engineering, inventory, pricing (w26)	product_id, master_id, dimension_values

Event name	Producer	Consumers	Payload summary
dev.passage.product.released.v1	product-service	inventory, purchasing, sales, costing	product_id, legal_entity_id, item_model_group_id, defaults
dev.passage.engineering.bom.version.activated.v1	engineering-service	planning (w22), production (w21), costing (w39)	bom_version_id, product_id, site_id, effective_from, line_count
dev.passage.engineering.route.version.activated.v1	engineering-service	planning, scheduling (w23), production	route_version_id, operations[]
dev.passage.engineering.eco.released.v1	engineering-service	notification, audit, PLM adapter	eco_id, affected_versions[], release_date
dev.passage.migration.run.completed.v1	migration-service	notification, reconciliation UI, audit	run_id, project_id, counts{staged, loaded, rejected}, hash_match

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Party dedup thresholds tuned to synthetic data miss real-world noise	Medium	High	Splink model trained on 10k fixture plus 2k hand-labeled real-shaped records; thresholds adjustable per tenant	Product
Effective-dated hierarchies become slow under as-of queries at 50k nodes	Medium	Medium	Closure table per version, materialized on activation; p95 target 20 ms enforced in CI	Engineering
Product dimension explosion makes variant counts unbounded	Low	High	Hard cap 10,000 variants per master; explosion is a Temporal job with checkpointing	Engineering
BOM cycles or phantom recursion break explosion	Medium	High	Cycle check on every line insert (closure table), depth cap 25	Engineering
AX/SAP extracts arrive with codepage and enum drift	High	Medium	Mapping DSL with explicit enum tables and reject-with-reason; dry-run before load	Implementation partner
Wave 10 depends on three prior waves landing on time	Medium	Medium	Connectors for DirParty/CustTable built against wave 7 as soon as it passes Proof	Product

1.6 Wave 6: Enterprise Structure & Organization Model

1.6.1 Scope and business context

AX scopes everything by `DataAreaId` (CompanyInfo) and organizes units in `OMOperatingUnit` with hierarchies in `OMHierarchyType`/`OMHierarchyRelationship`; SAP scopes by `BUKRS` (T001) with plants (T001W), storage locations (T001L), sales organizations (TVKO) and purchasing organizations (T024E) wired together in SPRO. Wave 6 builds the single organization model every later service filters by. At the end of the wave an administrator can create a tenant's legal entities, operating units, sites, warehouses and locations; build effective-dated hierarchies for reporting, procurement and security purposes; attach validated postal addresses; generate fiscal calendars; and maintain countries, currencies, units of measure and languages. The shell's legal-entity switcher re-scopes every list page.

Out of scope: ledger period *statuses* (wave 14 owns open/closed per module), warehouse layout detail such as zones and location directives (wave 31), intercompany relationships (wave 40), and any transactional data.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX organization hierarchy designer (OMHierarchyRelationship) only shows one effective date at a time; previewing "what changes on 2026-07-01" means publishing and re-opening; a mistaken publish cannot be undone	Hierarchy relationships are stored as flat ValidFrom/ValidTo rows with no version concept	Versioned trees with draft/published states, as-of slider and a diff view before publish (FR-6.3, FR-6.11)	Preview clicks 9→1; wrong-date publishes recoverable in 100% of cases via superseding version
SAP enterprise structure (OX02/OX10/OX09/OX18/OB29) requires 6+ IMG transactions plus assignments to define one plant; a missing assignment surfaces only when the first goods movement fails	Configuration spread across customizing tables (T001, T001W, T001K, T001L, T024E) with assignment tables between them	One `Site` create carries its legal entity, calendar and default warehouse; invariants are checked at save, not at first transaction (FR-6.5)	Transactions 6→1; structural errors found at save: 100% vs ~0%
AX company switch (DataAreaId) reloads the whole client and loses open forms; cross-company inquiries need `crossCompany` X++	Company is a session property baked into every table	Header-scoped entity context; the shell switches without reload; read-only `ALL` scope behind a duty (FR-6.10)	Switch time 8–15 s→<300 ms; open views preserved 100%
Fiscal calendar period creation in AX (FiscalCalendarPeriod) and SAP (OB29 period tables) is manual per year; 4-4-5 week-53 years are keyed by hand and frequently wrong	No pattern generator	Pattern-driven generation with contiguity proof and golden fixtures (FR-6.7)	Setup effort 45 min/year→<10 s; boundary errors 0 against fixture
UoM conversion rounding silent in AX (UnitOfMeasureConversion stores Numerator/Denominator with decimal drift) and SAP (T006 integer ratios)	Ratio stored as scaled integers, rounded at storage	numeric(28,12) factor+offset, rounding only at display (FR-6.9, BR-6.8)	Round-trip error 1e-6→0 at 12 decimals across 50 pairs

1.6.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Legal entity	CompanyInfo, `OMLegalEntity` form, DataAreaId	T001 company code (OX02)	org-service `LegalEntity`; Legal entities screen	full
Operating units	OMOperatingUnit (dept, cost center, BU, value stream)	Cost center CSKS (KS01), profit center CEPC, controlling area TKA01	`OperatingUnit` with `type` enum	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Org hierarchies	OMHierarchyType, OMHierarchyPurpose, OMHierarchyRelationship, Organization hierarchy designer	Standard hierarchy KSH1/KCH1, enterprise organization	`HierarchyType/HierarchyVersion/HierarchyNode`; Hierarchy designer	improved (multiple purposes with effective-dated versions and as-of API)
Sites	InventSite, Sites form	T001W plant (OX10)	`Site`	full
Warehouses / locations	InventLocation, WMSLocation	T001L storage location (OX09); LAGP bins	`Warehouse`, `Location` (bin-level detail deferred to wave 31)	partial
Addresses	LogisticsPostalAddress, LogisticsLocation, LogisticsLocationRole, DirPartyLocation	ADRC / ADRCT, address version	`PostalAddress`, `AddressPurpose`	full
Fiscal calendars	FiscalCalendar, FiscalCalendarYear, FiscalCalendarPeriod, Ledger calendar	T009 fiscal year variant (OB29), T009B periods	`FiscalCalendar`, `FiscalYear`, `FiscalPeriod`	full (status per ledger deferred to wave 14)
Currencies	Currency table, ISO 4217	TCURC, TCURX decimals	`Currency` reference with minor units	full
Units of measure	UnitOfMeasure, UnitOfMeasureConversion (standard + product-specific)	T006 (CUNI), T006A	`UnitOfMeasure`, `UomConversion`; product-specific in wave 8	full
Countries / regions / languages	LogisticsAddressCountryRegion, LogisticsAddressState, LanguageTable	T005, T005S, T002	`Country`, `Region`, `Language`	full
Company-level parameters	CompanyInfo fields, Ledger (currency, calendar)	T001 fields (WAERS, PERIV, KTOPL)	`LegalEntity` fields + ledger linkage in wave 11	partial (chart of accounts linkage arrives in wave 11)

1.6.3 Functional requirements

- **FR-6.1 Tenant and legal-entity creation** — Must. The system shall create legal entities under a tenant with a unique 4–8 character code, legal name, country, accounting currency, default language and time zone.
  - Code is immutable after creation; names are effective-dated (name history table).
  - A legal entity cannot be deleted; it can be set `inactive`, which hides it from the switcher but keeps all references.
  - Maximum 500 legal entities per tenant in v1.
- **FR-6.2 Operating units** — Must. The system shall maintain operating units of type department, cost_center, business_unit and value_stream, each with a code unique within tenant and type.
  - Operating units are tenant-scoped, not legal-entity-scoped (mirrors AX, where OMOperatingUnit is shared), with an optional default legal entity.
  - A cost center may carry an optional `manager_party_id` resolved after wave 7.
- **FR-6.3 Multi-purpose effective-dated hierarchies** — Must. The system shall support hierarchy types with one or more purposes (reporting, procurement, expenditure, security, centralized_payments, custom) and versioned trees with `effective_from`/`effective_to` ranges that cannot overlap for the same type.
  - A node is any LegalEntity or OperatingUnit; one node appears at most once per version.
  - Draft versions may be edited freely; published versions are immutable; a new version is created by copying the current one.
  - Maximum depth 15, maximum 50,000 nodes per version.
- **FR-6.4 As-of hierarchy queries** — Must. The system shall answer parent, ancestors, descendants and path-to-root for any node as of any date with p95 < 20 ms at 50,000 nodes.
  - Implemented with a closure table materialized at publish time.
- **FR-6.5 Sites, warehouses and locations** — Must. The system shall maintain sites per legal entity, warehouses per site, and locations per warehouse, each with address, time zone and type (`standard`, `transit`, `quarantine`, `subcontractor` for warehouses).
  - Warehouse code unique per legal entity (as in AX InventLocationId per company); site code unique per legal entity.
  - A warehouse cannot move between sites once referenced by an inventory transaction (enforced from wave 16 via a reference-count check).
- **FR-6.6 Postal addresses with purposes** — Must. The system shall store structured postal addresses (ISO 3166-1 alpha-2 country, region, city, postal code, street lines, building, PO box) and attach them to any owner with purposes (business, delivery, invoice, remit_to, service, home, other) and a primary flag per purpose.
  - Country-specific formatting rules for the first seven countries (US, UK, DE, FR, IN, MX, CA); postal code regex per country.
  - Geocoding hook: an adapter interface returning lat/long and a confidence; default adapter is a no-op.
  - Addresses are effective-dated; an address edit creates a new version and keeps the old one for documents already printed.
- **FR-6.7 Fiscal calendars** — Must. The system shall generate fiscal years and periods from patterns: calendar-year (12 periods), 4-4-5 / 4-5-4 / 5-4-4 (12 periods + 53rd week handling), 13-period, and custom (user-entered period boundaries).
  - Each year has exactly one opening period and one closing period in addition to operating periods (AX convention), configurable off for SAP-style special periods 13–16.
  - Periods within a year are contiguous and non-overlapping; the generator proves this before saving.
- **FR-6.8 Reference data** — Must. The system shall seed and maintain ISO 3166 countries and regions, ISO 4217 currencies with minor units (JPY 0, KWD 3, USD/EUR 2), ISO 639-1 languages, and UCUM-aligned units of measure in classes (quantity, mass, length, volume, area, time, temperature, each).
  - Reference data is tenant-visible but platform-owned; tenants may add custom units and deactivate rows, not delete.
- **FR-6.9 UoM conversions** — Must. The system shall store conversions as `factor` and `offset` with numeric(28,12) precision and compute any-to-any within a class by path search of at most two hops.
  - Round-trip of 1 unit through A→B→A must return exactly 1 to 12 decimal places for the 50-pair fixture.
  - Temperature uses the offset (°C→°F: factor 1.8, offset 32).
- **FR-6.10 Legal-entity context switch** — Must. The system shall carry `X-Passage-LegalEntity` on every request; generated list pages filter by it, and the OPA policy denies requests for entities the user has no role assignment in.

- **FR-6.11 Hierarchy designer UI** — Should. The system shall provide a drag-and-drop tree editor with an effective-date slider that previews the tree as of a chosen date and highlights nodes that change between versions.
- **FR-6.12 Seed and export** — Should. The system shall export and import the complete organization model as a versioned JSON package (used by `make demo-seed` and by the migration framework in wave 10).
- **FR-6.13 Address validation adapter** — Could. The system shall expose an address-validation adapter interface (normalize, suggest) with a null default; third-party adapters are out of scope for this wave.

1.6.4 Data model

LegalEntity

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS key
code	text(8)	yes	unique per tenant, immutable, `^[A-Z0-9]{2,8}$`
name	text(100)	yes	current name; history in legal_entity_name
country_code	char(2)	yes	FK country
accounting_currency	char(3)	yes	FK currency; immutable after first posting (wave 11)
default_language	text(5)	yes	BCP 47, default `en-US`
time_zone	text	yes	IANA zone
primary_address_id	uuid	no	FK postal_address
registration_number	text(40)	no	company registry number
status	enum	yes	active, inactive
created_at / updated_at / version	timestamptz, int	yes	optimistic lock on `version`

HierarchyVersion / HierarchyNode

Field	Type	Required	Notes
id	uuid	yes	PK
hierarchy_type_id	uuid	yes	FK; type carries purposes[]
version_no	int	yes	monotonic per type
effective_from	date	yes	inclusive
effective_to	date	no	exclusive; null = open-ended
status	enum	yes	draft, published, superseded
node.id	uuid	yes	PK of hierarchy_node
node.version_id	uuid	yes	FK
node.org_unit_id	uuid	yes	LegalEntity or OperatingUnit id; unique per version
node.parent_node_id	uuid	no	null for root; exactly one root per version
node.sort_order	int	yes	sibling order
closure(ancestor_id, descendant_id, depth)	table	yes	rebuilt on publish

PostalAddress

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
country_code	char(2)	yes	FK
region_code	text(10)	conditional	required for US, CA, IN, MX
city	text(60)	yes	
postal_code	text(12)	conditional	validated by country regex
street_1 / street_2 / street_3	text(100)	street_1 yes	
building / po_box	text(20)	no	
latitude / longitude	numeric(9,6)	no	from geocoding hook
valid_from / valid_to	date	yes/no	effective dating
formatted	text	computed	rendered per country template

FiscalCalendar / FiscalPeriod — calendar: id, tenant_id, code, name, pattern (enum), week_start; period: id, calendar_id, fiscal_year, period_no (0 = opening, 1–13 operating, 99 = closing), name, start_date, end_date, type (opening/operating/closing/adjustment).

UnitOfMeasure / UomConversion — uom: id, code (`^[A-Za-z0-9/%]{1,10}$`), name, class, decimal_precision (0–6), system_flag; conversion: from_uom_id, to_uom_id, factor numeric(28,12), offset numeric(28,12), product_id (null here, used in wave 8).

Invariants: exactly one root per published hierarchy version; no overlapping effective ranges per hierarchy type; fiscal periods of a year tile the year with no gap or overlap; a warehouse's legal entity equals its site's legal entity; every `primary` address purpose is unique per (owner, purpose). Indexing: `hierarchy_node(version_id, org_unit_id)` unique; closure table PK (ancestor_id, descendant_id); `postal_address` GIN trigram on `formatted` for search; all tables partitioned logically by tenant_id with RLS policies, no physical partitioning in this wave.

1.6.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/org/v1/legal-entities	Create legal entity	`Idempotency-Key` header, 24 h	Maintain legal entities
GET	/org/v1/legal-entities/{id}	Read	n/a	View organization
PATCH	/org/v1/legal-entities/{id}	Update name/addresses/status	version check (409 on mismatch)	Maintain legal entities
POST	/org/v1/operating-units	Create operating unit	key header	Maintain operating units
POST	/org/v1/hierarchy-types/{id}/versions	Create draft version (copy of current)	key header	Maintain organization hierarchies
PUT	/org/v1/hierarchy-versions/{id}/nodes	Replace the full node set of a draft	full replace, naturally idempotent	Maintain organization hierarchies
POST	/org/v1/hierarchy-versions/{id}:publish	Publish; sets effective_from; builds closure	key header	Publish organization hierarchies
GET	/org/v1/hierarchy-types/{id}/as-of?date=2026-06-01&node=...	Ancestors/descendants/path as of a date	n/a	View organization
POST	/org/v1/sites, /org/v1/warehouses, /org/v1/locations	Create physical structure	key header	Maintain sites and warehouses
POST	/org/v1/addresses	Create address; `owner_type`, `owner_id`, `purposes[]`	key header	Maintain addresses
POST	/org/v1/fiscal-calendars/{id}:generate	Generate years from pattern	idempotent per (calendar, year)	Maintain fiscal calendars
GET	/org/v1/uom/convert?from=KG&to=LB&qty=10	Convert quantities	n/a	any authenticated
POST	/org/v1/export, /org/v1/import	Org model package	import idempotent by code	Administer organization

Commands validate: codes against regex; address against country rules; hierarchy publish rejects cycles, duplicate nodes, multiple roots, overlapping ranges; fiscal generation rejects a year already generated unless `force=true` and the year has no postings (checked via a ledger callback that returns `true` until wave 11 exists).

Event (CloudEvents type)	Trigger	Payload fields	Consumers
dev.passage.org.legalentity.created.v1	commit of legal entity	id, code, name, country_code, accounting_currency, time_zone	party, product, ledger, shell switcher
dev.passage.org.legalentity.updated.v1	PATCH commit	id, changed_fields[], version	shell, search
dev.passage.org.operatingunit.created.v1	commit	id, type, code, name	reporting, ledger dimensions
dev.passage.org.hierarchy.version.activated.v1	publish	hierarchy_type_id, purposes[], version_id, effective_from, node_count	OPA bundle builder, reporting, procurement
dev.passage.org.site.created.v1 / warehouse.created.v1 / location.created.v1	commit	id, legal_entity_id, parent_id, type, address_id	product, inventory
dev.passage.org.address.changed.v1	new address version	address_id, owner_type, owner_id, purposes[], valid_from	document services
dev.passage.org.fiscalcalendar.generated.v1	generate	calendar_id, fiscal_year, periods[]	ledger, period close

1.6.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Organization administration workspace	Tiles: legal entities, operating units, hierarchies, sites/warehouses, calendars, reference data	counts, last-changed, pending drafts	View organization
Legal entities list/detail	New, edit, deactivate, addresses tab, name history	code, name, country, currency, status	Maintain legal entities
Operating units list	New, bulk import (CSV), type filter	type, code, name, default legal entity	Maintain operating units
Hierarchy designer	Drag-drop nodes, add from picker, effective-date slider, diff vs previous version, publish	tree, effective range, node count, change badges	Maintain / Publish organization hierarchies
Sites and warehouses	New site, new warehouse under site, locations grid	code, name, type, address, time zone	Maintain sites and warehouses
Fiscal calendars	New calendar, generate years, period grid	pattern, year, period no, start, end	Maintain fiscal calendars
Reference data (countries, currencies, UoM, languages)	Deactivate, add custom UoM, add conversion, test conversion	code, name, class, factor, offset	Maintain reference data

Interaction rules: all grids support inline edit with validation on blur and a single save per row; `Ctrl+S` saves, `Esc` cancels, `Alt+N` new; the hierarchy designer validates structurally on every drop and semantically (effective range) on publish only; personalization (column order, saved filters) is stored per user per screen by the wave-5 shell; the legal-entity switcher is in the top bar and persists the last choice.

1.6.7 Business rules and edge cases

- **BR-6.1** Legal entity codes are upper-cased on save and compared case-insensitively; `DAT`, `SYS` and `ALL` are reserved.
- **BR-6.2** Hierarchy version state machine: `draft → published → superseded`; `draft` may be deleted; `published` becomes `superseded` only when a later version publishes with `effective_from` greater than its own; a published version can be ended early by publishing a successor, never by editing.
- **BR-6.3** Publishing a version whose `effective_from` is earlier than today requires the duty "Backdate organization hierarchies" and is audited with a reason text (min 10 chars).
- **BR-6.4** Moving a node moves its subtree; dropping a node onto its own descendant is rejected with `ORG-HIER-CYCLE`.

- **BR-6.5** A security-purpose hierarchy triggers an OPA bundle rebuild within 60 s of activation; until the rebuild completes, the previous bundle stays in force (no window with no policy).
- **BR-6.6** Fiscal 4-4-5 years ending on a 53-week boundary add the extra week to the last period of the year; the generator records `week53 = true` on that period.
- **BR-6.7** A fiscal year may be regenerated only while no period of it has any posting; otherwise `ORG-CAL-LOCKED` (409).
- **BR-6.8** UoM conversions are stored once in a canonical direction; the inverse is computed as 1/factor with the offset negated and divided; results are rounded only at display, using the target unit's `decimal_precision`.
- **BR-6.9** Address edits are copy-on-write: documents hold the address version id, not the address id, so a printed invoice never changes retroactively.
- **BR-6.10** Concurrent edits use the `version` column; a stale PATCH returns 409 with the current representation; the UI offers merge or reload.
- **BR-6.11** Deactivating a legal entity is refused while any user has it as their default entity (`ORG-LE-DEFAULT-IN-USE`).

1.6.8 Acceptance criteria

- **AC-6.1** (FR-6.3, FR-6.4) Given 1,000 effective-dated nodes across 5 purposes with 3 versions each, when the as-of API is called for 10,000 random (node, date) pairs, then every parent matches the fixture oracle and p95 latency < 20 ms.
- **AC-6.2** (FR-6.10) Given a user assigned to legal entities USMF and DEMF, when they switch from USMF to DEMF, then all 20 generated list pages show only DEMF rows and a direct API call with `X-Passage-LegalEntity: FRMF` returns 403.
- **AC-6.3** (FR-6.9) Given the 50-pair conversion fixture, when each quantity is converted A→B→A, then the result equals the input to 12 decimal places.
- **AC-6.4** (FR-6.7) Given calendar patterns 4-4-5, 13-period and calendar-year for fiscal 2026, when generated, then period boundaries equal the committed fixture files exactly, including the 53rd week in the 4-4-5 variant for 2028.
- **AC-6.5** (FR-6.3, BR-6.2) Given a published version effective 2026-01-01, when a draft with `effective_from` 2025-12-01 is published without the backdate duty, then the request is rejected with 403 and no version changes state.
- **AC-6.6** (FR-6.6) Given a US address without `region_code`, when saved, then validation fails with field-level error `ORG-ADDR-REGION-REQUIRED`; with `region_code = CA` and postal code `9410`, then error `ORG-ADDR-POSTAL-FORMAT`.
- **AC-6.7** (FR-6.5) Given warehouse WH1 under site S1 in USMF, when a PATCH attempts to set `site_id` to a site in DEMF, then 422 `ORG-WH-LE-MISMATCH`.
- **AC-6.8** (FR-6.12) Given the demo seed package, when imported twice, then the second import reports 0 creates, 0 updates.
- **AC-6.9** (FR-6.11) Given the designer with the slider at 2026-06-01, when a node moved only in the draft version is viewed, then it renders with a "changed" badge and the previous parent shown in a tooltip.

1.6.9 Test plan and fixtures

Fixtures: `org/demo-manufacturer.json` (3 legal entities USMF/DEMF/FRMF, 2 sites, 5 warehouses, 40 operating units, 3 hierarchy types, generated); `org/hierarchy-1k.json` (1,000 nodes, 5 purposes, 3 versions each, generated with seed 6); `org/uom-pairs.csv` (50 pairs, hand-checked); `org/fiscal-patterns/*.csv` (4-4-5, 4-5-4, 5-4-4, 13-period, calendar for 2025–2028, hand-checked against the NRF retail calendar); `org/addresses-7-countries.csv` (140 valid, 70 invalid).

Test	Type	Target	Pass criterion
Hierarchy as-of oracle	property	as-of API	10k samples, 0 mismatches, p95 < 20 ms
Closure rebuild	unit	publish	closure row count = Σ depth over nodes
Entity re-scope	property (Playwright)	20 list pages	0 cross-entity rows
UoM round-trip	unit	convert	12-decimal equality
Fiscal generation	golden	generator	byte-equal CSV
Address validation	table-driven	validator	210/210 expected outcomes
Publish under load	performance	50k-node publish	< 30 s, as-of p95 < 20 ms after
Event contract	contract (AsyncAPI)	7 events	schema registry accepts; consumer stubs decode

Reconciliation: node count per published version = closure rows with depth 0; count of warehouses per legal entity in export = count in database; every legal entity has exactly one `primary` business address or none.

1.6.10 Deployment and operations

Helm chart `passage-org` (Deployment ×2 replicas, CloudNativePG cluster `org-db` with 3 instances, Debezium connector CR for the outbox table). Config: `ORG_HIERARCHY_MAX_NODES=50000`, `ORG_GEOCODER=null`, `ORG_SEED_PACKAGE=demo-manufacturer`. Migrations via Atlas from entity metadata, applied by an Argo CD pre-sync Job. Feature flags: `org.hierarchy_designer`, `org.address_validation_adapter`. Scheduled job: nightly closure-table integrity check (Temporal cron `0 2 * * *`). SLOs: read p95 < 50 ms, as-of p95 < 20 ms, availability 99.9%. Alerts: outbox lag > 60 s, closure integrity failure, OPA bundle rebuild older than 5 min after a security hierarchy activation. Runbook: "Hierarchy publish stuck" (check Temporal workflow `publish-hierarchy-{id}`), "Rebuild closure table" (`passagectl org rebuild-closure --version`). Rollback: Argo rollback to the previous digest; migrations in this wave are additive only, so no down-migration is needed.

1.6.11 Migration from AX / SAP

Source	Passage entity	Notes
AX CompanyInfo / OMLegalEntity (DataAreaId, Name, LanguageId, CurrencyCode via Ledger)	LegalEntity	DataAreaId → code (upper-cased); Ledger.AccountingCurrency → accounting_currency
AX OMOperatingUnit (OMOperatingUnitNumber, OMOperatingUnitType)	OperatingUnit	type enum 1..4 → department/cost_center/business_unit/value_stream

Source	Passage entity	Notes
AX OMHierarchyType, OMHierarchyPurpose, OMHierarchyRelationship (ValidFrom/ValidTo)	HierarchyType, HierarchyVersion, HierarchyNode	each distinct ValidFrom in AX becomes a version; ValidTo max date → null
AX InventSite, InventLocation	Site, Warehouse	InventLocation.InventSiteId → site FK; warehouse type from InventLocation.WMSLocationId usage
AX LogisticsPostalAddress, LogisticsLocation, LogisticsLocationRole, DirPartyLocation	PostalAddress, AddressPurpose	roles Business/Delivery/Invoice/RemitTo map 1:1; others → other
AX FiscalCalendar, FiscalCalendarYear, FiscalCalendarPeriod	FiscalCalendar, FiscalPeriod	Type Opening/Operating/Closing → period type
AX UnitOfMeasure, UnitOfMeasureConversion (non product-specific)	UnitOfMeasure, UomConversion	Factor/Numerator/Denominator → factor; InnerOffset/OuterOffset → offset
SAP T001 (BUKRS, BUTXT, LAND1, WAERS, SPRAS, PERIV)	LegalEntity	PERIV → fiscal_calendar assignment
SAP T001W (WERKS, NAME1, ADRNR) / T001L (LGORT)	Site + Warehouse	plant → site; a default warehouse per plant is created when T001L is empty
SAP CSKS / CEPC / T001K	OperatingUnit	cost center, profit center; KOKRS stored as attribute
SAP ADRC (ADDRNUMBER, NAME1, STREET, POST_CODE1, CITY1, COUNTRY, REGION)	PostalAddress	DATE_FROM → valid_from
SAP T009 / T009B	FiscalCalendar, FiscalPeriod	special periods 13–16 → adjustment periods when `sap_special_periods=true`
SAP T006 / T006A / TCURC / TCURX / T005	reference data	matched to ISO seed by code; unmatched custom units created

Validation: every migrated warehouse resolves to a site; every address passes country rules or lands in the exception queue; hierarchy versions must tile without gaps. Not migrated: AX `OMInternalOrganization` team records (no Passage equivalent in v1), SAP sales/purchasing organizations (modeled in waves 17 and 26 as procurement and sales hierarchies), SAP client (MANDT) which maps to tenant by configuration.

1.6.12 Dependencies, risks, open questions

Depends on: wave 2 (tenant model, RLS, OPA duties), wave 3 (entity metadata and generator), wave 4 (outbox, Temporal for publish jobs), wave 5 (shell, generated forms, audit, number sequences for operating unit numbers).

Risk	Likelihood	Impact	Mitigation
Closure-table rebuild blocks publishing large versions	Medium	Medium	Rebuild runs as a Temporal activity; version stays `publishing` until done
Country address rules incomplete beyond 7 countries	High	Low	Generic fallback rule; country packs in wave 48
Shell switcher semantics conflict with cross-entity screens (consolidation)	Low	Medium	Switcher supports `ALL` for read-only cross-entity views behind a duty

Open questions: (1) Should operating units be legal-entity-scoped as an option for SAP-style cost centers tied to a controlling area? Proposed default: tenant-scoped with optional default legal entity; decide by wave 11. (2) Do we store SAP special periods as adjustment periods or as flags on period 12? Proposed: adjustment periods; decide by wave 14. (3) Is UCUM the canonical UoM code set or ISO 80000? Proposed: UCUM codes with ISO display names; decide by wave 8.

1.7 Wave 7: Global Party & Business Partner Model

1.7.1 Scope and business context

AX keeps one `DirPartyTable` (subtyped into `DirPerson` and `DirOrganization`) under the global address book, with `CustTable` and `VendTable` as company-specific roles pointing at the party; SAP ECC keeps separate `KNA1`/`KNB1` (customer general/company-code) and `LFA1`/`LFB1` (vendor) masters, and S/4 forces them into Business Partner (`BUT000`, roles in `BUT100`, addresses in `BUT020`/`ADRC`) through Customer-Vendor Integration. Wave 7 builds the party model once: one Party per real-world organization or person, with Customer, Vendor, Employee, Contact, Prospect and Bank roles layered per legal entity. At the end of the wave a steward can create and govern parties and roles, run duplicate detection and merge, validate tax registrations, and import a legacy CustTable/VendTable extract. Posting profiles, payment terms and credit limits are captured now as data; the GL hooks that use them arrive in waves 11, 19 and 30.

Out of scope: customer/vendor transactions and balances, bank account verification (wave 36), employee HR attributes beyond the role stub (wave 42), CRM pipeline (non-goal).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
In ECC the same supplier who is also a customer exists as LFA1 and KNA1 with two addresses that drift; S/4 CVI (`CVI_FS_CHECK_CUSTOMIZING`, `MDS_LOAD_COCKPIT`) is a months-long synchronization project	Two masters designed independently, linked by KNA1-LIFNR / LFA1-KUNNR cross references	Party is the only identity; roles reference it; addresses are shared by purpose (FR-7.1, FR-7.2)	Duplicate identities per real-world entity 2+→1; address drift incidents 0
AX duplicate check on DirPartyTable is exact-name only (`DirPartyTable.checkDuplicate`); near-duplicates ("ACME Corp." vs "Acme Corporation") pass and are found at year-end	No probabilistic matching	Fellegi-Sunter model (Splink) on name/tax id/address/phone with blocking (FR-7.5)	Planted near-duplicates caught: recall ≥ 0.90 vs ≈0.30 for exact match
Merging two vendors in AX is manual: re-point every open PO, settlement, and address, or leave one blocked forever; SAP has no merge, only block (XK05)	No merge primitive; FKs scattered across modules	Merge workflow rewrites references atomically through a consumer contract and keeps full role data (FR-7.6)	Merge effort 2–4 h→<1 min; dangling references 0
VAT number entry (CustTable.VATNum / KNA1-STCEG) is free text; wrong numbers surface as EU sales list rejections months later	No validation at entry	Country-specific format checks plus VIES/TIN adapters at save (FR-7.4)	Invalid tax ids at save-time rejected 100%; downstream filing rejections →0

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Creating a customer in AX touches CustTable form, address tab, contact info, posting profile, payment terms: 12+ clicks across 4 tabs; SAP XD01 walks 3 views × 5 tabs	Role and party data mixed in one wide form	One create flow with inline role panels and defaults from customer group (FR-7.3, FR-7.7)	Clicks 12→3; time-to-create 4 min→30 s

1.7.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Party	DirPartyTable, DirPerson, DirOrganization, DirPartyTable form (Global address book)	BUT000 (BP), BUT100 roles; ECC KNA1/LFA1 as separate masters	party-service `Party` with `party_type` person/organization	full
Customer role	CustTable (AccountNum, CustGroup, PaymentTermId, CreditMax, InvoiceAccount), CustGroup, Customers list page	KNA1 (KUNNR), KNB1 (company code), KNVV (sales area), XD01/FD01	`CustomerRole` (party_id, legal_entity_id)	full (sales-area data KNVV arrives in wave 26)
Vendor role	VendTable (AccountNum, VendGroup, PaymentTermId), VendGroup, Vendors list page	LFA1, LFB1, LFM1 (purchasing org), XK01/FK01	`VendorRole`	full (LFM1 in wave 17)
Contacts / persons	ContactPerson, DirPersonName (first/middle/last), DirPersonUser	KNVK contact person, BP person category	`ContactRole`, `PersonName`	full
Employee / worker	HcmWorker, DirPerson link	PA0001/PA0002, BP employee role	`EmployeeRole` stub (personnel number, legal entity)	partial (HR data in wave 42)
Addresses and contact info	DirPartyLocation, LogisticsPostalAddress, LogisticsElectronicAddress (phone/email/URL)	ADRC, ADR2 (phone), ADR6 (email), BUT020	`PartyAddress` (wave-6 address + purposes), `ElectronicAddress`	full
Tax registrations	TaxRegistration, TaxRegistrationType, CustTable.VATNum, VendTable.VATNum	KNA1-STCEG/STCD1..5, LFA1-STCEG, DFKKBPTAXNUM (BP tax numbers)	`TaxRegistration` (type, country, number, validated_at, result)	improved (validation adapters)
Duplicate detection / merge	DirPartyTable duplicate check (exact), Merge parties (D365 "Merge" on DirPartyTable, limited)	None in ECC; S/4 MDG-BP matching (separate license)	`DuplicateCandidate`, `MergeCase`, Temporal merge workflow	improved
Groups, posting profiles, terms	CustGroup/VendGroup, CustLedger/VendLedger posting profiles, PaymentTerm, CustPaymMode, DlvTerm, DlvMode	KNB1-AKONT reconciliation account, T052 payment terms (ZTERM), TINC Incoterms, KNVV-VSBED	`CustomerGroup`, `VendorGroup`, `PostingProfile` (account fields filled in wave 11), `PaymentTerm`, `DeliveryTerm`, `DeliveryMode`	full (GL accounts bound in wave 11)
Credit management	CustTable.CreditMax, CreditRating, blocked status (CustBlocked)	KNKK credit limit (FD32), UKM in S/4	`CreditProfile` (limit, currency, rating, hold)	partial (credit checks run in wave 27)
Bank role	BankAccountTable per customer/vendor (CustBankAccount/VendBankAccount)	BP bank details BUT0BK, LFBK/KNBK	`BankRole` plus `PartyBankAccount` (IBAN/BIC format check only)	partial (verification in wave 36)
Governance	Workflow on vendor changes (VendTable change proposals in D365), DirPartyTable data owner	MDG change requests, SAP workflow	Stewardship queue, change-approval workflow on protected fields	improved

1.7.3 Functional requirements

- **FR-7.1 Party core** — Must. The system shall create parties of type organization (legal name, trade name, registration number, industry code NAICS/NACE) or person (given, middle, family name, name prefix/suffix, date of birth optional, gender optional) with a tenant-wide party number from a number sequence.
  - Name search uses trigram similarity on `search_name` (normalized: casefold, diacritics stripped, punctuation removed).
  - Party `status` ∈ {active, inactive, merged}; merged parties keep a `merged_into_id` and all reads redirect.
- **FR-7.2 Roles per legal entity** — Must. The system shall attach Customer, Vendor, Contact, Employee, Prospect and Bank roles to a party; Customer and Vendor roles are keyed by (party_id, legal_entity_id) and carry an account number from the legal entity's customer/vendor number sequence.
  - A party may hold Customer and Vendor roles in the same legal entity (the AX "one party, two accounts" case).
  - Role deactivation keeps the row and sets `blocked = all` with a reason.
- **FR-7.3 Group defaults** — Must. The system shall maintain customer groups and vendor groups carrying default payment term, delivery term, delivery mode, posting profile, price/discount group placeholders and tax group placeholders; creating a role copies group defaults into the role and records `defaulted_from_group`.
- **FR-7.4 Tax registrations with validation** — Must. The system shall store tax registration numbers by type (VAT, EIN, SSN-masked, GSTIN, ABN, CNPJ, RFC, UTR) with per-country format validation (regex plus check digit where defined: GSTIN mod-36, ABN mod-89, CNPJ/RFC algorithms, UK VAT mod-97) and optional online validation through adapters.
  - VIES adapter calls `checkVatService` with a 5 s timeout; result cached 30 days; outage yields `validation_status = unverified`, never a save failure.
  - IRS TIN matching adapter is an interface with a recorded fixture implementation in v1.
- **FR-7.5 Duplicate detection** — Must. The system shall score candidate pairs on every create/update and on demand, using blocking on (normalized name first 4 chars | tax id | postal code+street number | phone last 7 | email domain) and a Fellegi-Sunter model with comparison levels for name (Jaro-Winkler ≥ 0.92 / ≥ 0.85 / else), tax id (exact), address (street+postal exact / postal only / else), phone, email.
  - Match weight ≥ 12 (probability ≈ 0.99) auto-flags as `likely_duplicate`; 6–12 flags `possible`; below 6 ignored.
  - Create with a `likely_duplicate` candidate returns 409 with the candidates unless `X-Passage-Override-Duplicate: true` and the duty "Override duplicate check".
  - Scoring of a single party against 1M parties completes in < 300 ms using blocking indexes.
- **FR-7.6 Merge workflow and survivorship** — Must. The system shall merge N parties into one survivor through a Temporal workflow: pick survivor, resolve field conflicts by survivorship rules (most-recently-verified tax id, most-recently-updated address, non-empty wins, manual override), union roles (conflicting roles in the same legal entity require explicit choice), then rewrite references in every consuming service via `party.merged.v1` with acknowledgement tracking.
  - Merge is reversible for 30 days through a stored pre-merge snapshot.

    - The workflow does not complete until every registered consumer acknowledges rewrite; stragglers alert after 10 min.
- **FR-7.7 Electronic addresses** — Must. The system shall store phone (E.164 normalized), email (RFC 5322 syntax check), URL, fax, with purposes and a primary flag per type.
- **FR-7.8 Credit profile** — Should. The system shall store credit limit with currency, credit rating, hold flag, and hold reason on the Customer role; changes to limit are protected fields under FR-7.10.
- **FR-7.9 Payment, delivery terms and modes** — Must. The system shall maintain payment terms (net days, day-of-month, cash discount percent and days, up to 3 discount tiers), delivery terms (Incoterms 2020 list seeded), and delivery modes; terms compute due dates deterministically (`due = invoice_date + net_days`, or next day-of-month ≥ invoice_date + offset_months).
- **FR-7.10 Change-approval workflow on protected fields** — Must. The system shall route changes to protected fields (bank account, tax registration, credit limit, payment term on vendor, remit-to address) through an approval workflow configured per legal entity; the change is held in `pending_change` and not visible to consumers until approved.
    - SLA 2 business days then escalation; rejection records a reason.
- **FR-7.11 Stewardship queue and ownership** — Should. The system shall assign a data owner (user or team) per party and show stewards a queue of duplicates, unverified tax ids, pending changes and incomplete records (missing primary address or primary email), sorted by age.
- **FR-7.12 Legacy import** — Must. The system shall import CustTable/VendTable/DirPartyTable and KNA1/LFA1 extracts (CSV) through the wave-5 import tool with the wave-10 mapping specs pre-seeded; duplicates found during import go to the stewardship queue rather than blocking the load.
- **FR-7.13 Party search API** — Must. The system shall expose typeahead search returning up to 20 parties in < 100 ms p95 over 1M rows, filtered by role and legal entity.

1.7.4 Data model

Party

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
party_number	text(20)	yes	number sequence `PARTY`, unique per tenant
party_type	enum	yes	organization, person
name	text(200)	yes	display name; computed for persons
search_name	text(200)	yes	normalized; trigram index
org.legal_name / org.trade_name / org.registration_number / org.industry_code	text	legal_name yes	organization subtype
person.given_name / middle_name / family_name / prefix / suffix / dob	text, date	given+family yes	person subtype; PII-flagged
status	enum	yes	active, inactive, merged
merged_into_id	uuid	no	set when status = merged
owner_user_id / owner_team_id	uuid	no	stewardship
language_code / time_zone	text	no	
version, created_at, updated_at	int, timestamptz	yes	

CustomerRole (VendorRole is symmetric with `vendor_group_id`, `vendor_hold`)

Field	Type	Required	Notes
id	uuid	yes	PK
party_id	uuid	yes	FK party
legal_entity_id	uuid	yes	FK; unique (party_id, legal_entity_id)
account_number	text(20)	yes	sequence `CUST-{LE}`; unique per legal entity
customer_group_id	uuid	yes	FK
payment_term_id / delivery_term_id / delivery_mode_id	uuid	payment yes	FK
posting_profile_id	uuid	yes	FK; GL accounts attached in wave 11
invoice_account_party_id	uuid	no	bill-to party (AX InvoiceAccount)
currency_code	char(3)	yes	default accounting currency
credit_limit / credit_currency / credit_rating / credit_hold	numeric(18,2), char(3), text, bool	no	protected fields
blocked	enum	yes	none, invoice, all
sales_tax_group_id	uuid	no	bound in wave 13
version	int	yes	

TaxRegistration

Field	Type	Required	Notes
id	uuid	yes	PK
party_id	uuid	yes	FK
registration_type	enum	yes	VAT, EIN, GSTIN, ABN, CNPJ, RFC, UTR, OTHER
country_code	char(2)	yes	
number	text(30)	yes	stored normalized (no spaces); unique (type, country, number) per tenant
valid_from / valid_to	date	no	
validation_status	enum	yes	unverified, valid, invalid, format_only
validated_at / validation_source	timestamptz, text	no	VIES, IRS_TIN, manual

Field	Type	Required	Notes
is_primary	bool	yes	one per (party, country)

DuplicateCandidate / MergeCase — candidate: id, party_a_id, party_b_id (a < b), match_weight numeric(8,4), probability, level_detail jsonb, status (open, dismissed, merged), created_at; merge_case: id, survivor_id, merged_ids uuid[], survivorship jsonb, snapshot jsonb, workflow_id, status (draft, running, awaiting_ack, completed, reverted, failed), consumer_acks jsonb.

Invariants: a party with status `merged` has no active roles; `account_number` never reused after merge (merged role numbers are retained as aliases in `customer_alias`); at most one primary tax registration per (party, country); every role has a posting profile. Indexing: `party` GIN trigram on `search_name`; btree on `(tenant_id, party_type, status)`; blocking indexes on `left(search_name,4)`, `tax_registration.number`, `electronic_address.normalized`; `duplicate_candidate(status, match_weight desc)`.

1.7.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/party/v1/parties	Create party (runs dedup)	`Idempotency-Key`	Maintain parties
GET	/party/v1/parties/{id}	Read (follows merged_into)	n/a	View parties
PATCH	/party/v1/parties/{id}	Update; protected fields → pending change	version check	Maintain parties
GET	/party/v1/parties:search?q=&role=customer&legalEntity=	Typeahead	n/a	View parties
POST	/party/v1/parties/{id}/roles/customer	Add customer role in `X-Passage-LegalEntity`	key header	Maintain customers
POST	/party/v1/parties/{id}/roles/vendor	Add vendor role	key header	Maintain vendors
POST	/party/v1/parties/{id}/tax-registrations	Add and validate	key header	Maintain parties
POST	/party/v1/tax-registrations/{id}:validate	Re-run adapter	idempotent (cached 30 d)	Maintain parties
GET	/party/v1/duplicates?status=open&minWeight=6	Stewardship list	n/a	Steward master data
POST	/party/v1/merge-cases	Start merge workflow	key header	Merge parties
POST	/party/v1/merge-cases/{id}:revert	Revert within 30 days	key header	Merge parties
POST	/party/v1/pending-changes/{id}:approve / :reject	Workflow decisions	key header	Approve master data changes
GET	/party/v1/payment-terms/{id}/due-date?invoiceDate=2026-03-31	Compute due date	n/a	any authenticated

Validation: tax registration format per country before adapter call; role creation requires the legal entity header and an existing group; merge requires all parties active and in the same tenant; search `q` minimum 2 characters.

Event	Trigger	Payload fields	Consumers
dev.passage.party.created.v1	commit	party_id, party_number, party_type, name, primary_address_id	search index, migration, audit
dev.passage.party.updated.v1	commit of non-protected change	party_id, changed_fields[], version	search, consumers caching names
dev.passage.party.customer.created.v1	role commit	party_id, legal_entity_id, account_number, customer_group_id, payment_term_id, posting_profile_id, currency_code	AR, sales, pricing, credit
dev.passage.party.vendor.created.v1	role commit	party_id, legal_entity_id, account_number, vendor_group_id, payment_term_id	AP, purchasing
dev.passage.party.duplicate.flagged.v1	score ≥ 6	candidate_id, party_a_id, party_b_id, match_weight	stewardship queue, notification
dev.passage.party.merged.v1	merge workflow enters awaiting_ack	survivor_id, merged_ids[], role_map{old_account→new_account}, merge_case_id	every service with party FKs; each replies on `dev.passage.party.merge.ack.v1`
dev.passage.party.change.approved.v1 / rejected.v1	workflow decision	party_id, field, old, new, approver_id, reason	audit, notification
dev.passage.party.taxregistration.validated.v1	adapter result	registration_id, status, source, validated_at	tax (wave 13), audit

1.7.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Global address book (parties list)	New party, search, filter by role/entity, open duplicates	party number, name, type, roles, primary address, status	View parties
Party detail	Edit, add role panels (customer/vendor/contact), addresses, electronic addresses, tax registrations, relationships, audit tab	all party fields; role panels collapsed by default	Maintain parties
Customers workspace (per legal entity)	New customer (party picker or inline create), block, edit terms, credit tab	account, name, group, terms, credit limit, hold, blocked	Maintain customers
Vendors workspace	New vendor, block, bank accounts (pending-change aware)	account, name, group, terms, bank verified flag	Maintain vendors
Duplicate review	Side-by-side compare, dismiss, start merge, pick survivor per field	match weight, field-level comparison levels	Steward master data
Merge case monitor	Progress, consumer acks, revert	consumers acked / pending, elapsed	Merge parties
Stewardship queue	Claim, resolve, bulk dismiss	item type, party, age, owner	Steward master data
Pending changes (approvals)	Approve/reject with reason	field, old/new, requester, SLA remaining	Approve master data changes

Interaction rules: the create-party dialog runs dedup after name and country are entered (debounced 400 ms) and shows candidates inline before save; `Ctrl+Enter` saves and opens the role panel; bulk block/unblock from the list is limited to 500 rows per action; protected fields render with a lock badge and show "pending" with the proposed value until approval; personalization of role panels (which are expanded) is per user.

1.7.7 Business rules and edge cases

- **BR-7.1** Party status machine: `active ↔ inactive`; `active → merged` only through a merge case; `merged` is terminal except via revert within 30 days, which restores the snapshot and sets `active`.
- **BR-7.2** Merge case machine: `draft → running → awaiting_ack → completed`; `running` or `awaiting_ack → failed` on activity error after 5 retries; `completed → reverted` within 30 days only when no consumer reports a post-merge transaction referencing the survivor that cannot be split (consumer rejects revert with reason).
- **BR-7.3** Survivorship default order per field: manual override > most recently validated (tax ids) > most recently updated > non-empty > survivor's value.
- **BR-7.4** Account numbers of merged roles become aliases; lookups by alias resolve to the survivor role and are logged for 12 months to catch stale integrations.
- **BR-7.5** Dedup weights are recomputed for an updated party only against candidates from its blocking keys; a full re-score job runs weekly (Temporal cron, Sunday 01:00 tenant local).
- **BR-7.6** Tax registration numbers are normalized before uniqueness checks (strip spaces, hyphens, dots; uppercase); two parties with the same VAT number in the same country are auto-flagged with weight +15.
- **BR-7.7** Payment term due date: `net_days` adds calendar days; `day_of_month` moves to that day in the month `offset_months` ahead, clamped to the last day; cash discount date computed the same way per tier; if discount date > due date the term is invalid at save.
- **BR-7.8** Protected-field changes by a user holding "Approve master data changes" still require a second approver (no self-approval).
- **BR-7.9** A person party's `dob` and government ids are PII-flagged: masked in lists, visible in detail only to duties with `pii.read`, excluded from search index and from CloudEvent payloads.
- **BR-7.10** Deleting a party is never allowed; `inactive` parties cannot receive new roles; roles on an `inactive` party cannot be unblocked.
- **BR-7.11** Concurrent merges touching the same party are serialized by a Temporal workflow id `merge-{tenant}-{min(party_ids)}`; the second request receives 409 with the running case id.
- **BR-7.12** Import rows with a `likely_duplicate` candidate are loaded with status `active` and a candidate record; they are not rejected, because migrations must be complete and reviewed afterward.

1.7.8 Acceptance criteria

- **AC-7.1** (FR-7.5) Given 10,000 seeded parties with 500 planted near-duplicates (name typos, swapped address lines, formatted tax ids), when the full re-score job runs, then flagged pairs with weight ≥ 6 achieve precision ≥ 0.95 and recall ≥ 0.90 against the labels.
- **AC-7.2** (FR-7.6) Given parties A and B each with a Customer role in USMF and B with a Vendor role in DEMF, when merged into A, then A holds all three roles, B's account numbers are aliases, every registered consumer acks, and the referential-integrity audit reports 0 rows referencing B.
- **AC-7.3** (FR-7.4) Given the recorded VIES fixture of 200 VAT numbers, when validated, then results equal the fixture; given the adapter stub returns a timeout, then the registration saves with `unverified` and an alert is raised after 3 consecutive timeouts.
- **AC-7.4** (FR-7.10) Given a customer with credit limit USD 50,000, when a user raises it to USD 120,000, then the stored limit stays 50,000, a pending change exists, the Temporal workflow has an open approval task, and after approval the limit is 120,000 with an `approved` event.
- **AC-7.5** (FR-7.2) Given a party with a Customer role in USMF, when a Vendor role is created in USMF, then both exist with distinct account numbers from the `CUST-USMF` and `VEND-USMF` sequences.
- **AC-7.6** (FR-7.9) Given payment term "Net 30 EOM+1" (day_of_month 31, offset 1) and invoice date 2026-01-31, when due date is computed, then it equals 2026-02-28; with cash discount 2% at 10 days, discount date is 2026-02-10.
- **AC-7.7** (FR-7.5) Given a `likely_duplicate` candidate at create, when the client retries with the override header but without the override duty, then 403.
- **AC-7.8** (FR-7.12) Given a 1,000-row CustTable extract with 20 duplicates of existing parties, when imported, then 1,000 roles exist, 20 duplicate candidates are open, and 0 rows are rejected for duplication.
- **AC-7.9** (FR-7.13) Given 1M parties, when typeahead `q=acm` is called 1,000 times, then p95 < 100 ms and results are ordered by similarity then name.
- **AC-7.10** (BR-7.9) Given a person with `dob`, when the list page is rendered by a user without `pii.read`, then the column shows `••••` and the CloudEvent payload has no `dob` key.

1.7.9 Test plan and fixtures

Fixtures: `party/parties-10k.jsonl` (generated with Faker seed 7, 500 labeled duplicate pairs in `party/dup-labels.csv`); `party/vies-recorded.json` (200 numbers, 150 valid, 50 invalid, recorded 2026-09); `party/tax-format-vectors.csv` (300 valid/invalid across 8 types); `party/custtable-extract.csv` and `party/kna1-extract.csv` (1,000 rows each, AX/SAP column names verbatim); `party/payment-terms.csv` (25 terms with 100 invoice-date → due-date oracle rows).

Test	Type	Target	Pass criterion
Dedup precision/recall	property	Splink model	P ≥ 0.95, R ≥ 0.90
Single-party scoring latency	performance	scorer	< 300 ms at 1M
Merge integrity	E2E (Temporal test server)	merge workflow	0 dangling FKs, all acks
Merge revert	E2E	revert	snapshot restored byte-equal
Tax format vectors	table-driven	validators	300/300
VIES adapter	contract (recorded)	adapter	matches fixture; timeout path
Change approval	Temporal workflow test	protected fields	blocked until approve; SLA timer fires after 2 business days
Due-date oracle	unit	payment terms	100/100

Test	Type	Target	Pass criterion
PII masking	contract	API + events	no PII fields in list or events
Typeahead	performance	search	p95 < 100 ms

Reconciliation: count(roles) per legal entity in import = source row count; Σ open candidates + dismissed + merged = total pairs ever flagged; every merged party has exactly one merge case.

1.7.10 Deployment and operations

Helm chart `passage-party` (service ×2, CloudNativePG `party-db`, Debezium connector, Splink scoring worker as a Python Deployment with 2 replicas, Temporal namespace `party`). Config: `PARTY_DEDUP_LIKELY_WEIGHT=12`, `PARTY_DEDUP_POSSIBLE_WEIGHT=6`, `PARTY_VIES_TIMEOUT_MS=5000`, `PARTY_MERGE_ACK_TIMEOUT_MIN=10`, `PARTY_APPROVAL_SLA_BUSINESS_DAYS=2`. Migrations additive via Atlas. Feature flags: `party.online_tax_validation`, `party.merge_revert`, `party.pii_masking` (default on). Scheduled jobs: weekly full re-score; daily VIES re-validation of registrations older than 30 days (max 5,000 per run); hourly stale-merge sweeper. SLOs: create p95 < 400 ms including dedup; search p95 < 100 ms; merge completion p95 < 5 min. Alerts: merge awaiting ack > 10 min; VIES failure rate > 20% over 15 min; duplicate queue > 1,000 open. Runbook entries: "Consumer not acking merge" (list registered consumers, `passagectl party merge-status`), "Re-train Splink model" (export labeled pairs, run `party-train`, publish model version). Rollback: previous digest; the Splink model version is pinned in config so a rollback restores scoring behavior.

1.7.11 Migration from AX / SAP

Source	Passage entity	Notes
AX DirPartyTable (PartyNumber, Name, NameAlias, InstanceRelationType, LanguageId)	Party	InstanceRelationType DirPerson/DirOrganization → party_type
AX DirPerson (FirstName, MiddleName, LastName, Gender, BirthDate via DirPersonName)	Party.person	names from DirPersonName valid-time current row
AX DirOrganization (OrgNumber), DirOrganizationName	Party.org	OrgNumber → registration_number
AX CustTable (AccountNum, Party, CustGroup, PaymentTermId, DlvTerm, DlvMode, Currency, CreditMax, CreditRating, Blocked, InvoiceAccount, VATNum, DataAreaId)	CustomerRole, TaxRegistration	DataAreaId → legal_entity; Blocked enum No/Invoice/All → blocked; VATNum → VAT registration with country from primary address
AX VendTable (AccountNum, Party, VendGroup, PaymentTermId, Blocked, VATNum, DataAreaId)	VendorRole	same pattern
AX ContactPerson (ContactPersonId, Party, ContactForParty)	ContactRole	ContactForParty → relationship row
AX DirPartyLocation + LogisticsLocationRole	PartyAddress with purposes	reuses wave-6 address load; IsPrimary → primary flag
AX LogisticsElectronicAddress (Type, Locator)	ElectronicAddress	Type Phone/Email/URL/Fax; phones normalized E.164 using party country
AX CustGroup/VendGroup, PaymentTerm, DlvTerm, DlvMode	groups, terms, modes	PaymentTerm.NumOfDays/PaymMethod (Net/CurrentMonth/…) mapped to net_days/day_of_month
SAP KNA1 (KUNNR, NAME1, LAND1, STCEG, STCD1, SPERR, LOEVM, ADRNR) + KNB1 (BUKRS, AKONT, ZTERM, ZUAWA)	Party, CustomerRole, TaxRegistration	one Party per KUNNR unless matched to an LFA1 by STCEG or ADRNR; BUKRS → legal_entity; ZTERM → payment term by code
SAP LFA1 (LIFNR, NAME1, STCEG, SPERR, SPERM) + LFB1 (BUKRS, AKONT, ZTERM)	Party, VendorRole	KNA1-LIFNR / LFA1-KUNNR cross references pre-merge the party
SAP BUT000 (PARTNER, TYPE, NAME_ORG1, NAME_FIRST, NAME_LAST) + BUT100 (RLTYP FLCU01/FLVN01/BUP001) + DFKKBPTAXNUM	Party, roles, TaxRegistration	roles map FLCU01→customer, FLVN01→vendor, BUP001→contact
SAP ADRC / ADR2 / ADR6 / KNVK	PartyAddress, ElectronicAddress, ContactRole	ADR2 TEL_NUMBER + TELNR_CALL merged
SAP T052 (ZTERM, ZTAGG, ZTAG1..3, ZPRZ1..2), TINC, KNKK (KLIMK, CTLPC)	PaymentTerm, DeliveryTerm, CreditProfile	T052 day-limit variants produce one Passage term per variant

Transformation notes: AX `Party` RecId joins are resolved through DirPartyTable.PartyNumber; SAP customer and vendor numbers are kept as `account_number` when they fit 20 characters, else a new sequence number is assigned and the SAP number stored as alias. Validation: every role must resolve to a legal entity and a group; tax ids failing format go to the stewardship queue, not rejection. Not migrated: AX `DirPartyRelationship` types other than contact-for; SAP sales-area (KNVV) and purchasing-org (LFM1) data (waves 26 and 17); SAP dunning data KNB5 (wave 30); AX `CustTrans`/`VendTrans` open items (transactional packs in wave 49).

1.7.12 Dependencies, risks, open questions

Depends on: wave 6 (legal entities, addresses, countries, number sequences per entity), wave 4 (Temporal approval template, outbox), wave 5 (import tool, audit, PII attributes from wave-3 metadata).

Risk	Likelihood	Impact	Mitigation
Consumers added in later waves forget to implement merge ack	Medium	High	Consumer registry with contract test required for any service subscribing to party events
VIES rate limits or outages block daily re-validation	High	Low	Backoff and cap per run; unverified is a non-blocking state
Splink model drifts for non-Latin names	Medium	Medium	Name normalization with Unicode NFKD; per-tenant threshold override

Open questions: (1) Do Prospect roles need a number sequence or stay numberless until conversion to Customer? Proposed: numberless; decide by wave 26. (2) Is the employee role created here or only in wave 42 with HcmWorker migration? Proposed: stub here (personnel number only); decide by wave 10. (3) Which survivorship rules are tenant-configurable in v1? Proposed: order list editable, levels fixed; decide by wave 10.

1.8 Wave 8: Product Master, Variants & Catalogs

1.8.1 Scope and business context

AX defines a shared `EcoResProduct` (subtyped `EcoResDistinctProduct` and `EcoResProductMaster`) with variants in `EcoResDistinctProductVariant`, dimension values in `EcoResSize/Color/Style/Configuration`, and releases it per company into `InventTable` with `InventDimGroup`-style product/storage/tracking dimension groups, `InventItemGroup` and `InventModelGroup`. SAP keeps `MARA` (client-level), `MARC` (plant), `MARD` (storage location), `MVKE` (sales), `MBEW` (valuation), material types (`T134`), variant configuration in LO-VC (`CUKB`, `CUOB`, `CABN/CAWN` characteristics) and the classification system (`KLAH/KSML`). Wave 8 defines the product once at tenant level, explodes variants from dimension groups, releases per legal entity with site/warehouse defaults, and adds attributes, categories, item groups, item model groups, GS1 identifiers and a constraint-based configurator foundation. At the end of the wave the product information management workspace holds the demo manufacturer's 500 items and 50 masters.

Out of scope: on-hand, costing values and inventory postings (waves 16, 20), sales/purchase prices (waves 17, 26), BOMs (wave 9), batch/serial *instances* (wave 16; this wave defines only whether an item is tracked), quality specs (wave 33).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Releasing an AX product to 10 companies means 10 passes through "Release products" plus editing InventTable defaults (item group, model group, site, warehouse) in each company; a missed default blocks the first PO with "Item model group not specified"	Release is a per-company copy with no template	Release profiles: one call releases to N entities with defaults derived from item group and site; missing-default check at release (FR-8.4)	Clicks per 10-entity release 60→3; first-transaction failures from missing defaults →0
SAP MM01 walks 15–20 views (basic, purchasing, MRP 1–4, accounting, costing); a material is "created" but unusable in a plant until every view is extended (MM01 extend)	View-based material master with per-plant/storage segments (MARC/MARD) each needing separate maintenance	One product record plus released-product per entity; plant-level data derives from site defaults (FR-8.4)	Time to a usable material in a new plant 25 min→<1 min
AX variant creation (EcoResProductVariant "Variant suggestions") is a modal that times out above ~500 combinations; D365 batch job needed	Synchronous form-driven explosion	Explosion is a Temporal job with checkpoints; 1,000 variants in <5 s (FR-8.2)	10k variants: 40 min batch→<60 s with progress
GTIN entry in AX (InventItemGTIN) and SAP (MARM-EAN11) accepts any 13 digits; bad check digits break EDI 856/ASN and retailer onboarding	No check-digit validation at entry	GS1 check-digit and prefix validation, generation from a GS1 company prefix (FR-8.9)	Invalid GTINs stored 0; retailer rejections for GTIN →0
Product attributes in AX (EcoResProductAttribute) require defining attribute types, attributes, categories and category attributes across four forms before a value can be entered; SAP class/characteristic (CL02/CT04) is similar	Classification metadata modeled as a separate hierarchy from products	Attribute sets attached to categories with inline attribute creation and typed validation (FR-8.6)	Setup of a 10-attribute class 30 min→3 min

1.8.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Shared product	EcoResProduct, EcoResDistinctProduct, EcoResProductMaster, Product details form, ProductNumber	MARA (MATNR, MTART, MEINS, MATKL), MM01/MM02/MM03	product-service `Product` (product_type item/service, subtype product/master)	full
Variants and dimensions	EcoResProductDimensionGroup, EcoResSize/Color/Style/Configuration, EcoResDistinctProductVariant, EcoResProductVariantDimensionValue	Generic material + variants (MARA-SATNR, KZKFG), characteristics CABN/CAWN	`DimensionGroup`, `DimensionValue`, `ProductVariant`	full
Storage / tracking dimension groups	EcoResStorageDimensionGroup, EcoResTrackingDimensionGroup (batch/serial active, physical/financial inventory flags)	MARA-XCHPF batch management, serial profile MARC-SERNP, valuation level	`StorageDimensionGroup`, `TrackingDimensionGroup`	full
Released product per company	InventTable (ItemId, ItemGroupId, ModelGroupId, DataAreaId), InventTableModule, InventItemSetupFilter, default order settings (InventItemInventSetup/PurchSetup/SalesSetup)	MARC (plant), MARD (storage location), MVKE (sales org), MBEW (valuation)	`ReleasedProduct` with `ReleasedProductSiteDefault`	full
Item groups / model groups	InventItemGroup, InventModelGroup (InventModel FIFO/LIFO/WAvg/Std, PhysicalNegative, FinancialNegative, PostPhysical, PostFinancial)	Material type T134, material group T023, valuation class T025, price control MBEW-VPRSV	`ItemGroup`, `ItemModelGroup`	full
Units and conversions	UnitOfMeasureConversion product-specific, InventTable unit per module	MARA-MEINS, MARM alternative units	`ProductUomConversion`, unit per module (inventory/purchase/sales/BOM)	full
Attributes / classification	EcoResAttribute, EcoResAttributeType, EcoResCategory, EcoResCategoryHierarchy, EcoResProductCategory	Classification: KLAH, KSML, CABN, CAWN, AUSP (CL01/CL02/CT04)	`AttributeType`, `Attribute`, `AttributeSet`, `Category`, `CategoryHierarchy`, `ProductAttributeValue`	full
Configurator	Product configuration models (PCConfiguration, PCComponent, PCConstraint — D365 constraint-based configurator)	LO-VC: CU41 configuration profile, dependencies CUKB, object dependencies CUOB	`ConfigurationModel` solved by OR-Tools CP-SAT in a Python worker	partial (pricing and BOM/route generation from configuration in waves 26 and 9)
GS1 identifiers	InventItemGTIN, InventItemBarcode, BarcodeSetup	MARM-EAN11, EAN category T006M (NUMTP)	`ProductIdentifier` (GTIN-8/12/13/14, SSCC generation)	improved (validation and generation)
Images / documents	DocuRef attachments on EcoResProduct, product image	DMS (CV01N) links	`ProductDocument` (object storage key, type)	full
Product lifecycle	D365 product lifecycle state (EcoResProductLifecycleState)	MARA-MSTAE cross-plant status, MARC-MMSTA plant status	`lifecycle_state` on Product and ReleasedProduct	full

1.8.3 Functional requirements

- **FR-8.1 Product and product master** — Must. The system shall create tenant-level products (subtype `product`) and product masters (subtype `master`) with product number (sequence `PROD` or manual), name, description, search name, product type (item/service), base unit, and lifecycle state.
  - Product number regex `^[A-Z0-9][A-Z0-9._-]{0,39}$`; immutable after release.
  - Masters carry a `dimension_group_id` with 1–4 product dimensions chosen from size, color, style, configuration (named per tenant, e.g. "length").
- **FR-8.2 Variant explosion** — Must. The system shall generate variants for a master from the Cartesian product of selected dimension values, or from an explicit list, as a Temporal job with 500-row checkpoints.
  - Hard cap 10,000 variants per master; 1,000 variants in < 5 s.
  - Variant number is `{master}-{size}-{color}-{style}-{config}` with empty segments collapsed; override pattern per tenant.
  - Lookup by (master_id, dimension values) resolves with a unique composite index.
- **FR-8.3 Storage and tracking dimension groups** — Must. The system shall define storage dimension groups (site, warehouse, location, inventory status, license plate — each active, mandatory, primary stocking, physical/financial inventory flags) and tracking dimension groups (batch active/mandatory, serial active/mandatory, serial capture at sale/receipt), assigned to a product and immutable once a released product has inventory transactions (callback from wave 16).
- **FR-8.4 Released products with site/warehouse defaults** — Must. The system shall release a product to one or more legal entities, creating `ReleasedProduct` with item group, item model group, units per module, default site and warehouse, and per-site default warehouse and order settings (minimum/multiple/standard order quantities, lead time) for inventory, purchase and sales.
  - Release profiles: a named template of defaults per (legal entity, item group); release of 1,000 products to 3 entities completes in < 60 s.
  - Cascade: a per-site override wins over the entity default; absent both, the request is rejected with the missing field named.
- **FR-8.5 Item groups and item model groups** — Must. The system shall maintain item groups (posting hooks for GL arrive in wave 11) and item model groups with costing method (FIFO, LIFO, weighted_average, weighted_average_date, standard, moving_average), negative physical/financial inventory flags, post physical/financial inventory flags, stocked flag, and quarantine/registration/picking requirements.
- **FR-8.6 Attributes and categories** — Must. The system shall maintain attribute types (text, integer, decimal, boolean, date, enum, currency, with unit and min/max), attributes, attribute sets, and category hierarchies (procurement, sales, retail, reporting) where a category carries attribute sets; a product in a category inherits required attributes and is incomplete until they are filled.
  - Maximum 200 attributes per product; enum attributes have up to 500 values.
- **FR-8.7 Product-specific UoM conversions** — Must. The system shall store per-product conversions (e.g. 1 BOX = 12 EA for product X) that override class conversions; resolution order product → class → error.
- **FR-8.8 Configurator foundation** — Should. The system shall define configuration models per master: components, attribute domains, constraints (expression language with `==, !=, <, >, in, implies, allDifferent, sum`), default values, and a `solve` endpoint that returns a valid configuration or the minimal conflicting constraint set, using CP-SAT with a 2 s time limit.
  - A 50-constraint model solves in < 200 ms; the result is a variant or a configured-product instance with attribute values.
- **FR-8.9 GS1 identifiers** — Must. The system shall store GTIN-8/12/13/14 per (product/variant, unit) with check-digit validation (mod-10 weighting 3-1), allocate GTINs from a tenant GS1 company prefix with a next-number counter, generate SSCC-18 with extension digit, and render barcodes (EAN-13, GS1-128, DataMatrix) as SVG/PNG.
- **FR-8.10 Product lifecycle and blocking** — Must. The system shall enforce lifecycle states `draft → active → phase_out → obsolete` per product and per released product, with module-level blocks (purchase, sales, inventory, production) on released products.
- **FR-8.11 Images and documents** — Should. The system shall attach images (max 10 MB each, 20 per product) and documents to products in object storage with a primary image used in lists.
- **FR-8.12 Product search and catalogs** — Should. The system shall provide typeahead over product number, name, GTIN and attribute values, filtered by legal entity and category, p95 < 100 ms at 1M products, and allow catalogs as named category subtrees with a validity range.

1.8.4 Data model

Product

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
product_number	text(40)	yes	unique per tenant
subtype	enum	yes	product, master, variant
master_id	uuid	conditional	FK for variants
product_type	enum	yes	item, service
name / description / search_name	text(60)/text(1000)/text(60)	name yes	translations in product_translation
base_uom_id	uuid	yes	FK
dimension_group_id	uuid	masters only	product dimensions
storage_dimension_group_id / tracking_dimension_group_id	uuid	yes at release	immutable after first transaction
lifecycle_state	enum	yes	draft, active, phase_out, obsolete
dimension_values	jsonb	variants	{size, color, style, config}; unique with master_id
primary_image_key	text	no	object storage
version	int	yes	

ReleasedProduct

Field	Type	Required	Notes
id	uuid	yes	PK
product_id	uuid	yes	FK; unique (product_id, legal_entity_id)
legal_entity_id	uuid	yes	
item_number	text(40)	yes	defaults to product_number; unique per legal entity
item_group_id / item_model_group_id	uuid	yes	FK
inventory_uom_id / purchase_uom_id / sales_uom_id / bom_uom_id	uuid	inventory yes	convertible to base
default_site_id / default_warehouse_id	uuid	yes for stocked	warehouse must belong to site
lifecycle_state	enum	yes	same enum as product
blocked_purchase / blocked_sales / blocked_inventory / blocked_production	bool	yes	default false
net_weight / gross_weight / volume + units	numeric(18,6)	no	
site_defaults	1:N		ReleasedProductSiteDefault(site_id, warehouse_id, min_qty, multiple_qty, std_qty, lead_time_days per module)
version	int	yes	

ItemModelGroup

Field	Type	Required	Notes
id / tenant_id / code / name	uuid, uuid, text(10), text(60)	yes	code unique per tenant
costing_method	enum	yes	fifo, lifo, weighted_average, weighted_average_date, standard, moving_average
stocked	bool	yes	default true
physical_negative / financial_negative	bool	yes	default false
post_physical / post_financial	bool	yes	default true
registration_required / receiving_required / picking_required / deduction_required	bool	yes	
quarantine_required	bool	yes	default false
include_physical_value	bool	yes	used by costing (wave 20)

ConfigurationModel — id, master_id, version, status (draft/active/retired), components jsonb, attributes jsonb (name, domain), constraints jsonb (expression list, ≤ 2,000), defaults jsonb. **ProductIdentifier** — id, product_id, uom_id, type (GTIN8/12/13/14, SSCC, UPC, custom), value text(18), is_primary, check_digit_valid bool.

Invariants: a variant's dimension_values keys equal its master's dimension group; every released stocked product has item model group and default site/warehouse; GTIN values pass check digit; `obsolete` products have no released product in `active`. Indexing: unique `(master_id, dimension_values)` via a generated hash column; GIN trigram on `search_name`; `product_identifier(value)` unique per tenant; `released_product(legal_entity_id, item_number)` unique; attribute values in `product_attribute_value(product_id, attribute_id)` with a GIN index on a jsonb projection for filtered search.

1.8.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/product/v1/products	Create product or master	`Idempotency-Key`	Maintain products
GET	/product/v1/products/{id}	Read with variants summary	n/a	View products
POST	/product/v1/products/{id}/variants:explode	Start explosion job; returns job id	key header; job id keyed by request hash	Maintain products
GET	/product/v1/products/{id}/variants:lookup?size=M&color=RED	Resolve variant	n/a	View products
POST	/product/v1/products/{id}:release	Release to legal entities with profile	idempotent per (product, entity)	Release products
PATCH	/product/v1/released-products/{id}	Defaults, blocks, lifecycle	version check	Maintain released products
PUT	/product/v1/released-products/{id}/site-defaults/{siteId}	Per-site defaults	full replace	Maintain released products
POST	/product/v1/products/{id}/attributes	Set attribute values (validated against type)	full replace of set	Maintain products
POST	/product/v1/products/{id}/identifiers	Add GTIN; `allocate=true` draws from prefix	key header	Maintain products
POST	/product/v1/configuration-models/{id}:solve	Solve with given selections	pure function	Configure products
GET	/product/v1/products:search?q=&legalEntity=&category=	Typeahead	n/a	View products
GET	/product/v1/uom/convert?product=&from=&to=&qty=	Product-aware conversion	n/a	any authenticated

Validation: dimension values must exist in the dimension group; release rejects a `draft` product; identifier type must match length; `solve` rejects selections outside domains with the offending attribute.

Event	Trigger	Payload fields	Consumers
dev.passage.product.created.v1	commit	product_id, product_number, subtype, product_type, base_uom	engineering, search

Event	Trigger	Payload fields	Consumers
dev.passage.product.variant.created.v1	per explosion checkpoint (batched, ≤ 500 ids)	master_id, variant_ids[], dimension_values[]	engineering, pricing, inventory
dev.passage.product.released.v1	release commit	product_id, legal_entity_id, item_number, item_group_id, item_model_group_id, default_site_id, default_warehouse_id, uoms	inventory, purchasing, sales, costing, planning
dev.passage.product.releasedproduct.updated.v1	PATCH	released_product_id, changed_fields[], blocks, lifecycle_state	all transactional services
dev.passage.product.identifier.added.v1	commit	product_id, type, value, uom_id	WMS (wave 31), EDI (wave 35)
dev.passage.product.configuration.solved.v1	solve returning a variant	model_id, master_id, selections, result_product_id	sales (wave 27), engineering

1.8.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Product information management workspace	Tiles: products, masters, released per entity, incomplete attributes, pending releases	counts, incomplete badges	View products
All products and masters	New, release, explode variants, lifecycle change	number, name, subtype, type, dimension groups, state	Maintain products
Product detail	Tabs: general, dimensions/variants grid, attributes, identifiers, images, releases, translations		Maintain products
Variant explosion dialog	Select values per dimension, preview count, start job, progress bar	dimension values, count, job status	Maintain products
Released products (per legal entity)	Edit defaults grid (inline), block, site defaults sub-grid, bulk edit by item group	item number, item group, model group, units, site, warehouse, blocks	Maintain released products
Item groups / item model groups	Edit, usage count	code, name, costing method, flags	Maintain product setup
Categories and attributes	Hierarchy tree, attribute set editor, assign products	category path, attributes, required	Maintain product categories
Configurator model editor	Add components/attributes/constraints, test solve, activate version	constraints, domains, solve result	Configure products
GS1 identifiers	Allocate, validate, print barcode	type, value, uom, valid	Maintain products

Interaction rules: the released-products grid supports bulk edit of selected rows (max 2,000) with one save; attribute inputs validate on blur against type and range; the explosion dialog warns above 2,000 variants and requires confirmation; barcode print uses the browser print dialog with an SVG preview; the legal-entity switcher filters the released-product views only.

1.8.7 Business rules and edge cases

- **BR-8.1** Lifecycle machine for Product and ReleasedProduct: `draft → active`, `active → phase_out`, `phase_out → active | obsolete`, `obsolete` terminal; a released product cannot be more "alive" than its product (product `obsolete` forces released `obsolete`).
- **BR-8.2** A master cannot change its dimension group once any variant exists; a dimension value cannot be deleted while any variant uses it.
- **BR-8.3** Explosion is idempotent: existing variants for a combination are skipped and reported; the job result lists created/skipped counts.
- **BR-8.4** Release copies defaults at release time; later changes to the release profile do not propagate (explicit "re-apply profile" action does, with a diff preview).
- **BR-8.5** Changing `costing_method` on an item model group is refused once any released product in it has a financial inventory transaction (wave-16 callback; returns false before wave 16).
- **BR-8.6** UoM conversion quantities round half-even to the target unit's `decimal_precision`; conversions that would produce zero from a non-zero input are rejected (`PRD-UOM-UNDERFLOW`).
- **BR-8.7** GTIN check digit: weights 3,1,3,… from the right of the payload; GTIN-12 stored as GTIN-14 with leading zeros for uniqueness; the same GTIN may not belong to two products in one tenant.
- **BR-8.8** Configurator time limit 2 s; on timeout the response is 409 `PRD-CFG-TIMEOUT` with the partial assignment; infeasible selections return the irreducible infeasible subset (IIS) of at most 10 constraints.
- **BR-8.9** Required attributes from a category are enforced at `active` transition, not at create, so drafts can be saved incomplete.
- **BR-8.10** Translations fall back language → base language of the tenant → product name; never empty.
- **BR-8.11** Variant numbers longer than 40 characters are truncated to 32 plus an 8-character hash, and the full dimension values remain the lookup key.

1.8.8 Acceptance criteria

- **AC-8.1** (FR-8.2) Given a master with 3 dimensions × 10 values, when explosion runs, then 1,000 variants exist in < 5 s, lookup by any combination returns the right variant in < 10 ms, and a second run creates 0 and skips 1,000.
- **AC-8.2** (FR-8.8) Given the 50-constraint fixture model and 100 selection sets (60 valid, 40 invalid), when solved, then valid sets return a configuration consistent with every constraint and invalid sets return an IIS, each in < 200 ms.
- **AC-8.3** (FR-8.4) Given release profile P for item group "FG" in USMF with default site S1/warehouse W1 and a site default for S2/W3, when 1,000 products are released and 10,000 random (product, site) default lookups run, then each returns the site override if present, else the entity default (property test).
- **AC-8.4** (FR-8.9) Given the 40 GS1 published test vectors plus 40 mutated ones, when validated, then 40 accepted and 40 rejected; given a prefix 0614141 with next number 123457, when allocated, then GTIN-13 `0614141123457x` has the correct check digit and the counter increments.
- **AC-8.5** (FR-8.3, BR-8.5) Given a released product with a financial transaction (simulated callback true), when its tracking dimension group is changed, then 409 `PRD-DIMGRP-LOCKED`.
- **AC-8.6** (FR-8.10) Given a released product with `blocked_sales = true`, then `releasedproduct.updated.v1` carries `blocks.sales = true` and a later wave's contract stub refuses a sales line.

- **AC-8.7** (FR-8.6) Given category "Fasteners" requiring attributes thread_pitch (decimal, 0.2–6.0 mm) and material (enum), when a product in it is activated without `material`, then 422 lists the missing attribute; with `thread_pitch = 7`, then 422 names the range.
- **AC-8.8** (FR-8.7) Given product X with 1 BOX = 12 EA and class conversion 1 CASE = 24 EA, when converting 2 BOX → CASE, then result is 1 CASE; when converting for product Y with no product conversion, then the class conversion applies.
- **AC-8.9** (FR-8.12) Given 1M products, when searched by a GTIN fragment, then p95 < 100 ms.

1.8.9 Test plan and fixtures

Fixtures: `product/demo-catalog.json` (500 items, 50 masters, 12 item groups, 4 model groups, 3 category hierarchies); `product/master-3x10x10.json`; `product/configurator-50.yaml` (bicycle model: frame, wheel, gear, color with 50 constraints; 100 selection sets labeled); `product/gs1-vectors.csv` (40 valid from GS1 check-digit calculator documentation, 40 mutated); `product/uom-product.csv` (30 product conversions); `product/mara-extract.csv` and `product/inventtable-extract.csv` (10,000 rows each).

Test	Type	Target	Pass criterion
Explosion performance and idempotency	performance + unit	explode job	< 5 s for 1,000; second run 0 creates
Variant lookup	property	lookup index	100% hits, p95 < 10 ms
Configurator fixture	table-driven	CP-SAT worker	100/100 expected outcomes, each < 200 ms
Release cascade	property	defaults resolver	10k lookups, 0 mismatches
GTIN vectors	unit	validator/allocator	80/80
Lifecycle transitions	state-machine property	product/released	every illegal transition rejected
Attribute validation	table-driven	attribute types	120 cases
Search latency	performance	typeahead	p95 < 100 ms at 1M
Event contracts	contract	6 events	registry accepts; consumer stubs decode

Reconciliation: count(variants) = |size| × |color| × |style| × |config| for fully exploded masters; count(released products) per entity = count in demo catalog; count(GTINs) = count(products with identifiers) in fixtures.

1.8.10 Deployment and operations

Helm chart `passage-product` (Go service ×2, CloudNativePG `product-db`, Debezium connector, `product-configurator` Python worker with OR-Tools, 2 replicas, 1 CPU / 1 GiB each). Config: `PRODUCT_MAX_VARIANTS=10000`, `PRODUCT_EXPLODE_CHECKPOINT=500`, `PRODUCT_CFG_TIME_LIMIT_MS=2000`, `PRODUCT_GS1_PREFIX` per tenant in the tenant settings entity, object storage bucket `passage-product-media`. Feature flags: `product.configurator`, `product.gs1_allocation`, `product.catalogs`. Scheduled jobs: nightly incomplete-attribute report; weekly orphan media sweep. SLOs: read p95 < 50 ms; release of 1,000 products < 60 s; solve p95 < 500 ms. Alerts: explosion job failed; configurator worker queue depth > 100; media bucket errors. Runbook: "Explosion job stuck" (Temporal `explode-{master_id}` inspect, resume from checkpoint), "Re-apply release profile". Rollback: previous digest; additive migrations only.

1.8.11 Migration from AX / SAP

Source	Passage entity	Notes
AX EcoResProduct (DisplayProductNumber, ProductType, InstanceRelationType), EcoResProductTranslation (Name, Description)	Product, translations	InstanceRelationType EcoResProductMaster → subtype master
AX EcoResProductMaster + EcoResProductDimensionGroup, EcoResProductMasterDimensionValue (Size/Color/Style/Configuration tables)	DimensionGroup, DimensionValue	dimension names kept as AX names unless renamed in mapping
AX EcoResDistinctProductVariant + EcoResProductVariantDimensionValue	ProductVariant	dimension_values from the four value tables
AX EcoResStorageDimensionGroup / EcoResTrackingDimensionGroup + *DimensionGroupFldSetup	storage/tracking groups	field flags per dimension
AX InventTable (ItemId, Product, ItemGroupId, ModelGroupId, DataAreaId, NetWeight, UnitVolume), InventTableModule (UnitId per module), InventItemSetupFilter / InventItemInventSetup (default site/warehouse), InventItemPurchSetup, InventItemSalesSetup	ReleasedProduct, ReleasedProductSiteDefault	one row per DataAreaId; InventDim resolved for site/warehouse defaults
AX InventItemGroup, InventModelGroup (InventModel, PhysicalNegative, FinancialNegative, PostPhysical, PostFinancial, Stocked)	ItemGroup, ItemModelGroup	InventModel enum → costing_method
AX UnitOfMeasureConversion with Product ≠ 0	ProductUomConversion	
AX EcoResAttribute*, EcoResCategory*, EcoResProductCategory, EcoResProductInstanceValue	attributes, categories, values	typed values from EcoResTextValue/IntValue/FloatValue/BooleanValue/DateTimeValue
AX InventItemGTIN (GTIN, ItemId, UnitId)	ProductIdentifier	check digit validated; failures to exception queue
SAP MARA (MATNR, MTART, MATKL, MEINS, MSTAE, XCHPF, SATNR, KZKFG, BRGEW/NTGEW/VOLUM)	Product, dimension/tracking hints	MTART → item group suggestion; XCHPF → batch active; SATNR → master link
SAP MAKT (MAKTX by SPRAS)	translations	
SAP MARC (WERKS, DISMM, PLIFZ, BSTMI, BSTRF, MMSTA, SERNP), MARD (LGORT), MVKE (VKORG, VTWEG), MBEW (BWKEY, VPRSV, BKLAS)	ReleasedProduct per company code of the plant; site defaults per plant	PLIFZ → lead_time_days; BSTMI/BSTRF → min/multiple qty; VPRSV S/V → standard/moving_average
SAP MARM (MEINH, UMREZ, UMREN), MEAN/MARM-EAN11	ProductUomConversion, ProductIdentifier	factor = UMREZ/UMREN
SAP KLAH/KSML/CABN/CAWN/AUSP (class type 001)	AttributeSet, Attribute, values	characteristic data types CHAR/NUM/DATE mapped; multi-value → enum set

Source	Passage entity	Notes
SAP CUKB/CUOB/CUKN (object dependencies)	ConfigurationModel	only `precondition/selection condition` dependency types with simple syntax are translated; others logged to exception queue

Validation: every released product resolves an item group and model group; units per module convertible to base; variants resolve to a master. Not migrated: AX `InventSum`/on-hand and `InventTrans` (wave 16 opening balances); SAP `MBEW` prices (wave 20); SAP `MLAN` tax classification (wave 13); AX retail-specific product tables (non-goal); LO-VC procedures and functions (no equivalent; re-authored as constraints).

1.8.12 Dependencies, risks, open questions

Depends on: wave 6 (sites, warehouses, UoM classes, legal entities), wave 5 (number sequences, import tool, media attachments), wave 4 (Temporal jobs, outbox). Does not depend on wave 7 (parties) except for vendor-item references, deferred to wave 17.

Risk	Likelihood	Impact	Mitigation
Configurator expression language grows into LO-VC complexity	Medium	Medium	Freeze grammar at 8 operators in v1; extensions via sanctioned functions only
Customers with 5+ product dimensions (apparel) exceed the 4-dimension model	Medium	Medium	Extra dimensions modeled as required attributes; evaluate a 5th dimension slot by wave 26
Release profiles diverge from AX default order settings semantics	Low	Medium	Fixture comparison against InventItemInventSetup exports

Open questions: (1) Should `item_number` be allowed to differ from `product_number` per entity (AX allows it)? Proposed: allowed but defaults equal; decide by wave 10. (2) Is moving-average a separate costing method or a parameter of weighted average? Proposed: separate, matching AX; decide by wave 20. (3) Catalogs: product-level or released-product-level membership? Proposed: product-level with entity filter at read; decide by wave 26.

1.9 Wave 9: Engineering Data: BOMs, Routings, Resources & Change Management

1.9.1 Scope and business context

AX stores bills of material as `BOMTable` (BOM id) with `BOMVersion` (which product/site/date range uses which BOM) and `BOM` lines; routes as `RouteTable`/`RouteVersion`/`RouteOpr` with `RouteOprTable` operations, `WrkCtrTable`/`WrkCtrResourceGroup` resources, `WrkCtrCapability` and `WorkCalendarTable`; D365 adds engineering change management (ECR/ECO, product versions). SAP keeps `MAST` (material-to-BOM link), `STKO` (BOM header), `STPO` (BOM items), `STAS` (alternatives), routings in `PLKO`/`PLPO`/`PLAS` with `MAPL` assignment, work centers in `CRHD`/`CRCA`/`KAKO` (capacity), and change numbers in `AENR` (CC01). Wave 9 delivers BOM headers with versions and lines (effective dates, scrap, phantoms, by-/co-products, process formulas), routing versions with operations, relations and resource requirements by capability, resources with calendars, an ECR/ECO workflow with where-used impact analysis, and an explosion/implosion API. The cost roll-up only records the structure the wave-39 roll-up needs.

Out of scope: production orders (wave 21), capacity scheduling and Gantt (wave 23), potency/catch weight execution (wave 24), costing sheets and standard-cost roll (wave 39), PLM adapters beyond the event contract.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX BOM designer (BOMDesigner form) loads a 7-level, 2,000-component BOM in 30–90 s and the "Calculate" explosion runs as a batch (BOMCalcJob) that users wait minutes for	Recursive X++ explosion row by row over BOMVersion/BOM	Closure-table-backed explosion service; 2,000 components in < 500 ms (FR-9.9)	Explosion 60 s–5 min→<0.5 s
SAP CS03 shows one BOM alternative at one date; "what is active on 2026-06-01" means CS03 with a key date, CS80 where-used, and CC03 change number, three transactions and still no diff	BOM validity scattered across STKO/STPO DATUV with AENNR change numbers	Versioned BOMs with as-of resolution, side-by-side version diff and effective-date preview (FR-9.1, FR-9.3)	Transactions to answer "what is active when" 3→1; clicks 15→2
Engineering change in AX 2012 is an unaudited edit of BOM lines; two engineers editing the same BOM overwrite each other; D365 ECM exists but is a separate product-version model many customers never turn on	No change object in the core BOM model	ECR→ECO workflow is the only path to change an active version; optimistic locking on drafts (FR-9.7, BR-9.3)	Unaudited changes to active BOMs 100%→0; lost-update incidents 0
Where-used (AX BOMWhereUsed, SAP CS15) is a flat list; an engineer cannot see which finished goods, orders, and sites a component change impacts before releasing	No impact analysis in the change flow	ECO impact analysis: multi-level where-used across sites with affected product count and open-order hooks (FR-9.8)	Impact assessment effort 2 h→<10 s
Work center capacity in SAP (CR02 capacities, KAKO) and AX (WrkCtrTable calendar + efficiency %) is keyed in several places; routing lead time calculated by hand in spreadsheets	Calendar, efficiency and operation times modeled separately	One calendar model with shifts and efficiency; routing lead-time calculation API (FR-9.6, FR-9.10)	Lead-time calculation manual→API, fixture-verified

1.9.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
BOM header / version	BOMTable (BOMId, Name, ItemGroupId, SiteId), BOMVersion (ItemId, BOMId, FromDate, ToDate, FromQty, ToQty, Active, Approved)	STKO (STLNR, STLAL, BMENG, DATUV), MAST (MATNR, WERKS, STLNR, STLAL), STAS, CS01/CS02/CS03	engineering-service `BomHeader`, `BomVersion`	full
BOM lines	BOM (LineNum, ItemId, BOMQty, BOMQtySerie, UnitId, FromDate, ToDate, ScrapConst, ScrapVar, BOMType Item/Phantom/Pegged/Vendor, Position, RoundUp, Formula)	STPO (POSNR, IDNRK, MENGE, MEINS, AUSCH, DUMPS phantom, ALPOS, POSTP item category, DATUV)	`BomLine`	full
Phantoms, by-/co-products	BOMType Phantom; co-products/by-products in PmfCoByProduct (process industries)	DUMPS, co-products MARC-KZKUP / STPO negative qty by-products	`BomLine.line_type` ∈ {item, phantom, pegged_supply, vendor, co_product, by_product}	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Formulas (process)	PmfFormula, formula versions, scalable, yield percent	PP-PI master recipes (C201), STKO BMENG base qty	`BomHeader.kind = formula`, `yield_percent`, `scalable` flag	partial (batch execution in wave 24)
Routing header / version	RouteTable (RouteId), RouteVersion (ItemId, RouteId, FromDate, ToDate, Active, Approved)	PLKO (PLNNR, PLNAL, DATUV), MAPL (MATNR, WERKS, PLNNR), CA01/CA02/CA03	`RouteHeader`, `RouteVersion`	full
Operations and times	RouteOpr (OprNum, OprId, SetUpTime, ProcessTime, ProcessPerQty, TransitTime, QueueTimeBefore/After, OprPriority, NextOprNum, LinkType), RouteOprTable	PLPO (VORNR, ARBPL, VGW01..06 standard values, LAR01..06 activity types, PLIFZ), PLAS	`Operation` with times, `OperationRelation` (sequential/parallel via `next_operation_no` + priority)	full
Resources / work centers	WrkCtrTable (WrkCtrId, WrkCtrType Vendor/Machine/HumanResources/Tool/Location, Efficiency, CalendarId), WrkCtrResourceGroup, WrkCtrCapability, WrkCtrCapabilityResource	CRHD (ARBPL, VERWE category, KAPID), CRCA capacities, KAKO (capacity header, utilization), CRHD-STEUS control key	`Resource`, `ResourceGroup`, `Capability`, `ResourceCapability`	full
Calendars	WorkCalendarTable, WorkCalendarDate, WorkCalendarDateLine (shifts), WorkCalendarTemplate	Factory calendar TFACD/TFACS (SCAL), shift definitions	`WorkCalendar`, `CalendarDay`, `Shift`	full
Engineering change	D365 ECM: EngChgRequest, EngChgOrder, product versions, impact analysis, release to legal entities	ECM: AENR change master (CC01), ECR/ECO (CC31/CC32), effectivity	`EngineeringChangeRequest`, `EngineeringChangeOrder`, Temporal workflow	full
Where-used / explosion	BOMWhereUsed, BOM calculation (BOMCalcTable), BOM designer	CS11/CS12/CS13 (explosion), CS15 (where-used)	`/engineering/v1/bom:explode`, `:implode`, `:where-used`	improved (latency, as-of)
Cost roll-up	BOMCalcTable, costing versions	CK11N/CK40N	structure hooks only; roll-up in wave 39	partial by design

1.9.3 Functional requirements

- **FR-9.1 BOM headers and versions** — Must. The system shall create BOM headers (bom_id from sequence `BOM`, name, kind item/formula, site optional) and versions binding a header to a product (released product optional for site-specific), with `effective_from/effective_to`, `from_qty/to_qty` ranges, `approved` and `active` flags.
  - At most one active version per (product, site, overlapping date range, overlapping quantity range); activation checks the overlap.
  - Active versions are immutable; changes go through an ECO (FR-9.7) or by creating a new draft version.
- **FR-9.2 BOM lines** — Must. The system shall maintain lines with line number, component product/variant, quantity per `series` (per N parents), unit, line type (item, phantom, pegged_supply, vendor, co_product, by_product), position/reference designator, constant and variable scrap percent, `round_up` policy (none, quantity, measurement, consumption), effective dates per line, resource consumption flag and operation number.
  - Maximum 5,000 lines per version; quantity numeric(18,6); scrap 0–100% with 2 decimals.
  - Co-/by-product lines carry a cost allocation percent (sum ≤ 100 across co-products) used in wave 24.
- **FR-9.3 As-of resolution and diff** — Must. The system shall resolve the applicable BOM and route version for (product, site, date, quantity) and produce a line-level diff between any two versions.
- **FR-9.4 Routing headers, versions and operations** — Must. The system shall create routes (route_id from sequence `ROUTE`), versions bound to a product/site with effective dates, and operations with operation number, operation id (reusable definition), setup time, run time per `process_qty`, queue before/after, transit time, overlap quantity, `next_operation_no` and `priority` (primary, secondary1..3) for parallel branches, scrap percent, and link type (hard/soft).
  - Maximum 500 operations per version; times in hours with 4 decimals.
- **FR-9.5 Resource requirements by capability** — Must. The system shall express each operation's requirement as resource, resource group, capability (with minimum level), resource type, or skill, with quantity of resources (default 1); scheduling (wave 23) resolves requirements, this wave validates that at least one resource satisfies each requirement at activation.
- **FR-9.6 Resources, groups, capabilities, calendars** — Must. The system shall maintain resources (type machine, tool, labor, vendor, location), resource groups per site, capabilities with levels and effective dates, and work calendars with templates (7 day patterns), shifts (start, end, efficiency percent), closed days, and a generator for a date range.
  - Calendar generation for 3 years × 5 shifts completes in < 2 s; calendars are inherited group → resource unless overridden.
- **FR-9.7 ECR/ECO workflow** — Must. The system shall create engineering change requests (title, reason, priority, affected products, requester) that move through a Temporal workflow `draft → submitted → under_review → approved | rejected`, and change orders that bundle draft BOM/route versions, go through approval, and on release atomically expire old versions (`effective_to = release_date − 1`) and activate new ones with `effective_from = release_date`.
  - Release date may be future; the workflow holds a timer until then.
  - An ECO may affect up to 500 versions; release across them is one transaction per service and compensated on failure.
- **FR-9.8 Where-used and impact analysis** — Must. The system shall return single- and multi-level where-used for a component as of a date, across sites, with distinct top-level products; an ECO impact report lists affected products, sites, active versions, and counts from consumer hooks (open production and planned orders — stubs return 0 until waves 21–22).
- **FR-9.9 Explosion and implosion API** — Must. The system shall explode a product to N levels (default all, max 25) as of a date and quantity, resolving phantoms inline, applying series and scrap, returning a tree and a flattened summed-component list; and implode (where-used tree) in the reverse direction.
  - 7-level, 2,000-component explosion < 500 ms; results are cacheable by (version set hash, date, qty).
- **FR-9.10 Routing lead-time calculation** — Should. The system shall compute operation and total lead time for a quantity: setup + run × qty / process_qty, queue and transit times added, parallel branches taking the max, and efficiency applied from the calendar; results in hours and in calendar days against the site calendar.
- **FR-9.11 Cycle and integrity checks** — Must. The system shall reject a line whose component's own active BOM (recursively) contains the parent (`ENG-BOM-CYCLE`), using the product-level closure table, and reject activation when any component is `obsolete`.

- **FR-9.12 Copy and template** — Should. The system shall copy BOM and route versions to another product or site, with optional line filtering, and keep a `copied_from_version_id` lineage.
- **FR-9.13 PLM event contract** — Could. The system shall publish ECO lifecycle events in a shape that a PLM adapter can consume and accept inbound `eco.proposed` payloads through the import tool.

1.9.4 Data model

BomVersion

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id	uuid	yes	RLS
bom_header_id	uuid	yes	FK bom_header(bom_id, name, kind, site_id)
product_id	uuid	yes	FK product (variant allowed)
site_id	uuid	no	null = all sites
effective_from / effective_to	date	from yes	inclusive
from_qty / to_qty	numeric(18,6)	no	quantity-dependent versions
status	enum	yes	draft, active, expired, superseded
approved_by / approved_at	uuid, timestamptz	no	
base_qty / yield_percent	numeric(18,6), numeric(5,2)	yes / formulas	base_qty default 1
eco_id	uuid	no	ECO that activated it
copied_from_version_id	uuid	no	lineage
version	int	yes	optimistic lock

BomLine

Field	Type	Required	Notes
id / bom_version_id	uuid	yes	FK
line_no	int	yes	unique per version, step 10
component_product_id	uuid	yes	FK product
quantity / series / uom_id	numeric(18,6), numeric(18,6), uuid	yes	quantity per `series` parents
line_type	enum	yes	item, phantom, pegged_supply, vendor, co_product, by_product
scrap_const / scrap_var_pct	numeric(18,6), numeric(5,2)	yes	default 0
round_up	enum	yes	none, quantity, measurement, consumption
position / reference_designators	text(20), text[]	no	
operation_no	int	no	consumption at operation
effective_from / effective_to	date	no	line-level
cost_allocation_pct	numeric(5,2)	co/by only	
warehouse_id / resource_consumption	uuid, bool	no	

Operation

Field	Type	Required	Notes
id / route_version_id	uuid	yes	FK route_version (same shape as BomVersion minus qty range)
operation_no	int	yes	unique per version
operation_id	text(20)	yes	reusable operation definition
next_operation_no / priority / link_type	int, enum, enum	no	priority primary/secondary1..3; link hard/soft
setup_time_h / run_time_h / process_qty	numeric(10,4), numeric(10,4), numeric(18,6)	yes	run time per process_qty
queue_before_h / queue_after_h / transit_h	numeric(10,4)	no	
overlap_qty / scrap_pct	numeric(18,6), numeric(5,2)	no	
requirements	1:N	yes	OperationRequirement(kind resource/group/capability/type/skill, ref_id, level, quantity)
cost_category_setup_id / run_id	uuid	no	bound in wave 39

Resource / WorkCalendar — resource: id, code, name, type, site_id, group_id, calendar_id, efficiency_pct (default 100), capacity_per_day_h, exclusive bool, vendor_party_id (type vendor); calendar: id, code, template jsonb (7 days × shifts), generated_from/to; calendar_day: calendar_id, date, open bool, shifts jsonb [{start, end, efficiency_pct}], total_h.

EngineeringChangeOrder — id, eco_number (sequence `ECO`), title, reason, priority, status (draft, submitted, approved, scheduled, released, rejected, cancelled), release_date, affected_versions uuid[] (draft versions to activate, with the versions they supersede), impact jsonb snapshot, workflow_id.

Invariants: no two active BOM versions overlap for the same (product, site, date, qty); every active version's lines reference non-obsolete components; product closure table has no cycles; a released ECO's versions are all `active` and their predecessors `expired`; operation numbers referenced by `next_operation_no` exist in the same version. Indexing: `bom_version(product_id, site_id, effective_from, status)`; `bom_line(component_product_id)` for where-used; product closure (parent_product_id, child_product_id, depth) maintained on activation; `calendar_day(calendar_id, date)` PK.

1.9.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/engineering/v1/bom-headers	Create header	`Idempotency-Key`	Maintain BOMs
POST	/engineering/v1/bom-versions	Create draft version (optionally copy from)	key header	Maintain BOMs
PUT	/engineering/v1/bom-versions/{id}/lines	Replace line set of a draft	full replace	Maintain BOMs
POST	/engineering/v1/bom-versions/{id}:activate	Activate outside ECO (requires duty "Activate BOM without ECO")	key header	Activate BOMs
GET	/engineering/v1/bom:resolve?product=&site=&date=&qty=	As-of version	n/a	View engineering data
POST	/engineering/v1/bom:explode	Body: product, site, date, qty, levels	pure	View engineering data
GET	/engineering/v1/bom:where-used?component=&date=&levels=	Implosion	n/a	View engineering data
GET	/engineering/v1/bom-versions/{a}:diff?against={b}	Line diff	n/a	View engineering data
POST	/engineering/v1/route-versions, PUT …/{id}/operations	Routes	as BOM	Maintain routes
POST	/engineering/v1/route-versions/{id}:lead-time?qty=	Lead-time calc	pure	View engineering data
POST	/engineering/v1/resources, /resource-groups, /capabilities, /calendars/{id}:generate	Resources and calendars	key header; generate idempotent per range	Maintain resources
POST	/engineering/v1/ecrs, /ecos, /ecos/{id}:submit, :approve, :release	Change management	key header	Maintain / Approve / Release engineering changes
GET	/engineering/v1/ecos/{id}/impact	Impact analysis	n/a	View engineering data

Validation: line set replacement runs cycle, unit-convertibility and component-state checks; activation validates overlap and resource satisfiability; `:release` requires `approved` status and all affected drafts to pass activation checks in a dry run before the transaction.

Event	Trigger	Payload fields	Consumers
dev.passage.engineering.bom.version.activated.v1	activation (direct or ECO)	bom_version_id, product_id, site_id, effective_from, effective_to, line_count, supersedes_version_id	planning, production, costing
dev.passage.engineering.bom.version.expired.v1	expiry by ECO	bom_version_id, effective_to, eco_id	planning, production
dev.passage.engineering.route.version.activated.v1	activation	route_version_id, product_id, site_id, operations[{no, resource requirements}]	planning, scheduling, production
dev.passage.engineering.resource.changed.v1	resource/calendar commit	resource_id, site_id, calendar_id, efficiency_pct, changed_fields[]	scheduling, shop floor
dev.passage.engineering.eco.submitted.v1 / approved.v1 / released.v1	workflow transitions	eco_id, eco_number, affected_versions[], release_date, approver_id	notification, audit, PLM adapter
dev.passage.engineering.calendar.generated.v1	generate	calendar_id, from, to, open_days	scheduling

1.9.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Engineering workspace	Tiles: BOMs, routes, resources, open ECRs/ECOs, versions expiring in 30 days	counts	View engineering data
BOM designer	Tree with inline line grid, add/remove, drag to reorder, as-of date picker, explode/implode toggle, version diff	line no, component, qty, series, unit, type, scrap, position, operation	Maintain BOMs
BOM versions	List per product, activate (with ECO or direct), copy	version, site, dates, qty range, status, ECO	Maintain BOMs
Route editor	Operation grid, network view (parallel branches), lead-time panel	op no, op id, next, priority, setup, run, process qty, requirements	Maintain routes
Resources and groups	New, capabilities tab, calendar assignment, utilization preview	code, type, site, group, calendar, efficiency	Maintain resources
Work calendars	Template editor, generate range, closed days	day, shifts, hours	Maintain resources
ECR / ECO	Create, add affected products/versions, impact report, submit/approve/release, schedule release date	number, title, status, affected, release date	Maintain / Approve / Release engineering changes
Where-used	Component picker, as-of date, levels, export CSV	parent, qty, level, site, version status	View engineering data

Interaction rules: BOM designer supports keyboard line entry (`Tab` through component/qty/unit, `Enter` adds next line at +10); component lookup searches wave-8 products filtered by lifecycle `active`; validation of cycles runs on line commit, overlap on activation; designer shows an "as of" ribbon with a date slider; network view in the route editor validates `next_operation_no` references live; personalization persists tree expansion depth.

1.9.7 Business rules and edge cases

- **BR-9.1** BOM/route version machine: `draft → active` (direct activation or ECO release), `active → expired` (effective_to passes or ECO), `active → superseded` (newer version activated for the same range by ECO), `draft → cancelled`; `active` is immutable.
- **BR-9.2** ECO machine: `draft → submitted → approved → scheduled → released`; `submitted | approved → rejected`; `draft | submitted | approved | scheduled → cancelled`; `released` terminal. `scheduled` waits on a Temporal timer for `release_date`.
- **BR-9.3** Draft versions use `version` optimistic locking; two engineers editing the same draft get 409 on the second save and a line-level merge view.
- **BR-9.4** Scrap: consumed quantity = `ceil_policy((qty_parent / series) × quantity × (1 + scrap_var_pct/100) + scrap_const)`; `round_up = quantity` rounds to the next whole unit, `measurement` to the unit's `decimal_precision`, `consumption` rounds the total across the order, not per line.
- **BR-9.5** Phantom lines are replaced by their own active BOM at explosion time, as of the same date; a phantom with no active BOM is reported as an explosion error, not silently skipped.

- **BR-9.6** Explosion depth cap is 25; exceeding it returns 422 `ENG-BOM-DEPTH`; cycle detection on insert makes this reachable only for legitimately deep structures.
- **BR-9.7** Parallel operations: an operation with `priority = secondary1` runs alongside the primary with the same `next_operation_no`; lead time takes the max of the branches; a secondary without a primary at the same join is rejected.
- **BR-9.8** Lead time with efficiency: run hours ÷ (resource efficiency × calendar shift efficiency); setup is not scaled by quantity; queue and transit are calendar-elapsed, not capacity-consuming.
- **BR-9.9** ECO release is all-or-nothing within engineering-service; failure after partial activation triggers Temporal compensation that reverts activated versions to `draft` and restores expired ones to `active`, then marks the ECO `approved` again with an error note.
- **BR-9.10** A release date earlier than today requires the duty "Backdate engineering changes" and is audited; a release date more than 365 days ahead is rejected.
- **BR-9.11** Deleting a resource is refused while any active route version requires it directly; capability-based requirements are unaffected.
- **BR-9.12** Co-product allocation percents on a version must sum to ≤ 100; by-products carry no allocation and may have negative cost in wave 24.

1.9.8 Acceptance criteria

- **AC-9.1** (FR-9.9) Given the 7-level, 2,000-component golden product, when exploded for qty 10 as of 2026-03-01, then the flattened component list equals `engineering/golden-explosion.csv` and p95 over 100 runs < 500 ms.
- **AC-9.2** (FR-9.3) Given BOM versions V1 (2025-01-01..2026-05-31) and V2 (2026-06-01..), when resolved as of 2026-01-01, then V1; as of 2026-06-01, then V2; as of 2026-05-31 with qty 1,000 where V3 covers qty ≥ 500, then V3.
- **AC-9.3** (FR-9.7, BR-9.2) Given an approved ECO replacing V1 with draft V2 at release date 2026-07-01, when the Temporal test clock passes 2026-07-01, then V1 is `expired` with effective_to 2026-06-30, V2 is `active`, both events are published, and the ECO is `released`.
- **AC-9.4** (FR-9.10) Given the 12 routing fixtures (sequential, two-branch parallel, overlap), when lead time is computed for qty 100, then results match fixture hours to 4 decimals.
- **AC-9.5** (FR-9.11) Given product A whose active BOM contains B, when a line adding A to B's draft is saved, then 422 `ENG-BOM-CYCLE` naming the path A→B→A.
- **AC-9.6** (FR-9.8) Given component C used in 30 products across 2 sites, when where-used is run as of today with levels = all, then 30 distinct top-level products and the ECO impact report shows the same set.
- **AC-9.7** (FR-9.5) Given an operation requiring capability "welding ≥ level 3" and no resource with it, when the route version is activated, then 422 `ENG-ROUTE-UNSATISFIED` names the operation.
- **AC-9.8** (BR-9.9) Given an ECO with 3 versions where the third fails activation, when released, then none are active, the ECO returns to `approved` with the error, and the audit shows the compensation.
- **AC-9.9** (FR-9.6) Given a 3-year calendar with 2 shifts and 10 closed days, when generated, then open-day count equals the fixture and generation < 2 s.
- **AC-9.10** (BR-9.4) Given a line qty 3 per series 2 with scrap_var 5% and scrap_const 1, round_up quantity, when the parent qty is 7, then consumption = ceil(7/2×3×1.05 + 1) = 13.

1.9.9 Test plan and fixtures

Fixtures: `engineering/golden-product.json` (7 levels, 2,000 components, 40 phantoms, 3 sites) with `golden-explosion.csv`; `engineering/version-dates.json` (12 overlapping-version scenarios); `engineering/routes-12.yaml` with lead-time oracle; `engineering/calendars.csv`; `engineering/eco-scenarios.yaml` (8 workflows including compensation); `engineering/bom-extract-ax.csv` (BOMVersion/BOM 10k rows) and `engineering/stko-stpo-extract.csv`.

Test	Type	Target	Pass criterion
Golden explosion	golden + performance	explode	byte-equal, p95 < 500 ms
As-of resolution	table-driven	resolve	12/12
Cycle detection	property (random DAGs + injected cycles)	line insert	0 cycles accepted, 0 false rejections
ECO workflow	Temporal test server	ECO	8/8 scenarios incl. compensation
Lead time	unit	calc	12/12 to 4 decimals
Scrap rounding	table-driven	consumption	40 cases
Calendar generation	unit + performance	generate	fixture equal, < 2 s
Where-used scale	performance	implode	100k lines, < 300 ms
Event contracts	contract	7 events	registry accepts

Reconciliation: Σ flattened explosion quantities = Σ over tree leaves; active version count per (product, site, date) ≤ 1 (nightly check); ECO affected versions = activated + expired events emitted.

1.9.10 Deployment and operations

Helm chart `passage-engineering` (Go service ×2, CloudNativePG `engineering-db`, Debezium connector, Temporal namespace `engineering`). Config: `ENG_BOM_MAX_LINES=5000`, `ENG_BOM_MAX_DEPTH=25`, `ENG_ROUTE_MAX_OPS=500`, `ENG_ECO_MAX_VERSIONS=500`, `ENG_EXPLODE_CACHE_TTL_S=300`. Feature flags: `engineering.direct_activation` (default off in production tenants), `engineering.formulas`, `engineering.plm_inbound`. Scheduled jobs: nightly overlap check; nightly expiry job setting `expired` on versions whose `effective_to` passed; 30-day expiry warning digest. SLOs: explode p95 < 500 ms; resolve p95 < 30 ms; ECO release p95 < 60 s. Alerts: overlap check finds > 0 rows; ECO stuck in `scheduled` past release_date + 10 min; explosion errors > 1% of calls. Runbook: "ECO compensation ran" (review error note, re-validate drafts, resubmit), "Rebuild product closure" (`passagectl eng rebuild-closure`). Rollback: previous digest; migrations additive.

1.9.11 Migration from AX / SAP

Source	Passage entity	Notes
AX BOMTable (BOMId, Name, SiteId, ItemGroupId)	BomHeader	
AX BOMVersion (ItemId, BOMId, FromDate, ToDate, FromQty, ToQty, Active, Approved, ApprovedBy, InventDimId→site)	BomVersion	Active=Yes → active; Approved without Active → draft with approved_by; date 1900-01-01 → null
AX BOM (BOMId, LineNum, ItemId, BOMQty, BOMQtySerie, UnitId, BOMType, ScrapConst, ScrapVar, Position, RoundUp, FromDate, ToDate, OprNum, InventDimId)	BomLine	BOMType enum Item/Phantom/Pegged/Vendor; co/by-products from PmfCoByProduct
AX RouteTable, RouteVersion, RouteOpr (OprNum, OprId, SetUpTime, ProcessTime, ProcessPerQty, TransitTime, QueueTimeBefore/After, OprPriority, NextOprNum, LinkType, Overlap), RouteOprTable (WrkCtrIdCost, requirements)	RouteHeader, RouteVersion, Operation, OperationRequirement	OprPriority Primary/Secondary1..3; requirements from RouteOprTable WrkCtrId / WrkCtrGroupId / WrkCtrCapability
AX WrkCtrTable (WrkCtrId, WrkCtrType, Efficiency, CalendarId, WrkCtrGroupId, VendAccount), WrkCtrResourceGroup, WrkCtrCapability, WrkCtrCapabilityResource (Level, FromDate, ToDate)	Resource, ResourceGroup, Capability, ResourceCapability	VendAccount → vendor_party_id via wave-7 alias
AX WorkCalendarTable, WorkCalendarDate, WorkCalendarDateLine (FromTime, ToTime, Efficiency), WorkCalendarTemplate	WorkCalendar, CalendarDay, Shift	times in seconds → hh:mm
D365 EngChgRequest / EngChgOrder (if present)	ECR / ECO	historical ECOs loaded as `released` with their versions
SAP MAST (MATNR, WERKS, STLNR, STLAL, STLAN usage) + STKO (STLAL, BMENG, BMEIN, DATUV, STLST status, AENNR) + STAS	BomHeader, BomVersion	one header per STLNR; one version per STLAL × DATUV; STLAN 1 (production) only; base_qty from BMENG
SAP STPO (POSNR, IDNRK, MENGE, MEINS, AUSCH scrap, DUMPS phantom, ALPOS, POSTP L/N/R, DATUV, DATUB, SANKA costing relevance)	BomLine	POSTP N (non-stock) → vendor; negative MENGE → by_product
SAP PLKO (PLNTY N, PLNNR, PLNAL, DATUV, VERWE, STATU) + MAPL + PLAS (sequence) + PLPO (VORNR, ARBPL, STEUS, VGW01..06, VGE0x units, PLIFZ, ZMERH, UEMUS overlap)	RouteHeader, RouteVersion, Operation	VGW01 setup, VGW02 machine, VGW03 labor → setup_time_h / run_time_h (labor kept as attribute); PLAS alternative sequences → parallel branches
SAP CRHD (OBJID, ARBPL, WERKS, VERWE, KAPID), CRCA, KAKO (KAPAZ, NGRAD utilization), CRTX	Resource, ResourceGroup, capacity_per_day_h	NGRAD → efficiency_pct
SAP TFACD/TFACS factory calendar, shift definitions	WorkCalendar	holiday calendars expanded for the migration range
SAP AENR (AENNR, DATUV, AETXT, AESTA)	ECO (historical, released)	effectivity dates recreated on versions

Validation: every line's component exists as a product in wave 8 and is not obsolete; every version's product is released at the version's site; no overlapping active versions after load (overlaps go to the exception queue with both versions listed); route requirements satisfiable. Not migrated: AX `BOMCalcTable` results (recomputed in wave 39), SAP `PLPO` activity-type costing fields `LAR0x` (wave 39), SAP document BOMs (STLTY D) and sales-order BOMs (wave 27 configured products), classification-based variant BOMs that depend on LO-VC procedures.

1.9.12 Dependencies, risks, open questions

Depends on: wave 8 (products, variants, UoM conversions, lifecycle), wave 6 (sites, calendars reference data), wave 7 (vendor parties for vendor-type resources), wave 4 (Temporal, outbox), wave 5 (number sequences `BOM`/`ROUTE`/`ECO`, audit).

Risk	Likelihood	Impact	Mitigation
Customers with un-versioned AX BOMs rely on in-place edits; forcing ECOs slows them	High	Medium	`engineering.direct_activation` flag per tenant with audit; ECO required only for products flagged `controlled`
SAP alternative sequences and parallel sequences map imperfectly to priority branches	Medium	Medium	Fixture comparison; unmapped PLAS structures reported, not silently flattened
Explosion cache serves stale results after ECO release	Low	High	Cache key includes the set of active version ids; release invalidates by product

Open questions: (1) Should BOM lines reference product variants directly or master plus dimension expressions (AX "dimension-based" lines)? Proposed: direct variant with an optional "inherit parent dimension" flag; decide by wave 21. (2) Formula yield at header or per line? Proposed: header yield plus line scrap; decide by wave 24. (3) Do ECOs cover released-product changes (item model group) too? Proposed: no in v1; decide by wave 10.

1.10 Wave 10: Data Migration Framework & Legacy Connectors

1.10.1 Scope and business context

AX ships the Data Import Export Framework (DIXF/DMF: `DMFDefinitionGroup`, `DMFEntity`, staging tables such as `DMFCustomerEntity`, `DMFVendorEntity`, `DMFProductEntity`, `DMFBOMEntity`) and D365 the data management framework with data entities (`CustCustomerV3Entity`, `VendVendorV2Entity`, `EcoResReleasedProductV2Entity`, `BOMEntity`), data projects and packages; SAP ships LSMW (recording, batch input, BAPI, IDoc), the S/4HANA Migration Cockpit (LTMC/LTMOM with migration objects such as `Customer`, `Supplier`, `Material`, `Bill of material`, `Routing`, `Work center`) and BAPIs (`BAPI_MATERIAL_SAVEDATA`, `BAPI_CUSTOMER_CREATEFROMDATA1`). Wave 10 builds the pipeline every later wave reuses: extract from AX SQL (ODBC) or SAP (RFC/OData or flat files), stage into a Postgres staging schema, map with a declarative DSL, validate, load through the public APIs of waves 6–9, and reconcile with row counts, hash totals and sampled field compares. Dry-run and idempotent re-run, a run ledger with lineage, and an exception queue with re-run close the loop. Phase 1 exits when the demo manufacturer's master data is fully migrated from a synthetic AX extract.

Out of scope: transactional data packs (open items, on-hand, open orders — wave 49), parallel-run diff harness (wave 49), live coexistence adapters (wave 49), X++/ABAP code conversion (non-goal).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
DIXF staging-to-target fails at row 48,213 of 100,000 with "Function DMFEntityWriter... has been incorrectly called"; the partial load leaves half-created customers and the fix is manual SQL cleanup before re-run	Non-idempotent target writes; errors swallowed into DMFExecution log	Every load is idempotent by natural key and run id; failed rows go to the exception queue, successful rows stay; re-run touches only changed rows (FR-10.6, FR-10.7)	Re-run cost after failure: hours of cleanup→0; duplicate rows after re-run 0
LSMW requires recording a transaction (XD01) screen by screen; a screen layout change in a support pack breaks the recording silently	Batch-input emulation of GUI screens	Loads go through versioned public APIs with contract tests (FR-10.5)	Recording breakage per upgrade: recurring→0
Reconciliation in both tools is "count rows in source, count in target, eyeball"; field-level drift (truncated names, dropped leading zeros in account numbers) is discovered weeks later	No hash or sampled compare	Row counts + per-column hash totals + stratified sampled field compare with tolerance per type (FR-10.8)	Field-level drift found at load: 0%→100% of sampled columns; sign-off evidence auto-generated
Mapping lives in Excel and in the heads of consultants; the enum mapping for CustTable.Blocked or MARA-MTART is re-discovered on every project	Mappings not declarative or versioned	Mapping DSL in git with unit tests, enum tables and rejection reasons (FR-10.3)	Mapping reuse across projects: 0→100% of standard objects; mapping defects found by tests before load
A 100k-item load through DIXF takes 6–10 hours; SAP LTMC similar	Row-at-a-time batch processing	Partitioned parallel loaders (Go) against bulk endpoints with checkpoints (FR-10.6)	170k master records in < 30 min vs 6–10 h

1.10.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Migration project / definition	DMFDefinitionGroup, D365 data project, data package (manifest + package header)	LSMW project/subproject/object; LTMC migration project	migration-service `MigrationProject` with `MappingSpec` versions	full
Source extraction	DIXF ODBC/file source, AX SQL direct, D365 entity export	RFC (RFC_READ_TABLE, BAPIs), OData, SE16 extracts, IDoc outbound	`Connector` (ax-odbc, d365-odata, sap-rfc, sap-odata, csv/parquet)	full
Staging	DMF*Entity staging tables	LSMW read/convert data files; LTMC staging tables (DB staging)	Postgres schema `staging` with typed tables per source object + raw jsonb	full
Mapping / transformation	DMF mapping (source → staging → target), X++ transforms	LSMW field mapping and conversion rules (ABAP), LTMOM	Mapping DSL (YAML) with functions, lookups, enum tables	improved (tested, versioned)
Validation	Entity validation on target write	LTMC simulate; BAPI return messages	Validation stage with declarative expectations plus target-API dry run	improved
Load	DMFEntityWriter, data entity insert	Batch input / BAPI / IDoc posting	Parallel loaders against wave 6–9 bulk endpoints	full
Reconciliation	DMF execution log counts	LTMC monitoring counts	Row counts, hash totals, sampled compare, exception queue	improved
Run history / lineage	DMFExecution, job history	LSMW logs, LTMC run history	`MigrationRun` ledger with per-row lineage	improved
Standard objects	DMF customer, vendor, product, BOM, route entities	Migration Cockpit objects Customer, Supplier, Material, BOM, Routing, Work center	Connector packs for DirParty/CustTable/VendTable/InventTable/EcoResProduct/BOM/Route and KNA1/LFA1/BUT000/MARA/MAST-STKO-STPO/PLKO-PLPO/CRHD	full for Phase 1 scope
Scheduling / monitoring	Batch jobs, Data management workspace	SM37, LTMC monitor	Temporal workflows, migration workspace	full

1.10.3 Functional requirements

- **FR-10.1 Migration projects and runs** — Must. The system shall define projects (name, source system type ax2012/d365fo/sapecc/s4, target tenant, object list in dependency order, mapping spec versions) and execute runs with modes `dry_run`, `load`, and `reconcile_only`; each run records start/end, user, counts per stage, and the mapping and connector versions used.
  - Object dependency order for Phase 1 is fixed: reference data → org → parties → products → released products → resources/calendars → BOMs → routes → ECOs.
- **FR-10.2 Connectors** — Must. The system shall provide connectors: `ax-odbc` (read-only SQL over the AX 2012 database with DataAreaId and RecId-based paging), `d365-odata` (data entities with `$skiptoken` paging and change tracking), `sap-rfc` (RFC_READ_TABLE with 512-byte row limit handling, BAPI reads), `sap-odata` (API_BUSINESS_PARTNER, API_PRODUCT_SRV, API_BILL_OF_MATERIAL_SRV), and `file` (CSV, XLSX, Parquet with declared schema and codepage).
  - Each connector supports incremental extraction by `ModifiedDateTime`/`RecVersion` (AX) or `AEDAT`/`LAEDA` (SAP) when available; otherwise full extract.
  - Extraction throughput target 20,000 rows/s per connector worker on the dev cluster.
- **FR-10.3 Mapping DSL** — Must. The system shall express mappings as versioned YAML: source object, target entity, key definition (natural key columns), field rules (`copy`, `const`, `concat`, `lookup` against staged or target tables, `enum` with explicit table and default, `coerce` with type and format, `expr` in a restricted expression language, `default`), and `reject_when` conditions with reason codes.
  - Type coercions: AX `utcdatetime`/SAP `DATS`+`TIMS` to timestamptz with source zone; `NoYes` and `X`/blank to boolean; decimals with explicit scale; trim and codepage normalization.
  - Every mapping ships unit tests (input row → expected output or rejection) runnable in CI.
- **FR-10.4 Validation stage** — Must. The system shall validate staged, mapped rows with declarative expectations (not null, unique, FK-exists in target or staging, regex, range, set membership, row-count within ±N% of previous run) and a target-API dry run that calls the wave 6–9 endpoints with `X-Passage-Dry-Run: true` to collect business validation errors without writing.
  - Validation output: per-row status (ok, warn, reject) with reason codes; a run proceeds to load only when rejects ≤ the project's threshold (default 0.5%).
- **FR-10.5 Load through public APIs** — Must. The system shall load exclusively through versioned bulk endpoints (`POST /org/v1/import`, `/party/v1/parties:bulk`, `/product/v1/products:bulk`, `/engineering/v1/bom-versions:bulk`, …) with batches of 500 and `Idempotency-Key =`

`sha256(run_id, object, natural_key)`, never by direct SQL into domain databases.

- **FR-10.6 Parallel, checkpointed, resumable loads** — Must. The system shall partition each object by hash of the natural key into N workers (default 8), checkpoint every batch in the run ledger, and resume from the last checkpoint after worker or pod failure without duplicates.
  - Target: 50k customers + 20k vendors + 100k items + 10k BOMs in < 30 min including reconciliation.
- **FR-10.7 Idempotent re-run** — Must. The system shall compute a content hash per mapped row; on re-run, rows with an unchanged hash and a successful prior load are skipped, changed rows are upserted by natural key, and rows absent from the source are reported (never deleted) unless the project enables `retire_missing` which sets the target to `inactive`.
- **FR-10.8 Reconciliation** — Must. The system shall produce, per object and run: source count, staged, mapped, rejected, loaded, skipped; per-column hash totals (sum of `hashtext` over normalized values for 1–20 chosen columns) on source-mapped vs target-read; and a stratified random sample (default 1%, min 100, max 5,000 rows) with field-by-field compare and per-type tolerance (decimals to scale, strings after trim/casefold, dates exact).
  - A run is `reconciled` only when counts tie, hash totals match, and sample mismatches = 0; otherwise `reconciled_with_exceptions` with every mismatch in the exception queue.
- **FR-10.9 Exception queue and re-run** — Must. The system shall list rejected and mismatched rows with reason, source key, mapped payload and target response; a user may edit the staged row, add a mapping override, or mark `accepted_difference` with a reason, then re-run only the exception set.
- **FR-10.10 Run ledger and lineage** — Must. The system shall keep an append-only ledger: for every loaded target record, the run id, source object, source key, content hash and mapped payload hash, queryable from the target record's audit tab ("migrated from AX CustTable US-00123 in run 42").
- **FR-10.11 Migration workspace** — Must. The system shall provide a workspace with project list, run history, per-stage progress, reconciliation report (downloadable as XLSX/PDF for sign-off), and exception queue.
- **FR-10.12 Packaging** — Should. The system shall package connectors and standard mapping packs as Helm sub-charts and OCI artifacts (`passage-migration-ax`, `passage-migration-sap`) versioned independently of the core service.
- **FR-10.13 PII handling in staging** — Must. The system shall encrypt PII-flagged staging columns at rest with a per-project key, purge staging data after a configurable retention (default 30 days after the run is signed off), and exclude PII from exception-queue exports unless the user holds `pii.read`.

1.10.4 Data model

MigrationRun

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id / project_id	uuid	yes	FK project(name, source_type, objects[], thresholds)
run_no	int	yes	sequence per project
mode	enum	yes	dry_run, load, reconcile_only, exceptions_only
status	enum	yes	created, extracting, staging, mapping, validating, loading, reconciling, reconciled, reconciled_with_exceptions, failed, cancelled
mapping_versions / connector_versions	jsonb	yes	pinned at start
started_at / ended_at / started_by	timestamptz, uuid	yes	
counts	jsonb	yes	per object: source, staged, mapped, rejected, loaded, skipped, mismatched
parent_run_id	uuid	no	for exceptions_only re-runs
workflow_id	text	yes	Temporal
signed_off_by / signed_off_at	uuid, timestamptz	no	enables staging purge timer

StagingRow (one physical table per source object, common columns)

Field	Type	Required	Notes
run_id / object	uuid, text	yes	partition key
source_key	text(200)	yes	e.g. `USMF
raw	jsonb	yes	source columns verbatim (PII columns encrypted)
mapped	jsonb	no	output of mapping
content_hash	bytea(32)	no	sha256 of normalized mapped payload
status	enum	yes	staged, mapped, rejected, validated, loaded, skipped, failed, mismatched, accepted_difference
reason_code / reason_detail	text	no	e.g. `MAP-ENUM-UNMAPPED`, `VAL-FK-MISSING`, `API-422`
target_id	uuid	no	id returned by load
attempt	int	yes	increments per re-run

MappingSpec

Field	Type	Required	Notes
id / project_id	uuid	yes	
source_object / target_entity	text	yes	e.g. `CustTable` → `party.customer_role`
version	int	yes	immutable once a run pinned it
natural_key	text[]	yes	columns of the mapped payload

Field	Type	Required	Notes
rules	jsonb	yes	ordered field rules
enum_tables	jsonb	no	{field: {source_value: target_value}, default}
reject_when	jsonb	no	[{expr, reason_code}]
tests	jsonb	yes	at least 1 case; CI fails on 0
checksum	text	yes	sha256 of the YAML

LineageEntry — run_id, object, source_key, target_entity, target_id, content_hash, loaded_at, idempotency_key (unique). **ReconciliationResult** — run_id, object, counts jsonb, hash_totals jsonb {column: {source, target, match}}, sample jsonb {size, mismatches[]}, status.

Invariants: a run's mapping versions never change after `extracting`; `loaded` rows have a lineage entry and a target_id; idempotency keys are unique across all runs; a `reconciled` status requires mismatches = 0 and rejects = 0 or all `accepted_difference`. Indexing: staging tables partitioned by `run_id` (list partitioning, dropped on purge); `(run_id, object, status)` btree; `lineage(target_entity, target_id)` for the audit tab; `lineage(idempotency_key)` unique.

1.10.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/migration/v1/projects	Create project	`Idempotency-Key`	Administer migration
POST	/migration/v1/projects/{id}/mapping-specs	Upload YAML; returns version and test results	checksum-idempotent	Maintain migration mappings
POST	/migration/v1/projects/{id}/runs	Start run (`mode`, `objects[]`, `from_checkpoint`)	key header; one active run per project	Run migrations
GET	/migration/v1/runs/{id}	Status and counts per stage	n/a	View migrations
POST	/migration/v1/runs/{id}:cancel	Cancel after current batch	key header	Run migrations
GET	/migration/v1/runs/{id}/reconciliation	Report (JSON; `Accept: application/vnd.openxmlformats…` for XLSX)	n/a	View migrations
GET	/migration/v1/runs/{id}/exceptions?object=&reason=	Exception queue	n/a	View migrations
PATCH	/migration/v1/exceptions/{rowId}	Edit staged row, set accepted_difference	version check	Resolve migration exceptions
POST	/migration/v1/runs/{id}:rerun-exceptions	Child run over exception set	key header	Run migrations
POST	/migration/v1/runs/{id}:sign-off	Sign-off; starts purge timer	key header	Approve migration sign-off
GET	/migration/v1/lineage?entity=party&id=	Lineage for a target record	n/a	View migrations
POST	/migration/v1/connectors/{type}:test	Connectivity and schema probe	pure	Administer migration

Validation: run start rejects if any object's mapping tests fail or a dependency object has no reconciled run in the project (override with `allow_unreconciled_deps=true` and the Administer duty); mapping upload rejects unknown functions, unmapped enum values without a default, and natural keys not produced by the rules.

Event	Trigger	Payload fields	Consumers
dev.passage.migration.run.started.v1	run start	run_id, project_id, mode, objects[], mapping_versions	notification, audit
dev.passage.migration.run.stage.completed.v1	each stage	run_id, object, stage, counts	workspace progress, audit
dev.passage.migration.run.completed.v1	terminal status	run_id, status, counts{staged, loaded, rejected, skipped, mismatched}, hash_match bool	notification, reconciliation UI, audit
dev.passage.migration.exception.raised.v1	reject or mismatch	run_id, object, source_key, reason_code	stewardship queue (wave 7 style)
dev.passage.migration.run.signedoff.v1	sign-off	run_id, signed_off_by, purge_at	audit, purge job

1.10.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Migration workspace	Tiles: projects, active runs, exceptions open, last reconciliation status	counts, status chips	View migrations
Project detail	Edit objects/order, upload mapping, view test results, start run (mode picker), connector test	object, mapping version, tests passed, last run	Administer migration
Run monitor	Per-object stage pipeline with counts and throughput, cancel, logs link (Tempo trace)	stage, rows/s, ETA, errors	Run migrations
Reconciliation report	Counts table, hash totals per column, sample mismatches grid, download XLSX/PDF, sign-off	object, source/target counts, hash match, mismatch fields	View / Approve migration sign-off
Exception queue	Filter by reason, edit row inline, bulk accept difference with reason, re-run selection	source key, reason code, detail, mapped payload, target response	Resolve migration exceptions
Mapping editor (read-mostly)	View rules, enum tables, test cases; edit opens YAML with schema validation	rule, source, target, function	Maintain migration mappings
Lineage tab (on any migrated record)	Open run, open staged row	run no, source object, source key, loaded at	View migrations

Interaction rules: run monitor auto-refreshes every 2 s via SSE; exception grid supports keyboard navigation and inline JSON edit with schema hints; bulk accept limited to 1,000 rows per action and requires a reason of ≥ 10 characters; reconciliation report is read-only once signed off; PII columns render

masked unless `pii.read`.

1.10.7 Business rules and edge cases

- **BR-10.1** Run state machine: `created → extracting → staging → mapping → validating → loading → reconciling → reconciled | reconciled_with_exceptions`; any stage `→ failed` after exhausting retries (5 per batch, exponential backoff 1–60 s); `→ cancelled` from any non-terminal state after the current batch commits; `dry_run` stops after `validating` with a simulated load report.
- **BR-10.2** Only one active run per project; a second start returns 409 with the active run id.
- **BR-10.3** Natural keys are normalized before hashing: trim, casefold for codes that the target treats case-insensitively, leading-zero preservation for SAP numbers (never cast to integer).
- **BR-10.4** Enum mapping with no match and no default rejects the row with `MAP-ENUM-UNMAPPED` and the value; mapping with default records a `warn` and the default used.
- **BR-10.5** Decimal coercion: AX `real` values are read as numeric(32,16) and rounded half-even to the target scale; a row whose rounding changes the value by more than 0.5 units of the last target decimal is flagged `warn` with code `MAP-PRECISION-LOSS`.
- **BR-10.6** Dates: AX `1900-01-01` and SAP `00000000` / `99991231` are nulls or open-ended markers by rule, never literal dates.
- **BR-10.7** Load batches are atomic per batch on the target side; a batch with one failing row is retried as singletons so one bad row rejects only itself.
- **BR-10.8** Re-run skip rule: skip when `content_hash` equals the last `loaded` lineage hash and the target record's version has not changed since (target `updated_at` ≤ lineage `loaded_at`); otherwise upsert; a target changed by a user after the load is reported as `TARGET-DRIFT` and not overwritten unless `overwrite_drift=true`.
- **BR-10.9** Hash totals use the same normalization on both sides; a hash mismatch with matching counts triggers an automatic widened sample (10%) before the run is marked with exceptions.
- **BR-10.10** Dependency between objects: a party role row whose party failed in the same run is rejected with `VAL-DEP-FAILED` referencing the parent source key, not retried blindly.
- **BR-10.11** Staging purge runs 30 days after sign-off (configurable 7–365); purge drops the run's partitions; lineage and reconciliation results are never purged.
- **BR-10.12** Concurrency with live users: loads run as a service account under OPA with the duties of the project owner; records created carry `created_by = migration:{run_id}` for audit.

1.10.8 Acceptance criteria

- **AC-10.1** (FR-10.6, FR-10.8) Given the seeded AX-shaped dataset (50k CustTable, 20k VendTable, 100k InventTable/EcoResProduct, 10k BOMVersion/BOM), when a `load` run executes on the dev cluster, then it reaches `reconciled` in < 30 min with counts tied, all hash totals matching and 0 sample mismatches.
- **AC-10.2** (FR-10.7) Given the completed run, when the identical run executes again, then loaded = 0, skipped = 170,000, and no target `updated.v1` events are published.
- **AC-10.3** (FR-10.3) Given the mapping test suites for the 12 standard Phase 1 objects, when CI runs, then every test passes including type coercion (utcdatetime, NoYes, DATS), enum mapping (Blocked, MTART), default fill and rejection reasons.
- **AC-10.4** (FR-10.6) Given a load in progress, when a loader pod is killed at batch 137, then the resumed run continues from checkpoint 136, finishes, and the lineage table has no duplicate idempotency keys.
- **AC-10.5** (FR-10.4) Given a `dry_run` with 10 rows pointing at a non-existent customer group, when validated, then 10 rows are `reject` with `VAL-FK-MISSING` and nothing is written to party-db.
- **AC-10.6** (FR-10.9) Given 25 exceptions, when 20 are edited and 5 marked accepted_difference, then `rerun-exceptions` loads 20, skips 5, and the parent run moves to `reconciled`.
- **AC-10.7** (BR-10.8) Given a customer modified by a user after load, when the run is repeated with changed source data, then the row is `TARGET-DRIFT` and the user's change is preserved.
- **AC-10.8** (FR-10.2) Given the `sap-rfc` connector against the recorded RFC_READ_TABLE fixture with 600-byte rows, when extracting KNA1, then all columns are reassembled across two field groups with 0 truncation.
- **AC-10.9** (FR-10.13) Given a signed-off run, when 30 days pass on the Temporal test clock, then staging partitions are dropped and lineage rows remain.
- **AC-10.10** (FR-10.1) Phase 1 exit: given the synthetic AX extract of the demo manufacturer, when the full project runs, then every object is `reconciled`, the wave 6–9 workspaces show the migrated data, and the reconciliation XLSX is attached to the sign-off record.

1.10.9 Test plan and fixtures

Fixtures: `migration/ax-seed/` (generated AX-shaped CSVs with real column names: DirPartyTable, DirPerson, DirOrganization, LogisticsPostalAddress, CustTable 50k, VendTable 20k, EcoResProduct/InventTable 100k, EcoResProductMaster 2k, BOMVersion/BOM 10k, RouteVersion/RouteOpr 3k, WrkCtrTable 200; generator seed 10); `migration/sap-seed/` (KNA1/KNB1 10k, LFA1/LFB1 5k, BUT000/BUT100 15k, MARA/MARC/MARD/MARM 20k, MAST/STKO/STPO 5k, PLKO/PLPO/MAPL 2k, CRHD 100); `migration/mapping-tests/*.yaml` (≥ 10 cases per object); `migration/rfc-recorded.json`; `migration/demo-manufacturer-ax/` (the phase-exit extract).

Test	Type	Target	Pass criterion
Mapping DSL	unit	every rule function	100% branch coverage on functions; all object test cases pass
Full load performance	performance	end-to-end run	< 30 min, reconciled
Idempotent re-run	E2E	run twice	0 loaded second time
Pod-kill resume	chaos	loaders	no duplicate idempotency keys
Dry run isolation	contract	target APIs	0 writes on `X-Passage-Dry-Run`
Connector contracts	contract (recorded)	ax-odbc, d365-odata, sap-rfc, sap-odata, file	schema probes and paging match fixtures
Reconciliation math	property	hash totals and sampling	injected drift in 1 of 10k rows detected ≥ 99% with widened sample
Exception re-run	E2E	exceptions_only	parent reaches reconciled

Test	Type	Target	Pass criterion
PII at rest	security	staging	encrypted columns unreadable with DB role; purge verified
Event contracts	contract	5 events	registry accepts

Reconciliation checks: source count = staged; staged = mapped + rejected; mapped = loaded + skipped + failed; Σ per-object loaded = lineage rows for the run; target row count delta (before/after run) = loaded − upserts of existing keys.

1.10.10 Deployment and operations

Helm chart `passage-migration` (TS orchestration service ×2, Go loader workers as a scalable Deployment 2–16 replicas via HPA on queue depth, CloudNativePG `migration-db` with staging partitions on a separate tablespace, Temporal namespace `migration`); sub-charts `passage-migration-ax` (ODBC driver image with FreeTDS, mapping packs) and `passage-migration-sap` (RFC SDK not redistributable: image expects the customer-mounted NW RFC library; OData path works without it). Config: `MIG_WORKERS_DEFAULT=8`, `MIG_BATCH_SIZE=500`, `MIG_REJECT_THRESHOLD_PCT=0.5`, `MIG_SAMPLE_PCT=1`, `MIG_STAGING_RETENTION_DAYS=30`, `MIG_PII_KEY_RING` (Vault path), connector secrets in Vault. Feature flags: `migration.retire_missing`, `migration.overwrite_drift`, `migration.sap_rfc`. Scheduled jobs: hourly purge timer check; nightly lineage integrity check. SLOs: loader throughput ≥ 100 rows/s per worker sustained; run monitor update lag < 2 s; availability 99.5% (non-critical path). Alerts: run `failed`; batch retry rate > 5%; staging tablespace > 80%; connector auth failure. Runbook: "Resume a failed run" (`passagectl mig resume --run`), "Rotate PII key" (re-encrypt staging for active runs), "Widen sample manually". Rollback: previous digest; staging schema per run is unaffected; mapping packs are OCI artifacts pinned by digest in the project.

1.10.11 Migration from AX / SAP

This wave is the migration tool; its own "migration" is the catalog of standard objects it ships with, each a mapping pack with tests:

Source object (connector)	Target entity (wave)	Pack notes
AX CompanyInfo/OMLegalEntity, OMOperatingUnit, OMHierarchy*, InventSite, InventLocation, LogisticsPostalAddress, FiscalCalendar*, UnitOfMeasure* (ax-odbc)	wave 6 entities	natural keys DataAreaId, OMOperatingUnitNumber, SiteId, InventLocationId
AX DirPartyTable, DirPerson, DirOrganization, DirPartyLocation, LogisticsElectronicAddress, CustTable, VendTable, ContactPerson, CustGroup, VendGroup, PaymentTerm, DlvTerm, DlvMode (ax-odbc)	wave 7 entities	natural keys PartyNumber, `DataAreaId
AX EcoResProduct*, InventTable, InventTableModule, InventItemSetupFilter, InventItemGroup, InventModelGroup, UnitOfMeasureConversion, EcoResAttribute*/Category*, InventItemGTIN (ax-odbc)	wave 8 entities	natural keys DisplayProductNumber, `DataAreaId
AX BOMTable, BOMVersion, BOM, RouteTable, RouteVersion, RouteOpr, RouteOprTable, WrkCtrTable, WrkCtrResourceGroup, WrkCtrCapability*, WorkCalendar* (ax-odbc)	wave 9 entities	natural keys BOMId, `BOMId
D365 F&O entities CustCustomerV3Entity, VendVendorV2Entity, EcoResReleasedProductV2Entity, BOMEntity, RouteEntity, WorkCenterEntity (d365-odata)	waves 7–9	same natural keys via entity key fields
SAP T001, T001W, T001L, CSKS, ADRC, T009/T009B, T006 (sap-rfc / file)	wave 6	keys BUKRS, WERKS, `WERKS
SAP KNA1/KNB1, LFA1/LFB1, BUT000/BUT100/BUT020, ADR2/ADR6, KNVK, T052, KNKK (sap-rfc / sap-odata API_BUSINESS_PARTNER)	wave 7	keys KUNNR, LIFNR, PARTNER; CVI links honored
SAP MARA, MAKT, MARC, MARD, MARM, MEAN, KLAH/KSML/CABN/CAWN/AUSP (sap-rfc / API_PRODUCT_SRV)	wave 8	keys MATNR, `MATNR
SAP MAST/STKO/STPO/STAS, PLKO/PLPO/PLAS/MAPL, CRHD/CRCA/KAKO, AENR (sap-rfc / API_BILL_OF_MATERIAL_SRV)	wave 9	keys `STLNR

Transformation notes are owned by the waves above; this wave contributes the shared functions (`ax_datetime`, `ax_noyes`, `sap_dats`, `sap_lead_zero`, `codepage_normalize`, `lookup_party_alias`). Validation rules are the union of each target wave's rules plus cross-object dependency checks. Not migrated by this wave: any transactional table (CustTrans, VendTrans, InventTrans, InventSum, BKPF/BSEG, MSEG, VBAK/VBAP, EKKO/EKPO — wave 49 packs), AX security roles (re-modeled in wave 2's admin UI), DMF/LSMW definitions themselves (replaced by mapping packs), X++/ABAP customizations (non-goal).

1.10.12 Dependencies, risks, open questions

Depends on: waves 6, 7, 8 and 9 (bulk endpoints with `X-Passage-Dry-Run` support and natural-key upsert semantics), wave 4 (Temporal, outbox), wave 5 (audit tab integration for lineage, import/export tool), wave 2 (service-account duties, Vault secrets).

Risk	Likelihood	Impact	Mitigation
Bulk endpoints in waves 6–9 lack dry-run or upsert semantics when wave 10 starts	Medium	High	Contract requirement added to each wave's Definition of Done; wave 10 contract tests run against waves 6–9 in CI
SAP RFC SDK licensing blocks a self-contained connector image	High	Medium	OData and flat-file paths are first-class; RFC is a customer-mounted library
Synthetic AX fixture does not resemble real data quality	Medium	Medium	Fixture generator injects 2% noise (codepage, trailing spaces, invalid enums); first real customer extract (PRD open question 2) replaces it by wave 20
Reconciliation sampling misses rare drift	Low	Medium	Hash totals per column catch systematic drift; widened sample on mismatch

Open questions: (1) Is the mapping DSL YAML plus a restricted expression language enough, or do packs need a plug-in hook in TypeScript? Proposed: YAML only in v1, hooks behind a flag by wave 49; decide by wave 16. (2) Should lineage be stored in each domain database (next to the record) or only in migration-db? Proposed: migration-db with the audit tab querying it; decide by wave 11. (3) Who owns sign-off on reconciliation for the phase-exit demo — product or the implementation partner persona? Proposed: both sign; decide at the phase-1 exit review.

Part 2 — Phase 2: Financial Core (waves 11–15)

2.0 Phase overview

Business outcome at phase exit. At the end of wave 15 a controller closes a fiscal period for the demo manufacturer's three legal entities entirely inside Passage: every financial fact lives in one append-only universal journal (`ledger` service, Go + Postgres) that balances by construction; each line carries transaction, accounting and reporting currency amounts and is posted simultaneously to the leading, local-GAAP and tax ledgers; tax is determined and settled to the cent; periods are controlled per ledger × module; allocations, accruals and year-end carry-forward are automated; and the balance sheet, income statement and cash flow statement are generated from the journal with drill-down to the originating document. Every later module (inventory, AP, AR, assets, projects) posts only through the wave 11 posting API — there is no second path to the general ledger.

Phase-exit sign-off criteria

- Trial balance for each demo entity and each ledger sums to 0.00 in all three currencies after 1M synthetic lines; balance projection equals a full re-aggregation of the journal (zero rows differ).
- Posting API sustains 5,000 lines/s for 30 min on the dev cluster with p99 < 200 ms; duplicate `Idempotency-Key` yields a byte-identical 200 response.
- Parallel-ledger fixture (20 documents) posts per the matrix in §2.12.7 and each ledger's trial balance is zero.
- 500-scenario tax matrix (US/EU/IN/CA) matches expected results; settlement run ties to GL tax accounts to the cent.
- One simulated period closed across 3 entities with a 40-task checklist, dependencies honored; year-end carry-forward produces opening balances equal to closing balances per dimension combination.
- Statement pack (TB, BS, P&L, CF) ties to the trial balance to the cent; cash flow reconciles to the change in cash; finance stakeholders record sign-off in the wave 15 issue.

Cross-wave data model summary

Entity	Owning service	Key relationships
Ledger, LedgerGroup	ledger	LegalEntity (wave 6) 1:N Ledger; Ledger → FiscalCalendar, AccountingCurrency, ReportingCurrency
MainAccount, AccountStructure, DimensionAttribute, DimensionValue, DimensionCombination	ledger	ChartOfAccounts 1:N MainAccount; AccountStructure → ordered DimensionAttributes; DimensionCombination = hashed tuple of DimensionValues
JournalEntry, JournalLine, JournalBatch, JournalBatchLine	ledger	JournalEntry 1:N JournalLine; JournalEntry N:1 Ledger; JournalBatch (unposted) → JournalEntry (posted); SourceDocument reference (URI)
Balance (projection)	ledger	Keyed by ledger × account × DimensionCombination × fiscal period × currency type
ExchangeRateType, ExchangeRate, CurrencyPolicy	fx	ExchangeRate → (type, from, to, valid_from); Ledger → ExchangeRateType default
RevaluationRun, RevaluationLine	ledger	Produces JournalEntry pairs (posting + reversal)
TaxCode, TaxRate, TaxGroup, ItemTaxGroup, TaxJurisdiction, ExemptionCertificate, WithholdingTaxCode	tax	TaxGroup N:M TaxCode; ItemTaxGroup N:M TaxCode; TaxRate effective-dated per TaxCode × Jurisdiction
TaxTransaction, TaxSettlementRun	tax	TaxTransaction → JournalLine (posted tax line); SettlementRun → JournalEntry
PeriodStatus, CloseTemplate, CloseTask, CloseRun	close (TS/Temporal) + ledger (PeriodStatus)	PeriodStatus keyed ledger × period × module; CloseRun → CloseTasks → assignees (Party/Employee role)
AllocationRule, AllocationRun, AccrualScheme, RecurringJournal	ledger	Runs produce JournalEntry; rules reference AccountStructure filters
ReportDefinition, ReportRow, ReportColumn, ReportTree, StatementRun, StatementCell	reporting	ReportDefinition → MainAccount ranges + DimensionValue filters; StatementCell → Balance keys → JournalLines

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.ledger.journal.posted.v1	ledger	reporting, tax, audit, lakehouse, all subledgers (waves 16+)	entry id, ledger id, document id, posting date, period, line count, totals per currency type, source ref, trace id
dev.passage.ledger.journal.reversed.v1	ledger	reporting, tax, subledgers	original entry id, reversing entry id, reason code
dev.passage.ledger.balance.rebuilt.v1	ledger	reporting, ops alerting	ledger id, high-water sequence, rows, diffs
dev.passage.fx.rate.published.v1	fx	ledger, reporting, AP/AR (later)	rate type, pair, valid_from, rate, source, previous rate
dev.passage.fx.provider.degraded.v1	fx	ops alerting	provider, last success, fallback mode
dev.passage.ledger.revaluation.completed.v1	ledger	reporting, close	run id, ledger, period, unrealized gain/loss totals, entry ids
dev.passage.tax.calculated.v1	tax	document services (waves 17+, 26+)	document ref, lines, tax amounts, provider used, fallback flag
dev.passage.tax.settled.v1	tax	reporting, ledger	settlement period, payable/receivable totals, entry id
dev.passage.close.period.status-changed.v1	ledger	all posting services, close, reporting	ledger, period, module, old → new status, actor, approvals
dev.passage.close.task.completed.v1	close	close board, notifications	run id, task id, assignee, evidence link
dev.passage.ledger.allocation.completed.v1	ledger	reporting, close	run id, rule ids, iterations, entry ids
dev.passage.ledger.year-end.closed.v1	ledger	reporting, close	ledger, fiscal year, opening entry ids, retained earnings per dimension
dev.passage.reporting.statement.generated.v1	reporting	distribution, audit	run id, definition id, period, hash, output URIs

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Balance projection drifts from journal under concurrent posting	Medium	High	Projection updated in the same Postgres transaction as the entry; nightly rebuild-and-compare job; drift alert at 1 row	ledger lead
Rounding differences across three currencies and N ledgers accumulate	Medium	High	ISO 4217 minor-unit rules in one shared Go package `money`; property tests over JPY/KWD/USD/EUR; rounding-difference account with tolerance	ledger lead
Tax engine correctness outside fixture coverage	High	High	Provider adapters (Avalara/Vertex) as production default for US; internal engine for VAT with accountant-reviewed fixtures	tax lead
Period reopened silently	Low	High	Reopen requires two distinct approvers (Temporal human tasks) and emits an audit event; no API bypass	close lead
Reporting projection lags and statements are stale at close	Medium	Medium	Statement run records the journal high-water sequence it covered; close task blocks until lag = 0	reporting lead
Accountant review capacity (two-person team)	High	Medium	Fixtures reviewed per wave by an external practicing accountant; sign-off recorded in the wave issue	product

2.11 Wave 11: Universal Journal & Chart of Accounts

2.11.1 Scope and business context

AX stores financial truth across `GeneralJournalEntry/GeneralJournalAccountEntry` with `DimensionAttributeValueCombination` as the account key, and SAP consolidated FI/CO into `ACDOCA`. Both rely on a posting layer that every module calls. Wave 11 builds that layer first: one append-only, double-entry journal per ledger, financial dimensions validated by account structures, a posting API with idempotency keys, a balance projection, and a journal workspace for manual, recurring and template journals with approval and attachments.

At the end of the wave a controller can define a chart of accounts and dimensions, post a manual journal (with approval), reverse it, and see the trial balance update in real time. Subledgers (waves 16+) will call `POST /api/v1/ledger/journal-entries` and nothing else.

Out of scope: multiple currencies beyond a single accounting currency (wave 12), tax lines (wave 13), period status enforcement beyond open/closed defaults (wave 14), statements (wave 15), intercompany (wave 40), budget control (wave 38).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX `LedgerJournalTable` form: posting a 2,000-line journal runs `LedgerJournalCheckPost` synchronously; validation errors surface one at a time in the Infolog after minutes of processing, forcing repeated post attempts	Validation and posting interleaved in X++ with per-line dimension lookups against `DimensionAttributeValueCombination`	`POST /ledger/validate` returns every violation with `line_index` in one pass; posting is a single atomic transaction	Validation of 2,000 lines < 1 s (vs 2–5 min); all errors in one response (vs 1 per attempt)
SAP FB50/F-02: a document that fails mid-posting (dump, lock on `NRIV` number range) can consume a document number or leave a parked `VBKPF` that is re-keyed	Number assignment and insert into `BKPF`/`BSEG` are not one atomic unit with the application-level checks	Gap-free `posting_sequence` assigned inside the same Postgres transaction; idempotency key makes retries safe	0 lost numbers; 0 duplicate postings under crash test; retry needs 0 user action
Integrations double-post when a timeout hides a success (AIF/IDoc retries, Data management framework re-runs)	No idempotency contract on `LedgerJournalTrans` import or BAPI_ACC_DOCUMENT_POST	`Idempotency-Key` header with 400-day retention; same key + same payload → replay, different payload → 409	Duplicate-post incidents 1–3/month typical → 0
AX trial balance list page / SAP FAGLB03 are read from totals tables (`FAGLFLEXT`) that drift from line items after failed batches; reconciliation jobs (RFINDEX) run for hours	Totals maintained by separate update paths from line inserts	`Balance` upserted in the posting transaction; rebuild-and-compare job emits a diff count	Balance vs journal drift = 0 rows; rebuild of 10M lines < 15 min (vs multi-hour RFINDEX)
Reaching the source document from a GL line takes 6–10 clicks (AX voucher transactions → origin → subledger; SAP FB03 → environment → document flow)	Linkage stored in `SubledgerVoucherGeneralJournalEntry` / `BKPF-AWKEY` with per-module resolution logic	Every entry carries `source.uri`; UI resolves it as one link; `trace_id` on every entry	Clicks to source 8 → 1; any posting traceable through Tempo end-to-end
External systems (MES, BI, PLM) need custom X++/ABAP or middleware to learn that a posting happened	No event stream from the ledger; polling on `GeneralJournalEntry`/`ACDOCA` timestamps	`dev.passage.ledger.journal.posted.v1` on Kafka via outbox, AsyncAPI-described, plus REST/MCP	Integration lead time weeks → same day; event lag p95 < 2 s (vs 15-min+ polling)

2.11.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Chart of accounts and main accounts	`MainAccount`, `LedgerChartOfAccounts`, form MainAccountListPage	`SKA1` (chart), `SKB1` (company-code view), FS00, OB13	`ledger.MainAccount`, `ledger.ChartOfAccounts`; screen Chart of accounts	full
Financial dimensions	`DimensionAttribute`, `DimensionAttributeValue`, `DimensionFinancialTag`	CO objects (cost center `CSKS`, profit center `CEPC`), ACDOCA fields `RCNTR`, `PRCTR`, `RBUSA`	`ledger.DimensionAttribute`, `ledger.DimensionValue` (up to 20 per ledger)	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Account structures / dimension validation	`DimensionHierarchy` (account structure), advanced rules	Field status groups (OBC4), validation/substitution (OB28/GGB0)	`ledger.AccountStructure` with ordered constraint rows	full
Ledger account combination	`DimensionAttributeValueCombination` (`RecId` as LedgerDimension)	ACDOCA field combination (no single key)	`ledger.DimensionCombination` (SHA-256 of ordered tuple)	improved: content-addressed, immutable
Journal entry / lines	`GeneralJournalEntry`, `GeneralJournalAccountEntry`, `SubledgerVoucherGeneralJournalEntry`	`BKPF`/`BSEG`, `ACDOCA`	`ledger.JournalEntry`, `ledger.JournalLine`	full
Manual journals (unposted)	`LedgerJournalTable`/`LedgerJournalTrans`, LedgerJournalTable form, journal names `LedgerJournalName`	FB50/F-02 park (`VBKPF`/`VBSEG`), FV50	`ledger.JournalBatch`, `ledger.JournalBatchLine`; screen Journals	full
Recurring / periodic journals	Periodic journals (`LedgerJournalTable.JournalType = Periodic`)	FBD1 recurring entries (`BKDF`)	`ledger.RecurringJournal` (template + schedule)	full
Journal approval	Ledger journal workflow `LedgerJournalApproval`	Parked document workflow (WS10000051)	Temporal approval workflow template (wave 4) bound to JournalBatch	full
Reversal	`LedgerJournalTrans` reverse transaction, `TransactionReversalTrans`	FB08, FBRA (reset and reverse)	`POST /journal-entries/{id}/reverse` creating negating entry	full
Voucher numbering	`NumberSequenceTable` per journal name	Document number ranges per document type (FBN1, `NRIV`)	Number sequence service (wave 5), gap-free per ledger × document type	full
Trial balance / balances	`GeneralJournalAccountEntry` aggregation, Trial balance list page, `LedgerBalance...`	`FAGLFLEXT` (totals), `GLT0`, FAGLB03	`ledger.Balance` projection; Trial balance screen	full
Posting definitions / posting profiles	Posting definitions, `LedgerPostingSetup`	Account determination (OBYC, T030)	Deferred to the owning subledger waves; wave 11 exposes only account + dimension resolution hooks	partial (by design)

2.11.3 Functional requirements

- **FR-11.1 Ledger and chart of accounts** — Must. The system shall maintain one or more ledgers per legal entity, each bound to a chart of accounts, a fiscal calendar (wave 6), and an accounting currency.
  - A chart of accounts may be shared across legal entities (SAP `SKA1` semantics); company-specific attributes (`SKB1` semantics: suspended, posting allowed, default dimensions) live on `MainAccountLedgerSetting`.
  - Main account id is 1–20 characters, alphanumeric, unique per chart; type ∈ {asset, liability, equity, revenue, expense, profit_and_loss_total, balance_sheet_total, reporting}; totals and reporting types are non-postable.
  - Maximum 50,000 main accounts per chart; 500 charts per tenant.
- **FR-11.2 Financial dimensions** — Must. The system shall support up to 20 dimension attributes per ledger, each backed either by a Passage entity (department, cost center, site, project, customer, vendor, item) or by a custom value list.
  - Dimension values are effective-dated (`active_from`, `active_to`) and may be suspended; values from backing entities are synchronized by consuming that entity's CloudEvents.
  - Default dimensions on master data (party, product, site) are stored as a `DimensionCombination` id and merged at posting time by precedence: explicit line > source document header > master data default > account default.
- **FR-11.3 Account structures and validation** — Must. The system shall validate every line's (main account, dimensions) tuple against the ledger's active account structures before posting.
  - An account structure is an ordered list of segments (main account first) with per-segment constraint rows: value ranges, lists, wildcards (`*`), and "blank allowed". Advanced rules add extra segments when a filter matches (AX advanced rule semantics).
  - Validation is deterministic: the first structure whose main-account filter matches is applied; no match → `LEDGER_STRUCTURE_NOT_FOUND`.
  - Structures are versioned with `effective_from`; a posting uses the version effective on the posting date.
- **FR-11.4 Dimension combinations** — Must. The system shall intern each validated tuple as an immutable `DimensionCombination` whose id is the SHA-256 of the canonical tuple `(structure_version_id, account_id, dim1_value_id, …, dim20_value_id)`.
  - Combinations are never deleted; display name is derived (`1010-DEPT01-CC200`).
  - Lookup by id is a primary-key read; lookup by tuple is a hash computation plus one read.
- **FR-11.5 Posting API** — Must. The system shall expose `POST /api/v1/ledger/journal-entries` that posts a balanced entry atomically or rejects it entirely, with an idempotency key.
  - Request: `Idempotency-Key` header (1–128 chars, scoped to tenant × legal entity), body per §2.11.5. Maximum 10,000 lines per entry; larger documents must be split by the caller with a shared `document_id`.
  - Response 201 with the posted entry; 200 on replay of the identical payload; 409 `LEDGER_IDEMPOTENCY_CONFLICT` if the same key is reused with a different payload hash; 409 `LEDGER_IDEMPOTENCY_IN_PROGRESS` with `Retry-After: 1` while the first request is still in flight.
  - 422 with a `violations[]` array, each `{code, line_index, field, message}`; codes in BR-11.9. Posting never partially succeeds.
  - Idempotency records are retained 400 days; replays after expiry are treated as new requests (documented contract).
- **FR-11.6 Append-only journal** — Must. The system shall never update or delete a posted `JournalEntry` or `JournalLine`; corrections are new entries.
  - Enforced by Postgres: the application role has INSERT and SELECT only on `journal_entry`/`journal_line`; a trigger raises on UPDATE/DELETE for all roles.
  - Every entry receives a gap-free `posting_sequence` (bigint) per ledger from the number sequence service in gap-free mode, assigned inside the posting transaction.
- **FR-11.7 Balance projection** — Must. The system shall maintain `ledger.Balance` rows keyed by (ledger, account, dimension combination, fiscal period, currency type) and update them in the same database transaction as the posting.
  - Fields: `debit`, `credit`, `net`, `line_count`, `last_posting_sequence`; period-to-date only; year-to-date and opening balances are computed at query time by summing periods (opening balances live in period 0, see wave 14).

    - A rebuild job (§2.11.7 BR-11.7) reconstructs the table from the journal and reports differences.
- **FR-11.8 Reversal** — Must. The system shall create an exact negating entry for any posted entry via `POST /journal-entries/{id}/reverse`.
    - Reversal date defaults to the original posting date; may be any open date ≥ original. Each reversing line has `debit/credit` swapped and `reversal_of_line_id` set.
    - An entry can be reversed once; reversing a reversal is permitted (re-instating), each linked to its predecessor.
- **FR-11.9 Journal workspace (manual journals)** — Must. The system shall provide journal batches (AX `LedgerJournalTable` semantics) with lines, totals, validation on save, submit-for-approval, and post.
    - Journal names carry: voucher number sequence, default offset account, approval workflow id, `posting_restriction` (own journals only / by user group), and `lines_limit` (default 5,000).
    - Batch statuses: draft → validated → in_approval → approved → posting → posted | rejected | cancelled (state machine in §2.11.7).
- **FR-11.10 Recurring journals and templates** — Should. The system shall allow a posted or draft batch to be saved as a template (amounts or percentages) and a template to be scheduled (cron expression, fiscal-period-relative dates) to generate batches.
    - Generation runs on Temporal at the scheduled time; generated batches enter the approval workflow unless the template is flagged `auto_post` and the user who scheduled it holds duty `Post ledger journals without approval`.
- **FR-11.11 Attachments and notes** — Should. The system shall attach files (max 25 MB each, 50 per batch) and notes to batches and carry references to the posted entry.
- **FR-11.12 Source document linkage** — Must. The system shall record on each entry a `source` object `{system, type, id, uri}` and on each line an optional `source_line_ref`, enabling drill-down from statements (wave 15) to the originating document.
- **FR-11.13 Opening balances load** — Must. The system shall accept opening balances through the migration framework (wave 10) as a journal entry in period 0 of the first fiscal year, balanced against a conversion-suspense account that must be zero after load.
- **FR-11.14 Query API** — Must. The system shall expose line and balance queries with filters on ledger, account ranges, dimension values, posting date and period, paginated by keyset cursor (max 1,000 rows per page).

2.11.4 Data model

MainAccount

Field	Type	Required	Notes
id	uuid	yes	PK
chart_of_accounts_id	uuid	yes	FK ChartOfAccounts
account_number	text(20)	yes	unique per chart; `^[A-Za-z0-9._-]{1,20}$`
name	text(120)	yes	translatable via wave 3 metadata
account_type	enum	yes	asset, liability, equity, revenue, expense, profit_and_loss_total, balance_sheet_total, reporting
debit_credit_default	enum	no	debit/credit; UI default only
is_postable	bool	yes	false for totals/reporting
suspended_from	date	no	posting rejected on/after this date
default_dimension_combination_id	bytea(32)	no	FK DimensionCombination
reconciliation_account_type	enum	no	none, customer, vendor, bank, inventory, fixed_asset — subledger-only posting (SAP recon account semantics)
created_at / created_by	timestamptz / uuid	yes	audit from CDC

AccountStructure (header) and **AccountStructureConstraint** (rows)

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	FK Ledger
version	int	yes	unique (ledger_id, name, version)
name	text(60)	yes	
segments	jsonb	yes	ordered array of dimension_attribute_id; first is MAIN_ACCOUNT
effective_from	date	yes	
status	enum	yes	draft, active, retired
constraint rows: segment_index	int	yes	0 = main account
constraint rows: parent_path	jsonb	yes	the matched values of previous segments this row applies under
constraint rows: allowed	jsonb	yes	list of `{from,to}` ranges, exact values, `*`, `blank`

JournalEntry and **JournalLine**

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	FK Ledger
legal_entity_id	uuid	yes	denormalized for RLS
document_id	uuid	yes	groups entries across ledgers and split batches
voucher	text(30)	yes	from number sequence; unique (ledger_id, voucher)
posting_sequence	bigint	yes	gap-free per ledger; unique (ledger_id, posting_sequence)
posting_date	date	yes	
fiscal_period_id	uuid	yes	FK FiscalPeriod (wave 6) resolved from posting_date
document_date	date	no	

Field	Type	Required	Notes
description	text(255)	no	
source_system / source_type / source_id / source_uri	text	yes	e.g. `ledger`, `manual_journal`, batch id, `passage://ledger/journal-batches/{id}`
idempotency_key	text(128)	yes	unique (tenant_id, legal_entity_id, idempotency_key)
payload_hash	bytea(32)	yes	SHA-256 of canonical request
reversal_of_entry_id	uuid	no	FK JournalEntry
posted_at / posted_by	timestamptz / uuid	yes	
trace_id	text(32)	yes	OpenTelemetry trace
line: id	uuid	yes	PK
line: entry_id	uuid	yes	FK
line: line_number	int	yes	unique per entry, 1..10000
line: dimension_combination_id	bytea(32)	yes	FK DimensionCombination (includes account)
line: main_account_id	uuid	yes	denormalized for indexing
line: debit_credit	enum	yes	D or C; amounts always positive
line: amount_txn / currency_txn	numeric(23,6) / char(3)	yes	wave 11: equals accounting currency
line: amount_acct / amount_rpt	numeric(23,6)	yes	rounded to minor units (BR-11.3)
line: quantity / unit	numeric(18,6) / text	no	statistical quantity
line: text	text(255)	no	
line: source_line_ref	text(200)	no	
line: reversal_of_line_id	uuid	no	

Balance (projection): `ledger_id`, `main_account_id`, `dimension_combination_id`, `fiscal_period_id`, `currency_type` (txn/acct/rpt), `currency` char(3), `debit`, `credit`, `net`, `line_count`, `last_posting_sequence`; PK over the first five.

Invariants

- For every JournalEntry: Σ debit lines = Σ credit lines in each of amount_txn (per currency), amount_acct and amount_rpt, exactly, after rounding.
- Every JournalLine's `dimension_combination_id` was valid under the account structure version effective on `posting_date`.
- `posting_sequence` is dense (no gaps) per ledger; `Balance.last_posting_sequence` ≤ max posted sequence.
- Σ Balance.net over all accounts for a (ledger, period, currency_type) = 0.

Indexing / partitioning: `journal_line` is range-partitioned by `fiscal_year` (one partition per year per tenant shard); indexes on (ledger_id, main_account_id, posting_date), (dimension_combination_id, posting_date), (entry_id); `journal_entry` indexed on (ledger_id, posting_sequence), (document_id), (source_system, source_type, source_id). `balance` is unpartitioned, PK is the clustered access path. All tables carry `tenant_id` with RLS.

2.11.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/ledger/journal-entries	Post a balanced entry	`Idempotency-Key` required	Post journal entries (service accounts) / Post ledger journals
POST	/api/v1/ledger/journal-entries/{id}/reverse	Create negating entry	`Idempotency-Key` required	Reverse journal entries
GET	/api/v1/ledger/journal-entries/{id}	Entry with lines	n/a	View ledger transactions
GET	/api/v1/ledger/journal-lines?ledger=&account_from=&account_to=&dim.{attr}=&date_from=&date_to=&cursor=	Line query	n/a	View ledger transactions
GET	/api/v1/ledger/balances?ledger=&period_from=&period_to=&account_from=&account_to=&dim.{attr}=&group_by=	Balance query	n/a	View ledger balances
POST	/api/v1/ledger/validate	Dry-run validation of an entry (no post)	none (stateless)	Post ledger journals
GET/PUT	/api/v1/ledger/charts/{chart}/accounts[/{number}]	Main account CRUD	PUT is naturally idempotent	Maintain chart of accounts
GET/PUT	/api/v1/ledger/ledgers/{ledger}/account-structures[/{id}]	Structure CRUD and activation	PUT idempotent	Maintain account structures
POST	/api/v1/ledger/dimension-combinations/resolve	Tuple → combination id	stateless	View ledger transactions
POST/GET/PATCH	/api/v1/ledger/journal-batches[/{id}] and /{id}:submit, :approve, :post	Manual journal lifecycle	`Idempotency-Key` on :post	Maintain / Approve / Post ledger journals
POST	/api/v1/ledger/balances:rebuild	Trigger rebuild (admin)	run id returned	Administer ledger

Posting request (canonical JSON)

```
POST /api/v1/ledger/journal-entries
Idempotency-Key: ap-invoice-7F3A-post
{ "ledger_group": "ALL" | "ledger_id": "...",
  "document_id": "uuid", "posting_date": "2026-03-31", "document_date": "2026-03-28",
```

```
  "description": "Vendor invoice 7F3A",
  "source": {"system":"ap","type":"vendor_invoice","id":"7F3A","uri":"passage://ap/invoices/7F3A"},
  "lines": [
    {"line_number":1,"account":"6010","dimensions":{"DEPT":"D100","CC":"C200"},
     "debit_credit":"D","amount":{"currency":"USD","value":"1250.00"},"text":"Freight"},
    {"line_number":2,"account":"2100","dimensions":{},"debit_credit":"C",
     "amount":{"currency":"USD","value":"1250.00"}} ] }
```

Response 201: `{ "entries":[{"id","ledger_id","voucher","posting_sequence","fiscal_period_id","totals":{"txn":{...},"acct":"1250.00","rpt":"1150.75"},"lines":[...]}], "idempotency":{"key","replayed":false}, "trace_id" }`. Amounts are strings with exactly the currency's minor-unit scale.

Commands / validation: `PostJournalEntry` (validation order: schema → ledger/period resolution → account existence and postability → structure validation → combination interning → currency/rounding (wave 12) → period status (wave 14) → balance check → sequence assignment → insert → balance upsert → outbox). `ReverseJournalEntry` (original exists, not already reversed unless `allow_reinstate`, target date open). `SubmitJournalBatch` (every line valid, totals balanced, user owns batch or holds duty). `ApproveJournalBatch` (approver ≠ submitter unless journal name allows self-approval).

Event	Trigger	Payload fields	Consumers
dev.passage.ledger.journal.posted.v1	entry committed	entry_id, ledger_id, document_id, voucher, posting_sequence, posting_date, fiscal_period_id, line_count, totals{txn[],acct,rpt}, source, trace_id	reporting, tax, audit, lakehouse, subledgers
dev.passage.ledger.journal.reversed.v1	reversal committed	original_entry_id, reversing_entry_id, reason_code, posting_date	reporting, subledgers
dev.passage.ledger.batch.status-changed.v1	batch transition	batch_id, journal_name, from, to, actor, workflow_run_id	notifications, audit
dev.passage.ledger.balance.rebuilt.v1	rebuild finished	ledger_id, high_water_sequence, rows_written, rows_differing, duration_ms	ops alerting
dev.passage.ledger.account.changed.v1	account/structure change	chart_id, account_number, change_type, effective_from	reporting, dimension sync

2.11.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Chart of accounts	Create, edit, suspend, import CSV, view balances	number, name, type, postable, suspended_from, default dimensions	Maintain chart of accounts
Financial dimensions	Add attribute, bind to entity, maintain custom values	attribute, backing entity, active_from/to, suspended	Maintain financial dimensions
Account structures	Design segments, constraints grid, validate sample, activate version	segment, allowed ranges, effective_from, status	Maintain account structures
Journals (list)	New batch, filter by status/name/owner, bulk submit	voucher, journal name, status, total debit/credit, owner, posting date	Maintain ledger journals
Journal batch (detail)	Add/edit lines inline, validate, submit, approve, post, attach, save as template	line grid: account, dimensions (segmented entry), D/C, amount, text; totals bar	Maintain / Approve / Post ledger journals
Trial balance	Filter ledger/period/dimension, drill to lines, export	account, opening, debit, credit, closing, net	View ledger balances
Ledger transactions	Filter, drill to source document, trace id link	voucher, date, account, dims, D/C, amount, source	View ledger transactions

Interaction rules: segmented account entry accepts `1010-D100-C200` typed as one string with per-segment lookup (Tab moves segment); validation on blur per line and on save for the batch; grid supports paste from Excel (up to 5,000 rows) and bulk fill-down of dimensions; keyboard: Ctrl+Enter = save line, Alt+P = post (confirm dialog), Esc = cancel edit; saved views and column personalization persist per user (wave 5 shell).

2.11.7 Business rules and edge cases

- **BR-11.1 Balance by construction.** An entry is accepted only if Σ debits = Σ credits in every amount column; the check is performed on rounded amounts and repeated inside the database transaction (CHECK via deferred constraint trigger).
- **BR-11.2 Zero-amount lines** are allowed only when `quantity` ≠ 0 (statistical) or the line carries `text`; otherwise rejected with `LEDGER_LINE_EMPTY`.
- **BR-11.3 Rounding.** Amounts are stored with scale equal to the currency's ISO 4217 minor units (USD/EUR 2, JPY 0, KWD/BHD 3); requests with excess precision are rejected with `LEDGER_AMOUNT_SCALE` (no silent rounding of transaction amounts). Derived amounts (wave 12) are rounded half-away-from-zero.
- **BR-11.4 Posting date.** Must fall in a fiscal period of the ledger's calendar; date before the ledger's `first_open_date` or in a closed period → `LEDGER_PERIOD_CLOSED`. Future dates are allowed up to 366 days.
- **BR-11.5 Concurrency.** Posting takes a row lock on `ledger` only for sequence assignment (sub-millisecond); balance upserts use `INSERT … ON CONFLICT DO UPDATE` ordered by PK to prevent deadlocks; the service retries serialization failures up to 3 times with jitter.
- **BR-11.6 Idempotency key semantics.** Key is compared after trimming; scope is (tenant, legal entity). A replay returns the original response including the original `trace_id` and `replayed: true`. Keys are persisted before the entry insert in the same transaction, so a crash leaves either both or neither.
- **BR-11.7 Balance rebuild algorithm.** (1) Record `hwm = max(posting_sequence)` per ledger. (2) In a shadow table, `INSERT … SELECT ledger_id, main_account_id, dimension_combination_id, fiscal_period_id, currency_type, SUM(debit), SUM(credit), COUNT(*), MAX(posting_sequence) FROM journal_line JOIN journal_entry WHERE posting_sequence ≤ hwm GROUP BY …`, chunked by fiscal period (each chunk ≤ 5M lines, committed separately). (3) Catch-up loop: apply lines with sequence in (hwm, new_hwm] as deltas; repeat until the delta is < 1,000 lines. (4) Take a 2-second advisory lock on posting for the ledger, apply the final delta, compare shadow to live row by row (FULL OUTER JOIN on PK, any non-zero difference recorded), swap tables atomically, release. (5) Emit `balance.rebuilt` with `rows_differing`; > 0 raises a P1 alert and the live table is kept as the rebuilt copy (the journal is truth).
- **BR-11.8 Reconciliation account protection.** Accounts with `reconciliation_account_type ≠ none` reject manual batch lines (`LEDGER_RECON_ACCOUNT_MANUAL`) unless the journal name is flagged `allow_recon_accounts` (migration and correction use only).

- **BR-11.9 Error codes (422 unless noted).** `LEDGER_UNBALANCED`, `LEDGER_LINE_LIMIT`, `LEDGER_LINE_EMPTY`, `LEDGER_AMOUNT_SCALE`, `LEDGER_ACCOUNT_NOT_FOUND`, `LEDGER_ACCOUNT_NOT_POSTABLE`, `LEDGER_ACCOUNT_SUSPENDED`, `LEDGER_RECON_ACCOUNT_MANUAL`, `LEDGER_STRUCTURE_NOT_FOUND`, `LEDGER_DIMENSION_REQUIRED`, `LEDGER_DIMENSION_INVALID`, `LEDGER_DIMENSION_INACTIVE`, `LEDGER_COMBINATION_INVALID`, `LEDGER_PERIOD_NOT_FOUND`, `LEDGER_PERIOD_CLOSED`, `LEDGER_PERIOD_ON_HOLD` (403), `LEDGER_DATE_INVALID`, `LEDGER_CURRENCY_MISMATCH`, `LEDGER_RATE_MISSING`, `LEDGER_ROUNDING_EXCEEDED`, `LEDGER_REVERSAL_ALREADY_REVERSED`, `LEDGER_IDEMPOTENCY_CONFLICT` (409), `LEDGER_IDEMPOTENCY_IN_PROGRESS` (409), `LEDGER_FORBIDDEN` (403).
- **BR-11.10 Journal batch state machine.** `draft → validated` (all lines valid) → `in_approval` (submit; skipped if journal name has no workflow) → `approved | rejected` → `posting` (lock) → `posted` (entry ids recorded) or back to `approved` with error on failure; `draft|validated|rejected → cancelled`; any line edit in `validated|approved` returns the batch to `draft`. Posted and cancelled are terminal.
- **BR-11.11 Document split.** Entries sharing a `document_id` are independent for balancing; callers that split a document must balance each part (use a clearing account) — the API does not defer balancing across requests.

2.11.8 Acceptance criteria

- **AC-11.1** (FR-11.5, FR-11.6) Given a balanced two-line USD entry in an open period, when posted with a new key, then 201 is returned within 200 ms p99, a `posting_sequence` is assigned, and `journal.posted` is observed on Kafka within 2 s.
- **AC-11.2** (FR-11.5) Given the same request repeated with the same key, when posted, then 200 is returned with `replayed: true` and no new entry exists.
- **AC-11.3** (FR-11.5) Given the same key with a changed amount, when posted, then 409 `LEDGER_IDEMPOTENCY_CONFLICT`.
- **AC-11.4** (FR-11.3) Given account 6010 requires DEPT under the active structure, when a line omits DEPT, then 422 `LEDGER_DIMENSION_REQUIRED` with `line_index`.
- **AC-11.5** (BR-11.1) Given lines summing to debit 100.00 / credit 99.99, when posted, then 422 `LEDGER_UNBALANCED` and no rows are written.
- **AC-11.6** (FR-11.7) Given 1M random balanced entries posted by 64 concurrent workers, when complete, then Σ Balance.net = 0 for every period and a rebuild reports `rows_differing = 0`.
- **AC-11.7** (FR-11.8) Given a posted entry, when reversed, then a new entry exists whose lines equal the original with D/C swapped and `reversal_of_line_id` set, and a second reversal returns `LEDGER_REVERSAL_ALREADY_REVERSED`.
- **AC-11.8** (FR-11.9, BR-11.10) Given a journal name with a 2-level approval, when a clerk submits a batch, then it is `in_approval`, the approver sees a Temporal task, and after approval the clerk can post.
- **AC-11.9** (FR-11.13) Given the demo opening balance file, when loaded through the migration framework, then the conversion-suspense account balance is 0.00 and the trial balance equals the AX extract totals.
- **AC-11.10** (FR-11.14) Given 10M lines, when querying lines for one account and month, then the first page returns in < 300 ms p95 using the keyset cursor.

2.11.9 Test plan and fixtures

- Fixtures: `coa-demo` (1,200 accounts, 3 charts, from the demo manufacturer's AX `MainAccount` extract); `dims-demo` (8 attributes, 2,400 values); `structures-demo` (4 structures, 2 advanced rules); `entries-random-1m` (generator: seeded, 1M entries, 2–50 lines each, balanced by construction); `entries-negative-120` (120 hand-written rejection cases, one per error code × variant); `opening-balances-demo` (3 entities, CSV from AX `GeneralJournalAccountEntry` aggregation).

Test	Type	Target	Pass criterion
Balanced-entry property test	property (Go rapid)	posting service	any generated balanced entry posts; any mutated unbalanced entry is rejected
Error-code table	unit	validator	120 cases map to expected code and line_index
Idempotency contract	contract (OpenAPI + Pact)	posting API	replay/conflict/in-progress semantics as specified
Concurrent posting	integration	Postgres + service	64 workers × 1M entries: zero sequence gaps/duplicates, TB = 0
Rebuild equivalence	integration	rebuild job	rows_differing = 0 after concurrent load
Throughput	performance (k6)	dev cluster 3 nodes	5,000 lines/s for 30 min, p99 < 200 ms, error rate 0
Journal workspace E2E	Playwright + axe	React shell	create → submit → approve → post; WCAG 2.1 AA pass
Append-only enforcement	unit (SQL)	triggers/grants	UPDATE/DELETE on journal tables raise for app role and superuser

Reconciliation checks: Σ Balance per (ledger, period) = 0; Σ Balance per account = Σ journal lines per account; count of `journal.posted` events = count of entries (outbox audit); opening-balance conversion suspense = 0.

2.11.10 Deployment and operations

Helm chart `passage-ledger` (deployment, HPA 3–12 pods on CPU 60%, PodDisruptionBudget 2, CloudNativePG cluster `ledger-db` 3 instances, Kafka topics `ledger.journal`, `ledger.batch`, Debezium outbox connector). Config: `LEDGER_MAX_LINES=10000`, `LEDGER_IDEMPOTENCY_TTL_DAYS=400`, `LEDGER_REBUILD_CHUNK_LINES=5000000`, `LEDGER_FUTURE_DATE_DAYS=366`. Migrations via Atlas generated from entity metadata; partitions for the next fiscal year created by a monthly job. Feature flags: `ledger.recurring_journals`, `ledger.advanced_rules`. Scheduled jobs (Temporal): nightly `balance-verify` (rebuild into shadow, compare, no swap unless diff), monthly `create-partitions`, recurring journal generator. SLOs: posting availability 99.9%; posting p99 < 200 ms; `journal.posted` publish lag p95 < 2 s. Alerts: balance verify diff > 0 (P1), sequence gap detected (P1), outbox lag > 60 s (P2), posting error rate > 1% over 5 min (P2). Runbook: rebuild procedure, sequence gap investigation, idempotency table growth. Rollback: Argo CD previous digest; schema migrations are additive-only in this wave; no data rollback required because the journal is append-only.

2.11.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `MainAccount`, `LedgerChartOfAccounts`, `MainAccountLegalEntity`	MainAccount, ChartOfAccounts, MainAccountLedgerSetting	type mapping from `MainAccountType`; suspended → `suspended_from`

Source	Passage entity	Notes
AX `DimensionAttribute`, `DimensionAttributeValue`, `DimensionFinancialTag`	DimensionAttribute, DimensionValue	entity-backed attributes re-bound to Passage parties/sites; custom lists copied
AX `DimensionHierarchy` (account structures), `DimensionConstraintTree/Node`	AccountStructure(+Constraint)	advanced rules flattened into constraint rows; activated version 1
AX `DimensionAttributeValueCombination`	DimensionCombination	recomputed from tuples; `RecId` kept in `legacy_ref` for cross-reference
AX `GeneralJournalEntry/GeneralJournalAccountEntry` (closing balances only)	JournalEntry (period 0)	opening balances per combination; detail history not migrated in Phase 2 (wave 49 transactional packs)
SAP `SKA1/SKAT/SKB1`	ChartOfAccounts, MainAccount, MainAccountLedgerSetting	`XSPEB` (blocked) → suspended; `MITKZ` → reconciliation_account_type
SAP `CSKS`, `CEPC`, `TGSB`	DimensionValue (CC, PC, BA attributes)	validity dates preserved
SAP `FAGLFLEXT/ACDOCA` closing balances by period	JournalEntry (period 0)	per ledger (`RLDNR`) → Passage ledger

Validation: account count match; every migrated combination validates under the migrated structure (exceptions queued); opening TB per entity equals source to the cent. Not migrated: AX `LedgerJournalTable` unposted batches (users re-key), SAP parked documents, posting definitions (re-implemented per subledger wave).

2.11.12 Dependencies, risks, open questions

Depends on: wave 3 (metadata entity generation), wave 4 (outbox, Temporal approval template), wave 5 (gap-free number sequences, audit), wave 6 (legal entities, fiscal calendars, currencies), wave 7 (party dimension backing), wave 10 (opening balance load).

Risk	Likelihood	Impact	Mitigation
Gap-free sequence becomes a throughput bottleneck	Medium	Medium	Sequence lock held < 1 ms; batch allocation for multi-entry documents; measured in the 5k lines/s test
Dimension combination table growth (millions of tuples)	Medium	Low	32-byte hash key; combinations are reused; monitored
Structure change invalidates existing combinations	Medium	Medium	Combinations are tied to structure version; old versions stay queryable

Open questions: (1) Should `amount_txn` be stored as integer minor units instead of numeric(23,6)? Default: numeric with scale check; decide by wave 12 start. (2) Idempotency scope per legal entity vs per tenant? Default: per legal entity; decide by wave 11 mid-point. (3) Maximum dimensions 20 vs AX's practical 10? Default 20; decide by wave 11 exit.

2.12 Wave 12: Multi-Currency, Parallel Ledgers & Revaluation

2.12.1 Scope and business context

AX carries transaction, accounting and reporting currency on `GeneralJournalAccountEntry` and uses posting layers (current/operations/tax); SAP New GL carries currency types (00 document, 10 company code, 30 group) on `ACDOCA` and posts to leading and non-leading ledgers through ledger groups. Wave 12 adds to the journal: an exchange rate service (`fx`, Go) with rate types, provider adapters, triangulation and audit; three-currency amounts with ISO 4217 rounding; ledger groups so one document posts to N ledgers with per-ledger accounting differences; FX revaluation of GL balances (open-item revaluation for AP/AR uses the same engine when those waves arrive); and currency translation for reporting-currency restatement.

After this wave a user can post a EUR invoice into a USD-accounting entity with EUR reporting currency, see the amounts in all three currencies, run the month-end revaluation and see it auto-reverse. Out of scope: realized gain/loss on settlement (waves 19 and 30), group-currency consolidation translation (wave 40), hedge accounting.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP OB08 / `TCURR`: rates entered per 100 units with `TCURF` factors; a wrong factor silently converts amounts 100x; no deviation check	Factor table separate from rate table; no validation against the previous rate	Rates stored as plain `numeric(20,10)` "1 from = rate to"; > 10% deviation requires confirmation and emits a warning event	Factor-related misposting incidents → 0; deviation flagged in < 100 ms at entry
AX `ExchangeRate` import batch (`ExchangeRateProviderImport`) runs on a batch group; a failed run leaves no rate and postings fail hours later with "no exchange rate"	Batch framework with no fallback or alerting	Temporal schedule with retries; last-known rate stays valid; `fx.provider.degraded` alerts immediately	Posting failures from missing rates → 0; outage detected < 1 min (vs discovered at posting)
Rounding differences between `AccountingCurrencyAmount` and `ReportingCurrencyAmount` land in penny-difference accounts with no trace to the line (AX `LedgerParameters`); SAP rounding to OB00 account per document	Each line rounded independently; residual posted to a separate account without lineage	Residual allocated to the largest line and recorded as `rounding_adjustment` with the tolerance rule in BR-12.4	Unexplained rounding postings → 0; every residual traceable to its line
Parallel valuation needs SAP ledger groups + FB50L and separate runs per ledger, or AX posting layers that most reports ignore	Ledger-specific posting bolted onto a single-ledger document model	One request to `ledger_group: ALL` yields N balanced entries sharing `document_id`, atomically	Per-ledger reconciliation effort 2 days/close → automated check; ledger-specific adjustments 1 call
FAGL_FC_VAL / AX revaluation runs take hours on large `BSEG` sets and users rerun after rate corrections, producing duplicate adjustments	Open-item scan over unindexed valuation sets; no run idempotency	Revaluation selects from `Balance` keyed by currency; runs are idempotent per (ledger, period, rate type) and auto-reverse	Revaluation of 100k combinations < 5 min (vs 1–4 h); duplicate adjustment runs → 0

2.12.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Exchange rate types	`ExchangeRateType`, Currency exchange rate types form	`TCURV` (rate types M, EURX, P), OB07	`fx.ExchangeRateType`	full
Exchange rates	`ExchangeRate`, `ExchangeRateCurrencyPair`, OB08-style form	`TCURR`, OB08, `TCURF` (factors)	`fx.ExchangeRate` with `from`, `to`, `valid_from`, `rate` numeric(20,10); factor handled by rate precision	full
Rate providers	Exchange rate providers (OANDA, Central Bank of Europe)	Market data interface (datafeed)	`fx` provider adapters: ECB, OpenExchangeRates, manual, CSV	full
Triangulation	Euro triangulation	`TCURR` via reference currency (`TCURV-BWAER`)	`fx` triangulation through ledger reference currency	full
Three currencies per line	`TransactionCurrencyAmount`, `AccountingCurrencyAmount`, `ReportingCurrencyAmount`	`WSL` (document), `HSL` (local), `KSL` (group) in ACDOCA	`amount_txn`, `amount_acct`, `amount_rpt` on JournalLine	full
Parallel ledgers / posting layers	Posting layers Current/Operations/Tax (`LedgerPostingLayer`)	Leading ledger 0L, non-leading ledgers, ledger groups, `RLDNR`	`ledger.Ledger` × `LedgerGroup`; entries per ledger share `document_id`	improved: unlimited ledgers, per-ledger calendar and currency
FX revaluation (GL)	Foreign currency revaluation (General ledger), `LedgerExchAdj`	FAGL_FC_VAL (balances), valuation methods OB59, valuation areas	`ledger.RevaluationRun` with method per ledger	full
Unrealized gain/loss accounts	Currency revaluation accounts per currency / ledger	OBA1 (KDB/KDF), OB09 per recon account	`CurrencyPolicy` per ledger × currency: unrealized_gain, unrealized_loss, adjustment accounts	full
Currency translation (reporting currency)	Reporting currency on ledger, revaluation of reporting currency	Currency translation (FAGL_FCV / consolidation)	`ledger.TranslationRun` (monthly, average/closing/historical by account type)	partial: group consolidation in wave 40
Rounding differences	Penny difference accounts (`LedgerParameters`)	Rounding difference account (OB00), tolerance	`CurrencyPolicy.rounding_difference_account`, tolerance per BR-12.4	full
Rate audit / history	Database log on `ExchangeRate`	`TCURR` change documents	`fx.ExchangeRate` is append-only with `superseded_by`	improved

2.12.3 Functional requirements

- **FR-12.1 Currency reference data and policy** — Must. The system shall use ISO 4217 currencies from wave 6 (code, numeric code, minor units) and maintain a `CurrencyPolicy` per ledger × currency with gain/loss/adjustment/rounding-difference accounts.
  - Minor units default from ISO 4217 (JPY 0, USD/EUR/GBP/INR 2, KWD/BHD/OMR 3, CLF 4); a tenant may not override them.
  - Non-ISO currencies (e.g., XTS test, loyalty points) can be defined with explicit minor units and flagged `is_statistical`.
- **FR-12.2 Exchange rate types and rates** — Must. The system shall maintain rate types (default, average, closing, budget, custom) and rates by (type, from, to, valid_from) with precision numeric(20,10).
  - Rate direction is `from → to` as "1 from = rate to"; the reciprocal is computed only if no explicit inverse rate exists and `allow_inverse` is set on the type.
  - Rates are append-only; a corrected rate for the same key supersedes the previous row and both are kept with `superseded_by`.
  - Maximum 10,000 pairs per type; history retained indefinitely.
- **FR-12.3 Rate lookup and triangulation** — Must. The system shall resolve a rate for (type, from, to, date) by: exact pair valid on date → inverse pair (if allowed) → triangulation through the ledger's reference currency (two hops, both must resolve) → `LEDGER_RATE_MISSING`.
  - Lookup is served from an in-memory cache per pod refreshed from `fx.rate.published` events; cold lookup p99 < 5 ms.
  - Triangulated amounts are computed in one step `amount × r1 × r2` at full precision, then rounded once.
- **FR-12.4 Provider adapters** — Must. The system shall import rates from ECB (daily reference rates), OpenExchangeRates and CSV through a Temporal schedule per rate type.
  - Provider timeout 10 s, 3 retries with exponential back-off; on failure the last-known rate remains valid and `fx.provider.degraded` is emitted; rates older than `max_rate_age_days` (default 5 business days) cause posting warnings, not rejections, unless the ledger sets `reject_stale_rates`.
  - Imported rates are stored with `source`, `provider_timestamp` and raw payload hash for audit.
- **FR-12.5 Three-currency amounts on every line** — Must. The posting API shall accept `amount` in any currency, convert to accounting and reporting currency using the ledger's default rate type (or an explicit `rate_override` with rate type and value), and store all three amounts rounded per currency.
  - If a line's currency equals the accounting currency, `amount_acct = amount_txn` exactly (no rate applied); likewise for reporting.
  - Explicit `rate_override` requires duty `Override exchange rates` and is recorded on the line.
  - An entry may contain lines in multiple transaction currencies; balancing is checked per transaction currency as well as in accounting and reporting currency.
- **FR-12.6 Rounding and residual allocation** — Must. The system shall apply ISO 4217 minor-unit rounding (half away from zero) per derived amount, then allocate any residual so that each derived column balances.
  - Algorithm in BR-12.4; the residual goes to the line with the largest absolute amount on the side that needs it; if the residual exceeds tolerance the post is rejected with `LEDGER_ROUNDING_EXCEEDED`.
- **FR-12.7 Ledger groups and parallel posting** — Must. The system shall define ledger groups (e.g., `ALL`, `LEADING`, `LOCAL`, `TAX`) per legal entity; a posting request addressed to a group produces one `JournalEntry` per member ledger, all committed in one transaction and sharing `document_id`.
  - Each ledger has its own accounting principle code (IFRS, US-GAAP, HGB, TAX), fiscal calendar, accounting currency, reporting currency and default rate type; conversions are performed per ledger.

    - Per-ledger differences are expressed by the caller (subledgers send ledger-specific lines, e.g., different depreciation) or by `LedgerAccountMapping` rows that remap an account per ledger (e.g., IFRS 16 lease accounts to operating lease expense in local GAAP).
    - A request addressed to a single non-leading ledger (ledger-specific adjustment, SAP FB50L semantics) is allowed when the user holds duty `Post ledger-specific adjustments`.
- **FR-12.8 FX revaluation of GL balances** — Must. The system shall run a revaluation per ledger × period × rate type over balance-sheet accounts flagged `revalue = true`, computing unrealized gain/loss per account × dimension combination × transaction currency, posting it on the period-end date and auto-reversing it on the first day of the next period.
    - Method per ledger: `closing_rate` (default, FAGL_FC_VAL semantics with reversal) or `cumulative_no_reversal` (delta against prior revaluation, AX semantics); both are supported, the ledger chooses.
    - Simulation mode produces the proposal without posting; the proposal is retained 90 days.
- **FR-12.9 Currency translation for reporting currency** — Should. The system shall restate reporting-currency balances monthly: monetary balance-sheet accounts at closing rate, non-monetary at historical (the posted `amount_rpt`), P&L at average rate, with the difference posted to a cumulative translation adjustment (CTA) equity account.
- **FR-12.10 Exchange rates workspace** — Must. The system shall provide screens for rate types, rate entry/import, provider schedules, lookup tester and rate history.
- **FR-12.11 Rate audit** — Must. The system shall record who/what created each rate, the provider payload hash, and expose a history view by pair; a rate change that moves more than 10% from the previous value requires confirmation in the UI and emits a warning event.
- **FR-12.12 Posting-time currency validation** — Must. The system shall reject a transaction currency that is suspended or not enabled for the legal entity with `LEDGER_CURRENCY_MISMATCH`, and a missing rate with `LEDGER_RATE_MISSING` including the (type, pair, date) in the violation.

2.12.4 Data model

ExchangeRate (fx)

Field	Type	Required	Notes
id	uuid	yes	PK
rate_type_id	uuid	yes	FK ExchangeRateType
from_currency / to_currency	char(3)	yes	ISO 4217; unique (type, from, to, valid_from, version)
valid_from	date	yes	effective until next valid_from for the same key
rate	numeric(20,10)	yes	> 0
source	enum	yes	manual, ecb, openexchangerates, csv, migration
provider_timestamp	timestamptz	no	
payload_hash	bytea(32)	no	SHA-256 of provider record
superseded_by	uuid	no	FK ExchangeRate; non-null means not current
created_at / created_by	timestamptz / uuid	yes	

Ledger (extended) and **LedgerGroup**

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id	uuid	yes	FK
code	text(10)	yes	e.g. `0L`, `IFRS`, `HGB`, `TAX`; unique per legal entity
accounting_principle	enum	yes	IFRS, US_GAAP, LOCAL_GAAP, TAX, MANAGEMENT
is_leading	bool	yes	exactly one leading ledger per legal entity
accounting_currency / reporting_currency	char(3)	yes	
reference_currency	char(3)	no	triangulation pivot (default accounting currency)
default_rate_type_id / closing_rate_type_id / average_rate_type_id	uuid	yes	FK ExchangeRateType
fiscal_calendar_id	uuid	yes	FK (wave 6)
rounding_mode	enum	yes	half_away_from_zero (default), half_even
revaluation_method	enum	yes	closing_rate, cumulative_no_reversal
group: id / legal_entity_id / code / member_ledger_ids	uuid / uuid / text / uuid[]	yes	`ALL` auto-maintained

RevaluationRun and **RevaluationLine**

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	
fiscal_period_id	uuid	yes	
rate_type_id	uuid	yes	typically closing
valuation_date	date	yes	period end
mode	enum	yes	simulate, post
status	enum	yes	proposed, posted, reversed, cancelled
posting_entry_id / reversal_entry_id	uuid	no	FK JournalEntry
totals_gain / totals_loss	numeric(23,6)	yes	accounting currency
line: account_id, dimension_combination_id, currency_txn		yes	key
line: balance_txn, balance_acct_posted, rate_used, balance_acct_revalued, adjustment	numeric	yes	adjustment = revalued – posted

Invariants: `amount_acct = round(amount_txn × rate, minor_units(acct))` per line before residual allocation; after allocation Σ D = Σ C in every column; exactly one leading ledger per legal entity; every member of a ledger group belongs to the same legal entity; a revaluation run that is `posted` with method `closing_rate` has a reversal entry dated period_end + 1 day.

Indexing: `exchange_rate` index on (rate_type_id, from_currency, to_currency, valid_from DESC) WHERE superseded_by IS NULL; `journal_line` adds index on (ledger_id, currency_txn, posting_date) for revaluation selection; `revaluation_line` PK (run_id, account_id, dimension_combination_id, currency_txn).

2.12.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/PUT	/api/v1/fx/rate-types[/{code}]	Rate type CRUD	PUT idempotent	Maintain exchange rates
POST	/api/v1/fx/rates	Create rate(s), bulk up to 5,000	`Idempotency-Key`	Maintain exchange rates
GET	/api/v1/fx/rates?type=&from=&to=&date=	Resolve a rate (with triangulation path)	n/a	View exchange rates
GET	/api/v1/fx/rates/history?type=&from=&to=	Rate history incl. superseded	n/a	View exchange rates
POST	/api/v1/fx/providers/{provider}:import	Trigger import now	run id	Administer exchange rates
POST	/api/v1/fx/convert	Convert amount(s) with rounding (stateless)	none	View exchange rates
GET/PUT	/api/v1/ledger/ledgers/{id}/currency-policies/{currency}	Gain/loss/rounding accounts	PUT idempotent	Maintain ledger setup
GET/PUT	/api/v1/ledger/legal-entities/{le}/ledger-groups[/{code}]	Ledger group membership	PUT idempotent	Maintain ledger setup
POST	/api/v1/ledger/revaluations	Create run (simulate or post)	`Idempotency-Key`	Run foreign currency revaluation
GET	/api/v1/ledger/revaluations/{id}[/lines]	Inspect proposal/results	n/a	View ledger balances
POST	/api/v1/ledger/revaluations/{id}:post	Post a simulated proposal	`Idempotency-Key`	Run foreign currency revaluation
POST	/api/v1/ledger/translations	Reporting-currency translation run	`Idempotency-Key`	Run currency translation

Posting request extensions (wave 11 contract, additive): `"ledger_group": "ALL"`, per line `"amount": {"currency":"EUR","value":"1000.00"}`, optional `"rate_override": {"rate_type":"M","rate":"1.0850000000"}`, optional `"ledger_lines": {"HGB": [ … ]}` for ledger-specific substitutions. Response contains `entries[]` one per ledger with `amounts: {txn, acct, rpt}` and `rates_applied[]`.

Commands: `PublishRate` (rate > 0, currencies enabled, 10% deviation warning), `RunRevaluation` (period open for module GL, accounts flagged, closing rate exists for every currency with balance else `LEDGER_RATE_MISSING`), `RunTranslation`.

Event	Trigger	Payload fields	Consumers
dev.passage.fx.rate.published.v1	rate committed	rate_type, from, to, valid_from, rate, previous_rate, source, deviation_pct	ledger cache, reporting, AP/AR
dev.passage.fx.provider.degraded.v1	import failure after retries	provider, last_success_at, rate_type, error	ops alerting
dev.passage.ledger.revaluation.completed.v1	run posted	run_id, ledger_id, period, rate_type, gain_total, loss_total, posting_entry_id, reversal_entry_id	reporting, close
dev.passage.ledger.translation.completed.v1	translation posted	run_id, ledger_id, period, cta_amount, entry_id	reporting, close

2.12.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Exchange rate types	Create, set defaults per ledger	code, name, allow_inverse, provider	Maintain exchange rates
Exchange rates	Enter, import CSV, view history, compare to provider	type, from, to, valid_from, rate, source, deviation vs previous	Maintain exchange rates
Provider schedules	Configure provider, cron, test connection, run now	provider, rate type, pairs, last run, status	Administer exchange rates
Rate lookup tester	Enter type/pair/date, see resolution path	path (direct/inverse/triangulated), rate, valid_from	View exchange rates
Ledger setup	Ledgers, groups, currency policies, account mapping	code, principle, currencies, rate types, revaluation method, accounts	Maintain ledger setup
Foreign currency revaluation	Simulate, review proposal grid, post, view reversal	account, dims, currency, balance txn, posted acct, rate, revalued, adjustment	Run foreign currency revaluation
Journal line (extended)	Enter amount in any currency; see acct/rpt amounts live	currency, amount, rate (editable with duty), acct amount, rpt amount	Maintain ledger journals

Interaction rules: the line grid shows derived amounts as read-only cells that recompute on currency/amount/date change within 100 ms; a rate override cell is editable only with the duty and is highlighted; the revaluation proposal grid supports excluding rows (recorded as exclusions on the run) and exporting to XLSX; a rate deviation > 10% opens a confirmation dialog with the previous rate.

2.12.7 Business rules and edge cases

- **BR-12.1 Minor units.** `scale(currency)` is the ISO 4217 minor unit; a stored amount must satisfy `amount = round(amount, scale)`; CHECK constraint `scale(amount) ≤ scale(currency)` is validated in the service and by a trigger.
- **BR-12.2 Rounding mode.** Default half-away-from-zero (matches AX `round()` and SAP commercial rounding); `half_even` is selectable per ledger for statistical ledgers only. Rounding occurs once per derived amount, after full-precision multiplication (numeric(38,16) intermediate).
- **BR-12.3 Rate date.** The rate for conversion is the one valid on `posting_date` unless the request supplies `rate_date` (allowed for AP/AR documents using document date); rate_date must be ≤ posting_date + 0 days (no future rates).
- **BR-12.4 Residual allocation.** For each derived column (acct, rpt): compute `rounded_i` per line; `residual = Σ rounded_debits − Σ rounded_credits`. If `|residual| ≤ tolerance = 0.5 × unit × line_count` (unit = 10^−scale), add `−residual` to the rounded amount of the line with

the largest `|amount|` on the side that reduces the residual (ties: lowest line_number); the adjusted line records `rounding_adjustment`. If `|residual| > tolerance` reject with `LEDGER_ROUNDING_EXCEEDED`. When the entry has only one line per side and the residual is non-zero, post the residual to `rounding_difference_account` instead of adjusting the line, provided `|residual| ≤ 1 unit`.

- **BR-12.5 Multi-currency balancing.** Balancing is checked per transaction currency; an entry with EUR debit and USD credit must include explicit cross-currency lines (clearing) or is rejected with `LEDGER_UNBALANCED` listing the unbalanced currency.
- **BR-12.6 Parallel-ledger posting matrix.** For a request to ledger group `ALL` with leading `0L` (IFRS), `HGB` (local GAAP) and `TAX`:

Source document type	0L (IFRS)	HGB (local)	TAX	Rule
Manual journal, ledger_group ALL	post	post	post	identical lines, converted per ledger
Ledger-specific adjustment (single ledger)	post only if addressed	post only if addressed	post only if addressed	duty-gated
FX revaluation	post	post	skip if ledger.revalue = false	run per ledger
Account-mapped lines (`LedgerAccountMapping`)	original account	mapped account	mapped account	mapping keyed (ledger, account, effective_from)
Subledger with principle-specific lines (`ledger_lines`)	common lines + 0L lines	common lines + HGB lines	common lines + TAX lines	each ledger must balance independently
Year-end close (wave 14)	per ledger calendar	per ledger calendar	per ledger calendar	independent runs

- **BR-12.7 Stale rates.** A rate older than `max_rate_age_days` yields a `warnings[]` entry `FX_RATE_STALE` in the response; with `reject_stale_rates` it is a 422 `LEDGER_RATE_MISSING`.
- **BR-12.8 Revaluation state machine.** `proposed → posted → reversed` (automatic at period+1 for closing_rate method) | `proposed → cancelled`; a second run for the same (ledger, period, rate type) while one is `posted` is rejected unless the first is reversed; cumulative method runs post the delta versus the last posted run.
- **BR-12.9 Revaluation selection.** Only balance-sheet accounts with `revalue = true` and non-zero `balance_txn` in a currency ≠ accounting currency; reconciliation accounts for AP/AR are revalued at open-item level in waves 19/30 and are excluded here (`revalue_level = open_item`).
- **BR-12.10 Provider precedence.** If two providers deliver the same (type, pair, date), the one with higher `priority` wins and the other is stored as `superseded`.
- **BR-12.11 Rate precision in display.** Rates display 6 decimals by default with full 10 on hover/export; amounts never display more than the currency scale.

2.12.8 Acceptance criteria

- **AC-12.1** (FR-12.5, FR-12.6) Given a 3-line EUR entry posted to a USD ledger with rate 1.0850000000, when posted, then each `amount_acct` equals the rounded product, Σ D = Σ C in USD, and at most one line carries a `rounding_adjustment` of ≤ 0.01.
- **AC-12.2** (BR-12.1) Given 10,000 random amounts in JPY, KWD, USD and EUR converted in all directions, when summed per entry, then every entry balances to the minor unit and no stored amount has scale beyond the currency's.
- **AC-12.3** (FR-12.3) Given no GBP→CHF rate but GBP→EUR and EUR→CHF, when converting, then the response path is `triangulated` and the amount equals one-step rounding of `a × r1 × r2`.
- **AC-12.4** (FR-12.7, BR-12.6) Given the 20-document parallel-ledger fixture, when posted to `ALL`, then 0L, HGB and TAX each receive the expected lines (mapped accounts and ledger-specific lines) and each ledger's TB is 0.
- **AC-12.5** (FR-12.8) Given the revaluation fixture (12 accounts × 4 currencies), when run with closing rates, then unrealized gain/loss per line equals the workbook to the cent and a reversal entry exists dated period end + 1.
- **AC-12.6** (FR-12.4) Given the ECB adapter times out 3 times, when the schedule runs, then no rates change, `fx.provider.degraded` is emitted and a P2 alert fires.
- **AC-12.7** (FR-12.11) Given a manual rate 12% above the previous, when saved, then the UI requires confirmation and the event carries `deviation_pct = 12.0`.
- **AC-12.8** (FR-12.12) Given a line in a currency not enabled for the entity, when posted, then 422 `LEDGER_CURRENCY_MISMATCH`.
- **AC-12.9** (FR-12.9) Given the translation fixture, when run, then CTA equals total assets – liabilities – equity restated at the specified rates, to the cent.

2.12.9 Test plan and fixtures

- Fixtures: `iso4217-2026` (all active currencies with minor units); `rates-ecb-2025` (recorded ECB daily reference rates, 12 months); `amounts-random-10k` (seeded amounts across JPY/KWD/USD/EUR incl. extremes 0.001 and 9,999,999,999.999); `parallel-ledger-20` (20 documents with expected per-ledger lines, accountant-reviewed); `reval-workbook` (12 accounts × 4 currencies with XLSX expected values, FAGL_FC_VAL semantics); `translation-workbook`.

Test	Type	Target	Pass criterion
Rounding property test	property	`money` package	balance after allocation for all generated entries; residual ≤ tolerance
Rate resolution unit tests	unit	fx resolver	direct/inverse/triangulated/missing per table
Provider adapter contract	contract (recorded HTTP)	ECB, OXR adapters	parse recorded payloads; failure modes
Parallel-ledger fixture	integration	posting API	per-ledger lines and TB = 0
Revaluation workbook	integration	revaluation job	adjustment per line equals workbook
Reversal timing	integration (Temporal time-skip)	revaluation workflow	reversal posted on period+1
Rate cache consistency	integration	ledger pods	a published rate is used by all pods within 2 s
Conversion latency	performance	fx + posting	posting p99 stays < 200 ms with 3-currency conversion

Reconciliation: per ledger per currency type TB = 0; Σ revaluation adjustments = posted gain – loss; `fx.rate.published` count = rate rows.

2.12.10 Deployment and operations

Helm chart `passage-fx` (deployment 2–6 pods, CloudNativePG `fx-db`, topics `fx.rates`, `fx.providers`); `passage-ledger` upgraded with currency policy tables. Config: `FX_MAX_RATE_AGE_DAYS=5`, `FX_PROVIDER_TIMEOUT_S=10`, `FX_DEVIATION_WARN_PCT=10`, `LEDGER_ROUNDING_MODE=half_away_from_zero`. Migrations: add `amount_rpt`/`currency_txn` backfill for wave 11 entries (= accounting amounts), add `ledger_group`, `currency_policy`, `revaluation_*`. Feature flags: `fx.triangulation`, `ledger.parallel_ledgers`, `ledger.translation`. Scheduled jobs: ECB import 16:30 CET business days; OXR hourly (if licensed); month-end revaluation on the close checklist (wave 14). SLOs: rate lookup p99 < 5 ms; import success ≥ 99% of business days. Alerts: provider degraded (P2), stale rate used in posting > 100/h (P3), revaluation run failure (P2). Runbook: manual rate entry during provider outage; re-run revaluation after rate correction (reverse then re-run). Rollback: digest rollback; feature flags off disable parallel posting (entries then go only to the leading ledger — documented as a degraded mode, not for production).

2.12.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `ExchangeRateType`, `ExchangeRateCurrencyPair`, `ExchangeRate`	ExchangeRateType, ExchangeRate	AX stores rate per 100 units by default → divided; `ValidFrom` preserved
AX `Ledger` (accounting/reporting currency, posting layers)	Ledger, LedgerGroup	Current → leading; Operations/Tax layers → additional ledgers only if the customer used them
AX `LedgerExchAdj` history	not migrated	prior revaluations are embedded in migrated closing balances
SAP `TCURV`, `TCURR`, `TCURF`, `TCURX`	ExchangeRateType, ExchangeRate	rate = `UKURS × TFACT/FFACT`; `TCURX` decimals cross-checked against ISO 4217, exceptions reported
SAP `T882` (ledger assignment), `T881` (ledgers), ledger groups	Ledger, LedgerGroup	leading `0L` → is_leading; currency types 10/30 → accounting/reporting
SAP OBA1/OB09 gain/loss accounts	CurrencyPolicy	per ledger × currency

Validation: rate counts per type; sampled 1,000 conversions compared with source values to the cent; opening balances in all three currencies tie per ledger. Not migrated: provider credentials (re-entered in Vault), valuation run history.

2.12.12 Dependencies, risks, open questions

Depends on: wave 11 (journal, posting API), wave 6 (currencies), wave 4 (Temporal schedules), wave 5 (audit).

Risk	Likelihood	Impact	Mitigation
SAP `TCURX` non-ISO decimals conflict with ISO 4217	Medium	Medium	Migration report flags conflicts; ISO wins; amounts rescaled with sign-off
Residual allocation changes a user-visible line amount	Medium	Low	Adjustment recorded on the line and shown in UI; alternative rounding-difference account configurable
Provider licensing (OXR)	Low	Low	ECB and manual are license-free defaults

Open questions: (1) Should reporting currency be per ledger or per legal entity? Default per ledger; decide by wave 12 mid-point. (2) Historical rate for non-monetary items in translation: store rate on line at posting (default) or look up by date? Default store; decide by wave 12 exit. (3) Support SAP-style "second local currency" (currency type 40) as a fourth amount? Default no (use a parallel ledger); revisit at wave 40.

2.13 Wave 13: Tax Engine (Sales/Use, VAT/GST, Withholding)

2.13.1 Scope and business context

AX determines tax by intersecting a sales tax group (party side, `TaxGroupHeading`/`TaxGroupData`) with an item sales tax group (`TaxItemGroupHeading`/`TaxItemGroupData`) to find tax codes (`TaxTable`) with rates (`TaxData`), posting results to `TaxTrans`. SAP uses tax procedures with condition types, tax codes per country (`T007A`, FTXP), jurisdiction codes for US, and withholding tax types/codes. Wave 13 builds the `tax` service (Go) that any document line calls for determination, calculation and posting lines, with an internal rules engine as default and an Avalara/Vertex-compatible provider adapter interface, plus settlement to the GL and statutory extracts (VAT return, EU sales list, Intrastat, SAF-T).

After this wave a manual journal or any later document gets correct tax lines for US sales/use tax, EU VAT (standard, reduced, reverse charge, intra-community), Indian GST (CGST/SGST/IGST) and Canadian GST/HST/PST, and the tax settlement run posts the net payable and reconciles to the tax accounts. Out of scope: document-level invoicing (waves 18, 29), 1099/26AS statutory filing formats (hooks only; forms in wave 48 country packs), customs duty.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP FTXP: a tax code is a single rate per procedure step with no validity dates; a rate change (e.g., DE 19% → 16% in 2020) forces new codes and mass master-data updates in `KNVI`/`MLAN`	`T007A` has no effective dating	`TaxRate` rows are effective-dated under a stable `TaxCode`; groups and parties keep their assignments	Rate change rollout 2–5 days of config → 1 rate row; zero master-data touches
AX `TaxTrans` and SAP `BSET` show the result but not why: users cannot see which group intersection, exemption or jurisdiction produced the amount; support tickets replay the document in a test client	Determination logic in X++ `Tax` class / condition technique with no trace output	`POST /tax/calculate:explain` returns `determination_trace[]` step by step; tester screen renders it	"Why this tax?" investigation 30–60 min → < 1 min
US multi-jurisdiction via SAP external tax interface or AX ISV connectors: provider outage blocks invoicing (RFC timeout, document cannot save)	Synchronous dependency with no fallback	Provider adapter with 2 s timeout, internal-engine fallback, `tax_fallback` flag and recommit queue	Invoicing downtime during provider outage → 0; fallback decision < 2.5 s

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Sales tax settlement (AX "Settle and post sales tax", SAP F.12 / S_ALR_87012357) posts a total that does not match the GL tax accounts; controllers reconcile by hand in Excel	Tax transactions and GL postings written by different code paths with separate rounding	`TaxTransaction` linked to `JournalLine` by id; settlement blocks on mismatch (`TAX_SETTLEMENT_MISMATCH`)	Settlement-to-GL reconciliation 4 h/period → automated, 0.00 variance enforced
Price-inclusive rounding differs across SAP condition steps and AX "per line / per document" settings, producing 1-cent mismatches with customer invoices and credit notes	Multiple rounding points without a distribution rule	One algorithm (BR-13.3) with largest-remainder distribution; credit notes use the original bracket	1-cent invoice disputes → 0 in the 500-scenario matrix

2.13.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Tax codes and rates	`TaxTable`, `TaxData` (effective-dated values, limits)	`T007A` (tax codes), FTXP, condition records `KONP`/`A003`	`tax.TaxCode`, `tax.TaxRate` (effective-dated, bracketed)	full
Party-side tax group	`TaxGroupHeading`, `TaxGroupData`, Sales tax groups form	Customer/vendor tax classification (`KNVI`/`LFA1-STKZU`), tax category on BP	`tax.TaxGroup` (N:M TaxCode)	full
Item-side tax group	`TaxItemGroupHeading`, `TaxItemGroupData`	Material tax classification (`MLAN`)	`tax.ItemTaxGroup` (N:M TaxCode)	full
Determination	Intersection of group and item group	Tax procedure (TAXUSJ/TAXEUR), access sequences, condition technique	Intersection + jurisdiction resolution + rule overrides (§2.13.7 BR-13.1)	improved: explainable trace
Jurisdictions	`TaxJurisdiction` (US), sales tax authorities `TaxAuthorityAddress`	Jurisdiction codes (`TTXJ`), external tax interface (Vertex/Sabrix)	`tax.TaxJurisdiction` hierarchy (country > state > county > city > district)	full
Exemptions	Tax exempt numbers, `TaxExempt` codes	Exemption licenses (withholding), tax code 0%	`tax.ExemptionCertificate` (party × jurisdiction × validity × reason)	full
Calculation modes	Price includes tax, calculation method (whole amount/interval), tax on tax	Condition calculation types, `KSCHL` steps, from/to step references	Inclusive/exclusive, compound sequence, bracketed rates	full
Reverse charge / use tax	Reverse charge item groups, Use tax checkbox on tax code	Acquisition tax codes (EU), use tax (US) via tax procedure	`TaxCode.mechanism` ∈ {standard, reverse_charge, use_tax, withholding}	full
Partial recoverability	Non-deductible percentage on tax code	Non-deductible tax (NAV/NVV condition types)	`TaxCode.recoverable_pct` with expense distribution	full
Withholding tax	Withholding tax codes/groups (`TaxWithholdTable`), 1099 fields	Extended withholding tax types/codes (`T059P`, `T059Z`), `WITH_ITEM`	`tax.WithholdingTaxCode`, `tax.WithholdingTransaction`; trigger at invoice or payment	full
Posted tax	`TaxTrans`	`BSET`, ACDOCA tax fields	`tax.TaxTransaction` linked to JournalLine	full
Settlement and reporting	Settlement periods, "Settle and post sales tax", EU sales list, Intrastat	Tax payable posting (F.12/S_ALR_87012357), ACR (Advanced Compliance Reporting), Intrastat (VE01)	`tax.TaxSettlementRun`, VAT return, ESL, Intrastat, SAF-T extracts	full (formats in wave 48)
External provider	Avalara/Vertex ISV connectors	External tax interface (RFC_CALCULATE_TAXES_DOC)	Provider adapter interface (`TaxProvider`) with failover	full

2.13.3 Functional requirements

- **FR-13.1 Tax codes and effective-dated rates** — Must. The system shall define tax codes per legal entity with mechanism (standard, reverse_charge, use_tax, withholding), rate rows effective-dated with optional amount brackets (`from_amount`, `to_amount`), recoverable percentage, jurisdiction, and posting accounts (payable, receivable, non-recoverable expense, use-tax expense).
  - Rate precision numeric(9,6) in percent (e.g., 19.000000); brackets evaluated on the line base or document base per code setting (AX "calculation method": whole amount vs interval).
  - Maximum 2,000 codes per legal entity; rate history retained indefinitely.
- **FR-13.2 Tax groups and item tax groups** — Must. The system shall maintain party-side tax groups and item-side item tax groups, each a set of tax codes; defaults come from party roles (customer/vendor), delivery address jurisdiction, and product/category.
- **FR-13.3 Jurisdiction hierarchy** — Must. The system shall model jurisdictions as a tree (country > state/province > county > city > special district) with address-to-jurisdiction resolution by postal code tables or provider lookup, and compute stacked rates (e.g., CA state 7.25% + county + district).
- **FR-13.4 Determination and calculation API** — Must. The system shall expose `POST /api/v1/tax/calculate` taking a document (header: legal entity, document type, dates, party, addresses, currency; lines: amount, quantity, product, tax group, item tax group, overrides) and returning per-line and per-document tax lines with an explain trace.
  - Stateless, p99 < 50 ms for 100-line documents with the internal engine; provider calls add their latency (timeout 2 s).
  - Order of operations in BR-13.1. Results carry `engine: internal | avalara | vertex`, `fallback: bool`, and `determination_trace[]`.
- **FR-13.5 Inclusive and exclusive prices** — Must. The system shall support price-inclusive lines by backing out tax: `base = gross / (1 + Σ non-compound rates)`, iteratively resolving compound codes; rounding per BR-13.3.
- **FR-13.6 Compound tax, reverse charge, use tax, partial recoverability** — Must. The system shall calculate tax-on-tax by code sequence, generate equal input and output lines for reverse charge, accrue use tax as a liability with the expense to the line's account, and split recoverable vs non-recoverable amounts.
- **FR-13.7 Exemption certificates** — Must. The system shall apply certificates by (party, jurisdiction, validity, reason code, optional product category), zeroing the applicable codes and recording the certificate id on the tax transaction; expired certificates produce `TAX_EXEMPTION_EXPIRED` warnings and full tax.
- **FR-13.8 Withholding tax** — Must. The system shall calculate withholding (vendor and customer) with code, base (gross, net, tax-inclusive), rate, threshold and annual accumulation, triggered at invoice or payment per code; results are stored as `WithholdingTransaction` and exposed to AP/AR (waves 19/30) and to 1099/26AS reporting hooks.
- **FR-13.9 Tax posting** — Must. The system shall convert a calculated document into journal lines (posting API, wave 11) with the tax account per code, dimension inheritance from the taxed line, and create `TaxTransaction` rows linked to the posted `JournalLine` ids, within the caller's document

transaction via a prepared-lines contract (`POST /tax/calculate` → caller includes returned lines in its posting request → tax consumes `journal.posted` to link).

- **FR-13.10 Provider adapters and failover** — Must. The system shall implement a `TaxProvider` interface (`Calculate`, `Commit`, `Void`, `ValidateAddress`) with Avalara AvaTax REST v2 and Vertex O-Series adapters, selectable per legal entity × document type; on timeout or 5xx the internal engine is used and the document is flagged `tax_fallback = true` for later recommit.
  - Provider credentials in Vault; requests and responses hashed and stored 7 years for audit (payload itself retained 13 months).
- **FR-13.11 Settlement periods and settlement run** — Must. The system shall define settlement periods per tax authority (monthly, quarterly, annual) and run a settlement that sums tax transactions by code and period, posts the net payable/receivable to the authority's vendor/customer (party role) or a settlement account, and marks transactions settled.
  - Re-runs for the same period post only the delta (late-posted transactions) as a new settlement with `version` incremented.
  - Settlement totals must equal the GL tax account movements for the period; a difference > 0.00 blocks posting with `TAX_SETTLEMENT_MISMATCH`.
- **FR-13.12 Statutory extracts** — Should. The system shall produce VAT return boxes (configurable box mapping per country), EU sales list (ESL, by customer VAT id and transaction type), Intrastat lines (commodity code, weight, value, partner country) and SAF-T (OECD v2.0 XSD-valid XML) for a period.
- **FR-13.13 Tax configuration workspace** — Must. Screens for codes, rates, groups, jurisdictions, certificates, withholding, settlement periods, provider settings and a calculation tester.
- **FR-13.14 Tax matrix test harness** — Must. The system shall ship a fixture runner that executes the 500-scenario matrix against the engine in CI and reports diffs per scenario.

2.13.4 Data model

TaxCode and **TaxRate**

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id	uuid	yes	
code	text(20)	yes	unique per legal entity
name	text(120)	yes	
mechanism	enum	yes	standard, reverse_charge, use_tax, withholding
jurisdiction_id	uuid	no	FK TaxJurisdiction
authority_party_id	uuid	yes	FK Party (role tax authority)
settlement_period_id	uuid	yes	FK TaxSettlementPeriod
calc_basis	enum	yes	line, document (bracket evaluation)
compound_sequence	int	no	null = not compound; higher sequences apply on base + earlier tax
recoverable_pct	numeric(9,6)	yes	default 100
payable_account_id / receivable_account_id / nonrecoverable_account_id / use_tax_expense_account_id	uuid	conditional	FK MainAccount
rounding	enum	yes	per_line, per_document (default per_document for VAT, per_line for US)
rate: tax_code_id, valid_from, valid_to		yes	non-overlapping intervals
rate: from_amount / to_amount	numeric(23,6)	no	bracket
rate: rate_pct	numeric(9,6)	yes	
rate: min_tax / max_tax	numeric(23,6)	no	caps

TaxTransaction

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id / ledger_id	uuid	yes	leading ledger
tax_code_id	uuid	yes	
jurisdiction_id	uuid	no	
source (system, type, id, line_ref)	text	yes	document line
journal_line_id	uuid	no	set when `journal.posted` consumed
tax_direction	enum	yes	output (sales), input (purchase), use, withholding
base_amount_txn / tax_amount_txn / currency_txn	numeric / char(3)	yes	
base_amount_acct / tax_amount_acct	numeric(23,6)	yes	
recoverable_amount_acct / nonrecoverable_amount_acct	numeric(23,6)	yes	sum = tax_amount_acct
rate_pct_applied	numeric(9,6)	yes	
exemption_certificate_id	uuid	no	
counterparty_party_id / counterparty_tax_id	uuid / text	no	for ESL
engine / fallback / provider_ref	enum / bool / text	yes	
settlement_run_id	uuid	no	null = unsettled
posting_date / fiscal_period_id	date / uuid	yes	

TaxSettlementRun: `id`, `legal_entity_id`, `authority_party_id`, `settlement_period_id`, `period_from`, `period_to`, `version`, `status` (proposed, posted, cancelled), `output_tax_total`, `input_tax_total`, `net_amount`, `entry_id`, `transaction_count`, `created_by`, `posted_at`.

ExemptionCertificate: `id`, `party_id`, `jurisdiction_id`, `certificate_number`, `reason_code`, `valid_from`, `valid_to`, `product_category_id` (optional), `document_uri`, `status` (active, expired, revoked).

Invariants: Σ TaxTransaction.tax_amount_acct per (code, period) = movement on the code's payable/receivable account for that period (checked by settlement); recoverable + nonrecoverable = tax; rate intervals per code do not overlap; a settled transaction cannot change `settlement_run_id` except to null via a cancelled run. Indexing: `tax_transaction` on (legal_entity_id, tax_code_id, fiscal_period_id, settlement_run_id), (source_system, source_type, source_id), (journal_line_id); partitioned by fiscal year.

2.13.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/tax/calculate	Determine and calculate for a document	stateless (optional `Idempotency-Key` for provider commit)	Calculate tax (service accounts)
POST	/api/v1/tax/calculate:explain	Same with full determination trace	stateless	View tax configuration
POST	/api/v1/tax/commit	Commit provider transaction after document posting	`Idempotency-Key` required	Calculate tax
GET/PUT	/api/v1/tax/legal-entities/{le}/codes[/{code}] and /codes/{code}/rates	Codes and rates	PUT idempotent	Maintain tax codes
GET/PUT	/api/v1/tax/legal-entities/{le}/groups[/{id}] and /item-groups[/{id}]	Groups	PUT idempotent	Maintain tax groups
GET/PUT	/api/v1/tax/jurisdictions[/{id}] and POST /jurisdictions:resolve	Jurisdiction tree and address resolution	PUT idempotent	Maintain tax jurisdictions
GET/POST/PATCH	/api/v1/tax/exemption-certificates[/{id}]	Certificates	`Idempotency-Key` on POST	Maintain tax exemptions
GET/PUT	/api/v1/tax/legal-entities/{le}/withholding-codes[/{code}]	Withholding	PUT idempotent	Maintain withholding tax
GET	/api/v1/tax/transactions?le=&code=&period=&settled=	Posted tax transactions	n/a	View tax transactions
POST	/api/v1/tax/settlements	Propose or post a settlement run	`Idempotency-Key`	Settle tax
POST	/api/v1/tax/settlements/{id}:post	Post a proposal	`Idempotency-Key`	Settle tax
POST	/api/v1/tax/extracts/{vat-return	esl	intrastat	saf-t}

Calculation request/response (abridged): request `{legal_entity, document_type, document_date, tax_date, currency, party:{id, role, tax_id}, ship_from, ship_to, lines:[{line_ref, amount, quantity, product_id, tax_group, item_tax_group, price_includes_tax, account, dimensions, override:{tax_code?, exempt_reason?}}]}`; response `{engine, fallback, lines:[{line_ref, base, taxes:[{code, jurisdiction, rate_pct, tax_amount, recoverable, nonrecoverable, direction}]}], document_taxes_by_code:[…], posting_lines:[… wave 11 line objects …], trace_id, determination_trace[]}`.

Validation: every line has resolvable groups or an override; dates within code validity; currency enabled; party tax id format valid for reverse charge; totals of `posting_lines` balance against the caller's lines.

Event	Trigger	Payload fields	Consumers
dev.passage.tax.calculated.v1	calculate with commit intent	document ref, engine, fallback, totals per code, trace_id	document services, audit
dev.passage.tax.transaction.recorded.v1	TaxTransaction linked to journal line	transaction id, code, period, base, tax, direction, journal_line_id	reporting, lakehouse
dev.passage.tax.settled.v1	settlement posted	run id, authority, period, output, input, net, entry_id, version	reporting, ledger
dev.passage.tax.provider.fallback.v1	provider failure	provider, document ref, error, retry_after	ops, AP/AR recommit job
dev.passage.tax.rate.changed.v1	rate row created	code, valid_from, rate_pct, previous	document services cache

2.13.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Tax codes	Create, add rate rows, set accounts, mechanism	code, mechanism, jurisdiction, current rate, authority, settlement period	Maintain tax codes
Tax groups / item tax groups	Assign codes, set defaults for party/product	group, codes, exempt flags	Maintain tax groups
Jurisdictions	Tree view, import postal tables, resolve address tester	level, code, parent, rates stacked	Maintain tax jurisdictions
Exemption certificates	Register, upload document, expire	party, jurisdiction, number, validity, reason	Maintain tax exemptions
Withholding codes	Rates, thresholds, trigger point, reporting category	code, rate, base, threshold, 1099 box	Maintain withholding tax
Calculation tester	Enter a sample document, see tax lines and trace	line, code, rate, base, tax, explanation	View tax configuration
Tax transactions	Filter by code/period/document, drill to journal line and source	document, code, base, tax, direction, settled	View tax transactions
Settlement	Propose, review by code, post, view history	authority, period, output, input, net, status, version	Settle tax
Provider settings	Select provider per entity/document type, test connection	provider, environment, company code, last success	Administer tax

Interaction rules: the tester renders the determination trace as a step list (groups → intersection → jurisdiction → exemptions → brackets → compounding → rounding); rate grids validate non-overlapping intervals on save; settlement proposals show a reconciliation panel (transactions vs GL) and block the

post button on mismatch.

2.13.7 Business rules and edge cases

- **BR-13.1 Order of operations.** (1) Resolve `tax_date` (document date unless the document type specifies delivery date). (2) Resolve party tax group (override > party address jurisdiction default > party role default) and item tax group (override > product > category > legal entity default). (3) Candidate codes = TaxGroup.codes ∩ ItemTaxGroup.codes; if the ship-to jurisdiction requires stacked rates, expand jurisdiction codes. (4) Apply exemption certificates: remove exempted codes, record certificate. (5) Apply explicit line override (duty-gated). (6) Compute base per line: exclusive → amount; inclusive → back out (FR-13.5). (7) Evaluate brackets on line or document basis per code. (8) Calculate non-compound codes, then compound codes in `compound_sequence` order on `base + Σ earlier taxes`. (9) Apply min/max caps. (10) Reverse charge → emit input and output lines of equal amount; use tax → liability line with offset to the line's expense account. (11) Split recoverable/non-recoverable. (12) Round per BR-13.3. (13) Withholding is computed separately on the configured base and never alters the document tax. Every step writes to `determination_trace`.
- **BR-13.2 Provider precedence.** If a provider is configured for (legal entity, document type), steps 3–11 are delegated; the provider's results are normalized into the same structures; the internal engine still validates rounding and balancing of the returned lines.
- **BR-13.3 Rounding.** `per_document` (VAT default): sum unrounded tax per code across lines, round once to currency minor units, then distribute to lines proportionally with largest-remainder allocation so line taxes sum to the document tax. `per_line` (US default): round each line's tax per code. Rounding mode is the ledger's (wave 12).
- **BR-13.4 Inclusive back-out with compound taxes.** Solve `gross = base × Π(1 + r_i)` for compound stacks and `gross = base × (1 + Σ r_i)` for parallel codes; base rounded last; the sum of base + taxes must equal gross to the minor unit (residual to the largest tax line).
- **BR-13.5 Brackets.** Evaluated on the absolute base; credit notes (negative amounts) use the same bracket as the positive amount; `calc_basis = document` sums all lines of the same code before bracket lookup.
- **BR-13.6 Effective dates.** A code with no rate valid on `tax_date` → `TAX_RATE_NOT_FOUND` (422); a rate change mid-period is applied by date, never pro-rated.
- **BR-13.7 Reverse charge eligibility.** Requires a counterparty tax id passing the country format check (wave 7) and both parties in different EU member states (or domestic reverse-charge categories configured); otherwise falls back to standard rate with warning `TAX_REVERSE_CHARGE_INELIGIBLE`.
- **BR-13.8 Withholding thresholds.** Annual accumulation per (vendor, code, fiscal year); withholding applies to the amount exceeding the threshold in the document that crosses it; accumulation is updated only on posting, not calculation.
- **BR-13.9 Settlement state machine.** `proposed → posted | cancelled`; `posted` is terminal; a cancelled run releases its transactions; settlement for a period may not post while the period is `closed` for module TAX (wave 14); a posted run cannot include transactions dated after `period_to`.
- **BR-13.10 Provider commit idempotency.** `commit` with the same key is replayed; a provider `Void` is issued when a document is reversed (`journal.reversed` consumer) and failures queue for retry with alerting after 24 h.
- **BR-13.11 Negative and zero rates.** Rate 0 is valid (zero-rated vs exempt distinguished by `reason_code` on the code); negative rates are rejected (`TAX_RATE_INVALID`).
- **BR-13.12 Multi-currency.** Tax is calculated in transaction currency and converted with the document's rates (wave 12); settlement is in accounting currency; the ESL reports in EUR using the ECB rate on tax_date when the accounting currency is not EUR.

2.13.8 Acceptance criteria

- **AC-13.1** (FR-13.4, FR-13.14) Given the 500-scenario matrix, when run in CI, then 100% of scenarios match expected tax per line and per document to the minor unit.
- **AC-13.2** (FR-13.5, BR-13.4) Given a EUR 119.00 inclusive line at 19% VAT, when calculated, then base 100.00 and tax 19.00; given 3 inclusive lines of 33.33 each at 19%, then the document tax equals round(Σ unrounded) and line taxes sum to it.
- **AC-13.3** (FR-13.6) Given a Canadian QC line of CAD 100.00 with GST 5% and QST 9.975% (non-compound since 2013), when calculated, then taxes are 5.00 and 9.98; given a configured compound stack, then the second tax applies on base + first tax.
- **AC-13.4** (FR-13.6) Given an intra-EU purchase from a DE vendor into FR with a valid VAT id, when calculated, then an input and an output line of equal amount are produced and the net tax is 0.00.
- **AC-13.5** (FR-13.7) Given an expired exemption certificate, when calculated, then full tax applies and `TAX_EXEMPTION_EXPIRED` is in warnings.
- **AC-13.6** (FR-13.10) Given AvaTax times out, when calculating, then the internal engine responds within 2.5 s total, `fallback = true`, and `tax.provider.fallback` is emitted.
- **AC-13.7** (FR-13.11) Given the settlement fixture for March 2026, when the run is posted, then net payable equals output – input, the posting balances, and Σ settled transactions equals the GL tax account movement to the cent.
- **AC-13.8** (FR-13.12) Given the demo EU entity, when SAF-T is generated, then the XML validates against the OECD v2.0 XSD.
- **AC-13.9** (FR-13.8, BR-13.8) Given a vendor with USD 550 cumulative and a USD 600 threshold, when a USD 100 invoice is posted, then withholding applies to USD 50.
- **AC-13.10** (BR-13.9) Given the TAX module is closed for the period, when a settlement post is attempted, then 422 `LEDGER_PERIOD_CLOSED`.

2.13.9 Test plan and fixtures

- Fixtures: `tax-matrix-500` (US 200: multi-jurisdiction, exemptions, use tax; EU 180: standard/reduced/reverse charge/triangulation; IN 70: CGST/SGST/IGST, TDS; CA 50: GST/HST/PST/QST; each with expected results reviewed by an accountant); `jurisdictions-us-sample` (5 states' postal tables); `settlement-2026-03` (1,200 transactions with GL expectations); `saf-t-demo`; recorded AvaTax/Vertex HTTP fixtures.

Test	Type	Target	Pass criterion
Matrix runner	fixture (CI)	internal engine	500/500 match
Rounding distribution property test	property	per_document rounding	line taxes sum to document tax; each line within 1 unit of proportional
Inclusive back-out property	property	FR-13.5	base + taxes = gross exactly
Provider contract	contract (recorded)	AvaTax, Vertex adapters	request mapping and response normalization
Failover	integration (fault injection)	provider client	fallback within 2.5 s, event emitted

Test	Type	Target	Pass criterion
Settlement reconciliation	integration	settlement job	transactions = GL movement
SAF-T schema	unit	extract generator	XSD valid
Calculation latency	performance	tax API	100-line document p99 < 50 ms internal

Reconciliation: Σ TaxTransaction per code and period = GL tax account movement; settlement net = VAT return net box; ESL totals = Σ reverse-charge/intra-community output transactions.

2.13.10 Deployment and operations

Helm chart `passage-tax` (deployment 3–10 pods, CloudNativePG `tax-db`, topics `tax.calculations`, `tax.transactions`, `tax.settlements`; Kafka consumer of `ledger.journal`). Config: `TAX_PROVIDER_TIMEOUT_MS=2000`, `TAX_PROVIDER_RETRIES=1`, `TAX_PAYLOAD_RETENTION_DAYS=395`, `TAX_DEFAULT_ROUNDING=per_document`. Secrets: provider credentials in Vault paths `tax/providers/{le}`. Migrations: tax schema; partitions per fiscal year. Feature flags: `tax.provider.avalara`, `tax.provider.vertex`, `tax.extracts.saft`. Scheduled jobs: provider recommit retry (hourly), certificate expiry notifications (daily), settlement reminders per period. SLOs: calculate p99 < 50 ms (internal), availability 99.9%, provider fallback rate < 0.5%/day. Alerts: fallback rate > 2% over 1 h (P2), commit retry queue > 100 (P2), settlement mismatch (P2). Runbook: provider outage, re-commit queue drain, rate correction after posting (reverse documents, never edit transactions). Rollback: digest rollback; flags disable providers (internal engine only).

2.13.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `TaxTable`, `TaxData`, `TaxAuthorityAddress`, `TaxPeriodHead`	TaxCode, TaxRate, authority Party, TaxSettlementPeriod	`TaxData` intervals → brackets; ledger posting group `TaxLedgerAccountGroup` → accounts
AX `TaxGroupHeading/TaxGroupData`, `TaxItemGroupHeading/TaxItemGroupData`	TaxGroup, ItemTaxGroup	exempt flags preserved
AX `TaxExemptCodeTable`, customer `TaxExemptNumber`	ExemptionCertificate	validity defaulted to migration date → open; steward review queue
AX `TaxWithholdTable`, `TaxWithholdGroupHeading`, 1099 setup	WithholdingTaxCode	
AX `TaxTrans` (unsettled only)	TaxTransaction	linked to migrated opening balance lines by code
SAP `T007A/T007S`, `KONP/A003`, `T030K`	TaxCode, TaxRate, accounts	tax procedure steps → compound sequences
SAP `TTXJ/TTXD` jurisdictions	TaxJurisdiction	
SAP `T059P/T059Z`, `LFBW`	WithholdingTaxCode, party assignments	
SAP `BSET` (unsettled)	TaxTransaction	

Validation: code counts; a 200-document sample recalculated in Passage matches the source tax to the cent; unsettled totals per code match. Not migrated: settled history (kept in the legacy archive), provider transaction logs.

2.13.12 Dependencies, risks, open questions

Depends on: wave 11 (posting), wave 12 (currency conversion for tax amounts), wave 7 (party tax ids, VIES check), wave 8 (product categories), wave 6 (addresses).

Risk	Likelihood	Impact	Mitigation
Jurisdiction tables for US are vast and change monthly	High	Medium	Providers are the production default for US; internal engine ships sample states only
Country-specific VAT box mappings	High	Medium	Mapping is configuration (wave 48 packs); wave 13 ships DE/FR/UK/IN reference mappings
Rounding disputes with legacy figures	Medium	Low	Rounding mode configurable per code; migration validation sample

Open questions: (1) Settlement counterpart as tax authority Party (vendor role) vs settlement account? Default Party; decide by wave 13 mid-point. (2) Store provider raw payloads for 13 months or 7 years? Default 13 months payload + 7 years hash; decide by wave 13 exit. (3) Should `tax/calculate` be callable from the posting API automatically for manual journals with a tax group? Default yes behind flag `ledger.auto_tax`; decide by wave 14.

2.14 Wave 14: Period Close, Year-End & Allocations

2.14.1 Scope and business context

AX controls posting through `LedgerPeriod` statuses (open, on hold, closed) per module (`LedgerPeriodModuleAccessControl`) and offers the financial period close workspace, allocation rules (`LedgerAllocation`), accrual schemes and year-end close; SAP uses OB52 posting period intervals per account type with authorization groups, the Financial Closing Cockpit, assessment/distribution cycles (KSV5/KSU5), the accrual engine and balance carry-forward (FAGLGVTR). Wave 14 adds to Passage: a period-control matrix enforced inside the posting API; a close checklist run on Temporal with templates, dependencies and a cross-entity board; an allocation engine (fixed, basis, iterative) with reversal; accrual schemes and recurring journal scheduling (completing FR-11.10); and year-end close with carry-forward per dimension.

After this wave a controller can close March 2026 for three entities from one board, with every subledger path rejected after close, allocations and accruals posted and reversed on schedule, and year-end producing opening balances. Out of scope: subledger-specific close tasks (inventory close wave 20, AP/AR aging waves 19/30), consolidation (wave 40), budget vs actual (wave 38).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP OB52 (`T001B`): closing a period is an edit of interval rows per account type and company code, done by Basis/FI power users across many company codes; a typo opens the wrong range and postings land in a closed month	Period control is a maintenance view, not a controlled transition with validation	Period-control grid with state machine (BR-14.3), GL-closes-last rule, bulk close across ledgers with confirmation	Close of 3 entities × 10 modules: 30 OB52 edits → 1 bulk action; wrong-period postings → 0
Reopening a closed period in AX (Ledger calendar form) or SAP is a single-user action with no approval; auditors find silent reopenings in the database log weeks later	No workflow on period status	Reopen requires two approvers, a reason, an expiry timer and emits `close.period.status-changed`	Unapproved reopenings → 0; audit evidence available in < 1 s rather than at audit time
Close tracking lives in Excel or the AX financial period close workspace that does not block dependent tasks; SAP Closing Cockpit setup requires consultants	Task status is informational; automation hooks are limited to batch jobs	Temporal-driven CloseRun with DAG dependencies, automation hooks that complete on completion events, cross-entity board	Close coordination emails/period 50+ → 0; task status latency hours → seconds
AX allocation rules (`LedgerAllocation`) and SAP KSV5/KSU5 cycles post rounding residues to odd destinations and have no reciprocal (iterative) option without CO-OM workarounds; reruns double-post	Residual handling undefined; cycles not idempotent	Largest-share residual rule, iterative solver with convergence check, runs reversible as a unit and idempotent by key	Allocation rerun duplicates → 0; reciprocal allocations supported natively (vs manual spreadsheets)
FAGLGVTR / AX year-end close must be rerun after late postings; operators delete and regenerate opening entries, breaking the audit trail	Carry-forward implemented as mutable opening balances	Year-end run reverses its predecessor and regenerates (append-only), verifies opening = closing per combination before commit	Opening/closing mismatch after reruns → 0; audit trail unbroken

2.14.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Period status	`LedgerPeriod` (Open/On hold/Closed), Ledger calendar form	`T001B` intervals via OB52 (periods 1–12, 13–16 special)	`ledger.PeriodStatus` per ledger × period × module	full
Module access control	`LedgerPeriodModuleAccessControl`, user groups per module	OB52 authorization group (`BRGRU`), account type (+, A, D, K, M, S)	PeriodStatus.module × `allowed_group_id`	full
Special periods	Closing period type on fiscal calendar	Periods 13–16	Fiscal period `kind = adjustment` (wave 6) with own PeriodStatus	full
Close workspace	Financial period close workspace (templates, tasks, dependencies, calendar)	Financial Closing Cockpit / Advanced Financial Closing	`close` service: CloseTemplate, CloseRun, CloseTask (Temporal)	full
Allocations	`LedgerAllocation`, allocation rules (basis/fixed/equally/spread), process allocation request	KSV5 (distribution), KSU5 (assessment), cycles/segments, FAGLGA35	`ledger.AllocationRule`, `AllocationRun` with fixed, basis, equal, iterative methods	full
Accruals	Accrual schemes (`LedgerAccrualScheme`), ledger accruals	Accrual engine (ACE), FBS1 with reversal	`ledger.AccrualScheme`; reversal via wave 11 reversal	full
Recurring entries	Periodic journals	FBD1/F.14	`ledger.RecurringJournal` (wave 11) with calendar-relative schedules	full
Year-end close	General ledger > Year-end close (`LedgerFiscalYearClose...`), opening transactions, retained earnings per dimension	FAGLGVTR balance carry-forward, retained earnings account (OB53)	`ledger.YearEndCloseRun` producing period-0 opening entries	full
Reopen period	Change status back to Open	OB52 edit	Dual-approval reopen workflow	improved: four-eyes, audited
Close calendar / status board	Workspace tiles per entity, task list by due date	Cockpit task list, status overview	Close board across legal entities	full

2.14.3 Functional requirements

- **FR-14.1 Period-control matrix** — Must. The system shall maintain a status per (ledger, fiscal period, module) where module ∈ {GL, AP, AR, INV, FA, PROJ, TAX, BANK, PAYROLL, ALL} and status ∈ {never_opened, open, on_hold, closed}.
  - `ALL` is the default when no module row exists; a module row overrides `ALL`. `on_hold` carries `allowed_group_id` (user group that may still post).
  - The posting API resolves module from `source.system` (mapping table, default GL for manual journals) and rejects per BR-14.1. The check is inside the posting transaction (status row read with `FOR SHARE`), so a close committing concurrently wins or loses atomically.
- **FR-14.2 Period status transitions** — Must. The system shall allow `never_opened → open`, `open ↔ on_hold`, `open|on_hold → closed`, and `closed → open` only through the reopen workflow (FR-14.3).
  - Closing a period for `ALL` requires every module row to be closed or absent; closing GL requires all other modules closed (GL closes last).
  - Period 13–16 adjustment periods may be open while 12 is closed.
- **FR-14.3 Reopen with dual approval** — Must. The system shall reopen a closed period only via a Temporal workflow requiring two distinct approvers holding duty `Approve period reopen`, a reason, and an expiry (reopened period auto-closes after N days, default 3).
  - Each reopen increments `reopen_count` and emits `close.period.status-changed` with both approvers.
- **FR-14.4 Close templates and runs** — Must. The system shall define close templates (tasks with module, description, assignee role, due offset in business days from period end, dependencies, evidence requirement, automation hook) and instantiate a `CloseRun` per (template, period, set of legal entities).
  - Maximum 500 tasks per template; dependencies form a DAG (cycle rejected on save).
  - Automation hooks call Passage jobs (revaluation, allocation, accrual, settlement, year-end, statement pack) and mark the task complete when the job emits its completion event.
- **FR-14.5 Close board** — Must. The system shall show tasks across entities by status (not started, in progress, blocked, completed, skipped) with due dates, owners, overdue counts, and allow bulk reassignment and evidence attachment.
- **FR-14.6 Allocation rules** — Must. The system shall define allocation rules with source (account/dimension filter, period, ledger), destination (account/dimension targets), method ∈ {fixed_percentage, fixed_weight, equally, basis}, basis source (balances of statistical or financial accounts by dimension, e.g., headcount), offset account, and `allocate_from_zero` behavior.

    - Percentages must sum to 100.000000 (tolerance 0); weights normalized; basis computed from the Balance projection for the period.
    - Rounding per BR-14.4; destination lines inherit dimensions from the basis.
- **FR-14.7 Iterative (reciprocal) allocations** — Should. The system shall solve mutually dependent rules (service departments allocating to each other) iteratively until the residual per source < 1 minor unit or 100 iterations, then post one entry per rule.
- **FR-14.8 Allocation runs and reversal** — Must. Runs support simulate and post modes, record the basis snapshot, and are reversible as a unit (reversing every produced entry); a rule may be flagged `auto_reverse_next_period`.
- **FR-14.9 Accrual schemes** — Must. The system shall define schemes (debit/credit accounts, frequency, number of occurrences, spread method ∈ {equal, by_days, custom}) that split a source line into future-dated entries, each auto-reversed per scheme (e.g., monthly accrual reversed on day 1).
    - Generated entries are scheduled on Temporal and posted on their dates if the period is open; otherwise the task appears as blocked on the close board.
- **FR-14.10 Year-end close** — Must. The system shall close a fiscal year per ledger: (a) for each P&L account × dimension combination, post the net to retained earnings (per dimension when `retain_dimensions = true`, else to the ledger's retained earnings combination) in the closing period, and (b) create opening entries in period 0 of the next year for each balance-sheet account × combination equal to its closing balance.
    - Re-runnable: a subsequent run reverses the prior run's entries and regenerates (append-only).
    - Requires all periods of the year closed for all modules except `GL` in `on_hold` for the closing user group; year-end entries are flagged `is_closing_entry`/`is_opening_entry` and excluded from P&L statements.
- **FR-14.11 Recurring journal scheduling** — Must. Completes FR-11.10: schedules by fiscal period offset (e.g., "last business day of period"), with pause/resume, next-run preview and a generated-batch history.
- **FR-14.12 Close reporting** — Should. The system shall report close duration per period and entity (period end to GL closed), tasks completed late, and reopen history.

2.14.4 Data model

PeriodStatus

Field	Type	Required	Notes
ledger_id	uuid	yes	PK part
fiscal_period_id	uuid	yes	PK part
module	enum	yes	PK part; ALL, GL, AP, AR, INV, FA, PROJ, TAX, BANK, PAYROLL
status	enum	yes	never_opened, open, on_hold, closed
allowed_group_id	uuid	no	required when on_hold
reopen_count	int	yes	default 0
reopen_expires_at	timestamptz	no	
changed_at / changed_by / approvals	timestamptz / uuid / jsonb	yes	approvals = [{user_id, at}] for reopen

CloseTask (within CloseRun)

Field	Type	Required	Notes
id	uuid	yes	PK
run_id	uuid	yes	FK CloseRun (template_id, fiscal_period_id, legal_entity_ids[], status)
legal_entity_id	uuid	yes	
template_task_id	uuid	yes	
module	enum	yes	
name / description	text	yes	
assignee_user_id / assignee_role	uuid / text	conditional	
due_at	timestamptz	yes	period_end + offset business days
depends_on	uuid[]	no	task ids
status	enum	yes	not_started, in_progress, blocked, completed, skipped
evidence_uri / evidence_required	text / bool	conditional	
automation	jsonb	no	{job, params, completion_event}
completed_at / completed_by	timestamptz / uuid	no	
temporal_workflow_id	text	yes	

AllocationRule and **AllocationRun**

Field	Type	Required	Notes
id	uuid	yes	PK
ledger_id	uuid	yes	
name	text(60)	yes	unique per ledger
source_filter	jsonb	yes	account ranges + dimension filters + `include_opening`
method	enum	yes	fixed_percentage, fixed_weight, equally, basis
destinations	jsonb	yes	[{account, dimensions, pct
offset_account_id / offset_dimension_mode	uuid / enum	yes	mode: source_dimensions, fixed

Field	Type	Required	Notes
iterative_group	text	no	rules in the same group are solved together
auto_reverse_next_period	bool	yes	
run: id, rule_ids[], fiscal_period_id, mode (simulate/post), status (proposed, posted, reversed, cancelled), iterations, basis_snapshot (jsonb), entry_ids[], reversal_entry_ids[]			

AccrualScheme: `id`, `ledger_id`, `code`, `debit_account_id`, `credit_account_id`, `frequency` (monthly, quarterly, custom), `occurrences`, `spread` (equal, by_days, custom[]), `start_rule` (document_date, next_period_start), `auto_reverse` bool, `reverse_day` int. **YearEndCloseRun**: `id`, `ledger_id`, `fiscal_year_id`, `retained_earnings_combination_id`, `retain_dimensions`, `status`, `closing_entry_ids[]`, `opening_entry_ids[]`, `previous_run_id`.

Invariants: at most one PeriodStatus per (ledger, period, module); `closed` GL implies every other module closed; an AllocationRun that is `posted` has Σ destination amounts = source amount per rule (after rounding); YearEndCloseRun opening balances per combination = closing balances (verified by the run before commit). Indexing: `period_status` PK; `close_task` on (run_id, status), (assignee_user_id, due_at); `allocation_run` on (ledger_id, fiscal_period_id).

2.14.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/ledger/ledgers/{id}/periods?year=	Period-control matrix	n/a	View period status
PATCH	/api/v1/ledger/ledgers/{id}/periods/{period}/modules/{module}	Transition status (open/on_hold/closed)	naturally idempotent on target status	Maintain period status
POST	/api/v1/ledger/ledgers/{id}/periods/{period}:request-reopen	Start reopen workflow	`Idempotency-Key`	Request period reopen
POST	/api/v1/close/templates and GET/PUT /{id}	Templates	PUT idempotent	Maintain close templates
POST	/api/v1/close/runs	Instantiate run for period × entities	`Idempotency-Key`	Run period close
GET	/api/v1/close/runs/{id}/tasks?entity=&status=	Board data	n/a	View close board
POST	/api/v1/close/tasks/{id}:start, :complete, :skip, :reassign	Task actions	`Idempotency-Key`	Work close tasks
GET/PUT	/api/v1/ledger/ledgers/{id}/allocation-rules[/{rule}]	Rules	PUT idempotent	Maintain allocation rules
POST	/api/v1/ledger/allocations	Run (simulate/post)	`Idempotency-Key`	Run allocations
POST	/api/v1/ledger/allocations/{id}:post, :reverse	Post proposal / reverse run	`Idempotency-Key`	Run allocations
GET/PUT	/api/v1/ledger/ledgers/{id}/accrual-schemes[/{code}]	Schemes	PUT idempotent	Maintain accrual schemes
POST	/api/v1/ledger/year-end-closes	Run year-end close (simulate/post)	`Idempotency-Key`	Run year-end close
GET/PUT	/api/v1/ledger/recurring-journals[/{id}] and :pause, :resume	Schedules	PUT idempotent	Maintain ledger journals

Commands: `TransitionPeriodStatus` (validates FR-14.2 ordering; closing with unposted approved batches dated in the period → `CLOSE_PENDING_BATCHES` unless `force` with duty); `RequestReopen` (reason ≥ 10 chars, two approvers distinct from requester); `RunAllocation` (period open for GL, basis non-zero or `allocate_from_zero`); `RunYearEndClose` (all periods closed per FR-14.10, retained earnings account exists and is equity).

Event	Trigger	Payload fields	Consumers
dev.passage.close.period.status-changed.v1	PeriodStatus transition	ledger_id, fiscal_period_id, module, from, to, actor, approvals[], reason	all posting services (cache), close, reporting, audit
dev.passage.close.run.created.v1	run instantiated	run_id, template_id, period, legal_entity_ids, task_count	notifications
dev.passage.close.task.completed.v1	task completed/skipped	run_id, task_id, status, actor, evidence_uri, late (bool)	board, notifications, reporting
dev.passage.ledger.allocation.completed.v1	run posted/reversed	run_id, rule_ids, period, iterations, entry_ids, reversal	reporting, close
dev.passage.ledger.accrual.scheduled.v1	scheme applied	source_entry_id, scheme, occurrences[] (dates, amounts)	close board
dev.passage.ledger.year-end.closed.v1	year-end posted	ledger_id, fiscal_year_id, closing_entry_ids, opening_entry_ids, retained_earnings_total	reporting, close

2.14.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Ledger calendar / period control	Grid ledger × period × module; set status; bulk close; request reopen	period, module, status, allowed group, changed by, reopen count	Maintain period status
Close templates	Build task list, dependencies (DAG view), due offsets, automation	task, module, role, due offset, depends on, evidence, hook	Maintain close templates
Close board	Instantiate run, kanban/list across entities, filter overdue, reassign, attach evidence, complete	entity, task, owner, due, status, blocked by, evidence	View / Work close tasks
Allocation rules	Define source, method, destinations, basis; simulate	name, method, source filter, destinations, offset, iterative group	Maintain allocation rules
Allocation runs	Simulate, review proposal (source → destinations), post, reverse	rule, source amount, destination, basis value, pct, amount	Run allocations
Accrual schemes	Define, preview schedule on a sample amount	code, accounts, frequency, occurrences, spread, reverse day	Maintain accrual schemes
Year-end close	Select year/ledger, simulate, review retained earnings by dimension, post	account, combination, closing balance, opening balance, retained earnings	Run year-end close

Interaction rules: period grid supports keyboard range selection and bulk status change with a confirmation summarizing affected modules; close board tasks open in a side panel without losing filters; blocked tasks show the blocking task inline; allocation proposals support "what-if" edits to percentages that recompute live but require saving the rule to post.

2.14.7 Business rules and edge cases

- **BR-14.1 Period-control decision.** For a line with posting_date → period P, module M (from source mapping), user U: status = row(P, M) if present else row(P, ALL) if present else `never_opened`. `open` → allow; `on_hold` → allow iff U ∈ allowed_group else 403 `LEDGER_PERIOD_ON_HOLD`; `closed` or `never_opened` → 422 `LEDGER_PERIOD_CLOSED`. Service accounts for subledgers are members of no on-hold group by default.
- **BR-14.2 Period-control matrix (reference configuration for March close).**

Module	Day 1–3 after period end	Day 4–5	After GL close
INV, PROJ, PAYROLL	on_hold (ops group)	closed	closed
AP, AR, BANK	open	on_hold (AP/AR supervisors)	closed
TAX	open	open	closed after settlement
FA	open	closed	closed
GL	open	on_hold (controllers)	closed
ALL (default for unmapped)	open	on_hold (controllers)	closed

- **BR-14.3 Period status state machine.** `never_opened → open`; `open → on_hold`; `on_hold → open`; `open|on_hold → closed`; `closed → open` only via reopen workflow (two approvals) which sets `reopen_expires_at`; on expiry a Temporal timer sets `closed` again and emits the event. Transitions are rejected with `CLOSE_INVALID_TRANSITION`.
- **BR-14.4 Allocation rounding.** Destination amounts = round(source × share, scale); residual = source – Σ destinations; residual applied to the destination with the largest share (ties: first by definition order). Basis values of zero across all destinations → skip rule with `ALLOC_ZERO_BASIS` unless `allocate_from_zero` = equally.
- **BR-14.5 Iterative convergence.** Reciprocal groups solved by Gauss–Seidel iteration on the unrounded amounts; stop when max |Δ| < 10^–scale / 2 or at 100 iterations (then fail `ALLOC_NO_CONVERGENCE`); rounding applied once after convergence per rule.
- **BR-14.6 Allocation period lock.** A run posts only if GL is open or on_hold with the runner in the allowed group; the basis period may differ from the posting period (e.g., prior-period headcount) and is snapshotted on the run.
- **BR-14.7 Accrual spread by days.** Amount_i = round(total × days_i / total_days); residual to the last occurrence; occurrences dated the first day of each period (or scheme `post_day`) and reversed the first day of the following period.
- **BR-14.8 Year-end retained earnings per dimension.** When `retain_dimensions`, the retained earnings line keeps the P&L line's dimension combination (replacing only the main account); dimension values inactive on the opening date are remapped to the ledger's `default_closing_combination` and listed in the run report.
- **BR-14.9 Year-end re-run.** A new run for the same (ledger, year) first reverses all entries of `previous_run_id` (dated the same dates), then regenerates; opening period 0 is not subject to period control of the new year's period 1.
- **BR-14.10 Close task dependencies.** A task becomes `in_progress`-eligible only when all `depends_on` tasks are `completed` or `skipped`; skipping requires a reason and duty `Skip close tasks`; automation tasks cannot be completed manually unless marked `manual_override_allowed`.
- **BR-14.11 Concurrency on close.** Closing GL takes `FOR UPDATE` on the PeriodStatus row and waits for in-flight postings (which hold `FOR SHARE`) up to 5 s before failing with `CLOSE_BUSY`; retried by the UI.
- **BR-14.12 Calendar mismatch across ledgers.** Period-control is per ledger; a ledger group posting where any member ledger's period is closed is rejected as a whole with the offending ledger named.

2.14.8 Acceptance criteria

- **AC-14.1** (FR-14.1, BR-14.1) Given AP is closed for March and GL open, when a vendor-invoice posting (source.system = ap) dated 2026-03-31 is attempted, then 422 `LEDGER_PERIOD_CLOSED`; a manual GL journal on the same date posts.
- **AC-14.2** (FR-14.1) Given every subledger posting path in the contract suite (manual, ap, ar, inv, fa, proj, tax, bank stubs), when the period is closed for ALL, then every path is rejected (table-driven test, 0 exceptions).
- **AC-14.3** (FR-14.2) Given AP is still open, when GL close is requested, then `CLOSE_INVALID_TRANSITION` naming AP.
- **AC-14.4** (FR-14.3) Given a closed period, when a reopen is requested and approved by two distinct users, then status is `open`, `reopen_count = 1`, and after the expiry timer the period is `closed` again (Temporal time-skip).
- **AC-14.5** (FR-14.6, BR-14.4) Given the allocation fixture (rent USD 10,000.01 by headcount 3/5/2), when run, then destinations are 3,000.00, 5,000.01, 2,000.00 and Σ = source.
- **AC-14.6** (FR-14.7) Given two service departments allocating 10% to each other, when run iteratively, then the result converges within 100 iterations and matches the reciprocal workbook to the cent.
- **AC-14.7** (FR-14.9) Given a USD 1,200.00 annual insurance invoice with a 12-occurrence equal scheme, when applied, then 12 entries of 100.00 are scheduled on period starts with reversals on the next period start.
- **AC-14.8** (FR-14.10) Given a 12-period dataset, when the year-end close is posted, then every balance-sheet combination's opening balance equals its closing balance, P&L accounts open at 0.00, and retained earnings change equals net income.
- **AC-14.9** (FR-14.4, FR-14.5) Given a 40-task template across 3 entities with dependencies, when the run completes, then no task was completed before its dependencies and the board shows 120 tasks completed.
- **AC-14.10** (BR-14.11) Given 50 concurrent postings in flight, when GL close is requested, then it either waits ≤ 5 s and succeeds with no posting committed after the close timestamp, or returns `CLOSE_BUSY`.

2.14.9 Test plan and fixtures

- Fixtures: `period-matrix-demo` (3 ledgers × 16 periods × 10 modules); `close-template-40` (40 tasks, 3 entities, 12 dependencies, 6 automation hooks); `alloc-fixtures-25` (fixed, weight, equal, basis, 3 reciprocal groups; XLSX expected); `accrual-fixtures-10`; `fy2025-12-periods` (12 posted periods, 400 combinations, expected opening balances).

Test	Type	Target	Pass criterion
Period decision table	unit	BR-14.1	60 cases (status × module × group)
Subledger path contract	contract	all posting clients' stubs	closed period rejected on every path

Test	Type	Target	Pass criterion
Close race	integration	ledger + close	no post commits after close timestamp
Reopen workflow	Temporal test	close service	two approvals, expiry auto-close
Allocation workbook	integration	allocation engine	totals equal, residual rule honored
Reciprocal convergence property	property	iterative solver	converges for random 2–6 department matrices with shares < 50%
Year-end equivalence	integration	year-end job	opening = closing per combination
Close board E2E	Playwright + axe	React	40-task run completes; WCAG AA

Reconciliation: Σ allocation destinations = Σ sources per run; Σ accrual occurrences = source amount; opening balances per combination = closing; retained earnings Δ = net income per ledger.

2.14.10 Deployment and operations

Helm charts: `passage-close` (TypeScript service, 2–4 pods, CloudNativePG `close-db`, Temporal namespace `close`), `passage-ledger` upgraded (period status, allocation, accrual, year-end). Config: `CLOSE_REOPEN_DEFAULT_DAYS=3`, `CLOSE_LOCK_WAIT_MS=5000`, `ALLOC_MAX_ITERATIONS=100`, `ALLOC_CONVERGENCE_FRACTION=0.5`. Migrations: `period_status` seeded from fiscal calendars (`never_opened`), existing demo periods set `open`. Feature flags: `close.reopen_workflow` (always on in prod), `ledger.iterative_allocations`. Scheduled jobs: accrual and recurring journal generators (Temporal schedules), reopen expiry timers, close reminder notifications (daily 08:00 entity time zone). SLOs: period decision adds < 2 ms to posting p99; close board p95 < 500 ms for 1,000 tasks. Alerts: reopen without expiry (P2), posting rejected rate for closed periods > 50/h (P3, indicates a subledger misconfiguration), allocation non-convergence (P3). Runbook: emergency reopen, stuck close lock, re-running year-end. Rollback: digest rollback; period status rows are retained; disabling the flag never bypasses period control.

2.14.11 Migration from AX / SAP

Source	Passage entity	Notes
AX `LedgerPeriod`, `LedgerPeriodModuleAccessControl`, `LedgerFiscalCalendarPeriod`	PeriodStatus	statuses copied; modules mapped (Ledger→GL, VendInvoice→AP, CustInvoice→AR, Inventory→INV, Asset→FA, Project→PROJ, Sales tax→TAX, Bank→BANK)
AX `LedgerAllocation`, `LedgerAllocationRule*` (basis/destination/source)	AllocationRule	basis "statistical" → basis spec on statistical accounts
AX `LedgerAccrualScheme`	AccrualScheme	
AX financial period close templates/tasks (`FinancialPeriodCloseTemplate*`)	CloseTemplate, template tasks	dependencies and due offsets preserved
SAP `T001B` (OB52)	PeriodStatus	account type + → ALL, D → AR, K → AP, A → FA, M → INV, S → GL; authorization group → allowed_group
SAP KSV5/KSU5 cycles (`T811C`, `T811S`, `T811K`)	AllocationRule	segments → rules; tracing factor → basis spec
SAP `T030` retained earnings (OB53)	Ledger.retained_earnings_combination_id	per ledger
SAP closing cockpit templates	CloseTemplate	

Validation: period statuses per module equal the source at cutover; allocation rules re-executed on the last closed period in simulate mode and compared to source postings (tolerance 0.01 per destination). Not migrated: completed close task history, prior allocation run history, SAP accrual engine objects (re-keyed as schemes).

2.14.12 Dependencies, risks, open questions

Depends on: wave 11 (posting API hook, reversal), wave 12 (revaluation hook on checklist), wave 13 (settlement hook), wave 4 (Temporal human tasks and timers), wave 6 (fiscal calendars with adjustment periods), wave 2 (user groups).

Risk	Likelihood	Impact	Mitigation
Subledger services (future waves) bypass module mapping	Medium	High	Source mapping is mandatory; unmapped `source.system` → GL and rejected when GL on hold; contract test suite grows per wave
Iterative allocation performance with many dimensions	Low	Medium	Solver limited to 50 rules per group; measured in fixtures
Year-end with millions of combinations	Medium	Medium	Chunked generation (50k lines per entry, multiple entries per run); run measured on 1M-combination synthetic set

Open questions: (1) Should on_hold allow service accounts of the owning module by default? Default no; decide by wave 14 mid-point. (2) Retained earnings per dimension as default? Default yes for the demo; decide with accountant review by wave 14 exit. (3) Do adjustment periods 13–16 get their own close checklist? Default: same run, tasks flagged `adjustment_only`.

2.15 Wave 15: Financial Reporting, Statements & Audit Trail

2.15.1 Scope and business context

AX ships Management Reporter (row, column, reporting tree definitions over dimensions) plus trial balance and ledger transaction inquiries; SAP uses financial statement versions (OB58, `T011`/`T011T`, `FAGL_011*`), Fiori financial statement apps and document-flow drill-down. Wave 15 builds the `reporting` service (Go, with its own Postgres read model fed by ledger events) with a row/column/tree report designer, standard statements (trial balance, balance sheet, P&L, cash flow indirect, comparatives, by-entity consolidated view), drill-down to the journal line and source document, scheduled distribution with

row-level security on dimensions, and the audit trail report. This is the Phase 2 exit wave: finance stakeholders sign off a closed period produced entirely from Passage.

Out of scope: group consolidation with eliminations and translation (wave 40), budgets (wave 38), lakehouse analytics and dashboards (wave 46), XBRL/iXBRL (wave 48).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Management Reporter reads from its own data mart refreshed by an integration service; statements lag the GL by 15–60 min and the mart silently breaks after AX upgrades or dimension changes	Separate ETL (MR integration) with no "as of" stamping	Read model fed by `journal.posted` events with per-ledger HWM; every statement shows the sequence it reflects; nightly zero-diff verify	Statement lag p95 < 10 s (vs 15–60 min); "mart out of sync" incidents → 0
SAP F.01/RFBILA00 and FSV (OB58) drill to FAGLL03 but lose the FSV context; reaching the source document from a balance-sheet line is 7+ screens and often impossible for migrated items	Hierarchy, totals and line items are separate programs without a shared key	Drill-down contract (BR-15.4): cell → balance keys → lines → `source_uri` → trace id, with sums that must equal the cell	Clicks balance → document 7 → 3; drill sum mismatches → 0 (property-tested)
Report designer in MR requires a Windows client, separate security (tree security) and definitions that cannot be versioned or promoted between environments except by manual export	Desktop tooling, no API, no versioning	Web designer, JSON export/import with schema version, immutable published versions, OPA dimension security shared with the rest of Passage	Definition promotion dev → prod 1 h manual → < 1 min API call; security rule sets 2 → 1
Cash flow statement is hand-built in Excel from the trial balance because MR/FSV cash flow setups are fragile and do not reconcile	No tagging model linking balance movements to cash flow categories with an enforced check	`cash_flow_tag` on rows, `CF-STD` indirect derivation with `CF_UNRECONCILED` failure and a "not classified" check row	Cash flow preparation 1–2 days/period → generated in < 20 s with 0.00 reconciliation enforced
Audit requests ("who posted, who approved, when was the period reopened") require pulling `SysDatabaseLog` / `CDHDR` extracts and workflow history separately, usually by IT	Audit data spread across logs with no finance-facing report	Audit trail report over CDC audit + ledger events with hashed export	Audit evidence turnaround days → minutes; controller self-service, 0 IT tickets

2.15.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Report definitions	Management Reporter row/column/tree definitions, report definitions	Financial statement version (OB58, `T011`, `FAGL_011PC/ZC`), Report Painter/Writer	`reporting.ReportDefinition` (rows, columns, tree)	full
Row definitions	MR row: description, link to financial dimensions, format codes (TOT, CAL), normal balance	FSV nodes with account intervals, debit/credit split, net result	`ReportRow` with kind ∈ {account_range, dimension_filter, total, calculation, header}	full
Column definitions	MR column: FD (period), CALC, comparative, YTD, budget	Period/comparison columns in F.01 / Fiori	`ReportColumn` with period expressions (`P`, `P-1`, `YTD`, `PY`, `PYTD`, `CALC`)	full (budget columns in wave 38)
Reporting tree	MR tree of reporting units (dimension values)	Profit center hierarchy (KE5X), FSV per company	`ReportTree` over dimension hierarchies (wave 6)	full
Trial balance	Trial balance list page, `LedgerTrialBalanceListPage`	FAGLB03, S_ALR_87012277, F.08	Trial balance statement (standard definition)	full
Balance sheet / P&L	MR default reports	F.01 (RFBILA00), Fiori Balance Sheet / P&L	Standard definitions `BS-STD`, `PL-STD`	full
Cash flow	MR cash flow (manual definitions)	Cash flow via FSV + S/4 Cash Flow app	`CF-STD` indirect with cash-account tagging and non-cash adjustments	full
Drill-down	MR drill to account/transactions; ledger transaction inquiry; voucher transactions	Document flow (FB03), FAGLL03 line items, drill from Fiori statements	Statement cell → balance keys → lines → source document URI → trace id	improved: trace id
Distribution	MR report library, scheduled generation, email	Report scheduling (SM36), SAP Analytics Cloud publication	`StatementRun` schedules with PDF/XLSX/CSV, email, link	full
Security	MR tree security, AX dimension security	Authorization objects (F_BKPF_BUK, K_CSKS), FSV authorization	OPA dimension-value policies applied to every balance query	full
Audit trail	Database log, voucher audit, workflow history	Change documents (CDHDR/CDPOS), FB03 header changes, SE16 on BKPF	Audit trail report over CDC audit (wave 5) + ledger events	full

2.15.3 Functional requirements

- **FR-15.1 Reporting read model** — Must. The system shall maintain in `reporting-db` a `ReportingBalance` fact (ledger × account × dimension combination × period × currency type) built from `ledger.journal.posted/reversed/balance.rebuilt/year-end.closed` events, with the covered `posting_sequence` high-water mark per ledger.
  - Lag SLO: p95 < 10 s after posting; every statement records the HWM it reflects and shows "as of sequence N / time T".
  - A nightly comparison against `ledger.Balance` must show zero differences.
- **FR-15.2 Report designer: rows** — Must. Row definitions contain ordered rows of kind header, account_range (account from/to, optional dimension filter), dimension_filter (any dimension values with wildcards), total (sum of row references), calculation (expression over row references with + − × ÷ and constants), with format (bold, indent, sign flip for normal credit balance, print zero, page break).
  - Maximum 2,000 rows per definition; expressions validated on save (unknown references, division by zero handled → blank).
- **FR-15.3 Report designer: columns** — Must. Column definitions specify period expressions relative to the run period (`P`, `P-n`, `YTD`, `PY`, `PYTD`, `Q`, `FY`), currency type (acct/rpt/txn with currency filter), ledger, dimension filter, variance/percent calculations between columns, and formatting.
  - Maximum 60 columns; comparative column sets (actual vs prior year) are templates.
- **FR-15.4 Reporting trees** — Must. Trees list reporting units (dimension value or hierarchy node from wave 6 hierarchies) with roll-up; a statement run produces one report per unit plus the roll-up, honoring OPA dimension security per viewer.

- **FR-15.5 Standard statements** — Must. The system shall ship definitions `TB-STD` (trial balance: opening, debit, credit, closing, net), `BS-STD`, `PL-STD`, `CF-STD` (indirect method: net income + non-cash adjustments + working-capital changes derived from balance movements of tagged accounts + investing + financing = change in cash), `TB-DIM` (trial balance by dimension), each parameterized by ledger, period, currency type, and tree.
  - Statements mark `is_closing_entry` / `is_opening_entry` handling: P&L excludes closing entries; BS includes opening entries.
  - Consolidated-by-entity view: side-by-side columns per legal entity and a simple sum column in reporting currency (no eliminations; wave 40).
- **FR-15.6 Statement runs** — Must. A run evaluates a definition for parameters and stores `StatementCell` values with the balance keys used, the HWM, duration, and a SHA-256 of the result; outputs rendered as HTML (interactive), PDF, XLSX and CSV.
  - 2,000-row × 12-column statement over 200k balances completes in < 20 s.
- **FR-15.7 Drill-down** — Must. From any cell: (1) balances view listing the (account, combination, period) keys with amounts summing to the cell; (2) journal lines for a selected key or the whole cell, paginated, with `voucher`, date, D/C, amounts, source; (3) source document link (`source_uri` resolved to a Passage route or external URL) and `trace_id` link to Tempo.
  - Drill honors the viewer's dimension security; the line sum shown equals the cell value exactly or the UI displays a reconciliation warning with the filtered-out amount (never silently).
- **FR-15.8 Scheduling and distribution** — Must. Definitions can be scheduled (cron or close-checklist hook) to run per entity set and distribute via email (PDF/XLSX attachments ≤ 20 MB, else link), shared link with expiry, and a report library with retention (default 7 years for statements run against closed periods, 90 days for ad-hoc).
- **FR-15.9 Row-level security on dimensions** — Must. OPA policies grant dimension values (e.g., cost centers) per user; balance queries, statements, drill-down and exports are filtered server-side; a report run by a scheduler uses the recipient's permissions per rendered copy.
- **FR-15.10 Audit trail report** — Must. The system shall report for a period and ledger: entries posted (who, when, source, idempotency replay flag), batch approvals (submitter, approvers, timestamps), reversals, period status changes including reopen approvals, rate overrides, tax fallbacks, allocation and year-end runs; exportable to CSV/PDF with a hash.
- **FR-15.11 Definition import/export** — Must. Definitions round-trip as JSON (schema-versioned) for backup, environment promotion and Management Reporter conversion.
- **FR-15.12 Management Reporter and FSV converters** — Should. The migration framework shall convert MR row/column/tree definitions (from the MR database `ControlRowDetail`, `ControlColumnDetail`, `ControlTreeDetail`) and SAP FSV (`T011`, `FAGL_011PC`, `FAGL_011ZC`) into Passage definitions with a conversion report of unsupported constructs.

2.15.4 Data model

ReportDefinition, **ReportRow**, **ReportColumn**

Field	Type	Required	Notes
definition: id	uuid	yes	PK
definition: code / name / version	text(30) / text(120) / int	yes	unique (tenant, code, version)
definition: kind	enum	yes	trial_balance, balance_sheet, income_statement, cash_flow, custom
definition: row_set_id / column_set_id / tree_id	uuid	yes/yes/no	
definition: default_ledger_id / currency_type	uuid / enum	yes	
definition: status	enum	yes	draft, published, retired
definition: schema_version	text	yes	for JSON import/export
row: id, row_set_id, sequence	uuid, uuid, int	yes	
row: kind	enum	yes	header, account_range, dimension_filter, total, calculation
row: label	text(200)	yes	
row: account_from / account_to	text(20)	conditional	
row: dimension_filter	jsonb	no	{attr: [values or ranges or *]}
row: references / expression	uuid[] / text	conditional	for total / calculation
row: normal_balance	enum	yes	debit, credit (sign presentation)
row: format	jsonb	no	bold, indent, print_zero, page_break, underline
row: cash_flow_tag	enum	no	operating_adjustment, working_capital, investing, financing, cash
column: id, column_set_id, sequence, label		yes	
column: period_expr	text	yes	`P`, `P-1`, `YTD`, `PY`, `PYTD`, `Q`, `FY`, `CALC`
column: currency_type / currency	enum / char(3)	yes/no	
column: ledger_code_override	text	no	compare ledgers side by side
column: dimension_filter	jsonb	no	
column: calc_expression	text	no	`C2-C1`, `(C2-C1)/C1`
column: format	jsonb	no	decimals, percent, width

StatementRun and **StatementCell**

Field	Type	Required	Notes
run: id	uuid	yes	PK
run: definition_id / version	uuid / int	yes	
run: parameters	jsonb	yes	ledger(s), period, tree unit, currency type, entities
run: hwm_by_ledger	jsonb	yes	{ledger_id: posting_sequence}
run: status	enum	yes	queued, running, completed, failed
run: started_at / completed_at / duration_ms		yes	

Field	Type	Required	Notes
run: result_hash	bytea(32)	yes	SHA-256 over cells
run: outputs	jsonb	yes	[{format, uri, size}]
run: requested_by / security_principal	uuid	yes	whose dimension security applied
cell: run_id, row_id, column_id		yes	PK
cell: value	numeric(23,6)	yes	
cell: balance_keys	jsonb	yes	[{ledger, account, combination, period, currency_type}] (compressed; up to 50k keys, overflow stored as a query spec)
cell: line_count	int	yes	

AuditTrailEntry (materialized view over audit store + ledger events): `occurred_at`, `ledger_id`, `kind` (posted, replayed, reversed, batch_submitted, batch_approved, batch_rejected, period_changed, reopen_approved, rate_override, tax_fallback, allocation_run, year_end_run, rebuild), `actor_id`, `object_ref`, `details` jsonb, `trace_id`.

Invariants: for every cell, Σ ReportingBalance over `balance_keys` = `value` (property-tested); total rows equal Σ referenced rows; `TB-STD` closing per account = opening + debit – credit; `CF-STD` change in cash = Σ cash-tagged account movements; `result_hash` is reproducible for the same HWM and definition version. Indexing: `reporting_balance` PK (ledger, period, account, combination, currency_type) plus (ledger, account, period); `statement_cell` PK (run_id, row_id, column_id); audit view indexed on (ledger_id, occurred_at).

2.15.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET/POST/PUT	/api/v1/reporting/definitions[/{code}] and /{code}:publish	Definitions CRUD and publish	PUT idempotent	Maintain report definitions
GET/PUT	/api/v1/reporting/row-sets[/{id}], /column-sets[/{id}], /trees[/{id}]	Designer parts	PUT idempotent	Maintain report definitions
POST	/api/v1/reporting/definitions/{code}:validate	Validate expressions, references	stateless	Maintain report definitions
GET/POST	/api/v1/reporting/definitions/{code}/export, /import	JSON round-trip	import `Idempotency-Key`	Maintain report definitions
POST	/api/v1/reporting/runs	Run a statement	`Idempotency-Key`	Run financial statements
GET	/api/v1/reporting/runs/{id}[/cells] and /outputs/{format}	Results and files	n/a	View financial statements
GET	/api/v1/reporting/runs/{id}/cells/{row}/{col}/balances	Drill level 1	n/a	View financial statements
GET	/api/v1/reporting/runs/{id}/cells/{row}/{col}/lines?cursor=	Drill level 2 (lines)	n/a	View ledger transactions
GET	/api/v1/reporting/balances?ledger=&period=&account_from=&… &group_by=	Ad-hoc balance query (secured)	n/a	View ledger balances
GET/PUT	/api/v1/reporting/schedules[/{id}]	Distribution schedules	PUT idempotent	Schedule financial statements
GET	/api/v1/reporting/audit-trail?ledger=&period=&kind= and /audit-trail:export	Audit trail report	export run id	View audit trail
GET	/api/v1/reporting/lag	Read-model lag per ledger	n/a	View financial statements

Queries and validation: `RunStatement` requires a published definition, period exists for every selected ledger, viewer has at least one permitted dimension value when a tree is used, and read-model lag ≤ `max_lag_seconds` (default 30) unless `allow_stale = true` (result flagged). `ImportDefinition` validates schema version and rejects duplicate codes without `overwrite`.

Event	Trigger	Payload fields	Consumers
dev.passage.reporting.statement.generated.v1	run completed	run_id, definition code/version, parameters, hwm_by_ledger, result_hash, outputs[]	distribution, close (automation hook), audit
dev.passage.reporting.statement.failed.v1	run failed	run_id, error, stage	ops, requester notification
dev.passage.reporting.definition.published.v1	publish	code, version, actor, diff_summary	audit, cache
dev.passage.reporting.lag.exceeded.v1	lag > threshold for 60 s	ledger_id, lag_seconds, hwm_ledger, hwm_reporting	ops alerting, close board
dev.passage.reporting.distribution.sent.v1	email/link delivered	run_id, recipients (hashed), format, size	audit

2.15.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Report designer	Create definition, edit rows/columns/tree in split panes, preview with live data, validate, publish, export/import	row label, kind, accounts, dimension filter, expression, format; column label, period expr, currency type	Maintain report definitions
Financial statements	Run (ledger, period, tree, currency), view interactive statement, compare runs, export	row, columns, as-of HWM, run status	Run / View financial statements
Drill-down panel	Cell → balances → lines → source document, open trace	account, combination, period, amount; voucher, date, D/C, amounts, source link	View ledger transactions
Trial balance (standard)	Period range, dimension filters, drill, export	account, opening, debit, credit, closing	View ledger balances
Report library	Browse outputs by entity/period/definition, share link, retention	definition, period, entity, generated at, hash	View financial statements
Schedules	Define cron/close hook, recipients, formats	definition, parameters, cron, recipients, last run	Schedule financial statements
Audit trail	Filter period/ledger/kind/actor, export with hash	occurred at, kind, actor, object, details, trace id	View audit trail

Interaction rules: designer previews render within 3 s against the demo dataset and update on each save; row grid supports drag-reorder, multi-select indent and "insert total of selection"; statements are keyboard navigable cell-by-cell (arrow keys) with Enter to drill; drill panels stack as breadcrumbs (Statement › Cell › Balances › Lines › Document); exports preserve number formatting and include an "as of" footer with HWM, definition version and result hash; all screens pass axe WCAG 2.1 AA.

2.15.7 Business rules and edge cases

- **BR-15.1 Period expression evaluation.** `P` = run period; `P-n` = n periods earlier in the ledger calendar (crossing fiscal years, adjustment periods skipped unless `include_adjustment_periods`); `YTD` = periods from fiscal year start through P, including adjustment periods ≤ P's year when flagged; `PY` = same period prior year; `Q` = fiscal quarter containing P.
- **BR-15.2 Balance-sheet amounts** are cumulative: Σ ReportingBalance.net from period 0 (opening) through the column's period; P&L amounts are the sum over the column's period range only and exclude `is_closing_entry` lines.
- **BR-15.3 Sign presentation.** Stored net = debit – credit; rows with `normal_balance = credit` display `-net`; calculation rows use displayed values unless the expression is prefixed `raw:`.
- **BR-15.4 Statement drill-down contract.** Level 1 returns balance keys whose Σ equals the cell value exactly (including rounding of the cell to the currency scale — none is applied since balances are already at scale). Level 2 returns lines for one key or the whole cell with keyset pagination (page size 500, max 1,000) ordered by (posting_date, posting_sequence, line_number); a `sum_of_lines` field accompanies every page and the final page's cumulative sum must equal the cell value; when dimension security filters lines, the response includes `filtered_amount` and the UI shows a warning.
- **BR-15.5 Cash flow (indirect) derivation.** Net income = P&L total for the range; operating adjustments = movements of rows tagged `operating_adjustment` (depreciation, provisions); working capital = −Δ of asset rows and +Δ of liability rows tagged `working_capital`; investing and financing = movements of tagged rows; cash = Δ of rows tagged `cash`. The statement fails validation (`CF_UNRECONCILED`) if net cash flow ≠ Δ cash by more than 0.00; untagged balance-sheet accounts appear in a "not classified" check row that must be zero.
- **BR-15.6 Read-model consistency.** A run captures HWM per ledger at start and evaluates only lines with `posting_sequence ≤ HWM`; two runs with equal HWM and definition version produce equal `result_hash`.
- **BR-15.7 Security.** Dimension security is applied as a filter on `ReportingBalance` queries (OPA returns allowed value sets per attribute, cached 60 s); a scheduled run renders per recipient; totals never include values the viewer cannot see; an "unrestricted" copy is produced only for recipients holding `View all dimensions`.
- **BR-15.8 Definition versioning.** Publishing creates an immutable version; runs reference the version; editing a published definition creates a draft of version n+1; retiring prevents new runs but keeps history.
- **BR-15.9 Comparative columns across calendars.** If `PY` period does not exist (calendar changed), the column shows blank with a footnote, not zero.
- **BR-15.10 Multi-ledger columns.** Columns with `ledger_code_override` are allowed only when the ledgers share the chart of accounts; otherwise `REPORT_LEDGER_INCOMPATIBLE`.
- **BR-15.11 Large cells.** Cells with > 50k balance keys store a query spec instead of the key list; drill level 1 re-executes the query at the run's HWM.
- **BR-15.12 Audit trail completeness.** Every `journal.posted` event must have an audit row within 60 s; a daily job counts events vs rows and alerts on any difference.

2.15.8 Acceptance criteria

- **AC-15.1** (FR-15.5) Given the demo entities' closed March 2026, when `BS-STD` and `PL-STD` run, then total assets = liabilities + equity (including current-period net income) to the cent and the P&L net equals `TB-STD` net of P&L accounts.
- **AC-15.2** (FR-15.5, BR-15.5) Given `CF-STD` for the same period, when run, then net cash flow equals the change in cash-tagged accounts to the cent and the "not classified" row is 0.00.
- **AC-15.3** (FR-15.7, BR-15.4) Given any cell of a run, when drilled to lines across all pages, then the cumulative sum equals the cell value exactly (property test over 1,000 random cells).
- **AC-15.4** (FR-15.7) Given a line from an AP-sourced entry (stub), when the source link is opened, then it resolves to the document route and the trace id opens in Tempo.
- **AC-15.5** (FR-15.11) Given a 200-row, 12-column definition, when exported to JSON and re-imported under a new code, then the two definitions produce identical cell values for the same parameters.
- **AC-15.6** (FR-15.9) Given a user permitted only cost center C200, when they run `TB-DIM`, then only C200 rows appear and drill-down lines are limited to C200 with `filtered_amount` shown on mixed cells.
- **AC-15.7** (FR-15.1, BR-15.6) Given postings continue during a run, when the run completes, then its `result_hash` is reproducible by re-running at the same HWM.
- **AC-15.8** (FR-15.8) Given a nightly schedule for 3 entities, when it fires, then PDF and XLSX outputs are in the library and the email is sent with attachments < 20 MB.
- **AC-15.9** (FR-15.10) Given a reopened period with a posting and re-close, when the audit trail runs, then it lists the reopen with both approvers, the posting, and the re-close in order with trace ids.
- **AC-15.10** (FR-15.6) Given 200k balances, when a 2,000 × 12 statement runs, then it completes in < 20 s.
- **AC-15.11** (Phase exit) Given the finance stakeholders' review, when they compare the Passage statement pack to the AX/SAP-derived expectations for the demo period, then every line ties and sign-off is recorded in the wave 15 issue.

2.15.9 Test plan and fixtures

- Fixtures: `demo-fy2026-closed` (the dataset produced by waves 11–14 for the demo entities, three ledgers, EUR reporting currency); `definitions-std` (TB-STD, BS-STD, PL-STD, CF-STD, TB-DIM); `definition-200x12` (designer round-trip fixture); `mr-sample-definitions` (Management Reporter export of 10 definitions); `fsv-sample` (SAP `T011`/`FAGL_011PC` extract for 2 FSVs); `security-cases` (8 users with distinct dimension grants).

Test	Type	Target	Pass criterion
Statement tie-out	integration	standard definitions	BS balances; P&L = TB; CF reconciles

Test	Type	Target	Pass criterion
Drill-down sum property	property	drill API	Σ lines = cell for 1,000 random cells
Period expression unit tests	unit	evaluator	40 cases incl. adjustment periods and calendar change
Designer round-trip	contract	export/import	equal cell values
MR/FSV converters	unit	migration converters	10 MR + 2 FSV definitions convert; unsupported constructs listed
Security filtering	integration	OPA + reporting	8 users see only permitted values; totals exclude
Read-model lag	integration	event consumer	p95 < 10 s under 5k lines/s
Statement performance	performance	run engine	2,000 × 12 over 200k balances < 20 s
Reporting UI	Playwright + axe	designer, statements, drill	WCAG AA; keyboard drill works

Reconciliation: `ReportingBalance` = `ledger.Balance` nightly (0 diffs); TB closing = opening + movements; BS = L + E; CF net = Δ cash; audit rows = posted events.

2.15.10 Deployment and operations

Helm chart `passage-reporting` (Go service 3–8 pods, CloudNativePG `reporting-db` 3 instances with 1 read replica for statement runs, Kafka consumer group `reporting-balances`, renderer sidecar for PDF (Chromium headless) with 2 CPU limit). Config: `REPORTING_MAX_LAG_S=30`, `REPORTING_PAGE_SIZE=500`, `REPORTING_CELL_KEY_LIMIT=50000`, `REPORTING_RETENTION_CLOSED_YEARS=7`, `REPORTING_RETENTION_ADHOC_DAYS=90`. Migrations: reporting schema; backfill of `ReportingBalance` from the ledger via a one-time replay of `ledger.journal` topic (compacted snapshot + replay). Feature flags: `reporting.cash_flow`, `reporting.multi_ledger_columns`, `reporting.pdf`. Scheduled jobs: nightly `reporting-balance-verify`, nightly statement pack for demo entities (23:00 UTC), library retention sweep (weekly), audit completeness check (daily). SLOs: read-model lag p95 < 10 s; statement run p95 < 20 s (reference size); drill level 2 first page p95 < 300 ms; availability 99.9%. Alerts: lag > 60 s (P2), verify diff > 0 (P1), audit completeness diff (P1), PDF renderer failures > 5% (P3). Runbook: rebuild read model from ledger replay (≈ 15 min per 10M lines), stuck run cancellation, retention restore from object storage. Rollback: digest rollback; read model can be rebuilt from the journal at any time, so no data rollback is needed.

2.15.11 Migration from AX / SAP

Source	Passage entity	Notes
Management Reporter `ControlRowDetail`, `ControlColumnDetail`, `ControlTreeDetail`, `ControlReport`	ReportRow, ReportColumn, ReportTree, ReportDefinition	row link types (FD, dimension sets) → account_range/dimension_filter; CALC with +−×÷ converted; unsupported: XBRL tags, cross-report links (listed)
AX trial balance / inquiry saved queries	not migrated	standard definitions replace them
SAP `T011`, `T011T`, `FAGL_011PC`, `FAGL_011ZC`, `FAGL_011QT` (FSV)	ReportDefinition (BS/P&L), ReportRow	account intervals → account_range; debit/credit split nodes → two rows with sign rules; net result node → calculation row
SAP Report Painter libraries	not migrated	re-authored in the designer; conversion report lists them
AX database log / SAP CDHDR history	not migrated	legacy audit remains in the archive; Passage audit begins at cutover

Validation: converted definitions run against migrated opening balances and are compared to the legacy final statements for the cutover period (tolerance 0.00 after sign normalization); conversion report reviewed by the controller.

2.15.12 Dependencies, risks, open questions

Depends on: wave 11 (journal events, balances), wave 12 (reporting currency amounts), wave 13 (tax audit events), wave 14 (period events, closing/opening flags, close-checklist hooks), wave 6 (dimension hierarchies for trees), wave 5 (audit store, Tempo trace links), wave 2 (OPA).

Risk	Likelihood	Impact	Mitigation
Read model lag at month-end peak invalidates "as of" statements	Medium	Medium	HWM stamping; close task waits for lag = 0; replica for runs
MR definitions use constructs with no equivalent	High	Low	Conversion report; manual re-authoring budgeted in migration plan
PDF rendering resource spikes	Medium	Low	Renderer sidecar limits; queue with concurrency 4 per pod
Stakeholder sign-off depends on accountant availability	Medium	High	Sign-off session scheduled at wave start; fixtures reviewed incrementally

Open questions: (1) Should the reporting read model live in `reporting-db` (default) or query `ledger.Balance` directly with replicas? Default separate read model; decide by wave 15 start. (2) Cash flow direct method in v1? Default no (indirect only); revisit at wave 36 (cash and bank). (3) Retention of ad-hoc runs 90 days vs 1 year? Default 90 days; decide by wave 15 exit.

Part 3 — Phase 3: Inventory & Procure-to-Pay (waves 16–20)

3.0 Phase overview

Business outcome at phase exit. At the end of wave 20 the demo plant buys, receives, pays for and values physical goods on Passage with no AX or SAP system in the loop. Every quantity movement is an immutable inventory transaction carrying physical and financial status plus storage and tracking dimensions (the InventTrans / MSEG role). Buyers run requisition → RFQ → agreement → purchase order under policy-driven approval (PurchReqTable / EBAN through PurchTable / EKKO). Receiving posts inventory and a GR/IR accrual to the universal journal; vendor invoices match 2-way or 3-way inside tolerances or route to an exception queue (VendInvoiceInfoTable / RBKP-RSEG). Open vendor items settle against payment runs that emit ISO 20022 pain.001 and NACHA files (VendSettlement / F110-REGUH). Inventory is valued under FIFO, LIFO, weighted average or standard cost with a close/recalculation job whose value report ties to the GL inventory accounts (InventClosing / CKMLCP). Suppliers acknowledge POs, send ASNs and submit invoices through a portal.

Phase-exit sign-off criteria

- Inventory on-hand projection equals SUM(inventory transactions) for a 1M-transaction fixture after a full rebuild; zero rows differ.
- Each of the 11 journal/posting types in this phase (movement, adjustment, transfer, BOM, counting, product receipt, vendor invoice, payment, settlement, close adjustment, revaluation) produces a balanced universal-journal entry; the daily subledger-to-GL reconciliation for inventory, GR/IR and AP control accounts reports a difference of 0.00 in accounting currency for the demo entity.
- 30 requisitions in the approval matrix route to the expected chain; 200 matching scenarios yield the expected match status and postings; 300 settlement scenarios post the expected discount and FX gain/loss; 100 costing sequences per method tie to the reference workbook to the cent.
- pain.001.001.09 output validates against the ISO 20022 XSD and is accepted by the bank test harness; a payment run killed mid-flight and resumed pays each open item exactly once.
- Inventory close on 1M transactions completes in under 20 minutes on the stage cluster (4 vCPU / 16 GiB inventory pod, Postgres 8 vCPU).
- Recorded end-to-end walkthrough (buy → receive → match → pay → close) signed off by both owners and the reviewing accountant.

Cross-wave data model summary

Entity	Owning service	Key relationships
InventoryTransaction (InventTrans / MSEG)	inventory (Go, `inventory` Postgres)	→ InventoryTransactionOrigin, → Dimension set, → Product (wave 8), → Site/Warehouse (wave 6)
InventoryTransactionOrigin (InventTransOrigin / MKPF)	inventory	1:N InventoryTransaction; references source document id+type (journal line, PO line, receipt line, production order line)
OnHand (InventSum / MARD+MCHB)	inventory (projection)	Keyed by product × dimension hash; rebuilt from InventoryTransaction
InventoryJournal / JournalLine (InventJournalTable/Trans)	inventory	Lines → InventoryTransactionOrigin; posts via ledger `Post(JournalEntry)` (wave 11)
Reservation	inventory	→ InventoryTransaction (status change), → batch/serial
CountPlan / CountSheet	inventory	→ JournalLine (variance)
PurchaseRequisition / Line (PurchReqTable / EBAN)	procurement (Go, `procurement` Postgres)	→ PurchaseOrderLine on consolidation; → Temporal approval workflow
RFQ / RFQReply (PurchRFQTable / EKKO type AN)	procurement	→ PurchaseOrder on award
PurchaseAgreement (PurchAgreementHeader / EKKO type K/L)	procurement	→ PurchaseOrderLine releases
TradeAgreement (PriceDiscTable / A017-KONP)	procurement	Vendor pricing lookup
PurchaseOrder / Line / Version (PurchTable/PurchLine, EKKO/EKPO)	procurement	→ InventoryTransaction (OnOrder), → Receipt, → VendorInvoice
ProductReceipt / Line (VendPackingSlipJour/Trans, MKPF/MSEG 101)	receiving (Go, `procurement` Postgres)	→ PurchaseOrderLine, → InventoryTransaction (Received), → GR/IR journal entry
VendorInvoice / Line (VendInvoiceInfoTable/Line → VendInvoiceJour, RBKP/RSEG)	payables (Go, `payables` Postgres)	→ ReceiptLine (3-way), → PurchaseOrderLine (2-way), → VendorTransaction, → GR/IR clearing
MatchResult	payables	→ VendorInvoiceLine; stores price/qty/total variance and status
VendorTransaction (VendTrans / BSIK→BSAK)	payables	Open item; → Settlement
Settlement (VendSettlement / BSE_CLR)	payables	Links payment/credit/invoice transactions with discount and FX
PaymentProposal / Payment / PaymentFile (LedgerJournalTrans payment journal, REGUH/REGUP)	payables	→ VendorTransaction, → bank account (wave 36 placeholder: bank master lite)
CostTransaction (InventCostTrans / CKMLPP-CKMLCR)	costing (Go, `inventory` Postgres schema `costing`)	→ InventoryTransaction; produced by close/recalc
CostingVersion / ItemCost (CostingVersion/InventItemPrice, MBEW)	costing	Standard and planned costs per product × site
SupplierPortalUser / Acknowledgement / ASN	supplier-portal (TypeScript BFF + `supplier` Postgres)	→ PurchaseOrder, → ProductReceipt (expected)

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.inventory.transaction.posted.v1`	inventory	costing, planning (w22), analytics	txn id, origin, product, dims, qty, issue/receipt status, physical/financial dates
`dev.passage.inventory.onhand.changed.v1`	inventory	procurement (ATP for PO), sales (w27), warehouse (w31)	product, dim hash, avail_physical, avail_ordered, delta

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.inventory.journal.posted.v1`	inventory	ledger (reconciliation), costing	journal id, type, line count, ledger entry id
`dev.passage.procurement.requisition.approved.v1`	procurement	procurement consolidation, budget control (w38)	req id, lines, approver chain
`dev.passage.procurement.po.confirmed.v1`	procurement	inventory (OnOrder txns), supplier-portal, EDI (w35)	po id, version, lines, vendor
`dev.passage.procurement.po.version.created.v1`	procurement	supplier-portal, receiving	po id, version, diff summary
`dev.passage.receiving.receipt.posted.v1`	receiving	inventory, payables (match candidates), costing	receipt id, lines, po lines, qty, GR/IR amount
`dev.passage.payables.invoice.posted.v1`	payables	ledger, costing (PPV/invoice-cost settle), tax (w13)	invoice id, vendor, lines, match status, amounts
`dev.passage.payables.invoice.match.exception.v1`	payables	workflow (Temporal), notifications	invoice id, line, variance type, amounts
`dev.passage.payables.settlement.posted.v1`	payables	ledger, analytics	settlement id, items, discount, fx gain/loss
`dev.passage.payables.payment.sent.v1`	payables	bank connectivity (w36), supplier-portal, cash (w36)	payment id, file id, amount, method, vendor
`dev.passage.payables.payment.status.changed.v1`	payables	notifications, analytics	payment id, old/new status, pain.002 reason code
`dev.passage.costing.close.completed.v1`	costing	ledger, analytics	close id, period, adjustments, duration
`dev.passage.costing.cost.changed.v1`	costing	inventory, planning, product costing (w39)	product, site, version, old/new cost, effective date
`dev.passage.supplier.asn.submitted.v1`	supplier-portal	receiving, warehouse (w31)	asn id, po lines, expected qty, carrier

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
On-hand projection drifts from transactions under concurrency	Medium	High	Single-writer per product × dimension partition (advisory lock), nightly rebuild-and-compare job, alert on any diff	Engineering
Matching tolerance semantics differ between AX (per item/vendor %) and SAP (tolerance keys) and users expect their old behavior	High	Medium	Tolerance model supports both percentage and absolute limits at vendor, item, vendor×item and entity levels; migration maps both	Product
FX gain/loss and discount postings fail accountant review	Medium	High	300-scenario fixture set reviewed by accountant before wave 19 close; property test that settlement always balances	Product + accountant
Inventory close performance on large transaction history	Medium	Medium	Settlement by product partition in Temporal child workflows; incremental close from last closed date; 1M-txn perf gate	Engineering
Bank file rejection in production (schema or bank-specific rules)	Medium	High	XSD validation in CI, bank test harness, per-bank profile overrides (SEPA, CH, UK)	Engineering
Supplier portal becomes an attack surface	Medium	High	Separate Keycloak realm, OPA policies scoped to vendor id, rate limits 60 req/min, no inbound write to GL	Security

3.16 Wave 16: Inventory Ledger, Dimensions & On-Hand

3.16.1 Scope and business context

Wave 16 creates the inventory ledger that every later quantity-bearing module (purchasing, receiving, production, sales, warehouse) writes to. In AX this is InventTrans with InventTransOrigin, InventSum and InventDim; in SAP it is the material document (MKPF/MSEG) with stock segments in MARD (storage location), MCHB (batch) and MSKA/MSPR (special stock). Before this wave Passage has products with dimension groups (wave 8) and sites/warehouses (wave 6) but no quantities.

At the end of the wave a user can: load opening balances through a movement journal; move, adjust, transfer and count stock; reserve stock manually or by FEFO/FIFO policy; block or quarantine a batch; run a cycle-count plan; and see on-hand by any combination of site, warehouse, location, batch, serial, inventory status. Every posting produces a balanced universal-journal entry through the wave 11 `Post(JournalEntry)` API.

Out of scope: purchase/sales-driven transactions (waves 17–18, 27), production consumption (wave 21), warehouse work and license plates (wave 31), quality orders (wave 33), costing and value adjustments beyond the running cost price captured on receipt (wave 20). Inventory in this wave values at the item's current cost price on receipt and issues at the running average; wave 20 replaces that with the full costing engine.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX On-hand form (InventOnhand) requires choosing a dimension display (InventDimParm) and re-querying per combination; SAP MMBE shows plant/storage location/batch only, MB52 is a batch report; users export to Excel to get location × batch × status	InventSum is pre-aggregated per InventDimId and queried through fixed display sets; MARD/MCHB are separate tables with no common query	Single `GET /onhand` with `groupBy` any dimension subset, served from one projection; shell grid pivots client-side	Clicks to answer "how much of X in batch B at location L": 9 → 2; query p95 < 50 ms vs 3–10 s
InventSum drift from InventTrans after failed batch or kernel patch; fix is the "Check inventory on-hand" consistency check run at night with users off the system	Dual-write of InventTrans and InventSum under X++ ttsbegin with retry-on-deadlock; no automated compare	Projection updated in the same Postgres transaction; weekly rebuild-and-compare; diff event alerts	Reconciliation effort: 2 h/month manual → 0 (automated, 0 diffs); detection latency: weeks → < 24 h
SAP MIGO posts synchronously but stock overview in BW/analytics lags the nightly extract; AX entity store refresh is hourly at best	Analytics copies instead of event stream	`onhand.changed` CloudEvent via outbox within 5 s; any consumer (BI, MES, WMS) subscribes without middleware	Data latency to external systems: 1–24 h → < 5 s

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Negative inventory silently allowed per InventModelGroup; errors surface at inventory close as unsettled issues (InventClosing "Cannot settle") months later	Check done at posting with a flag, no alerting; financial negative evaluated only at close	Negative check atomic under lock; negative rows surfaced on workspace tile and in daily reconciliation	Time to detect unauthorized negative stock: 30–90 days → same day; close-time settlement errors from negatives: 0
Cycle counting in AX needs the Counting journal, InventCountGroup setup and the "Create lines" dialog with 14 fields; SAP MI01/MI04/MI07 is three transactions and a printed count list	Journal-centric design with no plan object; count execution not mobile	`CountPlan` generates sheets on a cron; PWA count screen with blind count and offline queue; one `:post`	Steps from plan to posted variance: 3 transactions / ~25 clicks → 1 plan + 1 tap per line; counts per hour per operator ×3

3.16.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Inventory transaction record	InventTrans, InventTransOrigin, InventTransPosting; form InventTrans	MSEG/MKPF (material document), MATDOC in S/4; MB03/MIGO display	`inventory.InventoryTransaction`, `InventoryTransactionOrigin`; screen Inventory transactions	full
On-hand by dimension	InventSum, InventOnhand form, InventDimParm	MARD, MCHB, MSKA; MMBE stock overview, MB52	`inventory.OnHand` projection; On-hand inquiry	improved (any dimension combination, real time)
Dimensions	InventDim (InventSiteId, InventLocationId, wMSLocationId, InventBatchId, InventSerialId, InventStatusId, LicensePlateId, ConfigId/Size/Color/Style)	Plant (WERKS), storage location (LGORT), batch (CHARG), serial (SERNR via SERI), special stock indicator, valuation type	`DimensionSet` (site, warehouse, location, batch, serial, status, config/size/color/style, license_plate reserved for w31)	full
Movement / adjustment journal	InventJournalTable/Trans types Movement, LossProfit; InventJournalCheckPost	MIGO movement types 501/502 (receipt w/o PO), 551/552 (scrapping), 561/562 (initial stock)	`InventoryJournal` type MOVEMENT, ADJUSTMENT	full
Transfer journal	InventJournalTrans type Transfer	301/302 (plant to plant), 311/312 (storage loc), MB1B	`InventoryJournal` type TRANSFER (paired issue/receipt txns)	full
BOM journal	InventJournalTrans type BOM (InventBOMJournal)	261/262 consumption to order, 201 cost center	`InventoryJournal` type BOM (explode wave 9 BOM, issue components, receive parent)	partial (no route/overhead until w21)
Counting	InventJournalTrans type Count, InventCountJournal, counting groups (InventCountGroup)	Physical inventory MI01/MI04/MI07, IKPF/ISEG, cycle counting indicator ABC (MIBC)	`CountPlan`, `CountSheet`, journal type COUNT	full
Item model group	InventModelGroup (InventModel FIFO/LIFO/WeightedAvg/Std, PhysNegative, FinancialNegative, PostPhysicalInvent)	MBEW price control (S/V), negative stock allowed per plant/storage location (OMJ1)	`ItemModelGroup` (wave 8 entity, enforced here)	full
Reservation	InventTrans status ReservPhysical/ReservOrdered, InventMovement, reservation form, FEFO date-controlled (InventBatch expiry, PdsFEFO)	Stock reservations (MB21, RESB), batch determination (FEFO via strategy), batch status	`Reservation` with policy FEFO/FIFO/MANUAL	full
Blocking / quarantine	Inventory blocking (InventBlocking), quarantine order (InventQuarantineOrder), inventory status (D365)	Blocked stock (movement 344/343), quality inspection stock, batch restricted status	Inventory status `AVAILABLE`, `BLOCKED`, `QUARANTINE`; `Blocking` record	full
Unit conversion on movement	UnitOfMeasureConversion (intra/inter-class, product-specific)	MARM alternative UoM, conversion on MIGO	Wave 6/8 UoM service; quantity stored in inventory unit plus entered unit and qty	full
Inventory closing/recalc	InventClosing	CKMLCP	Deferred to wave 20	dropped-with-reason (wave 20)

3.16.3 Functional requirements

- **FR-16.1 Immutable inventory transactions** — Must. The system shall record every quantity movement as an append-only `InventoryTransaction` row linked to an `InventoryTransactionOrigin`; updates only change status fields and dates, never quantity or identity.
    - Status changes are written as a new `InventoryTransactionStatusChange` audit row; the current status lives on the transaction for query speed.
    - A transaction has exactly one of `issue_status` or `receipt_status` set (sign determined by quantity).
    - Corrections are new transactions with opposite sign referencing `corrects_transaction_id`.
    - Partition key: tenant_id, product_id; up to 50M transactions per tenant supported before range partitioning by `physical_date` year is required.
- **FR-16.2 Transaction status model** — Must. The system shall implement the AX issue/receipt status model: issue statuses `ON_ORDER → RESERVED_ORDERED → RESERVED_PHYSICAL → PICKED → DEDUCTED → SOLD`; receipt statuses `ORDERED → ARRIVED → REGISTERED → RECEIVED → PURCHASED`.
    - Physical statuses (affect physical on-hand): PICKED, DEDUCTED, REGISTERED, RECEIVED. Financial statuses: SOLD, PURCHASED.
    - Transitions may skip forward (ORDERED → RECEIVED directly) but never go backward except via the explicit `Unpick`, `Unregister`, `CancelReservation` commands.
    - SAP mapping on migration: 101 → RECEIVED, 102 → reversal, 261 → DEDUCTED, 301 → DEDUCTED+RECEIVED pair, 561 → RECEIVED+PURCHASED opening.
- **FR-16.3 On-hand projection** — Must. The system shall maintain `OnHand` per product × dimension set with the columns `posted_qty, posted_value, received, registered, arrived, ordered, deducted, picked, reserved_physical, reserved_ordered, on_order` and computed `avail_physical`, `avail_ordered`, `physical_invent`.
    - `physical_invent = posted_qty + received + registered − deducted − picked`.
    - `avail_physical = physical_invent − reserved_physical`.

- `avail_ordered = avail_physical + ordered + arrived − on_order − reserved_ordered`.
- Projection is updated in the same Postgres transaction as the inventory transaction (not via event) to guarantee the negative-stock check; the event is emitted from the outbox afterward.
- Query at any dimension subset: aggregation over the finest-grain rows, p95 < 50 ms for ≤ 10k rows.

- **FR-16.4 Negative inventory policy** — Must. The system shall reject an issue that would make `avail_physical` (physical negative forbidden) or `avail_ordered` (financial negative forbidden) negative when the item model group forbids it.
  - Enforcement uses `SELECT … FOR UPDATE` on the OnHand row; the check and the write are atomic.
  - When allowed, negative on-hand is flagged in the On-hand inquiry and in the daily reconciliation.
  - Error code `INV-NEG-001` with product, dimension set and shortfall quantity.
- **FR-16.5 Inventory dimensions** — Must. The system shall store the dimension set as a normalized `DimensionSet` row keyed by a SHA-256 hash of its canonical values (AX InventDimId pattern), with active dimensions determined by the product's storage and tracking dimension groups.
  - Dimensions: site, warehouse, location, batch, serial, inventory_status, config, size, color, style, license_plate (reserved), owner (reserved for w41 consignment).
  - Site is mandatory on every transaction; warehouse mandatory if the storage dimension group marks it mandatory.
  - A serial-tracked product requires quantity 1 per transaction and a unique serial per product in physical stock (duplicate serial → `INV-SER-001`).
- **FR-16.6 Inventory journals** — Must. The system shall support journal types MOVEMENT, ADJUSTMENT, TRANSFER, BOM, COUNT, OPENING_BALANCE with header/lines, validate, approval workflow (Temporal template from wave 4), and post.
  - Posting creates transactions, updates on-hand and calls `Post(JournalEntry)` with the journal id as idempotency key; a ledger failure rolls back the inventory posting (saga compensating step with explicit reversal if the ledger acknowledged).
  - Journal offset accounts: MOVEMENT requires a user-entered offset account; ADJUSTMENT uses the posting profile's inventory adjustment account; TRANSFER posts no GL unless cost differs by site (wave 20).
  - Maximum 10,000 lines per journal; posting throughput target 2,000 lines/s.
- **FR-16.7 Reservations** — Must. The system shall reserve quantity against on-hand (RESERVED_PHYSICAL) or expected receipts (RESERVED_ORDERED) by manual choice or by policy FIFO, FEFO, LIFO, with batch/serial selection.
  - FEFO orders candidate batches by `expiry_date ASC, receipt_date ASC`; batches with status BLOCKED or QUARANTINE or `expiry_date < required_date + shelf_life_margin_days` are excluded.
  - Reservation holds for `reservation_ttl_hours` (default 0 = until released or consumed).
  - Over-reservation is rejected; partial reservation returns the shortfall.
- **FR-16.8 Blocking and quarantine** — Must. The system shall allow blocking a quantity, batch or serial with reason code; blocked quantity is excluded from `avail_physical` and from policy-based reservation.
  - Blocking is implemented as inventory status change (status dimension) producing a status-transfer transaction pair.
  - Quarantine requires a disposition (release, scrap, return) to exit.
- **FR-16.9 Cycle counting** — Must. The system shall generate count sheets from a `CountPlan` (ABC class frequency, location range, last-count age, random sample) and post counted variances as COUNT journal lines.
  - Counting freezes on-hand for the counted dimension set (`count_lock`) while a sheet is open unless the plan allows movement with recount.
  - Variance tolerance per plan (qty % and value); above tolerance requires supervisor approval before posting.
  - Blind count option hides expected quantity.
- **FR-16.10 Unit conversions on movements** — Must. The system shall accept quantity in any unit with a conversion to the product's inventory unit, storing `qty` (inventory unit), `entered_qty`, `entered_unit`.
  - Rounding to the product's inventory unit decimals (default 2, max 6); residual beyond precision is rejected (`INV-UOM-002`).
  - Catch-weight is deferred to wave 24.
- **FR-16.11 Physical and financial dates** — Must. The system shall store `physical_date` (goods movement) and `financial_date` (value recognition) separately; the ledger entry uses `financial_date` and is validated against period control (wave 14) for module INVENTORY.
- **FR-16.12 Daily reconciliation** — Must. The system shall run a job that compares SUM(OnHand.posted_value) by inventory account and dimension to the ledger balance projection and raises `dev.passage.inventory.reconciliation.diff.v1` on any non-zero difference.
- **FR-16.13 Inventory transaction inquiry** — Should. The system shall provide a filterable, exportable inquiry over transactions with origin, status, dates and GL voucher drill-down.
- **FR-16.14 Opening balance load** — Should. The system shall load opening balances from AX InventSum/InventTrans or SAP MARD/MCHB/MBEW extracts through the wave 10 migration framework into an OPENING_BALANCE journal.

3.16.4 Data model

InventoryTransaction

Field	Type	Required	Notes
id	uuid	yes	PK

Field	Type	Required	Notes
tenant_id	uuid	yes	RLS
origin_id	uuid	yes	FK InventoryTransactionOrigin
product_id	uuid	yes	FK product (wave 8)
dimension_set_id	bytea(32)	yes	FK DimensionSet hash
qty	numeric(28,6)	yes	signed; inventory unit; ≠ 0
entered_qty / entered_unit	numeric(28,6) / text	yes	as keyed
issue_status	enum	cond.	ON_ORDER, RESERVED_ORDERED, RESERVED_PHYSICAL, PICKED, DEDUCTED, SOLD; null for receipts
receipt_status	enum	cond.	ORDERED, ARRIVED, REGISTERED, RECEIVED, PURCHASED; null for issues
physical_date	date	yes	goods movement date
financial_date	date	no	set when financial status reached
cost_amount_physical	numeric(19,4)	no	accounting currency, at physical posting
cost_amount_posted	numeric(19,4)	no	accounting currency, financial
cost_amount_adjustment	numeric(19,4)	no	written by wave 20 close
ledger_entry_id	uuid	no	voucher in universal journal
corrects_transaction_id	uuid	no	reversal/correction link
marked_transaction_id	uuid	no	explicit issue→receipt marking (wave 20)
created_at	timestamptz	yes	append-only

InventoryTransactionOrigin

Field	Type	Required	Notes
id	uuid	yes	PK
source_type	enum	yes	INVENT_JOURNAL_LINE, PURCH_LINE, RECEIPT_LINE, SALES_LINE, PROD_LINE, COUNT_LINE, TRANSFER_LINE, OPENING
source_id	uuid	yes	polymorphic FK; unique (source_type, source_id)
product_id	uuid	yes	denormalized
reference_number	text	no	PO number, journal number
direction	enum	yes	ISSUE / RECEIPT
created_at	timestamptz	yes	

OnHand

Field	Type	Required	Notes
tenant_id, product_id, dimension_set_id	composite PK	yes	
posted_qty, posted_value	numeric	yes	financially posted
received, registered, arrived, ordered	numeric	yes	receipt-side buckets
deducted, picked, reserved_physical, reserved_ordered, on_order	numeric	yes	issue-side buckets
last_txn_seq	bigint	yes	monotonic per row for rebuild diff
updated_at	timestamptz	yes	

InventoryJournal / InventoryJournalLine

Field	Type	Required	Notes
id, journal_number	uuid, text	yes	number sequence `INVJ-` gap-free optional
journal_type	enum	yes	MOVEMENT, ADJUSTMENT, TRANSFER, BOM, COUNT, OPENING_BALANCE
status	enum	yes	DRAFT, SUBMITTED, APPROVED, POSTING, POSTED, REJECTED, CANCELLED
posting_date	date	yes	financial_date default
offset_account_id	uuid	cond.	MOVEMENT mandatory
line: line_no, product_id, dimension_set_id, qty, entered_unit, cost_price, counted_qty, expected_qty, to_dimension_set_id (TRANSFER), bom_id (BOM)			
ledger_entry_id	uuid	no	after posting

Reservation: id, product_id, dimension_set_id, qty, demand_type, demand_id, policy (MANUAL/FIFO/FEFO/LIFO), status (ACTIVE, RELEASED, CONSUMED), expires_at, created_by.

CountPlan / CountSheet: plan id, name, selection (abc_class[], warehouse_ids[], location_range, max_days_since_count, sample_pct), frequency (cron), tolerance_qty_pct, tolerance_value, blind (bool); sheet id, plan_id, status (OPEN, COUNTED, APPROVED, POSTED), lines (product, dims, expected_qty, counted_qty, variance_qty, counted_by, counted_at).

Invariants: Σ qty over transactions with physical status for a (product, dimension set) = OnHand.physical_invent; posted_value changes only through financial status transitions and wave 20 adjustments; a serial has at most one open physical receipt not matched by an issue; DimensionSet hash is deterministic across services (shared Go package `passage/invdim`). Indexing: InventoryTransaction (tenant, product, dimension_set, physical_date), (origin_id), partial index on open statuses; OnHand clustered by (tenant, product); monthly range partitions on InventoryTransaction by physical_date when a tenant exceeds 20M rows.

3.16.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/inventory/journals	Create journal header	client key	InventoryJournalMaintain
POST	/api/v1/inventory/journals/{id}/lines:bulk	Add up to 10,000 lines	per-line client key	InventoryJournalMaintain
POST	/api/v1/inventory/journals/{id}:validate	Dry-run validation	n/a	InventoryJournalMaintain
POST	/api/v1/inventory/journals/{id}:post	Post (starts Temporal `PostInventoryJournal`)	journal id	InventoryJournalPost
GET	/api/v1/inventory/onhand?product=&site=&warehouse=&batch=&groupBy=	On-hand inquiry	n/a	InventoryInquire
GET	/api/v1/inventory/transactions?product=&status=&from=&to=	Transaction inquiry	n/a	InventoryInquire
POST	/api/v1/inventory/reservations	Reserve (manual or policy)	client key	InventoryReserve
DELETE	/api/v1/inventory/reservations/{id}	Release	idempotent	InventoryReserve
POST	/api/v1/inventory/blockings	Block qty/batch/serial	client key	InventoryBlock
POST	/api/v1/inventory/count-plans/{id}:generate	Create count sheets	plan+date	CycleCountMaintain
POST	/api/v1/inventory/count-sheets/{id}:post	Post variance journal	sheet id	CycleCountPost
POST	/internal/v1/inventory/transactions:apply	Service-to-service status transition (used by waves 17–20)	origin+step	service account
POST	/internal/v1/inventory/onhand:rebuild	Rebuild projection from transactions	run id	InventoryAdmin

Commands: `CreateJournal`, `AddLines`, `PostJournal` (validates period control, negative policy, dimension mandatory flags, UoM conversion existence, offset account structure), `Reserve` (validates qty > 0, policy-eligible batches, status), `Block`, `ApplyTransactionStep` (validates monotonic status order, quantity ≤ remaining at previous status).

Event	Trigger	Payload fields	Consumers
`dev.passage.inventory.transaction.posted.v1`	any transaction insert or status change	transaction_id, origin, product_id, dimension_set, qty, issue_status, receipt_status, physical_date, financial_date, cost_amount_*	costing, analytics, planning
`dev.passage.inventory.onhand.changed.v1`	OnHand row update	product_id, dimension_set, avail_physical, avail_ordered, delta_bucket, delta_qty	procurement, sales, warehouse
`dev.passage.inventory.journal.posted.v1`	journal reaches POSTED	journal_id, type, lines, ledger_entry_id, posting_date	ledger reconciliation, costing
`dev.passage.inventory.reservation.created.v1` / `.released.v1`	reservation lifecycle	reservation_id, demand, product, dims, qty, policy	sales, production
`dev.passage.inventory.reconciliation.diff.v1`	daily job finds diff ≠ 0	account, dimension, inventory_value, ledger_balance, diff	alerting

3.16.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Inventory management workspace	Tiles: journals awaiting approval, negative on-hand, open count sheets, reconciliation diffs	counts, last run	InventoryInquire
On-hand inquiry	Filter, group by any dimension, drill to transactions, export	product, site, warehouse, location, batch, serial, status, physical_invent, avail_physical, avail_ordered, posted_value	InventoryInquire
Inventory journal (header + grid)	New, add lines, validate, submit, post, copy, import CSV	journal_number, type, status, line grid: product, dims, qty, unit, cost_price, offset account	InventoryJournalMaintain
Transaction inquiry	Filter by status/date/origin, drill to voucher	reference, status, qty, dates, cost amounts	InventoryInquire
Reservation dialog	Reserve by policy or pick batches/serials	candidate batches with expiry, available qty	InventoryReserve
Count plan / count sheet	Generate, assign, count (mobile-friendly), approve, post	expected (hidden when blind), counted, variance	CycleCountMaintain
Blocking	Block / release with reason	product, dims, qty, reason, expected release date	InventoryBlock

Interaction rules: grid supports keyboard row entry (Tab across cells, Ctrl+Enter to add line), paste from spreadsheet with dimension column mapping, inline validation on blur for product/dimension and deferred validation (`:validate`) for on-hand and period checks; column sets and filters personalized per user (wave 5 shell). Count sheet works on the PWA shell with offline queue of counted quantities (synced on reconnect, conflict = latest count wins with audit).

3.16.7 Business rules and edge cases

- **BR-16.1** Status transitions are monotonic per transaction: issue `ON_ORDER(0) < RESERVED_ORDERED(1) < RESERVED_PHYSICAL(2) < PICKED(3) < DEDUCTED(4) < SOLD(5)`; receipt `ORDERED(0) < ARRIVED(1) < REGISTERED(2) < RECEIVED(3) < PURCHASED(4)`. Backward moves only through named commands that create a status-change audit row and an on-hand delta.
- **BR-16.2** Splitting: applying a step for a quantity less than the transaction quantity splits the transaction into two rows with the same origin (AX InventTrans split); sum of split quantities equals the original.
- **BR-16.3** Transfers create an issue (DEDUCTED) at the from-dimension and a receipt (RECEIVED) at the to-dimension in one database transaction; cost of the receipt equals the cost of the issue at the time of posting.
- **BR-16.4** Negative-policy check reads `avail_physical` under row lock; two concurrent issues against the last unit serialize, the second fails with `INV-NEG-001`. Deadlock avoidance: locks acquired in ascending (product_id, dimension_set_id) order within a journal.
- **BR-16.5** Rounding: quantities rounded half-up to the inventory unit's decimals; values to the accounting currency's minor unit (wave 12 rounding rules). Transfer of 1/3 of a 1.000-unit batch under 3 decimals yields 0.333 and leaves 0.667.

- **BR-16.6** Physical date may be in a closed inventory period only if financial date is in an open period and the item model group allows backdated physical posting; otherwise `INV-PER-001`.
- **BR-16.7** FEFO tie-break: equal expiry → earliest receipt_date → lowest batch_id (lexical). Batches with no expiry sort last.
- **BR-16.8** Count sheet posting posts only lines where counted_qty differs from expected_qty at the time of posting (not generation); movements after generation trigger a recount flag when the plan has `freeze=false`.
- **BR-16.9** Serial-tracked products reject quantities other than ±1 and reject a receipt of a serial already in physical stock for the same product (across all warehouses in the legal entity).
- **BR-16.10** Blocking never changes physical_invent; it moves quantity between inventory statuses so `physical_invent` is unchanged and `avail_physical` decreases.
- **BR-16.11** Journal state machine: `DRAFT → SUBMITTED → APPROVED → POSTING → POSTED`; `SUBMITTED → REJECTED → DRAFT`; `DRAFT|REJECTED → CANCELLED`. POSTING is a crash-safe intermediate state; the Temporal workflow owns it and is the only writer.
- **BR-16.12** Reversal of a posted journal creates a new journal of the same type with negated quantities, referencing the original; original rows are never modified.

3.16.8 Acceptance criteria

- **AC-16.1** (FR-16.4) Given item model group forbids physical negative and on-hand 1,000 units, when 10,000 concurrent issues of 1 unit are submitted, then exactly 1,000 succeed, 9,000 return `INV-NEG-001`, and on-hand is 0.
- **AC-16.2** (FR-16.3) Given 1,000,000 posted transactions, when the rebuild job runs, then the rebuilt OnHand table equals the live table row-for-row and the job completes in < 10 minutes.
- **AC-16.3** (FR-16.6) Given a MOVEMENT journal of 100 lines with offset account 500100, when posted, then one ledger entry exists with Σ debits = Σ credits, inventory account per posting profile, and `ledger_entry_id` set on all 100 transactions.
- **AC-16.4** (FR-16.7) Given 100 batches with distinct expiry dates and a FEFO reservation of 50 units, when reserved, then the batch with the earliest non-blocked expiry is selected first and the reservation spans batches in expiry order until 50 units are reached.
- **AC-16.5** (FR-16.5) Given a serial-tracked product with serial S-001 in stock, when a receipt of S-001 is posted, then it is rejected with `INV-SER-001`.
- **AC-16.6** (FR-16.9) Given a count plan with 5% qty tolerance, when a sheet line variance is 8%, then posting requires an approver with CycleCountApprove and the posted journal line equals counted – expected.
- **AC-16.7** (FR-16.11, FR-16.6) Given inventory period 2026-09 closed for module INVENTORY, when a journal with financial_date 2026-09-30 is posted, then it is rejected with the period-control error from wave 14 and no transaction is written.
- **AC-16.8** (FR-16.2) Given a receipt transaction at REGISTERED, when `Unregister` is called, then status returns to ARRIVED, registered bucket decreases, and a status-change audit row exists.
- **AC-16.9** (FR-16.12) Given a manual ledger journal posted to the inventory account without an inventory transaction, when the daily reconciliation runs, then a diff event is emitted with the exact amount.
- **AC-16.10** (FR-16.10) Given product unit EA with conversion 1 BOX = 12 EA, when a line of 2.5 BOX is posted, then qty = 30 EA, entered_qty = 2.5, entered_unit = BOX.

3.16.9 Test plan and fixtures

Fixtures: `inv-demo-plant` (200 products, 3 sites, 6 warehouses, 400 locations, 5,000 opening lines; generated); `inv-1m-txn` (1M synthetic transactions over 12 months, Zipf-distributed across 2,000 products; generated by `scripts/fixtures/inv_txn_gen.py`); `inv-fefo-100` (100 batches with expiry grid, 10 blocked); `inv-ax-extract` (InventSum/InventTrans/InventDim CSV sample from AX 2012 demo company USMF, 50k rows); `inv-sap-extract` (MARD/MCHB/MBEW/MSEG sample, 50k rows).

Test	Type	Target	Pass criterion
Status transition monotonicity	property (Go rapid)	ApplyTransactionStep	no sequence of commands produces a backward step without a named reversal command
On-hand formula	unit	OnHand compute	formulas in FR-16.3 hold for 10k random bucket vectors
Concurrency negative	E2E (k6)	journals:post	AC-16.1
Rebuild equality	performance	onhand:rebuild on inv-1m-txn	zero diff, < 10 min
Journal GL balance	contract	ledger Post	every journal type → balanced entry
FEFO selection	unit + E2E	reservations	AC-16.4
Outbox event schema	contract (AsyncAPI)	all 5 events	schema registry compatibility BACKWARD
Period control	contract	journals:post	AC-16.7
PWA offline count	E2E (Playwright)	count sheet	counts queued offline sync without loss

Reconciliation checks: Σ OnHand.posted_value per inventory account = ledger balance; Σ transaction qty (physical) = OnHand.physical_invent; count sheet variance total = COUNT journal total; AX migration: InventSum.PostedQty + Received + Registered – Deducted – Picked = Passage physical_invent per InventDimId.

3.16.10 Deployment and operations

Helm chart `passage-inventory` (Deployment ×3, HPA 3–12 on CPU 70%, Postgres CloudNativePG cluster `inventory-db` 3 instances, outbox Debezium connector `inventory-outbox`). Config: `INV_NEGATIVE_DEFAULT=forbid`, `INV_RESERVATION_TTL_HOURS=0`, `INV_JOURNAL_MAX_LINES=10000`, `INV_RECON_CRON=0 2 * * *`. Migrations via Atlas from entity metadata (`inventory/v16.*`). Feature flags: `inv.fefo_policy`, `inv.count_pwa`, `inv.transfer_cost_diff` (off until w20). Scheduled jobs (Temporal cron): daily reconciliation 02:00 tenant local, count sheet generation per plan cron, weekly rebuild-and-compare Sunday 03:00. SLOs: journal post p95 < 2 s for 1,000 lines; on-hand query p95 < 50 ms; event lag p95 < 5 s. Alerts: reconciliation diff ≠ 0 (page), outbox lag > 60 s (ticket), negative on-hand where forbidden (page, indicates bug). Runbook: rebuild projection, replay outbox, reverse journal. Rollback: Argo CD digest rollback; migrations are additive in this wave (no destructive change), so downgrade is safe.

3.16.11 Migration from AX / SAP

Source	Passage entity	Notes
AX InventSum (per InventDimId)	OPENING_BALANCE journal lines → InventoryTransaction + OnHand	PostedQty+Received+Registered−Deducted−Picked → qty; PostedValue+PhysicalValue → cost; reserved/ordered buckets not migrated (open orders re-created in w17/w27)
AX InventDim	DimensionSet	map InventSiteId→site, InventLocationId→warehouse, wMSLocationId→location, InventBatchId→batch, InventSerialId→serial, InventStatusId→inventory_status
AX InventTrans (closed history)	InventoryTransaction (read-only history, flag `migrated=true`)	optional; last 24 months; StatusIssue/StatusReceipt mapped 1:1
AX InventBatch, InventSerial	Batch, Serial (wave 8 tracking entities)	expiry, shelf-life advice
AX InventModelGroup	ItemModelGroup (wave 8)	already migrated; verified here
SAP MARD (LABST unrestricted, INSME quality, SPEME blocked)	OpeningBalance per plant/storage location with inventory_status AVAILABLE / QUARANTINE / BLOCKED	
SAP MCHB (CLABS per batch)	OpeningBalance with batch dim	batch master from MCH1/MCHA
SAP MBEW (LBKUM, SALK3, VERPR/STPRS)	cost on opening lines	value = SALK3 split proportionally by qty
SAP MSEG/MKPF history	InventoryTransaction history	BWART → status mapping table (101, 102, 261, 262, 301, 311, 551, 561)

Transformation notes: dimension hash computed after mapping; quantities converted to Passage inventory unit via MARM/UnitOfMeasureConversion; opening journal posts with financial_date = cutover date − 1 and offset account = migration clearing account. Validation: Σ migrated qty per product/site equals source extract; Σ value equals source within 0.01 × product count; no serial duplicates. Not migrated: InventSum reservation buckets (recreated by open documents), AX quarantine orders (re-entered manually; typically < 50), SAP special stock K/O (consignment/subcontracting; wave 21/41), InventClosing history (wave 20 decides).

3.16.12 Dependencies, risks, open questions

Depends on: wave 6 (sites, warehouses, UoM), wave 8 (products, dimension groups, item model groups, batch/serial masters), wave 11 (`Post(JournalEntry)`, posting profiles), wave 12 (currency rounding), wave 14 (period control module INVENTORY), wave 4 (Temporal, outbox), wave 10 (migration framework).

Risk	Likelihood	Impact	Mitigation
Row-lock contention on hot SKUs (one dimension set)	Medium	Medium	Lock per OnHand row only; batch issues in a journal sorted; measured by k6 at 10k concurrent
Dimension hash collision semantics across services	Low	High	Shared package with golden test vectors; hash includes dimension names
Count PWA offline conflicts	Medium	Low	Last-write-wins with audit; recount flag

Open questions:

1. Store license plate as a dimension now or add in wave 31? Default: reserve the column, null until wave 31. Decide by wave 17 close.
2. Should RESERVED_ORDERED be supported in wave 16 or only when sales/production demand exists? Default: implement now with manual demand type. Decide by wave 16 mid-point.
3. Monthly vs yearly partitioning threshold? Default: monthly when > 20M rows. Decide by wave 20.

3.17 Wave 17: Purchasing: Requisitions, RFQs, Agreements & Purchase Orders

3.17.1 Scope and business context

Wave 17 gives buyers the procurement document chain: purchase requisition (AX PurchReqTable/PurchReqLine, SAP EBAN), request for quotation with supplier replies (PurchRFQCaseTable/PurchRFQTable/PurchRFQReplyTable, SAP EKKO document type AN with quotation maintenance ME47), purchase agreements with release tracking (PurchAgreementHeader/PurchAgreementLine, SAP contracts type MK/WK and scheduling agreements LP via ME31K/ME31L), vendor trade agreements (PriceDiscTable journal type PriceDiscAdmTrans relation PurchPrice/PurchLineDisc, SAP condition records A017/KONP via MEK1), and purchase orders with versioned change management (PurchTable/PurchLine with PurchTableHistory and change-management parameters, SAP EKKO/EKPO with change documents CDHDR/CDPOS and version management in EREV). Purchase order confirmation writes ON_ORDER/ORDERED inventory transactions from wave 16.

At the end of the wave a requester can raise a requisition that routes by amount band and category to the right approvers, a buyer can consolidate approved requisition lines into POs, run an RFQ to several suppliers and award, create blanket agreements that POs release against, and send a PO as PDF/email with an EDI 850/ORDERS hook stubbed for wave 35. Out of scope: receiving (wave 18), invoice (wave 18), budget control on commitments (wave 38), subcontracting POs from production (wave 21), landed cost (wave 34), supplier portal acknowledgement (wave 20), catalog punch-out (v1.x).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX purchase order confirmation runs PurchFormLetter with totals recalculation and PDF through SSRS; 20–90 s per PO on a loaded AOS; ME21N saves fast but output (NEU message, ME9F) depends on a scheduled RSNAST00 job	Synchronous document class chain plus report server; SAP output via NAST condition technique and batch	Confirmation is a state transition that enqueues rendering (Temporal activity) and emits `po.confirmed`; PDF rendered by a stateless worker within 3 s	Confirm latency: 20–90 s → < 2 s UI, PDF < 5 s; no batch wait
Price lookup in AX (PriceDisc class, PriceDiscTable search over 3 account codes × 3 item codes) is opaque; buyers cannot see why a price was chosen; SAP condition analysis (ME23N > Conditions > Analysis) exists but is 6 clicks deep	No explain output; access sequence hidden	Every PO line carries `price_source` (agreement id, trade agreement id, last price, manual) with an "explain" popover; `GET /pricing/explain`	Clicks to see why a price applied: 6–8 → 1; price disputes per 1,000 POs: tracked, target −80%

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Change management in AX creates a full PurchTableHistory copy only when the parameter is on; comparing versions is a manual form; SAP version management requires EREV config and a reason per change	Version stored as whole-document copies without diff	Versions are stored as JSON-patch diffs against the previous version with a rendered side-by-side diff; re-approval rule evaluated per diff	Time to identify what changed between PO versions: 5 min → 10 s
Requisition approval in AX workflow (PurchReqReview) evaluates in the batch workflow processor every 60 s; escalation and delegation are per-user configuration pages; SAP release strategy (CL20N/CL24N classification) is nearly impossible to change without a consultant	Batch-polled workflow runtime; classification-based release strategy	Temporal approval workflow started synchronously on submit; policy is a declarative matrix (amount band × category × org) editable by a procurement admin	Submit-to-first-approver latency: 60–120 s → < 2 s; policy change lead time: days (consultant) → minutes (admin)
PO total in AX recomputed by PurchTotals class on demand; header/line totals can disagree until "Totals" is opened; SAP EKKO has no stored total	Totals derived lazily	Totals persisted on every line change with a database constraint test `header.total = Σ lines + charges + tax`	Total mismatches: non-zero → 0 across 1,000 random POs

3.17.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Purchase requisition	PurchReqTable, PurchReqLine; forms PurchReqTable, PurchReqReview workflow; policies (PurchReqControlRule)	EBAN/EBKN; ME51N/ME52N/ME53N; release strategy ME54N; ME57 assign source	`procurement.PurchaseRequisition`, `RequisitionLine`; Requisition workspace; policy matrix	full
Approval policy	Purchasing policies (PurchPolicy), amount bands via workflow conditions, category access policy	Release strategy (release groups/codes, classification characteristics CEBAN)	`ApprovalPolicy` (bands, category, org unit, requester role) compiled to Temporal workflow input	improved
Consolidation into PO	PurchReqConsolidation (consolidation opportunities), release approved lines (PurchReqPurchaseOrderCreate)	ME57/ME58 (assign and process), automatic PO creation ME59N	`POST /requisitions:consolidate` grouping by vendor × currency × site × delivery date window	full
RFQ	PurchRFQCaseTable, PurchRFQTable (per vendor), PurchRFQReplyTable, PurchRFQCompare; vendor collaboration portal reply (D365)	RFQ ME41/ME42/ME43 (EKKO BSTYP A), quotation ME47, price comparison ME49, rejection ME9A	`RFQ`, `RFQInvitation`, `RFQReply`; compare screen; award creates PO or agreement	full
Purchase agreement	PurchAgreementHeader, PurchAgreementLine, AgreementLineReleasedLine (release tracking); commitment types (product qty/value, category value, value)	Contracts (EKKO BSTYP K, ME31K/ME32K/ME33K), release orders, scheduling agreements (L, ME31L) with delivery schedules ME38	`PurchaseAgreement`, `AgreementLine` (commitment types QUANTITY, VALUE, CATEGORY_VALUE, TOTAL_VALUE); release tracking	full (scheduling agreement delivery schedules partial until w22 MRP)
Vendor pricing	Trade agreements PriceDiscTable (PurchPrice, PurchLineDisc, PurchMultiLnDisc, PurchEndDisc), account/item code hierarchy, date validity, quantity breaks	Conditions A017 (material info record), A018, A019, A025; KONP; info record EINA/EINE; MEK1	`TradeAgreement` (vendor > vendor group > all; product > product group > all; qty breaks; validity; currency) with explain	full
Purchase order	PurchTable, PurchLine, PurchTableHistory/PurchLineHistory (versions), PurchOrderStatus, DocumentStatus, PurchFormLetter_PurchOrder (confirm)	EKKO/EKPO/EKET (schedule lines)/EKKN (account assignment), ME21N/ME22N/ME23N, EREV versions	`PurchaseOrder`, `PurchaseOrderLine`, `DeliverySchedule`, `PurchaseOrderVersion`	full
Charges	MarkupTrans (misc. charges header/line), MarkupTable, auto charges (MarkupAutoTable)	Pricing conditions FRA1 (freight), ZOA1 (customs), header/item conditions	`Charge` (header/line; fixed, per unit, percent; debit/credit to item, vendor, ledger)	full
Delivery schedules	PurchLine delivery schedule (multiple delivery lines), confirmed delivery date	EKET schedule lines	`DeliverySchedule` rows per PO line	full
PO output	Print management (PrintMgmt), SSRS PurchPurchaseOrder report, email, AIF service PurchPurchaseOrderService	Message output NAST, forms MEDRUCK, IDoc ORDERS05	PDF via render worker, email via wave 4 notification service; EDI hook emits `po.confirmed` for wave 35	partial (EDI transport in w35)
Category hierarchy procurement	EcoResCategory procurement hierarchy, category policy	Material groups (T023), product categories	Wave 8 category hierarchy with procurement purpose	full
Vendor on-hold / status	VendTable Blocked (invoice/payment/all)	LFA1 SPERR/LOEVM, purchasing block LFM1 SPERM	Vendor role status (wave 7) checked on PO create/confirm	full

3.17.3 Functional requirements

- **FR-17.1 Purchase requisition** — Must. The system shall let any user with RequisitionCreate raise a requisition with lines for catalog products, non-catalog items (description + category) or services, with requested date, quantity, unit, estimated price, site/warehouse, financial dimensions and attachments.
    - Statuses: DRAFT, IN_REVIEW, APPROVED, REJECTED, CLOSED (all lines consolidated), CANCELLED.
    - Estimated price defaults from TradeAgreement lookup for the preferred vendor, else last PO price, else zero with warning.
    - Maximum 500 lines; account structure validated at submit using wave 11 rules.
- **FR-17.2 Approval policy matrix** — Must. The system shall evaluate an `ApprovalPolicy` with rules (amount band in accounting currency, category, operating unit, requester role) producing an ordered approver chain, executed by the wave 4 Temporal approval template with 48 h SLA timers, escalation and delegation.
    - Bands default: 0–1,000 USD auto-approve, 1,000–10,000 manager, 10,000–100,000 manager + category owner, > 100,000 + CFO.
    - Policy versions are effective-dated; the version at submit time governs the whole instance.
    - Any line edit after submit restarts the workflow.

- **FR-17.3 Consolidation into purchase orders** — Must. The system shall group approved requisition lines into POs by vendor × currency × site × delivery window (default 7 days) with buyer override of vendor and the ability to split lines.
    - Lines with no vendor route to a "Needs sourcing" queue from which an RFQ can be started.
    - Consolidated lines keep `requisition_line_id` on the PO line for commitment tracking.
- **FR-17.4 RFQ cycle** — Must. The system shall create an RFQ with lines, invite N vendors, capture replies (price, lead time, quantity breaks, validity, notes, attachments) manually or via the supplier portal (wave 20), compare, and award whole or per line to create a PO or a purchase agreement.
    - Statuses: DRAFT, SENT, REPLIES_OPEN, CLOSED, AWARDED, CANCELLED; reply deadline enforced (late replies flagged, accepted only with override).
    - Compare view ranks by total landed estimate = price × qty + charges; ranking rule configurable (lowest price, shortest lead time, weighted score).
- **FR-17.5 Purchase agreements** — Must. The system shall support blanket agreements with commitment types QUANTITY (product × qty), VALUE (product × amount), CATEGORY_VALUE and TOTAL_VALUE, validity dates, price/discount terms, max enforcement (hard/soft) and release tracking.
    - A PO line referencing an agreement line draws down `released_qty` / `released_amount`; exceeding a hard limit rejects with `PRC-AGR-003`.
    - Agreement prices override trade agreements when `price_override=true`.
- **FR-17.6 Vendor pricing (trade agreements)** — Must. The system shall resolve unit price and line discount from `TradeAgreement` rows by most-specific match: vendor > vendor group > all vendors, product > product group > all products, with currency, unit, quantity break (from_qty) and validity (from/to dates).
    - Deterministic ordering: specificity score (vendor 4/group 2/all 0 + product 4/group 2/all 0), then latest from_date, then lowest price when still tied (configurable `tie_rule`).
    - Returns `explain` listing considered rows and the winner.
    - Lookup p95 < 20 ms (aligns with wave 26 pricing SLO; wave 26 replaces this implementation with the condition engine while keeping the API).
- **FR-17.7 Purchase order lifecycle** — Must. The system shall manage POs with approval status (DRAFT, IN_REVIEW, APPROVED, REJECTED, CONFIRMED, FINALIZED, CANCELLED) and document status (NONE, RECEIVED_PARTIAL, RECEIVED, INVOICED_PARTIAL, INVOICED) derived from downstream documents.
    - Confirmation requires approval status APPROVED and vendor not blocked; on confirm the service calls inventory `transactions:apply` to create ORDERED receipts (one per delivery schedule line) and renders output.
    - Lines carry product or category, qty, unit, unit price, discount %, line amount, delivery date, confirmed delivery date, site, warehouse, financial dimensions, tax group/item tax group (wave 13), agreement line, requisition line.
- **FR-17.8 Versioned change management** — Must. The system shall snapshot a `PurchaseOrderVersion` at each confirmation and, when change management is enabled for the vendor or legal entity, require re-approval of changes after confirmation according to the policy's `reapproval_rules` (any change / price or qty increase > X % / new line / delivery date change).
    - Version numbers monotonic from 1; diff stored as JSON Patch (RFC 6902); version document rendered and sent to vendor marked "Change order v{n}".
    - Received or invoiced quantities cannot be reduced below received/invoiced totals (`PRC-PO-007`).
- **FR-17.9 Charges** — Must. The system shall support header and line charges with calculation (FIXED, PER_UNIT, PERCENT), debit/credit sides (ITEM = capitalize into inventory cost; LEDGER = expense account; VENDOR = payable to a different vendor), and auto-charges by vendor/product group.
    - Header charges allocate to lines at invoice time by quantity, amount or line count (wave 18 consumes).
- **FR-17.10 Delivery schedules** — Must. The system shall allow one PO line to have multiple delivery schedule rows (qty, date, confirmed date) summing to line qty, each producing its own ORDERED inventory transaction.
- **FR-17.11 Totals and tax** — Must. The system shall persist header totals (`subtotal`, `discount_total`, `charges_total`, `tax_total`, `total`) recalculated on every line/charge change, calling the wave 13 tax engine, with `total = subtotal − discount_total + charges_total + tax_total` to the cent.
- **FR-17.12 PO output** — Must. The system shall render the PO/change order to PDF (template per legal entity, wave 15 document designer basics), email it to the vendor's purchasing contact, and emit `po.confirmed` for EDI/portal consumers; rendering failures do not block confirmation (retry with backoff, alert after 3 failures).
- **FR-17.13 Purchasing policies** — Should. The system shall provide entity-level policy settings: change management on/off, requisition required for PO creation (yes/no per category), PO required before invoice (consumed by wave 18), reapproval rules, consolidation window.
- **FR-17.14 Buyer workspace** — Should. The system shall show the buyer the queue of approved requisition lines, RFQs awaiting replies, POs awaiting confirmation, agreements near expiry (< 30 days) or near limit (> 90 %).

3.17.4 Data model

PurchaseRequisition / RequisitionLine

Field	Type	Required	Notes
id, requisition_number	uuid, text	yes	sequence `PR-`
requester_id, on_behalf_of_id	uuid	yes/no	wave 7 worker/user
legal_entity_id, operating_unit_id	uuid	yes	
status	enum	yes	DRAFT, IN_REVIEW, APPROVED, REJECTED, CLOSED, CANCELLED
policy_version_id	uuid	cond.	set at submit
total_estimated (acct ccy)	numeric(19,4)	yes	recomputed
line: line_no, product_id or (description, category_id), qty, unit, estimated_unit_price, currency, requested_date, site_id, warehouse_id, preferred_vendor_id, dimension_default_id, agreement_line_id, status (OPEN, CONSOLIDATED, CANCELLED), po_line_id			

PurchaseOrder / PurchaseOrderLine / DeliverySchedule / PurchaseOrderVersion

Field	Type	Required	Notes
id, po_number	uuid, text	yes	sequence `PO-`, gap-free optional
vendor_id, invoice_vendor_id	uuid	yes	wave 7 vendor role; invoice vendor defaults to vendor
legal_entity_id, site_id, warehouse_id	uuid	yes	
currency, exchange_rate_type	text	yes	wave 12
approval_status	enum	yes	DRAFT, IN_REVIEW, APPROVED, REJECTED, CONFIRMED, FINALIZED, CANCELLED
document_status	enum	yes	derived (FR-17.7)
version_no	int	yes	current confirmed version; 0 before first confirmation
change_management	bool	yes	copied from policy at creation
payment_terms_id, delivery_terms_id, mode_of_delivery	uuid/text	yes/no	wave 7
subtotal, discount_total, charges_total, tax_total, total	numeric(19,4)	yes	transaction currency
line: line_no, product_id or category_id, description, qty, unit, unit_price, price_unit (per 1/100/1000), discount_pct, discount_amount, line_amount, delivery_date, confirmed_delivery_date, site_id, warehouse_id, dimension_set_defaults, tax_group_id, item_tax_group_id, agreement_line_id, requisition_line_id, price_source, received_qty, invoiced_qty, remaining_qty, status (OPEN, RECEIVED, INVOICED, CANCELLED)			
schedule: po_line_id, seq, qty, delivery_date, confirmed_date, inventory_transaction_id			Σ qty = line qty
version: po_id, version_no, confirmed_at, confirmed_by, snapshot (jsonb), patch (jsonb RFC 6902), reapproval_required (bool)			

PurchaseAgreement / AgreementLine: id, agreement_number, vendor_id, legal_entity_id, currency, valid_from, valid_to, status (DRAFT, EFFECTIVE, ON_HOLD, CLOSED), default_commitment_type; line: commitment_type, product_id/category_id, committed_qty/committed_amount, unit, unit_price, discount_pct, max_enforced (HARD/SOFT/NONE), price_override (bool), released_qty, released_amount, remaining.

TradeAgreement: id, relation (PRICE, LINE_DISC, MULTILINE_DISC, TOTAL_DISC), account_code (VENDOR/GROUP/ALL), account_id, item_code (PRODUCT/GROUP/ALL), item_id, currency, unit, from_qty, to_qty, amount or percent, price_unit, valid_from, valid_to, site_id (optional), lead_time_days, status (DRAFT, ACTIVE, EXPIRED).

RFQ / RFQInvitation / RFQReply: rfq id, number, status, reply_deadline, lines (product/category, qty, unit, required_date); invitation (vendor_id, sent_at, channel EMAIL/PORTAL, status INVITED, DECLINED, REPLIED); reply (invitation_id, line_id, unit_price, currency, lead_time_days, qty_breaks jsonb, valid_until, notes, attachments, awarded (bool)).

Charge: id, owner_type (PO_HEADER, PO_LINE, RECEIPT, INVOICE), owner_id, charge_code_id, calc (FIXED, PER_UNIT, PERCENT), value, amount, currency, debit_type (ITEM, LEDGER, VENDOR), credit_type, debit_account_id, credit_account_id, allocation_rule.

Invariants: `po.total = Σ line_amount − discount + charges + tax` enforced by a deferred constraint trigger; Σ schedule qty = line qty; `received_qty ≤ qty` and `invoiced_qty ≤ max(received_qty, qty)`; agreement `released ≤ committed` when HARD; exactly one ACTIVE TradeAgreement wins per lookup (ties resolved by `tie_rule`). Indexes: PO (tenant, vendor, approval_status), line (po_id, line_no), schedule (po_line_id), TradeAgreement (tenant, relation, account_code, account_id, item_code, item_id, valid_from) for lookup; requisitions by requester and status.

3.17.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/procurement/requisitions	Create requisition	client key	RequisitionCreate
POST	/api/v1/procurement/requisitions/{id}:submit	Submit for approval	req id + version	RequisitionCreate
POST	/api/v1/procurement/requisitions:consolidate	Create POs from approved lines	line id set hash	PurchaseOrderMaintain
POST	/api/v1/procurement/rfqs	Create RFQ	client key	RFQMaintain
POST	/api/v1/procurement/rfqs/{id}:send	Send invitations	rfq id	RFQMaintain
POST	/api/v1/procurement/rfqs/{id}/replies	Record a reply	invitation+line	RFQMaintain / portal vendor
POST	/api/v1/procurement/rfqs/{id}:award	Award to vendor(s)	rfq id	RFQAward
POST	/api/v1/procurement/agreements	Create agreement	client key	AgreementMaintain

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/procurement/pricing/lookup?vendor=&product=&qty=&date=&currency=	Resolve price/discount with explain	n/a	PurchaseOrderMaintain
POST	/api/v1/procurement/purchase-orders	Create PO	client key	PurchaseOrderMaintain
PATCH	/api/v1/procurement/purchase-orders/{id}	Edit header/lines (creates pending change)	ETag	PurchaseOrderMaintain
POST	/api/v1/procurement/purchase-orders/{id}:confirm	Confirm (new version)	po id + version_no+1	PurchaseOrderConfirm
POST	/api/v1/procurement/purchase-orders/{id}:cancel	Cancel remaining	po id	PurchaseOrderMaintain
GET	/api/v1/procurement/purchase-orders/{id}/versions/{n}/diff	JSON Patch and rendered diff	n/a	PurchaseOrderInquire
GET	/api/v1/procurement/purchase-orders/{id}/document.pdf	Rendered PO	n/a	PurchaseOrderInquire

Commands: `SubmitRequisition` (account structure valid, lines ≥ 1, no closed category), `ConsolidateLines` (same vendor/currency/site/entity, vendor not blocked), `AwardRFQ` (reply not expired, award qty ≤ RFQ qty), `ConfirmPO` (approved, vendor active, agreement limits, delivery schedules sum, tax calculated, totals constraint), `ApplyChange` (post-confirmation changes evaluated against reapproval rules; sets approval_status back to IN_REVIEW if required).

Event	Trigger	Payload fields	Consumers
`dev.passage.procurement.requisition.submitted.v1` / `.approved.v1` / `.rejected.v1`	workflow transitions	requisition_id, lines (product, qty, amount), total, approver chain, policy_version	notifications, budget (w38), analytics
`dev.passage.procurement.rfq.sent.v1` / `.awarded.v1`	RFQ send/award	rfq_id, vendors, lines, awarded vendor per line	supplier-portal (w20), notifications
`dev.passage.procurement.po.confirmed.v1`	confirm	po_id, version_no, vendor_id, lines (product, qty, unit, price, delivery dates, site/warehouse), totals, output channel	inventory, receiving (w18), supplier-portal, EDI (w35), analytics
`dev.passage.procurement.po.version.created.v1`	any confirmed version > 1	po_id, version_no, patch, reapproval_required	supplier-portal, receiving
`dev.passage.procurement.po.cancelled.v1`	cancel	po_id, lines cancelled, remaining qty	inventory (cancel ORDERED), supplier-portal
`dev.passage.procurement.agreement.threshold.v1`	released ≥ 90 % or expiry < 30 d	agreement_id, line, pct, expiry	notifications

3.17.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Procurement workspace (buyer)	Tiles and lists: lines to source, RFQs open, POs to confirm, agreements expiring, change orders pending re-approval	counts, aging	PurchaseOrderInquire
Requisition (requester, self-service)	Add product from catalog search, non-catalog line, submit, track approval	product, qty, price est., date, dimensions, approval timeline	RequisitionCreate
Approval task (shell inbox)	Approve, reject, delegate, request change, comment	requisition summary, lines, policy band hit, attachments	RequisitionApprove
Consolidation board	Group suggestions, drag lines between POs, change vendor, create POs	vendor, currency, site, lines, estimated total	PurchaseOrderMaintain
RFQ	Create, invite vendors, record replies, compare, award	lines, invitations, replies grid (price, lead time, validity), ranking	RFQMaintain
Purchase agreement	Create, add lines, view releases	commitment type, committed, released, remaining %, validity	AgreementMaintain
Purchase order	Create, add lines (price explain), charges, delivery schedules, confirm, view versions diff, print/email	header, lines grid, totals panel live, version timeline	PurchaseOrderMaintain
Trade agreement journal	Create rows in grid, validate overlap, post (activate)	relation, account/item codes, qty break, price, validity	TradeAgreementMaintain

Interaction rules: line grid keyboard entry with product search-as-you-type (p95 < 100 ms via wave 8 search index); price lookup fires on product/qty/vendor/date change and shows source badge; totals panel updates within 200 ms of any edit (server computed, optimistic client); bulk edit on selected lines (delivery date, site, dimensions); validation timing: inline for required/format, on `:confirm` for agreement limits and tax; approvers can act from email deep links with Keycloak session; personalization: saved views per buyer.

3.17.7 Business rules and edge cases

- **BR-17.1** Requisition state machine: `DRAFT → IN_REVIEW → APPROVED → CLOSED`; `IN_REVIEW → REJECTED → DRAFT`; `DRAFT|REJECTED → CANCELLED`; `APPROVED` lines may be individually CANCELLED by the requester before consolidation.
- **BR-17.2** PO approval state machine: `DRAFT → IN_REVIEW → APPROVED → CONFIRMED`; `CONFIRMED –(change requiring re-approval)→ IN_REVIEW`; `CONFIRMED –(change not requiring)→ CONFIRMED (version+1)`; `CONFIRMED → FINALIZED` when document_status = INVOICED and remaining = 0; any non-finalized → CANCELLED if received_qty = 0 and invoiced_qty = 0.
- **BR-17.3** Price resolution date is the PO line creation date by default, switchable to delivery date per policy; a trade agreement with `valid_to` null is open-ended; overlapping rows with equal specificity and dates are rejected at activation (`PRC-TA-002`).
- **BR-17.4** Quantity breaks use `from_qty ≤ qty < to_qty` in the agreement's unit after conversion; conversion failure falls back to the next less-specific row and records the fallback in explain.
- **BR-17.5** Rounding: unit price stored to 4 decimals of currency minor units (e.g., 12.3456 USD per price_unit); line_amount = round_half_even(qty × unit_price / price_unit × (1 – discount_pct/100), minor units); header totals are sums of rounded line amounts (never re-rounded), matching AX PurchTotals and SAP KOMP netwr behavior.

- **BR-17.6** Confirmation in a different fiscal period than creation is allowed; commitment date for budget (wave 38) is the confirmation date.
- **BR-17.7** Reducing PO line qty below `received_qty` is rejected; reducing below `invoiced_qty` but above `received_qty` is allowed only when the policy flag `allow_invoice_before_receipt` is true.
- **BR-17.8** Concurrency: PO edits use optimistic concurrency via ETag (`row_version`); a stale PATCH returns 412 with the current version and diff.
- **BR-17.9** Deleting a line on a confirmed PO cancels its remaining ORDERED inventory transactions; the line remains with status CANCELLED in versions.
- **BR-17.10** Agreement validity: a PO line may reference an agreement line only if the PO's requested delivery date falls within `valid_from..valid_to`; SOFT limit overrun emits a warning and `agreement.threshold`; HARD rejects.
- **BR-17.11** RFQ replies received after `reply_deadline` are stored with `late=true` and excluded from ranking unless the buyer overrides with a reason.
- **BR-17.12** Vendor currency differing from PO currency: PO currency defaults from the vendor's purchase currency; overriding requires an exchange rate available on the PO date (wave 12), else `PRC-FX-001`.

3.17.8 Acceptance criteria

- **AC-17.1** (FR-17.2) Given the default band policy, when 30 requisitions spanning 6 amount bands × 5 categories are submitted, then each routes to the expected approver chain per the test matrix in `fixtures/procurement/approval-matrix.csv`.
- **AC-17.2** (FR-17.8) Given a confirmed PO with change management on and rule "qty increase > 5 % requires re-approval", when a line qty is raised from 100 to 110, then version 2 is created with a JSON Patch on `/lines/0/qty`, approval_status = IN_REVIEW, and the vendor is not notified until re-confirmation.
- **AC-17.3** (FR-17.8) Given the same PO, when the delivery date moves by one day and the rule excludes dates, then version 2 is created with approval_status CONFIRMED and the change order PDF is emailed.
- **AC-17.4** (FR-17.6) Given rows for vendor V (price 10), vendor group G containing V (price 9), and all vendors (price 8), all valid, when lookup for V runs, then price 10 is returned and explain lists all three rows (property test over random overlapping sets: most specific valid row always wins).
- **AC-17.5** (FR-17.11) Given 1,000 random POs with 1–50 lines, charges and tax, when totals are read, then `total = subtotal − discount_total + charges_total + tax_total` to the cent for all 1,000.
- **AC-17.6** (FR-17.7) Given an approved PO with 2 delivery schedule rows, when confirmed, then 2 inventory transactions at ORDERED exist with matching quantities and dates and on-hand `ordered` increases accordingly.
- **AC-17.7** (FR-17.5) Given a QUANTITY agreement line of 1,000 EA with HARD enforcement and 950 released, when a PO line of 100 EA references it, then confirmation fails with `PRC-AGR-003`.
- **AC-17.8** (FR-17.4) Given an RFQ sent to 3 vendors with 2 replies before deadline and 1 after, when compared, then the late reply is excluded and awarding to the lowest creates a PO with the reply price and lead time.
- **AC-17.9** (FR-17.3) Given 10 approved requisition lines across 2 vendors and 2 sites, when consolidated with the default window, then 4 POs are created and each requisition line references its PO line.
- **AC-17.10** (FR-17.12) Given the PDF renderer is down, when a PO is confirmed, then confirmation succeeds, `po.confirmed` is emitted, and the render activity retries 3 times before alerting.

3.17.9 Test plan and fixtures

Fixtures: `proc-demo-vendors` (40 vendors with groups, payment terms, currencies USD/EUR; 300 trade agreement rows); `proc-approval-matrix` (30 requisitions, expected chains); `proc-po-1000` (1,000 random POs generated with seed for totals test); `proc-rfq-cases` (12 RFQs); `proc-ax-extract` (PurchTable/PurchLine/PurchAgreementHeader/PriceDiscTable from USMF); `proc-sap-extract` (EKKO/EKPO/EKET/KONP/A017 samples).

Test	Type	Target	Pass criterion
Price specificity	property	pricing/lookup	most-specific valid row wins over 10k random sets
Totals identity	property + E2E	PO totals	AC-17.5
Approval routing	E2E (Temporal test server)	requisition workflow	AC-17.1
Version diff	unit	JSON Patch generation/apply	apply(patch, v_n) = v_{n+1} for all fixtures
Confirm → inventory	contract	inventory `transactions:apply`	AC-17.6
Agreement limits	unit	ConfirmPO	AC-17.7
PDF render	E2E	render worker	PDF/A-2b valid, < 5 s
Lookup latency	performance	pricing/lookup at 500 rps	p95 < 20 ms
Event schemas	contract	6 events	AsyncAPI BACKWARD compatible

Reconciliation checks: Σ requisition lines CONSOLIDATED = Σ PO lines with requisition_line_id; Σ agreement released = Σ confirmed PO line qty/amount referencing it; Σ inventory ORDERED qty per PO line = remaining open schedule qty.

3.17.10 Deployment and operations

Helm chart `passage-procurement` (API Deployment ×3, render worker ×2, Temporal worker ×2, CloudNativePG `procurement-db`, outbox connector). Config: `PRC_CONSOLIDATION_WINDOW_DAYS=7`, `PRC_CHANGE_MGMT_DEFAULT=true`, `PRC_RFQ_LATE_POLICY=flag`, `PRC_PDF_TIMEOUT_S=20`. Migrations `procurement/v17.*`. Feature flags: `prc.rfq_portal_replies` (on in w20), `prc.edi_output` (w35), `prc.delivery_date_pricing`. Jobs: agreement threshold scan daily 06:00; RFQ deadline closer hourly; approval SLA escalation (Temporal timers). SLOs: PO create/confirm API p95 < 500 ms; requisition submit to workflow start < 2 s; PDF < 5 s p95. Alerts: workflow task backlog > 100 for 10 min; render failures > 5/h; pricing lookup p95 > 50 ms. Runbook: re-render document, restart stuck approval (Temporal signal), re-emit `po.confirmed`. Rollback: additive migrations; flag off `prc.change_mgmt` if version logic misbehaves (POs remain confirmable without versions).

3.17.11 Migration from AX / SAP

Source	Passage entity	Notes
AX PurchTable/PurchLine (open: PurchStatus Backorder, open qty > 0)	PurchaseOrder/Line version 1, approval_status CONFIRMED	remaining qty = PurchQty – received; received history not migrated (opening on-hand covers stock)
AX PurchAgreementHeader/Line, AgreementLineReleasedLine	PurchaseAgreement/Line with released totals	released computed from source releases
AX PriceDiscTable (relation PurchPrice, PurchLineDisc)	TradeAgreement	AccountCode/ItemCode map 1:1; QuantityAmountFrom/To → from_qty/to_qty
AX PurchReqTable approved-not-consolidated	PurchaseRequisition APPROVED	in-review requisitions re-submitted after cutover
AX PurchRFQ*	not migrated	open RFQs re-issued
SAP EKKO/EKPO/EKET (BSTYP F, open per EKPO ELIKZ ≠ X)	PurchaseOrder/Line/DeliverySchedule	NETPR/PEINH → unit_price/price_unit; MENGE – delivered
SAP EKKO BSTYP K/L, EKAB release docs	PurchaseAgreement/Line	KTMNG target qty, KTWRT target value
SAP A017/A018 + KONP (PB00, RA01)	TradeAgreement	KBETR/KPEIN; KONM scales → qty breaks
SAP EBAN open (BANPR approved)	PurchaseRequisition	

Validation: count and Σ open amount per vendor equal source; sample 50 POs compared field-by-field; price lookup for 200 random vendor/product pairs matches source price. Not migrated: closed POs (reporting only, in lakehouse wave 46), PurchTableHistory versions, SAP change documents, NAST output logs.

3.17.12 Dependencies, risks, open questions

Depends on: wave 16 (ORDERED transactions, on-hand), wave 7 (vendors, groups, terms, blocking), wave 8 (products, categories, search), wave 11–13 (dimensions, currency, tax), wave 4 (approval workflow, notifications), wave 5 (shell inbox, document templates basics).

Risk	Likelihood	Impact	Mitigation
Wave 26 pricing engine replaces trade agreement lookup; behavior drift	Medium	Medium	Keep `pricing/lookup` contract and fixture set; wave 26 must pass wave 17 tests
Re-approval rule complexity grows	High	Low	Rules are a small DSL with fixtures; defaults cover 90 %
PDF template per legal entity before document designer (w15 basics only)	Medium	Low	One default template plus logo/address substitution

Open questions:

1. Should requisitions support punch-out catalogs in v1? Default: no, v1.x. Decide by wave 20.
2. Gap-free PO numbering per legal entity by default? Default: off (sequence with gaps allowed); on for countries requiring it. Decide by wave 18.
3. Scheduling-agreement delivery schedules from MRP: wave 22 or wave 17 manual only? Default: manual only now. Decide by wave 21.

3.18 Wave 18: Receiving, Product Receipts & Three-Way Match

3.18.1 Scope and business context

Wave 18 closes the physical and financial loop on a purchase order. Receiving moves a PO line's inventory transaction from ORDERED through ARRIVED (arrival journal, SAP inbound delivery VL31N), REGISTERED (registration, SAP GR to quality/blocked stock) to RECEIVED by posting a product receipt (AX VendPackingSlipJour/VendPackingSlipTrans, SAP MKPF/MSEG movement type 101 with EKBE history). The product receipt posts inventory value against the GR/IR accrual account (AX "Purchase, accrual" / "Accrued purchases" in the posting profile, SAP WRX GR/IR clearing via OBYC). A vendor invoice (AX VendInvoiceInfoTable/VendInvoiceInfoLine pending invoice → VendInvoiceJour/VendInvoiceTrans; SAP RBKP/RSEG via MIRO) is captured manually, from an OCR/AP-automation adapter, or (wave 35) as UBL/Peppol, and the matching engine compares invoice price, quantity and total to PO and receipt within tolerances (AX price tolerance per item/vendor, quantity tolerance, invoice totals tolerance; SAP tolerance keys PP, DQ, BD, AN/AP, DW, KW). Matched invoices post the vendor liability (VendTrans / BSIK) and clear GR/IR; exceptions route to a Temporal resolution workflow. Evaluated receipt settlement (ERS / MRRL) generates invoices from receipts for vendors flagged for self-billing.

Out of scope: payment (wave 19), warehouse-directed put-away and license plates (wave 31), quality orders on receipt (wave 33), landed cost (wave 34), Peppol transport (wave 35), invoice-cost adjustments to inventory beyond the direct posting (wave 20 settles invoice-receipt cost differences), non-PO expense invoices with GL distribution (included here as 0-way "invoice register" with coding, approval by wave 4 workflow).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX pending vendor invoice form (VendInvoiceInfoTable) runs "Match product receipts" as a dialog, then matching icons update only after "Update match status"; users re-key invoice lines from paper; SAP MIRO proposes lines but the tolerance log (RBKP block indicator, MRBR release) is a separate transaction	Matching is a form-triggered class (VendInvoiceMatch), not an engine; capture is manual	Matching engine runs on every invoice save and on every receipt posting (re-match candidates), status visible inline; OCR adapter pre-fills lines; `MatchResult` per line with variances	Invoice entry time: 4–6 min → < 60 s; manual re-match steps: 3 → 0
GR/IR reconciliation: AX "Accrued purchases" report and SAP MR11 (GR/IR maintenance) plus F.19 regrouping are month-end chores; differences hide for months	GR/IR balance is an account balance with no per-PO-line open-item ledger in AX; SAP EKBE requires MR11 to clear residuals	`GrirOpenItem` per PO line × receipt, cleared by invoice line; GR/IR report = Σ open items and must equal the account balance (daily reconciliation)	Month-end GR/IR effort: 4–8 h → < 15 min; unexplained residual: non-zero → 0
Receipt posting latency: AX product receipt via PurchFormLetter_PackingSlip takes 10–40 s per receipt; SAP MIGO posts in ~1 s but inventory valuation updates MBEW synchronously causing lock waits (table lock on MBEW/MARD during month-end)	Document class chain and serialized valuation updates	Receipt posting is one inventory `transactions:apply` + one ledger `Post`; value updated per transaction, no material-level lock	Receipt post p95: 10–40 s (AX) / lock waits at close (SAP) → < 1.5 s, no cross-document locks

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Tolerance configuration: AX has three separate forms (price tolerance per item/vendor/company, quantity tolerance on item, totals tolerance in parameters) with "Allow price tolerance" flags scattered; SAP OMR6 tolerance keys use cryptic codes and a company-code-level default	Settings scattered without a single evaluation order	One `MatchingPolicy` entity with ordered scopes (vendor×item > vendor > item > entity) and explicit percent/absolute limits, previewable with "test an invoice"	Time to configure and verify a tolerance change: 1–2 h with testing → 5 min with the policy tester
ERS/self-billing in AX requires a batch "Generate invoices from product receipts"; SAP MRRL is a batch run; both mis-handle partial receipts with price changes	Batch design; invoice creation decoupled from the receipt event	ERS invoice created by the `receipt.posted` consumer within seconds using the price at receipt date and locked PO price	ERS invoice latency: nightly batch → < 10 s; ERS price mismatches: tracked, target 0

3.18.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Arrival / registration	Arrival overview (WMSArrivalOverview), arrival journal (WMSJournalTable type Reception), registration (InventTransRegister)	Inbound delivery VL31N, GR into blocked stock (103/105), quality inspection stock	`Arrival` (ARRIVED), `Registration` (REGISTERED) steps on ReceiptLine; arrival overview screen	full
Product receipt	PurchFormLetter_PackingSlip → VendPackingSlipJour/Trans, InventTrans Received, ledger posting PurchPackingSlip/accrual	MIGO 101 → MKPF/MSEG, EKBE (VGABE 1), GR/IR via WRX, inventory BSX	`ProductReceipt` / `ReceiptLine`; RECEIVED transactions; ledger entry Dr inventory / Cr GR/IR	full
Receipt reversal / correction	Product receipt correction, cancel (VendPackingSlipJour cancel)	102 reversal, MBST cancel document	`:correct` (new receipt with delta qty), `:cancel` (full reversal)	full
Vendor invoice (PO)	VendInvoiceInfoTable/Line (pending), VendInvoiceJour/Trans (posted), invoice register/approval journals	MIRO → RBKP/RSEG, EKBE VGABE 2; invoice park MIR7	`VendorInvoice` / `VendorInvoiceLine` (DRAFT → POSTED)	full
Non-PO invoice	Invoice journal (LedgerJournalTrans type VendInvoiceRegister / Vend), invoice register + approval	FB60, MIRO without PO	`VendorInvoice` kind EXPENSE with GL distribution lines	full
Matching	VendInvoiceMatch, match status (VendInvoiceMatchingStatus), price/qty/totals tolerances (VendInvoicePriceTolerance, InventTable quantity tolerance, Accounts payable parameters)	LIV tolerance keys OMR6 (PP, DQ, BD, AN, AP, DW, KW, ST, VP), blocking indicator RBKP ZLSPR, MRBR release	`MatchingEngine`, `MatchingPolicy`, `MatchResult`; exception workflow	full
Accrued purchases / GR/IR	Accrued purchases report (PurchAccrual), posting types PurchaseAccrual/PurchasePackingSlip	GR/IR clearing WRX, MR11, F.19, MB5S	`GrirOpenItem`, GR/IR report	improved
ERS	Vendor self-billing: "Generate invoices from product receipts" (PurchFormLetter with VendInvoiceInfoTable auto)	MRRL evaluated receipt settlement, LFM1 XERSY	`ErsPolicy` on vendor; consumer creates invoice from receipt	full
Invoice capture adapters	Invoice capture (D365 add-in), AIF VendVendInvoiceService	OpenText VIM, SAP Invoice Management, IDoc INVOIC	`POST /invoices:capture` with OCR adapter interface (JSON), UBL mapping stub	partial (Peppol w35)
Charges allocation at invoice	MarkupTrans allocation on invoice	Unplanned delivery costs, planned conditions FRA1	`Charge` allocation rules applied to invoice lines	full
Period control AP	Ledger calendar module Accounts payable	OB52 posting periods account type K	wave 14 period control module AP at `:post`	full

3.18.3 Functional requirements

- **FR-18.1 Arrival and registration** — Must. The system shall record arrival (ARRIVED) with carrier, packing slip number, pallet count and expected PO lines, then registration (REGISTERED) with quantity, batch, serial, location and inventory status, for any confirmed PO line with remaining quantity.
  - Over-receipt beyond `qty × (1 + over_delivery_pct)` (item default 0 %, policy per vendor×item) is rejected with `RCV-OVR-001`; under-delivery below `under_delivery_pct` closes the line only if `close_short=true`.
  - Serial-tracked lines require one serial per unit; duplicate serial rejected by inventory (`INV-SER-001`).
- **FR-18.2 Product receipt posting** — Must. The system shall post a `ProductReceipt` for registered or directly received quantities, moving inventory transactions to RECEIVED, writing `cost_amount_physical = qty × PO unit price (converted to accounting currency at the receipt date rate) + ITEM charges`, and posting Dr inventory (posting profile) / Cr GR/IR accrual through `Post(JournalEntry)` with idempotency key = receipt id.
  - For category (non-stock) lines the debit goes to the line's expense account; GR/IR credit is unchanged.
  - A `GrirOpenItem` is created per receipt line with amount in transaction and accounting currency.
  - Receipt date must be in an open INVENTORY period; packing slip number unique per vendor (`RCV-DUP-001`).
- **FR-18.3 Receipt correction and cancellation** — Must. The system shall support correction (delta quantity, positive or negative) and cancellation (full reversal) of a posted receipt while the receipt line has not been invoiced; invoiced lines require an invoice credit first.
- **FR-18.4 Vendor invoice capture** — Must. The system shall create a `VendorInvoice` in DRAFT from manual entry, from the capture adapter (`:capture` JSON with header fields, lines, confidence scores, page images) or from ERS, with vendor, invoice number, dates, currency, totals, lines linked to PO lines and/or receipt lines, charges, tax groups.
  - Duplicate check: same vendor + invoice number (case-insensitive, whitespace-stripped) within 24 months → `AP-DUP-001`.
  - Lines may be proposed from open receipts (3-way) or open PO lines (2-way); proposal fills qty = uninvoiced received qty.
  - Expense invoices (no PO) carry GL distribution lines validated by account structure.

- **FR-18.5 Matching policy** — Must. The system shall evaluate a `MatchingPolicy` resolved by scope precedence vendor×item > vendor > item > legal entity with fields: `match_level` (NONE, TWO_WAY, THREE_WAY), `price_tol_pct` (default 2.0), `price_tol_abs` (default 25.00 acct ccy per line), `qty_tol_pct` (default 5.0), `total_tol_pct` (default 1.0), `total_tol_abs` (default 50.00), `small_diff_abs` (default 0.50, auto-posted to small-differences account), `price_tol_mode` (UPPER_ONLY / BOTH), `date_tol_days` (0 = off).
    - A line passes price if `|invoice_unit_price − po_unit_price| ≤ max(po_price × pct, abs)` (lower prices always pass in UPPER_ONLY).
    - A line passes quantity (3-way) if `invoice_qty ≤ received_uninvoiced_qty × (1 + qty_tol_pct/100)`; 2-way compares to PO remaining qty.
    - Total passes if `|invoice_total − Σ matched line totals − charges − tax| ≤ max(total_tol_pct, total_tol_abs)`.
- **FR-18.6 Matching engine and status** — Must. The system shall compute `MatchResult` per line and header with statuses `NOT_MATCHED`, `MATCHED`, `PRICE_VARIANCE`, `QTY_VARIANCE`, `TOTAL_VARIANCE`, `NO_RECEIPT`, `WAITING_RECEIPT`; header status is the worst line status; the engine re-runs automatically when a receipt posts for a referenced PO line.
    - Engine is pure (inputs → result) and exposed at `POST /matching:evaluate` for the policy tester.
    - Price variance within tolerance posts the difference to the purchase price variance account (standard-cost items, wave 20 PPV) or to inventory/expense (actual-cost items).
- **FR-18.7 Exception workflow** — Must. The system shall start a Temporal `InvoiceMatchException` workflow for any invoice with a variance status, assigning to the buyer of the PO (price) or receiving supervisor (quantity) with actions: accept variance (post with reason code), request credit note, adjust PO price (creates PO version via wave 17), wait for receipt (timer, re-evaluate), reject invoice.
    - SLA 3 business days then escalation to AP manager; all actions audited on the invoice.
- **FR-18.8 Invoice posting** — Must. The system shall post a matched or accepted invoice: Dr GR/IR (clearing `GrirOpenItem` to the invoiced amount) or Dr expense/inventory (2-way/expense), Dr tax recoverable (wave 13), Cr vendor control account (posting profile) creating a `VendorTransaction` open item with due date and cash discount date per payment terms, moving PO inventory transactions to PURCHASED for invoiced quantity and setting `cost_amount_posted` from the invoice.
    - Idempotency key = invoice id; AP period must be open; posting date ≠ invoice date allowed (both stored).
    - Differences between receipt value and invoice value for stock items post to inventory (actual cost methods) or PPV (standard), and are recorded on the transaction as `cost_amount_adjustment` for wave 20.
- **FR-18.9 GR/IR reconciliation** — Must. The system shall provide the GR/IR report listing open items (received not invoiced, invoiced not received) by vendor, PO, line, age, with totals that equal the GR/IR ledger balance; a daily job emits `reconciliation.diff` if they differ.
- **FR-18.10 Evaluated receipt settlement** — Must. The system shall, for vendors with `ers_enabled`, create and post an invoice from each receipt within 10 s using PO price and the vendor's ERS invoice number format (`ERS-{receipt_number}`), tax per PO line, and send the self-billing document to the vendor.
- **FR-18.11 Charges on receipt and invoice** — Must. The system shall allocate header charges to lines by the rule on the charge (QTY, AMOUNT, LINES) at invoice posting, with ITEM-type charges capitalized into `cost_amount_posted`.
- **FR-18.12 Invoice credit notes** — Should. The system shall create a credit note (negative invoice) referencing an invoice and optionally a receipt correction; it posts the reverse entries and creates a negative `VendorTransaction` for wave 19 settlement.
- **FR-18.13 Receiving workspace and mobile receipt** — Should. The system shall provide a receiving workspace (expected today, overdue, registered not posted) and a PWA receipt screen scanning PO number, product GTIN and batch/serial barcodes (GS1 parsing from wave 8).

3.18.4 Data model

ProductReceipt / ReceiptLine

Field	Type	Required	Notes
id, receipt_number	uuid, text	yes	sequence `RCV-`
vendor_id, po_id	uuid	yes	one PO per receipt (multi-PO receipts split)
packing_slip_number	text	yes	unique per vendor
receipt_date, posting_date	date	yes	inventory physical/financial dates
status	enum	yes	ARRIVED, REGISTERED, POSTED, CORRECTED, CANCELLED
ledger_entry_id	uuid	no	
exchange_rate	numeric(18,8)	yes	PO currency → accounting at receipt_date
line: po_line_id, schedule_id, qty, unit, dimension_set_id, batch_id, serial_ids[], inventory_status, unit_price (from PO), amount_txn, amount_acct, charges_item_amount, inventory_transaction_id, invoiced_qty, grir_open_item_id			

VendorInvoice / VendorInvoiceLine / MatchResult

Field	Type	Required	Notes
id, internal_number	uuid, text	yes	sequence `VI-`
vendor_id, invoice_vendor_id	uuid	yes	
vendor_invoice_number	text	yes	duplicate check
kind	enum	yes	PO, EXPENSE, CREDIT_NOTE, ERS

Field	Type	Required	Notes
invoice_date, posting_date, due_date, discount_date	date	yes/no	from payment terms
currency, exchange_rate	text, numeric	yes	
status	enum	yes	DRAFT, IN_MATCHING, EXCEPTION, APPROVED, POSTED, REJECTED, CANCELLED
match_status	enum	yes	header worst-of
subtotal, charges_total, tax_total, total, total_entered	numeric(19,4)	yes	`total_entered` keyed from paper; must equal computed total within small_diff
source	enum	yes	MANUAL, OCR, UBL, ERS, PORTAL
capture_confidence	numeric(5,2)	no	from adapter
ledger_entry_id, vendor_transaction_id	uuid	no	after post
line: po_line_id, receipt_line_id, product_id/description, qty, unit, unit_price, line_amount, tax_group_id, item_tax_group_id, gl_account_id (expense), dimension_default_id, charges_allocated			
match_result: line_id, policy_id, price_variance_amt, price_variance_pct, qty_variance, total_variance, status, evaluated_at, accepted_by, reason_code			

GrirOpenItem

Field	Type	Required	Notes
id	uuid	yes	
po_line_id, receipt_line_id	uuid	yes	
vendor_id	uuid	yes	
received_qty, invoiced_qty	numeric	yes	
received_amount_acct, invoiced_amount_acct	numeric(19,4)	yes	
status	enum	yes	OPEN, PARTIAL, CLEARED, WRITTEN_OFF
cleared_at	timestamptz	no	

MatchingPolicy: id, scope (ENTITY, ITEM, VENDOR, VENDOR_ITEM), vendor_id, product_id, match_level, price_tol_pct, price_tol_abs, qty_tol_pct, total_tol_pct, total_tol_abs, small_diff_abs, price_tol_mode, date_tol_days, valid_from, valid_to.

Invariants: Σ GrirOpenItem (received – invoiced) in accounting currency = GR/IR ledger balance; receipt line `invoiced_qty ≤ qty`; invoice POSTED implies every line has MATCHED or accepted status; VendorTransaction amount = invoice total in transaction currency. Indexes: receipts (tenant, vendor, packing_slip_number) unique; invoices (tenant, vendor_id, lower(vendor_invoice_number)) unique partial where status ≠ CANCELLED; GrirOpenItem (po_line_id, status).

3.18.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/receiving/arrivals	Record arrival	client key	ReceivingMaintain
POST	/api/v1/receiving/receipts	Create receipt (register lines)	client key	ReceivingMaintain
POST	/api/v1/receiving/receipts/{id}:post	Post product receipt	receipt id	ReceivingPost
POST	/api/v1/receiving/receipts/{id}:correct	Delta correction	receipt id + seq	ReceivingPost
POST	/api/v1/receiving/receipts/{id}:cancel	Reverse	receipt id	ReceivingPost
POST	/api/v1/payables/invoices	Create invoice (manual)	client key	VendorInvoiceMaintain
POST	/api/v1/payables/invoices:capture	Adapter/OCR intake	adapter doc id	service account / VendorInvoiceCapture
POST	/api/v1/payables/invoices/{id}:propose-lines	Propose from receipts/PO	n/a	VendorInvoiceMaintain
POST	/api/v1/payables/invoices/{id}:match	Run matching now	n/a	VendorInvoiceMaintain
POST	/api/v1/payables/invoices/{id}:post	Post	invoice id	VendorInvoicePost
POST	/api/v1/payables/matching:evaluate	Policy tester (pure)	n/a	MatchingPolicyMaintain
GET	/api/v1/payables/grir?vendor=&age_gt=	GR/IR open items	n/a	APInquire
POST	/api/v1/payables/invoices/{id}/exceptions/{line}:accept	Accept variance	line+reason	InvoiceExceptionResolve

Commands: `PostReceipt` (open period, over-delivery, serial uniqueness, PO confirmed, exchange rate exists), `PostInvoice` (AP period open, match status acceptable, totals within small_diff, tax calculated, duplicate check), `AcceptVariance` (reason code required, amount ≤ approver limit from policy).

Event	Trigger	Payload fields	Consumers
`dev.passage.receiving.receipt.posted.v1`	receipt POSTED	receipt_id, po_id, lines (po_line, qty, dims, batch, serials, amount_acct), ledger_entry_id	inventory, payables (re-match, ERS), costing, supplier-portal
`dev.passage.receiving.receipt.corrected.v1`	correction/cancel	receipt_id, delta lines	payables, costing
`dev.passage.payables.invoice.captured.v1`	DRAFT created from adapter	invoice_id, vendor, source, confidence	notifications
`dev.passage.payables.invoice.match.exception.v1`	variance status	invoice_id, line_id, status, variances, assignee	workflow, notifications
`dev.passage.payables.invoice.posted.v1`	POSTED	invoice_id, vendor_transaction_id, lines, totals, grir cleared amounts, ledger_entry_id	ledger reconciliation, costing, tax, analytics
`dev.passage.payables.grir.cleared.v1`	GrirOpenItem CLEARED/WRITTEN_OFF	open_item_id, amounts	analytics

3.18.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Receiving workspace	Expected today/overdue, arrivals, registered-not-posted, post	PO, vendor, line, expected qty, arrived qty	ReceivingMaintain
Receipt (desktop + PWA)	Scan PO/GTIN/batch/serial, register, post, print receipt label	lines with remaining qty, batch/serial entry, location, status	ReceivingMaintain
Vendor invoice	Enter/review captured header, propose lines, match status badges, resolve, post	header, lines with PO/receipt links, variance columns, totals vs entered	VendorInvoiceMaintain
Invoice exception task	Accept, request credit, adjust PO, wait, reject	variance detail, policy limits, history	InvoiceExceptionResolve
Matching policy + tester	Create policy rows, "test an invoice"	scopes, tolerances, sample invoice result	MatchingPolicyMaintain
GR/IR report	Filter by age/vendor, drill to PO/receipt/invoice, write-off	open items, totals vs ledger balance	APInquire

Interaction rules: captured fields with confidence < 0.85 are highlighted and require explicit confirmation; totals-entered vs computed difference shown live; keyboard: Ctrl+M match, Ctrl+P post; PWA receipt supports camera scan and offline registration queue; bulk post of registered receipts; personalization of invoice line columns.

3.18.7 Business rules and edge cases

- **BR-18.1** Receipt state machine: `ARRIVED → REGISTERED → POSTED`; `POSTED → CORRECTED` (new correction receipt, original unchanged); `POSTED → CANCELLED` only if invoiced_qty = 0 on all lines; `ARRIVED|REGISTERED → CANCELLED` releases inventory statuses back to ORDERED.
- **BR-18.2** Invoice state machine: `DRAFT → IN_MATCHING → (MATCHED→) APPROVED → POSTED`; `IN_MATCHING → EXCEPTION → APPROVED|REJECTED`; `DRAFT|EXCEPTION → CANCELLED`; POSTED is terminal (reverse via credit note).
- **BR-18.3** Price comparison uses PO unit price in PO currency normalized to invoice price unit; invoice in a different currency than the PO is compared after conversion at the invoice date rate and flagged `currency_mismatch=true` (always exception unless policy allows).
- **BR-18.4** Quantity tolerance applies to over-invoicing only; under-invoicing leaves the receipt line partially open (`PARTIAL`) for a later invoice.
- **BR-18.5** Small differences: if the header total variance ≤ `small_diff_abs` the difference posts to the small-differences account and the invoice is MATCHED; the amount is recorded on `MatchResult`.
- **BR-18.6** Rounding: variance percent computed on unrounded values and compared at 4 decimals; amounts rounded half-even to minor unit; the GR/IR clearing amount is the receipt-side accounting amount for the invoiced quantity (pro-rata, rounded), so GR/IR residual after full invoicing of a line is exactly 0.
- **BR-18.7** Receipt date rate vs invoice date rate: GR/IR is cleared at the receipt-side accounting amount; the difference between invoice accounting amount and that clearing amount posts to inventory/expense (not FX gain/loss), matching SAP KDM/AX exchange adjustment on purchase.
- **BR-18.8** Concurrency: posting a receipt and an invoice for the same PO line serializes on the PO line row lock (`SELECT … FOR UPDATE` on `purchase_order_line`); the invoice's matching always sees the committed receipt state.
- **BR-18.9** ERS invoices skip matching (status MATCHED by construction) and are blocked if the PO price has changed since receipt until a buyer confirms the price.
- **BR-18.10** Invoice posting date in a closed AP period is rejected (`PER-AP-001`) even when the invoice date is open; the user may change posting date to the first open day with an audit note.
- **BR-18.11** Invoice vendor ≠ ordering vendor (factoring) is allowed when the ordering vendor's `invoice_account` points to the invoice vendor; the VendorTransaction belongs to the invoice vendor.
- **BR-18.12** Credit note quantity cannot exceed invoiced quantity on the referenced line; posting a credit note re-opens the GR/IR item and reverts inventory transaction status from PURCHASED to RECEIVED for that quantity.

3.18.8 Acceptance criteria

- **AC-18.1** (FR-18.5, FR-18.6) Given the 200-scenario matching fixture, when each invoice is evaluated and posted, then every match status, variance amount and ledger posting equals the expected row (0 mismatches).
- **AC-18.2** (FR-18.2, FR-18.8) Given 1,000 random PO lines received then fully invoiced at PO price, when GR/IR open items and the ledger balance are compared, then both are 0.00.
- **AC-18.3** (FR-18.1) Given a serial-tracked PO line of 3 units, when serials S1, S2, S2 are registered, then registration fails on the duplicate and records S1, S2 only after correction.
- **AC-18.4** (FR-18.8) Given AP period 2026-08 closed, when an invoice with posting date 2026-08-15 is posted, then it is rejected with `PER-AP-001` and no VendorTransaction or ledger entry exists.
- **AC-18.5** (FR-18.5) Given policy price_tol_pct 2 and price_tol_abs 25 and a PO price 100.00 USD, when an invoice line at 101.50 is matched, then status MATCHED with price variance 1.50 posted to PPV/inventory; at 103.00 status PRICE_VARIANCE and an exception workflow starts.
- **AC-18.6** (FR-18.7) Given a PRICE_VARIANCE exception, when the buyer accepts with reason "AGREED", then the invoice posts, the variance posts to the variance account, and the audit shows actor, reason and amount.
- **AC-18.7** (FR-18.10) Given an ERS vendor, when a receipt of 50 units at 10.00 EUR posts, then an invoice `ERS-{receipt}` of 500.00 EUR plus tax is posted within 10 s and a self-billing PDF is emailed.
- **AC-18.8** (FR-18.3) Given a posted receipt of 100 with 40 invoiced, when a correction of –70 is attempted, then it is rejected; –60 succeeds and GR/IR open item received_qty becomes 40.
- **AC-18.9** (FR-18.4) Given an invoice captured with confidence 0.6 on the total field, when opened, then the field is flagged and posting is blocked until confirmed.
- **AC-18.10** (FR-18.9) Given a manual journal posted to the GR/IR account, when the daily GR/IR reconciliation runs, then a diff event with the journal amount is emitted.

3.18.9 Test plan and fixtures

Fixtures: `match-200` (200 invoice scenarios CSV with expected status/postings, authored with the reviewing accountant; covers over/under-delivery, price up/down, multi-receipt, partial, currency mismatch, small difference, ERS, credit note); `grir-1000` (1,000 PO lines generated); `rcv-serial-batch` (tracked items); `ap-period-closed`; `ocr-samples` (25 invoice JSON payloads with confidence scores).

Test	Type	Target	Pass criterion
Matching engine	unit (table-driven)	matching:evaluate	match-200 all pass
Tolerance monotonicity	property	engine	increasing invoice price never moves status from variance to matched
GR/IR zero	E2E	receipt+invoice flow	AC-18.2
Receipt → inventory	contract	inventory apply	RECEIVED qty and cost equal receipt
Invoice → ledger	contract	Post	balanced entry; accounts per posting profile
Period control	contract	invoices:post	AC-18.4
ERS latency	E2E	receipt.posted consumer	< 10 s p95
Receipt post latency	performance (k6, 200 receipts/min)	receipts:post	p95 < 1.5 s
PWA scan	E2E (Playwright)	receipt screen	GS1 barcode → line, batch, serial filled

Reconciliation checks: Σ GrirOpenItem = GR/IR ledger balance; Σ VendorTransaction created in period = Σ invoice totals posted; Σ receipt amount_acct = Δ inventory account from receipts; invoiced_qty on receipt lines = Σ invoice line qty referencing them.

3.18.10 Deployment and operations

Helm charts: `passage-procurement` gains the receiving module (same Postgres, separate Deployment `receiving-api` ×2); `passage-payables` (API ×3, matching worker ×2, Temporal worker ×2, CloudNativePG `payables-db`, outbox connector). Config: `AP_SMALL_DIFF_DEFAULT=0.50`, `AP_PRICE_TOL_PCT=2.0`, `AP_QTY_TOL_PCT=5.0`, `AP_DUP_WINDOW_MONTHS=24`, `AP_EXCEPTION_SLA_DAYS=3`, `RCV_OVER_DELIVERY_PCT=0`. Migrations `procurement/v18.*`, `payables/v18.*`. Flags: `ap.ocr_adapter`, `ap.ers`, `ap.ubl_intake` (w35), `rcv.pwa`. Jobs: GR/IR reconciliation daily 02:30; exception SLA timers (Temporal); ERS consumer (Kafka consumer group `payables-ers`). SLOs: receipt post p95 < 1.5 s; invoice post p95 < 2 s; matching evaluate p95 < 100 ms; event lag < 5 s. Alerts: GR/IR diff ≠ 0 (page); exception backlog > 50 (ticket); ERS consumer lag > 100 messages. Runbook: re-run match for invoice set, reprocess failed capture payloads from DLQ topic `payables.capture.dlq`, write off GR/IR residual with approval. Rollback: additive migrations; disable `ap.ers` consumer; invoices in EXCEPTION are unaffected by rollback of the matching worker.

3.18.11 Migration from AX / SAP

Source	Passage entity	Notes
AX VendPackingSlipJour/Trans for open PO lines (received not invoiced)	ProductReceipt POSTED + GrirOpenItem	amount from VendPackingSlipTrans value; creates GR/IR opening balance matched to GL migration
AX VendInvoiceInfoTable (pending invoices)	VendorInvoice DRAFT	re-matched on load
AX VendInvoicePriceTolerance, InventTable tolerance fields, AP parameters	MatchingPolicy rows	scope mapping: item+vendor → VENDOR_ITEM etc.
AX VendTable self-billing flags	Vendor `ers_enabled`	
SAP EKBE VGABE 1 without matching VGABE 2 (GR not IR)	ProductReceipt + GrirOpenItem	WRX balance per PO item
SAP RBKP/RSEG parked (RBSTAT A/B) or blocked (ZLSPR)	VendorInvoice DRAFT / EXCEPTION	
SAP OMR6 tolerance keys (T169G) PP, DQ, BD, AN/AP	MatchingPolicy ENTITY scope	PP → price_tol; DQ → qty_tol; BD → small_diff; AN/AP → total_tol
SAP LFM1 XERSY	ers_enabled	

Validation: GR/IR opening Σ equals GL migrated balance per legal entity; count of open receipt lines equals source; parked invoices re-matched produce the same block/no-block result as SAP for a 50-invoice sample. Not migrated: posted invoice history (becomes VendorTransaction open items in wave 19 migration; line detail stays in lakehouse), MRBR release logs, AX invoice register journals already approved (post before cutover).

3.18.12 Dependencies, risks, open questions

Depends on: wave 17 (PO lines, versions, charges), wave 16 (transaction steps, serial/batch), wave 11–14 (posting profiles, currency, tax, period control), wave 4 (Temporal workflows), wave 8 (GS1 parsing).

Risk	Likelihood	Impact	Mitigation
Accountant disagrees with GR/IR FX treatment (BR-18.7)	Medium	Medium	Confirm with accountant in match-200 review before posting logic is frozen
OCR adapter quality varies; low-confidence fields	High	Low	Confidence threshold gating; no auto-post below 0.95
Multi-PO invoices	Medium	Low	Supported by line-level PO links; receipts stay one PO per document

Open questions:

1. Auto-post invoices with 100 % match and confidence ≥ 0.95 without human review? Default: configurable per vendor, off by default. Decide by wave 19.
2. Should quantity tolerance also apply to under-invoicing close-out (auto-close residual GR/IR)? Default: no; manual write-off. Decide by wave 20.
3. Non-PO invoice approval: route by GL account owner or by amount only? Default: amount bands reuse of the wave 17 policy. Decide by wave 18 mid-point.

3.19 Wave 19: Accounts Payable, Settlement & Payments

3.19.1 Scope and business context

Wave 19 turns posted vendor invoices into paid liabilities. Invoices from wave 18 are open items (AX VendTrans with VendTransOpen; SAP BSIK) that settle against payments, credit notes and prepayments (AX VendSettlement / SpecTrans marking; SAP clearing to BSAK with BSE_CLR, F-53/F-44). A

payment proposal (AX LedgerJournalTrans payment journal with VendPaymProposal; SAP F110 proposal run producing REGUH/REGUP) selects items by due date and cash discount, applies holds, approves, generates payment media (ISO 20022 pain.001 for SEPA/ACH-via-ISO, NACHA for US ACH, check print, positive pay file; SAP DMEE / PMW, AX payment formats via ER), posts the payment to the bank clearing account and vendor control account, and ingests bank status (pain.002) to update payment status. Prepayments, credit notes, reversals and vendor aging complete the AP subledger.

Out of scope: bank statement reconciliation and cash positioning (wave 36, which consumes this wave's payment events), bank connectivity transport (wave 36; this wave writes files to an object store path and exposes them for download), customer-side settlement (wave 30), 1099/withholding statutory reporting beyond withholding calculation (wave 13 provides calculation; reporting in wave 48), intercompany payments (wave 40).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX payment proposal (VendPaymProposal) runs as a modal dialog from the payment journal, locks the journal while running, and must be re-run from scratch to change selection; SAP F110 parameters → proposal → edit proposal → payment run → print is five screens, with proposal logs (REGUP exceptions) read in a separate list	Proposal is a transient batch producing journal lines; no proposal entity	`PaymentProposal` is a persisted entity with selection criteria, line-level include/exclude, holds, and re-selection diff; approve → run is one action	Screens per payment run: 5–7 → 2; time from "select" to "approved file": 45 min → < 5 min
Settlement in AX via "Settle open transactions" (CustVendOpenTransManager) silently computes discount and FX differences; users discover realized gain/loss or discount-lost postings only in the ledger; SAP F-53 shows the clearing document but FX difference accounts (OB09) are hidden until posted	No preview of settlement postings	`POST /settlements:preview` returns the exact journal (discount, FX gain/loss, write-off) before commit; UI shows it inline	Unexpected settlement postings found at month-end: typical 5–20/month → 0; review time per settlement: 2 min → 10 s
Payment formats: AX requires Electronic Reporting configurations per bank (hundreds of nodes), SAP DMEE trees; a bank-specific tag change takes a consultant day	Format defined in a proprietary tree editor	Canonical pain.001.001.09 generator with per-bank profile overrides as JSON (≤ 20 keys), validated against the XSD in CI and a bank harness	Time to onboard a new bank profile: 2–5 days → < 2 h; file rejections at bank: tracked, target < 0.1 %
Payment run idempotency: AX journal posting interrupted mid-batch leaves half-posted lines and a generated file; re-posting can pay twice; SAP F110 "restart" needs SM37 analysis	No durable workflow; file generation and posting are separate steps without a shared idempotent record	Temporal `PaymentRun` workflow with idempotency keys per open item; file is derived from posted payments, never the reverse	Duplicate payments from crash/retry: non-zero historical → 0 by test
Vendor aging in AX (VendAgingReport) is an SSRS batch over VendTrans; SAP S_ALR_87012078 runs minutes for large vendors; neither reconciles to the control account automatically	Report computed from open items with no tie to GL	Aging served from the `VendorTransaction` projection with bucket config; daily job asserts Σ open = AP control balance per vendor group	Aging report time: 2–10 min → < 2 s; monthly AP-to-GL reconciliation: 2 h → automated, 0 diff

3.19.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Vendor open items	VendTrans, VendTransOpen; Vendor transactions form; VendTable balance	BSIK (open), BSAK (cleared), FBL1N	`VendorTransaction` with `open_amount`; vendor transactions screen	full
Settlement	VendSettlement, CustVendOpenTransManager (Settle open transactions), marking (SpecTrans), partial settlement, cross-rate	Clearing F-44/F-53/F-51, residual/partial payment, BSE_CLR, AUGBL	`Settlement`, `SettlementLine`; manual, during payment, automatic	full
Cash discount	CashDisc (terms), VendTrans CashDiscDate, discount taken/lost accounts, "use cash discount" normal/always/never	Payment terms T052 ZBD1T/ZBD1P, SKTO/SKE accounts, OBXU/OBXV	`PaymentTerms` (wave 7) extended with discount tiers; settlement algorithm	full
FX on settlement	Exchange adjustment realized gain/loss (LedgerParameters, CurrencyExchangeAdjustment)	OB09 realized gain/loss accounts, KDF	realized gain/loss in settlement journal (wave 12 accounts)	full
Payment journal / proposal	LedgerJournalTable type Payment, LedgerJournalTrans, VendPaymProposal (by due date / cash discount / both), VendPaymModeTable (method of payment), payment specification, CustVendPaymJournal	F110 (FBZP config), REGUH/REGUP, payment methods T042Z, house banks FI12	`PaymentProposal`, `PaymentMethod`, `Payment`, `PaymentRun` (Temporal)	full
Payment media	ER formats (ISO 20022 credit transfer, NACHA CCD/CTX, check layout), VendOutPaym classes	DMEE/PMW formats, F110 print program RFFOUS_T, check lot FCHI	`PaymentFile` generators: pain.001.001.09, NACHA (PPD/CCD/CTX), check PDF + MICR, positive pay CSV	full
Bank status ingestion	Return file import (ER pain.002), payment status	pain.002 via bank communication management (BCM)	`POST /payments/status:ingest` pain.002 parser	full
Positive pay	Positive pay format (BankPositivePayExport)	Not standard (custom/BCM)	positive pay CSV per bank profile	full
Prepayments	Prepayment invoice on PO (PurchPrepayTable), prepayment journal with posting profile	Down payment request F-47, down payment F-48 (special G/L A), clearing F-54	`Prepayment` (PO-linked or free) with separate posting profile, applied at invoice posting or settlement	full
Credit notes and reversal	Negative invoice, reversal (TransactionReversal), void check	Credit memo MIRO/FB65, reverse FB08, void FCH8/FCH9	wave 18 credit note; `Payment:reverse`, `Payment:void`	full
Vendor aging	Vendor aging report (VendAgingReport), aging period definitions	Aging S_ALR_87012078, due date analysis	`GET /aging` with bucket definitions; ties to control account	improved
Withholding tax at payment	Withholding tax (TaxWithhold*) calculated at payment	Extended withholding at payment (T059P)	wave 13 withholding engine invoked in settlement; posts withholding payable	full
Payment advice	Vendor payment advice (print management)	Payment advice RFFOAVIS	email/portal advice PDF with settled invoices	full

3.19.3 Functional requirements

- **FR-19.1 Vendor open items** — Must. The system shall maintain a `VendorTransaction` for every posted invoice, credit note, payment and prepayment with `amount_txn`, `amount_acct`, `open_amount_txn`, `open_amount_acct`, `due_date`, `discount_date`, `discount_amount`, `status` (OPEN, PARTIALLY_SETTLED, SETTLED), and a `hold` flag with reason.
    - Open amount is updated only by settlement rows; `Σ open_amount_acct` per posting profile account equals the AP control balance (daily assertion).
    - Vendor balance query p95 < 50 ms for vendors with ≤ 100k transactions.
- **FR-19.2 Settlement algorithm** — Must. The system shall settle a set of credit-side items (payment, credit note, prepayment) against debit-side items (invoices) per vendor with the following algorithm, previewable before commit:
    1. Order target invoices by `settlement_priority` (user marks first, then due_date ASC, then amount DESC).
    2. For each invoice, `discount = round(invoice_amount_txn × discount_pct, minor)` if `settlement_date ≤ discount_date` (or terms are `ALWAYS`), else 0; for partial settlement, discount is pro-rated by `settled/invoice_amount` unless policy `discount_on_partial=false`.
    3. `applied_txn = min(available_credit, open_amount_txn − discount)`; invoice open amount reduces by `applied_txn + discount` (discount closes the item only when `applied_txn = open − discount`).
    4. Accounting amounts: `applied_acct_invoice = applied_txn × invoice_rate`, `applied_acct_credit = applied_txn × credit_rate`; `fx_diff = applied_acct_credit − applied_acct_invoice`; positive = realized loss for a vendor payable, negative = realized gain (wave 12 accounts).
    5. If a residual `|open_amount_txn| ≤ write_off_limit` (default 1.00 in transaction currency) and `auto_write_off=true`, post the residual to the write-off account and close the item.
    6. Cross-currency settlement (credit currency ≠ invoice currency) converts both to accounting currency at their own rates; the settlement unit is accounting currency; residuals follow step 5.
    7. Withholding (wave 13) is computed on `applied_txn` and reduces the payment, posting withholding payable.
    - Worked example: invoice 1,000.00 EUR posted at 1.1000 USD/EUR (1,100.00 USD), terms 2 % 10 net 30, paid on day 8 at rate 1.0800. Discount = 20.00 EUR. Applied = 980.00 EUR. Invoice-side acct = 980 × 1.10 = 1,078.00 USD; discount acct = 20 × 1.10 = 22.00; payment acct = 980 × 1.08 = 1,058.40. Journal: Dr AP control 1,100.00; Cr bank clearing 1,058.40; Cr discount taken 22.00; Cr realized FX gain 19.60. Balanced to the cent.
- **FR-19.3 Payment terms and discounts** — Must. The system shall compute `due_date` and up to two discount tiers (e.g., 2/10, 1/20, net 30) from the wave 7 payment terms, with `discount_basis` (gross, net of tax) and `use_cash_discount` (NORMAL, ALWAYS, NEVER) per vendor.
- **FR-19.4 Payment proposal** — Must. The system shall create a `PaymentProposal` from criteria (legal entity, vendor range/group, method of payment, due date to, discount date to, selection rule BY_DUE_DATE / BY_DISCOUNT / BOTH, minimum/maximum payment, currency, bank account), listing candidate open items with proposed amount and discount, excluding held vendors/items, and allowing line-level include/exclude, amount change and reselection with diff.
    - Discount optimization: when `BY_DISCOUNT`, items whose discount date is within `discount_lookahead_days` (default 7) are included; annualized discount value vs `cost_of_capital_pct` (default 8 %) decides early payment when `optimize=true`.
    - Proposal approval uses the wave 4 workflow with amount bands (default: > 100,000 USD two approvers).
- **FR-19.5 Payment run** — Must. The system shall execute an approved proposal as a Temporal `PaymentRun` workflow: for each vendor group create `Payment` records (one per vendor × currency × bank account × method), settle items (FR-19.2), post the payment journal (Dr AP control, Cr bank clearing per method), generate the `PaymentFile`, and mark payments SENT.
    - Each activity is idempotent on `(proposal_id, vendor_transaction_id)`; a resumed run skips already-settled items and already-generated files; the file is regenerated from posted payments if missing.
    - Throughput target: 10,000 open items settled and 2,000 payments generated in < 5 minutes.
- **FR-19.6 ISO 20022 pain.001** — Must. The system shall generate pain.001.001.09 files validated against the XSD with the following mapping:
    - `GrpHdr/MsgId` = `PaymentFile.file_number` (≤ 35 chars); `CreDtTm` = generation timestamp; `NbOfTxs` = count of CdtTrfTxInf; `CtrlSum` = Σ InstdAmt; `InitgPty/Nm` = legal entity name; `InitgPty/Id/OrgId/Othr/Id` = bank-assigned customer id from bank profile.
    - `PmtInf` per (debtor account × execution date × method): `PmtInfId` = `{file_number}-{seq}`; `PmtMtd` = `TRF`; `BtchBookg` = profile `batch_booking` (default true); `PmtTpInf/SvcLvl/Cd` = `SEPA` for SEPA profiles else `NURG/URGP`; `ReqdExctnDt/Dt` = payment date; `Dbtr/Nm`, `Dbtr/PstlAdr` = legal entity; `DbtrAcct/Id/IBAN` (or `Othr/Id` for non-IBAN) = bank account; `DbtrAgt/FinInstnId/BICFI` = bank BIC; `ChrgBr` = `SLEV` (SEPA) or profile default `SHAR`.
    - `CdtTrfTxInf` per Payment: `PmtId/InstrId` = payment number; `PmtId/EndToEndId` = payment number (≤ 35 chars, alphanumeric); `Amt/InstdAmt @Ccy` = amount and currency; `CdtrAgt/FinInstnId/BICFI` = vendor bank BIC (omitted when IBAN-only SEPA and profile allows); `Cdtr/Nm` = vendor name (≤ 140 chars, transliterated to the Latin character set required by the bank profile); `Cdtr/PstlAdr/Ctry` plus structured address when profile requires; `CdtrAcct/Id/IBAN` or `Othr/Id` + `CdtrAgt/.../ClrSysMmbId` for routing-number countries; `RmtInf/Ustrd` = concatenated invoice numbers (≤ 140 chars) or `RmtInf/Strd/RfrdDocInf/Nb` + `RfrdDocAmt` per settled invoice when `structured_remittance=true`; `Purp/Cd` = `SUPP`.
    - AX/SAP lineage: REGUH ZBNKN/ZBNKL → CdtrAcct, REGUH VBLNR → EndToEndId, REGUP BELNR → RfrdDocInf/Nb; AX LedgerJournalTrans PaymReference → EndToEndId.
- **FR-19.7 NACHA, checks, positive pay** — Must. The system shall generate NACHA files (file header 1, batch header 5 with SEC code PPD/CCD/CTX, entry detail 6, addenda 7 for CTX remittance, batch control 8, file control 9, blocking factor 10, 94-char records, hash of routing numbers), check PDFs with MICR line and check number from a check lot, and positive pay CSV (bank account, check number, amount, payee, issue date, void flag) per bank profile.
- **FR-19.8 Payment status and pain.002** — Must. The system shall ingest pain.002.001.10 status reports and map `GrpSts`/`TxSts` (ACCP, ACSC, ACSP, PDNG, RJCT) with `StsRsnInf/Rsn/Cd` (AC01, AC04, AM04, …) to payment statuses, moving RJCT payments to REJECTED, reversing settlement and re-opening items; NACHA returns (R01–R85) map likewise.

- **FR-19.9 Payment reversal and void** — Must. The system shall reverse a posted payment (new journal, settlement undone, items re-opened, file not regenerated) and void a check (check status VOID, positive pay void record), with reason codes and audit.
- **FR-19.10 Prepayments** — Must. The system shall post prepayment invoices (PO-linked, with prepayment posting profile and tax) and prepayment journals; at vendor invoice posting or manual settlement the prepayment applies against the invoice, reclassifying from the prepayment account to AP control.
- **FR-19.11 Vendor aging** — Must. The system shall produce aging by vendor/group with configurable buckets (default 0–30, 31–60, 61–90, 91–180, > 180 by due date or document date), as of any date, in transaction and accounting currency; totals tie to the AP control account per vendor group (daily job + report footer).
- **FR-19.12 Payment advice** — Should. The system shall email (or publish to the portal, wave 20) a payment advice PDF per payment listing settled invoices, discounts and withholding.
- **FR-19.13 Holds** — Should. The system shall support vendor-level holds (INVOICE, PAYMENT, ALL, from wave 7 status) and item-level holds with release date; held items are excluded from proposals and listed on the AP workspace.
- **FR-19.14 Automatic settlement** — Could. The system shall optionally settle credit notes against the oldest open invoices of the same vendor at posting (`auto_settle_credit_notes`).

3.19.4 Data model

VendorTransaction

Field	Type	Required	Notes
id, voucher	uuid, text	yes	ledger entry voucher
vendor_id, legal_entity_id	uuid	yes	
type	enum	yes	INVOICE, CREDIT_NOTE, PAYMENT, PREPAYMENT, PREPAYMENT_INVOICE, WRITE_OFF, FX_ADJ
source_id	uuid	yes	invoice/payment id
document_date, posting_date, due_date, discount_date, discount_date_2	date	yes/no	
currency, exchange_rate	text, numeric(18,8)	yes	
amount_txn, amount_acct	numeric(19,4)	yes	signed (invoice negative = credit to vendor)
open_amount_txn, open_amount_acct	numeric(19,4)	yes	
discount_amount, discount_amount_2	numeric(19,4)	no	
status	enum	yes	OPEN, PARTIALLY_SETTLED, SETTLED
hold_reason, hold_release_date	text, date	no	
posting_profile_id, control_account_id	uuid	yes	
payment_method_id, bank_account_id (vendor)	uuid	no	default from vendor

Settlement / SettlementLine

Field	Type	Required	Notes
id, settlement_number	uuid, text	yes	
vendor_id, settlement_date	uuid, date	yes	
kind	enum	yes	MANUAL, PAYMENT_RUN, AUTO, REVERSAL
ledger_entry_id	uuid	yes	discount/FX/write-off/withholding postings
reversed_by_id	uuid	no	
line: credit_transaction_id, debit_transaction_id, applied_txn, applied_acct_debit, applied_acct_credit, discount_txn, discount_acct, fx_diff_acct, write_off_txn, withholding_txn, closes_debit (bool), closes_credit (bool)			

PaymentProposal / Payment / PaymentFile

Field	Type	Required	Notes
proposal: id, number, criteria (jsonb), status (DRAFT, SELECTED, IN_APPROVAL, APPROVED, RUNNING, COMPLETED, CANCELLED), payment_date, bank_account_id, method_id, total_by_currency (jsonb), approved_by[]			
proposal_line: proposal_id, vendor_transaction_id, proposed_txn, discount_txn, included (bool), hold (bool), exclusion_reason			
payment: id, payment_number, proposal_id, vendor_id, vendor_bank_account_id, method_id, currency, amount_txn, amount_acct, payment_date, status (CREATED, POSTED, SENT, ACCEPTED, SETTLED_AT_BANK, REJECTED, REVERSED, VOID), end_to_end_id, check_number, settlement_id, ledger_entry_id, file_id, status_reason_code			
file: id, file_number, format (PAIN001, NACHA, CHECK_PDF, POSITIVE_PAY), bank_profile_id, generated_at, payment_count, control_sum, sha256, storage_uri, status (GENERATED, DOWNLOADED, TRANSMITTED, ACKNOWLEDGED, REJECTED)			

PaymentMethod / BankProfile: method (id, code, type TRANSFER/ACH/CHECK/CARD/WIRE, format, bank_account_id, clearing_account_id, check_lot_id, grace_days, cutoff_time); bank profile (id, bank, format variant SEPA/CH/UK/US, overrides jsonb ≤ 20 keys: charge bearer, batch booking,

BIC required, structured remittance, char set, max Ustrd length).

Invariants: for every settlement line `applied_acct_credit − applied_acct_debit − fx_diff_acct = 0`; `Σ settlement lines applied on a transaction ≤ |amount|`; transaction SETTLED ⇔ open_amount = 0; `PaymentFile.control_sum = Σ payments in file`; one Payment per (proposal, vendor, currency, bank account, method). Indexes: VendorTransaction (tenant, vendor_id, status, due_date), (control_account_id, status); payments (proposal_id), (end_to_end_id) unique; settlements by transaction ids.

3.19.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/payables/vendors/{id}/transactions?status=open	Open items	n/a	APInquire
POST	/api/v1/payables/settlements:preview	Compute settlement journal	n/a	APSettle
POST	/api/v1/payables/settlements	Commit manual settlement	client key	APSettle
POST	/api/v1/payables/settlements/{id}:reverse	Reverse	settlement id	APSettle
POST	/api/v1/payables/payment-proposals	Create and select	client key	PaymentProposalMaintain
POST	/api/v1/payables/payment-proposals/{id}:reselect	Re-run selection with diff	proposal id + criteria hash	PaymentProposalMaintain
PATCH	/api/v1/payables/payment-proposals/{id}/lines/{lineId}	Include/exclude/amount	ETag	PaymentProposalMaintain
POST	/api/v1/payables/payment-proposals/{id}:submit	Start approval	proposal id	PaymentProposalMaintain
POST	/api/v1/payables/payment-proposals/{id}:run	Execute PaymentRun	proposal id	PaymentRunExecute
GET	/api/v1/payables/payment-files/{id}/content	Download file	n/a	PaymentFileDownload
POST	/api/v1/payables/payments/status:ingest	pain.002 / NACHA return intake	message id	service account / PaymentStatusIngest
POST	/api/v1/payables/payments/{id}:reverse	Reverse payment	payment id	APSettle
POST	/api/v1/payables/payments/{id}:void	Void check	payment id	APSettle
POST	/api/v1/payables/prepayments	Prepayment invoice/journal	client key	VendorInvoicePost
GET	/api/v1/payables/aging?asOf=&by=due	document&buckets=	Aging	n/a

Commands: `CommitSettlement` (same vendor, no held items, rates available, AP period open, Σ applied ≤ available credit), `RunPayment` (proposal APPROVED, bank account active, cutoff time, vendor bank account validated — IBAN mod-97, ABA checksum), `IngestStatus` (file matches known MsgId/OrgnlEndToEndId; unknown ids → DLQ).

Event	Trigger	Payload fields	Consumers
`dev.passage.payables.settlement.posted.v1` / `.reversed.v1`	settlement commit/reverse	settlement_id, vendor_id, lines (ids, applied, discount, fx_diff), ledger_entry_id	ledger reconciliation, analytics
`dev.passage.payables.proposal.approved.v1`	approval complete	proposal_id, totals by currency, payment_date	notifications
`dev.passage.payables.payment.posted.v1`	payment journal posted	payment_id, vendor, amount, method, bank account, settled items	cash (w36), analytics
`dev.passage.payables.payment.sent.v1`	file generated & marked sent	payment_id, file_id, format, end_to_end_id	bank connectivity (w36), supplier-portal (w20)
`dev.passage.payables.payment.status.changed.v1`	pain.002/NACHA return	payment_id, old_status, new_status, reason_code	notifications, supplier-portal
`dev.passage.payables.aging.reconciled.v1`	daily job	as_of, vendor_group, open_total, control_balance, diff	alerting

3.19.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
AP workspace	Due this week, discounts expiring, held items, proposals awaiting approval, rejected payments	counts, amounts by currency	APInquire
Vendor transactions	Filter open/all, settle (opens preview), hold/release, view voucher	document, dates, amounts, open, discount available, status	APInquire / APSettle
Settlement dialog	Mark items, set date, preview journal, commit	credit/debit items, applied, discount, FX diff, write-off, resulting journal	APSettle
Payment proposal	Criteria form, selection grid, include/exclude, reselect diff, submit, run	vendor, item, due, discount, proposed amount, hold reason	PaymentProposalMaintain
Payment run monitor	Progress, failures, retry, download files, advices	payments by status, file list, control sums	PaymentRunExecute
Payment files / bank profiles	Download, mark transmitted, configure profile overrides, test generate	format, bank, count, control sum, status	PaymentFileDownload / BankProfileMaintain
Aging	Bucket view, drill to items, export	vendor group, buckets, total, control balance, diff	APInquire

Interaction rules: settlement preview updates within 300 ms of marking an item; proposal grid supports multi-select include/exclude and keyboard navigation; amounts always shown with currency code; approvals from shell inbox; bank profile editor validates with a sample file generation and XSD check before save; personalization of buckets and columns.

3.19.7 Business rules and edge cases

- **BR-19.1** Payment state machine: `CREATED → POSTED → SENT → ACCEPTED → SETTLED_AT_BANK`; `SENT|ACCEPTED → REJECTED` (settlement reversed, items re-opened, payment stays for audit); `POSTED|SENT → REVERSED` (manual); `SENT (check) → VOID`. Proposal: `DRAFT → SELECTED → IN_APPROVAL → APPROVED → RUNNING → COMPLETED`; `IN_APPROVAL → SELECTED` on rejection; `DRAFT|SELECTED → CANCELLED`.

- **BR-19.2** Discount eligibility uses `settlement_date` (payment date), not run date; a proposal with payment_date ≤ discount_date takes the discount even if generated after it; `use_cash_discount=ALWAYS` takes it regardless of date, `NEVER` never.
- **BR-19.3** Partial payment discount: default `discount_on_partial=true` pro-rata; residual item keeps the remaining discount entitlement with the same discount date.
- **BR-19.4** Rounding: discount and FX amounts rounded half-even to the currency minor unit; the FX difference is computed as the balancing figure so the settlement journal always balances by construction; any rounding residual ≤ 0.01 in accounting currency posts to the rounding account (wave 12).
- **BR-19.5** Cross-currency: when credit and debit currencies differ, `applied` is expressed in accounting currency; the transaction-currency open amounts are reduced by `applied_acct / own_rate`; residual below `write_off_limit` closes the item.
- **BR-19.6** Settlement in a closed AP period is rejected; reversal posts in the current open period with a reversal voucher referencing the original.
- **BR-19.7** Concurrency: settlement locks VendorTransaction rows in ascending id order; a payment run and a manual settlement on the same item serialize; the loser sees `AP-SET-009 item no longer open` and the proposal line is marked excluded with reason.
- **BR-19.8** Idempotency: `PaymentRun` activities key on `(proposal_id, vendor_transaction_id)`; the file activity keys on `(proposal_id, bank_account_id, method_id, payment_date)` and is deterministic (same payments → same bytes except `CreDtTm`), so a regenerated file is byte-identical in content fields and carries the same `MsgId`.
- **BR-19.9** `EndToEndId` and `MsgId` are restricted to `[A-Za-z0-9/-?:().,'+ ]` ≤ 35 chars; vendor names are transliterated per bank profile (e.g., `ß → ss`); failure to transliterate rejects the payment line before file generation.
- **BR-19.10** Minimum payment amount (default 1.00) and negative vendor balance: a vendor whose included items net to ≤ 0 is excluded from the proposal with reason `NET_CREDIT`; credit notes are settled against invoices inside the proposal before amounts are computed.
- **BR-19.11** Prepayment applied at invoice posting cannot exceed the invoice total; the remainder stays open as a prepayment transaction.
- **BR-19.12** Rejected payment (pain.002 RJCT AC01 invalid account) re-opens items with a `hold_reason=BANK_DETAILS` on the vendor's bank account until corrected.

3.19.8 Acceptance criteria

- **AC-19.1** (FR-19.2) Given the 300-scenario settlement fixture, when each scenario is previewed and committed, then discount, FX gain/loss, write-off and withholding amounts and accounts equal expected values for all 300, and every settlement journal balances.
- **AC-19.2** (FR-19.2) Given the worked example in FR-19.2, when settled, then the journal lines are exactly Dr AP 1,100.00; Cr bank clearing 1,058.40; Cr discount 22.00; Cr FX gain 19.60.
- **AC-19.3** (FR-19.6) Given a proposal of 50 SEPA payments, when the run completes, then the pain.001.001.09 file validates against the XSD, `CtrlSum` equals Σ amounts, and the bank test harness returns ACCP for all 50.
- **AC-19.4** (FR-19.5) Given a run of 2,000 payments killed after 1,200 payments are posted, when the workflow resumes, then exactly 2,000 payments exist, no item is settled twice, and the file contains 2,000 transactions.
- **AC-19.5** (FR-19.11) Given open items across 3 vendor groups, when aging is run as of today, then Σ buckets per group equals the AP control account balance per posting profile and the daily job emits diff = 0.
- **AC-19.6** (FR-19.4) Given an invoice with 2 % discount expiring in 5 days and cost of capital 8 %, when a BY_DISCOUNT proposal with optimize=true runs, then the item is included with the discount (annualized 2 % × 365/20 ≈ 36.5 % > 8 %).
- **AC-19.7** (FR-19.8) Given a SENT payment, when a pain.002 with TxSts RJCT and Rsn AC01 is ingested, then the payment is REJECTED, its settlement reversed, the invoice re-opened with hold BANK_DETAILS, and `payment.status.changed` emitted.
- **AC-19.8** (FR-19.7) Given 10 US vendors paid by ACH CCD, when the NACHA file is generated, then record counts, block count (multiple of 10), entry hash and total debit/credit match the file control record and a NACHA validator passes.
- **AC-19.9** (FR-19.10) Given a PO prepayment of 300.00 USD and a later invoice of 1,000.00 USD, when the invoice posts, then 300.00 is applied, AP open = 700.00, and the prepayment account is reduced by 300.00.
- **AC-19.10** (FR-19.9) Given a posted check payment, when voided, then the check status is VOID, the positive pay file contains a void record, items are re-opened, and the payment journal is reversed.

3.19.9 Test plan and fixtures

Fixtures: `ap-settle-300` (300 scenarios authored with the accountant: full/partial, discount taken/lost/always, cross-currency, write-off, withholding, prepayment, credit note netting); `ap-pain001-bank-profiles` (SEPA, UK, CH, US-ISO) with expected XML; `ap-nacha-10`; `ap-aging-demo` (3 vendor groups, 5,000 items); `ap-pain002-samples` (ACCP, RJCT codes); `ap-crash-run` (2,000 payments with injected failure).

Test	Type	Target	Pass criterion
Settlement scenarios	unit (table-driven)	settlement engine	AC-19.1
Settlement balances	property	engine	random items/rates/dates: journal always balances; Σ applied ≤ credit
pain.001 XSD	contract	file generator	valid against `pain.001.001.09.xsd` for all profiles
Bank harness	E2E	sandbox bank	ACCP for fixture file
NACHA validator	unit	NACHA generator	AC-19.8
Run idempotency	E2E (Temporal test server with activity failure injection)	PaymentRun	AC-19.4
Aging ties	E2E	aging job	AC-19.5
Status ingest	unit	pain.002 parser	all codes map; unknown → DLQ
Proposal throughput	performance	10k items	< 5 min run, p95 settle activity < 50 ms

Reconciliation checks: Σ VendorTransaction.open_amount_acct per control account = ledger balance; Σ payments POSTED per bank clearing account = ledger clearing balance; Σ file control sums = Σ payments SENT; settlement discount Σ = discount account movement for the period.

3.19.10 Deployment and operations

Helm chart `passage-payables` extended: payment-run Temporal worker ×2, file generator worker ×2, object store bucket `passage-payment-files` (S3-compatible, server-side encryption, 7-year retention, write-once). Config: `AP_WRITE_OFF_LIMIT=1.00`, `AP_DISCOUNT_LOOKAHEAD_DAYS=7`, `AP_COST_OF_CAPITAL_PCT=8`, `AP_MIN_PAYMENT=1.00`, `AP_FILE_RETENTION_DAYS=2555`. Migrations `payables/v19.*`. Flags: `ap.nacha`, `ap.checks`, `ap.positive_pay`, `ap.auto_settle_credit_notes`, `ap.structured_remittance`. Jobs: aging reconciliation daily 03:00; discount-expiry digest daily 07:00; pain.002 poller (wave 36 replaces with bank connectivity). SLOs: settlement preview p95 < 300 ms; proposal selection of 10k items < 10 s; file generation < 30 s for 2,000 payments. Alerts: aging diff ≠ 0 (page); payment run failed activity > 3 retries (page); files generated but not downloaded/transmitted within 4 h (ticket); rejected payments > 1 % of a run (ticket). Runbook: resume stuck run (Temporal), regenerate file, reverse payment, correct vendor bank account and re-propose. Rollback: additive migrations; payment files already transmitted are not affected; disable `ap.*` format flags to stop generation.

3.19.11 Migration from AX / SAP

Source	Passage entity	Notes
AX VendTrans + VendTransOpen (open items)	VendorTransaction OPEN/PARTIALLY_SETTLED	AmountCur/AmountMST → amount_txn/acct; VendTransOpen.AmountCur → open; CashDiscDate, DueDate; vendor balance Σ must equal migrated AP control
AX VendSettlement history	not migrated	closed history in lakehouse
AX VendPaymModeTable, VendPaymModeSpec	PaymentMethod	ER format name → format enum
AX BankAccountTable (vendor bank accounts, VendBankAccount)	vendor bank accounts (wave 7 role data)	IBAN/BIC/ABA validated on load
AX CashDisc, PaymTerm	PaymentTerms discount tiers	
AX LedgerJournalTrans open payment journals	not migrated	post or delete before cutover
SAP BSIK (open vendor items)	VendorTransaction	DMBTR/WRBTR, ZFBDT+ZBD1T → due/discount dates, ZLSPR → hold; UMSKZ A → PREPAYMENT
SAP T042Z/T042E (payment methods), FBZP house banks	PaymentMethod, bank accounts	
SAP LFBK (vendor bank details)	vendor bank accounts	
SAP REGUH/REGUP of unexecuted F110 proposals	not migrated	complete or delete before cutover
SAP T052 payment terms	PaymentTerms	

Validation: Σ open items per vendor and per control account equals the source trial balance at cutover (0.00 difference); count of open items equals source; 100 random vendors' bank details re-validated. Not migrated: cleared items (BSAK, settled VendTrans), check registers (void history), positive pay history, payment files.

3.19.12 Dependencies, risks, open questions

Depends on: wave 18 (VendorTransaction creation), wave 12 (rates, realized gain/loss accounts, rounding), wave 13 (withholding), wave 14 (AP period control), wave 7 (vendor bank accounts, terms, holds), wave 4 (Temporal, approvals), wave 36 for bank transport (forward dependency; this wave uses file download).

Risk	Likelihood	Impact	Mitigation
Bank-specific pain.001 variants exceed the 20-key profile model	Medium	Medium	Profile model reviewed against 4 banks' implementation guides; escape hatch: XSLT post-processor per profile
Accountant rejects discount-on-partial default	Low	Low	Policy flag; fixture covers both
Payment file leakage (contains bank details)	Low	High	Object store with per-tenant prefix, OPA-gated download, audit on every download, no email attachment of files

Open questions:

1. pain.001 version: .09 (2019) as default or .03 for banks still on the older schema? Default: .09 with profile override to .03. Decide by wave 19 mid-point.
2. Should the proposal auto-net credit notes across currencies? Default: same currency only. Decide by wave 19 close.
3. Card payments (virtual cards) in v1? Default: method type exists, no provider adapter until v1.x. Decide by wave 35.

3.20 Wave 20: Inventory Costing & Supplier Collaboration

3.20.1 Scope and business context

Wave 20 makes inventory value correct. Waves 16–18 wrote `cost_amount_physical` and `cost_amount_posted` on every inventory transaction at the running estimate (receipt price, running average on issues). This wave adds the costing engine that settles issues against receipts under the item model group's method — FIFO, LIFO, weighted average (running or periodic) or standard cost — and the inventory close/recalculation job (AX InventClosing with InventSettlement and InventCostTrans; SAP material ledger period-end closing CKMLCP with CKMLPP/CKMLCR and MBEW price control S/V, MR21 price change, MR22 debit/credit) that posts `cost_amount_adjustment` rows and GL adjustments so that the inventory value report ties to the GL inventory accounts by site, warehouse, item group and financial dimension. Standard cost captures purchase price variance (PPV) on receipt and invoice. Costing versions (AX CostingVersion/InventItemPrice; SAP CK11N/CK24 marking and release for material-level prices) hold planned and standard costs by product × site with activation and revaluation of on-hand.

The supplier portal (D365 vendor collaboration; SAP Ariba Supplier Network / SNC) at `supplier.<env>.passage.dev` lets vendor users acknowledge POs (accept/reject/propose changes), send ASNs, reply to RFQs (wave 17 hook), submit invoices against POs (wave 18 `:capture` source PORTAL), view

payment status and advices (wave 19), and exchange documents.

Out of scope: BOM cost roll-up, costing sheets and production variance decomposition (wave 39), production order WIP settlement (wave 21 uses this engine for its receipts/issues), landed cost (wave 34), cost accounting ledger (wave 38), portal e-invoicing via Peppol (wave 35), supplier onboarding/qualification workflow (v1.x).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX inventory close (InventClosing) takes 4–12 h for large companies, blocks posting in the period, fails on unsettled negative issues with cryptic "cannot settle" messages, and must be cancelled and re-run end to end	Single-threaded settlement per item in X++ with helper threads; whole-period scope; no incremental	Close runs as a Temporal workflow with per-product child workflows (parallelism 32), incremental from the last closed date, resumable, with a pre-check report listing items that cannot settle and why	Close duration on 1M transactions: 4–12 h → < 20 min; re-run after failure: full → incremental from checkpoint
Inventory value does not tie to the GL: AX "Inventory value report" (InventValueReport) requires report configuration (InventValueReportSetup) and still differs from the GL because of manual journals and timing; SAP MB5L vs FAGLB03 differences investigated monthly	Value report computed from InventTrans; GL has postings from other sources; no per-dimension open-item relationship	Value report is a projection from `CostTransaction` rows with the GL voucher on each; daily reconciliation per inventory account × dimension; every difference row drills to the causing voucher	Monthly inventory-to-GL reconciliation: 1–3 days → < 1 h; unexplained differences: non-zero → 0
Weighted average in AX during the period uses running average per receipt, then close recalculates to periodic average, changing COGS retroactively with surprises at close; SAP MAP (VERPR) can go negative or spike on small stock with large invoice differences	Two-phase (running vs periodic) design; MAP recalculated on invoice with stock near zero	Method choice `WEIGHTED_AVG_RUNNING` (what you see during the period is final) or `WEIGHTED_AVG_PERIODIC` (explicit), with a preview of close adjustments by product before posting; guard that invoice differences on zero stock post to price difference, not to a residual unit	Unexpected COGS adjustments at close: typical 1–3 % of COGS → < 0.1 %; MAP spike incidents: tracked, target 0
Standard cost activation in AX requires costing version setup, pending prices, "Activate pending prices", then a revaluation posting that is hard to reconcile; SAP CK24 marking/release with CKM3 analysis is a multi-transaction month-end routine	Multi-step manual procedure	`CostingVersion` with pending/active state; `:activate` posts the revaluation of on-hand in one voucher and shows the per-site delta before activation	Steps to activate standard costs for a plant: 6–9 transactions → 2 (review, activate); activation latency: hours → < 2 min for 10k products
Vendors call or email for PO status, ASN entry is EDI-only or re-keyed by the warehouse, invoices arrive by email PDF; D365 vendor collaboration needs Azure AD B2B guest setup per vendor user; Ariba requires a separate subscription	Portal is an add-on with separate identity	Portal built on the same APIs/events with its own Keycloak realm; vendor admin invites users self-service; ASN pre-fills receiving (wave 18)	Vendor onboarding to portal: days → < 15 min; PO acknowledgement latency: 2–5 days → < 1 day median; receipts pre-filled from ASN: 0 % → > 80 % of portal vendors' receipts

3.20.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity level
Costing method per item	InventModelGroup.InventModel (FIFO, LIFO, LIFO date, Weighted avg, Weighted avg date, Standard cost, Moving average)	MBEW VPRSV (S standard / V moving average), material ledger actual costing for periodic	`ItemModelGroup.costing_method` (FIFO, LIFO, LIFO_DATE, WEIGHTED_AVG_RUNNING, WEIGHTED_AVG_PERIODIC, STANDARD, MOVING_AVG)	full
Inventory close	InventClosing, InventSettlement, InventCostTrans, InventClosingRecalculation; Close and adjustment form	CKMLCP (material ledger cockpit: selection, determine sequence, single/multi-level price determination, post closing), CKMLPP/CKMLCR	`InventoryClose` workflow, `CostSettlement`, `CostTransaction`	full
Recalculation	Recalculation (without closing), cancel recalculation	Material ledger price determination preview; CKM3 analysis	`:recalculate` (preview/post without closing period)	full
Adjustment	Adjustment of on-hand / transactions (InventAdjOnhand, InventAdjTrans)	MR22 debit/credit material, MR21 price change	`CostAdjustment` (on-hand or transaction level)	full
Standard cost / costing versions	CostingVersion, InventItemPrice (pending/active), activation, revaluation posting	Standard price STPRS, CK11N estimate, CK24 mark/release, price change document MR21, revaluation	`CostingVersion`, `ItemCost` (PENDING/ACTIVE), `:activate` with revaluation voucher	full (roll-up in w39)
Purchase price variance	Standard cost variance posting types PurchPriceVariance at receipt/invoice	PRD price difference account (OBYC PRD) on GR/IR	PPV posting at receipt and invoice (wave 18 hooks) with `CostTransaction` kind PPV	full
Inventory value report	InventValueReport, InventValueReportSetup, Inventory aging	MB5L (stock value), MC.9 / MB52, CKM3	`GET /costing/value-report` with dimension grouping and GL tie-out column	improved
Marking	InventTrans marking (MarkingRefInventTransOrigin)	Not standard (batch-specific valuation partial)	`marked_transaction_id` on transactions (wave 16 field) honored by settlement	full
Moving average	Moving average (InventModel MovingAverage) with product cost adjustment at invoice	MAP VERPR update on GR/IR	`MOVING_AVG` with invoice-difference settlement to stock proportionally	full
Vendor collaboration	D365 Vendor collaboration workspace (PO confirmation, consignment, invoices, RFQ replies)	Ariba Supplier Network, SNC/Supplier Portal	Supplier portal (PO ack, ASN, invoice, RFQ reply, payments, documents)	full
ASN	D365 vendor collaboration ASN; AIF ASN inbound	ASN via EDI DESADV, SNC	`AdvanceShipNotice` → receiving expected lines	full

3.20.3 Functional requirements

- **FR-20.1 Costing methods** — Must. The system shall value issues against receipts by the item model group's method:
  - **FIFO**: issues settle against the oldest unsettled receipts by `financial_date`, then `physical_date`, then id. Example: receipts 100 @ 10.00 (Jan 5), 100 @ 12.00 (Jan 20); issue 150 on Jan 25 → cost = 100 × 10.00 + 50 × 12.00 = 1,600.00; remaining on-hand 50 @ 12.00 = 600.00.
  - **LIFO**: issues settle against the newest receipts preceding the issue date (LIFO_DATE) or the newest in the period (LIFO). Same data, LIFO_DATE: 150 issue = 100 × 12.00 + 50 × 10.00 = 1,700.00; remaining 50 @ 10.00 = 500.00.

    - **WEIGHTED_AVG_PERIODIC**: all issues in the period settle at `(opening value + Σ receipts value) / (opening qty + Σ receipts qty)`. Same data: average = 2,200 / 200 = 11.00; issue cost 1,650.00; remaining 550.00.
    - **WEIGHTED_AVG_RUNNING**: each issue uses the average at its moment (in date order): issue on Jan 25 after both receipts = 11.00 → 1,650.00; an issue of 50 on Jan 10 would be at 10.00.
    - **MOVING_AVG**: like running average, but invoice-price differences on already-issued quantity post to price difference (not to stock); differences on remaining stock adjust its value.
    - **STANDARD**: all transactions at the active `ItemCost` for product × site; receipt at PO price 10.50 against standard 10.00 for 100 units posts PPV 50.00 Dr; invoice at 10.80 posts additional PPV 30.00; inventory stays 1,000.00.
    - Settlement rows (`CostSettlement`) link issue to receipt with qty and value; `cost_amount_adjustment` on the issue = settled value – previously posted value.

- **FR-20.2 Inventory close** — Must. The system shall run `InventoryClose` for a legal entity up to a close date as a Temporal workflow: pre-check (unsettled negative on-hand, missing invoices for RECEIVED-not-PURCHASED when `close_requires_invoice=true`, open count sheets), settlement per product (child workflows, parallelism configurable, default 32), adjustment postings to the ledger (one voucher per product group × account, idempotent on close id), and lock of transactions with `financial_date ≤ close_date` against cost changes.

    - Performance: 1M transactions, 2,000 products in < 20 min on the stage sizing.
    - Cancel close reverses adjustment vouchers and unlocks; a close cannot be cancelled if a later close exists.
    - Close requires the INVENTORY period for the close date to be OPEN; after close completes it may be set CLOSED (wave 14).

- **FR-20.3 Recalculation** — Must. The system shall run the same settlement as close without locking, posting adjustments with `kind=RECALC` reversible by the next recalculation or close; preview mode returns per-product adjustment totals without posting.

- **FR-20.4 Cost transactions and value report** — Must. The system shall write a `CostTransaction` for every value-affecting event (physical posting, financial posting, invoice difference, settlement adjustment, PPV, revaluation, manual adjustment) with the ledger voucher, and serve the inventory value report (quantity and value by product, site, warehouse, item group, dimension; posted vs physical value; opening, receipts, issues, adjustments, closing) whose totals equal the ledger inventory account balances per dimension on the report date.

    - Daily reconciliation job `costing.reconcile` emits `reconciliation.diff` on any difference.
    - Value report for 10k products p95 < 5 s.

- **FR-20.5 Costing versions and standard cost activation** — Must. The system shall maintain `CostingVersion` (type STANDARD, PLANNED; blocking flags; fallback version) with `ItemCost` rows (product, site, unit cost, cost breakdown placeholder for wave 39, effective_from, status PENDING/ACTIVE/SUPERSEDED). Activation sets ACTIVE, supersedes the previous, posts revaluation of on-hand `(new − old) × physical qty` per product × site to inventory / revaluation account, and emits `cost.changed`.

    - Pending prices can be imported (CSV/API) and copied from the previous version with a percent change.
    - Activation is atomic per product and idempotent per version id.

- **FR-20.6 Cost adjustments** — Must. The system shall allow manual adjustments of on-hand value (per product × dimension: new unit cost or amount) and of specific receipt transactions (amount), posting to inventory / adjustment account and flowing to issues on next close.

- **FR-20.7 Purchase price variance** — Must. The system shall, for STANDARD items, post PPV at receipt `(PO price − standard) × qty` and at invoice `(invoice price − PO price) × qty` (wave 18 hook), to the PPV account from the posting profile; for actual-cost methods the invoice difference adjusts the receipt's cost (`cost_amount_posted`) and flows to issues through settlement.

- **FR-20.8 Marking** — Should. The system shall honor explicit marking between an issue and a receipt (set by users or by wave 21/27 processes) so settlement pairs marked transactions first regardless of method.

- **FR-20.9 Supplier portal: PO acknowledgement** — Must. The portal shall show confirmed POs and versions to the vendor's users; they shall accept, reject (reason) or propose changes (date, qty, price per line) which create a `PurchaseOrderAcknowledgement`; proposed changes become a pending change on the PO for the buyer to accept (creating a new version via wave 17) or decline.

    - SLA tracking: acknowledgement due within `ack_due_days` (default 2); overdue listed on the buyer workspace.

- **FR-20.10 Supplier portal: ASN** — Must. The portal shall create an `AdvanceShipNotice` (carrier, tracking, ship date, expected arrival, packages with SSCC optional, lines with PO line, qty, batch, serials, expiry) that pre-creates ARRIVED expected lines in receiving; receipt registration from an ASN pre-fills quantities and tracking dimensions.

- **FR-20.11 Supplier portal: invoices and payments** — Must. The portal shall let vendors submit invoices against received PO lines (source PORTAL into wave 18 `:capture`, structured, no OCR), view invoice match/approval status (without internal variance amounts), and see payment status, advices and remittance details from wave 19.

- **FR-20.12 Supplier portal: identity and authorization** — Must. The portal shall use a separate Keycloak realm `suppliers` with vendor organizations as groups, vendor-admin self-service invitation of users (email verification, MFA optional per vendor), OPA policies scoping every query to the vendor ids the user belongs to, and rate limits (60 requests/min per user, 1,000/min per vendor).

- **FR-20.13 Supplier portal: RFQ replies and documents** — Should. The portal shall accept RFQ replies (wave 17 FR-17.4) and exchange documents (certificates, contracts) with virus scanning and 25 MB size limit.

- **FR-20.14 Phase-exit walkthrough** — Must. The system shall support the recorded end-to-end demo: requisition → PO → portal acknowledgement → ASN → receipt → invoice (portal) → match → payment run → close → value report tie-out, on the stage environment with the demo plant.

3.20.4 Data model

CostTransaction

Field	Type	Required	Notes
id	uuid	yes	
inventory_transaction_id	uuid	yes	FK
product_id, site_id, dimension_set_id	uuid/bytea	yes	denormalized for the report
kind	enum	yes	PHYSICAL, FINANCIAL, INVOICE_DIFF, SETTLEMENT_ADJ, PPV, REVALUATION, MANUAL_ADJ, RECALC
qty	numeric(28,6)	yes	0 for value-only
amount_acct	numeric(19,4)	yes	signed
close_id / recalc_id	uuid	no	
ledger_entry_id	uuid	yes	voucher
financial_date	date	yes	
created_at	timestamptz	yes	append-only

CostSettlement

Field	Type	Required	Notes
id	uuid	yes	
close_id	uuid	yes	or recalc id
issue_transaction_id, receipt_transaction_id	uuid	yes	
qty_settled	numeric(28,6)	yes	> 0
value_settled_acct	numeric(19,4)	yes	
method	enum	yes	method used
sequence	int	yes	order of settlement for determinism
cancelled (bool), cancelled_by_close_id		no	

InventoryClose

Field	Type	Required	Notes
id, close_number	uuid, text	yes	
legal_entity_id, close_date	uuid, date	yes	
kind	enum	yes	CLOSE, RECALC, PREVIEW
status	enum	yes	PRECHECK, PRECHECK_FAILED, SETTLING, POSTING, COMPLETED, CANCELLING, CANCELLED, FAILED
parallelism	int	yes	default 32
products_total, products_done, settlements_count, adjustment_total_acct	int/numeric	yes	progress
started_at, completed_at, duration_s	timestamptz/int		
temporal_workflow_id	text	yes	

CostingVersion / ItemCost: version (id, code, type STANDARD/PLANNED, status DRAFT/ACTIVE/ARCHIVED, fallback_version_id, block_activation (bool), site_scope[]); item cost (version_id, product_id, site_id, unit, unit_cost_acct, effective_from, status PENDING/ACTIVE/SUPERSEDED, activated_at, revaluation_ledger_entry_id, source MANUAL/IMPORT/COPY/ROLLUP (w39)).

PurchaseOrderAcknowledgement / AdvanceShipNotice / SupplierUser: ack (id, po_id, version_no, vendor_user_id, response ACCEPT/REJECT/PROPOSE, reason, proposed_lines jsonb, status PENDING_BUYER/ACCEPTED/DECLINED, created_at); asn (id, asn_number, vendor_id, po_id, carrier, tracking_number, ship_date, expected_arrival, packages jsonb (sscc, weight), lines (po_line_id, qty, batch, serials[], expiry), status SUBMITTED/ARRIVED/RECEIVED/CANCELLED); supplier user (keycloak_subject, vendor_ids[], role VENDOR_ADMIN/VENDOR_USER, invited_by, mfa_enabled, last_login).

Invariants: for each issue transaction Σ qty_settled over non-cancelled settlements ≤ |qty|; after a COMPLETED close every issue with `financial_date ≤ close_date` is fully settled or listed in the close exception report; Σ CostTransaction.amount_acct per inventory account = ledger balance; exactly one ACTIVE ItemCost per (STANDARD version chain, product, site); a vendor user can read only rows where vendor_id ∈ their vendor_ids (OPA + RLS policy on the `supplier` database). Indexes: CostTransaction (tenant, product_id, financial_date), (ledger_entry_id); CostSettlement (issue_transaction_id), (receipt_transaction_id); ItemCost (product_id, site_id, status).

3.20.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/costing/closes	Start close (kind CLOSE/RECALC/PREVIEW)	entity+close_date+kind	InventoryCloseRun
GET	/api/v1/costing/closes/{id}	Progress and exceptions	n/a	CostingInquire
POST	/api/v1/costing/closes/{id}:cancel	Cancel/reverse	close id	InventoryCloseRun
GET	/api/v1/costing/value-report?asOf=&groupBy=&site=	Inventory value with GL tie-out	n/a	CostingInquire
POST	/api/v1/costing/adjustments	On-hand or transaction adjustment	client key	CostAdjust
POST	/api/v1/costing/versions	Create costing version	client key	CostingVersionMaintain
POST	/api/v1/costing/versions/{id}/item-costs:import	Import pending costs (CSV/JSON)	file hash	CostingVersionMaintain
POST	/api/v1/costing/versions/{id}:activate	Activate pending costs with revaluation	version id	CostingVersionActivate

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/costing/products/{id}/cost?site=&date=	Current cost (used by waves 21, 27)	n/a	CostingInquire
POST	/internal/v1/costing/ppv	Hook from receiving/payables for STANDARD items	source doc id	service account
GET	/portal/v1/purchase-orders	Vendor's POs	n/a	portal: VendorUser
POST	/portal/v1/purchase-orders/{id}:acknowledge	Accept/reject/propose	po+version	portal: VendorUser
POST	/portal/v1/asns	Submit ASN	client key	portal: VendorUser
POST	/portal/v1/invoices	Submit invoice (→ payables capture)	vendor invoice number	portal: VendorUser
GET	/portal/v1/payments	Payment status and advices	n/a	portal: VendorUser
POST	/portal/v1/users:invite	Vendor admin invites user	email	portal: VendorAdmin

Commands: `StartClose` (period open, no running close for entity, close_date ≥ last close), `ActivateVersion` (no block flag, all pending costs > 0 unless `allow_zero_cost`, site in scope), `Acknowledge` (latest version only; proposals require at least one changed line), `SubmitASN` (PO confirmed, qty ≤ remaining × (1 + over_delivery_pct), serial count = qty for serial items).

Event	Trigger	Payload fields	Consumers
`dev.passage.costing.close.completed.v1` / `.cancelled.v1` / `.failed.v1`	close lifecycle	close_id, entity, close_date, kind, products, settlements, adjustment_total, duration_s, exceptions_count	ledger reconciliation, notifications, analytics
`dev.passage.costing.cost.changed.v1`	activation, manual adjustment affecting unit cost	product_id, site_id, version, old_cost, new_cost, effective_from, revaluation_amount	inventory (cost price cache), planning (w22), product costing (w39)
`dev.passage.costing.reconciliation.diff.v1`	daily tie-out diff ≠ 0	account, dimension, inventory_value, ledger_balance, diff, sample vouchers	alerting
`dev.passage.supplier.po.acknowledged.v1`	portal ack	po_id, version, response, proposed changes	procurement (pending change), notifications
`dev.passage.supplier.asn.submitted.v1`	ASN submit	asn_id, po_id, lines, expected_arrival, tracking	receiving (expected lines), warehouse (w31), notifications
`dev.passage.supplier.invoice.submitted.v1`	portal invoice	invoice_id (payables), vendor_id, po_id, total	payables (already captured), notifications

3.20.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Costing workspace	Last close status, reconciliation diffs, pending costing versions, products with negative value or unsettled issues	counts, amounts, durations	CostingInquire
Inventory close	Pre-check, start, monitor (progress bar per product), exceptions list, cancel	close date, kind, parallelism, exceptions (product, reason, qty)	InventoryCloseRun
Inventory value report	Group by, as-of date, GL tie-out column, drill to cost transactions and vouchers, export	product, site, qty, physical value, financial value, GL balance, diff	CostingInquire
Costing version / item costs	Create, import, copy with %, compare to active, activate with revaluation preview	product, site, pending cost, active cost, delta, on-hand qty, revaluation amount	CostingVersionMaintain
Cost adjustment	Adjust on-hand or receipt, preview voucher, post	product, dims, qty, current value, new unit cost, delta	CostAdjust
Supplier portal: home	Open POs needing ack, ASNs, invoices in progress, payments	counts, due dates	portal VendorUser
Supplier portal: PO	View versions, accept/reject/propose per line, print	lines, dates, prices, version diff	portal VendorUser
Supplier portal: ASN	Create from PO lines, add packages, batches/serials, submit	PO lines, qty, batch, expiry, serials, tracking	portal VendorUser
Supplier portal: invoices / payments	Submit invoice from receipts, status timeline, payment advices	invoice no, amount, status, payment date, reference	portal VendorUser
Supplier portal: users	Invite, deactivate, MFA	email, role, status	portal VendorAdmin

Interaction rules: the close screen refreshes progress via server-sent events every 2 s; value report defaults to the last close date; activation shows a mandatory revaluation preview with total and top-20 deltas; portal is WCAG 2.1 AA, mobile-friendly, English + German + French + Spanish strings in v1, no internal data (variance amounts, standard costs, approvals) exposed; portal file uploads scanned before storage.

3.20.7 Business rules and edge cases

- **BR-20.1** Close state machine: `PRECHECK → SETTLING → POSTING → COMPLETED`; `PRECHECK → PRECHECK_FAILED`; `SETTLING|POSTING → FAILED` (resumable); `COMPLETED → CANCELLING → CANCELLED` only when it is the latest close for the entity.
- **BR-20.2** Settlement determinism: receipts are ordered by (financial_date, physical_date, id) and issues by (financial_date, physical_date, id); the same inputs always produce the same `CostSettlement` sequence (property-tested), so re-runs after crashes are idempotent.
- **BR-20.3** Unsettled issues (negative on-hand at settlement time) are valued at the running average of the product × site (fallback: active standard, then last receipt cost) and listed as exceptions; a later receipt settles them at the next close with an adjustment.
- **BR-20.4** Rounding: settlement values are computed in full precision and rounded half-even to the minor unit per settlement row; the last settlement row of a receipt absorbs the rounding residual so Σ settled value = receipt value exactly.
- **BR-20.5** Periodic weighted average with transfers between sites: the average is per product × site (`costing_level=SITE`, default) or per product (`costing_level=ENTITY`) as configured on the item model group; transfers at site level settle like issues/receipts.
- **BR-20.6** Invoice difference after close: an invoice posted after the receipt's period was closed posts the difference to the invoice's period as `INVOICE_DIFF` and flows to issues in the next close (never reopens a closed period).
- **BR-20.7** Standard cost revaluation uses the physical on-hand quantity at activation time; transactions already financially posted are not restated; activation during an in-flight close is rejected (`CST-ACT-004`).

- **BR-20.8** Close date must be ≤ today and ≥ the last completed close; transactions with `financial_date ≤ close_date` are locked against new cost adjustments except through `:cancel`.
- **BR-20.9** Portal acknowledgement applies to the latest PO version only; a new version invalidates pending acknowledgements (status SUPERSEDED) and re-requests one.
- **BR-20.10** ASN quantity may exceed PO remaining only within the over-delivery tolerance (wave 18); duplicate tracking number per vendor is rejected; an ASN can be cancelled until any of its lines is RECEIVED.
- **BR-20.11** Portal invoice submission is limited to lines with received, uninvoiced quantity; the vendor cannot submit a price different from the PO price unless the buyer enabled `portal_price_override` for the vendor (creating a price-variance exception in wave 18).
- **BR-20.12** Concurrency: close child workflows take an advisory lock per product; wave 16 journal posting for a product whose close child is settling waits up to 5 s then fails with `CST-LOCK-001` (retry); costs read by waves 21/27 are the last committed values.

3.20.8 Acceptance criteria

- **AC-20.1** (FR-20.1) Given the 100 receipt/issue sequences per method fixture, when settled by close, then every issue cost, remaining on-hand value and adjustment equals the reference workbook to the cent for FIFO, LIFO, LIFO_DATE, WEIGHTED_AVG_PERIODIC, WEIGHTED_AVG_RUNNING, MOVING_AVG and STANDARD.
- **AC-20.2** (FR-20.1) Given the FR-20.1 example data, when closed under FIFO, LIFO_DATE and WEIGHTED_AVG_PERIODIC, then issue cost is 1,600.00, 1,700.00 and 1,650.00 and remaining value 600.00, 500.00 and 550.00.
- **AC-20.3** (FR-20.2) Given the 1M-transaction fixture, when a close runs on stage, then it completes in < 20 min, and the value report at close date equals the GL inventory balances per site and item group with diff 0.00.
- **AC-20.4** (FR-20.7) Given a STANDARD item at 10.00 with a PO at 10.50 and invoice at 10.80 for 100 units, when received and invoiced, then PPV postings are 50.00 and 30.00 Dr, inventory value is 1,000.00, GR/IR nets to zero.
- **AC-20.5** (FR-20.5) Given 10,000 pending item costs, when the version is activated, then all are ACTIVE within 2 min, one revaluation voucher per site balances to `Σ (new − old) × on-hand`, and `cost.changed` is emitted per product.
- **AC-20.6** (FR-20.2) Given a close killed during SETTLING at 40 %, when resumed, then it completes with the same settlement rows as an uninterrupted run (byte-equal export).
- **AC-20.7** (FR-20.9) Given a confirmed PO visible in the portal, when the vendor proposes a delivery date change, then a pending change appears on the buyer's PO and accepting it creates version 2 and emits `po.version.created`.
- **AC-20.8** (FR-20.10) Given an ASN for 2 batch-tracked lines, when the goods arrive and the receiver opens the receipt from the ASN, then quantities, batches and expiry dates are pre-filled and registration takes one confirmation.
- **AC-20.9** (FR-20.12) Given a user of vendor A, when they request vendor B's PO by id, then the portal returns 404 and the OPA decision log records the denial.
- **AC-20.10** (FR-20.14) Given the demo plant, when the phase-exit walkthrough is executed, then every step completes on stage and the final value report, GR/IR report and AP aging each reconcile to the GL with 0.00 difference.

3.20.9 Test plan and fixtures

Fixtures: `cost-methods-700` (100 sequences × 7 methods with reference workbook `cost_reference.xlsx` maintained by the accountant; includes negative on-hand, transfers, invoice differences, returns); `cost-1m` (reuses `inv-1m-txn` with receipts/invoices from waves 17–18 generators); `cost-std-activation-10k`; `portal-demo-vendor` (vendor with 3 users, 20 POs, 5 ASNs); `cost-ax-extract` (InventClosing, InventSettlement, InventItemPrice from USMF); `cost-sap-extract` (MBEW, CKMLHD/CKMLPP/CKMLCR samples).

Test	Type	Target	Pass criterion
Method correctness	unit (table-driven)	settlement engine	AC-20.1
Determinism/idempotency	property	engine	identical inputs → identical settlements; resume = uninterrupted
Close performance	performance (stage)	InventoryClose	< 20 min on cost-1m
Value report tie-out	E2E	value-report vs ledger	diff 0.00 after close
PPV	contract	receiving/payables → costing/ppv	AC-20.4
Activation	E2E	versions:activate	AC-20.5
Portal authz	security (OPA unit + E2E)	portal APIs	AC-20.9; no cross-vendor leakage across 1,000 random requests
Portal accessibility	E2E (axe)	portal screens	0 serious/critical violations
Phase-exit walkthrough	E2E (Playwright, recorded)	full flow	AC-20.10

Reconciliation checks: Σ CostTransaction per inventory account = ledger balance; Σ settlement value per receipt = receipt value; Σ PPV cost transactions = PPV account movement; revaluation voucher = Σ item deltas × on-hand; portal invoice totals = payables captured totals.

3.20.10 Deployment and operations

Helm charts: `passage-costing` (API ×2, Temporal close workers ×4 with CPU 2 each, schema `costing` in `inventory-db`), `passage-supplier-portal` (React app served by the shell at `supplier.<env>.passage.dev`, BFF ×2 in TypeScript, CloudNativePG `supplier-db`, Keycloak realm `suppliers` provisioned by Helm hook, Envoy Gateway route with rate limiting and WAF rules). Config: `CST_CLOSE_PARALLELISM=32`, `CST_LOCK_WAIT_S=5`, `CST_LEVEL_DEFAULT=SITE`, `CST_ALLOW_ZERO_COST=false`, `PORTAL_RATE_LIMIT_USER=60`, `PORTAL_UPLOAD_MAX_MB=25`, `PORTAL_ACK_DUE_DAYS=2`. Migrations `inventory/v20.*` (costing schema), `supplier/v20.*`. Flags: `cst.moving_avg`, `cst.marking`, `portal.rfq_replies`, `portal.invoice_submit`, `portal.mfa_required`. Jobs: reconciliation daily 02:15; ack-overdue scan daily 08:00; portal inactive-user cleanup weekly. SLOs: value report p95 < 5 s; cost lookup p95 < 20 ms; close < 20 min per 1M txns; portal page load p95 < 1.5 s; portal availability 99.9 %. Alerts: reconciliation diff ≠ 0 (page); close FAILED (page); close running > 60 min (ticket); portal auth failures > 100/min from one IP (security ticket, auto-block). Runbook: resume close, cancel close, re-run reconciliation, rotate portal realm client secret, revoke vendor user. Rollback: additive migrations; costing schema can be disabled (flag `cst.enabled=false` reverts to wave 16 running estimate without data loss); portal can be taken down independently (DNS) without affecting internal flows.

3.20.11 Migration from AX / SAP

Source	Passage entity	Notes
AX InventModelGroup.InventModel	ItemModelGroup.costing_method	FIFO→FIFO, LIFO→LIFO, LIFODate→LIFO_DATE, WeightedAvg→WEIGHTED_AVG_PERIODIC, WeightedAvgDate→WEIGHTED_AVG_RUNNING, StdCost→STANDARD, MovingAverage→MOVING_AVG
AX InventItemPrice (active, PriceType Cost) per CostingVersion	ItemCost ACTIVE in version `MIG-STD`	per site; pending prices migrated as PENDING
AX InventSum PostedValue / wave 16 opening	opening cost layers: one receipt layer per product × dimension at average cost	FIFO/LIFO layer detail from last InventClosing unsettled receipts (InventTrans with open qty) when available, else collapsed to one layer
AX InventClosing history	not migrated	reporting only
SAP MBEW (VPRSV, STPRS, VERPR, LBKUM, SALK3, PEINH)	costing_method (S→STANDARD, V→WEIGHTED_AVG_RUNNING or MOVING_AVG per decision), ItemCost (STPRS/PEINH), opening layer value SALK3	per plant (BWKEY) → site
SAP CKMLPP/CKMLCR (material ledger period values)	opening layer value when ML active	actual price of last closed period
SAP MLCD / CKMLHD history	not migrated	
D365 vendor collaboration users / Ariba supplier accounts	SupplierUser invitations (not credentials)	vendors re-register via invitation email

Validation: Σ opening layer value per product × site equals source value (0.01 tolerance per product, 0.00 total per account); standard costs for 200 random products equal source; costing method distribution equals source counts. Not migrated: settled cost history, SAP ML price analysis, AX recalculation logs, portal documents.

3.20.12 Dependencies, risks, open questions

Depends on: wave 16 (transactions, cost fields, marking field), wave 17 (PO prices, versions, pending changes), wave 18 (receipts, invoice differences, capture), wave 19 (payment status), wave 11–14 (posting, period control, vouchers), wave 4 (Temporal), wave 2–3 (Keycloak realms, OPA, metadata), wave 7 (vendor contacts).

Risk	Likelihood	Impact	Mitigation
Opening cost layers from AX/SAP are collapsed, so first FIFO/LIFO close differs from legacy	High	Medium	Document in cutover; parallel-run compare in wave 49; collapsed layer flagged in value report
Close parallelism contends with daytime posting	Medium	Low	Run off-hours by default; lock wait 5 s with retry; advisory locks per product only
Portal security incident	Low	High	Separate realm/DB, OPA deny-by-default, pen-test before phase exit, WAF, rate limits, audit
Accountant requires LIFO by period layers (US tax) beyond LIFO_DATE	Medium	Low	Periodic LIFO pools deferred; documented as v1.x

Open questions:

1. SAP MAP (V) maps to WEIGHTED_AVG_RUNNING or MOVING_AVG by default? Default: MOVING_AVG (closest to VERPR behavior on invoice differences). Decide by wave 20 mid-point.
2. Should the portal expose expected payment dates (from proposals) or only executed payments? Default: executed only. Decide by wave 20 close.
3. Costing level ENTITY vs SITE default for weighted average? Default: SITE. Decide by wave 20 mid-point with the accountant.
4. Dollar-value LIFO pools for US tax reporting: v1.x or wave 39? Default: v1.x. Decide by wave 39 planning.

Part 4 — Phase 4: Manufacturing & Planning (waves 21–25)

4.0 Phase overview

Business outcome at phase exit. A discrete or process manufacturer can run its plant on Passage: master planning nets demand against supply across a multi-level BOM and generates planned purchase, production and transfer orders; planners firm those into production or batch orders, schedule them against finite resource capacity on a Gantt, release them to the floor, and operators execute them from a kiosk while machines report counts over OPC-UA/MQTT. Every material issue, route consumption and report-as-finished posts WIP through the universal journal (wave 11) and the inventory ledger (wave 16); order end settles actual against planned cost and posts variances. Lean plants run kanban replenishment and production flows with periodic backflush costing. The phase-exit demo (wave 25 proof) manufactures a configured product from planned order to finished goods with full costing, and the WIP, inventory and variance accounts reconcile to the journal.

Phase-exit sign-off criteria

- 100% of production order status transitions in the wave-21 state machine are covered by a property test; every illegal transition is rejected and every legal one posts the fixture-defined inventory and GL entries.
- WIP control account balance = Σ(open production orders' posted consumption – posted RAF value) to the cent, verified by the nightly subledger reconciliation job for 30 consecutive days in `stage`.
- The 40 APICS-style MRP fixtures (wave 22) pass; a 50k-item × 12-period × 5-level plan completes in < 10 min on the 3-node reference cluster and produces byte-identical planned orders on re-run.
- Scheduler produces feasible, non-overlapping schedules on all 30 OR-Library job-shop instances; 5k operations scheduled in < 60 s; CP-SAT mode within 5% of known optimum on instances ≤ 10×10.
- Process fixtures: potency consumption, co/by-product allocation (sum equals batch cost exactly), catch-weight dual quantities, and the event-kanban simulation pass.
- Kiosk E2E: an 8-hour shift of 6 operators clocks jobs offline for 30 min, syncs without duplicates, and the production order's labor cost equals Σ(hours × pay rate) from the approved registrations.
- IoT: a simulated machine publishes 10,000 events/min for 60 min with a broker restart at minute 30; zero events lost (sequence check); OEE tile lag p95 < 5 s.
- A practicing cost accountant signs off the WIP/variance fixture set (50 orders) against a reference workbook.

Cross-wave data model summary

Entity	Owning service (Postgres DB)	Key relationships
ProductionOrder	`production` (production_db)	→ ReleasedProduct (w8), BomVersion/RouteVersion (w9), Site/Warehouse (w6), PlannedOrder (w22), SalesOrderLine (w27, pegged)
ProductionBomLine, ProductionOperation, ProductionJob	`production`	child of ProductionOrder; ProductionJob → Resource (w9)
ProductionJournal (+ lines)	`production`	→ ProductionOrder; posts to InventoryTransaction (w16) and JournalEntry (w11)
CostEstimate, WipPosting, Variance	`production`	→ ProductionOrder; → JournalEntry
BatchOrder, Formula, CoProductLine, BatchAttribute, CatchWeightQty	`production` (process extension)	BatchOrder is a ProductionOrder subtype; → InventoryBatch (w16)
PlanVersion, Requirement, PlannedOrder, Pegging, ActionMessage, CoverageGroup, ForecastLine	`planning` (planning_db)	Requirement ↔ Pegging ↔ Requirement; PlannedOrder → ProductionOrder/PurchaseOrder/TransferOrder on firming
ResourceCapacity, ScheduledJob, ScheduleRun	`scheduling` (scheduling_db)	ScheduledJob → ProductionJob; ResourceCapacity → Resource + WorkCalendar (w9)
ValueStream, ProductionFlow, FlowActivity, KanbanRule, Kanban, KanbanCard	`lean` (lean_db)	Kanban → ProductionOrder (manufacturing kanban) or TransferOrder (withdrawal)
Worker, TimeRegistration, PayLine, ProfileDay	`shopfloor` (shopfloor_db)	TimeRegistration → ProductionJob; PayLine → payroll export (w43)
MachineEvent, OeeBucket, DowntimeRecord	`shopfloor`	MachineEvent → Resource; OeeBucket aggregated per resource × 15 min

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`passage.production.order.status-changed.v1`	production	planning, scheduling, shopfloor, lean, analytics	orderId, from, to, qty, dates, actor
`dev.passage.production.journal.posted.v1`	production	inventory (w16), ledger (w11), planning	journalId, type, lines[item, qty, cost], journalEntryId
`dev.passage.production.order.settled.v1`	production	ledger, costing (w20), analytics	orderId, actualCost, plannedCost, variances[]
`dev.passage.planning.plan.completed.v1`	planning	production, purchasing (w17), scheduling	planVersionId, counts, duration, messages
`passage.planning.planned-order.firmed.v1`	planning	production, purchasing, inventory	plannedOrderId, targetType, targetId
`dev.passage.scheduling.job.scheduled.v1`	scheduling	production, shopfloor, planning	jobId, resourceId, start, end, runId
`passage.lean.kanban.state-changed.v1`	lean	production, inventory, shopfloor	kanbanId, from, to, handlingUnits
`dev.passage.shopfloor.registration.posted.v1`	shopfloor	production, payroll (w43)	registrationId, workerId, jobId, hours, payType
`dev.passage.shopfloor.machine.event.v1`	iot-edge → shopfloor	shopfloor (OEE), production, analytics	machineId, resourceId, kind, counts, state, ts

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
WIP postings drift from the journal under concurrent journal posting on one order	Medium	High	Per-order advisory lock; idempotency key per journal; nightly WIP reconciliation job with alert at any non-zero delta	Engineering
MRP non-determinism from map iteration / parallelism	Medium	High	Deterministic sort keys (item, date, source id) at every stage; plan hash in `plan.completed` event; CI compares hashes across 3 runs	Engineering
Scheduling heuristic produces infeasible schedules on calendar edge cases (DST, zero-capacity days)	Medium	Medium	Calendar materialized in UTC minutes per resource; property test "no job outside open capacity" over 10k jobs	Engineering
Process-industry semantics (potency, co-products) misunderstood	Medium	High	Fixtures reviewed by a process-industry consultant before wave 24 close	Product
IoT connector drops events on broker restart	Low	High	Edge buffer (SQLite ring, 24 h) + Kafka idempotent producer + per-machine sequence numbers	Engineering
Two-person team on five engine-heavy waves	High	High	Reuse OR-Tools, statsforecast, frePPLe comparison harness; cut scope per the open questions rather than slip proofs	Product

4.21 Wave 21: Production Orders, WIP & Order-Based Costing

4.21.1 Scope and business context

Wave 21 is where Passage first consumes components and produces finished goods. Before it, the demo plant buys, receives, pays and values inventory (phase 3) but cannot transform it. After it, a production supervisor can create a production order for a BOM/route item, estimate its planned cost, release it, start it (auto-consuming materials by backflush or manually via a picking list), report route time on route and job cards, report quantities as finished including by-/co-products and scrap, and end the order so that actual cost settles against planned cost with material, labor and overhead variances posted to the GL. Subcontracted operations create a linked purchase order to the vendor.

Out of scope: scheduling against capacity (wave 23; wave 21 schedules with infinite capacity only), MRP-generated orders (wave 22), formula/batch orders, co-product allocation methods beyond the basic by-quantity method (wave 24), operator kiosk and time registration (wave 25), standard-cost roll-up and activation (wave 39; wave 21 reads an active standard or weighted-average cost from wave 20).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX ProdTable status update dialogs (Estimate → Schedule → Release → Start) each open a modal with 8–15 parameters (ProdParmStartUp, ProdParmRelease); a supervisor clicks ~40 times to take one order to Started; SAP CO02 needs CO01 → release → CO11N → MIGO 261 across four transactions	Each status is a separate parameter table and batch class (ProdMultiStartUp); no sensible defaults persisted per site	One `:start` command with site-level defaults; parameter overrides only in an "advanced" drawer; bulk start on the list page	Clicks per order to Started 40 → 3; bulk 500 orders in < 5 min
WIP vs GL drift is found at month-end: AX ProdCalcTrans and the ledger disagree after a failed ProdUpd posting left half a journal; SAP KKAX WIP calculation runs as a batch (CO88 settlement) hours before close	Non-atomic multi-table posting (ProdJournal → InventTrans → LedgerTrans) without idempotency; WIP computed periodically rather than per posting	Journal post is one transaction with idempotency key; WipPosting rows per journal; nightly reconciliation job, delta alerts	WIP reconciliation effort 2 days/month → 0; drift detected < 24 h instead of at close
Backflush quantities silently rounded: AX BOM lines with `Round-up` = No and InventTable unit precision truncate 0.3333 to 0.33, accumulating variance; SAP RESB rounding profile (MARC.RDPRF) applied inconsistently to backflush	Rounding applied at explosion, not at posting; precision taken from the unit not the line	BR-21.2 rounds at posting with 6-decimal internal precision and explicit per-line round-up flag; variance of rounding reported	Rounding-induced variance < 0.01% of material cost (today 0.3–1%)
Cost analysis requires re-keying into Excel: AX "Price calculation" form (ProdCalcTrans) and SAP CO03 cost itemization have no drill to the GL line; auditors re-key order costs into spreadsheets	Cost itemization stored separately from the ledger posting; no shared trace id	Costs tab drills from variance → WipPosting → JournalEntry (w15 drill) with one trace id; `/costs` API for BI	Time to explain one order's variance 30 min → < 1 min; 100% of cost lines link to a journal entry
Epicor Kinetic and Infor LN require a separate MES (Plex, Infor MES) to see live order progress; AX users run the ProdTable list with manual refresh	Status and quantities updated only by batch posting classes	`order.status-changed` and `journal.posted` CloudEvents on Kafka within 1 s; any MES/BI consumes them without middleware	Order-status latency to external systems 15 min polling → < 2 s event

4.21.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Production order header	ProdTable, ProdStatus enum, form ProdTable, menu item ProdTable	AUFK/AFKO/AFPO, CO01/CO02/CO03	`production` service, entity ProductionOrder, screen Production orders	full
Order BOM and route snapshot	ProdBOM, ProdRoute, ProdRouteJob (ProdUpdCreate explosion)	RESB (components), AFVC/AFVV (operations), order read of BOM/routing	ProductionBomLine, ProductionOperation, ProductionJob	full
Estimation (planned cost)	ProdParmCostEstimation, ProdCalcTrans, status Estimated	Preliminary costing (CO01 → cost), KKF6N	CostEstimate entity, `:estimate` action	full
Picking list	ProdJournalTable (JournalType=Bom), ProdJournalBOM, backflush via BOM line flushing principle	Goods issue MIGO 261, backflush at CO11N	ProductionJournal type `picking-list`	full
Route card / job card	ProdJournalRoute (route card), ProdJournalRoute by job (job card), ProdParmReportFinished	CO11N confirmation (AFRU), CO15 order confirmation	ProductionJournal types `route-card`, `job-card`	full
Report as finished	ProdJournalProd, ProdParmReportFinished, accept error quantity	Goods receipt MIGO 101 for order, CO15 final confirmation	ProductionJournal type `report-as-finished`	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
WIP posting	Production posting profiles (ProdPosting), ledger accounts WIP issue/receipt, Picking list issue/offset	WIP calculation KKAX/KKAO, results analysis (RA key)	WipPosting rows, posting profile per item group × site	full
Order end and settlement	ProdParmCostUpdate (End), ProdCalcTrans actual, variance accounts	Settlement KO88/CO88, variance calculation KKS2	`:end` action, Variance entity, settlement to journal	full
Subcontracting	Vendor resource type, BOM line type Vendor, auto PO	External processing operation (PP02 control key), PO item category L	ProductionOperation.subcontract → PurchaseOrderLine (w17)	full
Production order inquiry	ProdTable listpage, ProdCalcTrans inquiry, ProdJournal inquiries	COOIS, CO03 cost analysis	Production order detail, Cost tab, Status log	improved (single trace id per transaction)
Scrap and error quantity	ProdJournalProd.QtyError, BOM line scrap %, route scrap %	Assembly scrap (MARC), component scrap (STPO), operation scrap	Scrap fields on BOM line, operation and RAF line	full
Reset status	ProdTable reset status (ProdUpdStatusDown)	Not native (reversal via CO13 cancel confirmation)	`:reset-status` with reversal journals	full
Production pools / groups	ProdPool, ProdGroup	Production scheduler profile, order type	OrderGroup (posting profile key)	partial — pools dropped; replaced by saved views

4.21.3 Functional requirements

- **FR-21.1 Create production order** — Must. The system shall create a ProductionOrder for a released product with a BOM version and route version resolved as of the order's delivery date and site, and explode them into ProductionBomLine, ProductionOperation and ProductionJob rows at creation.
  - Notes: Order number from the `production-order` number sequence (continuous not required). Default quantity from released-product default order settings; multiple/min/max enforced. Manual BOM/route version override allowed only before Estimated. Phantom BOM lines are exploded in place; line numbering keeps the parent's line number with a suffix (`30.1`, `30.2`). Explosion of a 7-level, 2,000-component product completes in < 1 s (wave 9 explosion API).
- **FR-21.2 Status lifecycle** — Must. The system shall enforce the state machine in BR-21.1 with statuses Created, Estimated, Scheduled, Released, Started, ReportedAsFinished, Ended and Cancelled.
  - Notes: Transitions are commands (`:estimate`, `:schedule`, `:release`, `:start`, `:report-finished`, `:end`, `:cancel`, `:reset-status`); each is idempotent by `Idempotency-Key`. Each transition writes a StatusLog row with actor, timestamp, trace id. SAP mapping on export: CRTD→Created, REL→Released, PCNF→Started, CNF/DLV→ReportedAsFinished, TECO/CLSD→Ended.
- **FR-21.3 Estimation** — Must. The system shall compute planned cost per order: material (Σ BOM line qty × (1 + scrap%) × unit cost from the active costing version), labor (Σ operation setup + run time × resource cost category rate), overhead (costing-sheet surcharges from wave 20; full costing sheets in wave 39), and subcontract (PO price).
  - Notes: Creates inventory transactions: receipt `ordered` for the finished item, issues `on-order` for BOM lines (w16 statuses). Re-estimate is allowed until Started and replaces the estimate atomically. Rounding: cost amounts to 4 decimals internally, 2 on posting.
- **FR-21.4 Release and printing** — Must. The system shall release an order, generate the picking list and job card documents (PDF), and set BOM-line reservations according to the order's reservation policy (none, estimated, released, started).
  - Notes: Release is blocked if any BOM line has an unreleased product in this legal entity. Release with shortages raises a warning listing short lines; a `strict` flag turns it into an error.
- **FR-21.5 Picking list journal** — Must. The system shall post picking-list journals that issue BOM-line quantities from inventory to WIP, with proposal modes: remaining quantity, consumption proportional to RAF quantity, or manual.
  - Notes: Flushing principle per BOM line: `manual`, `start` (consumed at Start), `finish` (consumed at RAF), `available-at-location` (consumed when job reports). Over-consumption beyond 100% + line tolerance (default 10%) requires duty *Post production over-consumption*. Posting creates one InventoryTransaction per line with status `deducted` and the GL entry Dr WIP / Cr Inventory.
- **FR-21.6 Route card and job card journals** — Must. The system shall post route consumption (setup hours, run hours, quantity good, quantity error) per operation (route card) or per job (job card), valued at the resource's cost category rate, posting Dr WIP / Cr Labor absorption (and Cr Overhead absorption for indirect costs).
  - Notes: Hours in decimals to 2 places; minimum postable 0.01 h. Reporting on an operation of a non-Started order auto-starts the order if the order's parameter `autoStartOnFeedback` is true (default true, AX parity). Operation `finished` flag closes it to further feedback unless reopened.
- **FR-21.7 Report as finished** — Must. The system shall post RAF journals that receive the finished item (and by-/co-product lines) into inventory at the estimated unit cost (standard cost if the item's costing method is standard), consume `finish`-flushed BOM lines, and post Dr Finished goods / Cr WIP.
  - Notes: `acceptError` flag posts error quantity as scrap without inventory receipt. RAF quantity > remaining + tolerance (default 0%) is rejected. The last RAF sets `endJob = true` on all jobs when the `endJobs` parameter is set. Tracking dimensions (batch/serial) required per the item's tracking dimension group; serials must be unique across the tenant (w16).
- **FR-21.8 End and settle** — Must. The system shall end an order by computing actual cost from posted journals, reversing WIP, posting actual cost to the consumption and inventory accounts, and posting variances per BR-21.6.
  - Notes: End date must be ≥ last journal date and in an open inventory and GL period (w14 period control). Remaining `on-order` and `ordered` inventory transactions are cancelled. Orders with RAF quantity 0 and posted consumption are ended with 100% of cost to variance (AX "end with zero quantity" behavior) and require duty *End production order with zero output*.
- **FR-21.9 Reset status** — Should. The system shall reset an order to an earlier status, reversing postings made by later statuses (reversal journals with negative quantities, linked to originals).
  - Notes: Reset from Ended is not allowed (AX parity); reset from ReportedAsFinished to Started reverses all RAF journals. Reset is blocked if any reversed inventory transaction has been consumed by a later issue (on-hand would go negative).
- **FR-21.10 Subcontracting** — Should. The system shall create a purchase order line (w17) for an operation whose resource is of type Vendor, linked to the operation, and shall post the vendor invoice amount as the operation's actual cost on product receipt.
  - Notes: PO line quantity = operation quantity in the service item's UoM; delivery date = operation end. Changes to order quantity after PO confirmation raise a PO change (w17 versioning).

- **FR-21.11 Order inquiry and cost view** — Must. The system shall show, per order, planned vs actual cost by cost group (material, labor, overhead, subcontract), the journal list, inventory transactions, GL entries with drill-down to the journal entry (w15 drill-down) and the status log.
- **FR-21.12 Bulk status update** — Should. The system shall apply a status transition to up to 500 selected orders in one batch job (Temporal), reporting per-order success/failure.
- **FR-21.13 Posting profiles** — Must. The system shall resolve ledger accounts for each posting type (picking list issue, picking list offset, WIP issue, WIP account, RAF receipt, RAF offset, variance accounts per variance type) by item group × site, falling back to All.
- **FR-21.14 Co-product RAF (basic)** — Could. The system shall accept RAF lines for co-/by-products declared on the BOM (w9), allocating cost by quantity only; allocation methods arrive in wave 24.

4.21.4 Data model

ProductionOrder

Field	Type	Required	Notes
id	uuid	yes	PK
tenant_id, legal_entity_id	uuid	yes	RLS
order_number	text	yes	unique per legal entity; number sequence
product_id, variant_id	uuid	yes/no	FK ReleasedProduct (w8)
site_id, warehouse_id	uuid	yes	FK (w6)
quantity	numeric(18,6)	yes	> 0, in inventory UoM
qty_started, qty_raf, qty_error	numeric(18,6)	yes	default 0
bom_version_id, route_version_id	uuid	yes	FK (w9), snapshot taken at creation
status	enum	yes	Created…Cancelled
delivery_date, start_date, end_date	date	yes/no/no	start/end set by scheduling
scheduling_method	enum	yes	`operations` or `jobs`
reservation_policy	enum	yes	none/estimated/released/started
posting_profile_key	text	yes	item group × site resolved key
origin_type, origin_id	enum, uuid	no	PlannedOrder, SalesOrderLine, Manual, Kanban
version	int	yes	optimistic concurrency

ProductionBomLine: id, order_id (FK), line_number (text), item_id, variant_id, bom_qty_per (numeric 18,6), bom_series_size (numeric), scrap_pct (numeric 7,4, 0–100), qty_required (numeric 18,6, computed), qty_consumed (numeric 18,6), flushing_principle (enum), line_type (enum: item, phantom, pegged-supply, vendor), warehouse_id, location_id, reservation_status (enum), unit_cost_est (numeric 18,6).

ProductionOperation: id, order_id, oper_number (int), next_oper_number (int, 0 = last), priority (enum: primary, secondary1–4), resource_group_id, resource_id (nullable until scheduled), setup_time_h, run_time_h_per (numeric 10,4), process_qty (numeric), queue_before_h, queue_after_h, transit_h, scrap_pct, cost_category_setup_id, cost_category_run_id, subcontract (bool), purchase_line_id (nullable), status (enum: open, started, finished), scheduled_start, scheduled_end (timestamptz).

ProductionJob: id, operation_id, job_type (enum: queue-before, setup, process, overlap, transport, queue-after), resource_id, planned_start, planned_end, planned_hours, reported_hours, qty_good, qty_error, status.

ProductionJournal: id, order_id, journal_type (enum: picking-list, route-card, job-card, report-as-finished), journal_number, posting_date, posted (bool), posted_at, posted_by, journal_entry_id (w11 FK), reversal_of (nullable self FK), idempotency_key (unique).

CostEstimate / WipPosting / Variance: CostEstimate(order_id, cost_group, planned_qty, planned_unit_cost, planned_amount, costing_version_id); WipPosting(order_id, journal_id, posting_type, amount_dr, amount_cr, ledger_account, journal_entry_id); Variance(order_id, variance_type enum {lot-size, production-quantity, production-price, production-substitution}, cost_group, amount, journal_entry_id).

Invariants:

- Σ WipPosting.amount_dr – Σ amount_cr for an order = 0 once status = Ended.
- qty_raf + qty_error ≤ quantity × (1 + raf_tolerance_pct/100).
- Every posted ProductionJournal has exactly one journal_entry_id and its inventory transactions sum to the journal's quantities.
- A ProductionOperation with subcontract = true has exactly one purchase_line_id once the order is Released.

Indexing: ProductionOrder (tenant_id, legal_entity_id, status, delivery_date); (tenant_id, product_id, status); ProductionJournal unique (tenant_id, idempotency_key); WipPosting (order_id); partition ProductionJournal and WipPosting by legal_entity_id hash (16) once rows > 50M.

4.21.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/production/orders	Create order (explodes BOM/route)	Idempotency-Key header	Maintain production orders
GET	/api/v1/production/orders/{id}	Read order with lines, ops, jobs, cost summary	n/a	View production orders
PATCH	/api/v1/production/orders/{id}	Edit header fields allowed for current status	ETag/version	Maintain production orders
POST	/api/v1/production/orders/{id}:estimate	Estimate	Idempotency-Key	Maintain production orders
POST	/api/v1/production/orders/{id}:release	Release (and :schedule, :start, :end, :cancel, :reset-status)	Idempotency-Key	Release production orders / End production orders
POST	/api/v1/production/orders/{id}/journals	Create journal with proposal (`proposal=remaining	proportional	none`)

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/production/journals/{id}:post	Post journal	Idempotency-Key	Post production journals
GET	/api/v1/production/orders/{id}/costs	Planned vs actual by cost group	n/a	View production costs
POST	/api/v1/production/orders:bulk-status	Batch transition up to 500 ids	Idempotency-Key	Release production orders
GET	/api/v1/production/posting-profiles	Resolve accounts for item group × site	n/a	View production setup

Commands: `CreateOrder` (validates product released, BOM/route active as of delivery date, qty multiples); `PostJournal` (validates order status allows journal type, period open, quantities within tolerance, tracking dims present); `EndOrder` (validates no unposted journals, all subcontract POs invoiced or explicitly accrued).

Event	Trigger	Payload fields	Consumers
`dev.passage.production.order.created.v1`	order committed	orderId, orderNumber, productId, qty, siteId, deliveryDate, originType, originId	planning, analytics
`passage.production.order.status-changed.v1`	any transition	orderId, from, to, qty, startDate, endDate, actor, occurredAt	planning, scheduling, shopfloor, lean
`dev.passage.production.journal.posted.v1`	journal posted	journalId, orderId, type, postingDate, lines[{itemId, qty, unitCost, operNumber, hours}], journalEntryId	inventory, ledger, planning
`dev.passage.production.order.settled.v1`	End posted	orderId, plannedCost{material,labor,overhead,subcontract}, actualCost{…}, variances[{type,costGroup,amount}]	ledger, costing, analytics
`passage.production.subcontract.po-requested.v1`	release of subcontract op	orderId, operNumber, vendorId, serviceItemId, qty, dueDate	purchasing

4.21.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Production orders (list)	Create, Estimate, Release, Start, RAF, End, bulk status, saved views by status/site	Order, Item, Qty, Remaining, Status, Delivery date, Site, Planned cost, Actual cost	View production orders
Production order detail	Tabs: Lines (BOM), Route, Jobs, Journals, Costs, Inventory transactions, Status log	Header fields; per-tab grids	Maintain production orders
Journal editor	Add lines, proposal, validate, post, print	Item/operation, qty, hours, error qty, reason, batch/serial	Post production journals
Cost analysis	Compare planned vs actual, drill to journal entry	Cost group, planned, actual, variance, variance %	View production costs
Posting profile setup	Maintain accounts by item group × site	Posting type, item group, site, account, dimensions	Maintain production setup

Interaction rules: `Ctrl+Shift+S` opens the status action menu; list page supports multi-select with inline status badge; validation runs on blur per field and fully on post; journal editor keeps unsaved lines in `localStorage` and warns on navigation; the Costs tab is read-only and recomputes on each journal post event (live update via SSE from the BFF).

4.21.7 Business rules and edge cases

- **BR-21.1 State machine.** Allowed transitions and their posting effects:

From → To	Preconditions	Posting
Created → Estimated	BOM/route exploded	Inventory transactions `ordered/on-order`; CostEstimate rows
Estimated → Scheduled	estimate exists	Operation/job dates (infinite capacity in w21)
Scheduled → Released	no unreleased components	Reservations per policy; documents
Released → Started	period open	Auto-post picking list for `start`-flushed lines; route card for ops with `autoStart`; Dr WIP / Cr Inventory
Started → ReportedAsFinished	qty_raf > 0	Dr FG inventory (est. cost) / Cr WIP; consume `finish`-flushed lines
ReportedAsFinished → Ended	no open journals; period open	Reverse WIP; post actual; post variances
Started → Ended	qty_raf = 0 and duty held	Same as above with 100% variance
Created/Estimated/Scheduled/Released → Cancelled	no posted journals	Cancel `ordered/on-order` transactions
Any non-Ended → earlier status (`:reset-status`)	reversal possible (BR-21.5)	Reversal journals

Any other transition returns HTTP 409 `PRODUCTION_ILLEGAL_TRANSITION`.

- **BR-21.2 Quantity rounding.** Required quantity = ceil-to-UoM-precision(qty_per × order_qty / series_size × (1 + scrap/100)); UoM precision from the unit definition (w6), default 6 decimals; "round up to whole unit" flag on the BOM line forces integer.
- **BR-21.3 Posting date.** A journal's posting date must be ≥ order creation date and within an open inventory period and GL period for the Production module; otherwise 422 `PERIOD_CLOSED`.
- **BR-21.4 Concurrency.** Journal posting takes `pg_advisory_xact_lock(hash(order_id))`; concurrent posts serialize; a stale `version` on PATCH returns 412.
- **BR-21.5 Reversal safety.** Reset status is refused when any inventory transaction created by a to-be-reversed journal has a downstream issue with status `deducted` or `picked`.
- **BR-21.6 Variance decomposition at End** (standard-cost items; weighted-average items post only a total variance to the item's inventory adjustment account):
  - Production quantity variance = Σ (actual qty – planned qty for actual output) × standard price.
  - Production price variance = Σ actual qty × (actual price – standard price).

    - Lot-size variance = planned fixed cost (setup, constant overhead) × (1 – actual output / planned output).
    - Production substitution variance = cost of consumed items/operations not on the estimate minus estimate of items not consumed.
    - Sum of the four equals actual cost – (qty_raf × standard unit cost); property-tested.
- **BR-21.7 Worked settlement example.** Item FG-100, standard cost USD 50.00/unit = material 30.00 + labor 15.00 + overhead 5.00. Order 100 units. Estimate: 300 kg RM @ 10.00 = 3,000.00; 50 h @ 30.00 = 1,500.00; overhead 10.00/h × 50 = 500.00; total 5,000.00. Actual: picking list 305 kg at weighted-average 10.20 = 3,111.00; route cards 46 h = 1,380.00; overhead 46 × 10.00 = 460.00; actual total 4,951.00. RAF 100 units posts Dr Finished goods 5,000.00 / Cr WIP 5,000.00. WIP before End: Dr 4,951.00, Cr 5,000.00, balance Cr 49.00. End posts: material quantity variance Dr 50.00 ((305 – 300) × 10.00), material price variance Dr 61.00 (305 × 0.20), labor quantity variance Cr 120.00 ((46 – 50) × 30.00), overhead quantity variance Cr 40.00; WIP cleared to 0.00; net variance Cr 49.00 (favorable). EUR equivalents are carried on each line at the posting-date rate (w12).
- **BR-21.8 Scrap.** Operation scrap compounds backward along the route: quantity to start at operation n = order qty / Π(1 – scrap_k) for k ≥ n, rounded up at each operation.
- **BR-21.9 Negative inventory.** Picking list posting that would drive on-hand negative is rejected unless the item model group allows physical negative inventory (w16 policy).
- **BR-21.10 Serial/batch at RAF.** A RAF line for a serial-tracked item must carry exactly qty_good serial numbers; duplicates across the tenant are rejected at 422 `SERIAL_DUPLICATE`.

4.21.8 Acceptance criteria

- **AC-21.1** (FR-21.1) Given a released product with a 3-level BOM and a 4-operation route, when I create an order for 100 units, then the order has the exploded lines with phantoms flattened and quantities per BR-21.2.
- **AC-21.2** (FR-21.2) Given an order in Created, when I call `:start`, then I get 409 `PRODUCTION_ILLEGAL_TRANSITION`.
- **AC-21.3** (FR-21.3) Given an estimated order, when I read `/costs`, then planned material equals Σ line qty × unit cost from the active costing version to 4 decimals.
- **AC-21.4** (FR-21.5) Given a Started order with `start`-flushed lines, when I RAF 100 units with proportional consumption, then consumed quantities equal BOM qty × 100 × (1 + scrap%) exactly (wave proof).
- **AC-21.5** (FR-21.7) Given a batch-tracked finished item, when I post RAF without a batch, then the post fails with 422 and no inventory transaction is created.
- **AC-21.6** (FR-21.8) Given the BR-21.7 fixture, when I end the order, then the four variance rows total –49.00 and the WIP account balance for the order is 0.00.
- **AC-21.7** (FR-21.8) Given a partially RAF'd order (60 of 100) ended with duty, then remaining WIP equals posted consumption – 60 × standard cost and reconciles to the GL WIP account.
- **AC-21.8** (FR-21.9) Given an order in ReportedAsFinished whose finished goods were already shipped, when I reset to Started, then the reset is refused with `REVERSAL_BLOCKED_DOWNSTREAM`.
- **AC-21.9** (FR-21.10) Given an operation on a Vendor resource, when I release the order, then a PO line exists for the vendor with the operation quantity and due date.
- **AC-21.10** (FR-21.12) Given 500 Released orders, when I bulk-start them, then the batch completes in < 5 min and the report lists per-order results.

4.21.9 Test plan and fixtures

Fixtures: `prod-orders-50` (50 orders with known consumptions, standard and weighted-average items, generated from the demo plant BOMs; expected WIP and variance postings in a reviewed XLSX); `prod-bom-7level` (from wave 9); `prod-state-machine` (table of all 9×9 transition pairs with expected result); `prod-subcontract-5`.

Test	Type	Target	Pass criterion
State machine	property (Go rapid)	transitions	every illegal transition rejected, every legal one posts expected entries
Settlement fixtures	unit	`settle()`	50/50 orders match workbook to the cent
Backflush	unit	RAF proportional	consumed = BOM × qty × (1+scrap)
Posting contract	contract	ledger/inventory APIs	journals produce balanced entries, idempotent on replay
Period control	contract	journal post	closed period rejected on every journal type
Order E2E	Playwright	create→end	demo order ends with 0 WIP
Journal post latency	performance	200-line picking list	p95 < 2 s

Reconciliation: Σ open-order WIP = GL WIP control account; Σ RAF value = inventory receipts from production (w16) for the period; Σ variances = GL variance accounts.

4.21.10 Deployment and operations

Helm chart `production` (Go service, 3 replicas, HPA on CPU 70%), `production-db` (CloudNativePG, 2 instances, 100 GB). Config: `PRODUCTION_RAF_TOLERANCE_PCT` (0), `PRODUCTION_OVERCONSUMPTION_PCT` (10), `PRODUCTION_AUTOSTART_ON_FEEDBACK` (true). Migrations via Atlas from entity metadata. Feature flags: `prod.subcontracting`, `prod.coproduct-basic`. Scheduled jobs: nightly `wip-reconcile` (Temporal, 02:00 tenant local), `cancel-stale-created-orders` (weekly). SLOs: create p95 < 300 ms; transition p95 < 500 ms; journal post p95 < 2 s; availability 99.9%. Alerts: WIP reconciliation delta ≠ 0 (page); journal post error rate > 1% 5 min (warn). Runbook: "WIP delta" — list orders with unbalanced WipPosting, check outbox lag, replay. Rollback: Argo CD to prior digest; migrations are additive within the wave.

4.21.11 Migration from AX / SAP

Source	Passage entity	Notes
ProdTable (AX) / AUFK+AFKO+AFPO (SAP)	ProductionOrder	status map per FR-21.2; only open orders (status < Ended / not TECO)
ProdBOM / RESB	ProductionBomLine	qty_required recomputed; consumed qty from InventTrans / MSEG 261

Source	Passage entity	Notes
ProdRoute, ProdRouteJob / AFVC, AFVV	ProductionOperation, ProductionJob	reported hours from ProdRouteTrans / AFRU
ProdJournalTable+Bom/Route/Prod / AFRU, MSEG	ProductionJournal (history)	posted journals loaded read-only, no re-posting
ProdCalcTrans / CO cost itemization	CostEstimate (planned), WipPosting (actual, as opening balance)	WIP opening = Σ actual – Σ RAF value, posted as one opening journal per order

Validation: per-order WIP opening = legacy WIP report (AX ProdCalcTrans actual, SAP KKAX) within 0.01; counts of open orders by status match. Not migrated: ended/TECO orders (go to the lakehouse via w46 history load), production pools, legacy posting-profile accounts (re-mapped to the new chart). Recommended cutover: end all orders in legacy that are ≥ 95% complete.

4.21.12 Dependencies, risks, open questions

Depends on: w9 BOM/route explosion and resources; w16 inventory transactions and statuses; w11 posting API; w14 period control; w17 PO creation; w20 costing versions and standard cost.

Risk	Likelihood	Impact	Mitigation
Variance semantics differ between AX (4 types) and SAP (price/quantity/resource-usage/lot-size/…)	High	Medium	Ship AX's four; document SAP mapping; w39 adds decomposition
Backflush under weighted-average produces cost at post time not at End	Medium	Low	Revalue at End using inventory close (w20) settlement

Open questions: (1) Should Cancelled be exposed for Released orders with reservations? Default yes with automatic un-reserve; decide by wave 21 mid-point. (2) Co-product basic allocation in w21 or defer entirely to w24? Default defer; decide at w21 kickoff.

4.22 Wave 22: Master Planning: MRP, MPS & Forecasting

4.22.1 Scope and business context

Wave 22 gives planners the engine that decides what to make, buy and move. Before it, orders are created by hand. After it, a nightly (or on-demand) plan nets all demand (sales orders, forecasts, safety stock, dependent demand, transfer demand) against all supply (on-hand, purchase orders, production orders, transfers, planned orders) level by level through the BOM, applying coverage groups and lead times, and produces planned purchase, production and transfer orders with action and futures/delay messages, each fully pegged to the demand it serves. Planners firm planned orders into real orders, and a statistical forecast (ETS/Croston) seeds demand with manual override and a consensus workflow. Out of scope: finite capacity (wave 23; wave 22 plans infinite), DDMRP buffers, multi-site network optimization, and ATP (wave 27 reuses the engine's supply picture).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Nightly MRP regeneration takes hours: AX ReqCalcTask on 50k items runs 3–6 h with helper threads contending on ReqTrans; SAP MD01 NEUPL regenerative run 2–4 h; Epicor Kinetic "MRP Full Regen" and Infor LN "Order Planning" routinely 4+ h, so planners work on yesterday's plan	Row-at-a-time netting with database round trips per item; planned orders deleted and recreated in place	In-memory level-by-level engine with sparse buckets, snapshot per PlanVersion, pointer swap; dynamic net-change within 60 s of an order event	50k items 4 h → < 10 min; net change 4 h → 60 s
No live pegging: AX ReqTransCov is only populated during the run and lost on re-run; SAP MD04 pegged requirements (MD09) require multi-level order report and are unavailable for planned orders mid-run; Plex "Where Used/Demand" pegging is single-level	Pegging stored as a transient by-product of the run	Pegging rows persisted per version; `/pegging` explain tree to any depth; "Explain this plan" in the UI	Time to answer "why is this planned order here" 20 min → < 5 s; 100% of qty explained
Exception noise: SAP MD06 exception groups and AX action messages generate tens of thousands of rows planners ignore; Infor LN exception messages cannot be filtered by cause	Messages generated for every supply without fences or root-cause grouping; no deduplication	Action/futures fences per coverage group; messages grouped by root cause (one delayed PO → one message with propagated effects)	Messages per planner per run 5,000 → < 300; 90% of messages actionable (survey)
Non-deterministic results: AX plans differ run to run under helper threads; SAP parallel MRP (OMIQ) ordering varies, so planners cannot diff two runs	Parallel processing without stable ordering	Deterministic sort keys and plan hash; version-to-version diff API	Identical input → identical hash 100% of runs; diff of two versions in < 2 s
Forecast re-keying: AX ForecastSales grid vs D365 demand forecasting Excel round-trips; SAP MD61 PIR upload by LSMW; Epicor forecast import by CSV	Forecasting done outside the ERP and pasted back	Statistical baseline in the `forecast` service, in-grid override, consensus workflow, published as a version	Forecast cycle 2 days → 2 h; zero CSV round-trips

4.22.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Plan definition	ReqPlan (static/dynamic), ReqPlanVersion	MRP area, planning run parameters (MD01/MD02 processing keys NETCH/NEUPL)	Plan, PlanVersion	full
Coverage group	ReqGroup (ReqPOType: Requirement/Period/MinMax/Manual)	MRP type (PD/VB/ND), lot-sizing procedure (T438A: EX, TB, FX, WB), MRP group	CoverageGroup	full
Item coverage overrides	ReqItemTable	MARC MRP views 1–4 (DISMM, DISLS, EISBE)	ItemCoverage	full
Requirement transactions	ReqTrans (ReqRefType), ReqPO (planned orders)	MDTB (MRP list), PLAF (planned orders), MD04 stock/requirements list	Requirement, PlannedOrder	full
Pegging	ReqTransCov	Pegged requirements (MD09), order report	Pegging	improved (explain tree API)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Action messages	ReqTrans action (Advance/Postpone/Decrease/Increase/Derived)	Exception messages (MD06, groups 1–8)	ActionMessage	full
Futures/delay messages	ReqTrans futures date	Exception 10/15 (reschedule in/out), 30 (plan process according to schedule)	DelayMessage	full
Low-level codes	BOMCalcLevel / ReqCalcTask level loop	Low-level code (MARA.DISST)	Item.low_level_code	full
Forecast	ForecastSales, ForecastPurch, reduction keys, D365 demand forecasting (Azure ML)	PIR (MD61), consumption mode (MARC.VRMOD), flexible planning/SOP	ForecastLine, forecast service	full
MPS	Master scheduling via item coverage + planning time fence	MPS (MD40), MPS items (MRP type M0–M4)	plan scope `mps` with time fence	full
Firming	ReqTransPO firming to ProdTable/PurchTable	MD14/MD15 convert planned order	`:firm`	full
Intercompany planned orders	Intercompany master planning	Special procurement key 40/70	planned transfer orders only; intercompany in w40	partial

4.22.3 Functional requirements

- **FR-22.1 Plan versions** — Must. The system shall run a plan as a batch job (Temporal) that snapshots inputs into a PlanVersion and writes outputs there, swapping the "current" pointer atomically on success.
  - Notes: Two plan kinds, `static` (nightly, full regeneration) and `dynamic` (net-change, triggered by order events within 60 s, limited to affected items). Regeneration deletes unfirmed planned orders of the version; firmed ones are supply. Max 20 retained versions per plan; older ones purged by nightly job.
- **FR-22.2 Coverage groups** — Must. The system shall support coverage codes Requirement (lot-for-lot), Period (coverage time fence in days, orders grouped per period), Min/Max (reorder to max when PAB < min) and Manual (no planned orders, only messages), with safety stock, positive/negative days, coverage time fence, firming time fence, freeze time fence, action and futures message settings.
  - Notes: Defaults: coverage time fence 100 days, negative days 1, positive days 0 (ignore existing supply earlier than requirement when 0 means unlimited; AX semantics documented in BR-22.4), firming fence 0, freeze fence 0. SAP mapping: PD→Requirement/Period; VB→Min/Max with reorder point = min; ND→Manual.
- **FR-22.3 Netting algorithm** — Must. The system shall compute net requirements per BR-22.1 and produce planned orders with order type by the item's default order settings (purchase, production, transfer, kanban).
- **FR-22.4 Lot sizing** — Must. The system shall apply minimum, maximum, multiple and standard order quantities from default order settings per site, and period grouping.
  - Notes: After rounding up to multiple, if the result exceeds maximum, split into ⌈qty/max⌉ orders. Max 1,000 splits per requirement.
- **FR-22.5 Lead time and calendars** — Must. The system shall offset dates by purchase lead time (vendor-specific from trade agreements, else item), production lead time (route-based: Σ operation times at infinite capacity, else item lead time), transfer lead time, plus safety margins (receipt, issue, reorder margins) using the site's working calendar.
- **FR-22.6 Pegging** — Must. The system shall record Pegging rows linking each unit of supply to the demand it covers, FIFO by requirement date then by priority (sales order > forecast > safety stock), and expose an explain tree.
- **FR-22.7 Action and delay messages** — Must. The system shall generate Advance, Postpone, Decrease, Increase, Cancel and Derived actions for existing supply, and Futures/Delay messages when a planned date falls before today (plus the delay propagates upward to the demand).
- **FR-22.8 Forecast and reduction** — Must. The system shall accept forecast lines by item/site/period (and allocation keys) and reduce them by sales orders using reduction principles: none, percent per period, transactions (forecast minus open orders in the period), transactions-dynamic period.
- **FR-22.9 Statistical forecast** — Should. The system shall generate a baseline forecast per item/site from 24+ months of history using ETS (seasonal/non-seasonal selected by AICc) or Croston/TSB for intermittent items (ADI > 1.32), via the Python `forecast` service, with MAPE/wMAPE reported per item.
  - Notes: Batch for 50k items < 15 min. Manual override per cell; consensus workflow (Temporal approval) locks the forecast version.
- **FR-22.10 Firming** — Must. The system shall firm a planned order into a ProductionOrder (`:estimate` optional), PurchaseOrder line (grouped by vendor/date per setting) or TransferOrder, preserving pegging to the firmed supply.
- **FR-22.11 Planned order maintenance** — Must. The system shall allow planners to change quantity/date/vendor on a planned order, approve it (becomes firmed on next run), or delete it.
- **FR-22.12 Net requirements view** — Must. The system shall show per item/site the MD04-style time-phased list: period, gross requirements, scheduled receipts, planned receipts, PAB, net requirement, with drill to pegging.
- **FR-22.13 Plan scope** — Should. The system shall support plan filters by site, item group, planner, low-level-code range, and `mps` scope (MPS items only; MRP uses MPS output as fixed supply).
- **FR-22.14 Determinism** — Must. The system shall produce byte-identical outputs for identical inputs; the plan hash is emitted in `plan.completed`.

4.22.4 Data model

CoverageGroup: id, tenant_id, code (text, unique), coverage_code (enum), coverage_time_fence_days (int, default 100), negative_days, positive_days (int), safety_stock_policy (enum: today, first-issue), firming_fence_days, freeze_fence_days, action_fence_days, futures_fence_days, period_days (int, Period only), rescheduling_margin_days, generate_actions (bool), generate_futures (bool).

ItemCoverage: id, item_id, site_id, warehouse_id (nullable), coverage_group_id, min_qty, max_qty, safety_margin_receipt_days, lead_time_days (override), vendor_id (default), planned_order_type (enum).

PlanVersion: id, plan_id, number (int), kind (static/dynamic), snapshot_at (timestamptz), status (running/complete/failed/current), items_planned, planned_orders_count, messages_count, duration_ms, plan_hash (bytea, sha256).

Requirement (ReqTrans): id, plan_version_id, item_id, variant_id, site_id, warehouse_id, ref_type (enum: sales-line, forecast, safety-stock, bom-line, transfer-issue, on-hand, purchase-line, production-order, transfer-receipt, planned-order), ref_id, direction (demand/supply), qty (numeric 18,6, +supply/–demand), requirement_date (date), futures_date (date, nullable), low_level_code (int), priority (int).

PlannedOrder (ReqPO): id, plan_version_id, number, item_id, site_id, warehouse_id, order_type (purchase/production/transfer/kanban), qty, order_date, delivery_date, vendor_id, from_warehouse_id, status (enum: unprocessed, approved, firmed), firmed_target_type, firmed_target_id, bom_version_id, route_version_id, lot_rule_applied (text).

Pegging (ReqTransCov): id, supply_requirement_id, demand_requirement_id, qty, is_marking (bool).

ActionMessage: id, requirement_id, action (enum), suggested_qty, suggested_date, reason, delay_days.

ForecastLine: id, model_id, item_id, site_id, period_start, period_end, qty, source (statistical/manual/consensus), reduced_qty, version.

Invariants: Σ Pegging.qty per supply ≤ supply qty; Σ Pegging.qty per demand = demand qty when fully covered (else a `Cancel`-free shortage message exists); PAB(t) = PAB(t–1) + Σ supply(t) – Σ demand(t) for every period; only one PlanVersion per plan has status `current`.

Indexing: Requirement (plan_version_id, item_id, site_id, requirement_date); Pegging (demand_requirement_id), (supply_requirement_id); PlanVersion rows partitioned by plan_version_id (one partition per version, dropped on purge — avoids 50M-row deletes).

4.22.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/planning/plans/{planId}:run	Start plan (static/dynamic, scope filter)	Idempotency-Key; one running version per plan	Run master planning
GET	/api/v1/planning/plans/{planId}/versions/{v}	Version status, counts, hash	n/a	View planning
GET	/api/v1/planning/planned-orders?planId&status&site&itemGroup	List/filter planned orders	n/a	View planning
PATCH	/api/v1/planning/planned-orders/{id}	Change qty/date/vendor; approve	ETag	Maintain planned orders
POST	/api/v1/planning/planned-orders:firm	Firm up to 1,000 ids with grouping options	Idempotency-Key	Firm planned orders
GET	/api/v1/planning/net-requirements?itemId&siteId&planId	MD04-style time-phased view	n/a	View planning
GET	/api/v1/planning/planned-orders/{id}/pegging	Explain tree (depth ≤ 20)	n/a	View planning
GET	/api/v1/planning/messages?planId&type	Action/delay messages	n/a	View planning
POST	/api/v1/planning/forecasts/{modelId}:generate	Statistical forecast batch	Idempotency-Key	Maintain forecasts
PUT	/api/v1/planning/forecasts/{modelId}/lines	Bulk upsert manual lines	full replace per item/site/period	Maintain forecasts
GET	/api/v1/planning/coverage-groups	Setup	n/a	View planning setup

Event	Trigger	Payload fields	Consumers
`dev.passage.planning.plan.started.v1`	run begins	planId, versionId, kind, scope	analytics
`dev.passage.planning.plan.completed.v1`	version becomes current	planId, versionId, itemsPlanned, plannedOrders{purchase,production,transfer}, messages, durationMs, planHash	production, purchasing, scheduling
`passage.planning.planned-order.firmed.v1`	firm	plannedOrderId, orderType, targetId, qty, dates	production, purchasing, inventory
`dev.passage.planning.message.raised.v1`	message created	requirementId, refType, refId, action, delayDays	purchasing, sales (w27), notifications
`dev.passage.planning.forecast.published.v1`	consensus approved	modelId, version, items, horizonPeriods	planning run

4.22.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Master planning workspace	Run plan, view last run, tiles: delayed orders, actions, planned orders by type	Plan, version, status, duration, counts	View planning
Planned orders	Filter, approve, firm (grouped), change, delete, bulk	Number, item, site, type, qty, order/delivery date, vendor, status, delay	Maintain planned orders
Net requirements (MD04)	Time-phased grid, pegging drill, simulate (what-if qty change)	Period, GR, SR, PR, PAB, net req, messages	View planning
Action messages	Apply action to the source order (advance/postpone PO date), dismiss	Action, order, qty, date, suggested date, delay	Maintain planned orders
Forecast grid	Edit cells, generate statistical, submit consensus	Item, site, period columns, source badge, MAPE	Maintain forecasts
Coverage groups / item coverage	Setup	per data model	Maintain planning setup

Interaction rules: planned-orders grid supports keyboard range selection and `F` to firm; cell edits in the forecast grid validate ≥ 0 immediately; the net-requirements grid is virtualized to 365 periods × 20 columns; pegging tree lazy-loads children.

4.22.7 Business rules and edge cases

- **BR-22.1 Netting algorithm.** Per plan run:
  1. Snapshot inputs (on-hand by dimension at physical-available, open orders, forecasts after reduction, safety stock, item coverage, BOM/route versions as of each requirement date, calendars) into the PlanVersion.
  2. Assign low-level codes: LLC(item) = max over all active BOMs of the depth at which the item appears (finished items 0). Recomputed when any BOM changes (w9 event). Cycles are rejected at BOM activation.
  3. For LLC = 0 … max: for each item/site in that level, sort requirements by date then priority then ref id (determinism).

4. Build time buckets by day; PAB_0 = on-hand – safety stock (policy `today`) or on-hand (policy `first-issue`, safety stock added as a demand on the first issue date).
5. For each day t: PAB_t = PAB_{t–1} + SR_t + PR_t – GR_t. If PAB_t < 0 (Requirement/Period) or PAB_t < min (Min/Max), net requirement N = –PAB_t (or max – PAB_t).
6. Lot-size N: Requirement → N; Period → aggregate N over the next `period_days`; Min/Max → max – PAB; then apply min/multiple/max/standard qty.
7. Create planned order with delivery date t – receipt margin; order date = delivery date – lead time (working days); if order date < today → futures_date = today + lead time; raise Delay message with delay_days and propagate the futures date to pegged demand.
8. Peg the planned order to the demand(s) that caused it, FIFO.
9. Explode planned production orders: for each BOM line, create dependent demand Requirement (ref_type bom-line) on the item at date = order start (or operation start when route-based), qty per BR-21.2; these appear at a higher LLC and are processed in step 3 of that level.
10. For existing supply earlier than its earliest pegged demand by more than positive_days (0 = unlimited) raise Postpone; later than demand by more than negative_days raise Advance; unpegged supply beyond the fence raises Cancel/Decrease.
11. Write outputs, compute plan hash over (planned orders, pegging, messages) sorted, mark version current.

- **BR-22.2 Worked example (3-level BOM, weekly buckets, today = Monday of week 1).** FG-100 (LLC 0) uses 2 × SA-200 (LLC 1) and 1 × RM-300; SA-200 uses 3 × RM-300, so RM-300 has LLC 2. Coverage: all Requirement; FG-100 lead time 1 wk, safety stock 10, on-hand 20; SA-200 lead time 1 wk, multiple 100, on-hand 50; RM-300 purchase lead time 2 wk, minimum 500, on-hand 200, open PO 300 due week 2. Demand: sales order 120 FG week 4; forecast 80 FG week 5.
  - FG-100: PAB_0 = 20 – 10 = 10. Week 4: 10 – 120 = –110 → planned production PP-1 110, due wk 4, start wk 3. Week 5: 0 – 80 = –80 → PP-2 80, due wk 5, start wk 4. Dependent demand: SA-200 220 wk 3, 160 wk 4; RM-300 110 wk 3, 80 wk 4.
  - SA-200: PAB_0 = 50. Week 3: 50 – 220 = –170 → multiple 100 → PP-3 200, due wk 3, start wk 2; PAB 30. Week 4: 30 – 160 = –130 → PP-4 200, due wk 4, start wk 3; PAB 70. Dependent RM-300: 600 wk 2, 600 wk 3.
  - RM-300: PAB_0 = 200. Week 2: 200 + 300 (PO) – 600 = –100 → min 500 → planned purchase PO-1 500 due wk 2, order date wk 0 < today → futures date wk 3, Delay message 7 days; PAB 400. Week 3: 400 – 600 – 110 = –310 → PO-2 500 due wk 3, order date wk 1 = today (no delay); PAB 190. Week 4: 190 – 80 = 110.
  - Propagation: PP-3 pegged to PO-1 inherits futures wk 3 → PP-3 cannot finish before wk 4 → PP-1 futures wk 5 → Delay message on sales order line (requested wk 4, expected wk 5, delay 7 days). Pegging explains 100% of each planned order: PP-1 → SO 110 of 120 (10 from on-hand after safety), PO-1 → PP-3 (100 of 500; 400 surplus unpegged, no Cancel because within coverage time fence).
- **BR-22.3 Forecast reduction.** `transactions` principle: reduced forecast for a period = max(0, forecast – Σ sales order qty in the period); periods are forecast-line periods, not calendar months.
- **BR-22.4 Fences.** Requirements inside the freeze fence generate no planned orders; inside the firming fence, planned orders are auto-firmed; action messages are generated only inside the action fence; futures messages only inside the futures fence.
- **BR-22.5 Safety stock replenishment date.** With policy `today`, the safety-stock planned order is dated today + lead time; with `first-issue`, the first demand date.
- **BR-22.6 Dynamic plan scope.** A dynamic run replans the changed item and all items whose LLC is greater and that are reachable through its BOM; it never changes planned orders outside that set.
- **BR-22.7 Transfer planning.** Warehouse-level coverage with `transfer` order type creates demand at the from-warehouse at delivery date – transfer lead time.
- **BR-22.8 Calendar edge cases.** Lead-time offsets skip closed days; if a date falls on a closed day it moves backward (supply) or forward (demand) to the nearest open day.
- **BR-22.9 Firming concurrency.** Firming is a saga: create target order → mark planned order firmed → on failure compensate by deleting the target; a planned order already firmed returns 409.
- **BR-22.10 Quantity precision.** Requirements carry the item's inventory UoM to 6 decimals; forecast lines in forecast UoM converted at load.

4.22.8 Acceptance criteria

- **AC-22.1** (FR-22.3, 22.4, 22.5) Given the BR-22.2 fixture, when the static plan runs, then the planned orders PP-1…PP-4, PO-1, PO-2 exist with the stated quantities and dates and a 7-day delay message on the sales line.
- **AC-22.2** (FR-22.6) Given any planned order in the 40-fixture set, when I read its pegging, then Σ pegged qty = planned qty.
- **AC-22.3** (FR-22.14) Given the 50k-item dataset, when the plan runs three times, then all three plan hashes are equal and each run completes in < 10 min.
- **AC-22.4** (FR-22.2) Given a Min/Max item with min 100, max 400 and PAB 90, when planned, then one planned order of 310 is created.
- **AC-22.5** (FR-22.7) Given a PO due 10 working days before its earliest pegged demand and positive_days = 5, when planned, then a Postpone message suggests the demand date – receipt margin.
- **AC-22.6** (FR-22.8) Given forecast 100 and sales orders 30 in a period with `transactions` reduction, then gross requirement from forecast is 70.
- **AC-22.7** (FR-22.9) Given 24 months of intermittent history (ADI 2.1), when the statistical forecast runs, then the Croston/TSB model is chosen and MAPE is reported.
- **AC-22.8** (FR-22.10) Given 3 approved planned purchase orders for one vendor on one date with grouping on, when firmed, then one PO with 3 lines is created and the planned orders show status firmed with target id.
- **AC-22.9** (FR-22.1) Given a running static version, when a second run is requested, then 409 `PLAN_ALREADY_RUNNING`.
- **AC-22.10** (FR-22.12) Given item RM-300 in the example, when I open net requirements, then the PAB column shows 400, 190, 110 for weeks 2–4.

4.22.9 Test plan and fixtures

Fixtures: `mrp-textbook-40` (APICS CPIM-style scenarios: lot-for-lot, FOQ, POQ, safety stock, scrap, phantom, multi-source, delays; expected planned orders as JSON); `mrp-3level` (BR-22.2); `mrp-50k` (generated: 50k items, 5-level BOMs, 12 monthly periods, 200k sales lines); `forecast-history-2k` (24 months, mix of smooth/intermittent); `frepple-compare` (shared dataset exported to frePPLe, expected within ±1 day and ±1% qty).

Test	Type	Target	Pass criterion
Textbook scenarios	unit	engine	40/40 match

Test	Type	Target	Pass criterion
PAB identity	property	engine	PAB recurrence holds for every item/day
Pegging completeness	property	pegging	Σ peg = planned qty for all planned orders
Determinism	integration	plan run	3 identical hashes
Scale	performance	50k plan	< 10 min, memory < 8 GB
Firming saga	contract	planning ↔ production/purchasing	compensations on injected failure
Forecast models	unit (Python)	statsforecast wrapper	model selection matches fixture
frePPLe comparison	integration (optional)	shared dataset	within tolerance

Reconciliation: Σ supply – Σ demand per item/site over the horizon = closing PAB; planned production orders firmed = ProductionOrders with origin PlannedOrder.

4.22.10 Deployment and operations

Helm charts `planning` (Go, 2 replicas API + 1 planner worker pool with 8 CPU / 16 GB request, Temporal worker), `planning-db` (CloudNativePG, 200 GB), `forecast` (Python, 2 replicas). Config: `PLANNING_MAX_VERSIONS` (20), `PLANNING_DYNAMIC_DEBOUNCE_SECONDS` (60), `PLANNING_HORIZON_DAYS` (365). Flags: `planning.dynamic`, `planning.statistical-forecast`, `planning.mps-scope`. Scheduled: nightly static plan 01:00 site-local; weekly forecast generation Sunday 22:00; version purge daily. SLOs: 50k-item plan < 10 min; planned-order list p95 < 300 ms; pegging read p95 < 500 ms. Alerts: plan failed or > 20 min (page); dynamic-plan lag > 5 min (warn). Runbook: "plan stuck" — inspect Temporal workflow, cancel, re-run; "hash mismatch" — compare sorted outputs between versions. Rollback: prior digest; versions are immutable so rollback never corrupts plans.

4.22.11 Migration from AX / SAP

Source	Passage entity	Notes
ReqGroup (AX) / T438A + MRP type config (SAP)	CoverageGroup	ReqPOType→coverage_code; SAP lot-size EX→Requirement, TB→Period, FX→fixed qty via ItemCoverage standard qty, WB→Period(7)
ReqItemTable / MARC MRP views	ItemCoverage	DISMM PD/VB/ND; EISBE→safety; DISLS; PLIFZ→lead time
ForecastSales / PIR (PBIM/PBED)	ForecastLine	periods normalized to weeks or months
ReqPO (firmed only) / PLAF (firmed, PLAF-AUFFX = X)	PlannedOrder status firmed	unfirmed planned orders are not migrated — regenerated by the first run
ReqTrans, ReqTransCov / MDTB	not migrated	regenerated

Validation: first Passage plan vs last legacy plan on identical snapshot: planned order count within ±5% and 95% of planned orders match item/date ±1 day. Not migrated: planning history, AX ReqCalcTask logs, SAP MRP lists.

4.22.12 Dependencies, risks, open questions

Depends on: w21 production orders (firming target, route-based lead times); w17 PO creation; w16 on-hand and inventory transactions; w9 BOM/LLC; w6 calendars; w27 sales orders (demand arrives later — until then sales demand is loaded via fixtures and the `sales-line` ref type).

Risk	Likelihood	Impact	Mitigation
Without sales orders until w27 the plan is demo-only	Certain	Low	Forecast + fixture demand; re-proof at w27
Memory blow-up on 50k × 365 daily buckets	Medium	Medium	Sparse buckets (only dates with movement); weekly buckets beyond 90 days

Open questions: (1) Daily vs weekly buckets beyond 90 days — default weekly; decide at w22 kickoff. (2) Whether to expose `positive_days = 0` as "unlimited" (AX) or 0 — default AX semantics; decide by w22 mid-point.

4.23 Wave 23: Capacity Planning & Finite Scheduling

4.23.1 Scope and business context

Wave 23 replaces the infinite-capacity dating of waves 21–22 with a scheduler that places operations and jobs on resources with calendars and finite capacity, forward or backward, honoring precedence, setup, queue and transit times, primary/secondary resources and capabilities. Planners see a resource × time Gantt, drag jobs, see capacity load and bottlenecks, and optionally run a CP-SAT optimization that minimizes weighted lateness and makespan. Schedule dates flow back to production orders (start/end, job dates) and to master planning (planned production orders use scheduled lead times). Out of scope: shop-floor dispatch lists on the kiosk (wave 25 reads this wave's jobs), tool/fixture management, multi-plant capacity.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX job scheduling on a 2,000-job plant takes 20–40 min (WrkCtrCapRes rebuilt per order in X++); SAP CM21 capacity leveling is manual dispatch per work center and PP/DS needs a separate APO/embedded install with CIF integration	Scheduling runs per order inside the status-update dialog; capacity reservations stored row-by-row; finite scheduling is an add-on product	Scheduler service with materialized UTC capacity intervals, GiST exclusion constraint, site-level run with checkpoints	5,000 operations 30 min → < 60 s; no separate product
Gantt is read-mostly: D365 Gantt drag does not validate secondary resources or calendars until save; SAP CM21 planning board shows overloads but cannot propagate to dependent orders; Epicor Kinetic Multi-Resource Scheduling requires the APS module and locks the board during save	Validation happens server-side after commit; no dependency graph in the board	`:move` with `dryRun` preview showing violations and dependent impact before commit; canvas Gantt	Reschedule-and-verify 15 min → < 30 s; zero committed double-bookings

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Load reports lag reality: AX "Capacity load" form (WrkCtrCapRes) and SAP CM01 are as-of the last scheduling run; Infor LN "Capacity Utilization" is a batch report	Load is not maintained incrementally	CapacityLoad recomputed per run and per move; bottleneck events	Load view staleness hours → < 5 s after any move
Optimization is unavailable or opaque: no optimizer in AX; SAP PP/DS optimizer configuration is a consulting project; Plex scheduling is rule-based only	Optimization tied to proprietary engines	CP-SAT via OR-Tools with a stated objective, time limit and reported solver status; heuristic kept if no improvement	Known-optimum gap ≤ 5% on small instances; optimization run self-service in 1 click
Scheduled dates do not reach planning: AX master planning re-derives lead times from the route rather than from the schedule; SAP MRP ignores PP/DS results unless CIF is configured	Planning and scheduling keep separate date models	`order.dates-changed` events feed planning; `planning.finite-capacity` flag uses scheduled dates	Date mismatch between plan and schedule 100% of orders → 0

4.23.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Operations scheduling	ProdTable scheduling (operations), WrkCtrCapRes (capacity reservations), scheduling direction	Lead-time scheduling (CA97), capacity requirements KBED/KBEZ	ScheduleRun mode `operations`, ResourceCapacity	full
Job scheduling	Job scheduling (ProdRouteJob), WrkCtrCapRes by job	Detailed scheduling (PP/DS), SAP APO	mode `jobs` (minute level)	full
Finite/infinite capacity	ReqGroup/parameters finite capacity, finite capacity on resource	Capacity leveling CM21/CM22, finite scheduling flag	per-resource `finite` flag; run option	full
Capacity load view	WrkCtrCapRes load inquiry, Capacity load graph	CM01 (load), CM05, CM07	Capacity load screen	full
Gantt	Gantt chart (D365)	Planning board CM21, PP/DS detailed scheduling board	Scheduling workspace Gantt	improved (drag with constraint check)
Calendars/efficiency	WorkCalendar, efficiency % on resource, capacity per calendar day	Factory calendar, capacity header (CR02), utilization %	WorkCalendar (w9), ResourceCapacity.efficiency_pct	full
Primary/secondary resources	Route operation resource requirements (capability/group/resource/type), priority	Alternative sequences, work center hierarchy, split (AFVV)	ProductionOperation.priority; capability matching	full
Optimization	none native	PP/DS optimizer (genetic), S/4 pMRP	CP-SAT mode	improved
Reschedule dependents	Reschedule with reference orders	Rescheduling from MRP, CM21 dispatch	`:reschedule` with dependency propagation	full
Bottleneck resource	Bottleneck resource flag (D365 scheduling)	Bottleneck work center (CM21)	utilization > threshold detection	full

4.23.3 Functional requirements

- **FR-23.1 Capacity model** — Must. The system shall materialize ResourceCapacity as open intervals (UTC minutes) per resource per day from its WorkCalendar (shifts), efficiency % and capacity-per-day override, for a rolling horizon of 365 days.
  - Notes: Rebuilt incrementally when a calendar changes (w9 event). Finite resources cannot be loaded beyond open interval minutes; infinite resources ignore the limit but still only schedule in open intervals.
- **FR-23.2 Operations scheduling** — Must. The system shall schedule operations in daily buckets, forward from a start date or backward from a delivery date, selecting a resource per operation by priority (named resource > resource group > capability) and finite availability.
- **FR-23.3 Job scheduling** — Must. The system shall schedule jobs (queue-before, setup, process, overlap, transport, queue-after) at minute granularity with the same direction and resource choice rules, splitting process jobs across shifts but not across resources.
- **FR-23.4 Constraints** — Must. The scheduler shall honor: precedence (next_oper_number), max one process job per finite resource at any minute, secondary resources booked concurrently with the primary, setup before process, overlap transfer batch (operation n+1 may start after `overlap_qty` units), transit time between resources, calendar open intervals, order start ≥ today (forward) and end ≤ delivery date (backward, else convert to forward and flag late), operation `finished` jobs frozen, firmed jobs frozen.
- **FR-23.5 Dispatch heuristic** — Must. The default algorithm is priority-list scheduling: orders sorted by (delivery date, priority, order number); each operation placed at the earliest feasible interval on the best resource (least finish time, tie → least load); complexity O(n log n) per resource.
- **FR-23.6 CP-SAT optimization** — Should. The system shall offer an optimization run via OR-Tools CP-SAT (Python `scheduler-opt` service) with objective minimize Σ weight × lateness + λ × makespan, time limit configurable (default 60 s), horizon ≤ 2,000 operations per run (larger sets are windowed).
- **FR-23.7 Gantt** — Must. The system shall render resources × time with jobs as bars colored by order, allow drag to move/reassign a job within its capability set, and show constraint violations (red outline + tooltip) before commit.
- **FR-23.8 Reschedule with propagation** — Must. Moving a job shall reschedule all successors of the same order forward and, optionally, the dependents (orders pegged to the moved order's output) with a preview of affected orders before commit.
- **FR-23.9 Capacity load** — Must. The system shall show load % = Σ booked minutes / Σ open minutes per resource/group/period (day/week), with bottleneck flag when load > 90% in ≥ 3 consecutive weeks (threshold configurable).
- **FR-23.10 Feedback to planning** — Must. The system shall publish scheduled dates so that production orders update start/end, and master planning uses scheduled lead times for firmed orders and route-based estimates for planned orders (optionally finite in the plan run via flag `planning.finite-capacity`).
- **FR-23.11 Schedule runs** — Must. Scheduling can run per order (on `:schedule`), per selection, or for the whole site nightly, as Temporal jobs with checkpoints every 500 operations.
- **FR-23.12 Capability matching** — Should. Resource selection shall honor capability requirements with minimum level and expiry dates (w9 capabilities).

4.23.4 Data model

ResourceCapacity: id, tenant_id, resource_id, day (date), interval_start (timestamptz), interval_end, open_minutes (int), efficiency_pct (numeric 5,2), finite (bool), source (calendar/override).

ScheduledJob: id, job_id (FK ProductionJob), order_id, oper_number, resource_id, start (timestamptz), end, minutes, run_id, firmed (bool), late_minutes (int), overlap_qty (numeric), secondary_resource_ids (uuid[]).

ScheduleRun: id, scope (order/selection/site), mode (operations/jobs), direction (forward/backward), finite (bool), algorithm (heuristic/cp-sat), time_limit_s, status, started_at, finished_at, operations_count, makespan_minutes, total_lateness_minutes, objective_value, solver_status (OPTIMAL/FEASIBLE/UNKNOWN).

CapacityLoad (materialized): resource_id, period_start, period_type, open_minutes, booked_minutes, load_pct, bottleneck (bool).

Invariants: no two ScheduledJob rows on the same finite resource overlap in [start, end); every ScheduledJob lies within ResourceCapacity open intervals of its resource; for jobs j, k in the same operation chain with k successor of j: k.start ≥ j.end + transit; a ScheduledJob with firmed = true is unchanged by any run without `force`.

Indexing: ScheduledJob GiST exclusion constraint `EXCLUDE USING gist (resource_id WITH =, tstzrange(start,end) WITH &&) WHERE (finite)`; ResourceCapacity (resource_id, day); CapacityLoad refreshed per run.

4.23.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/scheduling/runs	Start run (scope, mode, direction, finite, algorithm)	Idempotency-Key	Run scheduling
GET	/api/v1/scheduling/runs/{id}	Run status/metrics	n/a	View scheduling
GET	/api/v1/scheduling/gantt?siteId&from&to&resourceIds	Jobs for Gantt window (≤ 5,000 bars)	n/a	View scheduling
POST	/api/v1/scheduling/jobs/{id}:move	Move/reassign with `dryRun` preview	Idempotency-Key	Maintain schedule
POST	/api/v1/scheduling/jobs/{id}:firm	Firm/unfirm	Idempotency-Key	Maintain schedule
GET	/api/v1/scheduling/capacity-load?resourceIds&period=week	Load and bottlenecks	n/a	View scheduling
GET	/api/v1/scheduling/resources/{id}/capacity?from&to	Open intervals	n/a	View scheduling
POST	/api/v1/scheduling/resources/{id}/capacity-overrides	Add downtime/extra shift	Idempotency-Key	Maintain capacity

Event	Trigger	Payload fields	Consumers
`dev.passage.scheduling.run.completed.v1`	run done	runId, scope, operations, makespanMinutes, totalLatenessMinutes, solverStatus	production, planning
`dev.passage.scheduling.job.scheduled.v1`	job placed/moved	jobId, orderId, operNumber, resourceId, start, end, runId, firmed	production, shopfloor
`passage.scheduling.order.dates-changed.v1`	order start/end changed	orderId, start, end, lateMinutes	production, planning, sales (w27)
`dev.passage.scheduling.bottleneck.detected.v1`	load > threshold	resourceId, periods[], loadPct	notifications

4.23.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Scheduling workspace (Gantt)	Zoom day/week, drag job, reassign, firm, run heuristic/optimize, preview propagation	Resource rows, job bars (order, item, qty, start/end), late markers, open-interval shading	Maintain schedule
Capacity load	Filter resource/group/period, open bottleneck, drill to jobs	Resource, period, open h, booked h, load %, bottleneck	View scheduling
Schedule runs	Start, monitor, compare runs	Scope, mode, algorithm, status, makespan, lateness, duration	Run scheduling
Resource capacity	Overrides, view calendar materialization	Day, intervals, efficiency, finite	Maintain capacity

Interaction rules: drag snaps to 5-minute grid; `Shift+drag` moves the whole order; violations block commit unless the user holds *Override scheduling constraints*; Gantt virtualizes rows and uses canvas rendering for > 1,000 bars; undo stack of 20 moves per session.

4.23.7 Business rules and edge cases

- **BR-23.1 Duration.** Process minutes = (run_time_h_per × qty / process_qty) × 60 / (efficiency_pct/100); setup minutes = setup_time_h × 60 / efficiency; both rounded up to whole minutes.
- **BR-23.2 Backward fallback.** If backward scheduling places any job before now + 1 h, the run re-schedules the order forward from now and sets late_minutes = end – delivery date end-of-day.
- **BR-23.3 Shift splitting.** A process job crossing a closed interval is split into segments on the same resource; setup is never split (if it does not fit before close, it moves to the next open interval).
- **BR-23.4 Secondary resources.** Booked for the same interval as the primary; if no secondary is free, the primary slot is skipped.
- **BR-23.5 Overlap.** With overlap_qty q > 0, operation n+1 earliest start = start_n + setup_n + process minutes for q units + transit.
- **BR-23.6 Firmed and reported jobs.** Jobs with reported hours or `finished` status keep their actual start; remaining minutes = planned – reported, floor 0.
- **BR-23.7 DST and time zones.** All intervals stored in UTC; calendars defined in site time zone; a 23-hour DST day yields fewer open minutes — property test covers 2026-03-29 (EU) and 2026-11-01 (US).
- **BR-23.8 Resource selection tie-break.** Equal finish time → lower load % → lower resource code (deterministic).
- **BR-23.9 CP-SAT model.** Interval variables per job, `NoOverlap` per finite resource, `AddNoOverlap2D` not used; precedence as start ≥ end + transit; objective Σ w_i × max(0, end_i – due_i) + λ × makespan with λ = 0.1 default; `num_workers` = 8; result accepted if FEASIBLE; heuristic result kept if solver returns UNKNOWN.

- **BR-23.10 Propagation limit.** Dependent rescheduling stops after 500 orders or 5 pegging levels and reports the truncation.

4.23.8 Acceptance criteria

- **AC-23.1** (FR-23.2, 23.4) Given OR-Library instance `ft06`, when scheduled with the heuristic, then the schedule is feasible (no overlap, precedence held); when optimized with CP-SAT 60 s, makespan = 55 (known optimum).
- **AC-23.2** (FR-23.4) Given 10k random jobs across 50 finite resources, then the exclusion constraint never fires and the property test "every job inside open intervals" passes.
- **AC-23.3** (FR-23.8) Given an order whose output is pegged to a sales line, when I move its last job 2 days later, then the preview lists the sales line delay and after commit the order's end date and `order.dates-changed` event carry the new date.
- **AC-23.4** (FR-23.11) Given 5,000 operations on the demo plant, when the site run executes, then it completes in < 60 s.
- **AC-23.5** (FR-23.3) Given a 300-minute process job on a resource with a shift ending in 120 minutes, then two ScheduledJob segments of 120 and 180 minutes exist on the same resource.
- **AC-23.6** (FR-23.9) Given booked 150 h against 160 h open for 3 consecutive weeks, then the resource is flagged bottleneck and an event is emitted once (not per run).
- **AC-23.7** (FR-23.1) Given a calendar change closing next Friday, then the ResourceCapacity rows for that day are removed within 30 s and affected jobs show violations.
- **AC-23.8** (FR-23.10) Given a scheduled order, when planning runs with `planning.finite-capacity` on, then its supply date equals the scheduled end date.
- **AC-23.9** (FR-23.6) Given an optimization run that times out with UNKNOWN, then the heuristic schedule remains and the run reports solver_status UNKNOWN.

4.23.9 Test plan and fixtures

Fixtures: `orlib-jobshop-30` (ft06, ft10, ft20, la01–la20, abz5–abz9 with known optima); `sched-calendar-edge` (DST days, 24/7, zero-capacity holidays); `sched-5k-ops` (demo plant scaled); `sched-propagation` (pegged chains 5 deep).

Test	Type	Target	Pass criterion
Job-shop feasibility	unit	heuristic	30/30 feasible
CP-SAT optimality	integration	optimizer	within 5% on instances ≤ 10×10
No double-booking	property	ScheduledJob	0 violations over 10k jobs
Calendar edges	unit	capacity materialization	open minutes match fixture
Gantt move	Playwright	move + propagate	persisted and dependents updated
Scale	performance	5k ops	< 60 s, p95 Gantt load < 1 s

Reconciliation: Σ booked minutes per resource (CapacityLoad) = Σ ScheduledJob minutes; production order start/end = min/max of its scheduled jobs.

4.23.10 Deployment and operations

Helm: `scheduling` (Go, 2 API replicas + 2 worker replicas, 4 CPU / 8 GB), `scheduler-opt` (Python OR-Tools, 1 replica, 8 CPU / 16 GB, scale-to-zero via KEDA on queue depth), `scheduling-db` (CloudNativePG 50 GB). Config: `SCHED_BOTTLENECK_PCT` (90), `SCHED_OPT_TIME_LIMIT_S` (60), `SCHED_HORIZON_DAYS` (365). Flags: `scheduling.cp-sat`, `scheduling.propagate-dependents`, `planning.finite-capacity`. Jobs: nightly site run 00:30 (before planning); capacity materialization daily. SLOs: 5k-op run < 60 s; Gantt window p95 < 1 s; move p95 < 400 ms. Alerts: run failure (page), optimizer OOM (warn). Rollback: prior digest; ScheduledJob rows versioned by run_id so a bad run is reverted by re-pointing to the previous run.

4.23.11 Migration from AX / SAP

Source	Passage entity	Notes
WrkCtrCapRes (AX) / KBED+KBEZ (SAP)	ScheduledJob	only for open orders; re-scheduled on first run, legacy dates kept as `firmed` if the order is Started
WrkCtrTable capacity/efficiency, WorkCalendar / CR02 capacity header, factory calendar	ResourceCapacity (materialized)	from w9 resources and calendars
ProdRouteJob dates / AFVV dates	ProductionJob planned dates	preserved for reference

Not migrated: historical capacity reservations, PP/DS optimizer profiles, AX scheduling parameters (re-set in Passage).

4.23.12 Dependencies, risks, open questions

Depends on: w21 jobs/operations; w22 planned orders and pegging; w9 resources, capabilities, calendars.

Risk	Likelihood	Impact	Mitigation
Heuristic quality far from optimum on large instances	High	Low	CP-SAT windowing; planners can firm; measure lateness trend
Gantt performance in the browser	Medium	Medium	Canvas rendering, windowed fetch ≤ 5,000 bars

Open questions: (1) Should optimization run automatically nightly or only on demand? Default on demand; decide by w23 mid-point. (2) Tool/fixture resources as secondary resources in w23 or w44 (EAM)? Default w44.

4.24 Wave 24: Process Manufacturing & Lean (Formulas, Batches, Kanban)

4.24.1 Scope and business context

Wave 24 extends production to two further modes. Process manufacturers (food, chemicals, pharma) create batch orders from formulas whose lines scale with batch size, consume ingredients by potency, produce co- and by-products with cost allocation, track batch attributes, shelf life and disposition, and handle catch-weight items with a dual quantity. Lean plants define value streams and production flows with activities, replenish through kanban rules (manufacturing and withdrawal; fixed, scheduled and event replenishment), work kanban cards on a board, and cost production flows by periodic backflush. Formulas and recipes are version-controlled through engineering change (w9). After this wave a second demo plant (food/chem line) runs a batch order and a kanban-driven cell. Out of scope: PI sheets with electronic signatures (w33 quality covers signatures), process control system integration beyond the IoT connector of w25, recipe scale-up optimization.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Rescaling a batch order in AX (PmfProdTable "Rescale batch") recomputes lines but silently rounds to the inventory unit precision and does not preserve percent totals; SAP COR2 quantity change re-reads the recipe and drops manual line edits	Scaling implemented as re-explosion; rounding residual not tracked	BR-24.1 scaling with residual assigned to the largest scalable line, side-by-side preview, manual edits preserved	Percent-controlled Σ = 100.0000 on 100% of batches; rescale 12 clicks → 2
Potency adjustment is manual: AX requires the batch potency on InventBatch to be keyed before picking and the picking list does not recompute compensating lines automatically; SAP active-ingredient management needs batch characteristics plus custom proportion units	Potency not part of the consumption proposal	BR-24.2 proposal computes active and compensating quantities from recorded batch attributes	Potency re-keying 0 (attributes recorded once); consumption error < 0.1%
Co-product costing errors at End: AX TCA allocation misallocates when a co-product has zero standard cost; SAP apportionment structures (TKKAA) must be maintained per material and settlement KO88 fails silently with status messages	Allocation data in separate setup objects, no invariant check	Allocation normalized with residual rule and property test Σ = batch cost; preview before End	Allocation mismatches 0; End failures surfaced synchronously with reason
Kanban cards and boards are disconnected: AX kanban board (KanbanBoardScheduleView) refreshes manually and cannot be used on a tablet; SAP PK13N is a GUI-only status screen; Plex and Epicor Kinetic kanban rely on printed cards and scanner re-keying	Board is a list page over KanbanJob with no push updates	Touch board with SSE refresh, GS1 DataMatrix cards, event kanbans within 5 s	Card-to-board latency 5 min refresh → < 2 s; re-keying 0
Shelf-life and disposition rules are enforced only at picking in AX (PdsShelfLife) and by batch status in SAP (MSC2N), so expired or on-hold batches appear in proposals	Rules applied post hoc rather than in the proposal query	Proposals exclude expired/on-hold batches; override requires duty + reason	Expired-batch consumption incidents → 0; audit reason on 100% of overrides

4.24.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Formula	PmfFormula, BOMVersion with PmfFormulaSize, PmfFormulaLine (scalable, percent-controlled)	Master recipe (C201), material list in recipe, base quantity	Formula (BomVersion subtype), FormulaLine	full
Batch order	ProdTable with ProdType=Batch (PmfProdTable), batch order status = ProdStatus	Process order (COR1/COR2), AUFK with order category 40	BatchOrder (ProductionOrder subtype)	full
Co-/by-products	PmfCoByProd, cost allocation (TCA, manual, none), by-product cost type	Co-product settlement rule (apportionment structure), by-product with negative qty	CoProductLine, allocation_method	full
Potency	PdsBatchAttrib, PdsBatchAttribItem, base attribute, potency unit, active-ingredient line type	Active ingredient management (batch characteristic, proportion unit)	BatchAttribute with `potency` role; consumption rule	full
Batch attributes / disposition	PdsBatchAttrib, PdsBatchAttributes on InventBatch, PdsDispositionCode	Batch classification (MSC1N characteristics), batch status, usage decision	BatchAttribute, InventoryBatch.disposition (w16 batch)	full
Shelf life	InventBatch expiry, best-before, shelf-advice, PdsShelfLife	Shelf-life expiration date (SLED), minimum remaining shelf life	shelf_life_days, best_before, expiry on batch	full
Catch weight	PdsCW, CW unit, nominal/min/max qty	Catch-weight management (S/4 CWM, parallel UoM)	CatchWeightQty on product and transactions	full
PI sheet / process instructions	n/a (route card notes)	PI sheets (CORA/CO60), control recipes, process messages	operation instructions on kiosk (w25); messages via events	partial — no XSteps
Production flow / value stream	LeanProductionFlow, LeanProductionFlowActivity, ValueStream (OMOperatingUnit)	PP-REM production version, product cost collector, MF50 planning table	ProductionFlow, FlowActivity, ValueStream (w6 OperatingUnit)	full
Kanban rules and kanbans	KanbanRule (fixed/scheduled/event), Kanban, KanbanJob, KanbanJobSchedule, kanban board	Control cycle PK01, kanban board PK13N, PK05 demand source, PK17	KanbanRule, Kanban, KanbanJob	full
Backflush costing	Lean backflush costing, production flow costing (periodic)	Product cost collector (KKF6N), REM backflush MFBF, periodic settlement	FlowCostingPeriod	full
Repetitive planning table	n/a	MF50	kanban board takt view	partial

4.24.3 Functional requirements

- **FR-24.1 Formulas** — Must. The system shall define a Formula as a BomVersion with formula_size and FormulaLine fields: scalable (bool), percent_controlled (bool, lines sum to 100%), line_type (ingredient, active-ingredient, phantom, co-product, by-product), per_series/quantity.
  - Notes: Percent-controlled formulas validate Σ scalable line % = 100.00 ± 0.01. Formula version changes go through ECO (w9); a batch order snapshots the formula at creation.
- **FR-24.2 Batch orders** — Must. The system shall create BatchOrder from a formula, scaling scalable lines proportionally to batch size (non-scalable lines fixed), with the same state machine and journals as w21 plus co-/by-product RAF lines.
- **FR-24.3 Potency** — Must. The system shall consume active-ingredient lines by potency: consumed qty = planned qty × (standard potency / actual batch potency), with compensating lines (fillers) adjusted so the batch size is kept.
  - Notes: Potency attribute defined per item (base attribute, target value, min/max); batch potency recorded at receipt or by quality (w33) on the InventoryBatch. Batches outside min/max are not pickable for the line.

- **FR-24.4 Co-/by-product costing** — Must. The system shall allocate batch cost to co-products by method: none (all cost to main item), manual (fixed %), total cost allocation (TCA: Σ co-product standard cost shares), by quantity; by-products reduce cost by burden: per series, per quantity, percent, or recycled (negative issue).
    - Notes: Allocation percentages are normalized to sum 100.0000; rounding residual goes to the main product; property test Σ allocated = batch cost.
- **FR-24.5 Batch attributes and disposition** — Must. The system shall record BatchAttribute values per inventory batch (numeric/text/enum/date with min/max/target), block batches by disposition code (available, on-hold, rejected, quarantine) and restrict picking by customer-specific attribute ranges (sales linkage in w27).
- **FR-24.6 Shelf life** — Must. The system shall compute best_before = manufacturing date + best_before_days and expiry = manufacturing date + shelf_life_days at RAF, support shelf-advice (advice days before expiry) and FEFO picking on batch orders, and block consumption of expired batches unless duty *Consume expired batch* is held.
- **FR-24.7 Catch weight** — Must. The system shall support catch-weight items with inventory quantity in the CW unit (e.g. cases) and a nominal weight with min/max, carrying both quantities on every inventory and production transaction; cost is per inventory (weight) quantity.
    - Notes: RAF requires actual weight per CW unit; variance outside min/max rejected; sales/purchase price is per weight (w26 pricing uses nominal until actual known).
- **FR-24.8 Production flows** — Must. The system shall model ValueStream → ProductionFlow (versioned, with takt time, activation dates) → FlowActivity (process or transfer; cycle time, resource/work cell, input/output locations, picking activities) and validate connectivity (every process activity has an output).
- **FR-24.9 Kanban rules** — Must. The system shall define KanbanRule with type (manufacturing/withdrawal), replenishment strategy (fixed: number of kanbans and product quantity; scheduled: via planning as planned kanbans; event: generated from sales line, BOM line, kanban line, minimum stock), lead time, fixed quantity, activity sequence, and automatic planning/firming.
- **FR-24.10 Kanbans and board** — Must. The system shall create Kanban (card) with KanbanJobs per activity, states Not planned → Planned → Prepared → In progress → Completed → Emptied (card returns), print cards with barcodes (GS1 DataMatrix), and provide a board per work cell (columns by job status) with drag completion and quantity/scrap entry.
- **FR-24.11 Kanban events** — Must. Event kanbans shall be generated within 5 s of a triggering sales line (w27 or fixture) or BOM line of a production order, and withdrawal kanbans create transfer jobs that post inventory transfers on completion.
- **FR-24.12 Lean backflush costing** — Should. The system shall cost production flows periodically: per costing period, issue materials at standard/weighted-average on kanban completion, accumulate conversion cost by activity (cycle time × cost rate), and settle at period end with variances to the flow's posting profile.
- **FR-24.13 Kanban manufacturing orders** — Must. Manufacturing kanbans shall create a lightweight ProductionOrder of type `kanban` (no routing jobs; single process activity) that reports as finished on kanban completion.
- **FR-24.14 Takt and board metrics** — Could. The board shall show takt vs cycle, WIP count per column and kanbans overdue (planned < now).

4.24.4 Data model

FormulaLine: id, bom_version_id, line_number, item_id, line_type (enum), qty (numeric 18,6), percent (numeric 9,4), scalable (bool), percent_controlled (bool), potency_base_item (bool), unit_id, scrap_pct, is_co_product (bool), allocation_pct (numeric 7,4), by_product_cost_type (enum: per-series, per-quantity, percent, recycled).

BatchOrder (extends ProductionOrder): formula_size (numeric), batch_size (numeric), scale_factor (numeric 18,9, computed), planning_item_id (for co-product planning), potency_mode (enum), manufacturing_date, batch_number (text, generated at RAF via `batch` sequence).

CoProductLine: id, order_id, item_id, planned_qty, raf_qty, allocation_method (enum: none, manual, tca, quantity), allocation_pct, allocated_cost (numeric 18,4), is_by_product (bool), by_product_cost_type.

BatchAttribute: id, item_id, attribute_id, data_type (numeric/text/enum/date), target, min, max, role (enum: potency, generic), unit_id; **BatchAttributeValue**: batch_id (w16 InventoryBatch), attribute_id, value_numeric, value_text, value_date, recorded_at, recorded_by.

CatchWeightQty (on product): cw_unit_id, nominal_qty (numeric 18,6), min_qty, max_qty; on InventoryTransaction and journal lines: cw_qty (numeric 18,6) alongside qty.

ProductionFlow: id, value_stream_id (OperatingUnit), version, name, site_id, takt_time_s (int), status (draft/active/expired), active_from, active_to; **FlowActivity**: id, flow_id, name, kind (process/transfer), cycle_time_s, work_cell_resource_id, input_location_id, output_location_id, predecessor_ids (uuid[]), picking_required (bool).

KanbanRule: id, item_id, site_id, warehouse_id, location_id, rule_type (manufacturing/withdrawal), replenishment (fixed/scheduled/event), fixed_qty, number_of_kanbans (int), min_kanbans, max_kanbans, lead_time_h, event_source (sales/bom/kanban/min-stock), auto_plan (bool), auto_firm (bool), flow_activity_sequence (uuid[]), effective_from, effective_to.

Kanban: id, rule_id, card_number, qty, cw_qty, state (enum), handling_units (int), planned_start, due, production_order_id / transfer_id, event_ref_type, event_ref_id, printed_at; **KanbanJob**: id, kanban_id, activity_id, sequence, state (not-planned/planned/prepared/in-progress/completed), start, end, reported_qty, scrap_qty.

FlowCostingPeriod: id, flow_id, period_start, period_end, status (open/settled), material_cost, conversion_cost, output_value, variance, journal_entry_id.

Invariants: Σ FormulaLine.percent for percent-controlled scalable lines = 100.0000; Σ CoProductLine.allocated_cost + main-item cost = batch actual cost (after by-product burden); Kanban state transitions only forward except `Emptied → Not planned` on card reuse; fixed-rule active kanbans count ≤ number_of_kanbans; cw_qty present iff the item is catch-weight.

Indexing: Kanban (rule_id, state); KanbanJob (activity_id, state, planned_start) for the board; BatchAttributeValue (batch_id); FormulaLine (bom_version_id, line_number).

4.24.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/production/batch-orders	Create batch order from formula with batch_size	Idempotency-Key	Maintain batch orders
POST	/api/v1/production/batch-orders/{id}:rescale	Change batch size before Started	Idempotency-Key	Maintain batch orders
POST	/api/v1/production/batch-orders/{id}/co-products:allocate	Preview/apply allocation	Idempotency-Key	Post production journals
PUT	/api/v1/inventory/batches/{id}/attributes	Record attribute values	full replace	Maintain batch attributes
POST	/api/v1/inventory/batches/{id}:disposition	Change disposition	Idempotency-Key	Maintain batch disposition
POST	/api/v1/lean/flows	Create/version production flow	Idempotency-Key	Maintain production flows
POST	/api/v1/lean/flows/{id}:activate	Validate and activate	Idempotency-Key	Maintain production flows
POST	/api/v1/lean/kanban-rules	Create rule	Idempotency-Key	Maintain kanban rules
POST	/api/v1/lean/kanbans:generate	Generate kanbans for a rule (fixed/event)	Idempotency-Key	Maintain kanbans
POST	/api/v1/lean/kanban-jobs/{id}:complete	Complete job with qty/scrap/handling units	Idempotency-Key	Work kanban board
GET	/api/v1/lean/board?workCellId	Board view (jobs by column)	n/a	Work kanban board
POST	/api/v1/lean/flows/{id}/costing-periods/{p}:settle	Settle period	Idempotency-Key	Settle production flows

Event	Trigger	Payload fields	Consumers
`passage.production.batch-order.created.v1`	batch order created	orderId, formulaVersionId, batchSize, scaleFactor, coProducts[]	planning, analytics
`passage.production.batch.attributes-recorded.v1`	attribute values saved	batchId, itemId, values[{attributeId, value}], potency	quality (w33), sales (w27)
`passage.production.co-product.allocated.v1`	End of batch order	orderId, lines[{itemId, method, pct, amount}], byProductBurden	ledger, costing
`dev.passage.lean.kanban.created.v1`	kanban generated	kanbanId, ruleId, qty, eventRefType, eventRefId, due	production, shopfloor
`passage.lean.kanban.state-changed.v1`	job/kanban transition	kanbanId, jobId, from, to, qty, scrap, handlingUnits	production, inventory, shopfloor
`passage.lean.flow-period.settled.v1`	period settled	flowId, period, materialCost, conversionCost, outputValue, variance, journalEntryId	ledger, analytics

4.24.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Formula designer	Add lines, toggle scalable/percent, define co/by-products, validate 100%, submit ECO	Line, item, type, qty/%, scalable, potency base, allocation %	Maintain formulas
Batch order detail	Rescale, estimate, release, RAF with co-products, attributes tab	Batch size, scale factor, co-product lines, batch number, dates, attributes	Maintain batch orders
Batch attributes / disposition	Record values, change disposition, view history	Attribute, value, target/min/max, pass flag, disposition	Maintain batch attributes
Production flow designer	Canvas of activities, connect, set cycle times, activate	Activity, kind, cycle time, work cell, locations	Maintain production flows
Kanban rules	Create fixed/scheduled/event rules, compute number of kanbans	Item, type, replenishment, qty, kanbans, lead time	Maintain kanban rules
Kanban board (work cell)	Drag job between columns, complete with qty, print cards, filter	Columns Not planned/Planned/Prepared/In progress/Completed; card: item, qty, due, order	Work kanban board
Flow costing	Open/settle period, variance drill	Period, material, conversion, output, variance	Settle production flows

Interaction rules: formula percent cells show a running total with red when ≠ 100; board is touch-friendly (min 44 px targets) and refreshes via SSE; card print uses the w5 document service with GS1 DataMatrix encoding `(240)kanbanId`; rescale shows before/after line quantities side-by-side.

4.24.7 Business rules and edge cases

- **BR-24.1 Scaling.** scale_factor = batch_size / formula_size; scalable line qty = formula qty × scale_factor; non-scalable lines unchanged; percent-controlled lines qty = batch_size × percent/100; rounding to UoM precision with residual added to the largest scalable line so Σ = batch_size.
- **BR-24.2 Potency consumption.** For an active-ingredient line with standard potency P_s and picked batch potency P_a: qty = planned × P_s / P_a; the compensating line qty = planned_comp – (qty – planned); negative compensating qty is rejected with 422 `POTENCY_COMPENSATION_NEGATIVE`. Example: 100 kg at 50% standard, batch at 40% → 125 kg; filler reduced by 25 kg.
- **BR-24.3 Co-product allocation (TCA).** share_i = std_cost_i × raf_qty_i / Σ_j std_cost_j × raf_qty_j; allocated_i = round(batch_cost_after_by_products × share_i, 4); residual to the main product. Example: batch cost USD 10,000.00; main A (std 40, 200 units = 8,000), co-product B (std 20, 100 = 2,000): A gets 8,000.00, B 2,000.00. By-product C per-quantity burden 1.50 × 50 = 75.00 reduces batch cost first (9,925.00 allocated 7,940.00 / 1,985.00; C received at 75.00).
- **BR-24.4 Shelf life.** best_before/expiry computed at RAF from manufacturing_date (defaults to RAF date); if the formula has `inherit-shortest-shelf-life`, the output expiry = min(ingredient batch expiries) when earlier.
- **BR-24.5 Catch-weight RAF.** cw_qty integer ≥ 1; qty (weight) must satisfy min_qty × cw_qty ≤ qty ≤ max_qty × cw_qty; cost per weight unit.
- **BR-24.6 Kanban state machine.** Not planned → Planned (date set) → Prepared (picking complete) → In progress (first job started) → Completed (last job completed; manufacturing: RAF posted; withdrawal: transfer posted) → Emptied (consumed; card returns to Not planned for fixed rules, or is deleted for event rules). Backward moves only Planned → Not planned. Illegal moves return 409.
- **BR-24.7 Fixed rule kanban count.** number_of_kanbans = ⌈(daily demand × lead_time_days × (1 + safety%)) / fixed_qty⌉ as a suggestion; rule cannot have more active (non-Emptied) kanbans than number_of_kanbans.
- **BR-24.8 Event kanban quantity.** Event kanbans for demand D create ⌈D / fixed_qty⌉ kanbans; last kanban is full-size (overproduction allowed) unless `exact_qty` flag.

- **BR-24.9 Flow period settlement.** variance = material_cost + conversion_cost – output_value; settled at period end only if all kanbans with jobs in the period are Completed or Emptied, else 409 `FLOW_PERIOD_OPEN_KANBANS` listing them.
- **BR-24.10 Expired batch.** A picking line selecting a batch with expiry < posting date fails unless duty held, in which case the audit log records the override reason (required, ≥ 10 chars).

4.24.8 Acceptance criteria

- **AC-24.1** (FR-24.2, BR-24.1) Given a 1,000 kg formula with a 50 kg non-scalable line, when I create a 2,500 kg batch, then scalable lines are ×2.5 and the non-scalable line stays 50 kg.
- **AC-24.2** (FR-24.3) Given the BR-24.2 example, when the picking list proposes consumption, then the active line is 125 kg and the filler is reduced by 25 kg.
- **AC-24.3** (FR-24.4) Given the BR-24.3 fixture for each allocation method, when the order ends, then Σ allocated cost + by-product burden = batch actual cost exactly.
- **AC-24.4** (FR-24.6) Given an expired ingredient batch, when picked without the duty, then 422 `BATCH_EXPIRED`.
- **AC-24.5** (FR-24.7) Given a catch-weight item (nominal 10 kg, min 9, max 11), when I RAF 10 cases at 95 kg, then the post fails; at 102 kg it succeeds with qty 102 and cw_qty 10.
- **AC-24.6** (FR-24.11) Given an event rule with fixed_qty 20, when a sales line for 50 arrives, then 3 kanbans exist within 5 s and completing them through the board posts RAF of 60.
- **AC-24.7** (FR-24.10) Given a kanban in Planned, when I drag it to Completed, then 409 and the board reverts.
- **AC-24.8** (FR-24.12) Given a flow period with 10 completed kanbans, when settled, then the journal entry balances and the variance equals material + conversion – output.
- **AC-24.9** (FR-24.1) Given a percent-controlled formula summing to 99.5%, when I activate it, then validation fails.
- **AC-24.10** (FR-24.5) Given a batch with disposition on-hold, when a batch order picks it, then the batch is excluded from proposals.

4.24.9 Test plan and fixtures

Fixtures: `pmf-potency-20` (standard/actual potency pairs with expected consumption); `pmf-coby-12` (allocation methods × by-product cost types); `pmf-catchweight-10`; `lean-flow-demo` (3-activity flow, 2 fixed rules, 1 event rule); `lean-kanban-sim` (sales lines → event kanbans → board completion).

Test	Type	Target	Pass criterion
Potency	unit	consumption proposal	20/20
Allocation sums	property	allocation	Σ = cost for random costs/quantities
Catch weight	unit	RAF validation	min/max enforced
Kanban state machine	property	transitions	illegal rejected
Event kanban latency	integration	event→kanban	p95 < 5 s
Board E2E	Playwright	board	kanbans complete, RAF posted
Flow settlement	contract	ledger	balanced entry

Reconciliation: Σ co-product allocated cost = batch cost; Σ kanban RAF = inventory receipts from kanban orders; flow period variance = GL flow variance account.

4.24.10 Deployment and operations

Helm: `production` extended (flag `prod.process`), `lean` (Go, 2 replicas), `lean-db` (CloudNativePG 50 GB). Config: `LEAN_EVENT_KANBAN_SLA_S` (5), `PMF_POTENCY_TOLERANCE_PCT` (0.5). Flags: `prod.process`, `prod.catch-weight`, `lean.kanban`, `lean.flow-costing`. Jobs: shelf-advice notification daily 06:00; flow period auto-settle (optional) monthly. SLOs: board load p95 < 500 ms; kanban generation p95 < 5 s. Alerts: event-kanban backlog > 100 (warn); flow settlement failure (page). Rollback: prior digest; flags off disables the second plant's workspaces without data loss.

4.24.11 Migration from AX / SAP

Source	Passage entity	Notes
PmfFormula/BOMVersion+BOM lines (AX) / PLKO+PLPO+STPO recipe (SAP)	Formula, FormulaLine	formula size → base quantity; percent lines derived from quantities when SAP recipe is in base units
PmfCoByProd / co-product apportionment (TKKAA)	CoProductLine on formula	TCA percentages from apportionment structure
PdsBatchAttrib, PdsBatchAttribItem, PdsBatchAttributes / batch classification (AUSP, CABN)	BatchAttribute, BatchAttributeValue	characteristic → attribute; numeric/char/date types
InventBatch expiry/best-before / MCH1 VFDAT, HSDAT	InventoryBatch (w16) fields	already migrated in w16; this wave adds attributes
PdsCW / CWM units	CatchWeightQty	nominal/min/max
LeanProductionFlow, LeanProductionFlowActivity / production versions, REM profiles	ProductionFlow, FlowActivity	active versions only
KanbanRule, Kanban, KanbanJob / PKHD, PKPS	KanbanRule, Kanban (active cards)	completed/emptied cards not migrated

Validation: formula explosion of 100 sampled batches matches legacy consumption to 0.01%; active kanban count per rule equals legacy. Not migrated: process-order history, PI sheet archives, SAP XSteps.

4.24.12 Dependencies, risks, open questions

Depends on: w21 order engine; w16 batches and FEFO; w9 ECO; w6 operating units (value streams); w22 scheduled kanbans via planning.

Risk	Likelihood	Impact	Mitigation
Potency/attribute semantics vary by industry	High	Medium	Configurable role model; consultant review
Lean costing conflicts with order-based WIP reconciliation	Medium	High	Kanban orders are ProductionOrders — same WIP accounts; flow periods settle through the same End logic

Open questions: (1) Should by-product `recycled` (negative issue) be supported in v1? Default yes. (2) Board per work cell vs per flow? Default work cell; decide by w24 kickoff.

4.25 Wave 25: Shop Floor Execution, Time Registration & IoT

4.25.1 Scope and business context

Wave 25 puts production on the floor. Operators use a touch kiosk (PWA) to clock in and out, start and stop jobs (including bundling several jobs), report good and scrap quantities with reason codes, run quality checks and view instructions and drawings, and keep working for up to 30 minutes offline. Registrations are turned into pay lines (overtime, shift premiums, flex) through profiles, approved by supervisors and exported to payroll (w43). Machines publish counts, states and downtime over OPC-UA/MQTT through an edge connector into Kafka; the shopfloor service computes OEE per resource in 15-minute buckets and drives andon alerts. Labor and machine time flows into the production order's route-card postings. This is the phase-exit wave: the demo plant manufactures a configured product from planned order to finished goods with full costing. Out of scope: payroll calculation itself (w43), quality orders and usage decisions (w33; kiosk checks are simple pass/fail capture), maintenance work orders from machine events (w44 consumes the events), label printing beyond kanban cards (w32).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Job-card re-keying: AX JmgTermReg registrations are "calculated" and "transferred" by a supervisor before ProdJournalRoute lines exist, so route cards are re-keyed or posted a day late; SAP CO11N confirmations are typed by a clerk from paper tickets; Infor LN and Epicor Kinetic MES data collection posts labor in a nightly batch	Registration and production posting are separate modules joined by a transfer job	Registration stop generates and posts the job-card journal (option immediate); reversal on edit	Labor posting latency 24 h → < 5 s; re-keyed tickets 0
Kiosk needs the AOS: AX terminal registration (JmgTermReg form) is a thick client; D365 production floor execution requires connectivity; Plex is cloud-only with no offline mode	No client-side queue or idempotent replay	PWA with IndexedDB queue, client UUIDs, conflict flags	Lost registrations during outages → 0 over a 30-min disconnect
Machine data lives outside the ERP: AX has no native machine integration; SAP MII/PCo is a separate licensed stack with custom BLS transactions; Plex offers machine connectivity only through its own gateway	Proprietary connectors and point-to-point integration	`iot-edge` (OPC-UA/MQTT) → Kafka CloudEvents; any system subscribes to `shopfloor.machine-events`	Time to connect a new machine 2 weeks → 1 day; integration middleware 0
OEE is computed in spreadsheets or a BI tool from exported confirmations; SAP DM OEE requires its own master data; Epicor Advanced MES OEE updates per shift	No event-level counts in the ERP; ratios averaged incorrectly across shifts	OeeBucket per 15 min from events, ISO 22400 formula, roll-up by summing minutes/counts	OEE visibility per shift → per 15 min with p95 lag < 5 s
Pay calculation surprises: AX JmgPayTable overtime depends on profile day types supervisors do not see, so corrections happen after payroll export; SAP time evaluation (RPTIME00) is a batch run with cryptic log	Rules engine hidden from the people approving	PayLine per rule with source registrations shown on the approval screen; recalculation on edit	Post-export corrections 5%/period → < 0.5%; approval time per team 2 h → 15 min

4.25.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Job registration terminal	JmgTermReg (terminal registration form), JmgJobTable, JmgRegistration, D365 production floor execution	SAP MII / Digital Manufacturing shop floor UI, PP-PDC terminals, CO11N	Kiosk PWA, `shopfloor-bff`, TimeRegistration	full
Clock in/out, absence	JmgStampJournalTable (raw), JmgTmpRegistration, JmgAbsenceCode, JmgProfileTable	Time events (CATS/PT), HR time types	Registration kinds clock-in/out, absence; ProfileDay	full
Job bundling	Jmg bundle jobs (JmgJobBundle)	n/a (multi-order confirmation)	bundle_id on registrations, cost split by quantity/hours	full
Feedback (qty/scrap)	JmgJobTable feedback, ProdJournalRoute from Jmg, error cause codes	CO11N yield/scrap/rework, reason for variance (AFRU.GRUND)	job-card journals auto-generated; ScrapReason	full
Pay calculation	JmgPayTable, pay agreements (JmgPayAgreement), overtime/flex, pay-item transfer	HR time evaluation (RPTIME00), wage types	PayLine, PayRule engine	full
Approval and transfer	Jmg calculate → approve → transfer to payroll (JmgPayTransfer)	CATS approval, transfer to HR	Approve + export (w43 payroll)	full
Machine integration	n/a native (IoT Intelligence for D365, Sensor Data Intelligence)	SAP MII PCo/OPC, Plant Connectivity, SAP DM	`iot-edge` connector (OPC-UA/MQTT) → Kafka	improved
OEE	D365 Sensor Data Intelligence (partial)	SAP DM OEE module	OeeBucket per resource × 15 min; ISO 22400 KPIs	full
Downtime reasons	Jmg indirect activities	PM notification/downtime reasons	DowntimeRecord with reason code	full
Andon	n/a	SAP DM andon	AndonAlert via event + kiosk banner	improved
Instructions/drawings	Route notes, document attachments (DocuRef)	PI sheets, DMS documents on operation	Operation documents (w5 attachments) on kiosk	full
Offline kiosk	n/a (requires AOS)	n/a	PWA with IndexedDB queue	improved

4.25.3 Functional requirements

- **FR-25.1 Kiosk login** — Must. The system shall let a worker log in on a shared kiosk by badge scan (GS1 barcode/NFC) or PIN mapped to a Keycloak user, with session timeout 10 min idle and switch-user in ≤ 2 taps.
- **FR-25.2 Clock in/out and absence** — Must. The system shall record clock-in/out and absence registrations with the device id and server timestamp, applying profile rounding (BR-25.2).
- **FR-25.3 Job start/stop and bundling** — Must. The system shall list dispatchable jobs for the worker's resource/work cell (from w23 ScheduledJob, ordered by start), allow start/stop of one job or a bundle, and split bundled time by quantity reported (default) or equally.
  - Notes: A worker may have one open job/bundle at a time; starting another auto-stops the previous. Jobs in status `finished` are hidden unless the supervisor reopens.
- **FR-25.4 Feedback** — Must. The system shall capture good qty, scrap qty with reason code, and "finished" flag per job; on stop it generates a job-card journal (w21) with hours and quantities and posts it (configurable: immediately or at approval).
- **FR-25.5 Quality checks** — Should. The system shall present check items attached to the operation (text/numeric/pass-fail with limits) and record results; a failed mandatory check blocks `finished` until a supervisor override.
- **FR-25.6 Instructions and documents** — Must. The kiosk shall show operation instructions (markdown) and attachments (PDF, images, video ≤ 50 MB) cached for offline use.
- **FR-25.7 Pay calculation** — Must. The system shall calculate PayLines per worker-day from registrations using the worker's profile (day types, shift windows, breaks) and pay agreement rules: base, overtime after daily/weekly thresholds, shift premium, flex balance.
  - Notes: Defaults: overtime > 8.00 h/day at 1.5×, > 40.00 h/week at 1.5×, Sunday 2.0×; rounding per BR-25.2; max 24 h/day validation.
- **FR-25.8 Approval and export** — Must. Supervisors shall approve per worker-day (bulk by team), with edits audited; approved days are locked and exported as `dev.passage.shopfloor.payday.approved.v1` for w43 and as CSV.
- **FR-25.9 IoT ingestion** — Must. The `iot-edge` connector shall subscribe to OPC-UA nodes (via `gopcua`) and MQTT topics (Mosquitto/EMQX, QoS 1), normalize to the MachineEvent schema (BR-25.4), buffer locally (SQLite ring, 24 h), and publish to Kafka topic `shopfloor.machine-events` with idempotent producer and per-machine monotonic sequence.
- **FR-25.10 OEE** — Must. The system shall compute OEE per resource per 15-minute bucket and shift (BR-25.5) from machine events and planned time from the w23 schedule/calendar, with dashboard lag p95 < 5 s.
- **FR-25.11 Downtime and andon** — Must. The system shall open a DowntimeRecord when a machine reports DOWN or no count for > `idle_threshold_s` (default 300), require a reason code within the kiosk, and raise an AndonAlert (kiosk banner + notification) on downtime > 10 min, scrap rate > 5% in a bucket, or quality fail.
- **FR-25.12 Offline kiosk** — Must. The PWA shall queue registrations and feedback in IndexedDB while disconnected (≥ 30 min, ≤ 5,000 actions), replay in order on reconnect with client-generated UUIDs for idempotency, and resolve conflicts per BR-25.7.
- **FR-25.13 Machine-to-job linkage** — Should. Machine counts shall be attributed to the job currently running on that resource (from registrations or the schedule), and may auto-fill good/scrap quantities on the kiosk for operator confirmation.
- **FR-25.14 Supervisor view** — Must. A supervisor workspace shall show who is clocked in, current jobs, open downtimes, OEE tiles, and pending approvals.

4.25.4 Data model

Worker (shopfloor projection of Party/Employee w7): id, party_id, badge_id (unique), pin_hash, default_resource_id, work_cell_id, profile_id, pay_agreement_id, cost_rate (numeric 18,4), active.

TimeRegistration: id (client UUID), tenant_id, worker_id, kind (enum: clock-in, clock-out, job-start, job-stop, absence, indirect), job_id (nullable), bundle_id (nullable), qty_good, qty_scrap, scrap_reason_id, started_at, stopped_at (timestamptz), server_received_at, device_id, offline (bool), hours (numeric 10,4, computed), status (enum: raw, calculated, approved, exported), journal_id (w21 job card), note.

ProfileDay: id, profile_id, day_type, shift_start, shift_end, break_windows (jsonb), rounding_minutes (int), flex_allowed (bool), tolerance_before_min, tolerance_after_min.

PayLine: id, worker_id, work_date, pay_type (enum: base, overtime-1.5, overtime-2.0, shift-premium, flex-in, flex-out, absence-paid, absence-unpaid), hours, rate_multiplier, amount (numeric 18,4), currency, source_registration_ids (uuid[]), approved_by, approved_at, export_batch_id.

MachineEvent: id, machine_id, resource_id, seq (bigint), kind (enum: count, state, downtime, alarm, measurement), ts (timestamptz, device), received_at, good_count, reject_count, state (RUNNING/IDLE/DOWN/SETUP/OFF), reason_code, measurement_name, measurement_value, unit, job_id (attributed), payload (jsonb).

OeeBucket: resource_id, bucket_start (15 min), shift_id, planned_minutes, run_minutes, down_minutes, ideal_cycle_s, total_count, good_count, availability, performance, quality, oee (numeric 7,4 each), source_events (int).

DowntimeRecord: id, resource_id, started_at, ended_at, reason_code (nullable until coded), category (planned/unplanned), job_id, worker_id, andon_raised (bool).

Invariants: for each worker at most one open job-start without job-stop; PayLine hours per worker-day ≤ 24.00; OeeBucket.run_minutes + down_minutes ≤ planned_minutes; MachineEvent (machine_id, seq) unique; approved PayLines are immutable (new lines + reversal on correction).

Indexing: TimeRegistration (worker_id, started_at), (job_id), unique id (client UUID); MachineEvent partitioned by month (TimescaleDB-free: native range partitions), index (resource_id, ts); OeeBucket PK (resource_id, bucket_start); retention: raw MachineEvent 90 days in Postgres, forever in Iceberg (w46); OeeBucket 3 years.

4.25.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/shopfloor/sessions:badge	Kiosk login by badge/PIN	n/a	Kiosk login
POST	/api/v1/shopfloor/registrations	Create registration(s), batch of ≤ 500 (offline replay)	client UUID per item	Register time
GET	/api/v1/shopfloor/jobs?workerId&resourceId&status	Dispatch list	n/a	Register time

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/shopfloor/jobs/{id}:feedback	Qty good/scrap/finished, checks	client UUID	Register time
GET	/api/v1/shopfloor/operations/{id}/instructions	Instructions + attachment URLs (signed, cacheable)	n/a	Register time
POST	/api/v1/shopfloor/pay:calculate	Calculate pay lines for worker-days	Idempotency-Key	Calculate time registrations
POST	/api/v1/shopfloor/pay:approve	Approve worker-days (bulk)	Idempotency-Key	Approve time registrations
POST	/api/v1/shopfloor/pay:export	Export batch (CSV + event)	Idempotency-Key	Export time to payroll
GET	/api/v1/shopfloor/oee?resourceIds&from&to&bucket=15m	OEE series	n/a	View shop floor
POST	/api/v1/shopfloor/downtimes/{id}:code	Assign reason	Idempotency-Key	Register time
GET	/api/v1/shopfloor/andon?siteId	Open alerts	n/a	View shop floor
POST	/api/v1/shopfloor/machines	Register machine → resource mapping, tags	Idempotency-Key	Maintain shop floor setup

Event	Trigger	Payload fields	Consumers
`dev.passage.shopfloor.registration.posted.v1`	registration stored	registrationId, workerId, kind, jobId, hours, qtyGood, qtyScrap, offline	production (job card), analytics
`dev.passage.shopfloor.machine.event.v1`	edge publish	see BR-25.4	shopfloor OEE, production, maintenance (w44)
`passage.shopfloor.oee.bucket-updated.v1`	bucket recomputed	resourceId, bucketStart, availability, performance, quality, oee	dashboards (w46)
`dev.passage.shopfloor.andon.raised.v1`	alert condition	alertId, resourceId, kind, severity, jobId, startedAt	notifications, kiosk
`dev.passage.shopfloor.payday.approved.v1`	approval	workerId, workDate, payLines[{type, hours, multiplier, amount}], approvedBy	payroll (w43)

4.25.6 User experience

Screen	Primary actions	Key fields/columns	Duty required
Kiosk home	Badge in/out, my jobs, start/stop, bundle, report, downtime reason, andon	Worker, clock status, current job, elapsed, qty good/scrap, offline indicator	Kiosk login / Register time
Job detail (kiosk)	View instructions/drawings, checks, report	Operation, order, item, qty planned/reported, checks	Register time
Supervisor board	Who is in, current jobs, downtimes, OEE tiles, approve days	Worker, job, elapsed, resource state, OEE A/P/Q, pending	Approve time registrations
Time approval	Calculate, edit with reason, approve, export	Worker, date, registrations, pay lines, flags (missing clock-out)	Approve time registrations
OEE dashboard	Resource/period filter, drill to downtime and buckets	OEE, A, P, Q, top losses (Pareto)	View shop floor
Machine setup	Map machine → resource, tags, ideal cycle time, thresholds	Machine, resource, protocol, node ids/topics, cycle s	Maintain shop floor setup

Interaction rules: kiosk targets ≥ 48 px, no hover-only affordances, numeric keypad for quantities, high-contrast theme, every action confirmed in ≤ 300 ms (optimistic UI); offline banner shows queued count; supervisor edits require a reason; OEE dashboard uses the w46 dashboard type when available, otherwise the Grafana panel embedded via the shell.

4.25.7 Business rules and edge cases

- **BR-25.1 Time attribution.** Hours for a job registration = (stopped_at – started_at) minus breaks that fall inside per the profile; bundle hours split by Σ qty_good (fallback equal split); hours rounded to 0.01 h on the job-card journal.
- **BR-25.2 Rounding and tolerances.** Clock-in within tolerance_before_min (default 10) of shift start rounds to shift start; clock-out within tolerance_after_min (default 5) rounds to shift end; otherwise round to nearest rounding_minutes (default 1, max 15).
- **BR-25.3 Missing clock-out.** At shift end + 4 h a missing clock-out auto-closes at shift end with flag `missing-clock-out`; the day cannot be approved until a supervisor confirms or edits.
- **BR-25.4 Machine event schema.** CloudEvents envelope `type: dev.passage.shopfloor.machine.event.v1`, `source: urn:passage:iot-edge:<site>:<edgeId>`, `subject: <machineId>`, `id: <machineId>-<seq>`, `time: <device ts RFC 3339>`, `datacontenttype: application/json`, `data: { "machineId": "M-101", "resourceId": "uuid", "seq": 184233, "kind": "count|state|downtime|alarm|measurement", "ts": "2026-10-21T08:15:03.250Z", "goodCount": 12, "rejectCount": 1, "state": "RUNNING", "reasonCode": null, "measurement": {"name": "temp_c", "value": 71.4, "unit": "Cel"}, "jobId": null, "tags": {"line": "L1"} }`. Counts are deltas since the previous count event; absolute counters are converted at the edge with rollover detection (32-bit). Events older than 7 days or with seq ≤ last seen are dropped and counted in a metric.
- **BR-25.5 OEE formula (ISO 22400 aligned).** Availability = run_minutes / planned_minutes; Performance = (ideal_cycle_s × total_count / 60) / run_minutes, capped at 1.0 with an over-rate flag; Quality = good_count / total_count (1.0 if total = 0); OEE = A × P × Q. Planned minutes = scheduled open minutes minus planned downtime (category planned). Example shift: planned 480 min, unplanned down 60 → A = 0.875; ideal cycle 60 s, total 380 → P = 380 / 420 = 0.9048; good 361 → Q = 0.95; OEE = 0.7522. Buckets roll up to shift/day by summing minutes and counts, not by averaging ratios.
- **BR-25.6 Idle detection.** No count event for idle_threshold_s while state = RUNNING flips the bucket to IDLE from the last count; a later count closes the idle gap.
- **BR-25.7 Offline conflict resolution.** Replayed registrations are accepted in client timestamp order; a job-start for a job that was finished server-side while offline is accepted but flagged `job-finished-conflict` for supervisor review; a duplicate client UUID is a no-op; clock skew > 5 min between device and server is corrected by the offset measured at reconnect and recorded.
- **BR-25.8 Pay thresholds.** Weekly overtime counts hours after 40.00 across the pay week (profile-defined start day); daily and weekly overtime do not double-count: hours already paid at 1.5× daily are excluded from the weekly threshold calculation.
- **BR-25.9 Posting to production.** Job-card journals from registrations post at cost_rate of the worker (if set) else the resource's cost category; a registration edited after posting creates a reversing journal and a new one.
- **BR-25.10 Andon de-duplication.** One open AndonAlert per (resource, kind); re-raising extends it; auto-closes when the condition clears for 5 min.

4.25.8 Acceptance criteria

- **AC-25.1** (FR-25.3, 25.4, BR-25.1) Given 6 operators on a shift with the fixture schedule, when they clock jobs on the kiosk (Playwright) and the day is approved, then each production order's labor cost = Σ hours × worker cost rate to the cent.
- **AC-25.2** (FR-25.7) Given a worker registering 09:00–19:30 with a 30-min break on a Tuesday, then pay lines are base 8.00 h and overtime-1.5 2.00 h.
- **AC-25.3** (FR-25.9) Given a simulated machine publishing 10,000 events/min for 60 min with a Kafka broker restart at minute 30, then every seq is present in MachineEvent and no duplicates exist.
- **AC-25.4** (FR-25.10, BR-25.5) Given the BR-25.5 shift example, then the shift OEE reads 0.7522 and the tile updates within 5 s of the last event.
- **AC-25.5** (FR-25.12) Given a kiosk disconnected for 30 min with 40 actions queued, when it reconnects, then all 40 are stored once in order and conflicts are flagged, not lost.
- **AC-25.6** (FR-25.11) Given a machine DOWN for 11 min without a reason, then an andon alert is open, the kiosk shows a banner, and coding the reason closes the downtime but keeps the alert until cleared.
- **AC-25.7** (FR-25.2, BR-25.2) Given shift start 07:00 and clock-in 06:53, then the registration time is 07:00.
- **AC-25.8** (FR-25.8) Given an approved day, when a supervisor edits a registration, then a reversal and a new PayLine are created and the export batch includes both.
- **AC-25.9** (FR-25.5) Given a mandatory numeric check with limits 10–12 and a value 13, then `finished` is blocked until override with reason.
- **AC-25.10** (phase exit) Given the demo plant, when the configured product (w8 configurator) is planned (w22), scheduled (w23), produced through kiosk registrations and machine counts (w25) and ended (w21), then the order's actual cost, variances and WIP reconcile to the journal and the recorded walkthrough is signed off.

4.25.9 Test plan and fixtures

Fixtures: `shift-6-operators` (schedule, badge ids, profiles, expected pay lines and order costs); `pay-rules-30` (daily/weekly overtime, Sunday, flex, missing clock-out); `iot-sim-10k` (Go simulator producing counts/states with controllable gaps and rollover); `oee-shift-cases` (10 shifts with hand-computed A/P/Q); `offline-replay-40`.

Test	Type	Target	Pass criterion
Pay rules	unit	PayRule engine	30/30
OEE	unit	bucket calc	matches hand-computed to 4 decimals
Event ordering/loss	integration	iot-edge → Kafka → DB	0 lost, 0 dup over 600k events with broker restart
Kiosk shift E2E	Playwright	kiosk + production	labor cost ties
Offline sync	Playwright (offline mode)	PWA queue	40/40 once, in order
Throughput	performance	ingest	10k events/min sustained, consumer lag < 5 s
Accessibility	axe	kiosk screens	WCAG 2.1 AA

Reconciliation: Σ approved hours per order = Σ job-card hours posted; Σ PayLine hours = Σ registration hours per worker-day; Σ MachineEvent good counts attributed to a job = kiosk-confirmed good qty (variance report).

4.25.10 Deployment and operations

Helm: `shopfloor` (Go, 3 replicas; Kafka consumer group `shopfloor-oee`), `shopfloor-bff` (TS, 2 replicas, SSE), `kiosk` (static PWA served by the shell CDN path `/kiosk`), `iot-edge` (Go, deployed per plant; DaemonSet or single-node k3s at the edge; Mosquitto bundled optionally), `shopfloor-db` (CloudNativePG 200 GB, monthly partitions). Config: `SF_IDLE_THRESHOLD_S` (300), `SF_ANDON_DOWNTIME_MIN` (10), `SF_OFFLINE_MAX_ACTIONS` (5000), `IOT_BUFFER_HOURS` (24), `IOT_OPCUA_ENDPOINTS`, `IOT_MQTT_BROKER`. Flags: `shopfloor.iot`, `shopfloor.oee`, `shopfloor.pay-export`. Jobs: auto clock-out (hourly), OEE rollup (every 15 min), MachineEvent partition drop (daily, > 90 days). SLOs: kiosk action p95 < 200 ms server-side; ingest 10k events/min with consumer lag p95 < 5 s; availability 99.9%; edge buffer never exceeds 80% of 24 h. Alerts: consumer lag > 30 s (page), edge offline > 5 min (warn), seq gap detected (warn), andon storm > 50 open (warn). Runbook: "edge offline" — check MQTT/OPC-UA connectivity, buffer fill, restart connector; "seq gap" — replay from edge buffer by seq range. Rollback: prior digest; edge connector version pinned per plant with staged rollout.

4.25.11 Migration from AX / SAP

Source	Passage entity	Notes
JmgEmployee / HR mini-master (PA0001)	Worker	badge ids, default resource, profile
JmgProfileTable, JmgProfileDay / work schedule rules (T508A)	ProfileDay	shifts, breaks, rounding
JmgPayAgreement, JmgPayTable / wage types, time evaluation rules	PayRule config	overtime thresholds and multipliers
JmgAbsenceCode, Jmg error cause codes / absence types, AFRU.GRUND	reason codes	mapped to Passage code lists
JmgStampJournalTable, JmgRegistration (last 90 days) / CATSDB, AFRU	TimeRegistration (history, status exported)	read-only history for balances
Flex balances (JmgFlexTrans) / time accounts (PT_BAL00)	opening flex balance per worker	one opening line

Validation: flex balances per worker equal legacy at cutover; pay lines for the last closed week recomputed in Passage match legacy within 0.01 h per worker. Not migrated: raw terminal logs older than 90 days, machine data (none in AX/SAP core), MII configurations.

4.25.12 Dependencies, risks, open questions

Depends on: w21 job-card journals; w23 scheduled jobs (dispatch list); w24 kanban jobs on the kiosk; w7 employee party; w5 attachments and document service; w4 Kafka/Temporal; w43 payroll (consumer, later).

Risk	Likelihood	Impact	Mitigation
OPC-UA server diversity (security policies, node models)	High	Medium	Support None/Basic256Sha256 and companion-spec-agnostic tag mapping; certify against open62541 and a PLC simulator
Offline conflicts corrupt labor cost	Medium	High	Client UUID idempotency, flagged conflicts, supervisor review before posting option
Phase-exit demo depends on four prior waves' stability	High	High	Freeze w21–24 APIs two weeks before w25 close; rehearsal run weekly

Open questions: (1) Post job cards immediately or at approval? Default immediately with reversal on edit; decide by w25 kickoff. (2) Edge connector as DaemonSet in the plant cluster vs standalone binary? Default standalone k3s node per plant; decide by w25 mid-point. (3) Which OEE loss taxonomy (six big losses vs ISO 22400 element set)? Default six big losses mapped to ISO 22400 KPIs; decide by w25 kickoff.

Part 5 — Phase 5: Order-to-Cash (waves 26–30)

5.0 Phase overview

Business outcome at phase exit. At the end of wave 30 a manufacturer runs the complete buy → make → sell → collect loop on Passage. A sales rep quotes and takes an order at a deterministically resolved price; the order is promised against cumulative ATP and checked against credit; the warehouse picks, packs and ships it with a posted packing slip that moves inventory cost to the universal journal; billing posts the invoice with tax and an ASC 606 / IFRS 15 recognition schedule; cash is applied from lockbox, remittance or bank statement; collections runs dunning and disputes; returns flow through RMA to credit note; and every subledger reconciles to the journal by automated check. Wave 30 is the first go-live candidate.

Phase-exit sign-off criteria (each produces an artifact in `reconciliation/phase-5-exit/`):

- One simulated 31-day month (fixture period 2027-03) runs end-to-end on `stage` for demo entities DMF-US (USD) and DMF-DE (EUR): ≥ 200 purchase orders received (Phase 3), ≥ 150 production orders reported as finished (Phase 4), 1,000 sales orders, ≥ 900 shipments, ≥ 950 invoices including ≥ 40 bundles with revenue schedules, 600 customer payments across manual / BAI2 lockbox / camt.053, one dunning run, 30 RMAs, one rebate settlement, inventory close and period close for 2027-03.
- Daily reconciliation passes for all 31 days with zero unexplained differences: AR open items = GL 1200; deferred revenue roll-forward = GL 2400; delivered-not-invoiced = GL 1405; inventory subledger = GL 1400; WIP = GL 1350; rebate accrual = GL 2150; tax payable = tax engine settlement = GL 2300; unapplied cash = GL 1205.
- Trial balance for 2027-03 balances; GL 4000 = Σ posted invoice revenue lines – credit notes – deferrals; GL 5000 = Σ packing-slip cost – return receipts.
- Pricing: 500 scenario fixtures and 10k-request determinism property test green; p95 < 20 ms for a 10-line order.
- Order intake 500 orders/min at p99 < 500 ms; ATP never over-promises (property test over 100 fixtures).
- Cash application ≥ 90% auto-match on the 500-payment fixture with zero false matches; dunning run produces letters for exactly the expected customer set.
- Pick-pack-ship E2E for 1,000 orders with zero inventory mismatch; packing-slip reversal contract tests green.
- Recorded walkthrough (≤ 45 min) and a reconciliation pack signed by a practicing accountant, both linked from the wave 30 issue.

Cross-wave data model summary

Entity	Owning service (Postgres db)	Key relationships
condition_record, pricing_procedure, rebate_agreement, price_journal	pricing (pricing_db, Go)	→ customer/group (w7), product (w8); explain refs consumed by sales/billing
sales_quotation, sales_order(_line), sales_agreement, order_hold, credit_profile, atp_bucket	sales (sales_db, Go)	→ condition_record; supply/demand from w16/w17/w21/w22; production order link (w21)
shipment, container, packing_slip, carrier_shipment	fulfillment (fulfillment_db, Go) + carrier-gateway (TS)	→ sales_order_line; inventory_transaction (w16); journal posting (w11)
invoice(_line), free_text_invoice, credit_note, subscription	billing (billing_db, Go)	→ packing_slip; tax (w13); journal (w11)
revenue_contract, performance_obligation, revenue_schedule	revrec (revrec_db, Go)	→ invoice_line; journal (w11)
open_item, payment, settlement, lockbox_batch, dunning_run, dispute, interest_note, return_order	receivables (receivables_db, Go) + collections-workflows (TS, Temporal)	→ invoice, customer, camt.053 statement, journal
portal_user, payment_intent	customer-portal (portal_db, TS BFF) + payments-gateway (TS)	→ party contact (w7), open_item, Stripe/Adyen adapter

Cross-wave event catalog (CloudEvents on Kafka topics `passage.<domain>`, Avro in schema registry, Debezium outbox)

Event	Producer	Consumers	Payload summary
passage.pricing.condition-record.activated.v1	pricing	sales cache, analytics	id, type, keys, validity, version
passage.pricing.rebate-accrual.posted.v1	pricing	ledger, receivables	agreement, period, amount, journal ref
dev.passage.sales.order.confirmed.v1	sales	planning, fulfillment, credit, analytics	order, customer, lines with confirmed dates, totals
passage.sales.order-hold.applied.v1 / released.v1	sales	fulfillment, collections, portal	order, hold type, reason, actor
passage.fulfillment.packing-slip.posted.v1	fulfillment	billing, inventory, ledger, sales, portal	slip, lines, qty, cost snapshot, journal ref
dev.passage.billing.invoice.posted.v1	billing	receivables, revrec, ledger, pricing, portal	invoice, lines, tax, totals, due date
dev.passage.revrec.schedule.recognized.v1	revrec	ledger, analytics	schedule line, obligation, amount, period
dev.passage.receivables.payment.applied.v1	receivables	ledger, sales (credit), portal	payment, settlements[], residual
passage.receivables.customer-balance.changed.v1	receivables	sales credit, portal	customer, balance, overdue, oldest item days, dunning level
passage.receivables.return-order.dispositioned.v1	receivables	fulfillment, billing, inventory	RMA line, disposition, qty, restocking

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
AX "find next" vs. SAP access-sequence semantics confuse migrated users	High	Medium	Two reference procedures shipped; both in the 500-scenario fixtures	Product
ATP over-promises because a supply source is missed	Medium	High	Projection rebuilt nightly from events and diffed; property test gates release	Engineering
Delivered-not-invoiced drifts from GL 1405	Medium	High	Packing slip and invoice share one cost_snapshot_id; daily check from wave 28 day 1	Engineering

Risk	Likelihood	Impact	Mitigation	Owner
ASC 606 allocation disputed by auditors	Medium	High	40 fixtures reviewed by a practicing accountant; residual-cent rule documented	Product + accountant
False auto-match in cash application	Low	High	Single candidate ≥ 85 with 20-point margin; every auto-match reversible	Engineering
Go-live simulation exposes Phase 3/4 defects late	High	Medium	Dry-run at wave 28 exit with sell/collect stubs	Both owners

5.26 Wave 26: Pricing, Discounts & Trade Agreements

5.26.1 Scope and business context

Every order, quotation, invoice and rebate in Phase 5 depends on a price that is explainable and repeatable. Wave 26 ships the `pricing` service: condition records with validity and quantity breaks, access sequences resolving from the most specific key to the most general, line/multiline/total discounts, surcharges, exclusion groups, rounding rules, campaigns, rebate agreements with accrual, atomic price journals, a simulation/explain API and commission hooks. At wave end a pricing administrator maintains agreements in the Pricing workspace and any caller receives, for a customer/product/date/quantity/currency tuple, a net price plus the ordered list of conditions that produced it. Out of scope: rebate settlement (wave 30), vendor pricing (wave 17), configurator price components beyond attribute-keyed records, storefront promotions.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Nobody can say why a line priced at 93.40: AX "Find next" results are invisible on SalesLine; SAP needs the VA01 condition "Analysis" screen per line	Resolution is a side effect inside the order form (PriceDisc class / RV61A formulas), not a queryable function	`POST /pricing/resolve?explain=true` returns the access-by-access trace, stored with the line as `price_explain_ref`	"Why this price": 20–40 min consultant trace → < 10 s self-service; 100% of lines traced
Mass price change is a multi-hour, non-atomic job: PriceDiscAdmTrans posts row by row and leaves half-applied prices on failure; VK12 mass change has no undo	Row loop in X++/ABAP without a transaction boundary across the set	price_journal validates fully, posts in one transaction and reverses with one action	100k prices: 3–6 h plus cleanup → < 120 s; rollback 1 action
Overlapping agreements silently win by insertion order (first PriceDiscTable row found; A-table collisions blocked only per access)	No overlap validation at save	Activation rejects overlapping validity for the same key with 409 naming the conflict	Overlap-caused price disputes: ~10 per 1,000 agreements/month → 0
Rounding differs between order, packing slip and invoice (SalesLine.LineAmount re-rounded per form letter; KPEIN pricing-unit rounding)	Each consuming document re-rounds	Engine rounds once (BR-26.3); consumers reuse the stored net; invoice totals property-tested	Cent-level order/invoice differences: ~0.5% → 0
CPQ tools and storefronts re-implement price lists in middleware because pricing is unreachable outside the form call stack	No stateless pricing API	Resolve API p95 < 20 ms, condition-record events, MCP exposure via wave 46	External lookups: nightly export → live; zero duplicated price lists

5.26.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Sales price agreements	PriceDiscTable (Relation=PriceSales), PriceDiscAdmTable/Trans journal	VK11/VK12/VK13, KONH/KONP, A004/A005/A305, PR00	condition_record type PRICE	full
Line discount	PriceDiscTable (LineDiscSales), PriceDiscGroup	K004/K005/K007, KONP-KBETR	condition_record LINE_DISC	full
Multiline discount	PriceDiscTable (MultiLineDiscSales)	Group condition T685A-KZBZG, K020	MULTILINE_DISC with group key	full
Total discount	PriceDiscTable (EndDiscSales)	Header conditions HI01/RA01	TOTAL_DISC	full
Price/discount groups	PriceDiscGroup; CustTable.PriceGroup/LineDisc/MultiLineDisc/EndDisc; InventTableModule.PriceGroup	KNVV-KONDA/KDGRP, MVKE-KONDM	customer_pricing_group, product_pricing_group	full
Access order / exclusion	PriceDiscParameters, "Find next", "Search again"	V/07 access sequences (T682), V/08 procedure (T683S), exclusion T684	pricing_procedure, access_sequence, exclusion_group	improved (explain)
Smart rounding	PriceDiscRounding, journal rounding	DIFF condition, T001R	rounding_rule	full
Campaigns / promotions	Trade allowance agreements (TAMAgreement)	Sales deals VB21/VB22, promotions VB31 (KONA)	campaign	full
Rebates	PriceDiscRebate / TAMRebateAgreement, D365 Rebate management	VBO1/VBO2 (KONA, BO01–BO06, VBOX); S/4 condition contract WCOCO	rebate_agreement, rebate_accrual	partial (settlement w30)
Commissions	CommissionCalc, CommissionSalesRep, CommissionTrans	VBPA partner function; ICM	commission_rule + event hook	partial
Mass update	Price/discount agreement journals	VK12 mass change, MASS	price_journal (atomic, reversible)	improved

5.26.3 Functional requirements

- **FR-26.1 Condition types and procedure** — Must. The system shall define condition types (PRICE, LINE_DISC, MULTILINE_DISC, TOTAL_DISC, SURCHARGE, FREIGHT, REBATE_ACCRUAL, ROUNDING) ordered in a versioned pricing procedure per legal entity and sales channel. Notes: reference procedures `AX_STANDARD` and `SAP_RVAA01`; steps 10–990 with optional CEL requirement (≤ 512 chars); activated versions are immutable.
- **FR-26.2 Condition records** — Must. The system shall store records keyed by customer | customer group | all × variant | product | product group | all, with optional site/warehouse/channel/campaign, validity, currency, unit, ≤ 20 strictly increasing quantity breaks, amount and/or two percentages. Notes: overlapping ACTIVE records for the same key and currency are rejected with 409.
- **FR-26.3 Access sequence** — Must. The system shall resolve each condition type by trying accesses in configured order (BR-26.1) and stop at the first access yielding a candidate unless `search_next` is set, in which case it continues and keeps the lower net.

- **FR-26.4 Deterministic resolution with explain** — Must. The system shall return identical outputs for identical (procedure version, records as of pricing_date, request) and an explain trace (≤ 64 KB, content-addressed) listing every access tried, candidates, winner and reason. No wall-clock reads.
- **FR-26.5 Line, multiline and total discounts** — Must. The system shall apply line discounts per line, multiline discounts over summed quantity of lines sharing a group, and total discounts on order net after line discounts; two percentages apply sequentially (1–p1)(1–p2); amount before percentage.
- **FR-26.6 Surcharges** — Should. Fixed or percentage surcharges as separate conditions with their own revenue account; small-order threshold default 250 USD / 200 EUR.
- **FR-26.7 Rounding** — Must. Unit prices at NUMERIC(19,6); line amounts half-even to ISO 4217 minor units (wave 12); smart rounding bands for journals (0–10 → .99, 10–100 → .95, ≥ 100 → whole).
- **FR-26.8 Exclusion groups** — Must. BEST (lowest net) or FIRST (lowest step) among members; an `exclusive` winner suppresses later discount steps.
- **FR-26.9 Price journals** — Must. Mass create/update/delete through a journal that validates fully, posts atomically (100k lines < 120 s) and reverses by one action into a new journal referencing the original.
- **FR-26.10 Campaigns** — Should. Records grouped under a campaign with validity, target groups, budget and kill switch.
- **FR-26.11 Rebate agreements and accrual** — Must. Customer rebate agreements (tiered % or amount per unit; monthly/quarterly/annual; basis net invoiced or quantity) accrue on each posted invoice as Dr 4150 / Cr 2150 via the journal API; retroactive tier crossing recomputes the period and posts the delta.
- **FR-26.12 Commission hooks** — Could. Rules by sales rep × customer group × product group publish `commission.calculated` for Phase 8.
- **FR-26.13 Simulation** — Should. Simulate any tuple without a document and compare two procedure versions.
- **FR-26.14 Performance** — Must. p95 < 20 ms, p99 < 60 ms for a 10-line resolve; in-memory record cache per entity invalidated by `condition-record.activated`.

5.26.4 Data model

condition_record

Field	Type	Required	Notes
id, tenant_id, legal_entity_id	uuid	Y	PK (v7); RLS
condition_type, key_combination	enum, smallint	Y	FK access_sequence_step
customer_id / customer_group_id	uuid	N	exactly one or null (all)
product_id / product_group_id / variant_dims	uuid / jsonb	N	same rule; dims {color,size,config}
site_id, warehouse_id, sales_channel, campaign_id	uuid/text	N	optional dims
currency, unit	char(3), text	Y	ISO 4217; FK uom (w6)
from_quantity, amount	numeric(19,6)	Y/N	default 0
disc_pct_1, disc_pct_2	numeric(7,4)	N	0–100
valid_from, valid_to	date	Y	inclusive
search_next, exclusive	bool	Y	default false
status, version, journal_id	enum, int, uuid	Y	DRAFT/ACTIVE/EXPIRED/WITHDRAWN

pricing_procedure: id, legal_entity_id, name, version, status, steps[] (step_no, condition_type, requirement_cel, exclusion_group_id, statistical, account_key). **access_sequence_step**: condition_type, seq_no, key_combination, required_fields[].

rebate_agreement: id, customer_id | customer_group_id, product scope, basis (NET_AMOUNT | QUANTITY), period (MONTH | QUARTER | YEAR), tiers[] (from_value, rate_pct | amount_per_unit), currency, accrual_account 2150, expense_account 4150, valid_from/to, status, settlement_method (CREDIT_NOTE | PAYMENT). **rebate_accrual**: id, agreement_id, period_id, invoice_line_id, basis_value, accrued_amount, journal_posting_id, status (ACCRUED | ADJUSTED | SETTLED | REVERSED).

price_journal: id, name, lines[] (op, before/after snapshot), status (OPEN | VALIDATED | POSTED | REVERSED), posted_at, reversed_by_journal_id.

Invariants: no two ACTIVE records share (type, keys, currency, unit, from_quantity) with overlapping validity; every ACTIVE record has a journal_id; resolve is reproducible from (procedure version, pricing_date, request); Σ rebate_accrual per period = GL 2150 movement. Indexing: composite btree (tenant, entity, type, key_combination, customer_id, product_id, currency, valid_from) plus partial indexes per key combination; partition by legal_entity; cache snapshot per (entity, pricing_date), 60 s TTL plus event invalidation.

5.26.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/pricing/resolve	Resolve header + ≤ 500 lines; `?explain=true`	stateless	PricingResolve
POST	/api/v1/pricing/condition-records, …/{id}:activate, :withdraw	Record lifecycle with overlap check	Idempotency-Key	PricingMaintain
GET	/api/v1/pricing/condition-records?customer=&product=&asOf=	Query	—	PricingView
POST	/api/v1/pricing/journals, …/{id}:post, :reverse	Mass update	Idempotency-Key	PricingJournalPost
POST	/api/v1/pricing/procedures, …/versions/{v}:activate	Procedure maintenance	Idempotency-Key	PricingAdmin
POST/GET	/api/v1/pricing/rebates/agreements; /rebates/accruals?period=	Rebates	Idempotency-Key	RebateMaintain
POST	/api/v1/pricing/simulate	Simulation, version comparison	stateless	PricingView

Validation: resolve requires legal_entity_id, customer_id, currency, pricing_date, lines[].product_id/quantity/unit; quantity ≤ 0 or unknown unit → 422. Journal post requires status VALIDATED with zero errors.

Event	Trigger	Payload fields	Consumers
passage.pricing.condition-record.activated.v1	activate / journal post	id, type, keys, currency, validity, version, journal_id, retroactive	sales cache, analytics
passage.pricing.condition-record.withdrawn.v1	withdraw / reverse	id, version, reason	sales cache
dev.passage.pricing.journal.posted.v1	post / reverse	journal_id, line_count, reversed_journal_id	audit
passage.pricing.rebate-accrual.posted.v1	invoice.posted consumed	agreement_id, period_id, invoice_line_id, amount, currency, journal_posting_id	ledger, receivables
dev.passage.pricing.commission.calculated.v1	invoice.posted consumed	invoice_line_id, sales_rep_id, basis, rate, amount	Phase 8

5.26.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Pricing workspace	tiles: expiring in 30 d, draft journals, campaigns, accrual this period	counts	PricingView
Trade agreements grid	create, copy, activate, withdraw, export	type, customer/group, product/group, currency, from qty, amount, %, validity, status	PricingMaintain
Price journal	import CSV (≤ 100k rows), validate, post, reverse	op, before/after, errors	PricingJournalPost
Procedure designer	add step, requirement, exclusion, activate version	step, type, requirement, exclusive	PricingAdmin
Simulation / explain	enter tuple, run, compare versions	per step: access, candidate, winner, amount	PricingView
Rebate agreements	tiers, accruals, close period	customer, basis, tiers, accrued	RebateMaintain

Interaction rules: keyboard row navigation and Ctrl+D duplicate; bulk edit validity and percentage on multi-select; validation on blur and at activate; explain tree navigable by arrow keys; column layouts personalizable (wave 5 shell).

5.26.7 Business rules and edge cases

- **BR-26.1 Resolution algorithm.** For each procedure step in order: (1) evaluate requirement; (2) for the step's condition type iterate accesses by seq_no — PRICE default: 1 customer × variant, 2 customer × product, 3 customer × product group, 4 customer group × product, 5 customer group × product group, 6 customer × all products, 7 all customers × product, 8 all customers × product group, 9 all × all; (3) at each access select ACTIVE records matching keys, currency = document currency (no cross-currency fallback), unit = line unit or convertible via wave 6, valid_from ≤ pricing_date ≤ valid_to, from_quantity ≤ quantity; (4) pick greatest from_quantity, then most dims matched, then lowest net unit price; (5) stop unless winner has search_next, then continue and keep the minimum; (6) no candidate at any access → nothing for discounts, `PRICE_NOT_FOUND` for PRICE unless a manual price exists; (7) exclusion groups evaluated after member steps (BEST lowest net, FIRST lowest step); (8) `exclusive` winner suppresses later discount steps; (9) round per BR-26.3.
- **BR-26.2 Pricing date** = requested ship date by default (AX), configurable to order or delivery date (SAP KOMK-PRSDT).
- **BR-26.3 Rounding.** Net unit = amount × (1–p1/100) × (1–p2/100) at 6 decimals; line net = round_half_even(qty × net unit, minor units); totals are sums of rounded lines; a percentage total discount rounds once on the total.
- **BR-26.4 Multiline grouping** sums quantities of lines with the same group and unit on one document; adding a line re-resolves the group.
- **BR-26.5 Manual override** requires duty PricingOverride, keeps the trace with a reason code, and is excluded from rebate basis when the agreement says `exclude_manual`.
- **BR-26.6 Concurrency.** Activation locks the key hash (`SELECT … FOR UPDATE`); overlap → 409 with the conflicting id.
- **BR-26.7 Retroactive journals** never re-price posted invoices; the event carries `retroactive=true`.
- **BR-26.8 Rebate tiers** evaluate on cumulative period basis; crossing a tier recomputes the whole period and posts the delta on the next invoice or at period end.
- **BR-26.9 State machines.** Record: DRAFT → ACTIVE → EXPIRED (nightly) | WITHDRAWN. Journal: OPEN → VALIDATED → POSTED → REVERSED. No return to DRAFT.
- **BR-26.10 Negative prices** rejected; 100% discount allowed and yields 0.00 with trace.

5.26.8 Acceptance criteria

- **AC-26.1** (FR-26.3) Given C1 in group PG-A with access-2 record 95.00 USD and access-4 record 90.00, when resolved without search_next, then 95.00 and access 4 is shown as not evaluated.
- **AC-26.2** (FR-26.3) Given search_next on the access-2 record, then 90.00 and both accesses appear in the trace.
- **AC-26.3** (FR-26.2) Given breaks 0→10.00, 100→9.50, 500→9.00 EUR, when qty = 500 then 9.00; when 499 then 9.50.
- **AC-26.4** (FR-26.4) Given 10,000 random requests resolved twice on two replicas, then outputs are byte-identical.
- **AC-26.5** (FR-26.5) Given discounts 10%/5% on 100.00 USD × 3, then line net = 256.50.
- **AC-26.6** (FR-26.8) Given exclusion group BEST with LINE_DISC 8% and campaign 12%, then only 12% applies.
- **AC-26.7** (FR-26.9) Given a 100k-line journal with one invalid line, then nothing is written and the error names the line; after correction and post, reversal restores all prior versions.
- **AC-26.8** (FR-26.11) Given a 2% monthly rebate and a 10,000.00 USD invoice, then 200.00 posts Dr 4150 / Cr 2150.
- **AC-26.9** (FR-26.14) Given 50 concurrent callers over the 500 scenarios, then p95 < 20 ms.
- **AC-26.10** (FR-26.7) Given smart rounding ".99 below 10" and a journal raising 7.41 by 5%, then 7.99 is stored.

5.26.9 Test plan and fixtures

Fixtures: `pricing-500` (hand-built from AX trade-agreement and SAP RVAA01 examples, with expected price and trace); `pricing-random-10k` (seeded); `price-journal-100k`; `rebate-tiers-12` (accountant-reviewed).

Test	Type	Target	Pass criterion
Resolver scenarios	unit	resolve	500/500 price and trace
Determinism	property	resolve	identical across 10k runs and replicas

Test	Type	Target	Pass criterion
Overlap rejection	unit	activate	100% rejected with 409
Journal atomicity	integration	journals	failure leaves zero rows changed
Rebate accrual = GL	reconciliation	accrual vs ledger	Σ per period = GL 2150 to the cent
Latency	performance	10-line resolve, 50 callers	p95 < 20 ms, p99 < 60 ms
Sales ↔ pricing	contract (Pact)	resolve	green

Reconciliation: Σ rebate_accrual per (agreement, period) = journal postings tagged `source=rebate_accrual`; journal snapshots replay to current record versions.

5.26.10 Deployment and operations

Helm `passage-pricing` (Go, 3 replicas, HPA CPU 70%); CloudNativePG `pricing-db` (3 instances); Atlas migrations from metadata, additive-only. Config: `PRICING_CACHE_TTL=60s`, `PRICING_MAX_LINES=500`, `PRICING_EXPLAIN_MAX_BYTES=65536`. Flags: `pricing.search_next`, `pricing.sap_procedure`, `pricing.commission_hooks`. Temporal jobs: nightly expiry 00:10 entity-local; rebate recompute at period end. SLOs: 99.95% availability, p95 20 ms. Alerts: invalidation lag > 5 s, error rate > 0.5%, journal post > 300 s. Runbook: cache rebuild, journal reversal, procedure version rollback. Rollback: previous image digest via Argo CD.

5.26.11 Migration from AX / SAP

Source	Passage entity	Notes
PriceDiscTable (Relation, AccountCode/Relation, ItemCode/Relation, QuantityAmount, Amount, Percent1/2, Currency, FromDate/ToDate, SearchAgain)	condition_record	Table/GroupId/All → customer/group/all; Relation → type; SearchAgain → search_next
PriceDiscGroup; CustTable price/discount groups	customer_pricing_group, product_pricing_group	one group per AX type
KONH/KONP + A004/A005/A305 (KSCHL, DATAB/DATBI, KBETR, KONWA, KPEIN, KMEIN), KONM scales	condition_record	A-table → key_combination; KONM → breaks; KPEIN normalized to per-unit at 6 decimals
T683S / T682	pricing_procedure, access_sequence_step	map to nearest reference procedure; manual review
KONA (BO01–BO06), VBOX; TAMRebateAgreement	rebate_agreement, opening accrual to 2150	

Validation: 1,000 sampled customer/product pairs resolve to the legacy price (AX PriceDisc::findPrice export, SAP VK13 export) to the cent. Not migrated: records expired > 24 months (archived), commission history, pricing logs, posted PriceDiscAdmTrans (lakehouse).

5.26.12 Dependencies, risks, open questions

- Depends on: w6 units, w7 customers/groups, w8 products/variants/attributes, w11 journal, w12 currencies, w25 (phase order).
- Risks: KPEIN/per-unit mismatch on import → normalize at 6 decimals; cache stampede after a 100k journal → warm cache inside the journal transaction.
- Open questions: (1) Cross-currency fallback? Default no; decide by wave 27. (2) Margin-based commissions before Phase 8? Default no; decide by wave 29.

5.27 Wave 27: Sales Orders, Quotations, ATP & Credit

5.27.1 Scope and business context

Wave 27 ships the `sales` service: quotations with conversion, sales agreements, order types (journal, order, subscription, returned — completed in wave 30), delivery date control against cumulative ATP/CTP, credit management with limits, aging rules, holds and a Temporal release workflow, versioned change management, line-level status and a multi-channel intake API. A sales rep can quote, convert, promise a date and see holds; a credit manager releases orders. Out of scope: picking and shipping (wave 28), invoicing (wave 29), portal UI (wave 30), EDI intake (wave 35), aATP product allocation.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Promise dates go stale after a planning run: AX ATP is recomputed only when the "Available ship and receipt dates" dialog opens; SAP VBEP dates change only on VA02 re-check or the nightly V_V2 rescheduling batch	ATP computed on demand inside the form; supply changes do not re-promise open lines	atp_bucket projection from inventory/planning events (lag < 30 s); re-promise on `order.changed` and supply cancellation; backward propagation (BR-27.1)	Promise accuracy ~70% → ≥ 95%; re-promise latency nightly → < 30 s
Credit holds are opaque: SalesOrderHold shows a code; OVA8 writes only VBAK-CMGST and the VKM1 list	Rules evaluated inside posting classes; only the final status persists	`credit-check.evaluated` stores every rule, value and threshold; release queue shows the failing rule	Analyst lookup per release 5–8 min (FD32/CustTable) → < 30 s; 2-business-day SLA by timer
Order entry takes 12+ clicks across SalesTable → SalesLine → InventDim → ATP dialog (VA01 overview → item → schedule lines)	Price, ATP and dims live on separate tabs	One grid with inline price and ATP within 300 ms of product + quantity	Clicks per priced, promised line 12 → 2
Channel orders are re-keyed or loaded by nightly AIF/IDoc batches with no idempotency (duplicate ORDERS05 → duplicate VBAK)	Batch integration without a business-key dedup	Intake API with (channel, external_ref) idempotency at 500 orders/min, p99 < 500 ms; `order.confirmed` events for any consumer	Duplicates per 10k channel orders ~5 → 0; latency hours → < 500 ms
Changing a confirmed order loses history (SalesLine overwritten; CDHDR/CDPOS unreadable)	No versioned document	Versions with field-level diff and selective re-pricing/ATP/credit	"What changed and who": 15 min → < 5 s

5.27.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Quotations	SalesQuotationTable/Line, SalesQuotationStatus	VA21/VA22, VBAK VBTYP B; VA11 inquiries	sales_quotation, …:convert	full
Sales orders	SalesTable/SalesLine, SalesStatus, SalesLineStatus, SalesType	VA01/VA02, VBAK/VBAP/VBEP, item categories TAN/TAS	sales_order, sales_order_line, order_schedule_line	full
Sales agreements	SalesAgreementHeader/AgreementLine, release orders	VA31 scheduling agreements, VA41 contracts	sales_agreement with release tracking	full
ATP / delivery date control	InventDim ATP (DlvDateControlType ATP/CTP), ATP time fence, SalesLine.ConfirmedDlv	OVZ9 checking rule, VBEP confirmed lines, CO09	promising module, atp_bucket	full (CTP partial)
Credit limit	CustTable.CreditMax/MandatoryCreditLimit, CustParameters.CreditLimitType, D365 credit management	FD32/UKM_BP (KNKK), OVA8, VKM1 release	credit_profile, CREDIT hold, Temporal release	full
Order holds	SalesOrderHold / hold codes	VBAK-LIFSK delivery block, FAKSK billing block	order_hold (CREDIT, FRAUD, MANUAL, PRICE_REVIEW)	full
Change management	D365 order change approval	VA02 + CDHDR/CDPOS	sales_order.version + change_request	improved
Configured products	PCConfiguration on SalesLine	Variant configuration, KMAT	configuration_id (w8) + production link (w21)	partial
Backorders / alternative items	SalesBackorder; InventTable.AltItemId	V_RA; VB11 substitution	backorder workspace; alternative in ATP fallback	full
Multi-channel intake	AIF SalesSalesOrderService	IDoc ORDERS05, BAPI_SALESORDER_CREATEFROMDAT2	POST /sales/orders with idempotency	improved

5.27.3 Functional requirements

- **FR-27.1 Quotations** — Must. Priced quotations with expiry (default +30 d) and probability, convertible whole or by line to one order preserving explain refs.
- **FR-27.2 Sales orders** — Must. Header (customer, invoice account, currency, delivery address, terms, procedure) and lines (product, variant dims, qty, unit, site/warehouse, requested dates, price fields). Notes: types JOURNAL, ORDER, SUBSCRIPTION, RETURNED; ≤ 2,000 lines; sequence `SO-`.
- **FR-27.3 Delivery date control** — Must. Confirmed ship/receipt dates per line by NONE, SALES_LEAD_TIME, ATP or CTP, applying transport days and the customer receiving calendar.
- **FR-27.4 Cumulative ATP** — Must. Per (product, dims, site, warehouse): on-hand available + scheduled receipts − scheduled issues within the time fence, with backward propagation of negative periods (BR-27.1). Notes: supply = confirmed POs (w17), released/started production orders (w21), transfers, planned orders optional (default excluded); demand = confirmed sales lines, production BOM lines, transfer issues, optional safety stock.
- **FR-27.5 CTP** — Should. When ATP fails, request a capable-to-promise date from scheduling (w23) using BOM/route lead time and finite capacity, creating a planned order link.
- **FR-27.6 Backorders and partials** — Must. Split lines into schedule lines when partial delivery is allowed; backorder list by promised date.
- **FR-27.7 Credit profile and checks** — Must. Per customer per entity: limit, currency, limit type, tolerance, review date, max dunning level; rules in BR-27.3 evaluated at entry, confirmation, release and invoicing per policy.
- **FR-27.8 Holds and release** — Must. Holds block confirmation/release; Temporal release workflow with approvers by duty and a 2-business-day SLA timer.
- **FR-27.9 Change management** — Must. Version on every post-confirmation change with field-level diff; re-run pricing/ATP/credit for affected lines; optional approval per customer.
- **FR-27.10 Sales agreements** — Should. Blanket commitment by quantity or amount; released orders inherit price and decrement commitment atomically.
- **FR-27.11 Configured products** — Should. Accept a configuration id; on confirmation create the production/planned order linkage for make-to-order items.
- **FR-27.12 Intake API** — Must. External orders with idempotency keys at 500 orders/min, p99 < 500 ms, returning id and holds.
- **FR-27.13 Alternative items** — Could. Offer the configured alternative with its ATP when the ordered product fails inside the requested date.

5.27.4 Data model

sales_order (SalesTable / VBAK)

Field	Type	Required	Notes
id, order_number	uuid, text	Y	SO- sequence, unique per entity
tenant_id, legal_entity_id	uuid	Y	RLS
customer_id, invoice_account_id	uuid	Y	FK party (w7)
order_type, status	enum	Y	see BR-27.5
currency, pricing_procedure_version_id	char(3), uuid	Y	
delivery_address_id, delivery_terms, mode_of_delivery	uuid, text, text	Y	
payment_terms_id, cash_discount_id	uuid	Y/N	w7 terms
requested_ship_date, requested_receipt_date	date	Y	
credit_check_result	jsonb	N	last evaluation
version, etag	int, text	Y	optimistic concurrency
net/discount/tax/gross totals	numeric(19,4)	Y	derived, stored
channel, external_ref, idempotency_key	text	N	unique (tenant, channel, external_ref)

sales_order_line (SalesLine / VBAP): id, order_id, line_no, product_id, variant_dims, quantity, unit, site_id, warehouse_id, unit_price, disc_pct_1/2, line_net, tax_group, delivery_date_control, requested/confirmed ship and receipt dates, line_status (OPEN | RESERVED | PICKED | DELIVERED | INVOICED | CANCELLED), delivered_qty, invoiced_qty, remaining_qty, configuration_id, production_order_id, agreement_line_id, price_explain_ref, multiline_group, reservation_id. **order_schedule_line** (VBEP): line_id, seq, qty, confirmed_ship_date, source (atp_bucket id | planned order id).

credit_profile: customer_id, legal_entity_id, credit_limit numeric(19,4), currency, limit_type (NONE | BALANCE | BALANCE_PACKING_SLIP | BALANCE_ALL), tolerance_pct (5), tolerance_amount (1,000.00), oldest_item_days_max (30), max_dunning_level (3), review_date, risk_group, mandatory_check, exposure jsonb (open_orders, delivered_not_invoiced, open_ar, unapplied_cash, updated_at).

order_hold: id, order_id, hold_type, reason_code, status (ACTIVE | RELEASED | CLEARED), applied_by, released_by, workflow_run_id, applied_at, released_at. **atp_bucket** (projection): product_id, dims hash, site_id, warehouse_id, bucket_date, supply_qty, demand_qty, period_atp, cumulative_atp, source_refs, computed_at.

Invariants: Σ schedule_line.qty = line.quantity; delivered_qty ≤ quantity; remaining_qty = quantity – delivered_qty (0 after remainder cancellation); order status derives from line statuses; an order with an ACTIVE hold is never released; cumulative_atp[n] = min over k ≥ n of Σ period_atp[0..k]; every ATP-confirmed line references an atp_bucket as of confirmation. Indexing: lines (order_id, line_no); partial index on open line statuses for demand; atp_bucket PK (product, dims, site, warehouse, bucket_date); partition orders by legal_entity.

5.27.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/sales/quotations, …/{id}:send, :convert	Quotation lifecycle	Idempotency-Key	SalesQuotationMaintain
POST	/api/v1/sales/orders	Create (UI or channel)	Idempotency-Key + external_ref	SalesOrderMaintain / ChannelIntake
PATCH	/api/v1/sales/orders/{id}	Change, creates version	If-Match etag	SalesOrderMaintain
POST	/api/v1/sales/orders/{id}:confirm, :cancel; /lines/{l}:cancel-remainder	Price, ATP, credit, confirm; cancellation	Idempotency-Key	SalesOrderConfirm
POST/GET	/api/v1/sales/atp:check; /atp/buckets?product=&site=	Simulate ATP; bucket view (CO09)	stateless	SalesOrderView
POST	/api/v1/sales/orders/{id}/holds, …/holds/{h}:release	Apply / release hold	Idempotency-Key	SalesOrderHold / CreditRelease
GET/PUT	/api/v1/sales/credit-profiles/{customerId}	Credit profile	etag	CreditMaintain
POST	/api/v1/sales/agreements, …/{id}:release-order	Agreements	Idempotency-Key	SalesAgreementMaintain

Validation: confirm rejects unpriced lines (unless manual), quantity ≤ 0, customer status BLOCKED (w7); closed period on order date is a warning only (period control applies at posting).

Event	Trigger	Payload fields	Consumers
dev.passage.sales.order.confirmed.v1	confirm	order_id, customer_id, currency, lines[{line_id, product, dims, qty, site, warehouse, confirmed_ship_date}], totals, version	fulfillment, planning (w22), credit, analytics
dev.passage.sales.order.changed.v1 / cancelled.v1	PATCH / cancel	order_id, version, diff[] or line_ids, reason	fulfillment, planning, portal
dev.passage.sales.atp.promised.v1	confirm with ATP/CTP	line_id, requested, confirmed, bucket_date, source	planning, portal
passage.sales.order-hold.applied.v1 / released.v1	hold lifecycle	order_id, hold_type, reason, actor, workflow_run_id	fulfillment, collections, portal
passage.sales.credit-check.evaluated.v1	any check	customer_id, order_id, rules[{id, result, value, threshold}], outcome	audit, collections
dev.passage.sales.quotation.converted.v1	convert	quotation_id, order_id, lines	analytics, CRM

5.27.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Sales workspace	tiles: on hold, backorders, quotes expiring 7 d, intake errors	counts	SalesOrderView
Quotation detail	add line, price, send PDF, convert, mark lost	customer, lines, expiry, probability, totals	SalesQuotationMaintain
Order entry (header + grid)	add line with ATP badge, price, confirm, hold, cancel, copy	product, qty, unit, site/wh, requested/confirmed dates, price, disc, net, status	SalesOrderMaintain
ATP explorer	pick product/site, view buckets, simulate	bucket date, supply, demand, period ATP, cumulative ATP, sources	SalesOrderView
Credit release queue	view exposure, release/reject with comment	customer, order, amount, exposure vs limit, rule failed, age	CreditRelease
Backorders / change requests	re-promise, split, cancel remainder; approve diff	line, promised date, qty; field, old, new	SalesOrderMaintain / SalesOrderApprove

Interaction rules: Tab/Enter across the grid; product + quantity triggers inline price and ATP within 300 ms; validation at line commit and confirm; bulk re-promise on multi-select; default site/warehouse and column sets personalizable; changes to a confirmed order show the diff before save.

5.27.7 Business rules and edge cases

- **BR-27.1 Cumulative ATP worked example.** FG-100, site S1, warehouse W11, time fence 30 d, on-hand 100 on 2027-03-01. Supply: PO-2001 50 on 03-04; PRD-3001 80 on 03-08. Existing demand: SO-1001 60 on 03-02, SO-1002 40 on 03-05, SO-1003 70 on 03-09. Buckets open at each receipt: bucket 0 (03-01) supply 100, demand 60, period ATP 40; bucket 1 (03-04) 50 – 40 = 10; bucket 2 (03-08) 80 – 70 = 10. Cumulative ATP: 40, 50, 60. A new line of 45 requested 03-01: bucket 0 cumulative 40 < 45; first bucket with cumulative ≥ 45 is bucket 1 → confirmed ship date 2027-03-

04 for complete delivery, or schedule lines 40 on 03-01 and 5 on 03-04 if partial delivery is allowed. After confirming 45 on 03-04, bucket 1 period ATP = –35; negative periods propagate backward, so cumulative becomes 5, 5, 15. Had SO-1002 been 60, bucket 1 period ATP –10 would already have reduced bucket 0 cumulative to 30.

- **BR-27.2 Dates.** confirmed_receipt = confirmed_ship + transport days (mode of delivery) rolled to the customer's next receiving day; past requested dates become today with a warning; dates beyond the fence are promised at fence end with source `TIME_FENCE`.
- **BR-27.3 Credit check rules** (any FAIL → CREDIT hold; WARN displayed only): CR-1 exposure + order amount > limit × (1 + tolerance_pct/100) and > limit + tolerance_amount → FAIL, where exposure by limit_type is BALANCE = open AR + unapplied; BALANCE_PACKING_SLIP adds delivered-not-invoiced; BALANCE_ALL adds confirmed open orders; CR-2 oldest open item overdue > oldest_item_days_max → FAIL; CR-3 dunning level ≥ max_dunning_level → FAIL; CR-4 review_date passed or profile missing while mandatory_check → FAIL; CR-5 overdue > 25% of open AR → WARN; CR-6 order > 50,000 USD / 45,000 EUR for a customer younger than 90 days → FAIL. Amounts convert to the profile currency at the wave 12 spot rate on the check date.
- **BR-27.4 Hold precedence.** CREDIT and FRAUD block confirmation and release; MANUAL and PRICE_REVIEW block release only; a CREDIT release is valid 7 days; an amount increase > 5% re-triggers CR-1.
- **BR-27.5 State machines.** Order: DRAFT → OPEN → CONFIRMED → PARTIALLY_SHIPPED → SHIPPED → INVOICED → CLOSED; CANCELLED from DRAFT/OPEN/CONFIRMED. Line: OPEN → RESERVED → PICKED → DELIVERED → INVOICED; CANCELLED from OPEN/RESERVED. Quotation: DRAFT → SENT → CONVERTED | LOST | EXPIRED | CANCELLED.
- **BR-27.6 Concurrency.** PATCH requires If-Match; concurrent confirm and change → later call 409; ATP consumption serialized per (product, dims, site, warehouse) by Postgres advisory lock.
- **BR-27.7 Reservation.** ATP confirmation creates a wave 16 reservation when `reserve_on_confirm` (default true for ORDER); JOURNAL orders never reserve or consume ATP.
- **BR-27.8 Quantities** at 6 decimals; `whole_units` products reject fractions; unit conversion via wave 6 factors, half-even.
- **BR-27.9 Agreements** decrement commitment atomically and may not exceed it beyond overdelivery_pct (default 0); subscription/agreement lines re-price only when the agreement price changes.
- **BR-27.10 Intake duplicates.** Same (channel, external_ref) within 30 days returns the original id with 200; same key with a different payload → 409.

5.27.8 Acceptance criteria

- **AC-27.1** (FR-27.4) Given the BR-27.1 fixture, when a 45-unit line requests 2027-03-01 with complete delivery, then confirmed ship date = 2027-03-04 and the promised event references bucket 1.
- **AC-27.2** (FR-27.4) Given any `atp-100` fixture, when all orders confirm, then no cumulative ATP is negative after propagation and total promised ≤ on-hand + included supply (property).
- **AC-27.3** (FR-27.7, FR-27.8) Given limit 10,000 USD, open AR 8,500, tolerance 5% / 1,000, when a 2,000 USD order confirms, then no hold (10,500 within both tolerances) and a WARN; when a 3,000 USD order confirms, then a CREDIT hold applies (11,500 > 10,500 and > 11,000), and release by a CreditRelease user confirms the order and emits `order-hold.released`.
- **AC-27.4** (FR-27.11) Given a make-to-order product with configuration CFG-77, when confirmed, then a wave 21 production order references the line and the confirmed date = scheduled end + transport days.
- **AC-27.5** (FR-27.12) Given 500 orders/min for 10 min through intake, then p99 < 500 ms and all 5,000 orders exist exactly once.
- **AC-27.6** (FR-27.1) Given a sent 3-line quotation, when lines 1 and 3 convert, then one 2-line order exists with matching prices and explain refs and the quotation is CONVERTED.
- **AC-27.7** (FR-27.9) Given a confirmed order, when line 10 changes from 5 to 8, then version increments, the diff is recorded, only line 10 is re-promised and `order.changed` is emitted.
- **AC-27.8** (FR-27.6) Given partial delivery and ATP 40 of 45, then two schedule lines exist and the backorder list shows 5 units on 2027-03-04.
- **AC-27.9** (FR-27.10) Given an agreement of 1,000 units fully released, then the next release is rejected with 422 `AGREEMENT_EXHAUSTED`.

5.27.9 Test plan and fixtures

Fixtures: `atp-100` (from SAP OVZ9 and AX ATP examples with expected dates); `credit-40` (profiles plus migrated AR open items); `orders-5k-intake` (generated channel payloads); `quotes-50`; `demo-plant` (wave 25 plant with FG-100…FG-120 stock and open production orders).

Test	Type	Target	Pass criterion
ATP scenarios	unit	promising	100/100 expected dates
Never over-promise	property	promising	Σ promised ≤ supply in every bucket prefix
Credit rules	unit	credit	40/40 outcomes incl. tolerance edges
Hold release	E2E (Temporal)	workflow	released; SLA timer fires at 2 business days
Intake throughput	performance	POST /orders	500/min, p99 < 500 ms, 0 duplicates
Sales ↔ pricing/inventory/production	contract (Pact)	APIs	green
Order entry UI	E2E (Playwright + axe)	shell	flows pass, 0 serious a11y violations

Reconciliation: Σ open confirmed order amounts per customer = credit_profile.exposure.open_orders; wave 16 reservations = Σ RESERVED line quantities; nightly rebuilt ATP projection = incremental projection (diff 0).

5.27.10 Deployment and operations

Helm `passage-sales` (Go, 3 replicas, HPA on RPS); `sales-db`; Temporal namespace `sales`. Config: `ATP_TIME_FENCE_DEFAULT_DAYS=30`, `ATP_INCLUDE_PLANNED_ORDERS=false`, `CREDIT_CHECK_POINTS=confirm,release`, `INTAKE_RATE_LIMIT_PER_KEY=1000/min`. Flags: `sales.ctp`, `sales.alternative_items`, `sales.change_approval`. Jobs: ATP rebuild 01:00 entity-local; quotation expiry hourly; credit review warnings daily. SLOs: create p99 500 ms, 99.95% availability, projection lag < 30 s. Alerts: lag > 120 s, hold queue age > 2 business days, intake 5xx > 0.1%. Runbook: projection rebuild, stuck release workflow, re-promise after supply cancellation. Rollback: previous digest; additive migrations.

5.27.11 Migration from AX / SAP

Source	Passage entity	Notes
SalesTable (SalesId, CustAccount, InvoiceAccount, CurrencyCode, SalesStatus) + SalesLine (ItemId, InventDimId, SalesQty, SalesPrice, LinePercent, RemainSalesPhysical/Financial)	sales_order, lines	open orders only (Backorder, or Delivered with financial remainder); InventDim → dims + site/warehouse
SalesQuotationTable/Line (Sent); SalesAgreementHeader/Line	sales_quotation; sales_agreement	remaining commitment from released qty
CustTable.CreditMax, MandatoryCreditLimit, CustParameters.CreditLimitType	credit_profile	1:1 type mapping
VBAK/VBAP/VBEP (VBUK-GBSTK ≠ C), VBKD, VBPA	sales_order, lines, schedule lines	TAS → drop-ship flag; VBEP confirmed qty → schedule lines
KNKK (KLIMK, CTLPC), UKM_BP	credit_profile	risk class → risk_group
VBAK VBTYP B / E / G	sales_quotation / sales_agreement	

Validation: count and Σ open value per customer equal the legacy extract; every migrated line re-promises to ≤ legacy confirmed date + 1 day or is flagged. Not migrated: closed orders (lakehouse), CDHDR logs, SAP credit horizon settings (replaced by BR-27.3).

5.27.12 Dependencies, risks, open questions

- Depends on: w26 pricing; w16 on-hand/reservations; w17 PO receipts; w21 production orders; w22 planned orders and demand; w23 scheduling (CTP); w7 customers/terms; w8 configurator; w4 Temporal.
- Risks: projection staleness during planning runs (planning emits a batch-end marker; buckets flagged stale); credit exposure incomplete before waves 29/30 (migrated open AR loaded into `receivables_db.open_item`, schema created this wave).
- Open questions: (1) Default date control for migrated customers, ATP vs. lead time? Default ATP; decide by wave 28. (2) External fraud scoring? Default rule-only in v1; decide by wave 30.

5.28 Wave 28: Fulfillment: Picking, Packing & Shipping

5.28.1 Scope and business context

Wave 28 ships the `fulfillment` service and the `carrier-gateway` adapter. Confirmed orders release to the warehouse by rule, are picked (basic wave/batch picking; location directives and RF arrive in wave 31), packed into containers with SSCC labels, rated and labelled through a carrier abstraction, and shipped with a posted packing slip that deducts inventory and posts cost to the universal journal. Drop-ship and partial shipments are supported; tracking is published for the portal. Out of scope: location directives, license plates and mobile RF flows (wave 31), transportation planning and freight reconciliation (wave 34), hazmat and export compliance.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Delivered-not-invoiced never ties: AX "Cost of units delivered" via InventPosting drifts from CustPackingSlipJour after invoice cancellations; SAP PGI posts COGS at once so revenue and COGS land in different periods	Physical and financial postings decoupled across SalesFormLetter classes; reconciliation is a month-end report	Packing slip and invoice share one cost_snapshot_id; DNI projection reconciled to GL 1405 daily; AX-style and PGI-style matrices by policy	Month-end DNI reconciliation 4–8 h → 0 (daily, diff 0.00)
Packing-slip posting blocks the user 5–20 s and fails opaquely ("Update has been cancelled"; VL02N "posting period closed" after picking)	Synchronous multi-table update with late validation	Pre-validation at PACKED (period, holds, dims, negative stock) and a Temporal saga with compensation	Failures discovered after pick ~3% → < 0.1%; post p95 < 2 s
Each carrier is a project (AX ShipCarrier via desktop add-on; SAP XSI/TM)	No provider abstraction	carrier-gateway rate/label/track contract, reference adapters, `tracking-updated` events	Adding a carrier 4–8 weeks → < 2 days; tracking visible on 100% of shipments
Short picks and partials need manual SalesLine splits or VBEP edits	Schedule lines are not first-class in fulfillment	Schedule lines drive shipments; short pick auto-creates the remainder and releases its reservation	Manual splits per 1,000 orders ~40 → 0
SSCC errors at customer receiving (bad check digit, duplicate on re-print)	SSCC generated by label software, not validated by the ERP	Gap-free SSCC per GS1 prefix with mod-10 check, uniqueness and 12-month no-reuse in the container entity	Label rejections at customer DC ~1% of pallets → 0

5.28.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Release to warehouse	Release to warehouse, SalesLine.ReleaseStatus, InventPickingList	VL01N/VL10, LIKP/LIPS	shipment (release rules)	full
Picking	WMSPickingRoute, WMSOrderTrans, picking list registration	LT03 transfer order (LTAP); EWM warehouse task	pick_work	partial (advanced w31)
Packing / HU	WHSContainerTable, packing station	VEKP/VEPO handling units, HUMO	container (nested)	full
SSCC / labels	WHSContainerTable.LicensePlate	HU external id, GS1-128	container.sscc, PDF/ZPL label	full
Packing slip posting	CustPackingSlipJour/Trans, SalesFormLetter_PackingSlip, InventTrans StatusIssue=Deducted	VL02N post goods issue, MKPF/MSEG 601, LIKP-WBSTK	packing_slip saga	full
COGS posting	InventPosting: cost of units delivered / cost of goods delivered	OBYC GBB-VAX, BSX	matrix BR-28.1	full
Carrier interface	ShipCarrierTable, ShipCarrierShippingRequest	TM / Express Ship Interface	carrier-gateway	improved
Drop-ship	Direct delivery (PurchTable.DirectDelivery)	Third-party item category TAS, statistical GR	drop_ship flow	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Delivery schedules / ASN	SalesDeliverySchedule; AIF DESADV	VBEP; DESADV IDoc	schedule lines → shipments; ASN event (EDI w35)	partial

5.28.3 Functional requirements

- **FR-28.1 Release rules** — Must. Release confirmed, unheld lines by rule (confirmed ship date ≤ today + horizon, default 2 d; full ATP; priority) into one shipment per (order, warehouse, ship date) unless consolidation is enabled.
- **FR-28.2 Picking** — Must. Pick work by wave (≤ 200 lines or 50 orders) or batch; physical reservation (w16); confirm quantity, batch/serial and location; short picks create a remainder and release the unpicked reservation.
- **FR-28.3 Packing** — Must. Containers (type, tare, dims), nesting pallet → carton, SSCC from the GS1 company prefix with mod-10 check digit.
- **FR-28.4 Packing slip posting** — Must. Record delivered quantities, create inventory issues, post the journal per BR-28.1 at the current cost snapshot, atomically via a Temporal saga; sequence `PS-`.
- **FR-28.5 Cancellation** — Must. Reverse an uninvoiced packing slip with mirror inventory and journal entries referencing the original.
- **FR-28.6 Carrier abstraction** — Must. rate / label / manifest / track through a provider-agnostic contract with EasyPost-style, Shippo-style and `file` adapters; labels as PDF and ZPL; tracking by 30-min poll or webhook.
- **FR-28.7 Drop-ship** — Must. Linked PO (w17) for drop-ship lines; the vendor's ship confirmation triggers the packing slip; COGS at vendor cost without inventory.
- **FR-28.8 Partial shipments** — Must. Ship each schedule line separately; order stays PARTIALLY_SHIPPED until remaining_qty = 0.
- **FR-28.9 ASN** — Should. Emit an ASN with containers, SSCCs and quantities on post for the wave 35 EDI mapper.
- **FR-28.10 Tracking visibility** — Should. Publish tracking changes to portal and order detail.
- **FR-28.11 Mobile pick (basic PWA)** — Should. Pick confirmation with GS1-128 parsing (AI 00, 01, 10, 17, 21, 30, 37) and 5-minute offline tolerance.
- **FR-28.12 DNI reconciliation** — Must. delivered_not_invoiced projection = GL 1405, checked daily.

5.28.4 Data model

shipment

Field	Type	Required	Notes
id, shipment_number	uuid, text	Y	SH- sequence
legal_entity_id, warehouse_id	uuid	Y	one warehouse per shipment
customer_id, delivery_address_id	uuid	Y	
status	enum	Y	BR-28.6
planned_ship_date, actual_ship_date	date, timestamptz	Y/N	
carrier_code, service_level, mode_of_delivery	text	N	
lines[]	shipment_line	Y	order_line_id, schedule_line_id, qty_planned/picked/packed/shipped, dims
consolidation_key, packing_slip_id, carrier_shipment_id, version	text, uuid, uuid, int	N	

container: id, shipment_id, parent_container_id, container_type, sscc char(18), tare_weight, gross_weight, dimensions, uom, lines[] (shipment_line_id, qty, batch, serials[]), label_ref, status (OPEN | CLOSED | SHIPPED).

packing_slip (CustPackingSlipJour/Trans): id, packing_slip_number, shipment_id, order_id, posting_date, posted_at, status (PENDING | POSTED | CANCELLED | INVOICED), lines[] (order_line_id, qty, unit, cost_amount, cost_snapshot_id, inventory_transaction_ids[]), total_cost, journal_posting_id, reversal_of_id, document_ref.

carrier_shipment: id, shipment_id, provider, provider_ref, rate_quote, tracking_number, label_ref, status (QUOTED | LABELLED | MANIFESTED | IN_TRANSIT | DELIVERED | EXCEPTION), events[], cost_amount, currency.

Invariants: Σ container qty per line = qty_packed ≤ qty_picked ≤ qty_planned; packing slip qty = qty_shipped; one journal posting per slip plus one reversal if cancelled; SSCC check digit valid and unique per tenant; Σ cost of uninvoiced slips = GL 1405. Indexing: shipments (warehouse, status, planned_ship_date); packing slips (order_id), (posting_date), partitioned by month; containers unique sscc.

5.28.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/fulfillment/releases:run	Run release rules for a warehouse or orders	Idempotency-Key	WarehouseRelease
POST	/api/v1/fulfillment/waves, …/{id}:confirm-pick	Create wave; confirm picks	Idempotency-Key	WarehousePick
POST	/api/v1/fulfillment/shipments/{id}/containers, …/{cid}:close	Pack; close container (SSCC)	Idempotency-Key	WarehousePack
POST	/api/v1/fulfillment/shipments/{id}:rate, :label, :manifest	Carrier operations	Idempotency-Key	WarehouseShip
POST	/api/v1/fulfillment/shipments/{id}:post-packing-slip; /packing-slips/{id}:cancel	Post / reverse (saga)	Idempotency-Key	PackingSlipPost
GET	/api/v1/fulfillment/shipments/{id}/tracking	Tracking events	—	ShipmentView / Portal
POST	/api/v1/fulfillment/drop-ship/{orderLineId}:vendor-confirm	Vendor confirmation → packing slip	Idempotency-Key	PurchasingConfirm
POST	/webhooks/carriers/{provider}	Tracking webhook (HMAC)	provider event id	system

Validation: post requires status PACKED, no ACTIVE hold, posting date in an open inventory period (w14), and no negative on-hand unless the product policy allows (w16).

Event	Trigger	Payload fields	Consumers
dev.passage.fulfillment.shipment.released.v1	release	shipment_id, order_id, warehouse, lines	sales, analytics
dev.passage.fulfillment.pick.confirmed.v1	pick confirm	wave_id, lines[{order_line_id, qty, dims, location}]	inventory, sales
passage.fulfillment.packing-slip.posted.v1 / cancelled.v1	saga complete / cancel	packing_slip_id, order_id, lines[{order_line_id, qty, cost_amount, cost_snapshot_id}], journal_posting_id, posting_date	billing, sales, inventory, ledger, portal, credit
dev.passage.fulfillment.asn.issued.v1	post	shipment_id, containers[{sscc, lines}], carrier, tracking	EDI (w35)
passage.fulfillment.shipment.tracking-updated.v1	poll / webhook	shipment_id, tracking_number, status, event	portal, sales

5.28.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Fulfillment workspace	tiles: to release, picking, packed not shipped, carrier exceptions, DNI balance	counts	WarehouseView
Release board	run rules, release selected, hold	order, customer, ship date, ATP status, priority	WarehouseRelease
Wave / pick list	print, confirm, short pick, reassign	line, product, location, qty, dims, picker	WarehousePick
Packing station	scan line, add/close container, print label	container, type, weights, lines, SSCC	WarehousePack
Shipment detail	rate, label, manifest, post packing slip, cancel	carrier, service, cost, tracking, journal preview	WarehouseShip / PackingSlipPost
Mobile pick (PWA) / carrier exceptions	scan, confirm, skip; retry, switch provider, manual tracking	location, product, qty, batch/serial; provider error, age	WarehousePick / WarehouseShip

Interaction rules: scan field keeps focus; Enter confirms a line; GS1-128 auto-fills batch/serial; bulk release on multi-select; posting shows the journal preview before confirm; layouts per warehouse personalizable.

5.28.7 Business rules and edge cases

- **BR-28.1 Packing-slip COGS posting matrix** (accounts from the wave 16 posting profile; dimensions from the order line):

Scenario	At packing slip post	At invoice post (wave 29)
Stocked item, physical posting enabled (default, AX-style)	Dr 1405 Delivered-not-invoiced / Cr 1400 Inventory (physical) at cost snapshot	Reverse 1405/1400; Dr 5000 COGS / Cr 1400 Inventory (financial) at settled cost
Stocked item, policy `cogs_at_shipment` (SAP PGI-style)	Dr 5000 COGS / Cr 1400 Inventory	Revenue and AR only
Standard-cost item	Same, at standard; no variance at shipment	Same
Drop-ship line	No posting	Dr 5000 COGS / Cr 2110 GR/IR at PO cost when vendor confirmation posts
Non-stocked / service line	No posting	Revenue only
Catch-weight item	Cost on inventory quantity, not catch-weight quantity	Same
Cancellation	Mirror entries, negative amounts, same dims, reversal_of set	—

- **BR-28.2 Cost snapshot** = running average or FIFO estimate (w20) at posting, stored as cost_snapshot_id; inventory close adjusts 5000/1400, never 1405 after invoicing.
- **BR-28.3 Posting date** = actual ship timestamp in the warehouse time zone; closed inventory period → `PERIOD_CLOSED`, shipment stays PACKED.
- **BR-28.4 Saga.** (1) packing slip PENDING; (2) inventory issues; (3) journal posting with Idempotency-Key = packing_slip_id; (4) POSTED + event. Failure at 3 compensates 2; a retried 3 is a ledger no-op. PENDING > 10 min alerts.
- **BR-28.5 SSCC** = extension digit + GS1 company prefix + serial reference padded to 17 digits + mod-10 check (weights 3,1 from the right); serial from a gap-free sequence per prefix; no reuse for 12 months.
- **BR-28.6 Shipment state machine.** OPEN → RELEASED → PICKING → PICKED → PACKED → SHIPPED → DELIVERED; CANCELLED before SHIPPED (releases reservations); SHIPPED → PACKED on packing-slip cancellation.
- **BR-28.7 Over-pick** rejected; over-ship rejected unless the customer's overdelivery_pct allows.
- **BR-28.8 Consolidation** requires matching customer, address, warehouse, planned date and carrier and no `ship_separately` order.
- **BR-28.9 Carrier failures** retry 3× (2 s, 8 s, 32 s) then land in the exceptions queue; the slip may still post with a manual tracking number.
- **BR-28.10 Tracked items** must carry batch/serial at pick; the slip rejects lines missing tracking dims (w8 tracking dimension group).

5.28.8 Acceptance criteria

- **AC-28.1** (FR-28.4) Given 10 units at cost 12.50 USD with physical posting, when the slip posts, then on-hand –10, journal Dr 1405 125.00 / Cr 1400 125.00, and the event carries cost_amount 125.00.
- **AC-28.2** (FR-28.5) Given that slip cancelled before invoicing, then on-hand +10, a mirror posting exists and GL 1405 returns to its prior balance.
- **AC-28.3** (FR-28.3) Given prefix 0614141 and serial 0123456789, then SSCC = 006141410123456789 1 and the GS1-128 label scans back to it.
- **AC-28.4** (FR-28.2) Given a 100-unit line with 90 picked, then qty_picked = 90, 10 remain on backorder and their reservation is released.
- **AC-28.5** (FR-28.7) Given a drop-ship line, when the vendor confirms, then a slip posts with no inventory transaction and Dr 5000 / Cr 2110 at PO cost.
- **AC-28.6** (FR-28.6) Given recorded fixtures for both adapters, when rate/label/track run, then responses match the contract and labels are valid PDF and ZPL.
- **AC-28.7** (FR-28.1) Given 1,000 confirmed orders, when release → pick → pack → post runs, then all are SHIPPED and Σ inventory issues = Σ slip quantities with zero mismatch.
- **AC-28.8** (FR-28.12) Given 50 posted slips of which 20 invoiced, then DNI = cost of the 30 uninvoiced = GL 1405.
- **AC-28.9** (FR-28.11) Given scan `(01)00614141000012(10)LOT7(37)5`, then GTIN, batch LOT7 and qty 5 are filled.

5.28.9 Test plan and fixtures

Fixtures: `warehouse-sim-1k` (1,000 orders, 120 products, 3 warehouses, batch/serial mix); `sscc-200` (GS1 examples, valid/invalid); `carrier-recorded` (HTTP recordings); `dropship-30`; `cogs-matrix-14` (per cell × costing method, accountant-reviewed).

Test	Type	Target	Pass criterion
Posting matrix	contract (ledger)	post / cancel	14/14 balanced; cancel mirrors exactly
Saga compensation	integration (Temporal)	injected ledger failure	no inventory movement remains; slip not POSTED
SSCC / GS1-128	unit, property	check digit, parser	200/200; all AI combinations round-trip
Pick-pack-ship 1k	E2E	simulated warehouse	0 mismatch, all SHIPPED < 60 min
Carrier adapters	contract	recordings	green
DNI = GL 1405	reconciliation	daily job	0.00

Reconciliation: Σ inventory issue cost = Σ journal Cr 1400 from fulfillment; DNI projection = GL 1405; line delivered_qty = Σ slip qty.

5.28.10 Deployment and operations

Helm `passage-fulfillment` (Go), `passage-carrier-gateway` (TS; egress via Envoy, provider secrets in Vault), `fulfillment-db`; PWA served by the shell. Config: `RELEASE_HORIZON_DAYS=2`, `WAVE_MAX_LINES=200`, `GS1_COMPANY_PREFIX` per entity, `COGS_AT_SHIPMENT=false`. Flags: `fulfillment.consolidation`, `fulfillment.asn`, `fulfillment.mobile_pick`. Jobs: release every 15 min per warehouse; tracking poll 30 min; DNI check 02:00. SLOs: post p95 < 2 s, saga completion 99.9% within 60 s. Alerts: PENDING > 10 min, carrier error rate > 5%, DNI diff ≠ 0. Runbook: resume stuck saga, re-print label, manual tracking. Rollback: previous digest; Temporal workflow versioning lets in-flight sagas finish on old code.

5.28.11 Migration from AX / SAP

Source	Passage entity	Notes
CustPackingSlipJour/Trans with uninvoiced remainder	packing_slip, DNI opening balance	cost from InventTrans.CostAmountPhysical
WMSShipment, WMSPickingRoute (open); WHSContainerTable	shipment, pick_work, container	open picks reset to RELEASED; SSCC retained
LIKP/LIPS not goods-issued (WBSTK ≠ C)	shipment	A/B → RELEASED/PICKING
LIKP/LIPS goods-issued, not billed (FKSTK ≠ C)	packing_slip	cost from MSEG 601 DMBTR
VEKP/VEPO	container	EXIDV SSCC retained

Validation: DNI opening balance = legacy delivered-not-invoiced GL balance to the cent; open shipment quantities = legacy open deliveries. Not migrated: completed shipments, tracking history, carrier configuration (manual).

5.28.12 Dependencies, risks, open questions

- Depends on: w27 orders and holds; w16 inventory transactions/reservations; w20 cost snapshots; w17 POs; w11 journal; w14 period control; w8 tracking dimensions and GS1 ids; w5 PWA shell and document rendering.
- Risks: physical posting unfamiliar to SAP users (policy flag plus documented matrix); carrier sandbox instability (recorded fixtures gate CI; live sandbox nightly only).
- Open questions: (1) Default `cogs_at_shipment` for migrated SAP entities? Default true; decide by wave 29. (2) ASN EDI mapping owned by wave 28 or 35? Default 35; decide at wave 28 close.

5.29 Wave 29: Billing, Invoicing & Revenue Recognition

5.29.1 Scope and business context

Wave 29 ships the `billing` and `revrec` services. Posted packing slips and eligible order lines become invoices (one-off, consolidated, subscription, milestone); free-text invoices and credit notes post directly; layouts render to PDF with an e-invoice document for wave 35. Every posted invoice creates AR open items, tax postings (w13) and journal entries and, where the contract has multiple performance obligations or deferred terms, a revenue contract with standalone selling price allocation and recognition schedules. Out of scope: cash application (wave 30), intercompany invoicing (hook only; wave 40), project billing (wave 41), Peppol transmission (wave 35), metering sources beyond API-posted usage.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Bundle allocation lives in spreadsheets: D365 RevRecSchedule needs revenue prices per item and cannot explain allocation; SAP RAR (FARR_D_*) is a separate engine with its own posting run and FARR_RECON reconciliation	Rev rec bolted on after billing; allocation inputs not on the order	Revenue contract created at post from order lines with SSP order BR-29.2; allocation and schedules visible on the contract and in the invoice trace	Close-time rev rec effort 2–3 days → < 2 h review; 40/40 fixtures reproducible
Invoice totals differ from the order by cents (SalesLine rounding per form letter; MWST vs. BSEG rounding)	Each document re-rounds	Engine net reused; per-line tax; totals property-tested	Cent discrepancies ~0.5% of invoices → 0
Billing is batch: VF04 due list and AX summary update run nightly	Scheduled jobs over document-flow tables	`packing-slip.posted` creates the proposal at once; post p95 < 3 s; consolidation is policy, not batch	Shipment-to-invoice 12–24 h → < 5 min for non-consolidated accounts
Deferred revenue roll-forward rebuilt by hand from GL 2400 and exports	No system roll-forward that ties to GL	`GET /revrec/roll-forward` reconciled nightly to GL 2400 with drill-down	Preparation 1 day → 0; nightly diff 0.00

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Credit note on a partially recognized invoice needs manual journals (manual schedule edits; RAR contract change)	Credit notes ignore recognition state	BR-29.6 reverses recognized and deferred portions automatically	Manual rev rec journals per month 10–30 → 0

5.29.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Invoice from shipment	SalesFormLetter_Invoice, CustInvoiceJour/CustInvoiceTrans, summary update	VF01/VF04, VBRK/VBRP, copy control VTFL	invoice (SHIPMENT, CONSOLIDATED)	full
Free-text invoice	CustInvoiceTable/CustInvoiceLine	FB70; order-related billing (DR)	invoice type FREE_TEXT	full
Credit note	Credit note (negative CustInvoiceJour), free-text credit	Credit memo request G2 → VF01, VBRK VBTYP O	credit_note	full
Billing plans / subscriptions	D365 Subscription billing; milestone billing	FPLA/FPLT billing plans; BRIM	subscription, milestone_schedule	partial (usage basic)
AR open item creation	CustTrans / CustTransOpen	BSID / BSAD, BKPF/BSEG	receivables.open_item via invoice.posted	full
Tax	TaxTrans, TaxGroup/TaxItemGroup	FTXP, MWST, BSET	tax engine (w13) at post	full
Revenue recognition	RevRecSchedule, RevRecContract, revenue prices (D365)	RAR: FARR_D_CONTRACT / POBS / DEFITEM	revenue_contract, performance_obligation, revenue_schedule	full
Deferred revenue postings	Deferred revenue posting on item; schedule processing	RAR posting run, contract liability	recognition run to 2400/4000	full
Layout / e-invoice	SSRS SalesInvoice; Electronic reporting	Smart Forms/Adobe Forms; DRC	docgen template + UBL 2.1 document	partial (transmission w35)
Audit trace	CustInvoiceTrans.InventTransId	Document flow VBFA	GET /invoices/{id}/trace	improved

5.29.3 Functional requirements

- **FR-29.1 Invoice from shipment** — Must. Proposals from posted packing slips (and order lines for non-stocked items), one invoice per order by default, lines carrying qty, price, discounts, charges and tax codes. Notes: gap-free `INV-` per entity and fiscal year; invoice date ≥ packing slip date.
- **FR-29.2 Consolidated invoices** — Must. Combine slips for the same invoice account, currency, terms and cycle (daily/weekly/monthly).
- **FR-29.3 Posting** — Must. Atomic: journal (Dr 1200 gross; Cr 4000 per line account; Cr 2300 tax; Cr 2400 deferred obligations; COGS/DNI per BR-28.1), open item creation, `invoice.posted`. Totals = Σ line nets + charges + tax after per-currency rounding (property-tested).
- **FR-29.4 Free-text invoices** — Must. Account, dimensions, amount, tax code; recurring templates.
- **FR-29.5 Credit notes** — Must. Reference an original (whole or lines) or standalone; negative open item; reverse revenue, tax and recognition.
- **FR-29.6 Tax** — Must. Call the tax engine at proposal and post; reject post if the result moved > 0.01 without re-review; no silent zero tax when the engine is unavailable.
- **FR-29.7 Revenue contracts** — Must. Performance obligations per order (one per line or bundle templates per product), SSP (explicit, list, % of list, residual), proportional allocation, schedules.
- **FR-29.8 Recognition methods** — Must. POINT_IN_TIME (packing slip or invoice), RATABLE (daily or monthly), MILESTONE (manual/event), USAGE (posted usage); period-end run or event posts Dr 2400 / Cr 4000.
- **FR-29.9 Roll-forward** — Must. Opening, additions, recognized, adjustments, closing per period and contract, tied to GL 2400.
- **FR-29.10 Subscriptions** — Should. Monthly/quarterly/annual, in advance or arrears, daily proration on mid-period start, cancel and plan change, usage lines, daily billing job.
- **FR-29.11 Documents** — Must. PDF per entity and language; UBL 2.1 JSON for wave 35.
- **FR-29.12 Audit trace** — Should. invoice → packing slip → order line → price explain → schedules in one response.
- **FR-29.13 Intercompany hook** — Could. `invoice.posted` carries `intercompany=true` for intercompany parties (wave 40).

5.29.4 Data model

invoice (CustInvoiceJour / VBRK)

Field	Type	Required	Notes
id, invoice_number	uuid, text	Y	gap-free per entity + fiscal year
legal_entity_id, invoice_account_id, customer_id	uuid	Y	
invoice_type	enum	Y	SHIPMENT, CONSOLIDATED, FREE_TEXT, SUBSCRIPTION, MILESTONE, CREDIT_NOTE
status	enum	Y	PROPOSAL, POSTED, PARTIALLY_SETTLED, SETTLED, REVERSED
invoice_date, posting_date, due_date	date	Y	due from terms
currency, exchange_rate_id	char(3), uuid	Y	three-currency amounts (w12)
net_amount, charge_amount, tax_amount, gross_amount	numeric(19,4)	Y	gross = net + charges + tax
cash_discount_date, cash_discount_amount	date, numeric	N	
journal_posting_id, tax_document_id, original_invoice_id	uuid	Y/N	original for credit notes
document_ref, einvoice_ref	text	N	PDF, UBL

invoice_line (CustInvoiceTrans / VBRP): id, invoice_id, line_no, order_line_id, packing_slip_line_id, product_id, description, qty, unit, unit_price, disc_pct_1/2, line_net, tax_code, tax_amount, revenue_account, dimensions, cost_amount, performance_obligation_id, price_explain_ref.

revenue_contract: id, order_id, customer_id, currency, transaction_price, allocation_method, status (OPEN | COMPLETE | CANCELLED).
performance_obligation: id, contract_id, product_id | bundle_component, ssp_amount, ssp_source (EXPLICIT | LIST | PCT_OF_LIST | RESIDUAL),

allocated_amount, method, term_start, term_end, satisfied_at, deferred_account 2400, revenue_account 4000. **revenue_schedule**: id, obligation_id, period_id, scheduled_date, amount, status (SCHEDULED | RECOGNIZED | REVERSED), journal_posting_id.

subscription: id, customer_id, order_id, plan, billing_cycle, bill_in_advance, start_date, end_date, next_bill_date, proration (DAILY), status (ACTIVE | PAUSED | CANCELLED), lines[] (product, qty, unit_price, usage_based, usage_unit).

Invariants: gross = Σ line_net + charges + tax exactly after per-line rounding; Σ allocated = transaction_price (residual cents to the largest obligation); Σ schedules per obligation = allocated_amount; POSTED invoices are immutable; per period opening + additions − recognized ± adjustments = closing = GL 2400. Indexing: invoices (invoice_account_id, status, due_date), unique number per (entity, fiscal_year), partitioned by posting month; schedules (period_id, status).

5.29.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/billing/proposals	Build from slips / orders / subscriptions with consolidation	Idempotency-Key	InvoiceMaintain
POST	/api/v1/billing/invoices/{id}:post	Post (tax, journal, AR)	key = invoice id	InvoicePost
POST	/api/v1/billing/free-text-invoices, /credit-notes, …/{id}:post	Free-text and credit notes	Idempotency-Key	InvoiceMaintain / CreditNotePost
GET	/api/v1/billing/invoices/{id}/document?format=pdflubl; …/trace	Documents; audit trace	—	InvoiceView / Portal
POST	/api/v1/revrec/contracts/{id}:allocate; /milestones/{ob}:satisfy	Allocation; milestones	Idempotency-Key	RevRecMaintain
POST/GET	/api/v1/revrec/recognition-runs; /roll-forward?period=	Period run; roll-forward	key = period	RevRecPost / RevRecView
POST	/api/v1/billing/subscriptions, …/{id}:change, :cancel; /usage	Subscriptions and usage	Idempotency-Key	SubscriptionMaintain

Validation: post rejects closed AR/GL period, lines without revenue account, tax engine unavailable (503 `TAX_UNAVAILABLE`), credit beyond the original's remaining creditable amount.

Event	Trigger	Payload fields	Consumers
dev.passage.billing.invoice.posted.v1	post	invoice_id, number, type, customer, invoice_account, currency, dates, lines[{order_line_id, product, qty, net, tax, revenue_account, obligation_id}], totals, journal_posting_id, intercompany	receivables, revrec, ledger, pricing, portal, sales
passage.billing.credit-note.posted.v1	post	credit_note_id, original_invoice_id, lines, totals, journal_posting_id	receivables, revrec, ledger, portal
dev.passage.revrec.contract.allocated.v1	allocate	contract_id, obligations[{id, ssp, allocated, method}]	analytics, audit
dev.passage.revrec.schedule.recognized.v1	run / event	schedule_id, obligation_id, period_id, amount, journal_posting_id	ledger, analytics
dev.passage.billing.subscription.billed.v1	daily job	subscription_id, invoice_id, period, proration	portal, analytics

5.29.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Billing workspace	tiles: ready to invoice, proposals, failed posts, deferred balance, subscriptions due	counts, amounts	InvoiceView
Billing due list	filter, select, propose, consolidate	customer, slips, amount, cycle	InvoiceMaintain
Invoice proposal / detail	edit charges, review tax, post, print, credit	lines, tax breakdown, totals, obligations	InvoicePost
Free-text invoice	add lines, template, post	account, dims, amount, tax code	InvoiceMaintain
Revenue contract / roll-forward	view allocation, adjust SSP, satisfy milestone; run period, drill	obligation, SSP, allocated, recognized, deferred; opening… closing, GL 2400	RevRecMaintain / RevRecView
Subscriptions	create, change plan, pause, cancel	plan, cycle, next bill, proration preview	SubscriptionMaintain

Interaction rules: proposals show tax and journal preview before post; multi-select post ≤ 500 invoices with per-row results; roll-forward drill-down opens schedules by period; column sets personalizable.

5.29.7 Business rules and edge cases

- **BR-29.1 ASC 606 allocation worked example.** Bundle: hardware (SSP 8,000.00 USD, POINT_IN_TIME at packing slip), installation (SSP 1,500.00, MILESTONE), 12-month support (SSP 2,500.00, RATABLE). Σ SSP 12,000.00; contract price 10,800.00 (10% bundle discount). Factor 10,800 / 12,000 = 0.9 → hardware 7,200.00, installation 1,350.00, support 2,250.00. Invoice at shipment 2027-03-10: Dr 1200 10,800.00 (plus tax), Cr 4000 7,200.00, Cr 2400 3,600.00. Installation satisfied 2027-03-25: Dr 2400 / Cr 4000 1,350.00. Support 2027-03-10 to 2028-03-09 at 2,250.00 / 12 = 187.50 per month; first partial month 22/31 × 187.50 = 133.06; the last period absorbs the residual so Σ = 2,250.00. With a contract price of 10,799.99 the allocation 7,199.99 / 1,349.99 / 2,249.99 sums to 10,799.97; the 0.02 residual goes to the largest obligation (hardware 7,200.01).
- **BR-29.2 SSP determination order.** Explicit SSP per product × customer group → list price (PRICE access 7) → % of list per product group → residual (at most one obligation per contract).
- **BR-29.3 Rounding.** Line net half-even to minor units; tax per line (w13) then summed unless the engine returns document-level rounding; allocation at 2 decimals with residual rule; ratable schedules at 2 decimals, last period absorbs the residual.
- **BR-29.4 Dates.** invoice_date ≥ slip posting date; posting_date in an open AR period; due_date = invoice_date + terms (incl. end-of-month variants, w7); cash discount date per terms.

- **BR-29.5 Consolidation** groups by (invoice_account, currency, terms, cycle, address if `split_by_address`); ≤ 2,000 lines; a slip belongs to at most one invoice.
- **BR-29.6 Credit note on partially recognized revenue** reverses recognized revenue proportionally (Dr 4000) and unrecognized deferred (Dr 2400) in the credit period; later schedules become REVERSED.
- **BR-29.7 State machines.** Invoice: PROPOSAL → POSTED → PARTIALLY_SETTLED → SETTLED; POSTED → REVERSED only same posting date with no settlements. Schedule: SCHEDULED → RECOGNIZED → REVERSED (credit note only).
- **BR-29.8 Proration** is daily: amount × service days / period days, including 29 February; a mid-period plan change credits unused old-plan days and bills new-plan remaining days on one invoice; cancellation effective date ≥ today.
- **BR-29.9 Gap-free numbering.** Two posts of one proposal share the Idempotency-Key and return the same invoice; the number is drawn from a serialized sequence per (entity, fiscal_year) in the saga's final step, after tax and journal succeed, so a failed post consumes no number.
- **BR-29.10 Usage records** accepted up to 45 days after period end; later ones bill in the next period with `late=true`.

5.29.8 Acceptance criteria

- **AC-29.1** (FR-29.3) Given 1,000 random invoices in USD, EUR and JPY, then gross = Σ line_net + charges + tax for every invoice (property).
- **AC-29.2** (FR-29.7) Given the BR-29.1 bundle, then obligations are 7,200.00 / 1,350.00 / 2,250.00 and the first postings match the example to the cent.
- **AC-29.3** (FR-29.7, FR-29.8) Given the 40 ASC 606 fixtures, then allocation and recognition per period equal the accountant-reviewed workbook for 40/40.
- **AC-29.4** (FR-29.9) Given 2027-03 additions 3,600.00 and recognized 1,483.06, then closing 2,116.94 = GL 2400.
- **AC-29.5** (FR-29.2) Given 3 slips for one account on a weekly cycle, then one invoice is proposed and each slip becomes INVOICED on post.
- **AC-29.6** (FR-29.5) Given the posted 10,800.00 invoice with installation recognized, when a full credit note posts, then AR –10,800.00, recognized revenue reversed and remaining deferred reversed, with schedules REVERSED.
- **AC-29.7** (FR-29.10) Given a monthly 300.00 EUR subscription starting 2028-02-10 billed in advance, then the first invoice is 300 × 20/29 = 206.90 EUR.
- **AC-29.8** (FR-29.6) Given the tax engine is down, when posting, then 503 `TAX_UNAVAILABLE` and no journal or open item exists.
- **AC-29.9** (FR-29.4) Given a DMF-DE free-text line to 4900 with 19% VAT of 100.00 EUR, then Dr 1200 119.00 / Cr 4900 100.00 / Cr 2300 19.00.

5.29.9 Test plan and fixtures

Fixtures: `asc606-40` (bundles with SSPs, discounts, milestones, modifications; workbook `fixtures/revrec/asc606-40.xlsx`); `invoices-random-1k`; `subscription-edges-25` (mid-period, leap day, plan change, cancel); `consolidation-12`; `credit-notes-15`.

Test	Type	Target	Pass criterion
Totals	property	post	gross = Σ components, 3 currencies
ASC 606	unit	revrec	40/40 match workbook
Roll-forward = GL 2400	reconciliation	nightly	0.00
Proration edges	unit	subscription	25/25
Posting saga	integration	billing ↔ tax ↔ ledger ↔ receivables	no partial state on injected failures
Documents	snapshot	docgen templates	PDF text and UBL schema valid
Billing ↔ fulfillment/receivables	contract (Pact)	APIs	green

Reconciliation: Σ POSTED gross – credit notes = AR additions = GL 1200 debits from billing; recognized revenue = GL 4000 credits from billing + revrec; DNI reversals = slip costs invoiced.

5.29.10 Deployment and operations

Helm `passage-billing`, `passage-revrec` (Go); `billing-db`, `revrec-db`; docgen templates in a ConfigMap per entity. Config: `BILLING_MAX_LINES=2000`, `BILLING_BATCH_MAX=500`, `REVREC_RESIDUAL_RULE=largest`, `USAGE_LATE_DAYS=45`. Flags: `billing.consolidation`, `billing.subscriptions`, `revrec.usage`, `billing.einvoice`. Jobs: subscription billing 03:00; recognition run on period-close trigger (w14) plus nightly preview; roll-forward check 02:30. SLOs: post p95 < 3 s, 99.95% availability. Alerts: failed posts > 1% in 15 min, roll-forward diff ≠ 0, run > 30 min. Runbook: re-run proposal, same-day reversal, regenerate document. Rollback: previous digest; recognition run idempotent per period.

5.29.11 Migration from AX / SAP

Source	Passage entity	Notes
CustInvoiceJour/Trans and CustInvoiceTable/Line with open CustTransOpen	invoice (POSTED), lines	header amounts and due dates; lines for trace
RevRecSchedule / RevRecContract (D365)	revenue_contract, obligations, unrecognized schedules	unrecognized = opening 2400
VBRK/VBRP with open BSID	invoice, lines	VBTYP O → CREDIT_NOTE
FARR_D_CONTRACT / POBS / DEFITEM (RAR)	revenue_contract, obligations, schedules	deferred items per POB → schedules
FPLA/FPLT open billing plans	subscription / milestone_schedule	

Validation: Σ migrated open balances = legacy AR trial balance per customer; Σ unrecognized schedules = legacy deferred revenue balance. Not migrated: settled invoices before the current fiscal year (lakehouse), SSRS/Smart Forms layouts (re-templated), RAR posting history.

5.29.12 Dependencies, risks, open questions

- Depends on: w28 packing slips; w27 lines and terms; w13 tax; w11/w12 journal and currencies; w14 period control; w26 explain refs and rebate basis; w5 document rendering and gap-free sequences.

- Risks: gap-free numbering under saga failure (BR-29.9); missing SSP for migrated products (residual method with warning report).
- Open questions: (1) Document- vs. line-level tax rounding per country? Default line level unless the country pack says otherwise; decide by wave 30. (2) Subscription invoices gap-free per entity or per subscription? Default per entity; decide at wave 29 mid-point.

5.30 Wave 30: Accounts Receivable, Collections, Returns & Customer Portal

5.30.1 Scope and business context

Wave 30 closes the loop. The `receivables` service applies cash (manual, BAI2 lockbox, remittance advice, camt.053 with ML-assisted suggestions) and settles open items; `collections-workflows` (TS on Temporal) runs aging, dunning, interest notes, promises to pay, disputes and risk scoring; return orders flow through RMA dispositions to inventory and credit notes; wave 26 accruals settle; statements are produced; and the customer portal at `customer.<env>.passage.dev` exposes orders, invoices, payments (Stripe/Adyen sandbox), returns and documents. The wave ends with the Phase 5 exit: the demo manufacturer's one-month simulation with reconciled subledgers and stakeholder sign-off. Out of scope: non-customer bank reconciliation and cash forecasting (wave 36), collections-agency integration, portal self-registration beyond invited contacts.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Cash application is manual: AX settle open transactions (CustOpenTrans) makes the clerk find invoices per check; SAP F-28 plus FLB2 post-processing lands 30–50% of payments on account	Exact-reference matching only; no scoring, subset-sum or remittance parsing	BR-30.1 scored rules R1–R7, auto-apply ≥ 85, ML-ranked suggestions across lockbox, camt.053, remittance and portal	Auto-match 40–60% → ≥ 90%; clerk time per payment 3 min → < 20 s; false matches 0
Dunning produces wrong letters: F150 MHNK proposals ignore disputes without FSCM; CustCollectionLetterJour cannot skip promised items	Exclusion state not part of selection	Exclusions from open_item.dispute_id/promise_id at issue time; grace, interval and minimum rules (BR-30.2)	Retracted letters ~5% → 0; run preparation 2 h → < 10 min
Returns need three hand-keyed documents (ReturnTable → arrival journal → credit note; RE order → LR delivery → G2 request → VF01)	Separate forms with manual references	One RMA with dispositions (BR-30.3) driving receipt, scrap/repair/replace and credit note in a saga	Clicks per return ~45 → ≤ 8; cycle time 3–5 days → same day after receipt
Customers phone for statements, invoice copies and status; clerks re-send PDFs	No customer-facing surface on live open items	Portal on the same `open-items`, `invoices`, `shipments` APIs with card/ACH payment and return requests	"Copy of invoice/status" requests –80%; DSO –5 days via portal payments
AR-to-GL reconciliation is a month-end exercise (CustTrans vs. LedgerTrans; BSID vs. recon account via F.03)	Subledger and GL updated by different code paths	Every settlement posts through the journal API; Σ open_amount = GL 1200 daily, in the reconciliation pack	Effort 1–2 days/month → 0; diff 0.00 for 31/31 days

5.30.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Payments and settlement	CustPaymJournal (LedgerJournalTrans), CustTrans/CustSettlement, CustPaymModeTable	F-28, F-32, BSID → BSAD, OB41	payment, settlement, open_item	full
Lockbox	CustLockbox (BAI2 import)	FLB2 / FLB1 (BAI2)	lockbox_batch	full
Bank statement auto-match	Advanced bank reconciliation (camt.053)	FF_5 / EBS camt.053 auto-clearing	camt.053 import + matcher	improved (ML suggest)
Aging / collections	CustAgingLegalEntity, Collections workspace, CustCollectionsPool	FBL5N; FSCM Collections (UDM_SPECIALIST)	aging snapshot, worklist, risk score	full
Collection letters / dunning	CustCollectionLetterJour/Line, sequence, grace days	F150 (MHNK/MHND), FBMP procedure	dunning_run, dunning_letter	full
Interest notes	CustInterestJour/Trans	F.24 / F.2B	interest_note	full
Promise to pay / disputes	Collections promise; cases	FSCM dispute (UDM_DISPUTE)	promise_to_pay, dispute (Temporal)	full
Return orders / RMA	ReturnTable (SalesType=ReturnItem), ReturnDispositionCode, ReturnReasonCode	VA01 type RE, LR delivery, G2 credit memo request; S/4 advanced returns	return_order + return receipt + credit_note	full
Rebate settlement	Rebate settlement to credit note / AP	VBO1 settlement, B1/B2 credit memo	rebate_settlement → credit_note or AP payment	full
Statements / portal / card payments	CustAccountStatementExt; D365 Customer portal; Adyen connector	F.27; SAP Commerce; digital payments add-on	statement job; customer-portal; payments-gateway	improved

5.30.3 Functional requirements

- **FR-30.1 Open items and settlement** — Must. Open items from invoices, credit notes, payments and adjustments; full or partial settlement with cash discount, write-off within tolerance and FX gain/loss (w12) posted at settlement.
- **FR-30.2 Manual payments** — Must. Payment journal by mode (CHECK, ACH, WIRE, CARD, CASH): Dr 1000/1010 bank, Cr 1200; unapplied residual to 1205.
- **FR-30.3 Lockbox BAI2** — Must. Import record types 1–9, one payment per check with remittance detail, then match.
- **FR-30.4 Remittance advice** — Should. CSV/JSON/EDI 820 hook attached to payments for matching.
- **FR-30.5 camt.053 auto-match** — Must. Import statements, identify customer credits (CdtDbtInd=CRDT with RmtInf and debtor data), match per BR-30.1; a Python ranker reorders suggestions but never auto-applies on its own.
- **FR-30.6 Aging and worklist** — Must. Nightly aging (current, 1–30, 31–60, 61–90, 91–180, > 180 by due or invoice date) and a worklist ranked by risk score.
- **FR-30.7 Dunning** — Must. Levels, grace days, minimum amounts, blocks and intervals (BR-30.2); letters via docgen; fees posted; item level raised only at issue.

- **FR-30.8 Interest notes** — Should. Annual rate, ACT/365, minimum 5.00 USD / 5.00 EUR, posted as open items.
- **FR-30.9 Promises and disputes** — Must. Promises (amount, date) and disputes (reason, amount, owner, SLA) exclude items from dunning and interest while open; Temporal workflow to resolution (credit, paid, write-off, rejected).
- **FR-30.10 Risk scoring** — Should. 0–100 from overdue ratio, average days to pay, dunning level, disputes and broken promises; published to credit.
- **FR-30.11 Return orders (RMA)** — Must. RMA referencing invoice lines (or blind), 30-day validity, receipt to quarantine/inspection, dispositions (BR-30.3), inventory receipt at return cost, credit note with restocking charges.
- **FR-30.12 Rebate settlement** — Must. Settle wave 26 accruals per agreement period into a credit note (default) or AP payment (w19 hook), reversing 2150.
- **FR-30.13 Statements** — Should. Monthly and on-demand PDF statements and portal view.
- **FR-30.14 Customer portal** — Must. Invited contacts place orders, view invoices/documents, pay by card/ACH via payments-gateway, request returns and track shipments; Keycloak realm `portal`, OPA scoped to the contact's accounts; WCAG 2.1 AA.
- **FR-30.15 Phase-exit simulation** — Must. Harness `sim/phase5-month` drives Phases 3–5 end-to-end and produces the §5.0 reconciliation pack.

5.30.4 Data model

open_item (CustTrans/CustTransOpen, BSID)

Field	Type	Required	Notes
id, legal_entity_id, customer_id, invoice_account_id	uuid	Y	RLS
source_type, source_id	enum, uuid	Y	INVOICE, CREDIT_NOTE, PAYMENT, INTEREST, FEE, ADJUSTMENT, REBATE
document_number, document_date, due_date	text, date, date	Y	
currency, amount, amount_entity_currency	char(3), numeric(19,4) ×2	Y	debit +, credit −
open_amount	numeric(19,4)	Y	= amount − Σ settlements
cash_discount_date, cash_discount_amount	date, numeric	N	
dunning_level, last_dunned_at, dunning_block	smallint, date, bool	Y	defaults 0 / null / false
dispute_id, promise_id	uuid	N	exclusion flags
status	enum	Y	OPEN, PARTIALLY_SETTLED, SETTLED, REVERSED

payment: id, payment_number, customer_id (nullable until identified), payment_mode, amount, currency, received_date, value_date, bank_account_id, source (MANUAL | LOCKBOX | CAMT053 | PORTAL | REMITTANCE), source_ref (unique with source), payer_name, payer_account_hash, remittance[] (reference, amount), status (RECEIVED | SUGGESTED | APPLIED | ON_ACCOUNT | REVERSED), match_score, journal_posting_id. **settlement** (CustSettlement): id, payment_id | credit_item_id, open_item_id, settled_amount, cash_discount_taken, write_off_amount, fx_gain_loss, settled_at, journal_posting_id, reversal_of_id.

dunning_run: id, legal_entity_id, run_date, parameters, status (DRAFT | REVIEWED | ISSUED | CANCELLED). **dunning_letter**: id, run_id, customer_id, level, items[] (open_item_id, open_amount, days_overdue), fee_amount, document_ref, issued_at. **dispute**: id, customer_id, open_item_id, reason_code, disputed_amount, owner, status, sla_due, workflow_run_id.

return_order (ReturnTable): id, rma_number, customer_id, original_invoice_id, return_reason_code, status (BR-30.6), expires_at, lines[] (original_invoice_line_id, product, qty_expected, qty_received, disposition_code, return_cost_method (ORIGINAL_COST | CURRENT_COST | ZERO), restocking_pct, restocking_amount, inventory_receipt_ids[], credit_note_line_id).

Invariants: Σ settlements per item ≤ |amount|; open_amount = 0 ⇔ SETTLED; Σ OPEN open_amount per entity = GL 1200; Σ ON_ACCOUNT = GL 1205; a payment is APPLIED only when Σ settlements + residual = amount; dunning_level never decreases while open; disputed or promised items are excluded from dunning and interest; credited qty ≤ received qty × unit credit. Indexing: open items (customer_id, status, due_date), (document_number); payments (status, received_date), unique (source, source_ref); settlements partitioned by month.

5.30.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	/api/v1/receivables/open-items?customer=&status=&asOf=	Open items (FBL5N)	—	ARView / Portal (own)
POST	/api/v1/receivables/payments, …/{id}:match, :apply, :on-account, :reverse	Payment lifecycle; scored candidates	Idempotency-Key	ARPaymentPost
POST	/api/v1/receivables/lockbox/imports; /bank-statements/imports	BAI2 and camt.053 files	SHA-256 of content	ARImport
POST	/api/v1/receivables/settlements:batch	Settle with credits/payments, discount, write-off	Idempotency-Key	ARSettle
GET	/api/v1/receivables/aging?asOf=&bucketBy=	Aging snapshot	—	ARView
POST	/api/v1/receivables/dunning-runs, …/{id}:review, :issue, :cancel	Dunning	Idempotency-Key	DunningRun
POST	/api/v1/receivables/interest-notes:calculate, :post; /promises; /disputes, …/{id}:resolve	Interest, promises, disputes	Idempotency-Key	InterestPost / CollectionsMaintain
POST	/api/v1/sales/return-orders, …/{id}:authorize; /lines/{l}:receive, :disposition	RMA saga (sales + fulfillment + billing)	Idempotency-Key	ReturnMaintain / WarehouseReceive
POST	/api/v1/pricing/rebates/settlements	Settle accruals for a period	key = (agreement, period)	RebateSettle
GET/POST	/portal/api/v1/{orders,invoices,payments,returns,shipments,documents}	Portal BFF (scoped)	Idempotency-Key	PortalCustomer
POST	/api/v1/payments-gateway/intents; /webhooks/{provider}	Card/ACH intents; signed provider webhooks	provider id	system

Validation: apply requires Σ settlements + on-account residual = payment amount; cross-currency settlement only with FX settlement enabled; dunning issue requires REVIEWED; disposition requires qty_received > 0; portal payments require Σ selected open ≥ amount.

Event	Trigger	Payload fields	Consumers
dev.passage.receivables.payment.received.v1	import / manual / portal	payment_id, source, amount, currency, payer, remittance[]	matcher, analytics
dev.passage.receivables.payment.applied.v1	apply	payment_id, settlements[{open_item_id, amount, discount, write_off, fx}], residual, journal_posting_id	ledger, sales credit, portal, collections
passage.receivables.customer-balance.changed.v1	any open item change	customer_id, balance, overdue, oldest_item_days, dunning_level, risk_score	sales credit, portal
passage.receivables.dunning-letter.issued.v1	issue	run_id, customer_id, level, items[], fee, document_ref	docgen, portal, sales (hold at level 4)
dev.passage.receivables.dispute.opened.v1 / resolved.v1	dispute lifecycle	dispute_id, open_item_id, amount, reason, resolution	collections, billing, analytics
passage.receivables.return-order.authorized.v1 / dispositioned.v1	RMA	rma_id, line, disposition, qty, restocking, credit_note_id	fulfillment, inventory, billing, portal
dev.passage.pricing.rebate.settled.v1	settlement	agreement_id, period_id, amount, credit_note_id	ledger, receivables, analytics

5.30.6 User experience

Screen	Primary actions	Key fields/columns	Duty
AR workspace	tiles: unapplied cash, overdue by bucket, disputes, dunning due, RMAs awaiting inspection	amounts, counts	ARView
Cash application	import, accept/adjust suggestions, apply, on-account, split	payment, payer, amount, candidates with score and rule ids, residual	ARPaymentPost
Open items / settlement	select, settle, discount, write-off, reverse	document, due, open amount, discount available, level	ARSettle
Collections worklist / dispute detail	log contact, promise, dispute, hold order; assign, evidence, resolve	customer, risk score, overdue, oldest item, promises; amount, reason, SLA, timeline	CollectionsMaintain
Dunning run	create, preview letters, exclude, issue	customer, level, items, fee, blocks	DunningRun
Return order (RMA)	authorize, receive, inspect, disposition, credit	lines, disposition, restocking, credit preview	ReturnMaintain
Customer portal	place order, view invoices/documents, pay, request return, track	orders, invoices, balance, payments, RMAs, tracking	PortalCustomer

Interaction rules: cash application keyboard flow (J/K candidates, Enter accept, A apply) and bulk accept of all ≥ 85 suggestions; worklist saved filters; dunning preview opens the rendered letter; portal is mobile-first; provider hosted fields so no card data touches Passage.

5.30.7 Business rules and edge cases

- **BR-30.1 Cash application matching rules and scores.** Candidates are open items of the identified customer (all customers when unidentified). Score per candidate set: R1 exact document number in remittance/RmtInf/check memo +60; R2 amount = open amount of one item +30, or = Σ open of a subset of ≤ 5 items (0.00 tolerance) +25; R3 amount = open – available cash discount with discount date ≥ value_date +28; R4 payer bank account hash (MICR/IBAN) matches the customer's bank account +15; R5 payer name Jaro-Winkler ≥ 0.92 to customer or invoice account +10; R6 value_date within 7 days of due date +5; R7 short by < 1% or < 50.00 USD / 45.00 EUR +20 with a mandatory short-pay reason. Decision: auto-apply when exactly one set scores ≥ 85 and the runner-up ≤ winner – 20; 60–84 → SUGGESTED for one-click; < 60 → ON_ACCOUNT (1205) with top-3 suggestions kept. The ML ranker may reorder sets below 85 but cannot raise one above it. Every auto-application is reversible within the open period.
- **BR-30.2 Dunning levels and grace.** Level 1 Reminder: due date + grace days (default 5 per customer group) passed, no fee; Level 2: ≥ 15 days overdue, no fee; Level 3: ≥ 30 days, fee 25.00 USD / 20.00 EUR as open item FEE; Level 4: ≥ 45 days, fee 50.00 USD / 40.00 EUR, legal text, hold recommendation (sales applies CREDIT via CR-3). Rules: minimum letter amount 50.00 USD / 45.00 EUR per customer; ≥ 10 days since the last letter at any level; an item rises at most one level per run; letter level = max item level; items < 1.00 ignored; customers or items with dunning_block, open disputes or unbroken promises (promise date + 2 days) are excluded; a run is DRAFT until reviewed and issued, and only issue posts fees and levels; full payment resets the item, not the customer's history.
- **BR-30.3 RMA dispositions** (ReturnDispositionCode equivalents): CREDIT_ONLY — no physical return, credit at original price less restocking (default 0%); RETURN_TO_STOCK — receipt to the return warehouse at return cost (ORIGINAL_COST default), credit less restocking (default 15%, minimum 25.00 USD / 20.00 EUR); SCRAP — receipt then scrap issue to 5900, credit less restocking; REPAIR — receipt to quarantine, repair order hook (Phase 8), return to customer, no credit; REPLACE — receipt as RETURN_TO_STOCK or SCRAP plus a zero-price replacement order line referencing the RMA, credit offset by the replacement; REJECT_RETURN_TO_CUSTOMER — receipt to quarantine, ship back, no credit. Disposition is per line and immutable after credit; credit = qty received × original net unit price – restocking.
- **BR-30.4 Settlement accounting.** Cash discount within terms → Dr 4100; write-off within tolerance (≤ 5.00 or 0.5% of the item) → Dr 6100; FX difference on foreign-currency items → 7100/7200 at the settlement-date rate; partial settlement leaves PARTIALLY_SETTLED with reduced open_amount.
- **BR-30.5 Interest** = open_amount × annual_rate × days_overdue / 365 from due date + grace to the calculation date, once per item per period, excluded below the minimum and for disputed/promised items.
- **BR-30.6 State machines.** Payment: RECEIVED → SUGGESTED → APPLIED | ON_ACCOUNT → REVERSED; ON_ACCOUNT may later be applied. Dispute: OPEN → UNDER_REVIEW → RESOLVED_CREDIT | RESOLVED_PAID | RESOLVED_WRITE_OFF → CLOSED; REJECTED from OPEN/UNDER_REVIEW. RMA: REQUESTED → AUTHORIZED → RECEIVED → INSPECTED → DISPOSITIONED → CREDITED → CLOSED; REJECTED from REQUESTED; EXPIRED from AUTHORIZED after 30 days.
- **BR-30.7 Concurrency.** Settlement takes a row lock; a second clerk gets 409; imports deduplicated by SHA-256; camt.053 entries unique per (statement id, entry reference).
- **BR-30.8 Rebate settlement** runs after period close: amount = Σ accruals – prior settlements; a REBATE credit note posts Dr 2150 / Cr 1200; over-accrual posts a free-text invoice.

- **BR-30.9 Portal payments** create an intent for selected items; the signed webhook creates a PORTAL payment that auto-applies by R1 (score 100); refunds flow only through credit notes and the provider refund API with duty ARRefund.
- **BR-30.10 Statements** list items as of the statement date and never change balances; the portal reads the same `open-items` endpoint.

5.30.8 Acceptance criteria

- **AC-30.1** (FR-30.3, FR-30.5) Given `cash-app-500`, when matching runs, then ≥ 450 payments auto-apply and zero apply to the wrong items versus the answer key.
- **AC-30.2** (FR-30.1) Given a 1,000.00 USD invoice at 2% 10 net 30 and 980.00 paid on day 8, then the item settles fully with 20.00 to 4100.
- **AC-30.3** (FR-30.7) Given 60 customers (5 blocked, 4 disputed, 3 with promises), when the run issues, then letters exist for exactly the 31 expected customers at expected levels, level-3 fees are 25.00, and disputed items appear on no letter.
- **AC-30.4** (FR-30.7) Given level 2 issued 2027-03-05, when a run executes 2027-03-12, then no letter (10-day interval).
- **AC-30.5** (FR-30.11) Given an RMA of 4 units at 100.00 USD with RETURN_TO_STOCK and 15% restocking, when received and dispositioned, then an inventory receipt of 4 at original cost exists and a credit note posts Dr 4000 400.00 / Cr 4200 Restocking 60.00 / Cr 1200 340.00 plus tax lines.
- **AC-30.6** (FR-30.11) Given SCRAP, then the receipt is followed by Dr 5900 / Cr 1400 at return cost.
- **AC-30.7** (FR-30.12) Given 10,000.00 accrued for 2027-03 on a 2% agreement, when settled, then a REBATE credit note of 10,000.00 posts and GL 2150 for the agreement is 0.00.
- **AC-30.8** (FR-30.14) Given an invited contact paying two invoices totalling 1,250.00 EUR by card in the Stripe sandbox, then a PORTAL payment is created from the webhook and both items are SETTLED within 60 s.
- **AC-30.9** (FR-30.9) Given a dispute on a 500.00 item, then dunning excludes it; when resolved as credit, a credit note posts and the item settles.
- **AC-30.10** (FR-30.15) Given `sim/phase5-month` on stage, then every §5.0 check passes and the reconciliation pack is produced and signed.

5.30.9 Test plan and fixtures

Fixtures: `cash-app-500` (300 BAI2 checks, 150 camt.053 entries, 50 remittances; short pays, discounts, multi-invoice, unidentified payers; answer key); `dunning-60`; `rma-30` (per disposition × costing method); `disputes-20`; `rebate-settle-12`; `portal-e2e` (Stripe and Adyen sandbox recordings); `sim/phase5-month` (generated month for DMF-US and DMF-DE: 200 POs, 150 production orders, 1,000 orders, 600 payments, 30 RMAs).

Test	Type	Target	Pass criterion
Matching	unit	BR-30.1	≥ 90% auto-match, 0 false matches
Subset-sum bound	property	matcher	≤ 5 items, < 50 ms per payment
Dunning	unit / E2E	run	exact letter set and levels
RMA E2E	E2E	sales + fulfillment + billing + inventory + ledger	30/30 balanced
Settlement accounting	contract (ledger)	discount, write-off, FX	balanced
BAI2 / camt.053 parsers	unit / property	importers	spec samples parse; malformed rejected with line numbers
Portal	E2E (Playwright + axe)	order, pay, return	pass; 0 serious a11y violations
Month simulation	E2E + reconciliation	sim/phase5-month	all §5.0 checks; 31/31 daily reconciliations clean

Reconciliation: Σ open_amount = GL 1200; Σ ON_ACCOUNT = GL 1205; daily settlements = GL 1000/1010 debits from receivables; fees = GL 4300; interest = GL 4310; RMA receipts = wave 16 receipts = GL 1400 debits from returns; rebate settlements = GL 2150 debits; deferred revenue, DNI, inventory, WIP, AP and tax checks re-run on simulation data; trial balance balanced.

5.30.10 Deployment and operations

Helm `passage-receivables` (Go), `passage-collections-workflows` (TS, Temporal namespace `collections`), `passage-customer-portal` (React + TS BFF; ingress `customer.<env>.passage.dev`; Keycloak realm `portal`), `passage-payments-gateway` (TS; provider keys in Vault; PCI scope limited to tokens), matcher ranker (Python, CPU). Databases `receivables-db`, `portal-db`. Config: `MATCH_AUTO_THRESHOLD=85`, `MATCH_MARGIN=20`, `DUNNING_INTERVAL_DAYS=10`, `RMA_EXPIRY_DAYS=30`, `WRITE_OFF_TOLERANCE=5.00/0.5%`. Flags: `ar.ml_ranker`, `ar.camt053`, `collections.interest`, `portal.payments`, `portal.returns`. Jobs: aging and risk score 01:30; lockbox/camt.053 SFTP poll every 15 min; dunning proposal weekly (issue manual); statements on the 1st; rebate settlement after period close; interest monthly. SLOs: open-items read p95 < 100 ms, apply p95 < 1 s, portal 99.9%, webhook processing < 60 s. Alerts: unapplied cash > 2% of monthly receipts for 3 days, webhook failures, reconciliation diff ≠ 0, run stuck in DRAFT > 7 days. Runbook: reverse a mis-application, re-import a file, re-issue a letter, rotate provider keys. Rollback: previous digest per deployable; Temporal workflows versioned.

5.30.11 Migration from AX / SAP

Source	Passage entity	Notes
CustTrans + CustTransOpen (open); CustSettlement (current year)	open_item; settlement history	AmountCur/AmountMST → amounts; CollectionLetterCode → dunning_level
CustCollectionLetterJour (last), CustInterestJour (open)	dunning history; open_item INTEREST	
ReturnTable / SalesLine (ReturnStatus not Closed)	return_order	disposition codes → BR-30.3
BSID (open), BSAD (current year)	open_item; settlement history	MANSP → dunning_block; MSCHL/MANST → level
MHNK/MHND (last run); UDM_DISPUTE (open); VBAK RE/LR (open)	dunning history; dispute; return_order	
KONA/VBOX open accruals	rebate_accrual opening	settled in Passage

Validation: Σ open items per customer = legacy AR aging to the cent; open dispute and RMA counts equal; dunning levels per item equal. Not migrated: cleared items before the current fiscal year, collection notes (lakehouse), legacy statements, CustPaymModeTable/lockbox setup (manual).

5.30.12 Dependencies, risks, open questions

- Depends on: w29 invoices and credit notes; w28 receipts for RMAs; w27 credit profile and holds; w26 accruals; w19 AP payments (rebate payout); w16 inventory; w12 FX; w11 journal; w5 documents and Keycloak realms; Phases 3–4 for the simulation.

Risk	Likelihood	Impact	Mitigation
False auto-match in production	Low	High	threshold + margin, reversible, weekly audit sample of 50
Dunning sent to disputed customers	Low	High	exclusions evaluated at issue time
RMA credit without receipt	Medium	Medium	CREDIT_ONLY requires duty ReturnCreditOnly and reason code
PCI scope creep in the portal	Low	High	hosted fields only; tokens stored; annual SAQ-A
Sign-off slips because accountant review is late	Medium	High	accountant engaged from wave 29 fixtures onward

- Open questions: (1) Primary demo payment provider, Stripe or Adyen? Default Stripe sandbox; decide by wave 30 start. (2) Are dunning fees taxable in DMF-DE? Default untaxed in v1 pending the country pack (wave 48); decide at wave 30 mid-point. (3) Does the simulation include DMF-DE → DMF-US intercompany sales? Default no (wave 40); decide at wave 29 close.

Part 6 — Phase 6: Warehouse, Quality & Logistics (waves 31–35)

6.0 Phase overview

Business outcome at phase exit. At the end of wave 35 the demo manufacturer (live since wave 30) runs its warehouses on location-directed work executed from handheld scanners, inspects inbound, in-process and outbound inventory under rule-generated quality orders, plans and rates outbound freight against real carrier rate cards, reconciles carrier invoices and lands freight into inventory cost, and exchanges purchase orders, ASNs and invoices with a demo distributor over X12 EDI and Peppol. Every one of those flows posts into the wave-16 inventory ledger and the wave-11 universal journal through the existing posting APIs; Phase 6 adds no new financial truth, only new operational sources of it. The Phase 6 services are `wms` (Go, `wms_db`), `wms-mobile` BFF (TypeScript), `quality` (Go, `quality_db`), `tms` (Go, `tms_db`) with a Python `tms-optimizer` sidecar (OR-Tools), and `integration-hub` plus `edi` (TypeScript, `integration_db`). All run on the ADR-0001 stack; nothing in this phase introduces new infrastructure.

Phase-exit sign-off criteria

- Demo distributor ↔ demo manufacturer exchange PO (850/ORDERS) → ASN (856/DESADV) → invoice (810/INVOIC and Peppol UBL) end-to-end with zero manual re-keying; the invoice lands as a wave-19 vendor invoice at the distributor and matches 3-way.
- 100 location-directive fixtures resolve to the expected location; 500-order wave release creates work and replenishment with no deadlocks and p95 task assignment under 1 s.
- Mobile E2E inbound-to-outbound run on an emulated Zebra device using scanned GS1-128 labels passes in CI; 200 queued offline transactions sync without loss or duplicate.
- Quality property test: no pick, ship or consume succeeds against a quant with `quality_status = BLOCKED` across 10,000 randomized attempts; AQL sample sizes equal ANSI/ASQ Z1.4 Table I/II-A for every lot-size class.
- VRP solutions on Solomon C1/R1/RC1 instances are feasible and within 10% of best-known distance; 200 rating fixtures produce expected charges to the cent; landed cost allocation sums to the carrier invoice exactly.
- Daily reconciliation: `wms` on-hand by location sums to the wave-16 inventory ledger on-hand per item/warehouse; freight accruals in `tms` tie to journal account 2150 (accrued freight) to the cent.

Cross-wave data model summary

Entity	Owning service	Key relationships
Location, Zone, LocationProfile	wms	Warehouse (wave 6); capacity used by Work and Quant
LicensePlate (nested)	wms	Parent LP; Quant; Load; ShipmentContainer (wave 28)
Quant (on-hand by item × dims × location × LP)	wms	Projection of inventory ledger (wave 16); carries `quality_status`
LocationDirective, WorkTemplate, WaveTemplate, ReplenishmentTemplate	wms	Metadata; evaluated by the directive engine
Work, WorkLine, Wave, Load, DockAppointment	wms	Sales shipment (wave 28), purchase receipt (wave 18), production order (wave 21), TMS Load
MobileFlow, MobileMenu, DeviceSession, OfflineTransaction	wms-mobile	Work; WorkUser (Keycloak subject)
Test, TestGroup, SamplingPlan, QualityAssociation	quality	Item/product (wave 8); Party (wave 7)
QualityOrder, QualityResult, UsageDecision	quality	Quant (blocking), PurchaseReceipt, ProductionOrder, Shipment
Nonconformance, Correction (CAPA), Instrument	quality	QualityOrder; Vendor; Temporal workflow id
Carrier, CarrierService, RateCard, Accessorial, FuelIndex	tms	Party (wave 7)
Shipment, Route, Load (TMS view), Tender, TrackingEvent	tms	wms Load; Sales shipment (wave 28)
FreightBill, CarrierInvoice, LandedCostVoyage	tms	Vendor invoice (wave 19); inventory cost (wave 20); journal (wave 11)
TradingPartner, EDIProfile, Map, Interchange, Message, Acknowledgment	edi	Party; PurchaseOrder, SalesOrder, Shipment, Invoice
ApiClient, Subscription (webhook/Kafka), DeliveryAttempt, Quota	integration-hub	Keycloak client; Kafka topics

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
dev.passage.wms.work.created.v1	wms	wms-mobile, analytics	work_id, work_type, warehouse, priority, lines[]
dev.passage.wms.work.completed.v1	wms	inventory, sales-fulfillment, production	work_id, lines with from/to location, LP, qty
dev.passage.wms.load.shipped.v1	wms	tms, sales-fulfillment, edi (856)	load_id, shipments[], LPs/SSCCs, carrier, ship_date
dev.passage.wms.receipt.registered.v1	wms	purchasing (wave 18), quality	receipt_id, po lines, LPs, qty
dev.passage.quality.order.created.v1	quality	wms (block quant), analytics	quality_order_id, trigger, item, quant_ref, sample_size
dev.passage.quality.usagedecision.posted.v1	quality	wms, inventory, purchasing, production	quality_order_id, decision, accept_qty, reject_qty
dev.passage.quality.nonconformance.created.v1	quality	workflow, vendor-scorecard	ncr_id, source, severity, item, vendor
dev.passage.tms.shipment.rated.v1	tms	sales (freight charge), analytics	shipment_id, carrier, service, charges[], total
dev.passage.tms.tender.accepted.v1	tms	wms (dock), edi (204/990)	load_id, carrier, pickup_window
dev.passage.tms.freightbill.matched.v1	tms	accounts-payable (wave 19), costing (wave 20)	freight_bill_id, invoice_id, variance, landed_cost_lines[]
dev.passage.edi.message.received.v1	edi	purchasing, sales, ap	partner, standard, transaction_set, control numbers, document ref
dev.passage.edi.message.sent.v1	edi	analytics, integration-hub	partner, transaction_set, ack status
dev.passage.hub.webhook.delivered.v1	integration-hub	analytics	subscription_id, event_id, attempt, status

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Directive engine performance collapses on large location sets (50k bins)	Medium	High	Indexed candidate pre-filter per zone/profile; benchmark at 50k locations in wave 31 CI; p95 < 200 ms per line	wms lead
Scanner hardware variance (Zebra DataWedge vs Honeywell) breaks scan capture	Medium	Medium	Keyboard-wedge fallback plus intent API adapters; emulator E2E for both profiles	mobile lead
Quality blocking is bypassed by a path not covered by the property test	Low	High	Blocking enforced in `inventory` ledger reservation, not only in `wms`; property test enumerates all transaction types	quality lead
VRP solver time exceeds planning window on 1,000-stop problems	Medium	Medium	Time-limited local search (60 s) with feasible first solution guaranteed by sweep heuristic	tms lead
EDI partner specifics (implementation guides) diverge from base maps	High	Medium	Partner-level map overrides; simulator fixtures per partner; 997/CONTRL negative tests	integration lead
Peppol access point certification delays	Medium	Medium	Certified third-party AP (SMP/AS4) adapter first; own AS4 endpoint deferred to v1.x	integration lead

6.31 Wave 31: Advanced Warehouse Management

6.31.1 Scope and business context

Wave 28 ships sales orders from simple warehouses (item × location on-hand, manual pick lists). Wave 31 replaces that with the D365 WHS / SAP EWM model: a location hierarchy with capacity and profiles, license plates (LP) nested to any depth, and work that the system creates and directs by rule. After this wave a warehouse manager configures put-away and pick location directives, work templates and wave templates in metadata; releasing sales, transfer and production orders to the warehouse creates work with explicit from/to locations; replenishment runs on min/max, demand or wave triggers; loads are built and docks scheduled; receiving is ASN-driven with LPs. Out of scope: mobile execution (wave 32 — in wave 31 work is confirmed from the web workspace), quality blocking (wave 33), carrier rating (wave 34), cross-docking and catch-weight work (v1.x), labor standards (v1.x).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 location directives (WHSLocationDirective form) fail silently: work is created with blank put location or "no location found" only visible in the work details after wave processing; debugging requires the "Location directive failures" log (10.0.x) or SQL on WHSWorkLine	Directive evaluation is a batch-side X++ class with no trace; query ranges are opaque	`/location-directives:simulate` returns a rule-by-rule trace; blocked work has status `BLOCKED_NO_LOCATION` with the failing step on the wave	Time to diagnose a directive miss: 30–60 min (consultant with SQL) → < 1 min from the trace; 0 silent blanks
EWM put-away bin determination (/SCWM/T333 storage type search sequence + bin type + storage section) spans 6 IMG nodes; a change needs a transport and a test cycle	Configuration split across customizing tables with cross-dependencies	One LocationDirective entity with ordered lines/actions, versioned metadata, simulate-before-save	Setup clicks for a new put-away rule: ~40 across 6 transactions → ≤ 8 in one designer; change lead time days → minutes
D365 wave processing (WHSWaveProcessing batch) runs every 5–15 min; pickers wait; a failed wave leaves partial work and the wave in "Held" with a generic infolog	Batch framework polling; non-atomic processing per order	Temporal `WaveReleaseWorkflow` starts on release, per-order activities with per-line failure reasons	Order release to assignable work: 5–15 min batch → < 5 s p95 for a 50-order wave; failures itemized 100%
EWM /SCWM/MON and D365 "Work details" cannot show capacity or why a location is full; over-capacity bins happen because capacity check is advisory (D365 `Location stocking limits`)	Capacity is a soft check in some paths	Capacity is an invariant: creation and confirmation both enforce it; property-tested	Over-capacity incidents: tolerated today → 0 (test-enforced)
On-hand by LP in AX (WHSInventOnhand vs InventSum) drifts and is reconciled manually by inventory journals	Two on-hand tables maintained by separate code paths	Quants are a projection of the single inventory ledger; daily automated reconciliation with alert	Monthly reconciliation effort: 4–8 h → 0 h; drift detected within 24 h

6.31.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Location master	WMSLocation, WHSLocationProfile, WHSZone, WHSLocationFormat; form WHSLocation	/SCWM/LAGP storage bins, /SCWM/T301 storage types, sections /SCWM/T302, bin types /SCWM/T303	wms.Location, Zone, LocationProfile	full
License plates / HUs	WHSLicensePlate, WHSContainerTable	/SCWM/HUHDR, /SCWM/HUITM, packaging material	wms.LicensePlate (nested), Quant	full
Put-away/pick strategy	WHSLocationDirective, WHSLocationDirectiveLine, WHSLocationDirectiveAction; form WHSLocationDirective	Put-away strategies (/SCWM/T333 storage type search, bin determination), stock removal strategies (FIFO/LIFO/partial)	wms.LocationDirective with ordered rules	full
Work templates	WHSWorkTemplateTable, WHSWorkTemplateLine, WHSWorkTransType	Warehouse process types /SCWM/TPROCTY, warehouse order creation rules /SCWM/WOCR	wms.WorkTemplate	full
Work / tasks	WHSWorkTable, WHSWorkLine; form WHSWorkTable	/SCWM/ORDIM_O warehouse tasks, /SCWM/WHO warehouse orders	wms.Work, WorkLine	full
Wave management	WHSWaveTemplateTable, WHSWaveTable, WHSWaveLine; class WHSWaveProcessing	/SCWM/WAVE, /SCWM/WAVEHDR, /SCWM/WAVEITM; wave templates /SCWM/WAVETMP	wms.WaveTemplate, Wave; Temporal `WaveReleaseWorkflow`	full
Replenishment	WHSReplenishmentTemplate, WHSReplenishmentTemplateLine; class WHSReplenishmentProcess	Replenishment control (/SCWM/REPL, planned/order-related/automatic)	wms.ReplenishmentTemplate	full
Loads / dock	WHSLoadTable, WHSLoadLine, WHSDockAppointment (appointment scheduling)	Yard management, door/transportation units /SCWM/TU, /SCWM/DOOR	wms.Load, DockAppointment	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
ASN receiving	WHSASNItem, WHSUOMStructure, packing structure import	Inbound delivery /SCWM/INB (EWM delivery), expected goods receipt	wms.AsnPackingStructure → LP receiving	full
Cycle counting	WHSCycleCountPlan, WHSCycleCountThreshold, WHSCountingWork	Physical inventory /SCWM/PI_DOCUMENT, cycle counting indicator	wms.CountPlan, counting Work (web in 31, mobile in 32)	full
Slotting	WHSSlottingTemplate (D365 10.0.x)	EWM slotting (/SCWM/SLOT)	wms.SlottingRun (ABC by picks, recommendations)	partial — recommendations only, no auto-move
Labor management	n/a (ISV)	EWM labor management	—	dropped: v1.x

6.31.3 Functional requirements

- **FR-31.1 Location hierarchy with profiles** — Must. The system shall model locations as `warehouse / zone / aisle / rack / shelf / bin` with a LocationProfile that defines capacity (weight, volume, LP count, item count), mixing rules and allowed LP behavior.
  - Notes: capacity limits are checked when work is created and again at confirmation; a location with `allow_mixed_items = false` rejects a second item; location name is generated from a LocationFormat (`A-01-02-03` → aisle 1, rack 2, shelf 3); maximum 1,000,000 locations per warehouse.
- **FR-31.2 Nested license plates** — Must. The system shall track inventory on license plates that nest (pallet → case → inner) to depth 5, with a single move of a parent moving all children atomically.
  - Notes: LP id is an SSCC-18 when generated by Passage (extension digit 0, GS1 company prefix from wave 8 config); external LPs are accepted up to 40 characters; an LP can be at exactly one location or in transit on one Work.
- **FR-31.3 Location directives** — Must. The system shall resolve a destination (put-away) or source (pick) location for each work line by evaluating ordered directives, lines and actions as specified in BR-31.1.
  - Notes: directive scope = work order type × warehouse × optional query on item/customer/vendor; strategies: `consolidate`, `fifo`, `lifo`, `fefo`, `empty_location`, `match_packing_quantity`, `round_up_to_lp`; evaluation p95 < 200 ms per line at 50,000 locations.
- **FR-31.4 Work templates** — Must. The system shall create Work from templates whose lines define pick/put pairs, work class, stop-after-line and max lines/LPs per work header.
  - Notes: templates have a query (e.g., `shipping_method = LTL`); `work_break` rules split by customer, shipment or load; work class restricts which users may execute.
- **FR-31.5 Wave templates and release** — Must. The system shall group released order lines into waves by template (query, max lines 5,000, max weight, scheduled or manual release), create work and replenishment on release, and lock or allow late additions.
  - Notes: release runs as Temporal `WaveReleaseWorkflow` with per-order activities; automatic wave release on schedule (cron) or on order release; failed lines are listed on the wave with reason codes.
- **FR-31.6 Replenishment** — Must. The system shall create replenishment work by min/max (per location × item), by demand (wave shortage) and by load demand, from configured source zones.
  - Notes: min/max runs every 15 minutes by default; demand replenishment is created inside the wave before pick work; replenishment work carries priority 10 (higher than picks at 50) and must complete before the dependent pick is assignable.
- **FR-31.7 Work execution and exceptions** — Must. The system shall assign work by priority, work class and user filter; support short pick with reason, skip location, over-pick within tolerance and cancel-with-reversal.
  - Notes: short pick re-queues remaining qty as new work after a configurable retry; assignment is pessimistic (`SELECT … FOR UPDATE SKIP LOCKED`) so two users never receive the same work; p95 assignment < 1 s under 10k lines/day simulation.
- **FR-31.8 Loads and dock appointments** — Must. The system shall build loads from shipments (manual, rule, or by route), confirm loads, and schedule dock appointments with 15-minute slot granularity per door.
  - Notes: load confirmation posts packing slips via the wave-28 API; dock doors have calendars (wave 6); overlapping appointments on one door are rejected.
- **FR-31.9 ASN-driven receiving** — Must. The system shall import an ASN packing structure (from EDI 856 in wave 35, or JSON API now) and receive by scanning/selecting the pallet LP, creating quants per nested LP.
  - Notes: receipt without ASN remains supported (item + qty + LP); variance to ASN produces a receipt discrepancy with tolerance ±2% default.
- **FR-31.10 Cycle counting** — Should. The system shall generate count work by plan (location set, frequency), threshold (on-hand ≤ n) and ad hoc; count differences post via the wave-16 counting journal after supervisor approval above a tolerance.
  - Notes: tolerance default 1% of value or 5 units; blind counts hide expected qty.
- **FR-31.11 Slotting analysis** — Could. The system shall compute ABC velocity by picks/30 days and recommend item-to-location moves to forward pick zones.
  - Notes: output is a recommendation list; applying generates movement Work.
- **FR-31.12 Warehouse KPIs** — Should. The system shall expose lines picked per hour, work aging, dock utilization, replenishment lead time, short-pick rate via the Cube semantic layer.
  - Notes: materialized hourly; lag ≤ 60 s per PRD §8.

6.31.4 Data model

Location

Field	Type	Required	Notes
location_id	uuid	yes	PK
warehouse_id	uuid	yes	FK warehouse (wave 6)
name	text	yes	unique per warehouse, generated from LocationFormat
zone_id	uuid	yes	FK Zone
profile_id	uuid	yes	FK LocationProfile
aisle, rack, shelf, bin	text	no	parsed components; indexed for sort order

Field	Type	Required	Notes
max_weight_kg, max_volume_m3, max_lps, max_items	numeric(18,6)/int	no	null = unlimited
pick_sequence	int	yes	travel-path ordering within zone
status	enum	yes	ACTIVE, BLOCKED_IN, BLOCKED_OUT, BLOCKED, INACTIVE

LicensePlate

Field	Type	Required	Notes
lp_id	text(40)	yes	PK; SSCC-18 when generated
warehouse_id	uuid	yes	FK
parent_lp_id	text(40)	no	FK LicensePlate; depth ≤ 5 (trigger-enforced)
location_id	uuid	no	null only when on an in-progress Work
in_transit_work_id	uuid	no	FK Work; mutually exclusive with location_id
lp_type	enum	yes	PALLET, CASE, TOTE, INNER, CONTAINER
gross_weight_kg, volume_m3	numeric(18,6)	no	computed from children when nested
created_at	timestamptz	yes	

Work / WorkLine

Field	Type	Required	Notes
work_id	uuid	yes	PK; number sequence WRK-* (wave 5)
warehouse_id	uuid	yes	FK
work_type	enum	yes	PURCHASE_PUTAWAY, SALES_PICK, TRANSFER_ISSUE, TRANSFER_RECEIPT, PRODUCTION_PICK, RAF_PUTAWAY, REPLENISHMENT, MOVEMENT, COUNT, RETURN
status	enum	yes	see BR-31.3 state machine
priority	smallint	yes	1 (highest)–99; default by template
work_class_id	uuid	yes	FK; filters eligible users
wave_id, load_id, order_ref	uuid/text	no	FK Wave, Load; polymorphic order ref
assigned_user_id, locked_at	text/timestamptz	no	pessimistic lock
line_no, line_type (PICK/PUT), item_id, dims jsonb, qty, uom, from_location_id, to_location_id, lp_id, target_lp_id, line_status	—	yes	WorkLine; line_status OPEN/IN_PROGRESS/COMPLETED/SKIPPED/CANCELLED

LocationDirective

Field	Type	Required	Notes
directive_id	uuid	yes	PK
sequence	int	yes	evaluation order; unique per (work_type, warehouse, direction)
work_type, direction	enum	yes	direction PUT or PICK
warehouse_id	uuid	yes	
query	jsonb	yes	metadata query (wave 3 query model) against order line
lines	jsonb[]	yes	each: seq, from_qty, to_qty, uom, locate_packing_qty bool, restrict_by_uom
actions	jsonb[]	yes	per line: seq, strategy, fixed_location bool, location_query, allow_negative
multiple_sku	bool	yes	true = directive used for multi-SKU LPs

Invariants: Σ quant qty per (item, dims, warehouse) in `wms_db` equals inventory ledger physical on-hand for that warehouse at every ledger commit (eventual, reconciled daily and after each work completion); an LP is at exactly one location or on exactly one Work; location capacity ≥ Σ (quants + inbound in-progress work) at all times (property test); a WorkLine PUT always follows its PICK in `line_no` order within the same work. Indexing: `quant(warehouse_id, item_id, dims_hash, location_id)`; partial index `work(warehouse_id, priority, created_at) WHERE status = 'OPEN'`; `location(zone_id, pick_sequence)`; `license_plate(parent_lp_id)`. `wms_db` is partitioned by `tenant_id` list partitions over RLS.

6.31.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/wms/v1/locations:bulk	Create/update locations from format	Idempotency-Key header, 24 h	WMS_SETUP_MAINTAIN
POST	/wms/v1/location-directives	Create directive	key	WMS_SETUP_MAINTAIN
POST	/wms/v1/location-directives:simulate	Evaluate directives for a hypothetical line; returns trace	n/a (read)	WMS_SETUP_VIEW
POST	/wms/v1/waves/{id}:release	Release wave (starts Temporal workflow)	key; second call returns workflow id	WMS_WAVE_RELEASE
POST	/wms/v1/work:assign	Assign next work for user by filter	key per user+device	WMS_WORK_EXECUTE
POST	/wms/v1/work/{id}/lines/{n}:confirm	Confirm pick/put with actual qty, LP, location	key	WMS_WORK_EXECUTE
POST	/wms/v1/work/{id}:exception	Short pick, skip, cancel with reason	key	WMS_WORK_EXECUTE
POST	/wms/v1/receipts:from-asn	Register receipt from ASN packing structure	key = ASN id	WMS_RECEIVE
POST	/wms/v1/loads/{id}:confirm-shipment	Confirm load, post packing slips	key	WMS_SHIP_CONFIRM
GET	/wms/v1/quants?item=&location=&lp=	On-hand inquiry	—	WMS_INQUIRY
POST	/wms/v1/replenishment:run	Run min/max for a warehouse	key = warehouse+minute	WMS_REPLENISH

Commands (CQRS): `CreateWorkFromOrderLines` validates all lines belong to one warehouse and are released; `ConfirmWorkLine` validates qty ≤ open qty + over-pick tolerance, LP exists or is new at a PUT, location status allows direction, capacity; `ReleaseWave` validates template limits before starting the workflow.

Event	Trigger	Payload fields	Consumers
dev.passage.wms.work.created.v1	Work header committed	work_id, work_type, warehouse_id, priority, wave_id, lines[]	wms-mobile, analytics
dev.passage.wms.work.completed.v1	last line confirmed	work_id, lines[{item, dims, qty, from, to, lp}], user, duration_s	inventory (post transfer), sales-fulfillment, production
dev.passage.wms.wave.released.v1	workflow finished	wave_id, orders, work_count, replenishment_count, failures[]	analytics, sales-fulfillment
dev.passage.wms.load.shipped.v1	load confirmed	load_id, shipment_ids, lp_ids, carrier_id, ship_datetime	tms, edi, sales-fulfillment
dev.passage.wms.receipt.registered.v1	receipt committed	receipt_id, po_line_refs, lp_ids, qty by item	purchasing, quality

6.31.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Warehouse setup workspace	Locations grid, profiles, zones, formats; generate locations	name, zone, profile, capacity, status	WMS_SETUP_MAINTAIN
Location directive designer	Order directives, add lines/actions, simulate	sequence, work type, query, strategy, trace	WMS_SETUP_MAINTAIN
Wave workbench	Create/release/hold waves, view failures	wave id, template, lines, weight, status, failures	WMS_WAVE_RELEASE
Work list	Filter, reassign, change priority, cancel	work id, type, status, user, age, lines	WMS_WORK_SUPERVISE
Load planning workbench	Build loads from shipments, assign door, confirm	load, shipments, weight/volume, door, appointment	WMS_SHIP_CONFIRM
Receiving workspace	Receive from ASN/PO, create LPs, print	PO, ASN, LP, item, qty, discrepancy	WMS_RECEIVE
On-hand by location/LP	Inquiry, movement, LP tree	item, dims, location, LP, qty, status	WMS_INQUIRY

Interaction rules: grids support keyboard range selection and bulk priority change (max 500 rows per action); directive designer validates on save and shows a simulation trace (rule hit/miss per step); personalization (column sets, saved filters) per user from wave 5 shell; destructive actions (cancel work with inventory in transit) require a reason code and confirmation.

6.31.7 Business rules and edge cases

- **BR-31.1 Location directive evaluation algorithm.** For a work line (item, dims, qty, uom, order attributes), evaluate: (1) select directives where `work_type`, `direction`, `warehouse` match and `query` is true, ordered by `sequence`; (2) within a directive, select the first line where `from_qty ≤ qty_in_line_uom < to_qty` (uom converted via wave 6 conversions; `locate_packing_qty` uses the item's packing hierarchy); (3) evaluate actions of that line in `seq` order: each action produces candidate locations = locations matching `location_query` ∩ status allows direction ∩ (for PUT) capacity remaining ≥ line qty/weight/volume ∩ profile mixing rules; the strategy orders candidates (`consolidate`: locations already holding item+dims, then by pick_sequence; `empty_location`: empty only; `fifo`/`fefo` for PICK: quants by receipt date / expiry ascending, tie-break pick_sequence; `round_up_to_lp`: full-LP quantities first); (4) the first action yielding ≥ 1 candidate wins and the top candidate is returned; (5) if no action of the matched line yields a candidate, continue with the next *directive* (lines within a directive are mutually exclusive by quantity band, so no other line of the same directive is tried); (6) if no directive yields a location, the line is created with `to_location = null` and work status `BLOCKED_NO_LOCATION`, and the wave lists it as a failure. Worked example: PUT line, item `BOLT-M8`, 1,200 EA, packing 100/case, 12 cases/pallet. Directives for PURCHASE_PUTAWAY in `WH1`: seq 10 query `item_group = FASTENERS`; seq 20 catch-all. Directive 10 lines: L1 0–99 EA (actions: A1 consolidate in zone `FWD`), L2 100–1199 EA (A1 consolidate zone `CASE`, A2 empty_location zone `CASE`), L3 1200–∞ EA (A1 empty_location zone `RESERVE` fixed pallet locations, A2 empty_location zone `OVERFLOW`). Qty 1,200 falls in L3. A1 candidates: RESERVE locations with status ACTIVE, 0 LPs, max_weight ≥ 1,200 × 0.012 kg = 14.4 kg → `R-05-02-01` (pick_sequence 5021) wins. If RESERVE were full, A2 would choose the first empty OVERFLOW location; if none, directive 20 is tried; if it also fails, the line is BLOCKED_NO_LOCATION.
- **BR-31.2 Capacity arithmetic.** Remaining capacity = max – Σ(quant) – Σ(in-progress PUT lines targeting the location). Rounding: weight and volume to 6 decimals; LP count and item count are integers. A location with remaining weight 0.000001 kg does not accept a 0.000002 kg line.
- **BR-31.3 Work state machine.** `OPEN → ASSIGNED → IN_PROGRESS → COMPLETED`; `OPEN|ASSIGNED → CANCELLED` (only if no line confirmed); `IN_PROGRESS → PARTIALLY_COMPLETED` (short pick; remainder re-queued as new OPEN work with `parent_work_id`); `ASSIGNED → OPEN` on unassign or lock timeout (default 30 min idle); `BLOCKED_NO_LOCATION → OPEN` after a re-evaluation succeeds. No transition out of COMPLETED; reversal creates compensating MOVEMENT work.
- **BR-31.4 Wave creation state machine.** `CREATED → RELEASING → RELEASED` or `→ RELEASE_FAILED` (any line failed and template `fail_whole_wave = true`) or `→ RELEASED_WITH_ERRORS`; `RELEASED → HELD → RELEASED`; `CREATED|RELEASED_WITH_ERRORS → CANCELLED` reverses all unstarted work.
- **BR-31.5 Replenishment precedence.** Pick work whose source location has insufficient quant after replenishment demand is created carries `depends_on_work_id`; it is not assignable until the dependency is COMPLETED. Dependency cycles are impossible because replenishment sources are restricted to zones not used as pick zones in the same directive.
- **BR-31.6 FEFO with equal expiry** breaks ties by receipt timestamp then pick_sequence; expiry is compared as a date in the warehouse's time zone.
- **BR-31.7 Over-pick tolerance** default 0%; max configurable 10%; over-pick never applies to serial-tracked items or to PICK lines sourcing from a quant with `quality_status ≠ ACCEPTED` (wave 33).
- **BR-31.8 Concurrency.** Work assignment uses `FOR UPDATE SKIP LOCKED` on the open-work partial index; quant updates use optimistic `version` with retry 3×; two confirmations of the same line with the same idempotency key return the first result; with different keys the second fails `409 LINE_ALREADY_CONFIRMED`.
- **BR-31.9 Nested LP moves** are a single transaction updating `location_id` on the root only; children derive location from the root (`location_id` null on children with a parent). Un-nesting a child sets its own location to the parent's current location.
- **BR-31.10 ASN variance.** Received qty outside ±2% (configurable) of ASN qty creates a `ReceiptDiscrepancy` with reason; receipt still posts actual qty.

6.31.8 Acceptance criteria

- **AC-31.1** (FR-31.3) Given the 100-scenario directive fixture, when each line is simulated via `/location-directives:simulate`, then every result equals the expected location and the trace shows the expected winning directive/line/action.
- **AC-31.2** (FR-31.1, BR-31.2) Given a location with max_lps = 1 holding one LP, when a PUT line targets it, then the action skips it and the simulation trace states `capacity: lps 1/1`.
- **AC-31.3** (FR-31.5, FR-31.6) Given 500 released sales orders (2,300 lines) and the demo directives, when the wave is released, then 2,300 pick lines and the expected 41 replenishment works are created, no work is BLOCKED_NO_LOCATION, and the wave is RELEASED within 120 s.
- **AC-31.4** (FR-31.7, BR-31.8) Given 50 concurrent users calling `work:assign`, when 1,000 open works exist, then no work id is returned to two users and p95 latency < 1 s.
- **AC-31.5** (FR-31.7, BR-31.3) Given an IN_PROGRESS pick of 10 EA, when the user short-picks 6 with reason `EMPTY`, then the work becomes PARTIALLY_COMPLETED, a new OPEN work for 4 EA exists with parent reference, and the inventory ledger shows a 6 EA transfer only.
- **AC-31.6** (FR-31.2, BR-31.9) Given a pallet LP with 12 case LPs, when the pallet is moved, then all 13 LPs report the new location and one `work.completed` event is emitted.
- **AC-31.7** (FR-31.9) Given an ASN with 3 pallets, when the receiver confirms pallet 2 with qty 95 of 100, then a discrepancy of –5 (–5%) is recorded and quants exist for pallet 2 only.
- **AC-31.8** (FR-31.8) Given door D1 with an appointment 09:00–09:45, when a second appointment 09:30–10:00 is requested, then it is rejected with `DOOR_SLOT_OVERLAP`.
- **AC-31.9** (property) For 10,000 generated work and confirmation sequences, location capacity is never exceeded and LP location/transit exclusivity holds.

6.31.9 Test plan and fixtures

Fixtures: `wms-demo-warehouse` (1 warehouse, 4 zones, 8,000 locations, generated from format), `directive-scenarios-100.json` (hand-authored, reviewed against D365 demo data USMF behavior), `wave-500-orders` (generated from wave-28 order fixtures), `asn-3-pallet.json`, `sim-10k-lines-day` (synthetic order stream for the simulation harness).

Test	Type	Target	Pass criterion
Directive resolution	unit/table-driven	100 scenarios	100% expected location
Capacity invariant	property (rapid/Go)	10k sequences	zero violations
Work state machine	unit	all transitions	illegal transitions rejected
Wave release	E2E (Playwright + API)	500 orders	work counts, < 120 s
Assignment contention	performance (k6)	50 VUs	p95 < 1 s, no duplicates
Simulation	performance	10k lines/day × 3 days	no deadlocked work, no BLOCKED_NO_LOCATION
Directive perf	performance	50k locations	p95 < 200 ms/line
Nested LP	contract	API	depth 6 rejected

Reconciliation: Σ quant per item/warehouse = inventory ledger on-hand; Σ in-transit on Work = ledger "in transit" status; load shipped quantities = packing slip quantities (wave 28).

6.31.10 Deployment and operations

Helm chart `passage-wms` (deployments `wms-api`, `wms-directive-engine`, `wms-worker` for Temporal activities; HPA 2–10 on CPU 70%); CloudNativePG cluster `wms-db` (3 instances, 100 GB); Atlas migrations generated from entity metadata; config: `WMS_LOCK_TIMEOUT_MIN=30`, `WMS_REPLENISH_INTERVAL_MIN=15`, `WMS_MAX_WAVE_LINES=5000`. Feature flags: `wms.advanced` per warehouse (off = wave-28 simple mode), `wms.slotting`. Scheduled jobs (Temporal schedules): min/max replenishment, automatic wave release, count plan generation (daily 02:00 warehouse-local), KPI materialization hourly. SLOs: assignment p95 < 1 s; directive simulate p95 < 200 ms; wave release of 5,000 lines < 10 min; availability 99.9%. Alerts: `WmsWorkBlockedNoLocation > 50 in 15 min`, `WmsAssignmentP95 > 2 s`, `WmsQuantLedgerDrift ≠ 0 after daily reconciliation`, `WmsWaveWorkflowFailed`. Runbook: re-evaluate blocked work; release stuck locks; reconcile drift (compare `quant` vs ledger projection, apply correction journal with approval). Rollback: Argo CD to previous digest; migrations are additive within a wave; flag `wms.advanced=false` returns a warehouse to simple mode only if no OPEN work exists.

6.31.11 Migration from AX / SAP

Source	Passage entity	Transformation
WMSLocation, WHSLocationProfile, WHSZone, WHSLocationFormat	Location, LocationProfile, Zone, LocationFormat	parse name by format; capacity from profile; status from `WHSLocationStatus`
/SCWM/LAGP, /SCWM/T301, /SCWM/T302, /SCWM/T303	Location, Zone (storage type/section), LocationProfile (bin type)	LGTYP → zone, LGBER → sub-zone attribute; capacity from LAGP `MAX_WEIGHT`/`MAX_CAPA`
WHSLicensePlate, WHSInventOnhand by LP	LicensePlate, Quant	parent LP via `WHSLicensePlate.ParentLicensePlateId`
/SCWM/HUHDR, /SCWM/HUITM, /SCWM/AQUA	LicensePlate, Quant	HU nesting via `HUHDR.TOP`/`HIGHER_LEVEL`; quants from AQUA by LGPLA
WHSLocationDirective/Line/Action	LocationDirective	strategy enum mapped 1:1; AX queries translated to metadata query
WHSWorkTemplateTable/Line, WHSWaveTemplateTable, WHSReplenishmentTemplate	WorkTemplate, WaveTemplate, ReplenishmentTemplate	direct
EWM put-away/removal strategies (/SCWM/T333, /SCWM/T334)	LocationDirective	storage type search sequence → ordered actions
WHSWorkTable/Line (status Open/In process), /SCWM/ORDIM_O	Work, WorkLine	only open work migrated at cutover; completed work is history (not migrated)
WHSLoadTable/Line (not shipped)	Load	open loads only

Validation: location count and names match; on-hand by location/LP equals source on-hand snapshot (wave-16 reconciliation extended by location); open work count matches. Not migrated: closed work history (WHSWorkTable confirmed, /SCWM/ORDIM_C) — kept in the wave-46 lakehouse as archived source data; labor standards; EWM yard data (v1.x).

6.31.12 Dependencies, risks, open questions

Depends on: wave 30 (go-live baseline), wave 28 (shipments, packing slips), wave 18 (purchase receipts), wave 21 (production picking/RAF), wave 16 (inventory ledger, reservations, counting journal), wave 8 (GS1 company prefix, packing hierarchy), wave 6 (warehouses, calendars, UoM), wave 4 (Temporal, outbox).

Risk	Likelihood	Impact	Mitigation
Directive semantics differ subtly from D365 and confuse migrating customers	Medium	Medium	Simulation trace; parity fixtures from D365 USMF; documented differences
Quant/ledger drift under high concurrency	Low	High	Single posting path through inventory ledger; daily reconciliation alert

Open questions: (1) Should quants be owned by `wms` or remain a projection of `inventory`? Proposed: projection with `wms` as the only writer of location/LP dims; decide by wave 31 design review. (2) SSCC extension digit per warehouse or per tenant? Proposed: per warehouse; decide by wave 32.

6.32 Wave 32: Mobile Warehouse Execution (Scanners, RF, Voice)

6.32.1 Scope and business context

Wave 31 creates work; wave 32 lets warehouse workers execute it on handheld scanners. This replaces the D365 Warehouse Management mobile app (mobile device menus, menu items and steps) and the SAP EWM RF framework (/SCWM/RFUI, ITSmobile). After this wave a worker signs in on a Zebra or Honeywell Android device (or any PWA-capable browser), picks from a configured menu, scans GS1-128 labels that the system parses into LP/GTIN/lot/expiry/serial/weight, and completes receive, put-away, pick, pack, move, count and inquiry flows with offline tolerance. An administrator changes flows without an app release. Label printing (ZPL) ships through a print service. Out of scope: full voice-directed picking (adapter interface only; a voice vendor integration is v1.x), RFID portal reads beyond the EPC ingest endpoint, iOS native shell (PWA covers iOS).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 mobile menu items (WHSRFMenuItemTable) expose ~60 option fields per item; a flow change (e.g., add a lot confirmation step) needs a consultant, a config move across environments and a device re-login; D365 "mobile device steps configuration" cannot add a new step, only hide/rename	Flows are hard-coded X++ step classes parameterized by options	Flow designer with versioned step graphs, simulate, publish; devices pick up the version in ≤ 60 s	Flow change lead time: 1–3 days → < 30 min; 0 app releases per flow change
Both D365 mobile app and EWM RF (/SCWM/RFUI, ITSmobile) stop dead without Wi-Fi; workers in freezers, yards and trailers lose scans	Request/response screens with no client state	Offline queue with ordered, idempotent replay and a supervisor conflict queue	Lost transactions on disconnect: every one → 0; worker wait on reconnect: re-key → 0 keystrokes
GS1-128 labels must be scanned AI by AI in older D365 and ECC RF screens (one scan for GTIN, one for lot, one for expiry); mis-typed expiry dates reach the batch master	No composite-barcode parsing in the RF screen	GS1 parser (AI 00/01/10/17/21/310n…) fills all expected fields from one scan with check-digit verification	Scans per inbound case: 3–4 → 1; expiry/lot keying errors: ~1% of receipts → 0 (no keying)
Label printing via D365 document routing (WHSDocumentRoutingLayout) depends on the Document Routing Agent Windows service; failures are invisible on the device	Printing routed through a client-side agent with no status feedback	In-cluster print service with job status, retries and device-visible failure	Print failure detection: minutes to hours (someone notices) → < 2 s on device
EWM RF screen layouts (/SCWM/TPRES) and ITSmobile templates need ABAP/HTML work for scanner model differences	Screen rendering coupled to device generation	Responsive PWA from 320 px, DataWedge/Honeywell intent bridge, keyboard-wedge fallback	New scanner model onboarding: weeks of template work → same day (profile only)

6.32.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Mobile menu	WHSMobileAppMenu, WHSRFMenuTable, WHSRFMenuLine	/SCWM/TMENU, /SCWM/T_RF_MENU (RF menu manager /SCWM/RFUI)	wms-mobile.MobileMenu	full
Menu items / flows	WHSRFMenuItemTable (WHSRFMenuItem), work creation vs existing work modes	Logical transactions /SCWM/TLOGTR, steps /SCWM/T_RF_STEP	MobileFlow (step graph)	improved — visual designer, versioned
Detours / field config	Mobile device steps configuration (D365 10.0.x), WHSRFDetour	Presentation profile /SCWM/TPRES, personalization	FlowStep config, detours	full
Scanner integration	DataWedge via app; GS1 parsing (D365 10.0.21+)	RF barcode profiles, GS1 parsing in /SCWM/RFUI	GS1 parser service, DataWedge/Honeywell intents	full
Work user / session	WHSWorkUser, WHSWorkUserSessionState	Resource /SCWM/RSRC, /SCWM/TRSRC queue	DeviceSession, WorkUser (Keycloak)	full
Offline	n/a (online only)	n/a (online only)	OfflineTransaction queue	improved
Label printing	Document routing layouts (WHSDocumentRoutingLayout) to ZPL printers	Smart Forms / Adobe Forms via /SCWM/PRINT condition technique	PrintJob, ZPL templates	full
Voice	ISV	SAP EWM voice (ISV)	VoiceAdapter interface	partial — adapter only
RFID	n/a (ISV)	SAP AII / EWM RFID	EPC read endpoint	partial
Packing station	WHSPackingStation, container packing	Packing work center /SCWM/PACK	Pack flow + ShipmentContainer (wave 28)	full

6.32.3 Functional requirements

- **FR-32.1 Mobile app shell** — Must. The system shall provide a PWA and an Android native shell (WebView + scanner intent bridge) that run the same flow runtime, with login via Keycloak device flow and PIN re-auth after 8 h.
  - Notes: supports screen widths from 320 px; large-target mode; app updates are flow metadata, not binaries; shell release only for scanner bridge changes.
- **FR-32.2 Flow designer** — Must. The system shall let an administrator compose flows from step types (`scan`, `input`, `select`, `confirm`, `display`, `branch`, `call`), with validations, scan pattern expectations and error handling, versioned and published per warehouse.
  - Notes: a step declares expected scan classes (LP, GTIN, LOCATION, ANY); publishing validates the graph (no unreachable steps, every branch terminates); devices fetch new versions on next menu load (≤ 60 s).
- **FR-32.3 GS1 parsing** — Must. The system shall parse GS1-128 and GS1 DataMatrix element strings into application identifiers per BR-32.1, and route values to the expected step fields.
  - Notes: symbology identifiers `]C1`, `]d2`, `]Q3` accepted; FNC1 as GS (0x1D); passes the GS1 AI test suite in CI.
- **FR-32.4 Core flows** — Must. The system shall ship flows: Receive (PO/ASN/transfer), Put-away (system-directed and user-directed), Pick (work-directed, cluster to 8 orders, zone), Pack, Move (LP/qty), Count (planned/spot), Inquiry (item/location/LP).
  - Notes: all flows call the wave-31 work APIs; cluster picking assigns up to 8 works to positions on a cart with per-position LP.
- **FR-32.5 Offline queue** — Must. The system shall queue confirmations locally when offline (max 500 transactions or 24 h), replay them in order on reconnect with per-transaction idempotency keys, and surface conflicts to the user.
  - Notes: queue persisted in IndexedDB encrypted with a device key; replay uses `Idempotency-Key = device_id:seq`; conflict policy per BR-32.4.
- **FR-32.6 Device and session management** — Must. The system shall register devices, show active sessions, force sign-out, and bind a device to warehouses and menus.
  - Notes: device heartbeat every 30 s; session idle timeout 30 min aligned to the work lock.
- **FR-32.7 Label printing** — Must. The system shall render ZPL from templates (LP, location, shipping, item labels) and print via a print service supporting IPP/raw TCP 9100 printers and printer selection by step/user/location.
  - Notes: templates are Handlebars-style with GS1-128 (`^BC`) and DataMatrix (`^BX`) helpers; print service retries 3× and reports status.
- **FR-32.8 Exceptions on device** — Must. The system shall let users short pick, skip, report damaged, and request supervisor override by PIN on device.
  - Notes: override reasons configurable; all exceptions logged with user, device, location, timestamp.
- **FR-32.9 Voice and RFID integration points** — Could. The system shall expose a `VoiceAdapter` interface (prompt/response per step) and an EPC ingest endpoint mapping EPC SGTIN/SSCC to GTIN/LP.
  - Notes: EPC URI parsing for `urn:epc:id:sgtin` and `urn:epc:id:sscc`; reads deduplicated in a 2 s window.
- **FR-32.10 Scanner hardware integration** — Must. The system shall receive scans from Zebra DataWedge intents and Honeywell Data Collection intents in the native shell and from keyboard wedge in the PWA.
  - Notes: a scan is accepted only on the focused scan step; symbology prefix enabled in DataWedge profile provisioned by the shell.
- **FR-32.11 Telemetry** — Should. The system shall record per-step timing and error counts per device and user for the KPI layer.
  - Notes: OpenTelemetry spans per step; retention 90 days.

6.32.4 Data model

MobileFlow (versioned)

Field	Type	Required	Notes
flow_id	uuid	yes	PK
code	text	yes	unique per tenant (e.g., `PICK_CLUSTER`)
version	int	yes	increments on publish
status	enum	yes	DRAFT, PUBLISHED, RETIRED
steps	jsonb	yes	array of {id, type, prompt, expected_scan[], validations[], next{}}
entry_step_id	text	yes	
warehouses	uuid[]	yes	scope
published_at, published_by	timestamptz/text	no	

DeviceSession

Field	Type	Required	Notes
session_id	uuid	yes	PK
device_id	text	yes	FK Device (serial or generated)
user_id	text	yes	Keycloak subject
warehouse_id	uuid	yes	
menu_id	uuid	yes	
current_work_id	uuid	no	FK wms.Work
last_seen_at	timestamptz	yes	heartbeat
state	jsonb	yes	step pointer and field values

OfflineTransaction

Field	Type	Required	Notes
idempotency_key	text	yes	PK `device_id:seq`

Field	Type	Required	Notes
device_id, session_id	text/uuid	yes	
seq	bigint	yes	monotonic per device
command	text	yes	e.g., `ConfirmWorkLine`
payload	jsonb	yes	
captured_at	timestamptz	yes	device clock + server skew
replay_status	enum	yes	PENDING, APPLIED, CONFLICT, REJECTED
conflict_reason	text	no	

PrintJob: job_id, template_id, printer_id, zpl text, status (QUEUED/SENT/PRINTED/FAILED), attempts, created_by.

Invariants: a device has at most one active session; `seq` is strictly increasing per device and gaps are rejected; a PUBLISHED flow version is immutable. Indexing: `offline_transaction(device_id, seq)`; `device_session(user_id) WHERE state_active`.

6.32.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/mobile/v1/sessions	Start session (device, warehouse, menu)	key = device_id	WMS_WORK_EXECUTE
GET	/mobile/v1/menus/{id}	Menu with published flow versions	—	WMS_WORK_EXECUTE
POST	/mobile/v1/flows/{code}/steps/{id}:submit	Submit step input (scan or value)	key	WMS_WORK_EXECUTE
POST	/mobile/v1/scan:parse	Parse a barcode into AI map	—	WMS_WORK_EXECUTE
POST	/mobile/v1/offline:replay	Replay a batch of queued transactions	per-item keys	WMS_WORK_EXECUTE
POST	/mobile/v1/flows	Create/update draft flow	key	WMS_MOBILE_ADMIN
POST	/mobile/v1/flows/{id}:publish	Validate and publish	key	WMS_MOBILE_ADMIN
POST	/print/v1/jobs	Render template and print	key	WMS_PRINT
POST	/rfid/v1/reads	Ingest EPC reads	key = reader+epc+ts	WMS_RFID_INGEST

Event	Trigger	Payload fields	Consumers
dev.passage.mobile.flow.published.v1	publish	flow_id, code, version, warehouses	devices (push via SSE), analytics
dev.passage.mobile.session.started.v1 / ended.v1	session lifecycle	session_id, device_id, user_id, warehouse	analytics
dev.passage.mobile.offline.conflict.v1	replay conflict	idempotency_key, command, reason	supervisor queue
dev.passage.print.job.completed.v1	printer ack	job_id, printer, status	wms

6.32.6 User experience

Screen	Primary actions	Key fields	Duty
Device login / menu	Sign in, pick warehouse/menu	user, warehouse, menu items	WMS_WORK_EXECUTE
Flow runner (device)	Scan, enter, confirm, exception	prompt, expected scan class, context (work, location, LP)	WMS_WORK_EXECUTE
Flow designer (web)	Build graph, validate, simulate, publish	steps, branches, validations, versions	WMS_MOBILE_ADMIN
Device management	Register, assign, force sign-out	device id, model, last seen, session	WMS_MOBILE_ADMIN
Offline conflicts	Resolve, re-apply, reject	key, command, reason	WMS_WORK_SUPERVISE
Print management	Printers, templates, job log	printer, template, status	WMS_PRINT

Interaction rules: device UI uses one input per screen; Enter/scan advances; hardware back returns one step; audio and haptic feedback on success/error; font ≥ 16 px; no mouse-only actions; designer simulation runs a flow in a browser with a virtual scanner.

6.32.7 Business rules and edge cases

- **BR-32.1 GS1 AI parsing rules.** Input begins with an optional symbology identifier (`]C1` GS1-128, `]d2` DataMatrix, `]Q3` QR), then element strings. Parse left to right: read the AI (2–4 digits per the GS1 AI table), then its data per the table; fixed-length AIs have no separator, variable-length AIs end at FNC1 (GS, 0x1D) or end of string. Supported in this wave: **AI 00** SSCC, fixed 18 numeric, check digit (mod-10, weights 3/1) verified; **AI 01** GTIN, fixed 14 numeric, check digit verified, GTIN-12/13 zero-padded on lookup; **AI 10** batch/lot, variable ≤ 20 characters (GS1 AI encodable set 82); **AI 11** production date and **AI 17** expiry, fixed 6 numeric `YYMMDD` where `DD = 00` means last day of month, century by GS1 rule (year within −49/+50 of current year); **AI 21** serial, variable ≤ 20; **AI 310n** net weight kg, fixed 6 numeric with `n` implied decimal places (`3102000125` → 1.25 kg); **AI 37** count, variable ≤ 8 numeric; **AI 400** PO number, variable ≤ 30; **AI 410/412** GLN, fixed 13; **AI 91–99** company-internal, variable ≤ 90. Unknown AI → parse error `GS1_UNKNOWN_AI` with position. Example: `]C1010950123456789310123ABC` + GS + `17261231` + `21SN0007` → {01: 09501234567893, 10: 123ABC, 17: 2026-12-31, 21: SN0007}. A barcode without symbology identifier is parsed as GS1 only if it starts with a valid AI and the step expects a GS1 class; otherwise it is a plain value.
- **BR-32.2 Scan routing.** A `scan` step expecting `LP` accepts AI 00 or a plain value matching the LP format; expecting `GTIN` accepts AI 01 or an item barcode (wave 8 item barcodes); a mismatch shows "Expected location, scanned LP" and does not advance.
- **BR-32.3 Offline ordering.** Replay applies transactions in `seq` order; the first failure stops replay for that device and marks the rest PENDING until resolved, because later steps may depend on it.
- **BR-32.4 Conflict policy.** If a replayed `ConfirmWorkLine` targets a line that was cancelled or completed by another user, the transaction is CONFLICT; a supervisor may "apply as movement" (inventory moves as scanned, order link dropped) or reject. Timestamps use the captured device time corrected by the measured server skew at the last heartbeat.
- **BR-32.5 Cluster picking** assigns at most 8 works; a cart position is bound to one order's LP; scanning the wrong position is rejected; a short pick on one position does not release the others.

- **BR-32.6 Session lock.** A device session holds the work lock; heartbeat loss > 30 min releases the lock and sets the session to STALE; resuming re-locks if the work is still OPEN.
- **BR-32.7 PIN override** authorizes exactly one action within 60 s and is logged with the supervisor's subject.
- **BR-32.8 Print idempotency.** A print job with the same key within 10 min is not re-rendered; "print again" creates a new job with `reprint_of`.
- **BR-32.9 Flow publish** requires the simulation of each core path to succeed (designer runs a smoke script attached to the flow); devices mid-flow on the previous version finish on it.
- **BR-32.10 Expiry date** `DD=00` resolves to the last calendar day of the month; `YYMMDD` with an invalid month is a parse error.

6.32.8 Acceptance criteria

- **AC-32.1** (FR-32.3, BR-32.1) Given the GS1 AI test suite vectors (fixed and variable, with and without symbology identifiers), when parsed, then 100% match expected AI maps and check-digit failures are reported.
- **AC-32.2** (FR-32.4) Given the emulated-device Playwright suite, when a receive → put-away → pick → pack → ship flow is run with scanned GS1-128 labels from the ASN fixture, then the load ships and the inventory ledger reflects each move.
- **AC-32.3** (FR-32.5, BR-32.3) Given a device offline with 200 queued confirmations, when it reconnects, then all 200 are APPLIED exactly once, in order, and replaying the same batch again changes nothing.
- **AC-32.4** (FR-32.2) Given a published flow change adding a lot-confirmation step, when a device loads the menu, then the new version runs within 60 s with no app update.
- **AC-32.5** (FR-32.7) Given an LP label template, when a receive step completes, then a ZPL job with `^BC` SSCC is sent to the step's printer and status PRINTED is recorded.
- **AC-32.6** (BR-32.4) Given a queued confirmation for a line cancelled meanwhile, when replayed, then it is CONFLICT and appears in the supervisor queue.
- **AC-32.7** (FR-32.10) Given a Zebra DataWedge intent with `]C1` prefix, when received on a non-scan step, then it is ignored and logged.
- **AC-32.8** (FR-32.6, BR-32.6) Given a session with no heartbeat for 31 min, when another user requests work, then the stale session's work is assignable.

6.32.9 Test plan and fixtures

Fixtures: `gs1-ai-vectors.json` (GS1 test suite plus 60 Passage vectors for edge cases), `labels-asn-3-pallet.zpl` rendered to PNG for the emulator's camera-free scan injection, `offline-200.json`, `flows-core-v1` (seven flows), device profiles `zebra-tc52`, `honeywell-ct45`.

Test	Type	Target	Pass criterion
GS1 parser	unit	parser	100% vectors
Flow graph validation	unit	designer	unreachable/non-terminating graphs rejected
Offline replay	property	queue	exactly-once under random disconnects
Device E2E	E2E (Playwright, Android emulator)	7 flows	inbound-to-outbound passes on both device profiles
Replay throughput	performance	1,000 devices × 50 tx	all applied < 5 min
ZPL rendering	snapshot	templates	byte-identical to golden ZPL
Scan latency	performance	step submit	p95 < 300 ms

Reconciliation: Σ mobile confirmations per day = Σ work line confirmations in `wms`; print jobs PRINTED = labels confirmed on device.

6.32.10 Deployment and operations

Helm `passage-wms-mobile` (BFF, SSE push, 2–8 replicas), `passage-print` (print service with printer registry; one per site network, outbound TCP 9100/IPP), PWA assets served by the shell CDN path; Android shell signed APK distributed via MDM (SOTI/Workspace ONE) or sideload. Config: `MOBILE_OFFLINE_MAX_TX=500`, `MOBILE_SESSION_IDLE_MIN=30`, `PRINT_RETRY=3`. Flags: `mobile.offline`, `mobile.voice_adapter`. SLOs: step submit p95 < 300 ms; menu load < 1 s; print job dispatched < 2 s; offline replay 200 tx < 30 s. Alerts: offline conflicts > 10/h per warehouse; print failures > 5/15 min per printer; stale sessions > 20. Runbook: printer unreachable (check print service pod on site, re-queue jobs); device clock skew > 5 min (block replay, instruct re-sync). Rollback: flow versions are retained — republish the previous version; BFF rollback via Argo CD.

6.32.11 Migration from AX / SAP

Source	Passage entity	Transformation
WHSMobileAppMenu, WHSRFMenuTable, WHSRFMenuLine	MobileMenu	hierarchy preserved
WHSRFMenuItemTable (mode, work class, directed/user-directed, detours)	MobileFlow draft	generated from the menu item's options into the equivalent core flow with configured branches; requires manual review
WHSWorkUser	WorkUser attributes on Keycloak user	user id mapping from wave 2
WHSDocumentRoutingLayout, printers	Print templates, printers	ZPL layouts imported as-is where ZPL; others re-authored
/SCWM/TMENU, /SCWM/T_RF_MENU, /SCWM/TLOGTR	MobileMenu, MobileFlow draft	logical transaction → core flow mapping table
/SCWM/RSRC, /SCWM/TRSRC	WorkUser, device bindings	resource type → work class

Not migrated: RF screen layouts (/SCWM/TPRES) and D365 step field layouts — Passage flows are redesigned; mobile session state; print spool history.

6.32.12 Dependencies, risks, open questions

Depends on: wave 31 (work APIs, LPs), wave 25 (shop-floor PWA offline pattern and service-worker library), wave 8 (item barcodes, GS1 prefix), wave 2 (Keycloak device flow), wave 28 (containers/packing).

Risk	Likelihood	Impact	Mitigation
Offline replay creates subtle inventory errors	Medium	High	Strict ordering, conflict queue, property tests; offline off by default for count flows
Android WebView versions on older scanners	Medium	Medium	Minimum WebView 100; PWA fallback

Open questions: (1) Native iOS shell — proposed: no, PWA only; decide by wave 33. (2) Voice vendor for v1.x reference adapter (Honeywell Voice vs Lydia) — proposed: defer; decide by wave 35.

6.33 Wave 33: Quality Management & Nonconformance

6.33.1 Scope and business context

Wave 33 replaces AX quality management (test groups, quality associations, quality orders, nonconformances, corrections) and SAP QM (inspection plans, inspection lots, results recording, usage decisions, quality notifications, vendor evaluation). After this wave, quality orders are generated by association rules at purchase receipt, production report-as-finished and before shipment; samples are sized by AQL plans; results are recorded (variable and attribute); a usage decision releases or rejects inventory that was blocked from the moment the order was created; nonconformances drive corrections/CAPA through a Temporal 8D workflow; certificates of analysis print on shipment; supplier scorecards and SPC charts are computed from results; instruments have calibration schedules. Out of scope: stability studies, audit management (ISO 19011), FMEA, and the SAP QM-PM calibration order integration (Passage tracks calibration due dates and blocks results from overdue instruments).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX quality order blocking (InventQualityOrderTable → InventBlockingTable) blocks only the quantity registered on the receipt; production and sales paths can still pick inventory registered under a different dimension set, and "Inventory blocking" is bypassed by movement journals	Blocking is a reservation-like record checked by some, not all, transaction types	Hold lives in the inventory ledger reservation path and is checked by every transaction type; property test enumerates all	Movements of held stock: occasional (audit findings) → 0 across 10,000 randomized attempts
SAP usage decision (QA11) is a separate transaction from results recording (QE01), each 8–12 clicks with stock posting on a third tab; partial accept/reject needs manual quantity split and MIGO 321/322 follow-up	Inspection lot, results and stock posting are separate objects with separate screens	Results and decision on one order; quantity split posts both ledger legs automatically	Clicks per decision: ~25 across QE01/QA11/MIGO → ≤ 5; follow-up postings keyed manually: 2 → 0
AQL sampling in AX is fixed qty or % (InventTestItemSampling); Z1.4 tables are kept in spreadsheets and sample sizes keyed by hand; SAP QDP1 sampling procedures require master data per AQL/level combination	No standard sampling tables in the product	Z1.4 Tables I and II-A/B/C as versioned reference data with switching rules	Sample-size keying errors: ~2% of lots (spreadsheet) → 0; sampling setup per item: 15 min → 0 (plan reuse)
SAP quality notification 8D (QM01/QM02 tasks) has no timers; overdue corrective actions are found by report ZQM or not at all; effectiveness checks are forgotten	Notification tasks are static records without workflow timers	Temporal 8D workflow with due-date timers, escalations and an automatic 90-day effectiveness-check task	Overdue CAPA actions discovered: weeks later via report → escalated within 1 business day; effectiveness checks executed: ~50% → 100%
CoA generation (AX CoA report / SAP QC21) prints whatever results exist; amended results after shipment produce inconsistent certificates	Certificate is a report over mutable results, not a versioned document	CoA generated from the accepting decision, revisioned with `supersedes`	Inconsistent certificates reported by customers: tolerated → 0 (immutable revision chain)

6.33.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Tests and groups	InventTestTable, InventTestGroup, InventTestGroupTest (lines), InventTestVariable/Outcome	Master inspection characteristics QPMK (QS21), inspection plans QP01 (PLKO/PLPO/PLMK), catalogs QPCT	quality.Test, TestGroup, Characteristic catalog	full
Sampling	InventTestItemSampling (fixed qty/%)	Sampling procedures QDP1 (QDPS), sampling schemes QDS1, dynamic modification	SamplingPlan (fixed, %, AQL Z1.4)	improved — AQL built in
Triggers	InventTestAssociationTable (quality associations)	QMAT inspection setup per material/plant; lot origins 01/04/10/89	QualityAssociation	full
Quality orders	InventQualityOrderTable, InventQualityOrderLine, InventQualityOrderLineResults; form InventQualityOrderTable	Inspection lot QALS (QA01/QA02), results QAMR/QASR (QE01), QA11 usage decision (QAVE)	QualityOrder, QualityResult, UsageDecision	full
Inventory blocking	InventBlockingTable, InventQualityOrderTable blocking	Stock type Q (quality inspection), QM stock posting at UD	Quant `quality_status`; ledger status QUALITY_HOLD	full
Nonconformance	InventNonConformanceTable, problem/diagnostic types, InventNonConformanceCorrection, InventNonConformanceOperation	Quality notifications QMEL/QMFE/QMMA/QMSM (QM01), 8D report	Nonconformance, Correction (CAPA)	full
Certificates	Certificate of analysis (InventTestCertificateOfAnalysis report)	QC21/QC22 certificate profiles (QCVK)	CoA template + generation	full
Vendor evaluation	Vendor quality on nonconformance (reports)	ME61–ME65 vendor evaluation (ELBK/ELBM/ELBP)	SupplierScorecard	full
Calibration	InventTestInstrument (instrument master only)	QM-PM calibration (inspection lot type 14, equipment)	Instrument with calibration schedule	partial
SPC	n/a	QM control charts (QGC1)	SPC (X-bar/R, p-chart)	full
Cost of quality	n/a	QM-CO appraisal costs	CostOfQuality projection	improved

6.33.3 Functional requirements

- **FR-33.1 Tests, instruments and test groups** — Must. The system shall define tests (variable with UoM, target, min/max; attribute with outcome catalog), instruments, and test groups with sequence, sampling, and acceptance per test.

    - Notes: variable results compare to tolerances with 6-decimal precision; test group versions are effective-dated; a test marked `destructive` reduces sample qty on completion.
- **FR-33.2 Sampling plans** — Must. The system shall size samples by fixed quantity, percentage, or ANSI/ASQ Z1.4 single sampling (normal, tightened, reduced) at general inspection level II by default.
    - Notes: table excerpt and switching rules in BR-33.1/BR-33.2; sample qty is capped at lot qty; percentage rounding up.
- **FR-33.3 Quality associations** — Must. The system shall generate quality orders when a trigger matches: purchase receipt (registration or product receipt), production RAF (before/after put-away), sales order (at pick/before packing slip), transfer receipt, quarantine, and manual.
    - Notes: association scope = item / item group / all × vendor/customer / group / all × site; most specific wins (BR-33.3); one order per source line per event.
- **FR-33.4 Quality orders and results** — Must. The system shall record results per test per sample, compute pass/fail per test and order, and support retest and partial sampling.
    - Notes: results recorded with instrument id and user; a result from an instrument overdue for calibration is rejected; order status per BR-33.4.
- **FR-33.5 Inventory blocking** — Must. The system shall block the inspected quantity from pick, ship, consume and transfer-out when the quality order is created and release it only on an accepting usage decision; rejection moves qty to a quarantine location or blocked status.
    - Notes: enforcement lives in the inventory ledger reservation path (wave 16), not only in `wms`; blocked qty is visible as "quality hold" in ATP (wave 27) as unavailable.
- **FR-33.6 Usage decision** — Must. The system shall record a usage decision (accept, reject, accept-with-deviation, rework, return-to-vendor, scrap) with partial quantities, and post consequential transactions.
    - Notes: reject → nonconformance auto-created; return-to-vendor → wave-18 return order draft; scrap → wave-16 scrap journal with reason.
- **FR-33.7 Nonconformance** — Must. The system shall create nonconformances (sources: customer, vendor, internal, production, service request) with problem type, severity, quarantine action, related quality order and quantity.
    - Notes: statuses NEW → APPROVED → IN_CORRECTION → CLOSED, or REFUSED; attachments and photos from the mobile app.
- **FR-33.8 Correction / CAPA workflow** — Must. The system shall run an 8D correction workflow (D1 team … D8 close) on Temporal with root cause, containment, corrective action, due dates, approvals and an effectiveness check after a configurable period (default 90 days).
    - Notes: overdue actions escalate per wave-4 workflow escalation; effectiveness check task auto-created by a Temporal timer.
- **FR-33.9 Certificates of analysis** — Must. The system shall generate CoA documents per shipment/batch from accepted results using a template per customer/item, with PDF output and e-mail/portal delivery.
    - Notes: CoA shows result, specification, method, instrument; signature block from the usage-decision user.
- **FR-33.10 Supplier quality scorecard** — Should. The system shall compute per vendor: lots inspected, lot acceptance rate, PPM defective, NCR count by severity, on-time delivery (from wave 18), and a weighted score 0–100.
    - Notes: weights configurable; monthly materialization; exposed to the wave-20 supplier portal.
- **FR-33.11 SPC charts** — Should. The system shall compute X-bar/R (variable) and p/np (attribute) charts with Western Electric rules 1–4 flagged.
    - Notes: subgroup size from sampling plan; control limits from the last 25 subgroups.
- **FR-33.12 Cost of quality** — Could. The system shall tag inspection labor (wave 24 time), scrap, rework and returns to prevention/appraisal/internal/external failure categories for reporting.
    - Notes: amounts sourced from journal entries by dimension `coq_category`.

6.33.4 Data model

QualityOrder

Field	Type	Required	Notes
quality_order_id	uuid	yes	PK; number sequence QO-*
trigger	enum	yes	PURCHASE_RECEIPT, PRODUCTION_RAF, SALES_PICK, TRANSFER_RECEIPT, QUARANTINE, MANUAL
source_ref	jsonb	yes	{type, id, line} polymorphic
item_id, dims	uuid/jsonb	yes	batch/serial dims
site_id, warehouse_id	uuid	yes	
lot_qty, sample_qty, uom	numeric(18,6)/text	yes	
test_group_id, test_group_version	uuid/int	yes	
sampling_plan_id, inspection_severity	uuid/enum	yes	NORMAL/TIGHTENED/REDUCED
status	enum	yes	BR-33.4
block_ref	uuid	no	inventory ledger hold id
created_at, closed_at	timestamptz	—	

QualityResult

Field	Type	Required	Notes
result_id	uuid	yes	PK
quality_order_id, test_id, sample_no	uuid/uuid/int	yes	unique together with `retest_no`
value_numeric	numeric(18,6)	no	variable tests
outcome_id	uuid	no	attribute tests
passed	bool	yes	computed
instrument_id	uuid	no	must be in calibration
recorded_by, recorded_at	text/timestamptz	yes	
retest_no	int	yes	default 0

Nonconformance

Field	Type	Required	Notes
ncr_id	uuid	yes	PK; NCR-*
source	enum	yes	CUSTOMER, VENDOR, INTERNAL, PRODUCTION, SERVICE
item_id, dims, qty	—	yes	
problem_type_id, severity	uuid/enum	yes	severity MINOR/MAJOR/CRITICAL
quality_order_id, vendor_id, customer_id, production_order_id	uuid	no	
status	enum	yes	NEW, APPROVED, REFUSED, IN_CORRECTION, CLOSED
quarantine_order_id	uuid	no	
correction_workflow_id	text	no	Temporal workflow id

SamplingPlan: plan_id, method (FIXED/PERCENT/AQL), fixed_qty, percent, aql (numeric 3,2), inspection_level (S1–S4, I–III), severity default, switching_enabled. **Instrument**: instrument_id, calibration_interval_days, last_calibrated_at, due_at, status.

Invariants: a quant with an open QualityOrder referencing it has `quality_status ∈ {PENDING, BLOCKED}`; `Σ accept_qty + reject_qty + deviation_qty` of a usage decision = lot_qty; a QualityOrder is CLOSED only with a usage decision; a result cannot reference an instrument whose `due_at < recorded_at`. Indexing: `quality_order(status, site_id)`, `quality_result(quality_order_id, test_id, sample_no, retest_no)` unique, `nonconformance(vendor_id, created_at)`.

6.33.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/quality/v1/test-groups	Create/version test group	key	QM_SETUP_MAINTAIN
POST	/quality/v1/associations	Create association	key	QM_SETUP_MAINTAIN
POST	/quality/v1/sampling-plans:size	Compute sample size for lot qty/plan	—	QM_VIEW
POST	/quality/v1/quality-orders	Create manual order	key	QM_ORDER_MAINTAIN
POST	/quality/v1/quality-orders/{id}/results	Record results (batch)	key per batch	QM_RESULTS_RECORD
POST	/quality/v1/quality-orders/{id}:usage-decision	Post decision with quantities	key	QM_USAGE_DECISION
POST	/quality/v1/nonconformances	Create NCR	key	QM_NCR_MAINTAIN
POST	/quality/v1/nonconformances/{id}/corrections	Start CAPA workflow	key	QM_CAPA_MAINTAIN
POST	/quality/v1/coa:generate	Generate CoA for shipment/batch	key = shipment+batch	QM_COA
GET	/quality/v1/scorecards/vendors/{id}?period=	Scorecard	—	QM_VIEW

Commands: `CreateQualityOrderFromTrigger` validates association match and that the source line is in the required state; `PostUsageDecision` validates all required tests have results (or `allow_incomplete` with deviation reason) and quantity split sums to lot qty.

Event	Trigger	Payload fields	Consumers
dev.passage.quality.order.created.v1	order created	quality_order_id, trigger, source_ref, item, dims, lot_qty, sample_qty	inventory (hold), wms, analytics
dev.passage.quality.result.recorded.v1	result batch	quality_order_id, results[]	SPC projection
dev.passage.quality.usagedecision.posted.v1	decision	quality_order_id, decision, accept_qty, reject_qty, deviation_qty, user	inventory (release/hold), wms, purchasing, production
dev.passage.quality.nonconformance.created.v1	NCR	ncr_id, source, severity, item, vendor_id	vendor scorecard, workflow
dev.passage.quality.capa.closed.v1	8D closed	ncr_id, effectiveness_result	analytics

6.33.6 User experience

Screen	Primary actions	Key fields	Duty
Quality setup	Tests, groups, sampling plans, associations, instruments	test, type, tolerance, AQL, trigger scope	QM_SETUP_MAINTAIN
Quality order workbench	List by status/site, open, record, decide	order, item, trigger, sample, due, status	QM_ORDER_MAINTAIN
Results recording (web + mobile step)	Enter values per sample/test, attach	test, sample, value, outcome, instrument	QM_RESULTS_RECORD
Usage decision	Split quantities, choose decision, reason	accept/reject/deviation qty, disposition	QM_USAGE_DECISION
Nonconformance / 8D	Create, approve, run 8D, close	NCR, severity, D1–D8 tasks, due dates	QM_NCR_MAINTAIN
Supplier scorecard	Period filter, drill to NCRs	vendor, score, PPM, acceptance rate	QM_VIEW
SPC dashboard	Chart per item/test, rule violations	subgroups, limits, rule hits	QM_VIEW

Interaction rules: results grid is keyboard-first (Tab/Enter down a sample column); out-of-tolerance values highlight immediately on blur; usage decision disabled until required results exist; CoA preview before release.

6.33.7 Business rules and edge cases

- **BR-33.1 AQL table (ANSI/ASQ Z1.4, general inspection level II, single sampling, normal inspection) — excerpt used by the sizing function.**

Lot size	Code	Sample n	AQL 1.0 Ac/Re	AQL 2.5 Ac/Re	AQL 4.0 Ac/Re
2–8	A	2	use E (13): 0/1	use D (8): 0/1	use C (5): 0/1
9–15	B	3	use E: 0/1	use D: 0/1	use C: 0/1
16–25	C	5	use E: 0/1	use D: 0/1	0/1

Lot size	Code	Sample n	AQL 1.0 Ac/Re	AQL 2.5 Ac/Re	AQL 4.0 Ac/Re
26–50	D	8	use E: 0/1	0/1	use E (13): 1/2
51–90	E	13	0/1	use F (20): 1/2	1/2
91–150	F	20	use G (32): 1/2	1/2	2/3
151–280	G	32	1/2	2/3	3/4
281–500	H	50	2/3	3/4	5/6
501–1,200	J	80	3/4	5/6	7/8
1,201–3,200	K	125	5/6	7/8	10/11
3,201–10,000	L	200	7/8	10/11	14/15
10,001–35,000	M	315	10/11	14/15	21/22
35,001–150,000	N	500	14/15	21/22	21/22
150,001–500,000	P	800	21/22	21/22	21/22
500,001+	Q	1,250	21/22	21/22	21/22

Arrow cells ("use X") follow the Z1.4 rule: when the table points to another sample size, use that sample size and its Ac/Re; if it exceeds lot size, inspect 100%. The full Table I (code letters for levels S-1…S-4, I–III) and Tables II-A/B/C (normal/tightened/reduced) are stored as versioned reference data `z14_table` and tested cell by cell.

- **BR-33.2 Switching rules.** Normal → tightened when 2 of 5 consecutive lots are rejected; tightened → normal after 5 consecutive accepted; normal → reduced after 10 consecutive accepted (and switching enabled); reduced → normal on a rejection. Severity is tracked per (item, vendor, test group).
- **BR-33.3 Association specificity.** Score = 4·(item match) + 2·(party match) + 1·(site match); highest score wins; ties resolved by lowest `sequence`; `item group` counts as item match with weight 3. Exactly one order per source line per trigger event; a re-triggered event (e.g., second receipt line) creates a separate order.
- **BR-33.4 Quality order state machine.** `CREATED → OPEN (sample drawn) → RESULTS_IN_PROGRESS → PENDING_DECISION → CLOSED`; `CREATED|OPEN → CANCELLED` (only if source reversed; hold released); `PENDING_DECISION → RESULTS_IN_PROGRESS` on retest request; CLOSED is final; a decision can be *amended* only by a new NCR referencing the order.
- **BR-33.5 Blocking semantics.** The hold is placed synchronously in the same transaction as source line registration (receipt registration / RAF) through the inventory ledger `PlaceHold` API; if the hold cannot be placed the source transaction fails. Hold qty = lot qty (not sample qty). Destructive test consumption posts a scrap journal for the destroyed sample qty at decision time.
- **BR-33.6 Variable test evaluation** passes if `min ≤ value ≤ max` inclusive after rounding to the test's declared decimals (half-up); attribute passes if the outcome is flagged `pass`. Order passes if every test marked `required` passes on every sample within Ac/Re for AQL plans.
- **BR-33.7 Partial decisions.** Accept 90, reject 10: accepted qty released; rejected qty moved to the quarantine location with `quality_status = REJECTED`; both legs are ledger transactions with the quality order as reference.
- **BR-33.8 Instrument calibration.** `due_at = last_calibrated_at + interval`; results are rejected if `due_at < now` with error `INSTRUMENT_OVERDUE`; a 14-day warning is raised.
- **BR-33.9 CAPA timers.** D3 containment due in 2 business days, D5 corrective action in 30 days, effectiveness check at 90 days after D7 close; business days from the site calendar.
- **BR-33.10 CoA** includes only results from the accepting decision; amended results require a new CoA revision with `supersedes`.

6.33.8 Acceptance criteria

- **AC-33.1** (FR-33.3, BR-33.3) Given the trigger fixture of 60 receipts and 40 RAFs with overlapping associations, when posted, then exactly the expected 73 quality orders exist with the expected test groups.
- **AC-33.2** (FR-33.2, BR-33.1) Given lot sizes {5, 50, 51, 500, 501, 3200, 10001, 600000} at AQL 2.5 normal level II, when sized, then sample sizes are {5, 8, 20, 50, 80, 125, 315, 1250} (lot 51 → code E, whose AQL 2.5 cell points down to F, n = 20) and Ac/Re are {0/1, 0/1, 1/2, 3/4, 5/6, 7/8, 14/15, 21/22}, matching the Z1.4 reference cells.
- **AC-33.3** (FR-33.5) Given 10,000 random attempts to pick, ship, consume or transfer quants with open quality orders, when executed, then every attempt fails with `QUALITY_HOLD` and no inventory transaction is written.
- **AC-33.4** (FR-33.6, BR-33.7) Given a lot of 100 with results, when the decision is accept 90 / reject 10, then 90 are available, 10 sit in quarantine as REJECTED, and an NCR exists for 10.
- **AC-33.5** (FR-33.8, BR-33.9) Given an NCR with a started 8D workflow, when D7 closes and 90 days elapse (Temporal time-skipping test), then an effectiveness-check task is created and the workflow completes on its approval.
- **AC-33.6** (FR-33.4, BR-33.8) Given an instrument overdue since yesterday, when a result references it, then the request is rejected with `INSTRUMENT_OVERDUE`.
- **AC-33.7** (FR-33.9) Given an accepted batch shipped to a customer with a CoA template, when the shipment is confirmed, then a PDF CoA with every required test appears on the shipment and the customer portal.
- **AC-33.8** (BR-33.2) Given 5 consecutive lots of which 2 are rejected, when the 6th order is created, then its severity is TIGHTENED.
- **AC-33.9** (FR-33.10) Given 12 months of decisions and NCRs, when the scorecard is computed, then PPM = rejected units / inspected units × 10^6 and score matches the weighted formula fixture.

6.33.9 Test plan and fixtures

Fixtures: `z14-tables.json` (full Z1.4 Tables I, II-A/B/C transcribed and double-checked), `quality-triggers-100.json`, `results-variable-attribute.json`, `instruments-12.json`, `ncr-8d-sample`, `vendor-scorecard-12m`.

Test	Type	Target	Pass criterion
AQL sizing	unit	every cell of II-A/B/C	equals transcribed table
Association precedence	unit	100 triggers	expected orders

Test	Type	Target	Pass criterion
Blocking property	property	inventory ledger + wms	0 successful movements on held quants
8D workflow	Temporal replay/time-skip	CAPA	completes with effectiveness check
Usage decision quantities	unit	splits	sum invariant
CoA rendering	snapshot	PDF	matches golden
Results throughput	performance	1,000 results batch	< 2 s

Reconciliation: Σ quality-held qty in ledger = Σ lot_qty of open orders; rejected qty in quarantine = Σ reject_qty of decisions – dispositions.

6.33.10 Deployment and operations

Helm `passage-quality` (api, worker, 2–6 replicas), CloudNativePG `quality-db` (3 instances, 50 GB). Config: `QM_DEFAULT_INSPECTION_LEVEL=II`, `QM_EFFECTIVENESS_DAYS=90`, `QM_INSTRUMENT_WARN_DAYS=14`. Flags: `quality.enabled` per site, `quality.spc`. Scheduled: scorecard materialization monthly (1st, 03:00), instrument due scan daily, SPC recompute hourly. SLOs: order creation added to receipt latency < 100 ms p95; results batch p95 < 2 s; decision posting p95 < 500 ms. Alerts: orders PENDING_DECISION > 5 days; held qty vs open orders drift; instrument overdue with results attempted. Runbook: release orphan holds (order cancelled but hold remains); reprocess association for a trigger event. Rollback: `quality.enabled=false` stops new orders; existing holds remain until decided (never auto-release on rollback).

6.33.11 Migration from AX / SAP

Source	Passage entity	Transformation
InventTestTable, InventTestVariable, InventTestVariableOutcome	Test, outcome catalog	type Integer/Fraction → variable; Option → attribute
InventTestGroup + lines, InventTestItemSampling	TestGroup, SamplingPlan	fixed/percent plans direct; AQL plans created new
InventTestAssociationTable	QualityAssociation	event type mapping (Registration/Product receipt/RAF/Pick)
InventQualityOrderTable/Line/LineResults (open)	QualityOrder, QualityResult	open orders only; holds recreated via ledger API
InventNonConformanceTable, InventNonConformanceCorrection (open)	Nonconformance, Correction	statuses mapped; closed NCRs loaded as read-only history (last 24 months)
QPMK, PLKO/PLPO/PLMK	Test, TestGroup	one group per plan usage; characteristics by `MERKNR`
QDPS/QDS1, QMAT	SamplingPlan, QualityAssociation	SAP sampling procedure type → method; `QMAT` lot origin → trigger
QALS (open, no UD), QAMR/QASR	QualityOrder, QualityResult	lot origin 01→PURCHASE_RECEIPT, 04→PRODUCTION_RAF, 10→SALES_PICK, 89→MANUAL
QMEL/QMFE/QMMA/QMSM (open)	Nonconformance, Correction tasks	notification type Q1/Q2/Q3 → source
ELBK/ELBM/ELBP	SupplierScorecard history	last 12 months of scores only

Not migrated: closed inspection lots with UD (history to lakehouse), SAP control chart history (recomputed), certificate print history.

6.33.12 Dependencies, risks, open questions

Depends on: wave 32 (mobile results step), wave 31 (quarantine locations, work), wave 28 (shipment hooks for sales trigger and CoA), wave 21 (RAF hook), wave 18 (receipt hooks, return orders), wave 16 (ledger hold API — extended in this wave), wave 4 (Temporal).

Risk	Likelihood	Impact	Mitigation
Z1.4 transcription error	Low	High	Two independent transcriptions diffed; spot-check against the ASQ tables
Hold placement adds latency to receipts	Medium	Low	Synchronous but local DB call via ledger API; measured in CI

Open questions: (1) Should sales-pick quality orders block the *order line* or the *quant*? Proposed: quant; decide by wave 33 design review. (2) Cost of quality category as a journal dimension or a tag? Proposed: dimension `coq_category`; decide by wave 37 (cost accounting).

6.34 Wave 34: Transportation Management & Freight

6.34.1 Scope and business context

Wave 28 integrated parcel carriers for labels; wave 34 adds full transportation management replacing D365 TMS and SAP TM: shipments consolidated into loads and routes, rated against carrier rate cards with accessorials and fuel surcharges, rate-shopped, tendered to carriers with timeouts, tracked by EDI 214 or carrier APIs, and reconciled against carrier freight invoices with variance disputes and accruals. Freight and duties are allocated to inventory as landed cost feeding wave-20 costing. VRP route optimization runs on OR-Tools in the Python `tms-optimizer`. Out of scope: own-fleet driver apps and ELD, ocean booking, customs filing (hooks only), multi-leg international routing guides beyond two legs.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 rate route workbench (TMSRateRouteWorkbench) shows a total per carrier with no breakdown; a wrong rate (missing rate base assignment, fuel index not loaded) is found when the carrier invoice arrives weeks later	Rate engines (TMSRateEngineBase subclasses) return totals; no explain	Rating returns an explain trace per charge step; `NO_VALID_RATE_CARD` is explicit	Rate errors detected: at invoice (2–6 weeks) → at rating (< 300 ms); rating disputes caused by setup: ~5% of invoices → < 0.5%

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP TM carrier selection and tendering (freight order subcontracting) require TM configuration plus PI/PO for EDI 204/990; tender timeouts are monitored manually in the transportation cockpit	Tendering is a status field, not a durable process	Temporal `TenderWorkflow` with cascade, timeouts and carrier portal/EDI responses	Planner interventions per 100 tenders: ~30 (watching expiry) → < 3; tender-to-acceptance: hours of chasing → automatic cascade within the timeout
D365 freight reconciliation (TMSFreightBill ↔ TMSCarrierInvoice) is matched by hand per detail line; D365 lacks EDI 210 intake; accrual reversal needs a separate periodic job and often double-counts	Matching is a form, not a rule set; accrual lifecycle disconnected from the match	Rule-based line matching with tolerances, state machine to AP, accrual/reversal/invoice share one journal reference	Reconciliation effort: ~10 min per invoice → < 30 s for in-tolerance invoices (≥ 80% auto); account 2150 ties to zero per bill, 100%
Landed cost allocation (D365 Landed cost module voyages; SAP delivery-cost conditions FRA1/FRB1 on PO lines) rounds per line and leaves cent residuals in a variance account; multi-PO containers need manual apportionment	Per-line rounding without residual assignment; cost types not linked to receipts across POs	Allocation with residual to the highest-value line; voyage spans POs; estimated/actual variance rules by on-hand	Allocation residual: cents per voyage in variance → 0; time to land a 3-PO container: 1–2 h → < 5 min
Route planning in D365 is manual load building; SAP TM VSR optimizer requires PP/DS-style setup and licensing	No general VRP in D365; SAP optimizer is a separate engine with heavy configuration	OR-Tools VRP with explicit constraints, 60 s limit, planner override	Multi-stop planning time: 30–60 min per day of loads → < 1 min; route distance vs manual: –10% or better

6.34.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Carriers / services	TMSCarrier, TMSCarrierService, TMSCarrierServiceGroup	Business partner carrier role, /SCMTMS/ means of transport	tms.Carrier, CarrierService	full
Rate cards	TMSRateMaster, TMSRateBase, TMSRateBaseAssignment, TMSRateEngine classes (TMSRateEngineBase)	Freight agreements, rate tables, calculation sheets (/SCMTMS/TCM, /SCMTMS/TCCS)	RateCard, RateTable, CalcSheet	full
Accessorials / fuel	TMSAccessorialMaster, TMSAccessorialAssignment, TMSFuelIndex, TMSFuelIndexRegion	Charge types, fuel surcharge index tables	Accessorial, FuelIndex	full
Rate shopping / routing	Rate route workbench (TMSRateRouteWorkbench), TMSRoutePlan, TMSRouteGuide, TMSRouteSegment	Carrier selection (/SCMTMS/CAR_SEL), transportation cockpit	RateShop, RouteGuide	full
Loads / freight orders	WHSLoadTable (TMS fields), load building strategies	Freight units, freight orders /SCMTMS/TOR (TOR_ROOT/TOR_ITEM/TOR_STOP), freight bookings	Shipment, Load, Route	full
Tendering	Load tendering (manual/EDI 204, TMSLoadTender status)	Tendering (freight order subcontracting, EDI 204/990 via SAP TM)	Tender workflow (Temporal)	full
Tracking	n/a (ISV, driver check-in)	Event management (EM) / TM execution tracking	TrackingEvent (EDI 214/API)	improved
Freight reconciliation	TMSFreightBill, TMSFreightBillDetail, TMSCarrierInvoice, TMSCarrierInvoiceDetail, freight bill matching	Freight settlement document (/SCMTMS/SFIR), ERS/MIRO	FreightBill, CarrierInvoice, Match	full
Landed cost	Landed cost module (voyage, cost type, allocation)	Delivery costs on PO (FRA1/FRB1 conditions), actual costing	LandedCostVoyage, allocation to inventory	full
Route optimization	n/a (ISV)	PP/DS-based VSR optimizer	VRP (OR-Tools)	improved
Dock scheduling	WHSDockAppointment (wave 31)	Yard	wms.DockAppointment	full (wave 31)

6.34.3 Functional requirements

- **FR-34.1 Shipments and loads** — Must. The system shall create TMS shipments from sales/transfer/purchase (inbound freight) shipments and consolidate them into loads by rule (same ship-from, ship-to region, mode, date window) or manually.
  - Notes: load capacity by weight, volume, pallet positions; load types TL, LTL, PARCEL, INTERMODAL; inbound loads for vendor-collect freight.
- **FR-34.2 Route planning with VRP** — Must. The system shall plan multi-stop routes with OR-Tools under the constraints in BR-34.1 and return feasible routes within 60 s for ≤ 500 stops.
  - Notes: distance/time matrix from OSRM (self-hosted) or stored lanes; first solution by parallel cheapest insertion; guided local search until the time limit.
- **FR-34.3 Rating engine** — Must. The system shall compute charges per BR-34.2 from rate cards (lane-based, zone-based, distance-based, weight-break, per pallet, per container), accessorials and fuel surcharges, with an explain trace.
  - Notes: rate cards effective-dated with currency; minimum charge; rounding to cents half-up; deterministic given the same inputs.
- **FR-34.4 Rate shopping and route guides** — Must. The system shall rate a shipment across eligible carriers/services and rank by cost, transit time or a weighted score; route guides pre-select carriers by lane with fallbacks.
  - Notes: eligibility by service area, mode, hazmat flag, max weight; results cached 10 min per input hash.
- **FR-34.5 Tendering** — Must. The system shall tender loads to carriers (e-mail/portal, EDI 204 via wave 35, API) with accept/decline/timeout, cascading to the next carrier by route guide rank.
  - Notes: Temporal `TenderWorkflow` per load; default timeout 2 h; broadcast tendering option (first accept wins).
- **FR-34.6 Tracking and POD** — Must. The system shall ingest status events (EDI 214 codes AF/X1/X3/D1 etc., carrier APIs, driver check-in) and proof of delivery (signature/photo), and expose expected vs actual per stop.
  - Notes: late-delivery alerts when ETA > window end; POD attaches to the sales shipment for wave-29 invoicing holds if required.
- **FR-34.7 Freight bills and reconciliation** — Must. The system shall create a freight bill (expected charges) at ship confirmation, match carrier invoices (EDI 210 or manual/API) line by line with tolerances, and route variances to dispute.
  - Notes: tolerance default $5 / 2% per charge; match states per BR-34.4; matched invoices create a vendor invoice in wave 19 AP.
- **FR-34.8 Freight accrual** — Must. The system shall post freight accruals (Dr freight expense or inventory-in-transit, Cr accrued freight 2150) at ship confirmation and reverse on invoice match through the universal journal.
  - Notes: journal postings via the wave-11 posting API with idempotency key = freight bill id; customer-billed freight creates a wave-29 invoice charge line.

- **FR-34.9 Landed cost** — Must. The system shall allocate freight, duty, brokerage and insurance to purchase receipt lines by weight, volume, value, quantity or equal split, and post the allocation to inventory cost (wave 20) as a cost adjustment.
  - Notes: voyage groups multiple POs/containers; estimated cost at receipt, actual on invoice with variance posting; allocation residual cents go to the highest-value line.
- **FR-34.10 Carrier portal** — Should. The system shall let carriers accept tenders, update status and upload POD via the wave-20 supplier portal framework.
  - Notes: carrier users are Keycloak external identities scoped to their loads.
- **FR-34.11 Transportation KPIs** — Should. The system shall report cost per shipment/cwt/mile, on-time pickup/delivery, tender acceptance rate, invoice variance rate.
  - Notes: Cube metrics; daily materialization.

6.34.4 Data model

Load (TMS)

Field	Type	Required	Notes
load_id	uuid	yes	PK; shared id with wms.Load
mode, equipment	enum/text	yes	TL/LTL/PARCEL/INTERMODAL; `53FT_DRY`, `REEFER`
ship_from_site_id	uuid	yes	
carrier_id, service_id	uuid	no	set on tender accept
status	enum	yes	PLANNED, RATED, TENDERED, ACCEPTED, DISPATCHED, IN_TRANSIT, DELIVERED, CLOSED, CANCELLED
total_weight_kg, total_volume_m3, pallet_positions	numeric/int	yes	
planned_pickup_start/end	timestamptz	yes	window
route_id	uuid	no	FK Route
shipments	uuid[]	yes	via LoadShipment

RateCard / RateTable

Field	Type	Required	Notes
rate_card_id	uuid	yes	PK
carrier_id, service_id	uuid	yes	
basis	enum	yes	LANE, ZONE, DISTANCE, WEIGHT_BREAK, PER_PALLET, PER_CONTAINER, FLAT
currency	char(3)	yes	
valid_from, valid_to	date	yes	no overlap per carrier/service/basis
minimum_charge	numeric(18,2)	no	
rows	jsonb	yes	[{origin, destination, break_from, break_to, rate, unit}]
fuel_index_id	uuid	no	FK FuelIndex
accessorial_ids	uuid[]	no	

FreightBill / CarrierInvoice

Field	Type	Required	Notes
freight_bill_id	uuid	yes	PK
load_id, carrier_id	uuid	yes	
expected_lines	jsonb	yes	[{charge_code, amount, currency, basis_trace}]
expected_total	numeric(18,2)	yes	
accrual_journal_id	uuid	no	
match_status	enum	yes	UNMATCHED, MATCHED, VARIANCE, DISPUTED, APPROVED, SETTLED
carrier_invoice_id	uuid	no	FK CarrierInvoice (number, date, lines, total, source EDI210/MANUAL)
variance_total	numeric(18,2)	no	
ap_invoice_id	uuid	no	wave-19 vendor invoice

Tender: tender_id, load_id, carrier_id, rank, sent_at, expires_at, response (PENDING/ACCEPTED/DECLINED/EXPIRED), rate_quoted.
LandedCostVoyage: voyage_id, cost lines (type, amount, currency, allocation_method), receipt lines, status (ESTIMATED/ACTUAL/POSTED).
TrackingEvent: load_id, stop_no, code (EDI 214 status), occurred_at, location, source.

Invariants: a load has at most one ACCEPTED tender; Σ allocated landed cost per voyage cost line = cost line amount (to the cent); a freight bill's accrual is reversed exactly once; shipments on a load share ship-from site and mode. Indexing: `load(status, planned_pickup_start)`, `rate_card(carrier_id, service_id, valid_from, valid_to)`, `freight_bill(match_status)`, `tracking_event(load_id, occurred_at)`.

6.34.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/tms/v1/loads:build	Build loads from shipments by rule	key	TMS_PLAN
POST	/tms/v1/routes:optimize	VRP over loads/stops; returns routes	key = request hash	TMS_PLAN

Method	Path	Purpose	Idempotency	Auth duty
POST	/tms/v1/shipments/{id}:rate	Rate across carriers, return ranked with explain	— (cached)	TMS_RATE
POST	/tms/v1/loads/{id}:tender	Start tender workflow	key	TMS_TENDER
POST	/tms/v1/tenders/{id}:respond	Carrier accept/decline	key	TMS_CARRIER_RESPOND
POST	/tms/v1/tracking-events	Ingest events (214/API)	key = source+ref	TMS_TRACK_INGEST
POST	/tms/v1/carrier-invoices	Register carrier invoice; triggers match	key = carrier+invoice no	TMS_RECONCILE
POST	/tms/v1/freight-bills/{id}:approve	Approve variance / release to AP	key	TMS_RECONCILE_APPROVE
POST	/tms/v1/landed-cost/voyages/{id}:allocate	Allocate and post	key	TMS_LANDED_COST
GET	/tms/v1/loads/{id}/tracking	Timeline	—	TMS_VIEW

Event	Trigger	Payload fields	Consumers
dev.passage.tms.shipment.rated.v1	rating	shipment_id, options[{carrier, service, total, transit_days}], chosen	sales (freight charge), analytics
dev.passage.tms.tender.accepted.v1 / declined.v1 / expired.v1	tender response	load_id, carrier_id, rank, rate	wms, edi (204/990), analytics
dev.passage.tms.tracking.event.v1	ingest	load_id, stop_no, code, occurred_at	sales-fulfillment, customer portal
dev.passage.tms.freightbill.matched.v1	match	freight_bill_id, carrier_invoice_id, variance, lines[]	AP, costing
dev.passage.tms.landedcost.posted.v1	allocation posted	voyage_id, receipt_lines[{line, amount}]	costing (wave 20)

6.34.6 User experience

Screen	Primary actions	Key fields	Duty
Transportation workspace	Shipments to plan, loads, tenders pending, exceptions	shipment, weight, window, status	TMS_VIEW
Load planning workbench	Drag shipments to loads, optimize routes, map view	load, stops, capacity bars, route distance/time	TMS_PLAN
Rate shop	Compare carriers, explain, select	carrier, service, charges, transit, score	TMS_RATE
Tender board	Tender, cascade, override	load, carrier rank, expires, response	TMS_TENDER
Tracking	Timeline, map, POD	event code, time, location, ETA	TMS_VIEW
Freight reconciliation	Match, dispute, approve, release to AP	expected vs invoiced per charge, variance	TMS_RECONCILE
Landed cost voyages	Add costs, allocate, post	cost type, amount, method, allocation	TMS_LANDED_COST

Interaction rules: planning workbench supports keyboard move (Alt+arrow) and multi-select; capacity bars turn red above 100%; route map uses self-hosted tiles; reconciliation grid highlights lines outside tolerance.

6.34.7 Business rules and edge cases

- **BR-34.1 VRP constraints.** Model as capacitated VRP with time windows and pickup/delivery (OR-Tools `RoutingModel`): (1) vehicle capacity dimensions weight_kg, volume_m3, pallets (hard); (2) stop time windows [earliest, latest] with soft lateness penalty $50/min up to 60 min, hard beyond; (3) max route duration 11 h driving + service times, max 14 h on-duty; (4) depot start/end per vehicle; (5) pickup-before-delivery precedence for milk runs; (6) heterogeneous fleet (equipment type must match shipment requirement, e.g., reefer); (7) vehicle-location compatibility (dock height, hazmat ban); (8) max stops per route 25; (9) objective = Σ distance cost + Σ time cost + fixed vehicle cost + lateness penalties; (10) time limit 60 s, first solution strategy `PARALLEL_CHEAPEST_INSERTION`, metaheuristic `GUIDED_LOCAL_SEARCH`. Infeasible stops are dropped with penalty 10,000 and reported, never silently.
- **BR-34.2 Rating algorithm.** For each eligible (carrier, service): select the rate card valid on ship date; compute base charge by basis (weight-break: find row with `break_from ≤ chargeable_weight < break_to`, charge = max(rate × weight, next-break deficit rule: compare with next break's minimum and take the lower); zone: origin/destination zone lookup; distance: miles × rate); chargeable weight = max(actual, dimensional weight = volume_in³ / DIM factor 139 for parcel, class-based for LTL); apply minimum charge; add accessorials (flat or % of base); add fuel surcharge = base × index % effective on ship date (weekly index); convert to shipment currency at the wave-12 rate type `FREIGHT`; round each charge to cents half-up; total = Σ charges. Explain trace lists each step with inputs.
- **BR-34.3 Tender cascade.** Rank from route guide; send to rank 1; on DECLINED or EXPIRED (timeout) send to next; after the last, status `TENDER_FAILED` and a planner task is created; a late acceptance after expiry is rejected with `TENDER_EXPIRED`.
- **BR-34.4 Match state machine.** `UNMATCHED → MATCHED` (all lines within tolerance) `→ APPROVED → SETTLED` (AP invoice posted); `UNMATCHED → VARIANCE → DISPUTED → (MATCHED | APPROVED with adjustment)`; a carrier invoice without a freight bill creates an `UNEXPECTED_INVOICE` exception and is never auto-approved. Duplicate carrier invoice numbers per carrier are rejected.
- **BR-34.5 Accrual and reversal.** Accrual posts on `load.shipped` with amount = expected total; on APPROVED, post reversal of the accrual and the AP invoice for the invoiced amount; variance to freight variance account 5250. All three postings share the freight bill id in the journal reference.
- **BR-34.6 Landed cost allocation.** Allocate each cost line by chosen method; compute each receipt line's share = round(cost × weight_i / Σ weight, 2); residual = cost – Σ shares assigned to the line with the highest value (ties: lowest line number). Estimated vs actual differences post to inventory if the item is still on hand (proportional to remaining qty) else to a purchase price variance account 5260 (wave 20 rules).
- **BR-34.7 Time windows** are stored in UTC with the stop's time zone; VRP uses the stop's local calendar; a window spanning midnight is valid.
- **BR-34.8 Load capacity exceeded** on manual planning is a warning (planner override logged); on rule-based building it is a hard split into a second load.
- **BR-34.9 Tracking event ordering**: events are applied by `occurred_at`; an out-of-order delivery event (D1) after pickup (AF) missing sets stop status DELIVERED and backfills PICKED_UP as inferred.
- **BR-34.10 Rate card overlap** on save is rejected; a card with `valid_to < ship_date` is ignored and the explain trace states `NO_VALID_RATE_CARD`.

6.34.8 Acceptance criteria

- **AC-34.1** (FR-34.2, BR-34.1) Given Solomon C101, R101 and RC101 (100 customers), when optimized with a 60 s limit, then every route satisfies capacity and time windows and total distance is within 10% of best-known.

- **AC-34.2** (FR-34.3, BR-34.2) Given the 200-shipment rating fixture, when rated, then totals equal the expected charges to the cent and the explain trace lists weight-break, minimum, accessorial and fuel steps.
- **AC-34.3** (FR-34.5, BR-34.3) Given a route guide with 3 carriers and 10-minute timeouts (test clock), when carrier 1 ignores and carrier 2 declines, then carrier 3 receives the tender and, on acceptance, the load is ACCEPTED with carrier 3.
- **AC-34.4** (FR-34.7, BR-34.4) Given a freight bill of $412.50 and a carrier invoice of $418.00, when matched with tolerance $5/2%, then the bill is VARIANCE with $5.50 and no AP invoice exists until approval.
- **AC-34.5** (FR-34.8, BR-34.5) Given a load shipped and later matched, when postings are inspected, then accrual, reversal and AP invoice tie to zero on account 2150 for that freight bill.
- **AC-34.6** (FR-34.9, BR-34.6) Given a voyage with freight $1,000.00 allocated by weight over 3 receipt lines (weights 1, 1, 1 kg), when allocated, then shares are 333.33 / 333.33 / 333.34 and Σ = 1,000.00.
- **AC-34.7** (FR-34.6) Given a 214 with status D1, when ingested, then the stop is DELIVERED and the sales shipment shows delivered timestamp.
- **AC-34.8** (FR-34.1, BR-34.8) Given 30 shipments totaling 48,000 kg for a 20,000 kg TL rule, when loads are built, then 3 loads exist, none over capacity.

6.34.9 Test plan and fixtures

Fixtures: Solomon instances (C101, R101, RC101, C201, R201), `rating-200.json` (lanes, weights, accessorials, expected totals verified by hand in a spreadsheet committed alongside), `carriers-demo` (5 carriers, 12 services, 3 rate bases), `carrier-invoices-50.json` with 10 variance cases, `voyage-3.json`.

Test	Type	Target	Pass criterion
VRP benchmark	performance/quality	optimizer	feasible, ≤ 110% best-known
Rating determinism	unit/property	engine	same input → same output; totals exact
Tender cascade	Temporal time-skip	workflow	cascades per BR-34.3
Match tolerance	unit	50 invoices	expected statuses
Accrual tie-out	contract	journal API	zero balance per bill
Landed cost sums	property	allocation	Σ shares = cost for random inputs
214 ingest	contract	API	ordering per BR-34.9

Reconciliation: Σ accrued freight by open freight bill = journal account 2150 balance; Σ landed cost posted = Σ voyage actual costs; loads shipped in wms = loads DISPATCHED+ in tms.

6.34.10 Deployment and operations

Helm `passage-tms` (api, worker), `passage-tms-optimizer` (Python, OR-Tools, 1–4 replicas, 2 CPU/4 GB each), `passage-osrm` (optional self-hosted routing with regional extracts). CloudNativePG `tms-db` (3 instances, 50 GB). Config: `TMS_VRP_TIME_LIMIT_S=60`, `TMS_TENDER_TIMEOUT_MIN=120`, `TMS_MATCH_TOL_ABS=5`, `TMS_MATCH_TOL_PCT=2`, `TMS_DIM_FACTOR_PARCEL=139`. Flags: `tms.enabled`, `tms.landed_cost`, `tms.vrp`. Scheduled: fuel index import weekly (Monday 06:00), tender expiry sweep (Temporal timers), KPI daily. SLOs: rating p95 < 300 ms per shipment per carrier; VRP 500 stops < 60 s; match p95 < 1 s; availability 99.9%. Alerts: tender failures > 3/day; unexpected invoices > 10/day; accrual drift ≠ 0; optimizer queue depth > 20. Runbook: optimizer OOM (reduce batch or raise limits); OSRM down (fallback to stored lane distances with warning). Rollback: `tms.enabled=false` returns to wave-28 carrier labels; open tenders are cancelled with notification.

6.34.11 Migration from AX / SAP

Source	Passage entity	Transformation
TMSCarrier, TMSCarrierService, TMSCarrierServiceGroup	Carrier, CarrierService	linked to Party by vendor account
TMSRateMaster, TMSRateBase, TMSRateBaseAssignment, rate engine metadata	RateCard, RateTable	basis per rate engine type; effective dates from assignment
TMSAccessorialMaster/Assignment, TMSFuelIndex/Region	Accessorial, FuelIndex	direct
TMSRoutePlan, TMSRouteGuide, TMSRouteSegment	RouteGuide	segments → lanes
TMSFreightBill/Detail, TMSCarrierInvoice/Detail (open)	FreightBill, CarrierInvoice	unmatched/unapproved only
Landed cost voyages (open)	LandedCostVoyage	estimated costs only
/SCMTMS/TOR_ROOT/ITEM/STOP (open freight orders)	Load, Shipment, Route	TOR category FO → Load, FU → Shipment
Freight agreements, /SCMTMS/TCM rate tables, calc sheets	RateCard, CalcSheet	scales → weight breaks
/SCMTMS/SFIR (open settlements)	FreightBill	status mapping
PO delivery cost conditions (FRA1/FRB1)	LandedCostVoyage estimated	per PO line

Not migrated: historical freight invoices and tracking events (lakehouse); SAP EM event handlers; ISV route optimization data.

6.34.12 Dependencies, risks, open questions

Depends on: wave 33 (phase ordering), wave 31 (loads, docks), wave 28 (shipments, carrier labels), wave 29 (customer freight charges), wave 20 (costing adjustments, supplier portal), wave 19 (AP invoices), wave 18 (purchase receipts), wave 12 (FX rate type), wave 11 (journal), wave 7 (carrier parties).

Risk	Likelihood	Impact	Mitigation
OR-Tools solution quality on real (non-benchmark) data	Medium	Medium	Planner override; parameter presets; report gap vs heuristic baseline
Carrier invoice formats vary (EDI 210 vs PDF)	High	Low	Manual/API entry path; OCR is v1.x

Open questions: (1) OSRM self-hosted vs stored lane matrix only? Proposed: OSRM optional, lanes default; decide by wave 34 design review. (2) Should customer-billed freight use rated or actual cost? Proposed: rated at ship confirmation; decide by wave 34.

6.35 Wave 35: B2B Integration: EDI, e-Invoicing & Integration Hub

6.35.1 Scope and business context

Wave 35 closes Phase 6 by replacing AX AIF / D365 data entities, Dual-write and Electronic Reporting (ER) for e-invoicing, and SAP IDoc/ALE, PI/PO (Integration Suite) and Document and Reporting Compliance (DRC). It delivers an EDI engine (X12 850/855/856/810/820/214/204/990/997 and EDIFACT ORDERS/ORDRSP/DESADV/INVOIC/CONTRL) with AS2 and SFTP transport, partner profiles and acknowledgments; Peppol BIS Billing 3.0 UBL 2.1 e-invoicing through a certified access point, Factur-X/ZUGFeRD, and a country mandate framework with hooks for IT SDI and MX CFDI; and an integration hub with public REST/GraphQL APIs, webhooks, Kafka topic subscriptions, a Backstage API catalog and developer portal with keys and quotas, plus reference connectors for Salesforce, Shopify and Stripe. Out of scope: full implementation of country e-invoice mandates beyond the framework and hooks (wave 48 country packs), EDIFACT subsets beyond the listed messages, a self-operated Peppol AS4 gateway (v1.x).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP IDoc failures sit in status 51 (WE02/BD87); a 'MATERIAL not found' error on one E1EDP01 segment requires an ABAP-literate operator to read EDIDD SDATA, fix master data, then reprocess; D365 AIF (AifExceptionLog) and the data entity message processor are similar	Monitor shows raw segment data, no canonical view; reprocess is a separate transaction	Message monitor shows raw, canonical and the failing rule side by side; bulk reprocess with dry run	Mean time to resolve a failed inbound order: 30–90 min → < 5 min; operator needs ABAP/X++: yes → no
Outbound documents are produced by batch (SAP output determination NAST + RSNAST00 jobs; D365 batch jobs for outbound ports), so an ASN reaches the customer 15–60 min after shipment and 810s go out in nightly runs	Polling batch framework	Outbound documents generated from `load.shipped` / `invoice.posted` events via the outbox	Event-to-document latency: 15–60 min (batch) → < 60 s; nightly invoice runs → continuous
Every EDI partner requires a middleware project (PI/PO ICO + mapping, or an ISV EDI add-on for D365) with Java/graphical mappings nobody in the business can read	Mapping logic lives outside the ERP in a separate tool and skill set	Declarative maps in the product with partner overrides, test harness and simulator	New partner onboarding: 4–8 weeks → 1–3 days; middleware licenses for EDI: 1 → 0
External systems (MES, PLM, BI, agents) integrate through OData with no event feed (D365 business events cover few entities) or through BAPIs/RFCs needing SAP Gateway projects; nobody can see which APIs exist	No catalog, no subscription model, no quotas	Backstage catalog with OpenAPI/AsyncAPI for every entity, webhook/Kafka subscriptions with replay, keys and quotas, MCP (wave 47) over the same APIs	Time for an external team to first successful call: weeks (access request, gateway project) → < 1 h self-service; entities with event feeds: ~10% → 100%
ER-based e-invoicing (ERFormatMappingTable) and SAP DRC need a configuration import per country/version and validate only on submission; schematron failures come back from the network hours later	Validation happens at the authority or AP, not at creation	Schematron (CEN + Peppol) runs at validate time inside Passage; arithmetic rules checked first	Rejections from the Peppol network: ~3–5% of invoices → < 0.1%; feedback latency: hours → < 2 s

6.35.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Integration ports / endpoints	AIF: AifPort, AifInboundPort, AifOutboundPort, AifService, AifAction; D365 data entities (DMFEntity), OData	IDoc ports WE21, partner profiles WE20 (EDPP1, EDP12/EDP13/EDP21), distribution model BD64 (TBD05)	edi.TradingPartner, EDIProfile, Transport	full
Message log / monitor	AifDocumentLog, AifMessageLog, AifExceptionLog, AifGatewayQueue; D365 message processor	IDoc status EDIDS (WE02/WE05), BD87 reprocess, PI/PO message monitor	Message monitor with replay	improved — replay with state reproduction
EDI documents	ISV (partner EDI), data entity import/export	IDoc types ORDERS05, DELVRY03 (DESADV), INVOIC02, control record EDI_DC40 (EDIDC), data EDIDD (EDID4)	Map (canonical ↔ X12/EDIFACT)	full
e-Invoicing	Electronic reporting ER configurations (ERFormatMappingTable, ERModelMappingTable), Invoicing service	DRC eDocument framework (EDOCUMENT, EDOCUMENTFILE), Peppol exchange	Peppol UBL 2.1, Factur-X, mandate framework	full (Peppol), partial (country mandates: hooks)
CRM/commerce sync	Dual-write (entity maps to Dataverse)	SAP CPI prepackaged content	Reference connectors (Salesforce, Shopify, Stripe)	partial — reference connectors only
Business events / webhooks	BusinessEventsCatalog, BusinessEventsEndpoint	Event Mesh / enterprise event enablement	Webhook subscriptions, Kafka subscriptions	full
Public API catalog	OData $metadata, Open API via custom service	API Business Hub, Gateway (SEGW)	Backstage catalog, developer portal, OpenAPI/AsyncAPI	improved
API keys / quotas	Azure AD app registration only	API management (Integration Suite)	ApiClient, Quota	improved
Acknowledgments	n/a	STATUS/ALEAUD IDoc, CONTRL	997/999, CONTRL, MDN (AS2)	full

6.35.3 Functional requirements

- **FR-35.1 Trading partner profiles** — Must. The system shall define partners with identifiers (ISA sender/receiver qualifier+id, GS id, EDIFACT UNB id, Peppol participant id `0088:GLN/0192:orgnr`), transports (AS2, SFTP, HTTPS), maps per transaction set and direction, envelope options, and SLA (ack within n minutes).
    - Notes: a partner may have map overrides (per-segment rules) layered on the base map; test and production profiles; certificate rotation with overlap period.

- **FR-35.2 X12 and EDIFACT parsing/serialization** — Must. The system shall parse and generate X12 (versions 004010 and 005010) and EDIFACT (D96A, D01B) with configurable delimiters, validate against transaction-set schemas, and generate 997/999 and CONTRL acknowledgments.
  - Notes: control numbers (ISA13, GS06, ST02; UNB, UNH) are gap-free per partner per direction; duplicate inbound ISA13 within 30 days is rejected as a duplicate interchange and reported in the 997.
- **FR-35.3 Mapping engine** — Must. The system shall map between a canonical document model (order, order response, shipment notice, invoice, remittance, status, tender) and EDI segments using declarative maps (JSONata/TypeScript expressions), with lookups (partner item cross-reference, location GLN, UoM, qualifier tables).
  - Notes: base maps per BR-35.1–35.3 ship as fixtures; unit-tested; map versioning with effective dates.
- **FR-35.4 Document integration** — Must. The system shall convert inbound 850/ORDERS into wave-26 sales orders (with partner item cross-reference), outbound 855, 856/DESADV from `wms.load.shipped` with SSCC hierarchy, 810/INVOIC from wave-29 invoices, 820 remittance into wave-30 cash application, 214 into/out of TMS, 204/990 for tenders.
  - Notes: inbound order failures create a task with the raw and canonical payload; outbound documents are created from events via the outbox (no polling).
- **FR-35.5 Transport** — Must. The system shall send and receive via AS2 (RFC 4130: signed, encrypted, synchronous/asynchronous MDN) and SFTP (poll every 60 s, archive after pickup), with retries (5× exponential, max 1 h) and alerts on MDN timeout.
  - Notes: AS2 certificates stored in Vault; MDN required by default; SFTP file naming pattern per partner.
- **FR-35.6 Peppol e-invoicing** — Must. The system shall produce UBL 2.1 invoices and credit notes conforming to Peppol BIS Billing 3.0 (BR-35.4), validate against the CEN EN 16931 and Peppol schematron, and send/receive through a certified access point adapter (AS4 via the AP's API), handling the Peppol MLR/transport acks.
  - Notes: participant lookup via SML/SMP through the AP; invoices received create wave-19 vendor invoices; validation errors block sending and are shown on the invoice.
- **FR-35.7 Factur-X / ZUGFeRD and mandate framework** — Should. The system shall embed EN 16931 CII XML in PDF/A-3 (Factur-X profile EN 16931) and provide a country mandate framework (`MandateProvider` interface: validate, submit, poll status, store clearance id) with reference hooks for IT SDI (FatturaPA) and MX CFDI 4.0 (PAC stamping).
  - Notes: full IT/MX packs ship in wave 48; the framework stores clearance ids and statuses on the invoice.
- **FR-35.8 Public APIs and developer portal** — Must. The system shall publish versioned REST (OpenAPI 3.1) and GraphQL APIs for all entities exposed since wave 3, with API clients (Keycloak client credentials), scopes mapped to duties, per-client quotas (default 600 requests/min, burst 100) and a developer portal at `developers.<env>.passage.dev` generated from the Backstage catalog.
  - Notes: additive-change guarantee per PRD §8; deprecation ≥ 12 months with `Sunset` header; sandbox tenant per client.
- **FR-35.9 Webhooks and Kafka subscriptions** — Must. The system shall let clients subscribe to CloudEvents types by filter and receive them by webhook (HTTPS POST, HMAC-SHA256 signature, at-least-once, 8 retries over 24 h, dead-letter with replay) or by a dedicated Kafka topic with ACLs.
  - Notes: ordering per aggregate id; replay from a timestamp reproduces state when consumed idempotently; payload schema from the schema registry.
- **FR-35.10 Message monitor and replay** — Must. The system shall show every interchange/message with status, acks, errors and payloads; allow reprocessing inbound from raw, resending outbound, and SLA dashboards per partner.
  - Notes: payload retention 13 months (configurable); PII fields masked for non-admin duties.
- **FR-35.11 Reference connectors** — Should. The system shall ship connectors: Salesforce (accounts/opportunities ↔ parties/quotes), Shopify (orders/fulfillments/products), Stripe (payments/payouts into cash application), each built only on the public APIs and webhooks.
  - Notes: connectors are separate Helm charts; mapping configurable; they prove the hub, not full products.
- **FR-35.12 Partner simulator** — Must. The system shall include a partner simulator (AS2/SFTP endpoint plus document generator) used in CI and the demo distributor scenario.
  - Notes: generates 850/856/810 and ORDERS/DESADV/INVOIC with randomized but valid content.

6.35.4 Data model

TradingPartner / EDIProfile

Field	Type	Required	Notes
partner_id	uuid	yes	PK
party_id	uuid	yes	FK Party (wave 7)
isa_qualifier, isa_id, gs_id	text	no	X12 identifiers
unb_qualifier, unb_id	text	no	EDIFACT
peppol_participant_id	text	no	`scheme:id`
transports	jsonb	yes	[{type AS2/SFTP/HTTPS, url/host, cert_ref, mdn_mode, poll_interval}]
maps	jsonb	yes	[{transaction_set, direction, map_id, version, overrides}]
ack_sla_minutes	int	yes	default 60
environment	enum	yes	TEST, PROD

Interchange / Message

Field	Type	Required	Notes
interchange_id	uuid	yes	PK
partner_id, direction	uuid/enum	yes	IN/OUT
standard, version	enum/text	yes	X12 004010 / EDIFACT D96A / UBL 2.1
control_number	text	yes	unique per partner/direction/standard
raw_payload_ref	text	yes	object storage key (MinIO/S3-compatible, in-cluster)
received_at / sent_at	timestamptz	yes	

Field	Type	Required	Notes
ack_status	enum	yes	PENDING, ACCEPTED, ACCEPTED_WITH_ERRORS, REJECTED, TIMEOUT
messages	—	—	Message rows: message_id, transaction_set, st_control, canonical_ref (document id), status (RECEIVED, MAPPED, POSTED, FAILED, SENT, ACKED), error

Subscription / DeliveryAttempt

Field	Type	Required	Notes
subscription_id	uuid	yes	PK
client_id	text	yes	Keycloak client
kind	enum	yes	WEBHOOK, KAFKA
event_types	text[]	yes	CloudEvents type patterns
filter	jsonb	no	CESQL-style expression
endpoint_url, secret_ref	text	no	webhook
topic	text	no	Kafka `hub.<tenant>.<client>`
status	enum	yes	ACTIVE, PAUSED, DEAD_LETTERED
attempt: attempt_id, event_id, attempt_no, status_code, latency_ms, next_retry_at	—	—	DeliveryAttempt

ApiClient: client_id, scopes[], quota_per_min, burst, sandbox_tenant_id, owner contact, status. **EInvoice**: einvoice_id, invoice_id (wave 29/19), format (PEPPOL_UBL/FACTURX/CII/MANDATE), xml_ref, validation_report, transmission_id, clearance_id, status (DRAFT/VALIDATED/SENT/ACCEPTED/REJECTED).

Invariants: control numbers are strictly increasing without gaps per partner/direction; a Message in POSTED references exactly one Passage document; every outbound webhook event has ≥ 1 DeliveryAttempt or is DEAD_LETTERED; an EInvoice SENT has a passing validation report. Indexing: `interchange(partner_id, direction, control_number)` unique; `message(status, received_at)`; `delivery_attempt(subscription_id, next_retry_at) WHERE status = 'RETRY'`.

6.35.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/edi/v1/partners	Create partner profile	key	EDI_ADMIN
POST	/edi/v1/inbound/as2	AS2 receive endpoint (RFC 4130)	AS2 Message-ID	partner cert
POST	/edi/v1/interchanges/{id}:reprocess	Re-run mapping/posting from raw	key	EDI_OPERATE
POST	/edi/v1/messages/{id}:resend	Re-transmit outbound	key	EDI_OPERATE
POST	/edi/v1/maps:test	Run a map against a sample payload	—	EDI_ADMIN
POST	/einvoice/v1/invoices/{id}:validate	Build UBL/CII and run schematron	—	EINV_OPERATE
POST	/einvoice/v1/invoices/{id}:send	Transmit via AP or mandate provider	key = invoice id + format	EINV_SEND
POST	/hub/v1/clients	Register API client, scopes, quota	key	HUB_ADMIN
POST	/hub/v1/subscriptions	Create webhook/Kafka subscription	key	HUB_CLIENT
POST	/hub/v1/subscriptions/{id}:replay?from=	Replay events from timestamp	key	HUB_CLIENT
GET	/hub/v1/catalog	API/AsyncAPI catalog (Backstage-backed)	—	public (authenticated)
POST	/graphql	GraphQL endpoint	per mutation key	scopes

Event	Trigger	Payload fields	Consumers
dev.passage.edi.message.received.v1	inbound parsed	partner_id, standard, transaction_set, control numbers, canonical_ref	purchasing, sales, AP, tms
dev.passage.edi.message.posted.v1 / failed.v1	posting outcome	message_id, document_ref, error	monitor, tasks
dev.passage.edi.message.sent.v1 / acked.v1	transmit / ack	message_id, partner_id, ack_status	analytics
dev.passage.einvoice.status.changed.v1	validation/transmission	einvoice_id, invoice_id, status, clearance_id	AR/AP, customer portal
dev.passage.hub.webhook.delivered.v1 / deadlettered.v1	delivery	subscription_id, event_id, attempts	analytics, alerting

6.35.6 User experience

Screen	Primary actions	Key fields	Duty
Partner profiles	Create, transports, certificates, maps, test	identifiers, transport, map versions, SLA	EDI_ADMIN
Map designer	Edit expressions, test with samples, diff versions	segment, element, expression, lookup	EDI_ADMIN
Message monitor	Filter, view raw/canonical, reprocess, resend	partner, set, control no, status, ack, age	EDI_OPERATE
Partner SLA dashboard	Ack latency, failure rate, volume	partner, p95 ack, failures	EDI_OPERATE
e-Invoice workspace	Validate, send, view clearance, errors	invoice, format, status, schematron errors	EINV_OPERATE
Developer portal (external)	Browse APIs/events, create keys, test console, subscriptions, usage	OpenAPI, AsyncAPI, quotas, attempts	HUB_CLIENT
Hub admin	Clients, quotas, subscriptions, dead letters	client, scopes, quota, DLQ count	HUB_ADMIN

Interaction rules: monitor supports bulk reprocess (≤ 200) with a dry run; raw payloads open in a segment-tree viewer with element hover showing the X12 dictionary name; developer portal "try it" executes against the client's sandbox tenant only.

6.35.7 Business rules and edge cases

- **BR-35.1 X12 850 inbound mapping (004010) → canonical Order.**

Segment/element	Canonical field	Rule
ISA06/ISA08, GS02/GS03	partner resolution	must match profile; else reject interchange (TA1/997 R)
BEG01	purpose	00 original, 01 cancel, 04 change → create/cancel/change order
BEG02	order_type	SA standalone, BK blanket, RL release
BEG03	customer_po	required; duplicate per customer → reject with 997 AK5 R
BEG05	po_date	CCYYMMDD
CUR02	currency	ISO 4217; default partner currency
REF*DP / REF*IA	department / vendor id	optional
DTM*002 / DTM*010	requested_delivery / requested_ship	DTM02 CCYYMMDD
N1*ST, N3, N4	ship_to (GLN from N104 qualifier 92/UL)	resolve to party address; unknown → task
N1*BT	bill_to	
PO101	line_no	
PO102, PO103	qty, uom (EA, CA, LB…)	UoM via qualifier table
PO104, PO105	unit_price, basis (PE per each, TP per thousand)	price verified against wave-26 pricing; variance tolerance 1% else hold
PO106/07 pairs	item cross-reference	BP buyer part → partner item xref; VP vendor part → item; UP/EN → GTIN; first match wins in order BP, VP, UP, EN
PID*F	line description	informational
SAC (loop)	charges/allowances	C=charge, A=allowance; SAC05 amount N2 (implied 2 decimals)
CTT01	line count	must equal PO1 count else reject

- **BR-35.2 X12 856 outbound mapping (004010, structure S/O/P/I) from `wms.load.shipped`.**

Segment/element	Source	Rule
BSN01/02/03/04/05	00, shipment id (load+shipment), ship date, time, 0001	BSN05 hierarchy S-O-P-I
HL*1**S	shipment	
TD1	packaging code (PLT94), lading qty, weight (G, LB/KG)	from load totals
TD5	SCAC (TD503), mode (TD504 M motor)	from tms carrier
REF*BM, REF*CN	bill of lading, PRO number	from tms load
DTM*011	ship date	
N1*ST / N1*SF	ship-to / ship-from with GLN	qualifier 92 or UL
HL**1*O	order	PRF01 customer PO, PRF04 PO date
HL**2*P	pack (pallet LP)	MAN*GM SSCC-18 (AI 00 without AI)
HL**3*I	item	LIN (VP item, UP GTIN, BP buyer part), SN101 line, SN102 qty shipped, SN103 uom
CTT01	number of HL segments	

- **BR-35.3 X12 810 outbound mapping (004010) from the wave-29 customer invoice.**

Segment/element	Source	Rule
BIG01, BIG02, BIG03, BIG04	invoice date, invoice no, PO date, customer PO	BIG07 type DI (debit) / CR (credit note)
REF*BM, REF*CO	BOL, customer order	optional
N1*RE / N1*BT / N1*ST	remit-to / bill-to / ship-to	
ITD	terms (ITD01 code, ITD03 discount %, ITD05 discount days, ITD07 net days)	from wave-29 terms
DTM*011	ship date	
IT1	IT101 line, IT102 qty, IT103 uom, IT104 unit price, IT106/07 BP/VP	
TXI	tax type (TXI01 ST/VA/GS), amount, rate	per tax line (wave 13)
SAC	freight (SAC02 D240), allowances	charge lines
TDS01	total invoice amount, N2	equals invoice total × 100
CAD	carrier, PRO	optional
CTT01	line count	

EDIFACT ORDERS/DESADV/INVOIC maps follow the same canonical model (BGM/DTM/NAD/LIN/QTY/PRI/MOA/CPS/PAC/GIN SSCC) and are shipped as fixtures.

- **BR-35.4 Peppol BIS Billing 3.0 required elements (UBL 2.1 Invoice).** `CustomizationID = urn:cen.eu:en16931:2017#compliant#urn:fdc:peppol.eu:2017:poacc:billing:3.0`; `ProfileID = urn:fdc:peppol.eu:2017:poacc:billing:01:1.0`; `ID`; `IssueDate`; `DueDate` or `PaymentTerms`; `InvoiceTypeCode` (380 invoice, 381 credit note as `CreditNote` document); `DocumentCurrencyCode`; `BuyerReference` or `OrderReference/ID` (one required); `AccountingSupplierParty/Party`: `EndpointID @schemeID` (e.g., 0088), `PartyIdentification` (optional), `PartyName`, `PostalAddress/Country/IdentificationCode`, `PartyTaxScheme/CompanyID` with `TaxScheme/ID = VAT` when VAT applies, `PartyLegalEntity/RegistrationName`; `AccountingCustomerParty/Party` (same set, `EndpointID` required); `PaymentMeans/PaymentMeansCode` (30 credit transfer, 58 SEPA) with `PayeeFinancialAccount/ID` (IBAN); `TaxTotal/TaxAmount @currencyID`; `TaxTotal/TaxSubtotal` per category: `TaxableAmount`, `TaxAmount`, `TaxCategory/ID` (S, Z, E, AE, K, G, O), `Percent`, `TaxScheme/ID = VAT`; `LegalMonetaryTotal`: `LineExtensionAmount`, `TaxExclusiveAmount`, `TaxInclusiveAmount`, `PayableAmount` (and `AllowanceTotalAmount`/`ChargeTotalAmount` when present); ≥ 1 `InvoiceLine`: `ID`,

`InvoicedQuantity @unitCode` (UN/ECE Rec 20, e.g., C62, KGM), `LineExtensionAmount`, `Item/Name`, `Item/ClassifiedTaxCategory/ID` + `Percent` + `TaxScheme`, `Price/PriceAmount`. Arithmetic rules enforced before schematron: Σ line `LineExtensionAmount` = `LineExtensionAmount`; `TaxExclusiveAmount` = that – allowances + charges; `TaxInclusiveAmount` = exclusive + `TaxAmount`; `PayableAmount` = inclusive – prepaid + rounding; amounts to 2 decimals; VAT breakdown per category/rate equals Σ lines. Validation runs CEN-EN16931-UBL then PEPPOL-EN16931-UBL schematron; any `fatal` blocks sending.

- **BR-35.5 Control numbers and duplicates.** Outbound ISA13/GS06/ST02 per partner from a gap-free number sequence (wave 5); inbound duplicate ISA13 within 30 days → TA1 code 025 / reject; a 997 is generated within 60 s for every inbound functional group; missing partner 997 after `ack_sla_minutes` → alert and resend once.
- **BR-35.6 Idempotent posting.** Inbound 850 posting uses idempotency key `partner:BEG03:BEG01`; a re-sent original (BEG01 00) with the same PO number returns the existing order and is acknowledged accepted; a change (BEG01 04) applies as a wave-26 order change with version.
- **BR-35.7 Webhook delivery.** Retries at 10 s, 1 m, 5 m, 30 m, 2 h, 6 h, 12 h, 24 h; HTTP 2xx = success; 410 Gone pauses the subscription; signature header `Passage-Signature: t=<unix>,v1=<hmac>` over `t.body`; receiver must verify within 5 min of `t`. Events per aggregate are delivered in order; a failing event blocks later events for the same aggregate only (per-aggregate queue).
- **BR-35.8 Quotas.** Token bucket per client (rate/min, burst); exceeding returns 429 with `Retry-After`; quota consumption is counted per request and per 100 GraphQL complexity points.
- **BR-35.9 Replay.** Replay from timestamp T re-delivers events with `ce-passagereplay: true`; consumers must treat them idempotently; replay is capped at 1,000,000 events per request and runs as a Temporal workflow with progress.
- **BR-35.10 e-Invoice status.** `DRAFT → VALIDATED → SENT → ACCEPTED|REJECTED`; REJECTED allows correction and a new transmission with the same invoice id and a new `transmission_id`; an ACCEPTED invoice cannot be re-sent, only credited.
- **BR-35.11 AS2 MDN.** Async MDN awaited 1 h, then resend up to 5×, then FAILED with alert; MDN signature mismatch is FAILED immediately.
- **BR-35.12 PII masking** in the monitor hides N1/N3/N4 address lines and party contact elements for users without `EDI_PII_VIEW`.

6.35.8 Acceptance criteria

- **AC-35.1** (FR-35.2, FR-35.3, FR-35.4, FR-35.12) Given the partner simulator, when it sends 850, 855 is returned, 856 and 810 are produced after shipment and invoicing, and the simulator's 997s are matched, then every supported transaction set round-trips with no manual step and the message monitor shows ACKED for each.
- **AC-35.2** (FR-35.6, BR-35.4) Given 50 demo invoices (standard, reverse-charge, zero-rated, with allowances), when UBL is built, then all pass CEN and Peppol schematron with zero fatal errors and the amounts rules hold.
- **AC-35.3** (FR-35.9, BR-35.7) Given a webhook endpoint that fails 3 times then succeeds, when an event is published, then exactly one successful delivery is recorded after 3 retries and ordering per aggregate is preserved.
- **AC-35.4** (FR-35.9, BR-35.9) Given a consumer that rebuilds a projection from scratch, when events are replayed from the tenant's start, then the projection equals the current Passage state for the sales order aggregate fixture.
- **AC-35.5** (BR-35.5) Given an inbound interchange with an ISA13 already seen 10 days ago, when received, then it is rejected with TA1 025 and no order is created.
- **AC-35.6** (FR-35.8, BR-35.8) Given a client with quota 600/min, when 700 requests are sent in a minute, then ≥ 100 receive 429 with `Retry-After` and the portal usage chart shows the burst.
- **AC-35.7** (BR-35.1) Given an 850 with PO104 differing from the wave-26 price by 3%, when posted, then the sales order is created on price hold with a task.
- **AC-35.8** (BR-35.2) Given a shipped load with 2 pallets × 3 cases for 2 orders, when the 856 is generated, then HL levels are S(1) O(2) P(2) I(n) with MAN*GM SSCCs equal to the wms LP ids and CTT01 equals the HL count.
- **AC-35.9** (phase exit) Given the demo distributor and manufacturer tenants, when the distributor sends an 850, then the manufacturer ships and invoices and the distributor's AP shows a matched vendor invoice from the 810 and the Peppol UBL, both within 15 minutes of each event.

6.35.9 Test plan and fixtures

Fixtures: `x12-samples` (004010 850/855/856/810/820/214/204/990/997, 20 variants each from partner implementation guides), `edifact-samples` (ORDERS/ORDRSP/DESADV/INVOIC D96A), `peppol-invoices-50` (UBL built from wave-29 invoices, including all tax category codes), `factur-x-10.pdf`, `partner-simulator` config for the demo distributor, `webhook-chaos` (failing endpoint profiles), `replay-salesorder-projection`.

Test	Type	Target	Pass criterion
Parser/serializer	unit/property	X12/EDIFACT	round-trip byte-equal after normalization
Map tests	unit	each map	canonical equals expected JSON
Round-trip	E2E (simulator)	every transaction set	ACKED, documents posted
Schematron	contract	Peppol UBL	0 fatal on 50 invoices; known-bad fixtures produce expected rule ids
Webhook delivery	property	retries/ordering	at-least-once, ordered per aggregate
Replay	E2E	projection rebuild	equals current state
Quota	performance (k6)	gateway	429 behavior, p95 overhead < 5 ms
AS2 interop	contract	mendelson/OpenAS2	MDN exchanged signed+encrypted

Reconciliation: inbound 850 count = sales orders created + rejected + held; outbound 810 count = invoices posted with EDI partners; Peppol SENT = invoices flagged e-invoice for Peppol customers; webhook events published = delivered + dead-lettered.

6.35.10 Deployment and operations

Helm `passage-edi` (api, as2 receiver behind Envoy with client-cert passthrough, sftp poller, mapper workers), `passage-einvoice`, `passage-integration-hub` (gateway policies on Envoy Gateway for keys/quotas, webhook dispatcher, Kafka subscription provisioner using Strimzi `KafkaUser`/`KafkaTopic` CRs), `passage-devportal` (Backstage with the catalog plugin, OpenAPI/AsyncAPI docs rendered from the service registry), `passage-partner-simulator` (non-prod only), connector charts `passage-connector-{salesforce,shopify,stripe}`. CloudNativePG `integration-db` (3 instances, 200 GB; payloads in in-cluster object storage, 13-month retention). Config: `EDI_ACK_SLA_MIN=60`, `EDI_DUP_WINDOW_DAYS=30`, `HUB_DEFAULT_QUOTA=600`, `HUB_WEBHOOK_RETRIES=8`,

`EINV_AP_PROVIDER=<adapter>`. Flags: `edi.enabled` per partner, `einvoice.peppol`, `einvoice.facturx`, `hub.graphql`, `hub.replay`. Scheduled: SFTP poll 60 s, ack SLA sweep 5 min, DLQ report daily, certificate expiry check daily (alert 30 days before). SLOs: inbound 850 to sales order p95 < 30 s; outbound document generated within 60 s of the source event; 997 within 60 s; webhook first attempt within 5 s of event p95; gateway overhead p95 < 5 ms; availability 99.9%. Alerts: ack SLA breaches; AS2 MDN failures; DLQ depth > 100; certificate expiring; schematron failure rate > 5%. Runbook: partner certificate rotation; reprocess failed interchanges after a map fix; drain and replay DLQ; rotate client secrets. Rollback: per-partner flag off (queue inbound, do not process); map version pin to the previous version; hub gateway policies are versioned in Git with Argo CD rollback.

6.35.11 Migration from AX / SAP

Source	Passage entity	Transformation
AifPort/AifInboundPort/AifOutboundPort, AifService/AifAction	TradingPartner transports, map assignments	port → transport; service/action → transaction set/direction
AifDocumentLog, AifMessageLog (last 13 months)	Interchange/Message history (read-only)	payloads copied to object storage
ERFormatMappingTable/ERModelMappingTable (e-invoice configurations)	EInvoice format config	ER formats are not executable in Passage; only the country/format selection and customer flags migrate
Dual-write entity maps	Connector mappings (Salesforce reference)	field map table only
BusinessEventsEndpoint/Catalog	Subscription	endpoint type → WEBHOOK
EDPP1, EDP12/EDP13/EDP21 (WE20)	TradingPartner, maps	partner type LI/KU → vendor/customer party; message type ORDERS/DESADV/INVOIC → maps
EDIPOA/port definitions (WE21), TBD05 (BD64)	Transports, subscriptions	file ports → SFTP; tRFC ports → HTTPS webhook
EDIDC/EDID4/EDIDS (last 13 months, statuses 03/53/51)	Interchange/Message history	EDI_DC40 control fields (DOCNUM, IDOCTYP, MESTYP, SNDPRN, RCVPRN, STATUS) → message columns; EDIDD SDATA kept raw
EDOCUMENT/EDOCUMENTFILE (DRC)	EInvoice with clearance ids	status mapping; XML kept as reference
PI/PO integration directory (ICO)	Partner/transport inventory	exported as a spreadsheet for manual re-creation; routing logic not migrated

Not migrated: AIF transforms (XSLT/.NET), ABAP user exits in IDoc processing, PI/PO mappings (Java/graphical) — mapping logic is re-authored in Passage maps; IDoc history older than 13 months (lakehouse archive).

6.35.12 Dependencies, risks, open questions

Depends on: wave 34 (204/990/214 and freight invoices), wave 31 (SSCC/LPs for 856), wave 30 (cash application for 820), wave 29 (customer invoices), wave 26 (sales order intake and change), wave 19 (vendor invoices), wave 17 (purchase orders for outbound 850), wave 13 (tax categories for UBL), wave 7 (parties, GLNs), wave 5 (gap-free sequences), wave 4 (outbox, schema registry), wave 3 (entity APIs).

Risk	Likelihood	Impact	Mitigation
Peppol access point certification/onboarding timeline	Medium	Medium	Certified third-party AP adapter; two providers evaluated
Partner implementation guides require per-partner map work at every customer	High	Medium	Override layer, simulator, map test harness; estimate 1–3 days per partner
Public API surface freezes the data model early	Medium	High	Additive-only versioning; v1 scope limited to entities stable since wave 30
Licensing decision (PRD §12 Q1) affects developer portal terms	Medium	Low	Portal terms page templated; decision required before wave 35 close

Open questions: (1) Which certified Peppol AP provider for the reference adapter? Proposed: evaluate two, pick by wave 35 start. (2) GraphQL as first-class or REST-only for v1 public APIs? Proposed: both, GraphQL read-only at first; decide by wave 35 design review. (3) Where do connector products live long-term (core repo vs marketplace)? Proposed: separate repos under the hub; decide by wave 46.

Part 7 — Phase 7: Treasury, Assets & Controlling (waves 36–40)

7.0 Phase overview

Business outcome at phase exit. At the end of wave 40 the finance function runs entirely on Passage: treasury sees every bank balance intraday and a 13-week cash forecast built from the subledgers; fixed assets and leases depreciate under multiple books and post to the universal journal; controllers plan budgets, enforce budget control on commitments, and run a cost accounting ledger with allocations; product costs roll up from BOMs and routings with a cost component split, and production variances are decomposed by cause; intercompany trade posts mirror documents across legal entities and the group consolidates with eliminations and IAS 21 translation. This is the "finance parity" milestone in PRD §10 (wave 40): a practicing controller signs off that nothing they did in AX/SAP finance is missing.

Phase-exit sign-off criteria

- 1,000-line bank statement fixture auto-matches ≥ 95% of lines correctly with zero false positive matches (wave 36 proof).
- Cash forecast total per bucket equals the sum of source contributions to the cent, verified by a property test over 10,000 generated open items (wave 36).
- 100 asset × method × convention fixtures match IRS Publication 946 tables and the straight-line edge cases to the cent; the asset subledger ties to GL asset cost and accumulated depreciation accounts with zero difference (wave 37).
- IFRS 16 lease schedules match the accountant-reviewed workbook to the cent for 25 fixture leases (wave 37).
- Budget check blocks or warns every over-budget PO per policy; encumbrance liquidation on invoice is exact to the cent (wave 38).
- Every allocation cycle converges and `total cost out = total cost in` to the cent; cost accounting ledger primary costs reconcile to GL (wave 38).
- 50 multi-level cost roll-ups match the reference workbook across all cost components; variance decomposition sums to total variance for 200 ended production orders; mass roll of 50,000 items completes in < 15 minutes (wave 39).
- Intercompany mirror documents balance across entities for 1,000 random IC orders; IC reconciliation flags 100% of planted mismatches; the 3-entity, 2-currency demo group consolidates to accountant-reviewed statements (wave 40).
- Daily subledger-to-GL reconciliation jobs for bank, assets, cost accounting, costing and intercompany run green for 10 consecutive days in `stage`.

Cross-wave data model summary

Entity	Owning service (domain DB)	Key relationships
BankAccount, BankStatement, StatementLine, ReconciliationMatch, MatchingRule	`treasury-svc` (Postgres `treasury`)	BankAccount → LegalEntity, MainAccount (wave 11); StatementLine → payment (wave 19), receipt (wave 30)
CashForecastEntry, CashPosition	`cashforecast-svc` (Python, Postgres `treasury`)	Reads AP/AR open items, bank balances, recurring entries
BankConnection, OutboundFile, InboundFile	`bank-connect-svc` (TS, Postgres `treasury`)	Keys referenced by Vault path only
Asset, AssetBook, DepreciationProfile, AssetTransaction, Lease, LeaseSchedule	`asset-svc` (Postgres `assets`)	Asset → Project (wave 41 later), PurchaseLine (wave 17), InventTrans (wave 16); posts via journal API (wave 11)
BudgetRegisterEntry, BudgetPlan, BudgetControlRule, Encumbrance, BudgetCheckResult	`budget-svc` (Postgres `controlling`)	Encumbrance → Requisition/PO (wave 17), VendorInvoice (wave 18)
CostObject, CostElement, CostEntry, AllocationPolicy, AllocationRun, StatisticalMeasure	`costacct-svc` (Postgres `controlling`)	CostEntry mirrors journal lines by dimension (wave 11)
CostingVersion, CostingSheet, CostGroup, CostCalculation, CostComponentSplit, ItemCostRecord, VarianceEntry	`costing-svc` (Postgres `costing`, extends wave 20)	Reads BOM/Route versions (wave 9), production orders (wave 21), inventory cost trans (wave 20)
IntercompanyPartner, IcDocumentLink, IcJournal, IcReconciliationItem	`interco-svc` (TS, Postgres `group`)	Links SalesOrder (wave 27) ↔ PurchaseOrder (wave 17), invoices, payments
ConsolidationGroup, ConsolidationUnit, Ownership, EliminationRule, ConsolidationRun, ConsolidationEntry	`consol-svc` (Postgres `group`)	Reads balances per ledger (wave 11), rates (wave 12), statements (wave 15)

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.treasury.statement.reconciled.v1`	treasury-svc	journal (wave 11), cashforecast-svc, analytics (46)	statementId, bankAccountId, matchedCount, unmatchedCount, closingBalance
`dev.passage.treasury.cashposition.updated.v1`	cashforecast-svc	dashboards, interco-svc (netting)	legalEntityId, asOf, balancesByAccount[], forecastBuckets[]
`dev.passage.assets.depreciation.posted.v1`	asset-svc	costacct-svc, analytics	runId, bookId, period, assetCount, totalAmount, journalEntryId
`dev.passage.assets.asset.disposed.v1`	asset-svc	costacct-svc, tax (wave 13)	assetId, bookId, proceeds, nbv, gainLoss
`dev.passage.budget.check.completed.v1`	budget-svc	procurement (17), AP (18), workflow	sourceDocumentId, result (pass/warn/fail), availableBefore, amount, availableAfter
`dev.passage.controlling.allocation.completed.v1`	costacct-svc	budget-svc, analytics	runId, policyId, iterations, totalIn, totalOut, residual
`dev.passage.costing.standardcost.activated.v1`	costing-svc	inventory costing (20), production (21), budget-svc	costingVersionId, itemCount, revaluationJournalId, totalRevaluation
`dev.passage.costing.variance.posted.v1`	costing-svc	costacct-svc, analytics	productionOrderId, varianceByType{lotSize,quantity,price,substitution}, byComponent[]
`dev.passage.interco.mirror.created.v1`	interco-svc	sales (27), procurement (17), consol-svc	sourceDocId, sourceEntity, mirrorDocId, mirrorEntity, type
`dev.passage.consol.run.completed.v1`	consol-svc	reporting (15), analytics	runId, groupId, period, unitCount, eliminationCount, ctaAmount

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Bank connectivity (EBICS/Open Banking) certification differs per bank and takes weeks	High	Medium	Ship SFTP first; EBICS and PSD2 against public sandboxes; keep adapters behind one `BankChannel` interface	Engineering
Depreciation or lease math diverges from auditor expectation on edge cases (short years, mid-quarter, modifications)	Medium	High	Fixture library reviewed by the accountant; property tests that total depreciation always equals depreciable basis	Product
Budget control latency slows PO entry	Medium	Medium	p95 budget check < 150 ms SLO; per-dimension-combination balance cache rebuilt from events	Engineering
Reciprocal allocations fail to converge with circular self-references	Low	Medium	Exact linear solve for ≤ 2,000 senders; capped iteration with residual sweep rule (BR-38.6)	Engineering
Cost roll-up on cyclic or 20+ level BOMs	Medium	Medium	Cycle detection at BOM approval (wave 9) and at roll time; hard depth limit 20	Engineering
Two-person team across five accounting-heavy waves	High	High	Accountant review gate per wave; AI-assisted generation per PRD §11	Owners

7.36 Wave 36: Cash & Bank Management, Reconciliation & Treasury

7.36.1 Scope and business context

Wave 36 gives treasury a system of record for bank accounts and the daily mechanics around them. Before this wave, AP payments (wave 19) and AR receipts (wave 30) post to GL bank accounts, but nobody can import a bank statement, reconcile it, know the intraday position, or forecast cash. After it, a treasury analyst imports camt.053/BAI2/MT940 statements (or receives them automatically through SFTP/EBICS/Open Banking), the matching rule engine reconciles ≥ 95% of lines without touching them, exceptions are worked in a reconciliation workspace, and a cash position and 13-week forecast refresh from the subledgers every hour. Checks can be issued, voided and reprinted; letters of credit and bank guarantees are tracked with facility limits; intercompany netting and notional cash pooling basics are available for groups.

Out of scope: hedge accounting and derivative valuation (not in v1), in-house bank virtual accounts with interest calculation, lockbox OCR, treasury trading (money market, FX deals). Payroll (wave 43) and project (wave 41) forecast sources are consumed through the forecast source interface once those waves exist; until then recurring forecast entries stand in.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 advanced bank reconciliation: statement import via DMF (BankStatementFormat + XSLT transform) fails silently on bank camt dialects; the user re-imports after editing XSLT	Transform-first design; no schema validation surfaced to the user	Native camt/BAI2/MT940 parsers with per-bank profile overrides and line-level validation errors	Failed imports needing IT help: 1 in 5 → < 1 in 200
SAP FEBAN post-processing: each unmatched line is opened, the clearing document found in FBL5N, keyed and posted (≈ 14 clicks per line)	Posting rules (OT83) only test search strings; no candidate ranking	Scored candidates shown inline; `m` to match; N:1 batch matching	Clicks per exception 14 → 2; auto-match ≥ 95%
BankReconciliationMatchingRule in D365 is pass/fail; a 1-day value-date difference leaves the line unmatched	Boolean rules, no partial credit	Weighted scoring with tolerance and date-window decay	Unmatched lines after rules: 30% → < 5%
Cash position in AX = BankAccountTable.Balance updated at posting; SAP FF7A refreshes in a batch run; both ignore intraday camt.052	Batch-oriented, no intraday feed	camt.052 ingestion on arrival; position recomputed per event	Position latency 4 h batch → < 60 s after file arrival
EBICS keys held on a Windows share for a bank client tool; audit cannot prove who sent what	Connectivity outside the ERP	`bank-connect-svc` signs in Vault; every file hashed and acknowledged	100% of files with signer, hash and bank ack in audit

7.36.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Bank account master, signatories, approval	BankAccountTable, BankAccountTrans, BankGroup, form BankAccountTable; D365 bank account change workflow	FI12 house banks T012/T012K, BAM (FCLM_BAM) with signatories and workflow	`treasury-svc` BankAccount entity, workspace Bank accounts	full
Electronic statement import	BankStmtISOAccountStatement, BankStmtISOReportEntry, BankStatementFormat, DIXF ISO 20022 transform	FF_5 (FEBKO/FEBEP), FF67 manual statement, OT83 posting rules, external transaction codes	Statement import API, formats camt.053/BAI2/MT940/CSV	improved (native XSD validation, duplicate detection)
Advanced bank reconciliation	BankReconciliationHeader, BankReconciliationMatchingRule, BankReconciliationMatchingRuleSet, BankReconciliationWorkspace	FEBAN post-processing, FEB_BSPROC, search strings (OTPM), interpretation algorithms	Matching rule engine with scoring, reconciliation workspace	improved
Check management	BankChequeTable, BankChequeReprint, check void/reverse, positive pay BankPositivePayFormat	PAYR check register, FCH5/FCH8/FCH9 (void/reissue), FCHN, FCHX	Check entity with void/reprint/stop, positive pay (wave 19 export)	full
Letters of credit / guarantees	BankLCImport, BankLCExport, BankLGTable (letters of guarantee), BankFacility, BankFacilityAgreement	No native EBS equivalent; TRM/bank guarantee via FI-TRM or custom; S/4 Trade Finance	Facility, LetterOfCredit, Guarantee entities with limit tracking	full
Cash position (intraday)	Bank balance per account only (BankAccountTable.Balance), no intraday	FF7A cash position, FQM_FLOW (One Exposure), camt.052 intraday	camt.052 ingestion, CashPosition view by entity/currency/account	improved
Cash flow forecast	LedgerCov, LedgerCovDocument, Cash flow forecasting setup, workspace Cash overview	FF7B liquidity forecast, FQM planning levels/groups, Cash Management liquidity planner	`cashforecast-svc` buckets daily/weekly/monthly from AP/AR/recurring sources	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Intercompany netting, cash pooling	Not native (centralized payments only)	IHC In-House Cash, cash pooling (concentration) in S/4 Cash Management	Netting run creating IC settlement journals; notional pool structures	partial (no IHC virtual accounts)
Bank connectivity	None native (file-based via DMF)	Bank Communication Management (BCM), Multi-Bank Connectivity (MBC)	`bank-connect-svc` adapters SFTP/EBICS 3.0/PSD2 XS2A, Vault keys	improved

7.36.3 Functional requirements

- **FR-36.1 Bank account master** — Must. The system shall maintain bank accounts per legal entity with IBAN/BIC or routing/account numbers, currency, GL main account, bank transaction types, signatories and a change-approval workflow.
  - Notes: IBAN mod-97 and ABA routing checksum validated on save; account numbers stored encrypted at rest (pgcrypto, key in Vault) and shown masked except to the `TreasuryBankAccountMaintain` duty; status lifecycle Draft → Active → Inactive → Closed; changes to number/currency/GL account require 2-step approval (Temporal workflow); maximum 500 bank accounts per legal entity.
- **FR-36.2 Statement import** — Must. The system shall import bank statements in camt.053.001.02/08, BAI2, MT940 and configurable CSV, validate structure, detect duplicates and persist header, balances (OPBD/CLBD/CLAV) and entries.
  - Notes: files up to 50 MB and 200,000 entries; duplicate detection on (bankAccount, statementId or sequence, closingBalance); gap detection on statement sequence numbers raises a warning; each entry retains the raw source fragment for audit; import of 10,000 lines completes in < 30 s.
- **FR-36.3 Matching rule engine** — Must. The system shall match statement lines to Passage bank transactions using ordered rule sets, producing a 0–100 score per candidate and auto-matching above a configurable threshold.
  - Notes: criteria: amount (exact / tolerance absolute or %), value date window (±N days), reference regex capture (end-to-end id, check number, invoice number), counterparty name trigram similarity, bank transaction code map (BTC/domain-family-subfamily); weights sum to 100; default thresholds auto-match ≥ 90, suggest 60–89, unmatched < 60; 1:1, 1:N and N:1 matches with up to 50 lines per side; rules tested against historical statements with a precision/recall report before activation.
- **FR-36.4 ML match suggestions** — Should. The system shall rank unmatched lines' candidates with a model trained on the tenant's confirmed matches and present the top 3 suggestions with reasons.
  - Notes: Python gradient-boosted model in `cashforecast-svc`; retrained nightly if ≥ 200 new confirmed matches; suggestions never auto-post; model inputs and version logged per suggestion.
- **FR-36.5 Reconciliation workspace** — Must. The system shall let users confirm, split, create new transactions (fees, interest) from unmatched lines, mark lines as in-transit, and close the reconciliation when book and bank balances agree.
  - Notes: closing requires `|bank closing − (book balance + outstanding items)| = 0.00`; new transactions created from lines post through the wave-11 journal API with the statement line as source document; reversed matches keep history; reconciliation of 10,000 lines closes in < 60 s.
- **FR-36.6 Check management** — Must. The system shall issue, void, reprint and stop checks with gap-free check numbers per bank account.
  - Notes: void keeps the number consumed; reprint issues a new number and voids the old; status Created → Printed → Sent → Cleared / Voided / Stopped; positive pay file (wave 19 format) regenerated on void.
- **FR-36.7 Letters of credit and guarantees** — Must. The system shall record import/export letters of credit and bank guarantees against bank facilities with limits, expiry and amendments, and link them to purchase or sales orders.
  - Notes: facility utilization = Σ open instruments; new instrument blocked when utilization + amount > limit unless duty `TreasuryFacilityOverride`; expiry alerts 30/7/1 days; margin deposits post via the journal.
- **FR-36.8 Cash position** — Must. The system shall present intraday position per legal entity, currency and bank account from the latest camt.052 or statement balance plus same-day unposted payments and receipts.
  - Notes: refreshed on every inbound file and at least hourly; converted to entity reporting currency at the wave-12 spot rate; age of each balance displayed; stale (> 24 h) balances flagged.
- **FR-36.9 Cash flow forecast** — Must. The system shall compute forecast buckets (daily 30 days, weekly 13 weeks, monthly 12 months) from registered sources.
  - Notes: sources: AP open items (due date adjusted by vendor average days-late), AR open items (due date + customer DSO delta), open POs/SOs not yet invoiced (expected invoice date + terms), recurring entries, manual adjustments, later payroll (43) and projects (41); each source implements `ForecastSource.Contribute(asOf) []Contribution`; every bucket total equals the sum of its contributions (property test); 1M open items forecast in < 5 min.
- **FR-36.10 Intercompany netting** — Should. The system shall run a netting cycle across participating entities, net IC receivables/payables per pair and currency, and create settlement journals.
  - Notes: cycle statuses Proposed → Approved → Settled; netting rate = wave-12 rate on settlement date; residual below 1.00 in currency written off to IC rounding account.
- **FR-36.11 Cash pooling** — Could. The system shall model notional and physical (zero-balancing) pool structures and generate sweep transfers.
  - Notes: target balance per sub-account; sweep creates payment (wave 19) with pool header as counterparty.
- **FR-36.12 Bank connectivity** — Must. The system shall send payment files and receive statements through SFTP, EBICS 3.0 (H005) and PSD2/Berlin Group XS2A adapters with keys and certificates in Vault.
  - Notes: signing via Vault transit engine; private keys never leave Vault; every outbound file hashed (SHA-256) and logged with user, time, channel and bank acknowledgment; retries with exponential backoff up to 24 h; sandbox profiles per adapter.

7.36.4 Data model

BankAccount

Field	Type	Required	Notes
id	uuid	yes	PK
legal_entity_id	uuid	yes	FK org (wave 6)

Field	Type	Required	Notes
account_code	text(20)	yes	unique per entity
iban / account_number	text, encrypted	yes	one of the two; masked in API
bic / routing_number	text	no	validated
currency	char(3)	yes	ISO 4217
main_account_id	uuid	yes	FK ledger account (wave 11)
status	enum	yes	Draft/Active/Inactive/Closed
next_check_number	bigint	no	gap-free sequence
signatories	jsonb	no	[{partyId, limit, currency}]

BankStatementLine

Field	Type	Required	Notes
id	uuid	yes	PK
statement_id	uuid	yes	FK BankStatement
line_no	int	yes	unique in statement
booking_date / value_date	date	yes	
amount	numeric(19,4)	yes	signed, statement currency
btc_code	text	no	ISO 20022 domain/family/subfamily
references	jsonb	no	EndToEndId, InstrId, Ustrd, chequeNumber
counterparty_name	text	no	normalized for trigram index
status	enum	yes	Unmatched/Suggested/Matched/Created/InTransit
raw_fragment	text	yes	source XML/text slice

ReconciliationMatch

Field	Type	Required	Notes
id	uuid	yes	PK
reconciliation_id	uuid	yes	FK ReconciliationHeader
statement_line_ids	uuid[]	yes	1..50
bank_transaction_ids	uuid[]	yes	1..50
score	smallint	yes	0–100
rule_id	uuid	no	null when manual
matched_by	text	yes	user or `rule-engine`
matched_at	timestamptz	yes	
reversed_at	timestamptz	no	

CashForecastEntry: id, legal_entity_id, source_type (AP/AR/PO/SO/Recurring/Manual/Payroll/Project), source_id, expected_date, amount, currency, confidence (0–1), bucket_key, computed_at.

Invariants: Σ amounts of matched statement lines = Σ amounts of matched bank transactions ± configured tolerance; a bank transaction belongs to at most one non-reversed match; statement closing balance = opening balance + Σ lines; a closed reconciliation is immutable. Indexes: statement lines by (bank_account_id, value_date), GIN trigram on counterparty_name, partial index on status = Unmatched; CashForecastEntry partitioned monthly by expected_date, retention 24 months.

7.36.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/treasury/bank-accounts	Create account	Idempotency-Key	TreasuryBankAccountMaintain
POST	/api/v1/treasury/statements/import	Upload statement file (multipart)	file SHA-256	TreasuryStatementImport
POST	/api/v1/treasury/reconciliations/{id}/run-rules	Execute rule set	runId	TreasuryReconcile
POST	/api/v1/treasury/reconciliations/{id}/matches	Manual match	Idempotency-Key	TreasuryReconcile
POST	/api/v1/treasury/reconciliations/{id}/close	Close	state check	TreasuryReconcileApprove
POST	/api/v1/treasury/checks/{id}/void	Void check	state check	TreasuryCheckMaintain
GET	/api/v1/treasury/cash-position?entity=&asOf=	Position	n/a	TreasuryView
GET	/api/v1/treasury/cash-forecast?entity=&horizon=	Forecast buckets	n/a	TreasuryView
POST	/api/v1/treasury/netting-cycles	Propose netting	Idempotency-Key	TreasuryNetting
POST	/api/v1/treasury/connections/{id}/send	Send file via channel	file hash	TreasuryBankSend

Commands: `ImportStatement` (validates schema, rejects duplicates), `RunMatching` (rule set must be Active), `ConfirmMatch` (amount tolerance ≤ rule tolerance), `CloseReconciliation` (difference = 0). Queries served from read models.

Event	Trigger	Payload fields	Consumers
`dev.passage.treasury.statement.imported.v1`	import committed	statementId, bankAccountId, lineCount, openingBalance, closingBalance	cashforecast-svc, treasury-svc matcher

Event	Trigger	Payload fields	Consumers
`dev.passage.treasury.statement.reconciled.v1`	reconciliation closed	statementId, matchedCount, createdCount, closingBalance	journal, analytics
`dev.passage.treasury.check.voided.v1`	void	checkId, bankAccountId, number, reason	AP (19) positive pay
`dev.passage.treasury.cashposition.updated.v1`	position refresh	legalEntityId, asOf, balances[]	dashboards, interco-svc
`dev.passage.treasury.file.sent.v1`	outbound file acknowledged	connectionId, fileHash, channel, ackStatus	audit

7.36.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Bank accounts	Create, edit (workflow), close, view balances	code, IBAN (masked), currency, GL account, status, last statement	TreasuryBankAccountMaintain
Reconciliation workspace	Run rules, confirm, split, create transaction, close	line date/amount/reference, candidate score, difference, unmatched count	TreasuryReconcile
Checks	Print, void, reprint, stop	number, payee, amount, status	TreasuryCheckMaintain
Facilities and instruments	Create LC/guarantee, amend, close	facility, limit, utilization, expiry	TreasuryFacilityMaintain
Cash position	Filter entity/currency, drill to statement	account, balance, age, same-day pending	TreasuryView
Cash forecast	Change horizon, add manual entry, drill to source	bucket, inflow, outflow, net, cumulative	TreasuryView
Bank connections	Test, send, view log	channel, status, last file, ack	TreasuryBankSend

Interaction rules: the reconciliation grid supports keyboard `m` (match selected), `u` (unmatch), `n` (new transaction); multi-select up to 50 lines each side with running difference shown; rule changes require re-run before close; column layouts and filters persist per user via the wave-5 shell personalization.

7.36.7 Business rules and edge cases

- **BR-36.1** Score = Σ (criterion weight × criterion result ∈ [0,1]); amount tolerance criterion result = 1 − |Δ|/tolerance clipped at 0; date window result = 1 − |days|/window.
- **BR-36.2** Ties above the auto threshold with more than one candidate are downgraded to Suggested; auto-match requires a unique top candidate at least 5 points above the next.
- **BR-36.3** Reconciliation state machine: Open → InProgress → Closed; Closed → Reopened only by `TreasuryReconcileApprove` within the open fiscal period; reopening reverses nothing, it only allows new matches.
- **BR-36.4** Statement currency ≠ bank account currency is rejected; FX differences between book and bank on foreign receipts post to realized gain/loss (wave 12) with the statement line as source.
- **BR-36.5** A void after Sent and before Cleared triggers a positive pay void record; void after Cleared is rejected.
- **BR-36.6** Forecast bucket boundaries use the legal entity's calendar; a contribution whose expected date falls on a non-banking day moves to the next banking day.
- **BR-36.7** Concurrency: two users confirming matches on the same line receive 409 on the second write (optimistic version on StatementLine).
- **BR-36.8** Duplicate statement file with identical hash returns 200 with the existing statement id; same statement id with different closing balance returns 409.
- **BR-36.9** Facility utilization uses the instrument's original currency converted at the daily rate; limit breaches recomputed nightly and on rate change > 2%.

7.36.8 Acceptance criteria

- **AC-36.1** (FR-36.2) Given a camt.053 file with 1,000 entries, when imported, then 1,000 lines persist, opening + Σ lines = closing, and a re-upload returns the same statement id.
- **AC-36.2** (FR-36.3) Given the 1,000-line reconciliation fixture, when rules run, then ≥ 950 lines are auto-matched correctly and 0 are matched to the wrong transaction.
- **AC-36.3** (FR-36.5) Given a line for a USD 12.50 bank fee, when the user creates a transaction, then a journal entry Dr Bank fees 12.50 / Cr Bank 12.50 posts with the line as source document.
- **AC-36.4** (FR-36.5) Given a difference of 0.01, when closing, then the close is rejected with the difference displayed.
- **AC-36.5** (FR-36.6) Given check 1001 Printed, when voided and reprinted, then 1001 is Voided and 1002 is Printed with the same payee and amount.
- **AC-36.6** (FR-36.7) Given a facility of EUR 1,000,000 with EUR 950,000 utilized, when an LC of EUR 100,000 is created, then it is blocked without the override duty.
- **AC-36.7** (FR-36.9) Given 10,000 generated open items, when the forecast computes, then every bucket equals the sum of its contributions to the cent.
- **AC-36.8** (FR-36.12) Given an EBICS sandbox, when a pain.001 is sent, then the file hash, signature reference and bank acknowledgment are in the audit log and no private key bytes appear in any log or DB.

7.36.9 Test plan and fixtures

Fixtures: `recon-1000` (1,000 statement lines across camt.053/BAI2/MT940 with 60% exact, 25% reference-only, 10% N:1 batch receipts, 5% fees, generated from wave 19/30 payment fixtures); `forecast-10k` (generated open items with due dates across 12 months); `ebics-sandbox` (vendor sandbox recordings for contract tests); `facility-20` (LCs and guarantees with amendments).

Test	Type	Target	Pass criterion
Scoring function	Unit	rule engine	deterministic scores per criterion table
Statement parsers	Contract	camt.053 v02/v08, BAI2, MT940	100% of fixture files parse; schema errors reported with line
Match uniqueness	Property	reconciliation	no transaction in two active matches

Test	Type	Target	Pass criterion
Forecast additivity	Property	cashforecast-svc	bucket = Σ contributions
Reconcile-and-close	E2E	workspace	recon-1000 closes with difference 0
Import throughput	Performance	import API	10,000 lines < 30 s; 200,000 lines < 10 min
Adapter contracts	Contract	SFTP/EBICS/XS2A	sandbox round-trip with acknowledgment

Reconciliation checks: Σ BankAccount book balance = GL bank main account balance per entity nightly; Σ outstanding checks = positive pay open items; cash position = latest statement closing + pending.

7.36.10 Deployment and operations

Helm charts `treasury-svc`, `bank-connect-svc`, `cashforecast-svc`; config: `TREASURY_AUTO_MATCH_THRESHOLD=90`, `TREASURY_MAX_FILE_MB=50`, Vault role `bank-connect`. Atlas migrations generate tables from entity metadata. Feature flags: `treasury.ml-suggestions`, `treasury.cash-pooling`. Scheduled jobs (Temporal): `statement-poll` every 15 min per connection, `cash-position-refresh` hourly, `forecast-rebuild` hourly and on demand, `facility-expiry-check` daily 06:00. SLOs: import p95 < 30 s per 10k lines, rule run p95 < 20 s per 10k lines, forecast API p95 < 300 ms. Alerts: unreconciled lines > 7 days, connection failure > 3 retries, key rotation due. Runbook: resend file, rebuild forecast, rotate EBICS keys. Rollback: Helm rollback; matches are append-only with reversals, so no data rollback.

7.36.11 Migration from AX / SAP

Source	Passage entity	Notes
BankAccountTable, BankAccountTrans (open items)	BankAccount, BankTransaction	opening balance as of cutover; unreconciled items migrated as InTransit
BankChequeTable	Check	status map; next number continues
BankLCImport/Export, BankLGTable, BankFacility	LetterOfCredit, Guarantee, Facility	open instruments only
T012/T012K, BAM accounts	BankAccount	house bank + account id → account code
FEBKO/FEBEP (last 90 days)	BankStatement, StatementLine	for matching history and ML training
PAYR	Check	
LedgerCov / FQM_FLOW	not migrated	forecast is recomputed from sources

Validation: Σ migrated bank balances = GL bank accounts; IBAN checksums pass. Not migrated: closed reconciliations older than 90 days, matching rule configurations (re-authored, since scoring semantics differ).

7.36.12 Dependencies, risks, open questions

Depends on: wave 11 journal posting, wave 12 rates, wave 19 payment files and positive pay, wave 30 cash application, wave 35 integration hub for SFTP transport, wave 5 Vault integration.

Risk	Likelihood	Impact	Mitigation
Bank-specific camt dialects	High	Medium	Per-bank profile overrides; fixture per bank
ML suggestions mislead users	Medium	Low	Suggestions only, reasons shown, precision tracked

Open questions: (1) Default auto-match threshold 90 or 95? Proposed 90; decide by wave 36 build start. (2) EBICS versions to support (2.5 and 3.0)? Proposed 3.0 only; decide by wave 36. (3) Cash pooling in v1? Proposed Could; decide at phase 7 exit.

7.37 Wave 37: Fixed Assets & Leases

7.37.1 Scope and business context

Wave 37 adds the asset subledger. Capital spend currently lands on expense or clearing accounts; after this wave an accountant acquires assets from AP invoices, projects (CIP) or inventory, depreciates them monthly under several books (local GAAP, IFRS, tax) with different methods and conventions, transfers, splits, revalues, impairs and disposes of them with correct gain/loss, and accounts for leases under IFRS 16 / ASC 842 with right-of-use assets and lease liabilities. A physical inventory with barcode scanning reconciles the register to what exists.

Out of scope: enterprise asset maintenance (wave 44), tax depreciation reports beyond MACRS/ADS tables for the US and straight-line for other v1 countries (country packs in wave 48), component accounting for infrastructure-grade asset hierarchies deeper than 3 levels.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX depreciation proposal → LedgerJournalTrans → post: three forms, and one bad asset aborts the whole journal with an infolog	Journal-centric batch without per-asset isolation	Temporal run with per-asset checkpoints; failing assets reported, others post	Failed runs per month 2 → 0; run restart time 1 h → < 1 min
SAP AFAB runs in background only, errors in SM37 spool; test run then real run, 2–4 h for 100k assets	Batch job architecture	Online proposal preview with formula inputs; 1M asset-periods/h	Run time for 100k assets 2 h → < 6 min
AssetBook convention changes in D365 silently recalculate history; auditors cannot see why a prior charge changed	No immutable transaction log per book	Append-only AssetTransaction; prospective changes only, with explanation	Unexplained changes in audit sample 100% traceable
Mid-quarter test performed in Excel because neither AX nor SAP computes the 40% Q4 test	Tax rules outside the engine	Automatic test per tax book per year with delta posting	Manual tax workbooks: 1 per entity per year → 0
Acquisition from AP needs the invoice line fixed-asset fields, then an asset journal; SAP needs F-90 with asset account assignment keyed again	Two-step with re-keying	AP invoice line with asset flag posts directly to the book	Clicks 18 → 3; re-keyed acquisitions 0

7.37.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Asset master and groups	AssetTable, AssetGroup, AssetGroupBookSetup, form AssetTable	ANLA, ANLZ (time-dependent), asset classes ANKA/ANKT (OAOA), AS01/AS02/AS03	Asset, AssetClass entities	full
Books / depreciation areas	AssetBook (D365 books; AX 2012 value models + AssetDepBook), posting layer per book	ANLB per depreciation area (OADB), T093 chart of depreciation, ledger assignment	AssetBook per asset × book definition, posts to ledger layers (wave 12)	full
Depreciation profiles	AssetDepreciationProfile (StraightLine, ReducingBalance, Manual, Factor, Consumption, StraightLineLifeRemaining, 200%/175%/150%/125% DB, RB/SL), conventions	Depreciation keys T090NA/T090NAZ (AFAMA), multilevel methods T090NP, period controls T090NR	DepreciationProfile with method, convention, factor, tables	full
Depreciation run	AssetDepreciationProposal → LedgerJournalTrans (Asset journal), AssetTrans	AFAB / FAA_DEPRECIATION_POST, ANLP, ANLC	Depreciation run (Temporal) → journal entry per book/period	improved (restartable, per-asset diagnostics)
Acquisition from AP / project / inventory	PurchLine fixed asset fields, ProjTable capitalize, InventJournal to fixed asset	F-90, ABZON, MIGO with asset account assignment (A), AIAB/AIBU AuC settlement	AcquisitionSource links from wave 18 invoice, wave 41 project, wave 16 inventory journal	full
Transfer / split	AssetTransfer, "Split fixed asset" (D365)	ABUMN (intra/inter-company), partial transfer	Transfer and Split transactions preserving history	full
Disposal, revaluation, impairment	AssetTrans types DisposalSale, DisposalScrap, WriteUpAdj, WriteDownAdj, Revaluation	ABAON, ABAVN, F-92, ABAW revaluation, impairment via write-down	Disposal (sale/scrap, partial), Revaluation, Impairment	full
Leases	D365 Asset leasing (AssetLeaseLease, AssetLeaseBook, AssetLeasePaymentSchedule, AssetLeaseLiabilityAmortization)	RE-FX Contract and Lease Management (RECN, RECEEP valuation), IFRS 16 valuation rules	Lease, LeaseSchedule, ROU asset auto-created	full
Physical inventory	Not native (AX 2012 lacked); D365 none; partner add-ons	ANLA-INVNR only; third-party scanning	Count sessions with PWA barcode scanning	improved

7.37.3 Functional requirements

- **FR-37.1 Asset master** — Must. The system shall maintain assets with class, location, responsible worker, serial/barcode, parent/child (≤ 3 levels), and default dimensions.
  - Notes: asset number from wave-5 number sequences; status NotYetAcquired → Open → Sold / Scrapped / Transferred; barcode unique per legal entity.
- **FR-37.2 Books and profiles** — Must. The system shall attach N books per asset (default from class), each with a depreciation profile, service life, salvage value, convention, posting layer and ledger.
  - Notes: methods: straight-line (service life and life-remaining variants), declining balance (factor 125/150/175/200%, with or without switch to SL), units of production, manual, MACRS tables (GDS 3/5/7/10/15/20-year, ADS); conventions: none, half-year, mid-month (1st or 15th), mid-quarter, full-month, half-year next year; up to 10 books per asset.
- **FR-37.3 Straight-line formula** — Must. The system shall compute period depreciation as `(cost − salvage − accumulated adjustments) / remaining periods` with the convention applied to the first and last periods.
  - Notes: worked mid-month example: cost USD 12,000, salvage 0, 5-year life, in service 2026-03-10, mid-month (15th) convention. Monthly charge = 12,000 / 60 = 200.00. March 2026 takes half a month: 100.00. FY2026 = 100 + 9 × 200 = 1,900.00; FY2027–FY2030 = 2,400.00 each (9,600.00); FY2031 = Jan 200 + Feb 200 + Mar 100 = 500.00. Total 12,000.00. The last period absorbs any rounding residual so Σ depreciation = depreciable basis exactly.
- **FR-37.4 MACRS** — Must. The system shall compute MACRS using the declining-balance-with-switch formula and verify it against the IRS Publication 946 percentage tables.
  - Notes: worked 5-year GDS, half-year convention, cost USD 25,000: rate = 200%/5 = 40%. Y1 = 25,000 × 40% × ½ = 5,000.00 (20.00%). Y2 = 20,000 × 40% = 8,000.00 (32.00%). Y3 = 12,000 × 40% = 4,800.00 (19.20%). Y4: DB = 7,200 × 40% = 2,880.00; SL over remaining 2.5 years = 7,200 / 2.5 = 2,880.00 → switch (11.52%). Y5 = 2,880.00 (11.52%). Y6 = 1,440.00 (5.76%). Σ = 25,000.00. Mid-quarter applies when > 40% of the year's basis is placed in Q4; the system computes the test per tax book per year and alerts. Bonus depreciation percentage and Section 179 amounts recorded as first-year adjustments.
- **FR-37.5 Depreciation run** — Must. The system shall propose and post depreciation per book, period and legal entity as one journal entry per book with lines per asset (or summarized per account/dimension by option).
  - Notes: Temporal workflow with per-asset checkpoints; rerun for the same book/period is rejected unless the prior run is reversed; 1,000,000 asset-book-periods per hour; proposal preview shows per-asset amount and formula inputs.
- **FR-37.6 Acquisition sources** — Must. The system shall create acquisitions from AP invoice lines flagged as asset, project CIP capitalization, inventory journal issue to asset, and manual journal.
  - Notes: AP invoice with asset flag posts Dr Asset cost / Cr AP (instead of expense); CIP collects to the asset's under-construction book value and capitalizes on a date with transfer from CIP account; inventory issue uses the item's current cost (wave 20).
- **FR-37.7 Transfer and split** — Must. The system shall transfer assets between dimensions, locations or legal entities and split by percentage or amount into new assets with proportional cost and accumulated depreciation.
  - Notes: intercompany transfer posts disposal in the source and acquisition in the target at NBV or agreed price (wave 40 IC due-to/due-from); split percentages must sum to 100.00%.
- **FR-37.8 Disposal** — Must. The system shall dispose assets by sale (to a customer invoice, wave 29, or free text) or scrap, including partial disposal, posting cost reversal, accumulated depreciation reversal and gain/loss.
  - Notes: gain/loss = proceeds − NBV of the disposed portion; partial disposal uses the disposed share of cost and accumulated depreciation; depreciation up to the disposal date is posted first when `depreciate-on-disposal` is set per book.
- **FR-37.9 Revaluation and impairment** — Must. The system shall post revaluation (IAS 16 revaluation model) to OCI reserve or P&L per policy and impairment (IAS 36) as write-down with reversal limits.
  - Notes: impairment reversal cannot exceed the carrying amount that would have existed without impairment.

- **FR-37.10 Leases (IFRS 16 / ASC 842)** — Must. The system shall record leases with payment schedules, incremental borrowing rate, options, and generate the liability amortization and ROU depreciation schedules, posting initial recognition, monthly interest, payments and depreciation.
  - Notes: worked example, annual payments in arrears USD 10,000 for 5 years, IBR 5%: liability = 10,000 × (1 – 1.05^{-5})/0.05 = 43,294.77. Schedule: Y1 interest 2,164.74, closing 35,459.51; Y2 1,772.98 → 27,232.49; Y3 1,361.62 → 18,594.11; Y4 929.71 → 9,523.82; Y5 476.18 (plugged from 476.19 to clear the 0.01 residual) → 0.00. Total interest 6,705.23 = 50,000 – 43,294.77. ROU asset 43,294.77 depreciated straight-line 8,658.95 for Y1–Y4 and 8,658.97 in Y5. Monthly schedules use periodic rate = nominal/12 by default (option: effective (1+r)^(1/12) – 1). Modifications remeasure the liability with the ROU adjusted by the same amount; short-term (≤ 12 months) and low-value (< USD 5,000 / EUR 5,000) leases are expensed under election.
- **FR-37.11 Physical inventory** — Should. The system shall run count sessions with the PWA scanner, record found/missing/moved assets and create transfer or scrap proposals.
- **FR-37.12 Asset budgeting** — Should. The system shall forecast depreciation for existing assets and planned acquisitions and publish it as a budget source to wave 38.
- **FR-37.13 Subledger reconciliation** — Must. The system shall reconcile Σ asset cost and Σ accumulated depreciation per book to the GL accounts nightly and list differences by asset.

7.37.4 Data model

Asset: id, legal_entity_id, asset_number (unique per entity), class_id (FK AssetClass), name, status, location_id, responsible_worker_id, barcode, parent_asset_id, default_dimensions jsonb, acquired_date, quantity numeric(19,4).

AssetBook

Field	Type	Required	Notes
id	uuid	yes	PK
asset_id	uuid	yes	FK Asset
book_definition_id	uuid	yes	FK BookDefinition (ledger, posting layer, currency)
profile_id	uuid	yes	FK DepreciationProfile
service_life_periods	int	yes	1..1200
salvage_value	numeric(19,4)	yes	default 0
convention	enum	yes	None/HalfYear/MidMonth1/MidMonth15/MidQuarter/FullMonth/HalfYearNextYear
in_service_date	date	yes	
cost, accumulated_depreciation, nbv	numeric(19,4)	yes	projections from AssetTransaction
last_depreciated_period	text	no	fiscal period key
status	enum	yes	NotYetAcquired/Open/Closed/Sold/Scrapped

AssetTransaction

Field	Type	Required	Notes
id	uuid	yes	PK, append-only
asset_book_id	uuid	yes	FK
type	enum	yes	Acquisition/AcquisitionAdj/Depreciation/DepreciationAdj/Transfer/Split/Revaluation/Impairment/DisposalSale/DisposalScrap/CIPCapitalization
transaction_date	date	yes	
amount	numeric(19,4)	yes	signed
journal_entry_id	uuid	yes	FK wave-11 journal
source_type, source_id	text, uuid	no	AP invoice, project, inventory journal, lease
reversal_of_id	uuid	no	
run_id	uuid	no	depreciation run

Lease: id, legal_entity_id, lease_number, lessor_party_id, commencement_date, term_months, payment_frequency, payment_amount, payment_timing (arrears/advance), ibr_rate numeric(9,6), classification (Finance/Operating for ASC 842), rou_asset_id, status (Draft/Active/Modified/Terminated), election flags.

LeaseScheduleLine: lease_id, period_no, opening_liability, interest, payment, closing_liability, rou_depreciation, posted_journal_id.

Invariants: Σ AssetTransaction amounts by type = projected cost / accumulated depreciation; NBV ≥ 0 unless revaluation policy allows; Σ depreciation over life = cost – salvage; LeaseScheduleLine closing of period n = opening of n+1 and final closing = 0.00. Indexes: AssetTransaction by (asset_book_id, transaction_date), by run_id; AssetBook by (book_definition_id, status).

7.37.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/assets	Create asset	Idempotency-Key	AssetMaintain
POST	/api/v1/assets/{id}/books	Add book	Idempotency-Key	AssetMaintain
POST	/api/v1/assets/depreciation-runs	Propose run (book, period)	(book, period) unique	AssetDepreciate
POST	/api/v1/assets/depreciation-runs/{id}/post	Post proposal	state check	AssetPostJournal
POST	/api/v1/assets/{id}/transactions	Acquisition/adjustment/revaluation/impairment	Idempotency-Key	AssetPostJournal
POST	/api/v1/assets/{id}/dispose	Sale/scrap, partial %	Idempotency-Key	AssetDispose

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/assets/{id}/transfer	Transfer / split	Idempotency-Key	AssetMaintain
GET	/api/v1/assets/{id}/schedule?book=	Projected depreciation	n/a	AssetView
POST	/api/v1/leases	Create lease and compute schedule	Idempotency-Key	LeaseMaintain
POST	/api/v1/leases/{id}/modify	Remeasure	Idempotency-Key	LeaseMaintain
POST	/api/v1/assets/count-sessions	Physical inventory	Idempotency-Key	AssetCount

Validation: book must be Open and period open per ledger × module (wave 14); disposal percentage in (0,100]; lease IBR in (0, 50%].

Event	Trigger	Payload fields	Consumers
`dev.passage.assets.asset.acquired.v1`	acquisition posted	assetId, bookIds[], cost, sourceType, sourceId	budget-svc, analytics
`dev.passage.assets.depreciation.posted.v1`	run posted	runId, bookDefinitionId, period, assetCount, totalAmount, journalEntryId	costacct-svc, analytics
`dev.passage.assets.asset.disposed.v1`	disposal posted	assetId, bookId, percentage, proceeds, nbv, gainLoss	tax, analytics
`dev.passage.assets.lease.recognized.v1`	lease activated	leaseId, liability, rouAsset, termMonths, rate	analytics

7.37.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Fixed assets workspace	New asset, run depreciation, exceptions, reconciliation tile	assets by status, pending acquisitions, unreconciled difference	AssetView
Asset detail	Books tab, transactions tab, schedule, dispose, transfer	number, class, NBV per book, location	AssetMaintain
Depreciation run	Propose, review, post, reverse	book, period, asset count, total, exceptions	AssetDepreciate
Lease detail	Schedule, modify, terminate	liability, ROU, interest, next payment	LeaseMaintain
Count session (PWA)	Scan, mark found/missing, submit	barcode, location, found at	AssetCount

Interaction rules: schedule preview recalculates on any book field change before save; bulk class change via grid multi-select with validation at submit; depreciation run proposals paginate 500 assets per page with per-asset formula tooltip.

7.37.7 Business rules and edge cases

- **BR-37.1** Period depreciation rounds half-up to currency minor unit; the final period of the life is a plug so Σ = depreciable basis.
- **BR-37.2** Asset state machine: NotYetAcquired → Open (first acquisition) → Sold | Scrapped | Transferred (full IC transfer); Open → Open on partial disposal; no transaction on Sold/Scrapped except reversal of the disposal.
- **BR-37.3** Depreciation for a period cannot post if a prior period for the same book is unposted (sequential), except when the book's `catch-up` option is set, which posts the arrears into the current period.
- **BR-37.4** Change of service life or salvage after acquisition applies prospectively: new charge = (NBV – salvage) / remaining periods.
- **BR-37.5** Mid-quarter test: if Q4 in-service basis / full-year basis > 40% for a tax book, all assets of that year switch to mid-quarter; assets already depreciated are recomputed and the delta posts in the current period.
- **BR-37.6** Disposal date before last depreciated period requires reversal of later depreciation first; the system offers the reversal set.
- **BR-37.7** Partial disposal percentage p: disposed cost = cost × p, disposed accumulated depreciation = acc × p, rounded; residual cents stay on the remaining asset.
- **BR-37.8** Lease liability remeasurement on modification: new PV at revised rate (if scope change) or original rate (if only payment change under ASC 842 for operating leases); ROU adjusts by the same amount; if ROU would go below zero the excess hits P&L.
- **BR-37.9** Concurrency: the depreciation run holds a per-(book, period) Postgres advisory lock; a second run request returns 409.

7.37.8 Acceptance criteria

- **AC-37.1** (FR-37.3) Given the USD 12,000 mid-month example, when the schedule is generated, then FY2026 = 1,900.00, FY2027–2030 = 2,400.00 each, FY2031 = 500.00.
- **AC-37.2** (FR-37.4) Given the USD 25,000 MACRS 5-year example, when computed, then the yearly amounts are 5,000.00 / 8,000.00 / 4,800.00 / 2,880.00 / 2,880.00 / 1,440.00.
- **AC-37.3** (FR-37.4) Given 100 fixture assets across methods and conventions, when depreciated for full lives, then every amount matches the reference table to the cent.
- **AC-37.4** (FR-37.5) Given 10,000 assets, when the monthly run posts, then one journal entry per book exists, Σ lines = Σ proposal, and a second run for the same period is rejected.
- **AC-37.5** (FR-37.8) Given cost 10,000, accumulated 4,000, proceeds 7,000, 50% disposal, when disposed, then cost –5,000, accumulated +2,000, gain 4,000 posts and the remaining NBV is 3,000.
- **AC-37.6** (FR-37.10) Given the 5-year USD 10,000 lease at 5%, when activated, then liability = 43,294.77, Y5 interest = 476.18 and closing = 0.00.
- **AC-37.7** (FR-37.10) Given 25 fixture leases, when schedules compute, then every line matches the accountant workbook to the cent.
- **AC-37.8** (FR-37.13) Given the demo entity after a month, when reconciliation runs, then Σ asset cost and Σ accumulated depreciation equal the GL accounts with 0.00 difference.

7.37.9 Test plan and fixtures

Fixtures: `dep-100` (100 assets × SL, DB 150/200%, UoP, MACRS 3/5/7-year, conventions, short first years, salvage > 0), `lease-25` (accountant workbook: arrears/advance, monthly/annual, modifications, terminations), `disposal-30` (full/partial, before and after period end), `migration-nbv` (opening NBVs for the demo entity).

Test	Type	Target	Pass criterion
Method formulas	Unit	depreciation engine	match tables to the cent
Life totals	Property	engine	Σ depreciation = cost − salvage for random inputs
Lease PV and schedule	Unit/Property	lease engine	closing = 0.00; Σ interest = Σ payments − PV
Run posting	Contract	journal API	one entry per book; idempotent
Disposal flows	E2E	workspace	gain/loss per fixture
Run throughput	Performance	run	1M asset-book-periods/hour

Reconciliation checks: asset subledger to GL per book nightly; lease liability schedule to GL lease liability; ROU asset to GL ROU accounts.

7.37.10 Deployment and operations

Helm chart `asset-svc`; env `ASSET_RUN_BATCH_SIZE=500`, `ASSET_MAX_BOOKS=10`; Atlas migrations; flags `assets.physical-inventory`, `assets.asc842-operating`. Jobs: `asset-gl-reconcile` nightly 02:00, `lease-schedule-post` monthly on period open, `mid-quarter-test` on tax year end. SLOs: schedule API p95 < 200 ms; run throughput as above; reconciliation job < 10 min for 1M assets. Alerts: reconciliation difference ≠ 0, run failure, lease schedule unposted after period close. Rollback: Helm rollback; posted runs reversed through the reversal API, never deleted.

7.37.11 Migration from AX / SAP

Source	Passage entity	Notes
AssetTable, AssetGroup	Asset, AssetClass	
AssetBook / AssetDepBook, AssetDepreciationProfile	AssetBook, DepreciationProfile	method/convention mapped; opening NBV as acquisition + accumulated depreciation transactions dated cutover
AssetTrans	AssetTransaction (summarized)	history summarized to opening balances; detail retained in archive
AssetLease*	Lease, LeaseScheduleLine	remaining schedule recomputed; variance to legacy shown
ANLA, ANLZ, ANKA	Asset, AssetClass	
ANLB, ANLC, T090NA	AssetBook, DepreciationProfile	depreciation key → profile map table; ANLC year values → opening balances
ANLP, ANEP	AssetTransaction (summarized)	
RE-FX contracts	Lease	

Validation: Σ opening NBV per book = GL; count of active assets matches; recomputed next-period depreciation within 0.5% of legacy (differences listed). Not migrated: fully depreciated and retired assets older than 7 years (archive), AX value model posting profiles (re-authored).

7.37.12 Dependencies, risks, open questions

Depends on: wave 11 journal and dimensions, wave 12 posting layers, wave 14 period control, wave 18 AP invoice, wave 16 inventory journals, wave 36 for IC transfer settlement; forward-references wave 41 projects (CIP) and wave 44 EAM.

Risk	Likelihood	Impact	Mitigation
Legacy NBV differs from recomputed schedules	High	Medium	Load NBV as balances, not recomputed; variance report
ASC 842 operating lease single-cost rule complexity	Medium	Medium	Flag-gated; IFRS 16 first

Open questions: (1) Posting summarization default per account or per asset? Proposed per asset; decide by wave 37 start. (2) Support 175%/125% DB in v1? Proposed yes (cheap); decide by wave 37. (3) Bonus depreciation handled as profile or adjustment? Proposed adjustment; decide by wave 37.

7.38 Wave 38: Cost Accounting, Budgeting & Budget Control

7.38.1 Scope and business context

Wave 38 gives controllers a planning and control layer separate from statutory accounting. Before it, the universal journal records actuals by dimension but there is no budget to compare against, no way to stop a purchase order that exceeds a department's funds, and no management view that reclassifies and allocates costs the way the business is run. After it, budget register entries and planning scenarios flow through workflow into budget models; budget control checks requisitions, POs and invoices against available funds, recording encumbrances and liquidating them exactly; a cost accounting ledger mirrors primary costs from the journal into cost objects and cost elements, runs assessment/distribution allocations that converge and reconcile, supports activity-based costing and internal orders, and feeds the management reporting cubes consumed by wave 46.

Out of scope: public-sector fund accounting (appropriations, apportionments, year-end encumbrance carry-forward beyond the simple option), profitability analysis segment reporting (CO-PA style, wave 46 semantic layer), project budgets (wave 41).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 budget check on PO happens in a batch-like synchronous call that takes 3–8 s per line on large BudgetSourceTracking tables; buyers disable it	Available-funds query aggregates BudgetSourceTrackingDetail at check time	Per dimension-combination balance cache rebuilt from events; check is one indexed read	Check latency 5 s → p95 < 150 ms

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP FM availability control (FMAVCR01) rebuilds the AVC ledger (FMAVCT) when budget structure changes; FMAVCREINIT takes hours	Derived ledger with offline rebuild	Balance cache with incremental event replay	Rebuild for 1M documents 3 h → < 10 min
Encumbrance liquidation on invoice in AX leaves orphan amounts when the PO was partially received at a different price (BudgetSourceTracking residual)	Relief computed from invoice amount, not from PO reservation	Relief = min(invoice-matched share of PO reservation, remaining reservation); property-tested to the cent	Orphan encumbrances per month 20–50 → 0
KSU5/KSV5 cycles in SAP: a cycle with a circular segment fails with "iteration not converging" and the controller must restructure segments	Fixed 1,000-iteration loop with no exact solve	Exact linear solve ≤ 2,000 senders; residual sweep rule	Failed cycles per close 2–3 → 0
CAM cost accounting ledger in D365 requires "Process data" batch and loads GL into CAMCostEntry hours after posting	Separate ledger filled by batch import	Cost entries mirrored from journal events within seconds	Lag 4–24 h → < 5 s

7.38.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Budget register entries	BudgetTransactionHeader, BudgetTransactionLine, BudgetTransactionCode, BudgetModel, BudgetCycle	FMBB budget entry documents, FMBH; KP06/KP26 plan versions	BudgetRegisterEntry, BudgetModel	full
Budget planning	BudgetPlanHeader/Line, BudgetPlanningProcess, BudgetPlanStage, BudgetPlanScenario, BudgetPlanWorkflow, BudgetPlanTemplate (Excel)	KP06 plan versions, BPC/SAC planning	BudgetPlan with scenarios, stages, Temporal workflow	full
Budget control configuration	BudgetControlConfiguration, BudgetControlRule, BudgetGroup, BudgetControlUserGroup, budget threshold, funds available calculation	FM: FMDERIVE, funds centers FMFCTR, commitment items FMCI, tolerance profiles (OFUP/OF11), AVC ledger FMAVCT	BudgetControlRule with dimension subset, time span, tolerances, overrides	full
Encumbrances / commitments	BudgetSourceTracking, BudgetSourceTrackingDetail, BudgetCheckResult, LedgerSourceDocument encumbrance posting (purchase order encumbrance posting type)	COOI commitments, FMIOI open items, KSB2	Encumbrance entity with reservation/relief lines	full
Cost accounting ledger	CAMCostAccountingLedger, CAMCostObject, CAMCostElement, CAMCostElementDimension, CAMDimensionHierarchy, CAMCostEntry	CO area, KS01/KS02 CSKS, KSH1 hierarchy, CSKA/CSKB cost elements (KA01/KA06), COEP/COBK line items, COSP/COSS totals	CostObject, CostElement, CostEntry, DimensionHierarchy	full
Allocations	CAMCostAllocationPolicy, CAMCostDistributionPolicy, CAMAllocationBase, CAMCostBehaviorPolicy	KSV5 distribution, KSU5 assessment, cycles T811C/T811S/T811K, KSC6 indirect activity allocation	AllocationPolicy (assessment/distribution/reciprocal), AllocationBase	improved (exact solve)
Statistical measures / activity types	CAMStatisticalDimension, statistical measure provider templates	KL01 CSLA activity types, KP26 prices, KB21N	StatisticalMeasure, ActivityType with rates	full
Internal orders	Not native (projects used)	KO01 AUFK internal orders, KO88 settlement	InternalOrder cost object type with settlement rules	full
Cost control workspace	CAMCostControlWorkspace, overhead calculation, variance by cost object	S_ALR_87013611 (cost centers actual/plan/variance), KSB1	Cost control workspace with plan/actual/variance drill	full

7.38.3 Functional requirements

- **FR-38.1 Budget models and register entries** — Must. The system shall record budget register entries (Original, Revision, Transfer, Carry-forward, Pre-encumbrance, Encumbrance reservation) per budget model, cycle, dimension combination and period, with workflow approval.
  - Notes: a transfer must net to zero across its lines; entries post to a budget ledger separate from the universal journal; budget model hierarchies (sub-models) roll up; 2M register lines per model supported.
- **FR-38.2 Budget planning** — Must. The system shall run planning processes with stages, scenarios (up to 10 per plan), top-down and bottom-up aggregation, distribution keys, and Excel import/export, ending in a register entry.
  - Notes: scenarios compared side-by-side; comments per cell; stage transitions by Temporal workflow; budget plan line count up to 500,000 per plan.
- **FR-38.3 Budget control configuration** — Must. The system shall define control rules per legal entity: budget control dimensions (subset of ledger dimensions), budget cycle time span (year / quarter / period / custom), funds-available formula, documents under control (requisition, PO, vendor invoice, GL journal, expense, project), thresholds (warn %, block %), and user-group overrides.
  - Notes: rule activation versioned with effective date; disabling control does not delete history.
- **FR-38.4 Budget check order of operations** — Must. The system shall evaluate a budget check in this fixed order: (1) determine whether the document type and main account range are under control; (2) derive the budget control dimension combination from the line's ledger dimension; (3) resolve the budget cycle and time span containing the line's accounting date; (4) relieve the document's own prior reservation (revision of a PO, invoice matching a PO) so only the net change is checked; (5) compute available = original + revisions + transfers in – transfers out + carry-forward – actuals – encumbrances – pre-encumbrances (– unposted documents if enabled); (6) compare the net change against available with the rule's thresholds: ≤ warn% → pass, between warn% and block% → warn, > block% → fail; (7) apply user-group override (fail → warn for groups with `BudgetOverride`); (8) record BudgetCheckResult and, on pass/warn, reserve the amount atomically.
  - Notes: evaluation uses a per-combination balance row with `SELECT … FOR UPDATE` so two concurrent POs cannot both pass against the last dollar; p95 < 150 ms; results retained 7 years.
- **FR-38.5 Encumbrance lifecycle** — Must. The system shall create pre-encumbrances for approved requisitions, convert to encumbrances on PO confirmation, adjust on PO change, relieve on invoice posting or PO cancellation, and optionally carry forward at year end.
  - Notes: relief = min(invoice quantity × PO unit price, remaining reservation) per PO line; price differences on invoice create a new check for the excess; closed POs release residuals.
- **FR-38.6 Cost accounting ledger** — Must. The system shall maintain cost objects (cost centers, internal orders, products, projects as references), cost elements (primary mapped from main accounts, secondary for allocations), dimension hierarchies, and cost entries mirrored from journal

events within 5 s.
    - Notes: cost behavior (fixed/variable) classified by policy; one ledger per legal entity and currency; versioned hierarchies effective-dated.
- **FR-38.7 Allocation policies** — Must. The system shall run distribution (primary elements retained) and assessment (secondary element) allocations by fixed percentage, statistical measure, or actual cost basis, in ordered steps with reciprocal support.
    - Notes: iteration convergence rule: for reciprocal steps with n senders, when n ≤ 2,000 solve `(I − A) x = b` exactly (LU with partial pivoting); otherwise iterate Gauss–Seidel until max absolute unallocated residual on any sender ≤ 0.01 currency units or 100 iterations; after loop, any residual is swept to that sender's largest receiver; invariant `Σ cost in = Σ cost out` to the cent; rounding residual per step posted to the largest receiver.
- **FR-38.8 Activity-based costing** — Should. The system shall define activity types with planned rates, record activity quantities from production (wave 21) and manual entries, and allocate at rate × quantity with rate variances.
- **FR-38.9 Internal orders** — Should. The system shall create internal orders with budgets, collect costs, and settle to cost centers, assets (wave 37) or P&L by settlement rule.
- **FR-38.10 Cost control workspace** — Must. The system shall show plan vs actual vs committed per cost object and element, with variance %, drill to cost entries and to journal lines, and period comparison.
- **FR-38.11 Reconciliation** — Must. The system shall reconcile Σ primary cost entries per element and period to the journal balances and list differences.
- **FR-38.12 Management cubes** — Should. The system shall publish cost ledger facts and hierarchies to the lakehouse via CDC with a documented star schema for wave 46.

7.38.4 Data model

BudgetRegisterEntry: id, legal_entity_id, budget_model_id, entry_type (Original/Revision/Transfer/CarryForward/PreEncumbrance/Encumbrance), budget_cycle_id, status (Draft/Submitted/Approved/Posted/Rejected), lines[] (dimension_combination_id, period_key, amount numeric(19,4), currency), workflow_id, posted_at.

BudgetControlBalance

Field	Type	Required	Notes
id	uuid	yes	PK
rule_version_id	uuid	yes	FK BudgetControlRule version
control_dimension_key	text	yes	hashed dimension subset
time_span_key	text	yes	e.g. FY2026, 2026-Q3, 2026-09
budget_total	numeric(19,4)	yes	original + revisions + transfers + carry-forward
actuals	numeric(19,4)	yes	from journal events
encumbrances	numeric(19,4)	yes	
pre_encumbrances	numeric(19,4)	yes	
version	bigint	yes	optimistic + row lock on check

Encumbrance

Field	Type	Required	Notes
id	uuid	yes	PK
source_type	enum	yes	Requisition/PurchaseOrder/Invoice/Journal
source_line_id	uuid	yes	
balance_id	uuid	yes	FK BudgetControlBalance
reserved	numeric(19,4)	yes	
relieved	numeric(19,4)	yes	Σ relief lines
status	enum	yes	Open/PartiallyRelieved/Relieved/Released
lines	jsonb	yes	[{type: Reserve/Adjust/Relieve/Release, amount, documentId, at}]

CostEntry

Field	Type	Required	Notes
id	uuid	yes	PK, append-only
ledger_id	uuid	yes	cost accounting ledger
cost_object_id	uuid	yes	FK CostObject
cost_element_id	uuid	yes	FK CostElement (primary/secondary)
period_key	text	yes	
amount	numeric(19,4)	yes	
behavior	enum	yes	Fixed/Variable/Unclassified
origin	enum	yes	Journal/Allocation/Activity/Statistical/Manual
source_id	uuid	yes	journal line, allocation run, etc.
allocation_run_id	uuid	no	

AllocationPolicy: id, ledger_id, name, type (Distribution/Assessment/Reciprocal), steps[] (order, sender filter, receiver filter, basis {Fixed%/StatisticalMeasure/ActualCost}, secondary_element_id), schedule, version.

Invariants: a BudgetControlBalance's available = budget_total – actuals – encumbrances – pre_encumbrances; Σ Encumbrance.reserved – relieved for a balance = balance.encumbrances; Σ CostEntry(origin=Allocation) per run = 0 (senders negative, receivers positive). Indexes: BudgetControlBalance unique (rule_version_id, control_dimension_key, time_span_key); CostEntry partitioned by period_key, index (cost_object_id, cost_element_id, period_key).

7.38.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/budget/register-entries	Create entry	Idempotency-Key	BudgetMaintain
POST	/api/v1/budget/register-entries/{id}/submit	Start workflow	state check	BudgetMaintain
POST	/api/v1/budget/plans/{id}/scenarios	Add scenario	Idempotency-Key	BudgetPlan
POST	/api/v1/budget/checks	Perform check (document lines)	documentId+version	BudgetCheck (system)
GET	/api/v1/budget/available?rule=&dims=&span=	Available funds	n/a	BudgetView
POST	/api/v1/budget/encumbrances/{id}/relieve	Relief from invoice	documentId	BudgetCheck (system)
POST	/api/v1/controlling/allocation-runs	Run policy for period	(policy, period, version)	CostAccountingRun
GET	/api/v1/controlling/cost-objects/{id}/entries?period=	Entries	n/a	CostAccountingView
POST	/api/v1/controlling/statistical-measures/{id}/values	Load measure values	Idempotency-Key	CostAccountingMaintain
POST	/api/v1/controlling/internal-orders/{id}/settle	Settle	(order, period)	CostAccountingRun

Validation: check requests must carry accounting date, ledger dimension and amount per line; allocation run rejects when period is closed in the cost ledger or a prior step's run for the period is missing.

Event	Trigger	Payload fields	Consumers
`dev.passage.budget.register.posted.v1`	entry posted	entryId, modelId, type, lines summary	balance projector, analytics
`dev.passage.budget.check.completed.v1`	check	sourceDocumentId, lineResults[{result, available, amount}]	procurement, AP, workflow
`dev.passage.budget.encumbrance.relieved.v1`	relief	encumbranceId, amount, invoiceId, remaining	analytics
`dev.passage.controlling.allocation.completed.v1`	run complete	runId, policyId, iterations, totalIn, totalOut, residual	budget-svc, analytics
`dev.passage.controlling.costentry.created.v1`	mirror or allocation	entryId, costObjectId, elementId, amount, origin	lakehouse

7.38.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Budgeting workspace	New entry, approvals, available funds by rule	model, cycle, approved/pending, over-budget alerts	BudgetView
Budget plan grid	Edit cells, distribute, compare scenarios, submit stage	dimension rows × period columns, scenario columns, comments	BudgetPlan
Budget control rules	Configure dimensions, thresholds, user groups, activate	rule version, effective date, documents in control	BudgetConfigure
Budget check result (inline in PO)	View reason, request override	available, amount, result, threshold	Buyer
Cost accounting workspace	Run allocations, process status, reconciliation	ledger, period, steps status, in/out totals	CostAccountingRun
Cost control	Filter object/hierarchy node, drill	plan, actual, committed, variance, variance %	CostAccountingView

Interaction rules: plan grid supports copy/paste from Excel, keyboard fill-down, and distribute-by-key; budget check result renders inside the PO line without leaving the form; allocation runs show step progress and residual live via server-sent events.

7.38.7 Business rules and edge cases

- **BR-38.1** Budget check amounts convert to the ledger's accounting currency at the document's rate; a later rate change does not re-check.
- **BR-38.2** Register entry state machine: Draft → Submitted → Approved → Posted; Submitted → Rejected → Draft; Posted entries are immutable and corrected by a Revision.
- **BR-38.3** A PO revision that lowers the amount never fails a check; it releases the difference.
- **BR-38.4** Invoice exceeding the PO reservation: the matched part relieves; the excess is checked as a new amount under the invoice's document rule.
- **BR-38.5** Time span resolution for a date on a period boundary uses the legal entity's fiscal calendar; dates outside any open cycle fail the check with reason `NoBudgetCycle`.
- **BR-38.6** Allocation convergence: iteration stops at residual ≤ 0.01 or 100 iterations; residual sweep goes to the sender's largest receiver by basis share; step totals must satisfy `Σ out = Σ in` to the cent or the run fails and rolls back.
- **BR-38.7** Allocation runs for a period are versioned; rerun reverses the prior run's entries (origin Allocation) and creates new ones; actuals-based runs after new journal postings require rerun, flagged by the reconciliation job.
- **BR-38.8** A sender with zero basis in all receivers keeps its cost and is reported as unallocated; the run does not fail.
- **BR-38.9** Internal order settlement cannot exceed collected cost; over-settlement is rejected.
- **BR-38.10** Year-end carry-forward of open encumbrances creates a CarryForward register entry in the new cycle and releases the old reservation atomically.

7.38.8 Acceptance criteria

- **AC-38.1** (FR-38.4) Given budget 100,000.00 with actuals 60,000.00 and encumbrances 30,000.00, when a PO line of 12,000.00 is checked with warn 0% and block 5%, then the result is `fail` (available 10,000.00 < 12,000.00, over by 20%).
- **AC-38.2** (FR-38.4) Given the same balance, when a PO of 10,400.00 is checked, then the result is `warn` (over by 4%) and 10,400.00 is reserved.

- **AC-38.3** (FR-38.5) Given a PO line of 10 × 50.00 encumbered, when an invoice for 10 × 52.00 posts, then relief = 500.00 and a new check runs for 20.00.
- **AC-38.4** (FR-38.4) Given two concurrent POs of 8,000.00 against 10,000.00 available, when checked simultaneously, then exactly one passes.
- **AC-38.5** (FR-38.7) Given a reciprocal step with 3 service centers serving each other, when run, then Σ out = Σ in to the cent and the solution matches the analytic solve.
- **AC-38.6** (FR-38.11) Given a month of journal postings, when reconciliation runs, then Σ primary cost entries per element equal the journal balances with 0.00 difference.
- **AC-38.7** (FR-38.2) Given a plan with 3 scenarios and 2 stages, when approvers act, then the workflow completes and a register entry posts from the selected scenario.
- **AC-38.8** (FR-38.6) Given a journal entry posted, when 5 s elapse, then matching cost entries exist for every line with a primary element mapping.

7.38.9 Test plan and fixtures

Fixtures: `budget-check-200` (200 PO/invoice scenarios across thresholds, overrides, revisions, cycles); `alloc-reciprocal-10` (service centers with circular references and a 2,500-sender case for the iterative path); `plan-3-scenarios` (demo entity planning process); `costledger-month` (one month of journal postings mapped to elements).

Test	Type	Target	Pass criterion
Check order of operations	Unit	budget-svc	each step observable in result trace
Relief exactness	Property	encumbrance	Σ reserved − relieved ≥ 0, never negative; relief ≤ matched share
Concurrent checks	E2E	API	no double pass
Allocation conservation	Property	costacct-svc	Σ in = Σ out for random matrices
Convergence	Unit	iterative solver	residual ≤ 0.01 within 100 iterations for diagonally dominant cases
Check latency	Performance	API	p95 < 150 ms at 50 checks/s
Allocation throughput	Performance	run	1M cost entries, 5 steps < 10 min

Reconciliation checks: cost ledger primary costs = journal by element/period; Σ encumbrances = Σ open PO lines under control; budget balances = Σ posted register entries.

7.38.10 Deployment and operations

Helm charts `budget-svc`, `costacct-svc` (shared Postgres `controlling`); env `BUDGET_CHECK_TIMEOUT_MS=500`, `ALLOC_MAX_ITER=100`, `ALLOC_EXACT_SOLVE_MAX_N=2000`; flags `budget.unposted-in-available`, `controlling.abc`; jobs: `cost-mirror-lag-check` every 5 min, `cost-gl-reconcile` nightly, `allocation-schedule` per policy, `budget-balance-rebuild` on demand. SLOs: check p95 < 150 ms, mirror lag p95 < 5 s, allocation 1M entries < 10 min. Alerts: mirror lag > 60 s, reconciliation difference ≠ 0, check timeouts > 0.1%. Rollback: Helm; balance cache rebuilt from events if corrupted.

7.38.11 Migration from AX / SAP

Source	Passage entity	Notes
BudgetTransactionHeader/Line (current and next cycle)	BudgetRegisterEntry	posted entries summarized per combination/period
BudgetControlConfiguration, BudgetControlRule, BudgetGroup	BudgetControlRule	re-validated; user groups mapped to Keycloak groups
BudgetSourceTracking (open)	Encumbrance	open POs only, tied to migrated POs (wave 17)
CAMCostObject, CAMCostElement, CAMDimensionHierarchy	CostObject, CostElement, DimensionHierarchy	
CSKS, CSKT, SETHEADER/SETNODE (KSH1)	CostObject, DimensionHierarchy	
CSKA/CSKB	CostElement	primary mapped by main account
T811C/T811S/T811K cycles	AllocationPolicy	segment → step; tracing factors → basis
FMAVCT, COOI, FMIOI	Encumbrance	open commitments only
COSP/COSS plan versions, KP06	BudgetRegisterEntry (Original)	current year plan
COEP history	not migrated	cost entries mirrored from migrated journal balances

Validation: Σ migrated budget = legacy budget report per combination; open commitments count and total match. Not migrated: closed-cycle budget history older than 2 years (lakehouse archive), AX budget plan Excel templates.

7.38.12 Dependencies, risks, open questions

Depends on: wave 11 dimensions and balances, wave 14 fiscal calendars and allocation engine (reused for GL), wave 17 requisitions/POs, wave 18 invoices, wave 21 activity quantities, wave 37 depreciation forecasts.

Risk	Likelihood	Impact	Mitigation
Balance cache drift vs journal	Medium	High	Nightly reconcile; rebuild job; alert
Procurement UX slowed by checks	Medium	Medium	SLO; async pre-check on requisition

Open questions: (1) Include unposted documents in available funds by default? Proposed off; decide by wave 38 start. (2) Reciprocal exact solve threshold 2,000? Proposed yes; revisit at phase exit. (3) Internal orders as cost object type or separate module? Proposed cost object type; decide by wave 38.

7.39 Wave 39: Product Costing, Cost Roll-Up & Variance Analysis

7.39.1 Scope and business context

Wave 20 delivered the inventory costing engine with a standard cost method but no way to derive the standard: standards were keyed. Wave 39 computes them. Cost accountants define cost groups, costing sheets with indirect cost nodes, and costing versions; roll up costs from BOMs and routings per site and quantity with explosion modes and fallback rules; compare estimates; activate pending costs, which revalues on-hand inventory and posts the revaluation; and analyze purchase price and production variances decomposed into lot size, quantity, price and substitution by cost component. At the end of the wave the demo plant runs on standard cost with a monthly variance report.

Out of scope: actual costing / material ledger periodic unit price (wave 20 covers FIFO/weighted average), mixed costing across procurement alternatives, co-product cost allocation for process manufacturing beyond fixed percentage (wave 24 rules reused), sales-order costing for configured items.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX BOMCalcTable "Calculate items" for a costing version runs single-threaded; 50k items take 6–10 h and lock InventItemPriceSim	Sequential explosion per item re-exploding shared subassemblies	Topological roll-up computes each node once; parallel by level	50k items 8 h → < 15 min
CK40N costing run in SAP: selection, structure explosion, costing, analysis, marking, release as separate steps with logs in CKAPP; errors found at release	Multi-step batch with late validation	One run with upfront validation (missing prices, cycles, inactive routes) and per-item diagnostics	Rework runs per cycle 3 → 1
Cost component split lost in AX when a subassembly rolls into its parent (only cost group breakdown if "cost breakdown" enabled, which doubles InventCostTrans volume)	Breakdown as optional duplicate storage	Split is the primary representation; parent carries child split natively	Items with full split 40% → 100%
Production variance report in AX shows lot size/quantity/price/substitution totals that do not sum to the posted variance when rounding occurs across InventCostTrans	Independent rounding per variance type	Decomposition with residual assignment rule; property-tested	Unexplained variance per order 0.00
Standard cost activation (InventItemPrice activate / CK24 release) revalues inventory in a batch without preview; controllers discover the P&L hit after posting	No simulation of revaluation	Activation preview shows revaluation by item/site before commit	Unplanned revaluation surprises 0; preview < 60 s

7.39.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Costing versions	CostingVersion (type Standard cost / Planned cost), blocking, fallback principle, InventItemPrice, InventItemPriceSim	Costing variant OKKN, valuation variant, costing version (OKKN), KEKO header	CostingVersion with status, fallback, site scope	full
Cost groups and categories	CostGroup (direct material / direct manufacturing / indirect / undefined; fixed/variable), CostCategory with cost category prices, BOMCalcGroup	Cost component structure OKTZ (TCKH1–3), cost elements per component, activity types/prices KP26	CostGroup, CostComponent, CostCategory rate tables	full
Costing sheet	CostSheetNodeTable, CostSheetCalculationBasis, CostSheetCalculationFactor (surcharge/rate nodes, absorption basis)	Costing sheet (KZS2): base rows, overhead rows (percentage/quantity), credit rows; T683	CostingSheet with Basis/Surcharge/Rate/Total nodes	full
BOM calculation	BOMCalcTable, BOMCalcTrans, BOMCalcItem, explosion mode (multilevel / single level / make-to-order / according to BOM line type), profit settings	CK11N single item, CK40N run, CKIS items, KEPH component split, CK13N display	CostCalculation with explosion mode, split retained	full
Mass roll and activation	Costing version "Calculate items", "Activate pending prices", InventSettlement cost change variance	CK40N marking (CK24 mark/release), MBEW STPRS update, MR21	Mass roll (Temporal), activation with revaluation preview	improved
Purchase price variance	InventCostTrans variance type PurchPrice posted on receipt for standard cost items	Price difference account PRD on GR/IR (OBYC PRD)	VarianceEntry type PurchasePrice from wave 18 receipts/invoices	full
Production variances	ProdCalcTrans estimate, InventCostTransVariance (ProdLotSize, ProdQuantity, ProdPrice, ProdSubstitution) at ending	KKS1/KKS2 variance calculation categories (input price, input quantity, resource usage, remaining input, lot size, output price, remaining), CO88 settlement	Variance decomposition on production order end	full
Inventory cost revaluation	Cost change variance (InventSettlement), inventory value report	Price change document (MR21/CKMPRPUNMARK), ML revaluation	Revaluation journal posted on activation	full

7.39.3 Functional requirements

- **FR-39.1 Cost groups and components** — Must. The system shall classify item costs, cost categories (labor/machine) and indirect nodes into cost groups with type and behavior, and map cost groups to cost components for the split.
  - Notes: ≤ 50 cost groups, ≤ 20 components per tenant; an item has exactly one cost group per site override.
- **FR-39.2 Costing sheet** — Must. The system shall define a costing sheet tree with Basis nodes (cost groups selected), Surcharge nodes (% on a basis), Rate nodes (amount per routing hour or per unit), and Total nodes, per site and costing version.
  - Notes: surcharge basis evaluates only the current level's direct lines unless the node is flagged `include-rolled-subassemblies`; absorption posts to indirect accounts via posting profiles at production end.
- **FR-39.3 Costing versions** — Must. The system shall maintain costing versions (Standard, Planned, Simulation) with blocking of pending prices, fallback principle (None / Active version / Specified version), site scope, and effective date.
- **FR-39.4 Cost roll-up algorithm** — Must. The system shall compute item costs by: (1) building the BOM graph for the selection and detecting cycles (reject with path); (2) ordering nodes topologically, leaves first, ≤ 20 levels; (3) for each node computing direct material = Σ child quantity × (1 + scrap%) × child cost (purchased: price from version, else fallback; manufactured: rolled value with its split); direct manufacturing = Σ operations (setup time / costing lot size + run time per unit) × cost category rate; indirect = costing sheet nodes evaluated on this level's basis; (4) retaining the cost component split of every child inside the parent (not collapsed); (5) storing the result as a pending price in the version with explanation lines.
  - Notes: worked 2-level example (USD). Subassembly SA-200: 3 × RM-301 at 2.50 = 7.50; 1 × RM-302 at 1.00 with 10% scrap = 1.10; machining 0.25 h × 60.00 = 15.00; sheet: material overhead 10% on direct material = 0.86; labor overhead 50% on direct labor = 7.50. SA-

200 = 31.96 (DM 8.60, DL 15.00, IND-MAT 0.86, IND-LAB 7.50). Finished item FG-100: 2 × SA-200 = 63.92 carried as split (DM 17.20, DL 30.00, IND-MAT 1.72, IND-LAB 15.00); 1 × RM-300 at 4.00 → DM 4.00; assembly 0.5 h × 40.00 = 20.00 plus setup 1 h × 40.00 / costing lot size 10 = 4.00 → DL 24.00; material overhead 10% on own direct material purchased lines = 0.40; labor overhead 50% × 24.00 = 12.00. FG-100 = 104.32 with split DM 21.20, DL 54.00, IND-MAT 2.12, IND-LAB 27.00 (sums to 104.32). Explosion modes: multilevel (as above), single level (children at their active cost), make-to-order (explode only phantom and MTO lines), by BOM line type.

- **FR-39.5 Mass roll** — Must. The system shall roll all items in a version/site selection in parallel by topological level, with per-item diagnostics (missing price, inactive BOM/route, cycle), completing 50,000 items in < 15 minutes.
- **FR-39.6 Activation and revaluation** — Must. The system shall activate pending prices as of a date, compute on-hand revaluation = Σ on-hand quantity × (new – old standard) per item/site/dimension, show a preview, and post the revaluation journal (Dr/Cr Inventory vs cost change variance account).
    - Notes: activation is atomic per version/site; wave-20 standard cost engine switches at the activation timestamp; open production orders estimated at the old standard keep their estimate but variance at end uses the new standard for output (configurable).
- **FR-39.7 Purchase price variance** — Must. The system shall post PPV = (receipt unit cost – standard) × quantity at product receipt and adjust at invoice for standard-cost items, by cost component DM.
- **FR-39.8 Production variance decomposition** — Must. The system shall, when a production order ends, compute total variance = actual cost – standard cost of reported-as-finished (RAF) output, decomposed per cost component as: lot size variance LSV = Σ over fixed-per-lot elements (setup, constant scrap) of (standard per unit × Q_raf – estimated per lot ÷ Q_order × Q_raf); quantity variance QV = Σ_i (actual qty_i – standard qty_i for Q_raf) × standard unit cost_i for planned inputs i (inputs planned but not consumed give negative QV); price variance PV = Σ_i actual qty_i × (actual unit cost_i – standard unit cost_i) for inputs whose cost differs from standard (labor rate differences, components not on standard cost at that site, subcontract invoices); substitution variance SV = Σ_j actual qty_j × actual unit cost_j for inputs j not in the estimate. Identity: LSV + QV + PV + SV = total variance to the cent; residual from rounding assigned to QV of the largest planned line.
    - Notes: unfavorable is positive; output quantity variance for co-products handled by wave 24 percentages; variances post to GL by posting profile per type and component.
- **FR-39.9 Variance analysis workspace** — Must. The system shall present variances by order, item, site, period, type and component with drill to consumption and route transactions, and the monthly variance report.
- **FR-39.10 Cost estimate comparison and what-if** — Should. The system shall compare two versions or dates item-by-item by component, and allow a simulation version with overridden prices, rates and routes.
- **FR-39.11 Standard cost reconciliation** — Must. The system shall reconcile Σ on-hand quantity × active standard to GL inventory accounts per site nightly (extends the wave-20 inventory value report).

7.39.4 Data model

CostingVersion: id, legal_entity_id, code, type (Standard/Planned/Simulation), status (Open/Blocked/Active/Closed), fallback_principle, fallback_version_id, site_ids[], effective_from, activated_at.

CostCalculation

Field	Type	Required	Notes
id	uuid	yes	PK
costing_version_id	uuid	yes	FK
item_id, site_id	uuid	yes	
calc_quantity	numeric(19,4)	yes	costing lot size
explosion_mode	enum	yes	Multilevel/SingleLevel/MakeToOrder/ByLineType
bom_version_id, route_version_id	uuid	yes	wave 9
unit_cost	numeric(19,6)	yes	6-decimal unit cost
status	enum	yes	Pending/Active/Superseded/Error
diagnostics	jsonb	no	[{code, message, ref}]
run_id	uuid	no	mass roll

CostComponentSplit

Field	Type	Required	Notes
calculation_id	uuid	yes	FK, part of PK
component_id	uuid	yes	FK CostComponent, part of PK
level	enum	yes	Own/Rolled (child-carried)
amount	numeric(19,6)	yes	per calc_quantity
fixed_amount, variable_amount	numeric(19,6)	yes	behavior split

CostingSheetNode: id, sheet_id, parent_id, type (Basis/Surcharge/Rate/Total), cost_group_ids[], percentage numeric(9,4), rate numeric(19,6), rate_unit (Hour/Unit), include_rolled bool, absorption_account_id, order.

VarianceEntry

Field	Type	Required	Notes
id	uuid	yes	PK
source_type, source_id	enum, uuid	yes	ProductionOrder/Receipt/Invoice/Activation
type	enum	yes	LotSize/Quantity/Price/Substitution/PurchasePrice/CostChange
component_id	uuid	yes	
item_id, site_id	uuid	yes	

Field	Type	Required	Notes
amount	numeric(19,4)	yes	unfavorable positive
journal_entry_id	uuid	yes	
detail	jsonb	no	per input line actual/standard qty and cost

Invariants: Σ CostComponentSplit.amount per calculation = unit_cost × calc_quantity (6 decimals); exactly one Active calculation per (version, item, site); Σ VarianceEntry per production order = actual – standard of RAF to the cent. Indexes: CostCalculation unique partial (version, item, site) where status = Active; VarianceEntry by (site_id, period, type).

7.39.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/costing/versions	Create version	Idempotency-Key	CostingMaintain
POST	/api/v1/costing/sheets	Create/version sheet	Idempotency-Key	CostingMaintain
POST	/api/v1/costing/calculations	Roll one item (explosion mode, qty)	(version, item, site, bom, route, qty)	CostingCalculate
POST	/api/v1/costing/versions/{id}/roll	Mass roll selection	runId	CostingCalculate
GET	/api/v1/costing/calculations/{id}/explain	Explanation tree with split	n/a	CostingView
POST	/api/v1/costing/versions/{id}/activate	Preview (dryRun=true) or activate	(version, date)	CostingActivate
GET	/api/v1/costing/variances?site=&period=&type=	Variances	n/a	CostingView
POST	/api/v1/costing/variances/compute	Compute for ended order (system)	productionOrderId+version	CostingSystem
GET	/api/v1/costing/compare?a=&b=	Version comparison	n/a	CostingView

Event	Trigger	Payload fields	Consumers
`dev.passage.costing.calculation.completed.v1`	item rolled	calculationId, itemId, siteId, unitCost, split[]	analytics
`dev.passage.costing.rollrun.completed.v1`	mass roll finished	runId, itemCount, errorCount, durationMs	workspace
`dev.passage.costing.standardcost.activated.v1`	activation	costingVersionId, itemCount, revaluationJournalId, totalRevaluation	inventory costing, production, budget
`dev.passage.costing.variance.posted.v1`	variance computed	sourceId, type totals, byComponent[], journalEntryId	costacct-svc, analytics

7.39.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Costing workspace	Roll, activate, compare, diagnostics tile	version, pending count, errors, last activation	CostingView
Costing sheet designer	Add node, set basis/%/rate, validate	tree, cost groups, percentage, rate, absorption account	CostingMaintain
Calculation explain	Expand levels, view split, export	level, line, qty, unit cost, component amounts	CostingView
Activation preview	Filter item/site, approve	item, old, new, on-hand, revaluation	CostingActivate
Variance analysis	Pivot by type/component/order, drill	order, item, standard, actual, LSV, QV, PV, SV	CostingView

Interaction rules: the explain tree lazy-loads levels; keyboard `e` expands all; activation preview requires an explicit confirmation of the total revaluation amount; variance pivot persists per user.

7.39.7 Business rules and edge cases

- **BR-39.1** Unit costs are stored with 6 decimals; postings round to the currency minor unit at the transaction level; split rounding residual goes to the DM component.
- **BR-39.2** Version state machine: Open → Blocked (no edits of pending) → Active (per site activation) → Closed; Simulation versions cannot activate.
- **BR-39.3** A cyclic BOM (item appears in its own explosion path) fails the calculation with the path listed; phantoms are exploded in place.
- **BR-39.4** Fallback: missing price with fallback Active uses the active standard; with None, the calculation errors; fallback use is recorded in diagnostics.
- **BR-39.5** Setup cost per unit = setup time × rate ÷ costing lot size; costing lot size defaults from the item's default order settings (wave 8) and can be overridden per calculation.
- **BR-39.6** Scrap: BOM line scrap % multiplies quantity; route operation scrap % compounds backward through preceding operations.
- **BR-39.7** Activation while a mass roll is running for the same version/site is rejected (advisory lock).
- **BR-39.8** Revaluation uses on-hand at the activation timestamp from the wave-16 projection; inventory transactions timestamped after activation use the new standard.
- **BR-39.9** Variance identity: if ILSV + QV + PV + SV – totall > 0.00 after residual assignment, the computation fails and the order end is blocked with an error.
- **BR-39.10** Ended orders reopened (wave 21) reverse their variance entries; re-ending recomputes.

7.39.8 Acceptance criteria

- **AC-39.1** (FR-39.4) Given the worked SA-200/FG-100 fixture, when rolled multilevel, then FG-100 = 104.32 with split DM 21.20 / DL 54.00 / IND-MAT 2.12 / IND-LAB 27.00.
- **AC-39.2** (FR-39.4) Given 50 multi-level fixture products, when rolled, then every component amount matches the reference workbook to the cent.
- **AC-39.3** (FR-39.5) Given 50,000 items across 8 levels, when mass rolled, then the run completes in < 15 min with diagnostics for every error.

- **AC-39.4** (FR-39.6) Given 1,000 units on hand at 10.00 and a new standard 10.50, when activated, then a journal Dr Inventory 500.00 / Cr Cost change variance 500.00 posts and the inventory value report ties.
- **AC-39.5** (FR-39.8) Given an order for 100 planned, 90 RAF, consuming 290 of a component standard 3/unit at 2.00 plus 5 units of an unplanned part at 1.50, with setup 40.00 per lot and costing lot size 100, then QV = (290 – 270) × 2.00 = 40.00, SV = 7.50, LSV = 90 × (0.40 – 0.40) = 0.00 and total = 47.50 (plus any labor PV), and the identity holds.
- **AC-39.6** (FR-39.8) Given 200 ended fixture orders, when variances compute, then Σ types = total for every order.
- **AC-39.7** (FR-39.7) Given a receipt of 10 at 5.20 for an item standard 5.00, when posted, then PPV = 2.00 posts to DM.
- **AC-39.8** (FR-39.3) Given a cyclic BOM, when rolled, then the calculation errors with the cycle path and the run continues for other items.
- **AC-39.9** (FR-39.11) Given month end, when reconciliation runs, then Σ on-hand × standard equals GL inventory per site.

7.39.9 Test plan and fixtures

Fixtures: `rollup-50` (50 products, up to 6 levels, phantoms, scrap, setup, reference workbook); `rollup-50k` (generated 50,000 items, 8 levels, for performance); `variance-200` (ended orders with each variance type, reference workbook); `activation-demo` (demo plant on-hand with old/new standards).

Test	Type	Target	Pass criterion
Node cost formulas	Unit	roll-up	worked example exact
Split conservation	Property	roll-up	Σ split = unit cost × qty for random BOMs
Cycle detection	Unit	graph	every planted cycle reported
Mass roll time	Performance	run	50k items < 15 min on 4 workers
Activation revaluation	E2E	activation + journal	journal = preview; value report ties
Variance identity	Property	decomposition	LSV + QV + PV + SV = total for random orders

Reconciliation checks: inventory at standard = GL inventory per site; Σ variance journals per period = Σ VarianceEntry; activation revaluation journal = Σ preview rows.

7.39.10 Deployment and operations

Helm chart `costing-svc` (extends wave 20 chart, Postgres `costing`); env `COSTING_MAX_LEVELS=20`, `COSTING_ROLL_WORKERS=4`; flags `costing.whatif`, `costing.variance-new-standard-for-open-orders`; jobs: `variance-compute` on production order end events, `standard-cost-reconcile` nightly, `monthly-variance-report` on period close. SLOs: single calculation p95 < 500 ms for ≤ 6 levels; explain API p95 < 300 ms; mass roll as above. Alerts: roll error rate > 2%, variance identity failures > 0, reconciliation difference ≠ 0. Rollback: Helm; activation reversal is a new activation of the prior version (reversing journal), never a delete.

7.39.11 Migration from AX / SAP

Source	Passage entity	Notes
CostingVersion, InventItemPrice (active standard)	CostingVersion, CostCalculation (Active, no explanation)	active standards loaded as calculations with status Active and diagnostics `Migrated`
CostGroup, CostCategory, CostSheetNodeTable/CalculationBasis/Factor	CostGroup, CostCategory, CostingSheet	
BOMCalcTable/BOMCalcTrans	not migrated	re-rolled from migrated BOMs; comparison report vs legacy
InventCostTransVariance history	lakehouse archive	
MBEW (STPRS, PEINH), KEKO/KEPH released estimates	CostCalculation, CostComponentSplit	KEPH components mapped to CostComponent
OKTZ cost component structure, KZS2 sheets, KP26 rates	CostComponent, CostingSheet, CostCategory rates	
COSS/CO88 settled variance history	lakehouse archive	

Validation: re-rolled standard within 1% of legacy active standard for 95% of items (exceptions listed with explanation); Σ on-hand × migrated standard = migrated GL inventory. Not migrated: simulation versions, legacy calculation logs.

7.39.12 Dependencies, risks, open questions

Depends on: wave 9 BOM/route versions, wave 16 on-hand, wave 20 standard cost engine and inventory value report, wave 21 production order end and consumption, wave 24 co-product rules, wave 38 cost elements for absorption.

Risk	Likelihood	Impact	Mitigation
Legacy standards unreproducible	High	Medium	Load active standards as-is; re-roll comparison
Deep BOMs exceed memory in parallel roll	Low	Medium	Level batching; worker memory limit 2 GiB

Open questions: (1) Surcharge basis includes rolled subassemblies by default? Proposed no (AX behavior); decide by wave 39 start. (2) Variance for open orders at new standard? Proposed flag off; decide by wave 39. (3) Store split at 6 or 4 decimals? Proposed 6; decide by wave 39.

7.40 Wave 40: Intercompany & Financial Consolidation

7.40.1 Scope and business context

Wave 40 closes phase 7 and delivers finance parity. Groups with several legal entities trade with each other, share costs and must report as one. Before this wave each entity is an island: an intercompany sale is keyed twice, due-to/due-from balances drift, and consolidated statements are assembled in

spreadsheets. After it, pairing a customer in entity A with a vendor in entity B makes every sales order, packing slip, invoice and payment create its mirror in the partner automatically (including direct-delivery chains); intercompany journal lines post due-to/due-from entries in both entities; a reconciliation workspace matches both sides and flags mismatches; and the consolidation service builds group structures with ownership percentages, collects balances per unit, translates under IAS 21, eliminates intercompany balances, transactions, unrealized profit in inventory and investments, computes non-controlling interest, and produces consolidated statements through the wave-15 reporting designer with drill-down to the entity journal.

Out of scope: equity-method accounting for associates (v1.x), complex multi-tier ownership changes mid-year (step acquisitions), statutory group tax, transfer-pricing calculation itself (hooks only: documented IC prices and margin evidence export).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 intercompany: the mirror PO/SO is created by a synchronous X++ call; if the partner entity's number sequence or dimension validation fails, the user sees an infolog and the pair is left half-created (SalesTable without InterCompanyPurchId)	No saga; no compensation	Temporal saga with compensating cancel; both documents or neither	Orphan IC documents per month 5–20 → 0
SAP cross-company FB01 posts via OBYA clearing accounts; FBICR3 (ICR) reconciles weeks later by downloading both ledgers	Reconciliation is a separate batch extract	Every IC posting carries a pair id; reconciliation is continuous per posting	Mismatch detection latency 30 days → < 1 min
Intercompany billing in SAP SD (RD04 output → INVOIC IDoc → FB01 vendor invoice) fails silently in WE02 when a tax code is missing; AP sees nothing	Message-based integration inside one system	Mirror vendor invoice created in the same saga with validation before commit	IDoc-style failures to triage per month 10–30 → 0
AX consolidation (LedgerConsolidate online) copies balances into a consolidation company; eliminations (LedgerEliminationRule) are posted journals with no drill to source entity; reruns duplicate unless reversed manually	Copy-based consolidation into a ledger	Consolidation run is a derived, versioned dataset with elimination entries linked to source balances; rerun supersedes	Rerun effort 2 h → < 2 min; drill to entity 100%
Currency translation differences computed in Excel because AX "Consolidate online" translates all accounts at one rate type unless per-account overrides are maintained	Rate type per account instead of per statement category	IAS 21 methods by account category with CTA derived automatically	Manual CTA journals per period 1–3 → 0

7.40.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Trading partner setup	InterCompanyTradingPartner, InterCompanyEndpoint, InterCompanyEndpointActionPolicy, InterCompanyTradingValueMap, CustTable/VendTable intercompany flags	Customer/vendor with company code assignment, VBAK-VKORG IC sales org, OBYA clearing	IntercompanyPartner (customer in A ↔ vendor in B) with action policies and value maps	full
Mirror sales/purchase orders	SalesTable.InterCompanyOriginalPurchId, PurchTable.InterCompanyOriginalSalesId, InterCompanySalesOrder/PurchaseOrder classes, chains	Cross-company stock transport order (ME21N NB) with delivery and IC billing (VF01 IV)	Mirror document saga	full
Direct delivery chains	InterCompany direct delivery, SalesLine.InterCompanyInventDim, chain of orders	Third-party order processing, cross-company drop-ship	DirectDeliveryChain entity (3 orders linked)	full
Intercompany invoices	InterCompany invoice posting (sales invoice → vendor invoice), InterCompanyPurchInvoice	SD IV intercompany billing, RD04 → INVOIC → FB01	Mirror invoice creation in same saga	improved
Intercompany accounting (journals)	LedgerJournalTrans.OffsetCompany, LedgerIntercompany (due-to/due-from accounts), intercompany journal	FB01 cross-company (BVORG), OBYA, FAGL cross-company postings	IcJournal with due-to/due-from derivation	full
Intercompany reconciliation	Intercompany reconciliation report (D365 10.0.x)	ICR FBICR3 / FBIC010 (GL open items), FBICS3	IcReconciliationItem continuous matching	improved
Group structure and ownership	Consolidation legal entity, LedgerConsolidate, LedgerConsolidateLegalEntity (share %)	Consolidation units/groups (CXCUNIT), ownership in COI, ECCS CX5x	ConsolidationGroup, ConsolidationUnit, Ownership	full
Eliminations	LedgerEliminationRule, LedgerEliminationRuleLine, elimination journal (LedgerJournalType Elimination), ConsolidationAccount	Group Reporting reclassification/elimination tasks (ACDOCU document types 2x), IC profit in inventory, COI	EliminationRule engine, ConsolidationEntry	full
Currency translation	Consolidate online rate types per account, LedgerConsolidateExchangeRate	Translation methods (closing, average, historical), CTA	IAS 21 per account category, CTA derived	full
Consolidated statements	Management Reporter consolidated reporting tree	Group Reporting reports, ACDOCU-based analytics	Wave 15 designer over consolidated dataset with drill	full

7.40.3 Functional requirements

- **FR-40.1 Trading partners** — Must. The system shall pair a customer in entity A with a vendor in entity B (and vice versa) with action policies (auto-create, auto-confirm, auto-post packing slip, auto-post invoice), value maps (item numbers, units, dimensions, charges) and an IC pricing policy (agreed price list, cost plus %, transfer price table).
  - Notes: one active pairing per entity pair and direction; policies versioned; mirror creation requires both entities on the same tenant.
- **FR-40.2 Mirror-document matrix** — Must. The system shall create mirror documents per the matrix below in a Temporal saga with compensation.
  - Notes: matrix (source in selling/holding entity → mirror in buying/receiving entity): sales order → purchase order; sales order change (qty/date/price) → PO change with re-approval if beyond policy tolerance; packing slip → product receipt; sales invoice → vendor invoice (matched 3-way to the mirror PO and receipt); customer payment received → vendor payment settled (optional, when netting is not used); sales return (RMA) → purchase return; free-text invoice → vendor invoice journal; IC journal line with offset entity → due-to/due-from entries both sides; fixed asset IC transfer (wave 37) → disposal + acquisition; reverse direction (PO created in buyer first) → sales order in seller. Saga steps: validate partner and policy → reserve number sequences in both entities → create source → create mirror → link (`IcDocumentLink`) → publish; on any failure compensate by canceling created documents.
- **FR-40.3 Direct delivery chains** — Must. The system shall support customer order in A, IC PO from A to B, IC SO in B shipping directly to A's customer, with B's packing slip posting A's product receipt and A's customer packing slip in one saga.

- **FR-40.4 Intercompany pricing and transfer-pricing hooks** — Should. The system shall price IC lines from the policy, record the pricing basis on the link, and export a transfer-pricing evidence file (item, qty, price, basis, cost, margin) per period.
- **FR-40.5 Intercompany accounting** — Must. The system shall derive due-to/due-from accounts per entity pair from the IC accounting setup and post balanced entries in both entities for IC journal lines, allocations (wave 14/38) and expense re-charges, each carrying a shared `ic_pair_id`.
    - Notes: both entries post atomically through the journal API per entity; if the second post fails the first is reversed; amounts in the counterparty entity convert at the document rate with FX difference booked to IC FX accounts.
- **FR-40.6 Intercompany reconciliation** — Must. The system shall match IC balances and transactions per entity pair, currency and period, classifying each as Matched (difference ≤ 0.01 or FX tolerance), Timing (counterpart exists in the next period), or Mismatch, with a workspace to resolve.
    - Notes: FX tolerance default 0.5% of amount; period-end summary required to be free of Mismatch before consolidation unless overridden by `ConsolidationApprove`.
- **FR-40.7 Group structure** — Must. The system shall define consolidation groups and units (legal entities or sub-groups) with effective-dated ownership %, consolidation method (full / proportional — v1 full only), group currency and chart mapping (entity main account → consolidation account).
- **FR-40.8 Data collection and translation** — Must. The system shall collect trial balances per unit and period from the wave-11 balance projection, translate under IAS 21: assets/liabilities at closing rate, income/expense at average rate (or transaction-date rates by option), equity at historical rates, and derive CTA = translated closing net assets – (translated opening net assets + translated net income); CTA posts to the OCI consolidation account.
- **FR-40.9 Eliminations** — Must. The system shall apply elimination rules in order: (E1) IC revenue/COGS and other P&L; (E2) IC receivables/payables and loans with translation differences to CTA; (E3) unrealized profit in inventory (upstream shared with NCI, downstream parent only); (E4) investment in subsidiary against equity at acquisition with goodwill; (E5) NCI share of current-year profit and equity; (E6) dividends within group. Each produces ConsolidationEntry lines linked to the source balances.
    - Notes: worked 2-entity example (group currency USD; P owns 80% of S, S in EUR; closing rate 1.10, average 1.08, historical at acquisition 1.05). S sold goods to P for EUR 100,000 (cost EUR 70,000); 40% remain in P's inventory; S's receivable EUR 100,000 open; P's payable recorded at USD 108,500. E1: Dr Revenue 108,000.00 / Cr COGS 108,000.00 (100,000 × 1.08). E2: Dr AP 108,500.00, Dr CTA 1,500.00 / Cr AR 110,000.00 (100,000 × 1.10). E3: unrealized profit = 30,000 × 40% = EUR 12,000 × 1.08 = 12,960.00: Dr COGS 12,960.00 / Cr Inventory 12,960.00; upstream so NCI shares 20%: Dr NCI (equity) 2,592.00 / Cr NCI share of profit 2,592.00. E4: S share capital EUR 500,000 and pre-acquisition retained earnings EUR 200,000 at 1.05 = 735,000.00; P investment USD 620,000.00; Dr Share capital 525,000.00, Dr Pre-acquisition RE 210,000.00, Dr Goodwill 32,000.00 / Cr Investment in S 620,000.00, Cr NCI 147,000.00 (20% × 735,000). E5: S net income EUR 150,000 × 1.08 = 162,000.00; Dr Profit attributable to NCI 32,400.00 / Cr NCI (equity) 32,400.00. Each entry balances; the run reports Σ debits = Σ credits = 1,032,952.00 across eliminations E1–E5.
- **FR-40.10 Consolidation run** — Must. The system shall execute collection → translation → eliminations → NCI → validation as one versioned run per group/period with partial recompute for changed units, producing a consolidated trial balance and statements.
    - Notes: run of 50 units × 10,000 accounts in < 5 min; validation: consolidated balance sheet balances; Σ eliminations per rule balanced; no unresolved Mismatch items.
- **FR-40.11 Consolidated reporting and drill** — Must. The system shall expose the consolidated dataset to the wave-15 designer with drill from a consolidated line to unit balances, elimination entries and entity journal lines.

7.40.4 Data model

IcDocumentLink

Field	Type	Required	Notes
id	uuid	yes	PK
ic_pair_id	uuid	yes	shared by all documents in a chain
partner_id	uuid	yes	FK IntercompanyPartner
source_entity_id, source_doc_type, source_doc_id	uuid, enum, uuid	yes	
mirror_entity_id, mirror_doc_type, mirror_doc_id	uuid, enum, uuid	yes	
chain_id	uuid	no	direct delivery chain
saga_id	text	yes	Temporal workflow id
status	enum	yes	Creating/Linked/Compensated/Failed
pricing_basis	jsonb	no	policy, price list, markup

IcReconciliationItem: id, entity_a_id, entity_b_id, period_key, currency, ic_pair_id, amount_a, amount_b, difference, fx_tolerance_applied, status (Matched/Timing/Mismatch/Resolved), resolution_note, resolved_by.

ConsolidationUnit: id, group_id, legal_entity_id (or child_group_id), functional_currency, ownership[] ({from, to, percent numeric(7,4)}), method (Full), acquisition_date, historical_rate numeric(19,8), chart_mapping_id.

ConsolidationRun

Field	Type	Required	Notes
id	uuid	yes	PK
group_id	uuid	yes	
period_key	text	yes	
version	int	yes	increments per rerun
status	enum	yes	Collecting/Translating/Eliminating/Validating/Completed/Failed/Superseded
rates	jsonb	yes	closing, average, historical by currency
cta_amount	numeric(19,4)	no	

Field	Type	Required	Notes
validation	jsonb	no	checks and results
started_at, completed_at	timestamptz	yes/no	

ConsolidationEntry: id, run_id, rule_code (E1–E6/Translation/Collection), unit_id, consolidation_account_id, debit, credit, group_currency, source_refs jsonb ({unitId, accountId, amount, rate}), nci_share numeric(19,4).

Invariants: every IcDocumentLink with status Linked has both documents existing and amounts equal at document rate; Σ ConsolidationEntry debit = Σ credit per run and per rule_code; ownership percentages per unit never exceed 100.00% at any date; a Completed run is immutable (rerun creates a new version and marks the old Superseded). Indexes: IcDocumentLink by ic_pair_id, by (mirror_entity_id, mirror_doc_id); ConsolidationEntry by (run_id, consolidation_account_id); IcReconciliationItem by (entity_a_id, entity_b_id, period_key, status).

7.40.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	/api/v1/interco/partners	Create pairing and policy	Idempotency-Key	IntercompanyConfigure
POST	/api/v1/interco/mirrors	Start mirror saga for a source document (system)	sourceDocId+version	IntercompanySystem
GET	/api/v1/interco/links?pairId=	Chain view	n/a	IntercompanyView
POST	/api/v1/interco/journals	IC journal with offset entity	Idempotency-Key	IntercompanyPost
POST	/api/v1/interco/reconciliations/run?period=	Run matching	(period, version)	IntercompanyReconcile
POST	/api/v1/interco/reconciliations/items/{id}/resolve	Resolve	state check	IntercompanyReconcile
POST	/api/v1/consol/groups	Create group/units/ownership	Idempotency-Key	ConsolidationConfigure
POST	/api/v1/consol/runs	Start run (group, period)	(group, period) new version	ConsolidationRun
GET	/api/v1/consol/runs/{id}/trial-balance	Consolidated TB	n/a	ConsolidationView
GET	/api/v1/consol/runs/{id}/entries?rule=	Elimination entries with source refs	n/a	ConsolidationView
POST	/api/v1/consol/imports	External unit TB import	file hash	ConsolidationConfigure

Validation: mirror saga rejects when the partner policy is inactive or the mirror entity's period is closed; consolidation run rejects unresolved Mismatch items unless override; rates must exist for every unit currency.

Event	Trigger	Payload fields	Consumers
`dev.passage.interco.mirror.created.v1`	saga linked	icPairId, sourceDocId, sourceEntity, mirrorDocId, mirrorEntity, type	sales, procurement, analytics
`dev.passage.interco.mirror.compensated.v1`	saga failed	icPairId, failedStep, reason	workflow, alerts
`dev.passage.interco.journal.posted.v1`	IC journal both sides posted	icPairId, entityA, entityB, amount, currency, journalIds[]	reconciliation, consol-svc
`dev.passage.interco.reconciliation.flagged.v1`	mismatch	itemId, entities, difference	workspace, workflow
`dev.passage.consol.run.completed.v1`	run completed	runId, groupId, period, version, unitCount, eliminationCount, ctaAmount	reporting, analytics

7.40.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Intercompany workspace	Pending sagas, failed mirrors, mismatches, netting link	pair, source, mirror, status, age	IntercompanyView
Partner setup	Pair, policies, value maps, pricing	entity A/B, customer/vendor, actions, tolerances	IntercompanyConfigure
Chain view	Open any document in chain, retry/compensate	SO, PO, receipts, invoices, payments with status	IntercompanyView
IC reconciliation	Run, filter status, resolve, drill both sides	pair, currency, amount A, amount B, difference, status	IntercompanyReconcile
Consolidation workspace	Define group, start run, view validation, open statements	group, period, version, status, CTA, checks	ConsolidationRun
Elimination entries	Filter by rule/unit, drill to source	rule, unit, account, debit, credit, NCI share	ConsolidationView

Interaction rules: chain view renders as a graph with keyboard navigation between nodes; reconciliation grid pairs rows side by side with difference highlighted; consolidation run progress streams per step; statements open in the wave-15 viewer with a "consolidated" badge and drill menu.

7.40.7 Business rules and edge cases

- **BR-40.1** Mirror saga state machine: Creating → Linked; Creating → Compensated (all created documents canceled) → Failed when compensation itself fails (alert, manual); Linked documents cannot be deleted, only canceled through the chain.
- **BR-40.2** Source and mirror amounts are equal in document currency; each entity's accounting currency conversion uses its own rate at the document date; differences surface in reconciliation as FX-tolerated.
- **BR-40.3** A change to a linked sales order beyond policy tolerance (default qty ±0, price ±0.5%, date ±2 days) puts the mirror PO into re-approval (wave 17 workflow); within tolerance it updates silently.
- **BR-40.4** Due-to/due-from derivation: entity A line with offset entity B posts A: Dr/Cr account, Cr/Dr "Due from B"; B: Dr/Cr "Due to A", Cr/Dr the offset account; accounts resolved from the IC accounting matrix (A,B); missing matrix entry fails the post.
- **BR-40.5** Reconciliation Timing status requires the counterpart within the next period; after that it becomes Mismatch.
- **BR-40.6** Translation: historical rate per equity movement date; CTA computed per unit, then NCI share of CTA (ownership %) reclassified to NCI.
- **BR-40.7** Unrealized profit in inventory uses the seller's margin on the remaining quantity (FIFO assumption on IC receipts) and the average rate of the sale period.

- **BR-40.8** Ownership changes mid-period apply from the effective date; income is split pro rata by days unless a measured sub-period balance is provided.
- **BR-40.9** Consolidation rounding: entries computed at 6 decimals, rounded to the group currency minor unit per line; a per-rule rounding line keeps each rule balanced.
- **BR-40.10** Rerun with no changed units and same rates returns the existing version (idempotent); any change creates version +1 and supersedes.

7.40.8 Acceptance criteria

- **AC-40.1** (FR-40.2) Given a sales order of 100 × 25.00 EUR in A to the IC customer, when confirmed, then a PO 100 × 25.00 EUR exists in B linked by ic_pair_id, and both documents balance.
- **AC-40.2** (FR-40.2) Given 1,000 random IC orders including planted number-sequence failures, when sagas run, then every pair is either Linked with equal amounts or Compensated with no orphan document.
- **AC-40.3** (FR-40.3) Given a direct-delivery chain, when B posts the packing slip, then A's product receipt and A's customer packing slip post in the same saga with matching quantities.
- **AC-40.4** (FR-40.5) Given an IC journal line of USD 1,000.00 expense in A offset to B, when posted, then A shows Dr Expense 1,000.00 / Cr Due from B 1,000.00 and B shows Dr Due to A 1,000.00 / Cr Cash 1,000.00 with one ic_pair_id.
- **AC-40.5** (FR-40.6) Given 50 planted mismatches (missing mirror, wrong amount, wrong period), when reconciliation runs, then all 50 are flagged and no matched pair is flagged.
- **AC-40.6** (FR-40.8) Given unit S with opening net assets EUR 700,000 at 1.05, net income EUR 150,000 at 1.08, closing net assets EUR 850,000 at 1.10, when translated, then CTA = 935,000 − (735,000 + 162,000) = 38,000.00 USD.
- **AC-40.7** (FR-40.9) Given the worked P/S example, when the run executes, then entries E1–E5 post with the stated amounts and each rule balances.
- **AC-40.8** (FR-40.9) Given the 3-entity, 2-currency demo group fixture, when consolidated, then the consolidated balance sheet and P&L match the accountant-reviewed eliminations to the cent.
- **AC-40.9** (FR-40.11) Given a consolidated revenue line, when drilled, then unit balances, E1 entries and entity journal lines are reachable within 3 clicks.

7.40.9 Test plan and fixtures

Fixtures: `ic-orders-1000` (generated across 3 entities, 2 currencies, with planted failures and direct-delivery chains); `ic-recon-planted-50` (mismatch types); `consol-group-3` (P in USD, S1 in EUR 80%, S2 in USD 100%, IC sales, inventory, loans, dividends; accountant workbook); `translation-cases` (rate scenarios, mid-year acquisition).

Test	Type	Target	Pass criterion
Saga compensation	Property	interco-svc	no orphan documents under random failures
Mirror amount equality	Property	links	source = mirror in document currency
Due-to/due-from derivation	Unit	IC accounting	matrix cases
Reconciliation classifier	Unit	matcher	Matched/Timing/Mismatch per table
Translation and CTA	Unit	consol-svc	worked example exact
Elimination rules	Unit/E2E	run	P/S example; demo group workbook
Run performance	Performance	run	50 units × 10k accounts < 5 min
Drill-down	E2E	reporting	3 clicks to journal

Reconciliation checks: Σ due-from in A = Σ due-to in B per pair and currency (after FX tolerance); consolidated TB balances; Σ eliminations per rule = 0; consolidated equity = parent equity + NCI + CTA.

7.40.10 Deployment and operations

Helm charts `interco-svc` (TS, Temporal workers), `consol-svc` (Go), Postgres `group`; env `IC_SAGA_TIMEOUT_S=120`, `IC_FX_TOLERANCE_PCT=0.5`, `CONSOL_MAX_UNITS=500`; flags `interco.payment-mirroring`, `consol.external-import`; jobs: `ic-reconcile` hourly and at period close, `ic-saga-sweeper` every 5 min for stuck sagas, `consol-run-scheduled` on close checklist task (wave 14). SLOs: mirror saga p95 < 5 s, reconciliation run < 2 min for 100k items, consolidation as above. Alerts: compensated sagas > 0 per hour, stuck sagas > 10 min, run failures. Runbook: retry saga, manual compensation, rerun consolidation version. Rollback: Helm; runs are versioned datasets, links are append-only.

7.40.11 Migration from AX / SAP

Source	Passage entity	Notes
InterCompanyTradingPartner, InterCompanyEndpoint*, value maps	IntercompanyPartner, policies	
SalesTable/PurchTable IC links (open orders)	IcDocumentLink	open chains only, status Linked
LedgerIntercompany (due-to/due-from setup)	IC accounting matrix	
LedgerConsolidate*, LedgerEliminationRule/Line, ConsolidationAccount mapping	ConsolidationGroup, Unit, EliminationRule, chart mapping	
OBYA clearing accounts, customer/vendor IC assignments	IC accounting matrix, IntercompanyPartner	
Open cross-company STOs and IC billing documents	IcDocumentLink	
Group Reporting units/groups/FS item mapping, ECCS CX hierarchy	ConsolidationGroup, Unit, chart mapping	
ACDOCU / ECMCT history	lakehouse archive	prior consolidated results kept for comparison, not re-run

Validation: IC balances per pair equal legacy after migration; first Passage consolidation for the last closed legacy period matches legacy consolidated statements within 0.01% per line (differences explained). Not migrated: closed IC chains, legacy elimination journals (replaced by derived runs).

7.40.12 Dependencies, risks, open questions

Depends on: waves 11–15 (journal, rates, period control, reporting), wave 17 PO and approval, wave 18 vendor invoice matching, wave 27–29 sales orders, shipping and invoicing, wave 30 payments, wave 36 netting, wave 37 IC asset transfers, wave 39 cost for unrealized profit.

Risk	Likelihood	Impact	Mitigation
Cross-domain saga partial failure leaves stuck state	Medium	High	Sweeper job, compensation tests, manual runbook
Accountant disagreement on NCI/CTA presentation	Medium	Medium	Workbook-driven fixtures; sign-off gate
Group chart mapping effort per customer	High	Medium	Mapping import template; validation report

Open questions: (1) Mirror customer payments by default or rely on netting? Proposed netting; decide by wave 40 start. (2) Proportional consolidation in v1? Proposed no; decide at phase exit. (3) Transfer-pricing evidence format (CSV vs OECD XML)? Proposed CSV; decide by wave 40.

Part 8 — Phase 8: Projects, People & Service (waves 41–45)

8.0 Phase overview

Business outcome at phase exit. At the end of wave 45 the demo group (manufacturer plus the engineering-services entity added in wave 41) runs project operations, HR core, payroll, expense management, plant maintenance and field service on Passage, with every financial effect posted through the universal journal (wave 11) and reconciled daily. A project manager plans a WBS, books resources, captures time and expense, bills by T&M, fixed-price milestone or retainer, and recognizes revenue by percentage-of-completion. An HR administrator manages workers, positions and jobs with effective dating; employees self-serve absence and benefits. A payroll administrator runs a US gross-to-net cycle that matches IRS Publication 15-T to the cent and posts to the GL. A maintenance planner keeps plant machines on preventive and counter-based plans whose work orders cost to the asset and GL. A service coordinator dispatches field technicians under SLA with OR-Tools, and technicians execute offline on a PWA. This phase is where Passage reaches parity with AX Project management and accounting, HR, Payroll, Expense management, D365 Asset Management and Service management, and with SAP PS, HCM PA/OM/PT, PY, Concur, PM and CS/FSM.

Phase-exit sign-off criteria

- Percentage-of-completion estimate fixtures (12 projects, 4 periods each) match the accountant-reviewed workbook to 0.01 USD/EUR per period.
- Property test over 10,000 generated transaction sets: every invoice proposal contains exactly the billable transactions selected by its billing rule, no duplicates, no omissions.
- Effective-dated position/manager as-of queries return the correct answer for 100% of 50,000 randomized dates across the demo workforce (1,200 workers).
- 200 US payroll fixture scenarios match Pub 15-T percentage-method computations to the cent; payroll posting reconciles to GL accounts 6100–6199 and 2300–2349 with zero variance.
- Expense policy engine flags violations exactly per rule set on a 500-report fixture; reimbursements post to AP and settle in a wave-19 payment run.
- Maintenance plan fixtures (40 plans, time and counter) generate work orders on the expected dates/readings; work order cost (parts + labor) reconciles to asset cost and GL maintenance accounts.
- IoT threshold breach creates a maintenance request in p95 < 5 s; recorded downtime reduces wave-23 scheduling capacity.
- Dispatch property test: 1,000 generated schedules are feasible (skills, time windows, shifts) with a solver time p95 < 30 s for 200 work orders × 25 technicians.
- Recorded phase-exit walkthrough: projects, HR, payroll, maintenance and field service end-to-end on `stage`, signed off by both owners.

Cross-wave data model summary

Entity	Owning service	Key relationships
Project, WbsNode, ProjectTransaction, InvoiceProposal, ProjectEstimate	`projects` (Go, Postgres `projects`)	Customer (wave 7 party), Worker (wave 42), Journal (wave 11), Fixed asset CIP (wave 37)
Worker, Employment, Position, Job, Department, AbsencePlan, BenefitPlan	`hr` (TS, Postgres `hr`)	Legal entity/org hierarchy (wave 6), Time registrations (wave 25), User (Keycloak)
PayCycle, PayRun, PayStatement, EarningCode, DeductionCode, TaxRuleSet	`payroll` (Go, Postgres `payroll`)	Worker/Employment (wave 42), Journal (wave 11), Bank/payment file (wave 36)
ExpenseReport, ExpenseLine, ExpensePolicyRule, CardTransaction	`expense` (TS, Postgres `expense`)	Worker, Project (wave 41), AP invoice (wave 18), Tax (wave 13)
FunctionalLocation, Asset, MaintenancePlan, MaintenanceRequest, MaintWorkOrder, Counter	`eam` (Go, Postgres `eam`)	Resource (wave 22), Inventory item (wave 16), Fixed asset (wave 37), IoT reading (wave 25)
ServiceObject (installed base), ServiceAgreement, Case, ServiceWorkOrder, DispatchAssignment	`service` (TS) + `dispatch` (Python/OR-Tools), Postgres `service`	Customer, Sales shipment serial (wave 28), Worker (technician), Truck stock location (wave 16), Invoice (wave 29)

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.projects.transaction.posted.v1`	projects	journal, analytics, expense (link-back)	projectId, wbsId, txType, qty, cost, salesPrice, currency, journalId
`dev.passage.hr.employment.changed.v1`	hr	payroll, projects (resource validity), service (technician), Keycloak sync	workerId, employmentId, action, effectiveFrom, effectiveTo
`dev.passage.hr.absence.approved.v1`	hr	time (wave 25), projects (availability), payroll (paid-leave earnings)	workerId, planId, from, to, hours
`dev.passage.payroll.run.posted.v1`	payroll	journal, bank (wave 36), analytics	payRunId, payCycleId, grossTotal, netTotal, employerTaxTotal, journalId
`dev.passage.expense.report.approved.v1`	expense	ap (wave 18), projects	reportId, workerId, totalAmount, currency, lines[]
`dev.passage.eam.request.created.v1`	eam (or IoT bridge)	service desk UI, notifications	requestId, assetId, source, severity, readingValue
`dev.passage.eam.workorder.completed.v1`	eam	journal (cost settlement), fixed assets, scheduling (wave 23) capacity release	workOrderId, assetId, partsCost, laborCost, downtimeMinutes
`dev.passage.service.sla.breached.v1`	service	notifications, analytics	caseId or workOrderId, slaType, dueAt, breachedAt
`dev.passage.service.dispatch.published.v1`	dispatch	technician PWA, service	scheduleDate, assignments[] (workOrderId, technicianId, start, end)

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Payroll tax legal exposure (wrong withholding, late forms)	Medium	High	Table-driven rule sets versioned per tax year; 200-scenario Pub 15-T fixture gate; ADP/Gusto-style provider adapter as alternative path	Engineering/security
PII breach or over-exposure of HR/payroll data	Medium	High	PII fields in a separate vaulted schema with column-level encryption, OPA duty `HcmPIIRead`, masked by default, every read audited	Engineering/security
POC revenue disputes with auditors	Low	High	Accountant-reviewed workbook fixtures; estimate reversal/re-run is idempotent; loss-provision rule explicit	Product
Dispatch optimizer infeasible or slow at scale	Medium	Medium	CP-SAT with 30 s cap and greedy fallback; property test over generated instances	Engineering
Offline technician PWA conflicts	Medium	Medium	Per-work-order append-only action log with server-side merge; last-writer-wins only for free-text	Engineering
Scope of five large modules in five waves	High	High	Each wave ships the thin vertical slice in its deliverable list; talent/recruitment, subscriptions and predictive maintenance are explicitly basic	Product

8.41 Wave 41: Project Operations: WBS, Time, Expense, Billing & Accounting

8.41.1 Scope and business context

Project-centric businesses (engineering services, contract manufacturing, construction-adjacent, professional services) need a project ledger that AX delivers through Project management and accounting (ProjTable, ProjGroup, ProjWBS, hour/expense/item journals, ProjProposalJour, ProjEstimate) and SAP through PS (PROJ/PRPS WBS, AUFK networks, CJ20N, DP90 resource-related billing, KKA2 results analysis, CJ88 settlement). At the end of this wave a user can create a project hierarchy and WBS with scheduled tasks and resource bookings, record hour/expense/item/fee transactions that cost the project and post to the GL, generate invoice proposals by T&M, fixed-price milestone or retainer, run percentage-of-completion estimates that post WIP and accrued revenue, and read committed cost, EVM and profitability. A demo engineering-services legal entity (`ES-US`) is added alongside the manufacturer.

Out of scope: grant management beyond customer funding sources with limits; project quotations (use wave 26 quotes with a project link); multi-currency funding-rule splitting beyond percentage splits; resource scheduling optimization (wave 23 engine is reused for capacity only); D365 Project Operations "Project for the web" scheduling parity.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX invoice proposal (ProjProposalJour via "Create invoice proposal" dialog) takes 9–12 clicks per project and silently skips transactions whose line property changed after posting; SAP DP90 requires a DIP profile (ODP1) that most sites never get right, so billing runs produce missing lines nobody notices until the customer disputes	Selection is a batch class with no explainable predicate; eligibility state lives on the transaction and is mutated by other processes	Declarative billing rules with a stated predicate (BR-41.3), a "why not billed" explain endpoint per transaction, and proposals built in one action from the project workspace	Clicks 12 → 2 per proposal; omitted-billable-transaction rate 0 (property test); dispute rate attributable to billing errors −90%
Fixed-price estimates (ProjEstimate, "Create estimate" periodic job; SAP KKA2 + CJ88) run as month-end batch taking 1–4 h on 2,000 projects; reversal (ProjEstimate "Reverse") and re-run is manual and frequently leaves orphan WIP	Batch framework; estimate and settlement are two separate steps with separate postings	Estimate run as a Temporal workflow per project with idempotent reverse-and-rerun in one call; WIP projection reconciled nightly to GL	Estimate run 4 h → < 10 min for 2,000 projects; orphan WIP incidents 0; re-run effort 30 min → 1 click
Committed cost (ProjCommittedCost) refreshed by batch ("Update committed costs") so budget checks are hours stale; SAP availability control fires only at PO save and cannot be queried live	Polling batch over PurchLine; no event stream	Committed cost projection updated from PO/receipt events within 10 s and queryable per WBS node	Staleness 4 h batch → < 10 s; PO-over-budget surprises at month-end −95%
WBS scheduling in ProjWBS form recalculates the whole plan synchronously (30–90 s on 1,000 activities) and ignores worker calendars unless "Scheduling with resources" is enabled; SAP CJ20N networks require AFVC maintenance per activity	Form-bound X++ scheduler; calendars optional	CPM scheduler over role calendars, < 500 ms for 2,000 nodes, baseline versioning	Reschedule 60 s → < 0.5 s; calendar-violating task dates 0
Hour journal → ledger reconciliation (ProjEmplTrans vs LedgerTrans) done in Excel each close; SAP CATS → CO → PS needs three transfer steps (CAT5/CAT7)	Separate posting paths with no shared id	Every transaction carries journal_id; subledger-to-GL reconciliation is an automated nightly check with zero tolerance	Reconciliation effort 2 person-days per close → 0; variance 0.00
External systems (PLM, MES, BI) read project cost via nightly SQL extracts of ProjTable/ProjCostTrans	No supported API/event surface	REST/GraphQL endpoints plus CloudEvents on Kafka and MCP exposure (wave 47)	Integration lead time weeks → same day; data latency 24 h → < 60 s

8.41.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Project master and hierarchy	ProjTable, ProjGroup, ProjType (TimeMaterial, FixedPrice, Internal, Cost, Investment, Time), form ProjTable	PROJ (project definition), PRPS (WBS elements), CJ20N Project Builder	`projects` service, `Project` entity, screen Projects › Project	full
WBS and scheduling	ProjWBS, ProjPlanVersion, ProjWBSActivityEstimate	PRPS, AUFK networks, AFVC activities, CJ20N	`WbsNode`, `WbsDependency`, CPM scheduler	full
Hour/expense/item journals	ProjJournalTable/ProjJournalTrans → ProjEmplTrans, ProjCostTrans, ProjItemTrans, ProjRevenueTrans	CATS (CATSDB) → CO postings; MM goods issue to WBS; KB21N activity allocation	`ProjectJournal`, `ProjectTransaction` (type HOUR/COST/ITEM/FEE)	full
Invoice proposals	ProjProposalJour, ProjProposalEmpl/Cost/Item/OnAcc, ProjInvoiceJour	DP90/DP91 resource-related billing, billing request → VF01	`InvoiceProposal` → wave-29 `Invoice`	full
Fixed-price estimates	ProjEstimate, ProjPeriodTable, completed-contract / percentage-of-completion, cost template	KKA2 results analysis (RA keys, RA version), CJ88 settlement to CO-PA/GL	`ProjectEstimate` with method COMPLETED_CONTRACT / POC_COST / POC_MILESTONE	full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Ledger posting profiles	ProjPosting (ProjLedgerPostingSetup), ProjLinePropertySetup, ProjCategory	Cost elements, OKB9 default account assignment, settlement profile	`ProjectPostingProfile` keyed by (projectGroup, category, lineProperty)	full
Customer funding / grants	ProjFundingSource, ProjFundingRule, ProjFundingLimit, ProjGrant	Not native (grants management GM module)	`FundingSource`, `FundingRule` (percentage split, limits)	partial — grant compliance reporting dropped; v1.x
Resource roles and bookings	ResResource, ResRole, ResResourceRole, PSAProjectTeam; D365 PO bookable resources	HR mini-master in PS, capacity planning CMP2	`ResourceRole`, `ResourceBooking` (soft/hard)	full
Committed cost	ProjCommittedCost, ProjCommitmentTmp	Commitment items, CJI5	`CommittedCost` projection from PO/requisition events	full
EVM and reporting	Project statements, cash flow, invoice control	CJI3, S_ALR_87013532 cost/revenue reports	`/reports/projects/evm` (PV/EV/AC, CPI, SPI)	improved — live rather than batch

8.41.3 Functional requirements

- **FR-41.1 Project master and hierarchy** — Must. The system shall maintain projects with type (TIME_MATERIAL, FIXED_PRICE, INTERNAL, COST, INVESTMENT, TIME), group, customer, contract, currency, dates, stage and an unlimited parent/child hierarchy.
  - Stages: CREATED → IN_PROCESS → FINISHED (and ON_HOLD from IN_PROCESS); transitions gated by duty `ProjManage`.
  - Sub-projects inherit group, customer and contract unless overridden; a sub-project cannot change customer.
  - Project ID from wave-5 number sequence `PROJ`, 20 chars max.
- **FR-41.2 WBS with scheduling** — Must. The system shall maintain WBS nodes per project plan version with effort, duration, dependencies (FS/SS/FF/SF with lag), constraints and a CPM scheduler respecting resource calendars.
  - Plan versions: one BASELINE (immutable after approval), one WORKING; copying creates a new version.
  - Scheduler recomputes early/late start/finish, slack and critical path in < 500 ms for 2,000 nodes.
  - Calendars come from wave-8 resource calendars; a task cannot be scheduled on a non-working day of its assigned role calendar.
- **FR-41.3 Resource roles and bookings** — Must. The system shall define roles with cost and sales prices and allow soft/hard bookings of workers (wave 42) or generic resources to WBS nodes.
  - Hard booking over 100% utilization on a day is rejected; soft booking is warned.
  - Price precedence for hours: worker-specific → role × project → role × customer → role.
- **FR-41.4 Project transactions** — Must. The system shall record HOUR, COST (expense), ITEM, FEE and ON_ACCOUNT transactions through journals or integrated documents (timesheets wave 25, expense wave 43, PO/receipt wave 17, inventory journal wave 16).
  - Each transaction carries cost price, sales price, line property (chargeable/non-chargeable), category, WBS node, quantity, three-currency amounts.
  - Posting is atomic with the universal journal through the wave-11 posting API with idempotency key `projects:txn:{journalId}:{lineNo}`.
- **FR-41.5 Posting profiles** — Must. The system shall resolve GL accounts for cost, WIP, accrued revenue, invoiced revenue and on-account by (projectGroup, category, lineProperty) with fallbacks to (group, *, *) and (*, *, *).
  - Resolution failure blocks posting with error `PROJ-POST-001` naming the missing key.
- **FR-41.6 Invoice proposals** — Must. The system shall build invoice proposals selecting billable transactions per billing rule (T&M, fixed-price milestone, retainer/on-account, unit-of-delivery) and hand them to wave-29 invoicing.
  - Selection filters: project set, date range, transaction types, line property CHARGEABLE, not previously proposed, funding-rule allocation.
  - A proposal is editable until POSTED; removing a line returns it to the pool.
- **FR-41.7 Fixed-price estimates** — Must. The system shall run periodic estimates with methods COMPLETED_CONTRACT, POC_COST (cost-to-cost) and POC_MILESTONE, posting WIP, accrued revenue and loss provisions, with reversal and re-run.
  - Period source: wave-14 fiscal periods; one estimate per project per period; re-run reverses the prior posting first.
- **FR-41.8 Committed cost** — Should. The system shall project committed costs from requisitions, POs and item requirements in near-real time (< 10 s lag) and surface remaining budget.
- **FR-41.9 Customer funding sources** — Should. The system shall split project costs/revenue among funding sources by percentage with per-source limits and a limit-exceeded hold.
- **FR-41.10 EVM and profitability reporting** — Must. The system shall compute PV, EV (POC × BAC), AC, CPI, SPI, EAC and gross margin per project and WBS node as of any date.
- **FR-41.11 Project purchase orders and subcontracts** — Must. The system shall allow wave-17 POs to reference project/WBS so that receipts create ITEM/COST transactions and open POs appear as committed cost.

8.41.4 Data model

Project

Field	Type	Required	Notes
id	text(20)	yes	PK; number sequence PROJ
tenant_id	uuid	yes	RLS
legal_entity_id	uuid	yes	FK wave 6
parent_project_id	text(20)	no	FK self
project_group_id	text(10)	yes	FK ProjectGroup (type, posting profile)
project_type	enum	yes	TIME_MATERIAL, FIXED_PRICE, INTERNAL, COST, INVESTMENT, TIME
customer_party_id	uuid	cond	required for external types
contract_id	uuid	cond	FK ProjectContract (billing rules, funding)
currency	char(3)	yes	project currency
stage	enum	yes	CREATED, IN_PROCESS, ON_HOLD, FINISHED

Field	Type	Required	Notes
start_date, end_date	date	yes/no	end_date ≥ start_date
dimension_set_id	uuid	yes	default financial dimensions
version	int	yes	optimistic concurrency

WbsNode

Field	Type	Required	Notes
id	uuid	yes	PK
project_id, plan_version_id	text, uuid	yes	FK; plan version BASELINE/WORKING
wbs_code	text(40)	yes	unique per plan version, e.g. `1.2.3`
parent_id	uuid	no	FK self
name	text(120)	yes	
effort_hours	numeric(12,2)	no	≥ 0
duration_days	numeric(8,2)	no	working days
early_start, early_finish, late_start, late_finish	date	computed	scheduler output
constraint_type, constraint_date	enum, date	no	ASAP, SNET, FNLT, MSO
role_id	uuid	no	default resource role
is_billable_milestone	bool	yes	default false

ProjectTransaction

Field	Type	Required	Notes
id	uuid	yes	PK
project_id, wbs_node_id	text, uuid	yes/no	
txn_type	enum	yes	HOUR, COST, ITEM, FEE, ON_ACCOUNT
category_id	text(20)	yes	FK ProjectCategory (type-bound)
line_property	enum	yes	CHARGEABLE, NON_CHARGEABLE
txn_date	date	yes	
quantity	numeric(14,3)	yes	hours, units
cost_amount, sales_amount	numeric(18,4)	yes	in project currency; plus accounting and reporting currency columns
currency	char(3)	yes	
worker_id, item_id, vendor_party_id	uuid	cond	by txn_type
source_doc_type, source_doc_id	text, uuid	yes	TIMESHEET, EXPENSE, PO_RECEIPT, INV_JOURNAL, MANUAL
journal_id	uuid	yes	universal journal reference
proposal_id, invoice_id	uuid	no	set when billed
estimate_id	uuid	no	set when captured in POC

InvoiceProposal: id, project_id (or contract_id for multi-project), billing_rule_id, customer_party_id, currency, status (DRAFT, APPROVED, POSTED, CANCELED), from_date, to_date, total_cost, total_sales, tax_amount, lines[] (transaction_id, qty, unit_price, amount, funding_source_id).

ProjectEstimate: id, project_id, period_id, method, cost_to_date, estimated_cost_to_complete, estimated_total_cost, contract_value, poc_percent numeric(7,4), revenue_to_date, revenue_this_period, billed_to_date, accrued_revenue, deferred_revenue, loss_provision, status (DRAFT, POSTED, REVERSED), journal_id.

Invariants: Σ ProjectTransaction.cost_amount per project equals the project cost subledger balance and the GL WIP+cost accounts for the posting profile; a transaction appears on at most one POSTED proposal; one POSTED estimate per (project, period); WBS codes unique per plan version. Indexing: `ProjectTransaction (tenant_id, project_id, txn_date)`, partial index on `proposal_id IS NULL AND line_property = 'CHARGEABLE'` for proposal selection; `WbsNode (plan_version_id, parent_id)`; table partitioned by tenant hash (16).

8.41.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/projects`	Create project	`Idempotency-Key` header	ProjMaintain
PUT	`/api/v1/projects/{id}/plans/{ver}/wbs`	Replace WBS (bulk)	version ETag	ProjPlan
POST	`/api/v1/projects/{id}/plans/{ver}/schedule`	Run CPM scheduler	pure	ProjPlan
POST	`/api/v1/projects/journals`	Create hour/cost/item/fee journal	key	ProjJournalMaintain
POST	`/api/v1/projects/journals/{id}/post`	Post journal to project and GL	key	ProjJournalPost
POST	`/api/v1/projects/invoice-proposals`	Build proposal from selection criteria	key	ProjBillingMaintain
POST	`/api/v1/projects/invoice-proposals/{id}/post`	Hand to invoicing	key	ProjBillingPost
POST	`/api/v1/projects/estimates/run`	Run estimates for period	key per (project, period)	ProjEstimatePost
POST	`/api/v1/projects/estimates/{id}/reverse`	Reverse estimate	key	ProjEstimatePost
GET	`/api/v1/projects/{id}/evm?asOf=`	EVM metrics	n/a	ProjInquire

Commands validate: journal lines must reference a project in IN_PROCESS; category type must match txn_type; posting date within open period for module PROJECTS (wave 14); proposal selection requires a billing rule whose currency equals project currency.

Event	Trigger	Payload fields	Consumers
`dev.passage.projects.transaction.posted.v1`	journal/document posted	see 8.0	journal, analytics, expense
`dev.passage.projects.invoice-proposal.posted.v1`	proposal → POSTED	proposalId, invoiceId, lines[]	invoicing (wave 29), analytics
`dev.passage.projects.estimate.posted.v1`	estimate posted or reversed	estimateId, projectId, periodId, pocPercent, revenueThisPeriod, journalId, reversed	journal, analytics
`dev.passage.projects.wbs.scheduled.v1`	scheduler run	projectId, planVersionId, criticalPath[], finishDate	resource planning, notifications

8.41.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Projects workspace	tiles: proposals to approve, estimates due, over-budget, unposted journals	project, stage, budget remaining, CPI/SPI	ProjInquire
Project detail	stage, copy, open WBS, forecasts, statements	header fields, funding, posting profile	ProjMaintain
WBS Gantt	add/indent/outdent, link dependencies, schedule, baseline	code, name, effort, dates, slack, role, % complete	ProjPlan
Resource board	book, split, resolve over-allocation	worker/role × week heatmap	ProjPlan
Journals (hour/cost/item/fee)	add line, validate, post	project, WBS, category, qty, cost, sales, line property	ProjJournalMaintain
Invoice proposals	build, review, remove line, approve, post	transaction rows, totals, funding split	ProjBillingMaintain
Estimates	run, review, reverse, post	cost to date, ETC, POC, revenue, WIP	ProjEstimatePost
Project statements / EVM	drill to transaction and journal	PV/EV/AC, CPI, SPI, EAC, margin	ProjInquire

Interaction rules: Gantt supports keyboard indent (Tab), outdent (Shift+Tab), link (Ctrl+L); bulk edit of category/line property on journal lines; validation on line commit with inline errors; proposal grid personalization (column set saved per user).

8.41.7 Business rules and edge cases

- **BR-41.1 Cost-to-cost POC** — poc = cost_to_date ÷ estimated_total_cost, rounded to 4 decimals; revenue_to_date = contract_value × poc, rounded to currency minor unit; revenue_this_period = revenue_to_date − prior revenue_to_date. Worked example (USD, fixed-price, contract 500,000.00, original estimate 400,000.00): period 1 cost to date 120,000.00, ETC 280,000.00 → EAC 400,000.00, poc 0.3000, revenue to date 150,000.00, billed to date 100,000.00 → accrued revenue 50,000.00 (debit accrued revenue, credit revenue). Period 2 cost to date 260,000.00, ETC revised to 190,000.00 → EAC 450,000.00, poc 0.5778 (260,000/450,000 = 0.57777…), revenue to date 288,888.89, period revenue 138,888.89, billed to date 300,000.00 → deferred (billed in excess) 11,111.11 as liability; gross profit to date 28,888.89. A EUR project uses the same arithmetic in EUR with accounting-currency amounts at the period-end rate.
- **BR-41.2 Loss provision** — if EAC > contract_value, the full expected loss (EAC − contract_value) is provisioned in the period discovered; later improvements release the provision, never below zero.
- **BR-41.3 Invoice proposal selection** — a transaction is selected when: line_property = CHARGEABLE, proposal_id IS NULL, txn_date ∈ [from, to], project stage ∈ {IN_PROCESS, FINISHED}, txn_type ∈ rule.types, and for FIXED_PRICE rules only ON_ACCOUNT milestones with is_billable_milestone WBS nodes whose % complete = 100 or whose date ≤ to. Transactions under a funding source whose limit is reached are selected on the next source by priority; if none, the line is HELD with reason `FUND-LIMIT`.
- **BR-41.4 Retainers** — on-account (retainer) invoices create ON_ACCOUNT transactions with negative sales; T&M proposals apply open retainer balance before invoicing net, oldest first.
- **BR-41.5 Price precedence and rounding** — hour sales price resolved per FR-41.3; amounts = qty × price rounded half-even to minor unit per line, never per total.
- **BR-41.6 Period control** — a transaction with txn_date in a closed PROJECTS period is rejected (`PROJ-PERIOD-001`); estimates run only after the period's projects sub-ledger is closed for transactions.
- **BR-41.7 Stage FINISHED** — allowed only when no open journals, no DRAFT/APPROVED proposals, WIP balance zero or an estimate with method COMPLETED_CONTRACT posted for the final period.
- **BR-41.8 Concurrency** — proposal building takes a `SELECT … FOR UPDATE SKIP LOCKED` on candidate transactions; two concurrent builds for overlapping criteria never select the same transaction.

Estimate state machine: DRAFT → POSTED → REVERSED; REVERSED → (new DRAFT for same period). Proposal: DRAFT → APPROVED → POSTED; DRAFT/APPROVED → CANCELED.

8.41.8 Acceptance criteria

- **AC-41.1** (FR-41.7, BR-41.1) Given the fixture project in BR-41.1, when estimates run for periods 1 and 2, then revenue_this_period is 150,000.00 and 138,888.89 and the posted journal balances per the posting profile.
- **AC-41.2** (FR-41.7, BR-41.2) Given EAC revised to 520,000.00 in period 3, when the estimate runs, then loss_provision = 20,000.00 and the GL loss-provision account carries 20,000.00.
- **AC-41.3** (FR-41.6, BR-41.3) Given 10,000 generated transactions and a T&M rule for 2026-09, when a proposal builds, then every selected line satisfies the predicate and no qualifying line is omitted (property test).
- **AC-41.4** (FR-41.2) Given a WBS with FS dependency and lag 2 days on a role calendar closed on weekends, when scheduled, then successor early start is the next working day after lag.
- **AC-41.5** (FR-41.4, FR-41.5) Given an hour journal line with category LABOR and line property CHARGEABLE, when posted, then the journal debits WIP-cost and credits payroll allocation per profile and the transaction's journal_id is set.
- **AC-41.6** (FR-41.11) Given a PO line referencing WBS 1.2, when the receipt posts, then an ITEM transaction exists and committed cost decreases by the received amount within 10 s.

- **AC-41.7** (FR-41.9) Given two funding sources 70/30 with a 10,000.00 limit on source B, when proposals exceed the limit, then the excess line is HELD with reason FUND-LIMIT.
- **AC-41.8** (FR-41.10) Given BAC 400,000, poc 0.30, AC 120,000 and PV 100,000, when EVM is queried, then EV = 120,000, CPI = 1.00, SPI = 1.20.

8.41.9 Test plan and fixtures

Fixtures: `proj-estimates-12x4` (12 projects × 4 periods, accountant workbook `fixtures/projects/poc-workbook.xlsx`); `proj-txn-gen` (Hypothesis generator, 10,000 transactions, seeds committed); `proj-wbs-2000` (2,000-node WBS with 3,500 dependencies); `es-us-demo` (demo entity with 25 live projects derived from wave-29 customers).

Test	Type	Target	Pass criterion
POC arithmetic	unit	estimate engine	matches workbook to 0.01
Proposal selection	property	selection predicate	0 false positives/negatives over 10k
CPM scheduling	unit/perf	scheduler	correct dates; 2,000 nodes < 500 ms
Posting profile resolution	contract	posting API	every (group, category, property) in fixture resolves
Journal → GL	E2E	journals, journal service	balanced entries; event emitted
EVM	unit	report service	CPI/SPI per formulas
Proposal build concurrency	integration	two parallel builds	disjoint selections

Reconciliation: Σ transactions cost by project = project cost subledger = GL cost/WIP accounts; Σ POSTED proposal sales = invoiced revenue per wave-29; accrued – deferred revenue = GL accrued/deferred accounts.

8.41.10 Deployment and operations

Helm chart `passage-projects` (API, scheduler worker, estimate worker), Postgres cluster `projects` (CloudNativePG, 3 replicas), Atlas migration `projects/0001_initial`. Config: `PROJ_SCHEDULER_MAX_NODES=5000`, `PROJ_PROPOSAL_BATCH=500`. Feature flags: `projects.intercompany-hours`, `projects.funding-limits`. Temporal schedules: nightly `projects-reconcile` 02:00 tenant-local, `committed-cost-refresh` every 60 s. SLOs: journal post p95 < 400 ms for 100 lines; proposal build p95 < 5 s for 5,000 candidates; availability 99.9%. Alerts: reconciliation variance ≠ 0; estimate run failure; outbox lag > 60 s. Runbook: re-run estimate (reverse then run), rebuild committed cost projection from events. Rollback: Argo digest rollback; migrations are additive within the wave.

8.41.11 Migration from AX / SAP

Source	Passage entity	Notes
ProjTable, ProjGroup, ProjType	Project, ProjectGroup	Status map: Created→CREATED, InProcess→IN_PROCESS, Finished→FINISHED
ProjWBS, ProjPlanVersion	WbsNode, PlanVersion	AX activity numbers → wbs_code
ProjEmplTrans, ProjCostTrans, ProjItemTrans, ProjRevenueTrans, ProjOnAccTrans	ProjectTransaction	open (unbilled) only; billed history as read-only archive
ProjPosting, ProjCategory, ProjLinePropertySetup	ProjectPostingProfile, ProjectCategory, line property	
ProjEstimate, ProjPeriodTable	ProjectEstimate	last posted estimate per project as opening balance
PROJ, PRPS, AUFK/AFVC	Project, WbsNode	PRPS.POSID → wbs_code; networks flattened to nodes
CATSDB (released, unbilled)	ProjectTransaction HOUR	
KKA2 RA results (COSB)	ProjectEstimate opening	RA version 0 only

Validation: per-project cost to date, billed to date and WIP must equal source within 0.01; count of unbilled chargeable transactions equal. Not migrated: billed history detail (archived in lakehouse, wave 46), ProjForecast models older than current year, SAP settlement rules (CJ88) — replaced by posting profiles.

8.41.12 Dependencies, risks, open questions

Depends on: wave 11 posting API, wave 14 periods, wave 17 PO project references, wave 25 timesheets, wave 29 invoicing, wave 37 CIP, wave 38 budget control, wave 40 intercompany.

Risk	Likelihood	Impact	Mitigation
POC method disputes	Low	High	Workbook sign-off before wave close
Proposal performance on large backlogs	Medium	Medium	Partial index + batch 500

Open questions: (1) Support POC by effort (hours) in v1? Default no; decide by wave 42. (2) Grant compliance reporting (AX ProjGrant)? Default v1.x; decide by wave 46. (3) Intercompany resource hours at transfer price (wave 40 mirror documents)? Default v1.x; decide by wave 42.

8.42 Wave 42: Human Resources Core: Workers, Positions, Absence & Talent

8.42.1 Scope and business context

HR core replaces AX Human resources (HcmWorker, HcmEmployment, HcmPosition, HcmJob, HcmPositionWorkerAssignment, LeaveRequest/LeavePlan, HcmBenefit) and SAP HCM PA/OM (PA20/PA30 infotypes 0000 Actions, 0001 Org Assignment, 0002 Personal Data, 0008

Basic Pay, 2001 Absences; HRP1000/HRP1001 objects and relationships; PT time types; benefits PA-BN). After this wave, an HR administrator manages workers, employments, positions, jobs and departments with effective dating; employees request absence and enroll in benefits; managers approve and run simple reviews. The wave also establishes the PII vault used by payroll.

Out of scope: full recruitment (ATS), learning management, compensation cycles beyond fixed/variable plans, SuccessFactors-style succession, country-specific HR legal fields beyond the framework.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX HcmPositionWorkerAssignment edits in the Position form require "Maintain versions" dialogs; as-of reporting needs SQL over ValidFrom/ValidTo and routinely returns two managers for a day because ranges are inclusive-inclusive; SAP PA30 infotype splits (BEGDA/ENDDA) produce the same overlap at delimit time	Date-effective framework with inclusive end dates and no exclusion constraints	`[from, to)` ranges with Postgres exclusion constraints; one `as-of` endpoint; org chart with a date picker	Duplicate/overlapping manager answers 0 (property test over 50,000 dates); as-of query p95 < 50 ms vs 2–5 s SSRS reports
Hire in AX takes 25+ clicks across Worker, Employment, Position assignment, Compensation and User relation forms; SAP PA40 hire action walks 8–12 infotype screens and aborts midway on validation, leaving a partial employee	Action not modeled as a unit; each screen commits separately	One `PersonnelAction` with a diff, workflow approval, and atomic apply (future-dated via Temporal); SCIM provisions the user automatically	Clicks 25 → 4; partial-hire records 0; user provisioning 1–2 days manual → < 60 s
Leave accruals (LeaveBankTransaction) in AX and SAP time evaluation (PT60, schema TM04) run as batch and silently truncate proration to whole days; carry-over expiry needs custom PCRs	Accrual arithmetic hidden in batch/schema code; no stated formula	Formula stated in BR-42.3 with proration, caps, carry-over buckets and expiry; balance change events to ESS in real time	Accrual disputes –90%; balance visible to employee at T+0 instead of next payroll run
Benefits eligibility in HcmBenefit relies on static "Eligibility rules" with a query form that administrators cannot test; SAP PA-BN eligibility groupings need feature trees (PESTA)	Rule engine not simulatable	CEL eligibility expressions with a simulate endpoint over the whole workforce	Eligibility errors at open enrollment 0 on 100-employee fixture; rule authoring 1 day consultant → 10 min admin
PII (national ID, bank) is readable by any user with HR tables access; DB log auditing (AX) off by default, SAP PA20 display not logged	PII stored inline with no masking layer	Vaulted `hr_pii` schema, column encryption, masked default, every unmasked read audited with reason	Unaudited PII reads 0; fields exposed to non-HR roles 0

8.42.2 AX / SAP parity map

Capability	AX 2012 / D365	SAP ECC / S/4	Passage equivalent	Parity
Worker/person	HcmWorker, DirPerson, HcmPersonPrivateDetails	PA0002 Personal Data, PA0006 Address	`Worker` (links wave-7 party person), `WorkerPII` vault	full
Employment	HcmEmployment (ValidFrom/To), HcmEmploymentDetail	PA0000 Actions, PA0001 Org Assignment	`Employment` (effective-dated)	full
Positions and assignments	HcmPosition, HcmPositionDetail, HcmPositionWorkerAssignment, HcmPositionHierarchy	HRP1000 (S objects), HRP1001 A/B 008 holder, A 002 reports-to	`Position`, `PositionAssignment`, `PositionHierarchy`	full
Jobs	HcmJob, HcmJobDetail, HcmJobType, HcmJobFunction	HRP1000 C (job)	`Job`	full
Departments/org	OMOperatingUnit (Department), HcmPositionHierarchyType	HRP1000 O (org unit), PPOME	wave-6 org hierarchy, type DEPARTMENT	full
Compensation	HcmCompensationFixedPlan, HcmCompensationVariablePlan, HcmEmploymentFixedCompensation	PA0008 Basic Pay, wage types	`CompensationPlan`, `FixedCompensation` (effective-dated)	partial — no grade/step matrices in v1
Leave and absence	LeavePlan, LeaveBankTransaction, LeaveRequest; D365 HR leave types	PA2001 Absences, PT quota types (PA2006), accrual via time evaluation	`AbsencePlan`, `AbsenceBalance`, `AbsenceRequest`	full
Benefits	HcmBenefit, HcmBenefitPlan, HcmBenefitOption, HcmWorkerEnrolledBenefit	PA0167/0168/0169 benefit infotypes, PA-BN plans	`BenefitPlan`, `BenefitEnrollment`, `LifeEvent`	full
Skills/certifications	HcmSkill, HcmPersonSkill, HcmPersonCertificate	PD qualifications (Q objects), HRP1001 A 032	`Skill`, `WorkerSkill`, `Certification`	full
Performance	HcmDiscussion, HcmGoal, HcmRatingModel	Performance management (PMP), appraisals	`Review`, `Goal`	partial — basics only
Personnel actions	HcmPersonnelAction, HcmPersonnelActionType	PA40 actions, PA0000	`PersonnelAction` with workflow	full

8.42.3 Functional requirements

- **FR-42.1 Worker records** — Must. The system shall maintain workers (EMPLOYEE, CONTRACTOR) linked to one wave-7 person party, with personnel number from sequence `WRK`, and PII in a vaulted schema.
  - PII fields (national ID, date of birth, bank details, home address) encrypted at column level (AES-256-GCM, tenant key in Vault); masked in all reads unless duty `HcmPIIRead`.
- **FR-42.2 Effective-dated employment** — Must. The system shall maintain employments per legal entity with validity ranges and derived status (ACTIVE, ON_LEAVE, TERMINATED).
  - Multiple employments allowed across entities; at most one ACTIVE per entity at any instant.
- **FR-42.3 Positions, jobs, departments** — Must. The system shall maintain positions (FTE, job, department, reports-to position, cost dimensions) and jobs (title, type, function) with effective dating.
  - Position FTE default 1.00, range 0.05–2.00 (job-share); position assignments cannot exceed FTE in total unless `allowOverfill`.
- **FR-42.4 Position hierarchy as-of** — Must. The system shall answer manager, chain-of-command and direct reports for any worker as-of any date within p95 < 50 ms.
- **FR-42.5 Personnel actions** — Must. The system shall model hire, rehire, transfer, promotion, compensation change, leave of absence and termination as personnel actions with workflow approval and future-dated effect.

- **FR-42.6 Absence plans and accruals** — Must. The system shall accrue balances per plan (frequency MONTHLY/BIWEEKLY/ANNUAL/PER_HOUR_WORKED), with proration, caps, carry-over limits and expiry, and reconcile balances to approved requests and wave-25 time registrations.
- **FR-42.7 Absence requests** — Must. The system shall let employees request absence in hours or days, validate against balance and blackout periods, route to manager, and publish to time and payroll.
- **FR-42.8 Benefits** — Must. The system shall define plans (MEDICAL, DENTAL, VISION, LIFE, RETIREMENT_401K, HSA, FSA, OTHER) with options, eligibility rules, employee/employer rates, open enrollment windows and life events.
- **FR-42.9 Employee and manager self-service** — Must. The system shall provide ESS (personal data, absence, benefits, pay statements from wave 43) and MSS (approvals, team calendar, team positions).
- **FR-42.10 Skills and certifications** — Should. The system shall maintain skill profiles with rating scales and certifications with expiry, feeding service technician skills (wave 45).
- **FR-42.11 Performance basics** — Should. The system shall support goals, review templates and a review cycle with manager/employee sign-off.
- **FR-42.12 Compliance framework** — Must. The system shall support consent records, retention policies per data class and a country-specific field framework (extension slots per country code).
- **FR-42.13 Identity sync** — Must. The system shall provision/deprovision Keycloak users via SCIM on hire/termination actions.

8.42.4 Data model

Worker

Field	Type	Required	Notes
id	uuid	yes	PK
personnel_number	text(20)	yes	unique per tenant; sequence WRK
party_person_id	uuid	yes	FK wave-7 person
worker_type	enum	yes	EMPLOYEE, CONTRACTOR
preferred_name, legal_name	text	yes	legal_name from party
user_id	uuid	no	Keycloak subject
pii_ref	uuid	cond	FK WorkerPII (vault schema)
status	enum	computed	from employments

Employment

Field	Type	Required	Notes
id	uuid	yes	PK
worker_id	uuid	yes	FK
legal_entity_id	uuid	yes	FK
validity	tstzrange	yes	`[start, end)`; exclusion constraint on (worker_id, legal_entity_id) && validity
employment_type	enum	yes	FULL_TIME, PART_TIME, TEMP, CONTRACTOR
hire_date, seniority_date, termination_date	date	yes/no/no	
termination_reason	text(20)	cond	code table
pay_group_id	uuid	no	FK payroll (wave 43)
work_calendar_id	uuid	yes	FK wave 8

Position

Field	Type	Required	Notes
id	uuid	yes	PK
position_number	text(20)	yes	unique; sequence POS
job_id	uuid	yes	FK Job
department_id	uuid	yes	FK org unit type DEPARTMENT
legal_entity_id	uuid	yes	
fte	numeric(4,2)	yes	0.05–2.00
validity	tstzrange	yes	position existence
reports_to_position_id	uuid	no	effective-dated via PositionHierarchy
dimension_set_id	uuid	no	default dims for payroll/projects
allow_overfill	bool	yes	default false

PositionAssignment: id, position_id, worker_id, employment_id, validity tstzrange, fte_share numeric(4,2), is_primary bool. Exclusion constraint: (position_id && validity) unless allow_overfill; (worker_id, is_primary = true && validity).

AbsencePlan: id, code, unit (HOURS/DAYS), accrual_frequency, accrual_rate numeric(8,4), accrual_day (LAST_DAY/FIRST_DAY/PAYDATE), proration (DAYS_IN_PERIOD/NONE), max_balance, carry_over_max, carry_over_expiry_months, plan_year_start (MM-DD), negative_balance_allowed, min_service_days. **AbsenceBalance**: id, worker_id, plan_id, bucket (CURRENT/CARRY_OVER), period_start, accrued, used, adjusted, expired, balance. **AbsenceRequest**: id, worker_id, plan_id, from, to, quantity, status (DRAFT, SUBMITTED, APPROVED, REJECTED, CANCELED), approver_id, reason.

BenefitPlan: id, code, type, provider_party_id, options[] (code, coverage level, employee_rate, employer_rate, rate_basis PER_PAY_PERIOD/MONTHLY), eligibility_rule (CEL expression), enrollment_window, waiting_days, pre_tax bool, payroll_deduction_code. **BenefitEnrollment**: worker_id, plan_id, option_code, validity, dependents[], life_event_id.

Invariants: every ACTIVE employment has exactly one primary position assignment or an explicit `unassigned` flag; balances = accrued − used + adjusted − expired per bucket; hierarchy is acyclic at every instant (checked on write with a recursive CTE as-of the new validity range). Indexing: GiST on validity ranges; `PositionHierarchy (tenant_id, child_position_id, validity)`; PII schema `hr_pii` with separate role and no RLS bypass.

8.42.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/hr/workers`	Create worker (+ party)	key	HcmWorkerMaintain
POST	`/api/v1/hr/actions`	Submit personnel action (hire/transfer/terminate…)	key	HcmActionMaintain
GET	`/api/v1/hr/workers/{id}/as-of?date=`	Employment, position, manager as-of	n/a	HcmWorkerInquire
GET	`/api/v1/hr/positions/{id}/reports?asOf=&depth=`	Direct/indirect reports	n/a	HcmOrgInquire
POST	`/api/v1/hr/absence/requests`	Create request	key	EssAbsence
POST	`/api/v1/hr/absence/requests/{id}/decide`	Approve/reject	key	MssApprove
POST	`/api/v1/hr/absence/accrue`	Run accrual for plan/period	key per (plan, period)	HcmAbsenceAdmin
POST	`/api/v1/hr/benefits/enrollments`	Enroll (open enrollment or life event)	key	EssBenefits
GET	`/api/v1/hr/workers/{id}/pii`	Unmasked PII (audited)	n/a	HcmPIIRead

Validation: action effective date may be future-dated up to 365 days; termination date ≥ hire date; absence request quantity ≤ balance + negative allowance; benefit enrollment only inside window or within 30 days of a life event.

Event	Trigger	Payload	Consumers
`dev.passage.hr.employment.changed.v1`	action applied	see 8.0	payroll, projects, service, SCIM
`dev.passage.hr.position.assignment.changed.v1`	assignment start/end	workerId, positionId, validity, managerPositionId	authorization (manager roles), workflow routing
`dev.passage.hr.absence.approved.v1`	request approved	workerId, planId, from, to, hours	time, projects, payroll
`dev.passage.hr.benefit.enrollment.changed.v1`	enrollment start/end	workerId, planId, option, validity, employeeRate	payroll deductions

8.42.6 User experience

Screen	Primary actions	Key fields/columns	Duty
HR workspace	tiles: pending actions, expiring certifications, open enrollment		HcmWorkerInquire
Worker detail	tabs: employment, positions, compensation, absence, benefits, skills, documents	masked PII with reveal (audited)	HcmWorkerMaintain
Org chart	navigate as-of date, drag worker to position (creates action)	position, incumbent, FTE, vacancies	HcmOrgInquire
Personnel actions	create, approve, apply	type, effective date, changes diff	HcmActionMaintain
Absence admin	run accrual, adjust, view balances	plan, bucket, accrued/used/balance	HcmAbsenceAdmin
ESS portal	request absence, view balances, update address, enroll benefits, view pay statements		EssBase
MSS portal	approve requests, team calendar, team positions		MssBase
Benefits enrollment wizard	select options, add dependents, confirm	per-pay-period cost preview	EssBenefits

Rules: as-of date picker persists per session in Org chart; PII reveal requires reason text (≥ 10 chars) and logs to audit; all ESS screens pass axe WCAG 2.1 AA and work at 360 px width.

8.42.7 Business rules and edge cases

- **BR-42.1 Effective-dated position logic** — assignments, hierarchy edges and position attributes are stored as `[valid_from, valid_to)` ranges; a change on date D closes the current row at D and opens a new row from D; as-of queries use `validity @> asOf::timestamptz`. Manager of worker W as-of D = incumbent (primary assignment) of `reports_to_position` of W's primary position as-of D; if that position is vacant as-of D, walk up until an incumbent is found, max 10 levels, else NULL (AX "vacant position" behavior).
- **BR-42.2 Future-dated actions** — applied by a Temporal timer at 00:00 tenant-local on the effective date; the action can be withdrawn until then; applied rows are immutable (corrections create new actions).
- **BR-42.3 Accrual formula** — per period: `accrual = rate × proration_factor`, proration_factor = days_employed_in_period ÷ days_in_period when proration = DAYS_IN_PERIOD, rounded half-up to 2 decimals; balance capped at max_balance (excess discarded, logged). Example plan `US-PTO-15` (DAYS, MONTHLY, rate 1.25 on LAST_DAY, max 22.50, carry_over_max 5.00, expiry 3 months, plan year 01-01): hire 2026-03-16 → March accrual 1.25 × 16/31 = 0.65; April–December 9 × 1.25 = 11.25; total accrued 11.90; used 4.00; balance 7.90 at 2026-12-31. Year-end: carry-over bucket = min(7.90, 5.00) = 5.00; 2.90 expired. 2027-01-31: CURRENT accrues 1.25, total 6.25; consumption takes CARRY_OVER first; unused carry-over expires 2027-03-31 23:59:59.
- **BR-42.4 Negative balances** — allowed only if plan.negative_balance_allowed, down to −(rate × 2); termination with negative balance raises a payroll deduction suggestion.
- **BR-42.5 Overlapping absence** — a request overlapping an APPROVED request for the same worker is rejected; overlapping a SUBMITTED one warns.
- **BR-42.6 Benefit eligibility** — evaluated as-of enrollment effective date with CEL expression over {employment_type, fte, service_days, job.type, legal_entity, age}; waiting_days counted from hire_date; a terminated employment ends enrollments at the end of the month of termination (COBRA handling is v1.x).
- **BR-42.7 Hierarchy cycles** — rejected on write (`HCM-ORG-001`).
- **BR-42.8 Certifications** — expiry generates a notification at 60/30/7 days; expired certification removes skill match from wave-45 dispatch.

Absence request state machine: DRAFT → SUBMITTED → APPROVED | REJECTED; SUBMITTED/APPROVED → CANCELED (approved cancellation reverses usage if start date is in the future or within 30 days past with manager approval).

8.42.8 Acceptance criteria

- **AC-42.1** (FR-42.4, BR-42.1) Given 1,200 workers with 50,000 randomized as-of dates, when manager is queried, then results match the reference model 100%, p95 < 50 ms.
- **AC-42.2** (FR-42.3) Given position POS-100 with FTE 1.00 and an active assignment, when a second overlapping assignment is created, then it is rejected unless allow_overfill.
- **AC-42.3** (FR-42.6, BR-42.3) Given plan US-PTO-15 and hire 2026-03-16, when accruals run through 2027-01-31, then balances equal 0.65, 11.90, 7.90, 5.00 carry-over, 2.90 expired, 6.25 as computed.
- **AC-42.4** (FR-42.7) Given a balance of 2.00 days and a request for 3.00 on a plan that forbids negatives, when submitted, then it is rejected with `HCM-ABS-002`.
- **AC-42.5** (FR-42.8, BR-42.6) Given 100 fixture employees and eligibility rule `employment_type in ['FULL_TIME'] && service_days >= 30`, when evaluated, then eligible set matches the fixture expectation exactly.
- **AC-42.6** (FR-42.5, BR-42.2) Given a transfer action effective 2026-12-01, when the date arrives, then the old assignment closes at 2026-12-01T00:00 tenant-local and the new one opens.
- **AC-42.7** (FR-42.1) Given a user without HcmPIIRead, when reading a worker, then national ID is `***-**-1234` and the audit log records the masked read.
- **AC-42.8** (FR-42.13) Given a hire action applied, when SCIM sync runs, then a Keycloak user exists within 60 s; on termination it is disabled within 60 s.

8.42.9 Test plan and fixtures

Fixtures: `hr-workforce-1200` (generated workers, employments, positions across 3 entities with 5 years of actions); `hr-accrual-cases` (36 hire-date/plan combinations with expected balances); `hr-benefit-100` (100 employees, 6 plans, expected eligibility); `hr-pii-contract` (role × field masking matrix).

Test	Type	Target	Pass criterion
As-of manager/position	property	hierarchy queries	100% match reference, p95 < 50 ms
Accrual engine	unit	accrual formula	all 36 cases exact
Eligibility CEL	unit	rule evaluator	100/100
PII masking	security contract	OPA + API	matrix holds; every unmasked read audited
Action workflow	E2E	Temporal + API	future-dated transfer applies on date
SCIM	integration	Keycloak	user lifecycle within 60 s

Reconciliation: Σ absence usage = approved request hours = wave-25 absence time registrations; headcount by department as-of = Σ primary assignments.

8.42.10 Deployment and operations

Chart `passage-hr` (API, workflow worker, SCIM worker, ESS BFF), Postgres `hr` with schema `hr_pii` under a separate role and Vault-managed key. Flags: `hr.performance`. Schedules: `hr-apply-actions` 00:00 tenant-local, `hr-accrue` per plan frequency, `hr-cert-expiry` daily. SLOs: as-of query p95 < 50 ms; ESS page p95 < 300 ms; availability 99.9%. Alerts: action application failures; PII decrypt errors; SCIM lag > 5 min. Runbook: replay actions from event log; rotate tenant PII key (re-encrypt job). Rollback by digest; `hr_pii` migrations never drop columns.

8.42.11 Migration from AX / SAP

Source	Passage entity	Notes
HcmWorker, DirPerson, HcmPersonPrivateDetails	Worker, WorkerPII	PII to vault at load time
HcmEmployment, HcmEmploymentDetail	Employment	ValidFrom/ValidTo → validity range
HcmPosition, HcmPositionDetail, HcmPositionDuration	Position	
HcmPositionWorkerAssignment	PositionAssignment	
HcmPositionHierarchy	PositionHierarchy	type Line only
HcmJob, HcmJobDetail	Job	
LeavePlan, LeaveBankTransaction, LeaveRequest	AbsencePlan, AbsenceBalance (opening), AbsenceRequest (future only)	
HcmBenefit*, HcmWorkerEnrolledBenefit	BenefitPlan, BenefitEnrollment	
PA0000/0001/0002/0006/0008	Employment, Position assignment, Worker, PII, FixedCompensation	infotype BEGDA/ENDDA → ranges
HRP1000/HRP1001 (O, S, C; A/B 002, 003, 007, 008)	org units (wave 6), Position, Job, hierarchy, assignments	
PA2006 quotas, PA2001	AbsenceBalance opening, historical absences (archive)	

Validation: active headcount per entity equal; per-worker current position/manager equal; quota balances within 0.01. Not migrated: closed historical actions older than 5 years (archive), AX HRM recruiting history, SAP PD development plans.

8.42.12 Dependencies, risks, open questions

Depends on: wave 6 org hierarchy, wave 7 person party, wave 8 calendars, wave 2 Keycloak/OPA, wave 4 workflow, wave 25 time registration.

Risk	Likelihood	Impact	Mitigation
Effective-dating bugs	Medium	High	Range types + exclusion constraints + property tests
Country HR field explosion	High	Medium	Extension slot framework; first countries via wave 48

Open questions: (1) Should contractors live in HR or only in the party model? Default HR with worker_type CONTRACTOR; decide by wave 43. (2) ESS as separate PWA or shell module? Default shell module; decide at wave 42 start.

8.43 Wave 43: Payroll Engine & Expense Management

8.43.1 Scope and business context

This wave delivers a country-pluggable gross-to-net payroll engine replacing AX 2012 Payroll (PayrollPayStatement, PayrollEarningCode, PayrollTaxCode, PayrollBenefit, PayrollPayCycle) and SAP PY (schemas, PCRs, wage types, RT results table), and expense management replacing AX Travel and expense (TrvExpTable, TrvExpTrans, TrvPolicy, TrvPerDiem, TrvCreditCardTrans) and SAP Concur. After the wave, a payroll administrator runs a pay cycle from HR and time data, reviews gross-to-net, posts to the GL and produces a payment file; employees submit expense reports with receipts, policy checks, per diems and card transactions, and get reimbursed through AP.

Out of scope: country packs beyond US and the EU reference (DE assumed, see open question 3 in the master PRD); year-end form filing (structures only); garnishment court-order interfaces; Concur-style travel booking.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX 2012 payroll tax updates arrive as hotfix-delivered PayrollTaxCode data (Ceridian engine) with no diff view; SAP PY tax changes ship as HR support packs (BSI TUBs) that require a Basis transport cycle and regression of schemas/PCRs; a wrong table version is discovered only when net pay looks wrong	Rules embedded in code/transports, not versioned data	Rule sets as signed, versioned JSON packs with diff, publish approval and a frozen binding on each run (PayStatement.rule_version per line)	Tax-table deployment 1–2 weeks transport cycle → < 1 h; mis-versioned runs 0 (binding recorded on every line)
Gross-to-net results are opaque: AX pay statement lines do not show basis or bracket; SAP RT (PC_PAYRESULT) shows wage types only decipherable by a payroll consultant	No explain path from result to rule	Each `PayResultLine` stores basis_amount, rule_version and bracket; statement viewer drills to the rule	"Why is my tax X?" tickets resolved by admin in < 2 min vs consultant escalation; 200/200 fixtures explainable
Retro pay: AX PayrollRetroactiveEarning only handles earning-rate changes; SAP retro recalculation (`/551`, `/552` wage types) reruns whole periods and produces deltas nobody can reconcile	Recalculation designed around the batch, not around the delta	Retro engine replays original rule-set versions and books explicit RETRO_* delta lines	Retro reconciliation effort 4 h per run → 0; delta fixtures 25/25
Payroll GL posting (AX "Generate ledger journal" → LedgerJournalTrans; SAP PC00_M99_CIPE) is a separate step, often weeks late, and rounds per-worker amounts so the journal differs from statements by cents	Separate batch with its own rounding	Posting in the same workflow as finalize, per-line rounded amounts summed (no re-rounding) through the wave-11 posting API	Statement-to-GL variance 0.00; posting lag days → same run
Expense policy in TrvPolicy uses X++ expressions a consultant must change; violations appear only at submit; Concur audit rules are a separate SaaS admin console with no API to test	Policy not data, evaluated late	CEL rule schema (BR-43.5) evaluated at line commit and submit with a simulate-against-history endpoint	Rule change lead time 1–2 weeks → 10 min; employee rework (returned reports) –80%; 500/500 fixture exactness
Card transactions (TrvCreditCardTrans) imported by batch and matched by hand; receipts re-keyed	No matching logic, no OCR	Hourly feed import, tolerance matching, OCR draft lines	Manual line keying –90%; auto-match ≥ 95%

8.43.2 AX / SAP parity map

Capability	AX 2012 / D365	SAP ECC / S/4	Passage equivalent	Parity
Pay cycles and periods	PayrollPayCycle, PayrollPayPeriod	Payroll area, period parameters (T549A/T549Q)	`PayCycle`, `PayPeriod`	full
Earnings	PayrollEarningCode, PayrollEarningCodeRate, earning statements	Wage types (T512W), PCR rules	`EarningCode` (rule set versioned)	full
Deductions/benefits	PayrollBenefit, PayrollDeductionCode (AX via benefits)	Deduction wage types, PA0014/0015	`DeductionCode` (pre/post-tax, limits)	full
Taxes	PayrollTaxCode, PayrollTaxEngine (Ceridian), PayrollWorkerTaxCode	Tax schemas (US: UTAX via BSI TaxFactory)	`TaxRuleSet` per country/year; US federal/state tables	full for US; EU reference
Gross-to-net	PayrollPayStatement, PayrollPayStatementLine	Schema U000/D000, RT results table, CRT	`PayStatement` with `PayResultLine`	full
Retro pay	PayrollRetroactiveEarning	Retroactive accounting (recalculation)	`RetroCalculation`	full
GL posting	PayrollPayStatement distribution → LedgerJournal	PC00_M99_CIPE posting run	journal via wave-11 posting API	full
Pay statement/payment	PayrollPayStatement, direct deposit	Pay slip (PE51), DME (PC00_M99_FPAYM)	`PayStatement` PDF; NACHA/SEPA via wave 36	full
Year-end	W-2 data (PayrollW2)	Tax reporter (PU19)	`YearEndForm` structures W-2, 1099, DE Lohnsteuerbescheinigung	partial — structures, not filing
Expense reports	TrvExpTable, TrvExpTrans, TrvCategory	Concur expense reports; PR05 (classic)	`ExpenseReport`, `ExpenseLine`	full
Policies and per diems	TrvPolicy, TrvPolicyRule, TrvPerDiem	Concur audit rules, per diem tables (T706*)	`ExpensePolicyRule`, `PerDiemRate`	full
Card feeds	TrvCreditCardTrans import	Concur card feeds	`CardTransaction` (CSV/OFX/Visa VCF)	full

8.43.3 Functional requirements

- **FR-43.1 Pay groups, cycles, periods** — Must. The system shall define pay groups per legal entity with frequency (WEEKLY, BIWEEKLY, SEMIMONTHLY, MONTHLY), generated pay periods and pay dates, and a cutoff for time/absence.
- **FR-43.2 Versioned rule sets** — Must. The system shall express earnings, deductions, taxes and employer contributions as rule sets versioned by country and effective date; a pay run binds to the rule-set versions active on its period end.
- **FR-43.3 Earnings** — Must. The system shall compute REGULAR, OVERTIME (1.5×, 2.0×), PTO, HOLIDAY, BONUS, COMMISSION and SHIFT_DIFF from salary (FixedCompensation ÷ periods) and wave-25 time registrations.
- **FR-43.4 Deductions and contributions** — Must. The system shall compute pre-tax (401(k), Section 125 medical, HSA/FSA) and post-tax deductions with per-period, annual and percent-of-gross limits, and employer contributions (match with cap).
- **FR-43.5 US tax engine** — Must. The system shall compute federal income tax withholding per Pub 15-T percentage method (automated payroll systems), Social Security (6.2% to the annual wage base), Medicare (1.45% + 0.9% additional over 200,000), FUTA, and state income tax/SUI for the states covered by the year's table pack.
- **FR-43.6 EU reference pack (DE)** — Should. The system shall compute Lohnsteuer (via the BMF programmablauf tables), Solidaritätszuschlag, Kirchensteuer and social insurance contributions (KV/PV/RV/AV with Beitragsbemessungsgrenzen) for a demo employee set.
- **FR-43.7 Gross-to-net and pay statements** — Must. The system shall produce a pay statement per worker per period with earnings, pre-tax, taxes, post-tax, net, employer costs and YTD; statements are immutable after the run is FINALIZED.
- **FR-43.8 Retro pay** — Must. The system shall recalculate prior periods when backdated changes arrive (rate, hours, absence) and post deltas in the current period with an explanatory line.
- **FR-43.9 GL posting and payments** — Must. The system shall post the run to the GL by (earning/deduction/tax code × dimension from position) and generate NACHA/SEPA payment batches through wave 36.
- **FR-43.10 Provider adapter** — Should. The system shall export a provider payroll file (ADP/Gusto-style CSV/JSON) and import results for entities that outsource payroll, posting the same GL structure.
- **FR-43.11 Expense reports** — Must. The system shall let workers create reports with lines (category, date, amount, currency, merchant, project/WBS, attendees, receipt), split lines, itemize hotels and submit for approval.
- **FR-43.12 Policy engine** — Must. The system shall evaluate rule sets at line commit and submit, producing WARN, BLOCK, REQUIRE_JUSTIFICATION or REQUIRE_RECEIPT outcomes, with per-diem computation by location/date.
- **FR-43.13 Card feeds and receipts** — Must. The system shall import card transactions (CSV, OFX, Visa VCF 4.0) into a worker's unassigned pool, match receipts by amount/date/merchant (tolerance ±1% or 2 days), and OCR receipts into draft lines.
- **FR-43.14 Reimbursement** — Must. The system shall post approved reports as wave-18 vendor invoices against the worker's employee-vendor account (company-paid card lines against the card issuer vendor), with tax recovery lines where VAT is recoverable.

8.43.4 Data model

PayRun

Field	Type	Required	Notes
id	uuid	yes	PK
pay_group_id, pay_period_id	uuid	yes	FK
run_type	enum	yes	REGULAR, OFF_CYCLE, CORRECTION
status	enum	yes	DRAFT, CALCULATED, REVIEWED, FINALIZED, POSTED, PAID, CANCELED
rule_set_binding	jsonb	yes	{country, taxYear, versions{}} frozen at calculate
gross_total, net_total, employer_cost_total	numeric(18,2)	computed	
journal_id, payment_batch_id	uuid	no	
calculated_at, finalized_by	timestamptz, uuid	no	

PayStatement

Field	Type	Required	Notes
id	uuid	yes	PK
pay_run_id, worker_id, employment_id	uuid	yes	
gross, taxable_wages_fed, fica_wages, net	numeric(14,2)	yes	
lines[]	PayResultLine	yes	(code, kind EARNING/PRETAX/TAX/POSTTAX/EMPLOYER, hours, rate, amount, ytd_amount, basis_amount, rule_version)
w4	jsonb	cond	{filingStatus, multipleJobs, dependentsCredit, otherIncome, deductions, extraWithholding}
pdf_ref	text	no	object store key
hash	text	yes	SHA-256 of lines; immutability check

TaxRuleSet: id, country, jurisdiction (FED, state code, locality), tax_year, version, effective_from, brackets[] (filing_status, over, not_over, base_tax, rate), wage_base, rates{}, status (DRAFT, PUBLISHED). **EarningCode**: code, name, kind (HOURLY/SALARY/FLAT), multiplier, taxable flags (fed, fica, futa, state), gl_account_rule. **DeductionCode**: code, pre_tax bool, basis (FLAT/PERCENT_GROSS), limits {per_period, annual}, employer_match {percent, cap_percent}, gl_rule, priority.

ExpenseReport: id, worker_id, legal_entity_id, purpose, status (DRAFT, SUBMITTED, APPROVED, REJECTED, POSTED, PAID), total_amount, reimbursable_amount, company_paid_amount, currency, approver_chain[], ap_invoice_id. **ExpenseLine**: id, report_id, category_id, txn_date, amount, currency, exchange_rate, amount_accounting, merchant, city/country, project_id, wbs_node_id, card_txn_id, receipt_id, attendees[], itemizations[], policy_results[] (rule_id, outcome, justification), tax_code, personal bool. **ExpensePolicyRule** (schema in BR-43.5). **CardTransaction**: id, worker_id, card_last4, txn_date, post_date, merchant, mcc, amount, currency, status (UNASSIGNED, ASSIGNED, DISPUTED, PERSONAL).

Invariants: Σ PayStatement.net = PayRun.net_total = payment batch total; Σ GL debits = gross + employer cost; a FINALIZED statement's hash never changes; a card transaction belongs to at most one line; report total = Σ lines (reimbursable + company-paid). Indexing: `PayStatement (worker_id,`

`pay_run_id)` unique; `CardTransaction (worker_id, status)`; payroll tables partitioned by tax_year.

8.43.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/payroll/runs`	Create run for pay period	key	PayrollRunMaintain
POST	`/api/v1/payroll/runs/{id}/calculate`	Gross-to-net calculation (Temporal workflow)	key + status guard	PayrollRunMaintain
POST	`/api/v1/payroll/runs/{id}/finalize`	Freeze statements	key	PayrollRunFinalize
POST	`/api/v1/payroll/runs/{id}/post`	GL post + payment batch	key	PayrollRunPost
PUT	`/api/v1/payroll/rule-sets/{id}/publish`	Publish rule-set version	ETag	PayrollRuleAdmin
POST	`/api/v1/expense/reports`	Create report	key	EssExpense
POST	`/api/v1/expense/reports/{id}/lines`	Add line (policy evaluated)	key	EssExpense
POST	`/api/v1/expense/reports/{id}/submit`	Submit	key	EssExpense
POST	`/api/v1/expense/card-transactions/import`	Import feed	key per file hash	ExpenseAdmin
POST	`/api/v1/expense/receipts`	Upload + OCR	key	EssExpense

Validation: calculate requires HR cutoff passed and all time registrations for the period approved (or policy allows unapproved as 0 hours); finalize requires zero BLOCK validation results; expense submit requires every REQUIRE_* satisfied.

Event	Trigger	Payload	Consumers
`dev.passage.payroll.run.posted.v1`	GL post	see 8.0	journal, bank, analytics
`dev.passage.payroll.statement.finalized.v1`	finalize	statementId, workerId, net, payDate	ESS, provider adapter
`dev.passage.expense.report.approved.v1`	approval	see 8.0	ap, projects
`dev.passage.expense.policy.violated.v1`	BLOCK outcome at submit	reportId, lineId, ruleId, outcome	analytics, audit

8.43.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Payroll workspace	tiles: runs by status, retro pending, exceptions		PayrollInquire
Pay run detail	calculate, review exceptions, compare to prior run (variance %), finalize, post	worker, gross, taxes, net, Δ vs prior	PayrollRunMaintain
Statement viewer	drill line → rule version and basis	earnings, pre-tax, taxes, post-tax, net, YTD	PayrollInquire / EssPayStatement
Rule-set editor	edit brackets/rates, diff versions, publish	jurisdiction, tax year, brackets	PayrollRuleAdmin
Worker payroll profile	W-4 (2020+ fields), state forms, direct deposit (vaulted), garnishments		PayrollWorkerMaintain
Expense report (ESS, mobile)	add line, attach receipt (camera), match card, itemize, submit	category, amount, policy badges	EssExpense
Expense approval (MSS)	approve/reject/return, view receipt side-by-side		MssApprove
Policy admin	rule list, simulate against past reports		ExpenseAdmin

Rules: pay run variance view highlights workers whose net changed > 20% vs prior period; rule-set publish requires a second approver (SoD); expense mobile works offline for draft lines and receipt capture.

8.43.7 Business rules and edge cases

- **BR-43.1 US gross-to-net worked example** (rule set `US-FED-2025.1`, biweekly, 26 periods; Colorado flat 4.40% on federal taxable wages; W-4 2020+, Single, no Step 2/3/4 entries; 401(k) 6% pre-tax; employer match 50% of first 6%). Gross REGULAR 4,000.00. Pre-tax 401(k) 240.00. Federal taxable wages 3,760.00; FICA wages 4,000.00 (401(k) does not reduce FICA). Pub 15-T Step 1: adjusted annual wage = 3,760.00 × 26 = 97,760.00. Step 2 (standard withholding schedule, Single, 2025 table): bracket 54,875–115,125 → 5,578.50 + 22% × (97,760.00 – 54,875.00) = 5,578.50 + 9,434.70 = 15,013.20; ÷ 26 = 577.43. Step 3 credits 0; Step 4 extra 0 → federal withholding 577.43. Social Security 4,000.00 × 6.2% = 248.00 (YTD below wage base 176,100.00). Medicare 4,000.00 × 1.45% = 58.00. Colorado 3,760.00 × 4.40% = 165.44. Net = 4,000.00 – 240.00 – 577.43 – 248.00 – 58.00 – 165.44 = 2,711.13. Employer: match 120.00, SS 248.00, Medicare 58.00, FUTA 0.6% × 4,000 = 24.00 until YTD 7,000. GL: Dr salary expense 4,000.00, Dr employer tax expense 330.00, Dr 401(k) match expense 120.00; Cr net pay clearing 2,711.13, Cr 401(k) payable 360.00, Cr federal WH payable 577.43, Cr FICA payable 612.00, Cr FUTA payable 24.00, Cr CO WH payable 165.44.
- **BR-43.2 Rounding** — each tax computed on unrounded basis and rounded half-up to 0.01 per line; annualized federal amount divided by periods then rounded; YTD caps applied on the rounded cumulative.
- **BR-43.3 Wage base caps** — SS stops when YTD FICA wages reach the wage base, applying partial tax on the crossing period; additional Medicare starts on the crossing period for wages over 200,000.00 regardless of filing status.
- **BR-43.4 Retro** — a retro trigger is any change with effective date < current period start affecting a FINALIZED statement; recalculation re-runs the original rule-set versions, computes delta per line, and books deltas as RETRO_* lines in the current period; taxes on retro use current-period method (no amended returns).
- **BR-43.5 Expense policy rule schema** — JSON: `{ "id": "EXP-MEAL-075", "version": 3, "scope": {"legalEntities": ["ES-US"], "categories": ["MEALS"], "workerGroups": ["*"]}, "basis": "PER_DAY" | "PER_LINE" | "PER_REPORT" | "PER_TRIP", "condition": "line.amountAccounting > 75.00 && !line.attendees.size() > 0" (CEL over line, report, worker, perDiem), "threshold": {"amount": 75.00, "currency": "USD"}, "outcome": "WARN" | "BLOCK" | "REQUIRE_JUSTIFICATION" | "REQUIRE_RECEIPT", "message": "Meals over 75 USD/day need attendees", "effectiveFrom": "2026-01-01" }`. Example set: receipts required for any line ≥ 25.00 USD/EUR; lodging per night ≤ GSA/per-diem table × 1.5 else justification; mileage at IRS rate from table; alcohol category BLOCK; submission later than 60 days after txn_date BLOCK; duplicate (same merchant, amount, date within worker) BLOCK.

- **BR-43.6 Per diem** — rate by (country, city tier, date); first/last day 75%; meals provided reduce by 20/30/50% (breakfast/lunch/dinner); computed in rate currency then converted at line date.
- **BR-43.7 Finalization** — FINALIZED statements are immutable; corrections are OFF_CYCLE or CORRECTION runs that reference the original.
- **BR-43.8 Separation of duties** — the user who calculates cannot finalize; the user who publishes a rule set cannot be its author (enforced by wave-48 SoD engine, interim OPA check).

Pay run state machine: DRAFT → CALCULATED → REVIEWED → FINALIZED → POSTED → PAID; CALCULATED/REVIEWED → DRAFT (recalculate); DRAFT…REVIEWED → CANCELED. Expense report: DRAFT → SUBMITTED → APPROVED → POSTED → PAID; SUBMITTED → REJECTED | RETURNED (→ DRAFT).

8.43.8 Acceptance criteria

- **AC-43.1** (FR-43.5, BR-43.1) Given the worked-example worker, when the run calculates, then federal WH = 577.43, SS = 248.00, Medicare = 58.00, CO = 165.44, net = 2,711.13.
- **AC-43.2** (FR-43.5, BR-43.3) Given YTD FICA wages 174,000.00 and gross 4,000.00, when calculated, then SS = 130.20 (6.2% × 2,100.00).
- **AC-43.3** (FR-43.5) Given 200 fixture scenarios across filing statuses, Step 2 checkbox, Step 3/4 values and 6 states, when calculated, then all match Pub 15-T expectations to 0.01.
- **AC-43.4** (FR-43.8, BR-43.4) Given a rate increase effective two periods back, when retro evaluates and the next run calculates, then RETRO_REGULAR equals the summed delta and taxes recompute on the new gross.
- **AC-43.5** (FR-43.9) Given a posted run, when reconciliation runs, then GL payroll expense and liability accounts tie to Σ statements with 0.00 variance.
- **AC-43.6** (FR-43.12, BR-43.5) Given the 500-report fixture, when submitted, then policy outcomes equal the expected matrix exactly (0 false flags).
- **AC-43.7** (FR-43.13) Given a card feed with 1,000 transactions and 800 receipts, when matched, then ≥ 95% auto-match within tolerance and no cross-worker matches.
- **AC-43.8** (FR-43.14) Given an approved report with 300.00 USD reimbursable and 200.00 company-paid, when posted, then an AP invoice of 300.00 exists for the employee vendor and 200.00 for the issuer vendor, and the next payment run pays 300.00.

8.43.9 Test plan and fixtures

Fixtures: `payroll-us-200` (200 scenarios, expected values in `fixtures/payroll/us-2025-expected.csv` derived from Pub 15-T worksheets); `payroll-de-20` (20 DE employees against BMF test cases); `payroll-retro-25`; `expense-policy-500` (reports × rules with expected outcomes); `card-feed-1000` (VCF and OFX samples).

Test	Type	Target	Pass criterion
Federal/state withholding	unit	tax engine	200/200 to 0.01
Caps and crossing periods	property	FICA/FUTA	never over-withholds; crossing period correct
Retro deltas	unit	retro engine	25/25
GL posting	contract	posting API	balanced; account per code rule
Policy engine	unit	CEL evaluator	500/500
Receipt OCR	integration	OCR service	amount/date extracted ≥ 90% on sample set
Pay run perf	performance	calculate workflow	10,000 workers < 10 min

Reconciliation: Σ statement net = payment batch; Σ tax lines by jurisdiction = liability accounts; Σ reimbursable = AP employee-vendor invoices.

8.43.10 Deployment and operations

Charts `passage-payroll` (Go API, Temporal calc workers, statement renderer) and `passage-expense` (TS API, OCR Python worker, feed importer); Postgres `payroll` (encrypted bank columns via `hr_pii` key pattern) and `expense`. Rule-set packs delivered as signed JSON bundles (`rulesets/us/2025.1.json`) loaded by migration job. Flags: `payroll.de-pack`, `payroll.provider-adapter`, `expense.ocr`. Schedules: `payroll-cutoff-reminder`, `expense-card-import` hourly (SFTP pull), `expense-aging` daily. SLOs: calculate 10k workers < 10 min; statement fetch p95 < 200 ms; policy evaluation p95 < 100 ms per line. Alerts: calculation failures; tax table pack missing for period end; card import failures. Runbook: re-run calculation from DRAFT; reissue payment batch; rotate SFTP keys. Rollback by digest; FINALIZED data is never altered by migrations.

8.43.11 Migration from AX / SAP

Source	Passage entity	Notes
PayrollPayCycle, PayrollPayPeriod	PayCycle, PayPeriod	
PayrollEarningCode, PayrollTaxCode, PayrollBenefit	EarningCode, TaxRuleSet mapping, DeductionCode	AX tax engine codes mapped to jurisdictions
PayrollWorkerTaxCode, PayrollWorkerTaxRegion, W-4 data	worker payroll profile	
PayrollPayStatement (current year)	PayStatement (YTD opening balances only)	historical statements archived
TrvExpTable/TrvExpTrans (open)	ExpenseReport/ExpenseLine	DRAFT/SUBMITTED only
TrvPolicy/TrvPolicyRule, TrvPerDiem	ExpensePolicyRule, PerDiemRate	X++ expression rules rewritten to CEL (manual)
T512W wage types, T549*	EarningCode/DeductionCode, PayCycle	
RT (PCL2 cluster) current year	YTD opening balances	
PA0210 (W-4), PA0207/0208/0209	worker payroll profile	
Concur exports	ExpenseReport (open)	via CSV

Validation: YTD gross/tax per worker equal to source within 0.01; count of open expense reports equal. Not migrated: prior-year statements (lakehouse archive), SAP schemas/PCRs (rebuilt as rule sets), closed expense reports.

8.43.12 Dependencies, risks, open questions

Depends on: wave 42 HR, wave 25 time, wave 18 AP, wave 19 payments, wave 36 bank files, wave 13 tax codes, wave 12 FX.

Risk	Likelihood	Impact	Mitigation
Tax table maintenance each year	High	High	Signed rule-set packs; provider adapter path
OCR accuracy	Medium	Low	Draft lines always user-confirmed

Open questions: (1) EU reference DE vs UK (master PRD Q3). Default DE; decide before wave 43 start. (2) Store direct-deposit details in payroll or wave-36 bank domain? Default payroll with vault; decide by wave 43.

8.44 Wave 44: Enterprise Asset Management (Maintenance)

8.44.1 Scope and business context

EAM replaces D365 Asset Management (EntAssetObject, EntAssetFunctionalLocation, EntAssetMaintenancePlan, EntAssetWorkOrder, EntAssetCounter, EntAssetMaintenanceRequest, EntAssetBOM) and SAP PM (IFLOT functional locations, EQUI equipment, MPLA/MPOS plans with IP10 scheduling, QMEL notifications IW21, AUFK orders IW31/IW32, IMPTT measuring points, AFIH order header). After this wave a maintenance planner models the plant as functional locations and assets, sets time/counter/condition-based plans that generate work orders, dispatches technicians against wave-23 resource capacity, consumes spares from wave-16 inventory, records labor and downtime, and sees costs settle to the asset and GL. IoT readings from wave 25 trigger requests.

Out of scope: predictive ML models (hooks only), calibration management, fleet management, linear assets.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
SAP IP10/IP30 deadline monitoring is a batch job (RISTRA20) run daily or weekly; calls are generated late or in bursts and counter-based plans need IK11 readings entered by hand; D365 "Schedule maintenance plans" is likewise a batch with a fixed horizon	Scheduling is periodic recalculation, not event-driven	Scheduler runs on every completion/reading event plus nightly; forecast from rolling usage; algorithm stated in BR-44.1	Call latency 1–7 days batch → < 60 s after triggering event; 40/40 plan fixtures exact
Creating a corrective work order in D365 Asset Management from a request takes 8–10 clicks across Maintenance request → Work order → Job → Lines; SAP IW21 → IW31 → components → IW32 release is 4 transactions	Request and order are separate forms with re-keyed fields	Request triage converts to a WO in one action carrying asset, location, symptom, parts and checklist	Clicks 10 → 2; re-keyed fields 0
IoT condition data lives in a separate product (D365 IoT Intelligence, SAP ASPM/IoT) and raises alerts in another UI; maintenance requests are created by people copying values	No in-ERP condition engine	Condition rules on the wave-25 stream create deduplicated requests in < 5 s	Reading-to-request 1 h–1 day manual → p95 < 5 s; duplicate requests 0
Maintenance downtime never reaches production scheduling: SAP PM order has no capacity link to PP work-center scheduling except manual CM01 entries; D365 AM downtime is reporting-only	Separate modules without a shared resource model	WOs reference wave-22 resources; downtime emits a capacity exception consumed by the wave-23 scheduler	Planned-downtime scheduling conflicts –90%; capacity reflects downtime within 10 s
WO cost reconciliation: SAP KO88 settlement to asset/cost center with settlement rules per order; D365 AM costs post via project and need "project invoice/estimate" steps	Order cost is a separate settlement object	Cost rolls to the asset and GL on completion through the posting API; nightly reconciliation	Settlement step eliminated; WO-cost-to-GL variance 0.00

8.44.2 AX / SAP parity map

Capability	AX / D365	SAP ECC / S/4	Passage equivalent	Parity
Functional locations	EntAssetFunctionalLocation, types, structure	IFLOT, IFLOTX, IL01/IL02	`FunctionalLocation` (ltree path)	full
Assets/equipment	EntAssetObject, EntAssetObjectType, asset lifecycle states	EQUI, EQKT, IE01/IE02, equipment categories	`Asset` with `AssetType` and lifecycle	full
Counters/measuring points	EntAssetCounter, EntAssetObjectCounter, counter registration	IMPTT measuring points, IMRG measurement documents (IK11)	`Counter`, `CounterReading`	full
Maintenance plans	EntAssetMaintenancePlan, plan lines (time/counter), plan job types	MPLA/MPOS, IP10 scheduling, IP30 deadline monitoring, strategy plans (IP11)	`MaintenancePlan`, `PlanLine`, scheduler	full
Requests/notifications	EntAssetMaintenanceRequest, request lifecycle	QMEL/QMIH notifications M1/M2/M3, IW21/IW22	`MaintenanceRequest`	full
Work orders	EntAssetWorkOrder, EntAssetWorkOrderLine, job types, lifecycle states, checklists	AUFK/AFIH orders PM01/PM02/PM03, IW31/IW32, operations AFVC	`MaintWorkOrder`, `WorkOrderJob`, `Checklist`	full
Scheduling	Work order scheduling against resources (production resources)	Capacity planning CM01, work centers	wave-23 scheduler with `MAINTENANCE` job class	full
Parts and labor	Work order item requirements → inventory journal; hour journal	Component reservation (RESB), goods issue MIGO 261, confirmation IW41 (AFRU)	`WorkOrderPart` → wave-16 reservation/issue; `WorkOrderLabor`	full
Costing	Work order costs to asset (project-based in D365 AM)	Order settlement KO88 to cost center/asset, cost analysis	cost rollup to asset and GL via posting API; optional wave-37 capitalization	full
Condition monitoring	Counter thresholds; IoT Intelligence	PM-PRM, IoT via SAP IoT/ASPM	`ConditionRule` on IoT readings (wave 25)	improved
KPIs	Asset KPIs (D365 AM)	MCI7 MTTR/MTBF	`/reports/eam/kpis` MTBF, MTTR, availability, cost/asset	full

8.44.3 Functional requirements

- **FR-44.1 Functional locations** — Must. The system shall maintain a functional-location hierarchy per site (max depth 12) with types, with assets installable at locations and location history.
- **FR-44.2 Assets** — Must. The system shall maintain assets (type, manufacturer/model, serial, criticality 1–5, lifecycle state, parent asset, production resource link, fixed asset link, warranty) with install/remove history.
  - Lifecycle: CREATED → ACTIVE → INACTIVE → DISPOSED; only ACTIVE assets can receive work orders (override duty).
- **FR-44.3 Counters** — Must. The system shall maintain counters (unit, cumulative/reset, source MANUAL/IOT) with readings validated monotonic for cumulative counters and an average daily usage rate (rolling 30 days).
- **FR-44.4 Maintenance plans** — Must. The system shall define plans with lines of type TIME (interval + unit, calendar), COUNTER (interval + counter), or TIME_OR_COUNTER (whichever first) / TIME_AND_COUNTER, with job type, checklist, estimated hours and parts, early/late tolerances and generation horizon.
- **FR-44.5 Plan scheduling and generation** — Must. The system shall compute next-due dates/readings per BR-44.1 and generate work orders within the horizon, merging lines due within a merge window into one work order per asset.
- **FR-44.6 Maintenance requests** — Must. The system shall capture requests from users, mobile, IoT condition rules and wave-32 quality, with triage to work order, reject or defer.
- **FR-44.7 Work orders** — Must. The system shall manage work orders with jobs (asset, job type, trade, variant), lifecycle, checklists, planned/actual parts and hours, downtime, failure codes (symptom/cause/remedy) and completion notes.
- **FR-44.8 Scheduling** — Must. The system shall schedule work order jobs against wave-23 resources (maintenance teams, technicians via wave-42 workers) with finite capacity and record planned downtime on the production resource.
- **FR-44.9 Parts** — Must. The system shall reserve, pick and issue parts through wave-16 inventory journals with cost flowing to the work order.
- **FR-44.10 Labor** — Must. The system shall record labor via hour registrations (wave 25 kiosk or mobile) at trade cost rates.
- **FR-44.11 Costing and GL** — Must. The system shall accumulate parts, labor, external services (wave-17 PO) and overhead per work order, post to maintenance expense accounts by (asset type, job type, cost dimension of the functional location) and roll up to the asset; CAPITAL job types create wave-37 CIP postings.
- **FR-44.12 Condition rules** — Must. The system shall evaluate IoT readings against thresholds (above/below/rate-of-change over window) and create requests within 5 s of breach, with deduplication.
- **FR-44.13 KPIs** — Should. The system shall compute MTBF, MTTR, availability, cost per asset and backlog hours by asset, location and type.

8.44.4 Data model

Asset

Field	Type	Required	Notes
id	uuid	yes	PK
asset_number	text(30)	yes	unique; sequence AST
asset_type_id	uuid	yes	FK
name	text(120)	yes	
functional_location_id	uuid	no	current location
parent_asset_id	uuid	no	
site_id	uuid	yes	wave 6
serial_number, manufacturer, model	text	no	
criticality	smallint	yes	1–5
lifecycle_state	enum	yes	
production_resource_id	uuid	no	wave 22
fixed_asset_id	uuid	no	wave 37
warranty_until	date	no	
cost_dimension_set_id	uuid	no	

MaintenancePlan / PlanLine

Field	Type	Required	Notes
id	uuid	yes	PK (line)
plan_id	uuid	yes	FK plan (name, asset or asset type/location scope, active)
line_type	enum	yes	TIME, COUNTER, TIME_OR_COUNTER, TIME_AND_COUNTER
interval_value, interval_unit	int, enum	cond	DAYS, WEEKS, MONTHS, YEARS
calendar_id	uuid	no	wave 8; default 7-day
counter_id, counter_interval	uuid, numeric(14,2)	cond	
job_type_id, trade_id, variant	uuid, uuid, text	yes/no/no	
checklist_id	uuid	no	
est_hours, parts[]	numeric, jsonb	no	
early_tolerance_pct, late_tolerance_pct	smallint	yes	default 10/10 (SAP tolerance analog)
shift_factor_early, shift_factor_late	smallint	yes	0–100%, default 0/100
horizon_days, merge_window_days	int	yes	default 90 / 7
last_completed_at, last_completed_reading	timestamptz, numeric	no	updated on completion
next_due_at, next_due_reading, next_due_forecast_at	computed		

MaintWorkOrder

Field	Type	Required	Notes
id	uuid	yes	PK; number WO-{seq}
asset_id, functional_location_id	uuid	yes/no	
work_order_type	enum	yes	PREVENTIVE, CORRECTIVE, INSPECTION, CAPITAL, EMERGENCY
status	enum	yes	CREATED, SCHEDULED, RELEASED, IN_PROGRESS, ON_HOLD, COMPLETED, CLOSED, CANCELED
priority	smallint	yes	1 (emergency)–5
request_id, plan_line_id	uuid	no	provenance
due_at, planned_start, planned_end, actual_start, actual_end	timestamptz		
jobs[]	WorkOrderJob	yes	(job_type, trade, est_hours, actual_hours, checklist results)
parts[]	WorkOrderPart	no	(item, qty_planned, qty_issued, reservation_id, cost)
downtime_minutes	int	no	
failure_symptom, failure_cause, remedy	text(20)	no	code tables
cost_parts, cost_labor, cost_external, cost_overhead	numeric(14,2)	computed	
journal_ids[]	uuid[]		

Counter: id, asset_id, name, unit, kind (CUMULATIVE/RESETTABLE), source, last_reading, last_reading_at, avg_daily_usage numeric(12,4). **CounterReading**: id, counter_id, reading_at, value, source, iot_message_id. **MaintenanceRequest**: id, asset_id, source (USER, MOBILE, IOT, QUALITY), severity, description, reading_value, status (NEW, TRIAGED, CONVERTED, REJECTED, DEFERRED), work_order_id, dedupe_key.

Invariants: cumulative counter readings are non-decreasing in time; a COMPLETED work order has actual_start ≤ actual_end; Σ work order costs = Σ linked journal lines to maintenance accounts; one open (NEW/TRIAGED) IoT request per (asset, rule) dedupe_key. Indexing: `FunctionalLocation.path ltree` GiST; `PlanLine (next_due_at)` and `(next_due_forecast_at)`; `MaintWorkOrder (site_id, status, due_at)`; `CounterReading` partitioned monthly.

8.44.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/eam/assets`	Create asset	key	EamAssetMaintain
POST	`/api/v1/eam/counters/{id}/readings`	Register reading	key (source + message id)	EamCounterRegister / IoT bridge
POST	`/api/v1/eam/plans/schedule`	Recompute next due and generate WOs	key per (plan, run date)	EamPlanAdmin
POST	`/api/v1/eam/requests`	Create request	key (dedupe_key)	EamRequestCreate
POST	`/api/v1/eam/requests/{id}/convert`	Convert to work order	key	EamPlanner
POST	`/api/v1/eam/work-orders`	Create WO	key	EamPlanner
POST	`/api/v1/eam/work-orders/{id}/status`	Transition status	key + version	EamPlanner / EamTechnician
POST	`/api/v1/eam/work-orders/{id}/parts/issue`	Issue parts (inventory journal)	key	EamTechnician
POST	`/api/v1/eam/work-orders/{id}/labor`	Record labor	key	EamTechnician
GET	`/api/v1/eam/kpis?scope=&from=&to=`	MTBF/MTTR/cost	n/a	EamInquire

Validation: readings for cumulative counters must be ≥ last_reading else `EAM-CTR-001` (reset requires explicit reset flag); COMPLETED requires all checklist mandatory items answered and all reserved parts issued or released; CAPITAL work orders require a fixed_asset_id or CIP project.

Event	Trigger	Payload	Consumers
`dev.passage.eam.request.created.v1`	request created	see 8.0	planner UI, notifications
`dev.passage.eam.workorder.status.changed.v1`	transition	workOrderId, from, to, at, actor	mobile, analytics
`dev.passage.eam.workorder.completed.v1`	COMPLETED	see 8.0	journal, fixed assets, scheduling
`dev.passage.eam.counter.reading.registered.v1`	reading	counterId, assetId, value, at, source	plan scheduler, condition rules
`dev.passage.eam.downtime.recorded.v1`	downtime on resource	resourceId, from, to, workOrderId	scheduling (wave 23), OEE (wave 25)

8.44.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Asset management workspace	tiles: requests to triage, WOs due in 7 days, overdue, backlog hours		EamInquire
Plant structure tree	expand locations, install/move assets, open asset	location path, asset, state, criticality	EamStructureMaintain
Asset detail	counters, plans, WO history, costs, KPIs, documents		EamAssetMaintain
Maintenance plan editor	add line, simulate schedule 12 months, activate	type, interval, counter, next due, forecast	EamPlanAdmin
Request triage	convert, reject, defer, merge	severity, source, asset, age	EamPlanner
Work order board (kanban + Gantt)	schedule, release, assign, print	status, priority, due, assigned team	EamPlanner
Technician mobile (PWA)	start/stop, checklist, parts scan, labor, photos, complete		EamTechnician
KPI dashboard	filter by site/type/period	MTBF, MTTR, availability, cost	EamInquire

Rules: board drag changes planned_start with conflict warning; checklist items support numeric with tolerance (auto-flag out of range); mobile offline for up to 72 h with action replay.

8.44.7 Business rules and edge cases

- **BR-44.1 Plan scheduling algorithm** — run nightly and on each completion/reading event for the affected plan lines. For each active line: (a) base = last_completed_at (or plan start if none); time due `T = calendar.add(base, interval)` — if shift factors apply and completion was late/early by Δ versus the previous due, shift base by `Δ × shift_factor_late/early` (SAP IP10 analog); (b) counter due `R = last_completed_reading + counter_interval`; forecast date `F = now + (R − counter.last_reading) ÷ avg_daily_usage` (if avg_daily_usage = 0, F = NULL); (c) next_due_at = T for TIME; = F for COUNTER (next_due_reading = R); = min(T, F) for TIME_OR_COUNTER; = max(T, F) for TIME_AND_COUNTER; (d) generate a work order when `next_due_at − early_tolerance_pct × interval ≤ now + horizon_days` and no open WO exists for the line; (e) lines for the same asset due within merge_window_days are merged into one WO with multiple jobs; (f) a counter-based WO is released only when the actual reading ≥ R − tolerance or the forecast date arrives. Example: line TIME_OR_COUNTER, 90 days or 500 hours; last completed 2026-08-01 at 12,000 h; counter 12,300 h on 2026-10-21 with avg usage 6.5 h/day → T = 2026-10-30, R = 12,500, F = 2026-10-21 + 30.8 days = 2026-11-21 → next_due_at = 2026-10-30; generated on 2026-10-21 (horizon 90) with due 2026-10-30.
- **BR-44.2 Completion resets** — completion sets last_completed_at = actual_end and last_completed_reading = reading at completion (prompted if counter-based); the line reschedules immediately.
- **BR-44.3 Status transitions** — CREATED → SCHEDULED → RELEASED → IN_PROGRESS → COMPLETED → CLOSED; IN_PROGRESS ↔ ON_HOLD; any pre-COMPLETED → CANCELED (releases reservations). CLOSED requires all costs posted (no pending inventory or hour journals) and is final; cost corrections after CLOSED use a CORRECTION WO.
- **BR-44.4 Condition rules** — rule {counter or tag, operator >, <, RATE_OVER(window), value, severity, cooldown_minutes}; breach creates one request with dedupe_key `(asset, rule)` while a request is NEW/TRIAGED or within cooldown (default 60 min); latency target from reading ingest to request event p95 < 5 s.
- **BR-44.5 Downtime** — recording downtime on an asset linked to a production resource creates a capacity exception on wave-23 for [from, to]; planned downtime from SCHEDULED work orders is tentative and becomes firm at RELEASED.
- **BR-44.6 Parts costing** — issued at wave-16 running cost at issue time; returns credit at the same cost; partial issues allowed.
- **BR-44.7 Emergency WOs** — priority 1 skips SCHEDULED and may be IN_PROGRESS before RELEASED; must be regularized within 24 h (alert).
- **BR-44.8 KPIs** — MTBF = Σ uptime ÷ number of failures (CORRECTIVE/EMERGENCY WOs with failure codes) over the window; MTTR = Σ (actual_end − actual_start) of those WOs ÷ count; availability = 1 − downtime ÷ scheduled time.

8.44.8 Acceptance criteria

- **AC-44.1** (FR-44.5, BR-44.1) Given the BR-44.1 example, when the scheduler runs on 2026-10-21, then a WO exists with due 2026-10-30 and next_due_reading 12,500.
- **AC-44.2** (FR-44.5) Given 40 plan fixtures, when scheduled across a simulated year, then generated WO dates/readings match expected within 0 days / 0 units.
- **AC-44.3** (FR-44.11) Given a completed WO with 350.00 USD parts and 8 h labor at 45.00/h, when posted, then the GL maintenance expense is 710.00 and the asset cost history shows 710.00.
- **AC-44.4** (FR-44.12, BR-44.4) Given a bearing temperature rule > 85 °C, when a reading of 87 °C arrives, then a request exists within 5 s and a second reading of 88 °C within 60 min creates none.
- **AC-44.5** (FR-44.8, BR-44.5) Given a RELEASED WO with planned downtime 4 h on resource M-12, when wave-23 schedules, then M-12 capacity for that window is zero.
- **AC-44.6** (FR-44.3) Given a cumulative counter at 12,300, when a reading of 12,250 is posted without reset flag, then it is rejected with EAM-CTR-001.
- **AC-44.7** (FR-44.7, BR-44.3) Given a WO with an unanswered mandatory checklist item, when COMPLETED is requested, then it is rejected with `EAM-WO-004`.
- **AC-44.8** (FR-44.13, BR-44.8) Given 3 failures and 2,700 h uptime in a window, when KPIs are queried, then MTBF = 900 h.

8.44.9 Test plan and fixtures

Fixtures: `eam-plant-demo` (3 sites, 120 locations, 400 assets, 60 counters from wave-22 resources); `eam-plans-40` with a 365-day expected generation calendar; `eam-iot-stream` (10 readings/s replay from wave-25 fixture); `eam-wo-cost-50`.

Test	Type	Target	Pass criterion
Scheduler	unit/property	next-due computation	matches calendar; monotonic under completions
WO generation	E2E	nightly job	40/40 plans
IoT → request	performance	condition engine	p95 < 5 s at 10 readings/s
Cost posting	contract	posting API	balanced; asset rollup equal
Downtime → capacity	integration	scheduling API	capacity reflects downtime
Mobile offline	E2E (Playwright, offline mode)	PWA	72 h replay without loss

Reconciliation: Σ WO costs by account = GL maintenance accounts; Σ parts issued = inventory transactions with reference WO.

8.44.10 Deployment and operations

Chart `passage-eam` (Go API, scheduler worker, condition engine consumer, mobile BFF); Postgres `eam` with ltree and monthly partitions on readings. Flags: `eam.capital-wo`, `eam.predictive-hooks`. Schedules: `eam-plan-schedule` nightly 01:00 site-local plus event-driven; `eam-kpi-rollup` hourly. SLOs: condition engine p95 < 5 s; WO status API p95 < 200 ms; availability 99.9%. Alerts: scheduler run failure; condition consumer lag > 30 s; emergency WO not regularized in 24 h. Runbook: replay readings from Kafka by offset; re-run scheduler for a plan. Rollback by digest.

8.44.11 Migration from AX / SAP

Source	Passage entity	Notes
EntAssetFunctionalLocation	FunctionalLocation	
EntAssetObject, EntAssetObjectType	Asset, AssetType	lifecycle state map
EntAssetCounter, EntAssetObjectCounter	Counter, last reading	
EntAssetMaintenancePlan (+ lines)	MaintenancePlan, PlanLine	last completed from WO history
EntAssetWorkOrder (open)	MaintWorkOrder	closed as history
EntAssetMaintenanceRequest (open)	MaintenanceRequest	
IFLOT/IFLOTX	FunctionalLocation	TPLNR structure indicator → path
EQUI/EQKT/EQUZ	Asset	EQUZ for location/time slices
MPLA/MPOS/MHIS	MaintenancePlan, PlanLine, last call	
IMPTT/IMRG	Counter, CounterReading (last 12 months)	
QMEL (open), AUFK/AFIH PM orders (open)	MaintenanceRequest, MaintWorkOrder	

Validation: asset count per site; open WO count and planned cost equal; next due dates within tolerance of source IP30 output. Not migrated: closed orders older than 3 years (archive), SAP strategy packages (converted to explicit lines).

8.44.12 Dependencies, risks, open questions

Depends on: wave 16 inventory, wave 17 PO, wave 22 resources, wave 23 scheduler, wave 25 IoT, wave 37 fixed assets, wave 42 workers.

Risk	Likelihood	Impact	Mitigation
Counter forecast inaccuracy	Medium	Low	Rolling usage; manual override
IoT volume	Medium	Medium	Partitioned readings; downsampled retention 13 months

Open questions: (1) Capitalize CAPITAL WOs directly to fixed assets or always via CIP project? Default CIP project; decide by wave 44. (2) Calibration in v1? Default no.

8.45 Wave 45: Service Management & Field Service

8.45.1 Scope and business context

Service management replaces AX Service management (SMAServiceAgreementTable, SMAServiceOrderTable, SMAServiceObjectTable, SMADispatchBoard, SMARepair, service subscriptions), D365 Field Service (work orders, bookings, agreements, IoT alerts, mobile) and SAP CS/FSM (service notifications, service orders, IBase, warranties, SAP Field Service Management). After this wave: an installed base is built from shipped serials; agreements define entitlements, SLAs and pricing; cases become service work orders; a dispatcher schedules technicians with OR-Tools respecting skills, territories, time windows and travel; technicians execute offline on a PWA with photos, signatures and truck-stock parts; billing posts to the customer or warranty. This is the phase-exit wave.

Out of scope: customer-facing self-scheduling portal (v1.x), subscription billing beyond wave-29 recurring invoices, remote assist/AR, depot repair RMA beyond wave-30 RMA linkage.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
AX SMADispatchBoard is a manual Gantt: a dispatcher drags each SMAServiceOrderTable row by hand, with no skill or travel check; SAP FSM/RSO optimization is a separate licensed product with its own master data sync	No optimizer in the ERP; duplicated technician master	OR-Tools CP-SAT dispatch inside the service domain using wave-42 skills and wave-16 truck stock (BR-45.3)	Dispatch time per territory-day 2–3 h → < 30 s solve + 10 min review; skill/window violations 0
Entitlement: AX service orders pick the agreement the user chooses (SMAServiceOrderTable.ServiceAgreementId) with no SLA clock; SAP CS warranty check (IW31 "warranty" pop-up) is informational and SLA profiles live in CRM only	No deterministic entitlement or SLA engine in ERP	Resolver (BR-45.1) and business-hour SLA clocks with pause states and breach events (BR-45.2)	Wrong-agreement billing 0 on fixtures; SLA breaches discovered after the fact → warning at 75%
Installed base built by hand: AX service objects created manually after shipment; SAP IBase needs IB51/IB52 maintenance or a custom user exit from delivery	No event from shipment to service	Service objects created from wave-28 serialized shipment events with warranty start	Manual IBase upkeep 100% → 0; serial-to-object lag days → < 60 s
Technician execution on paper or a disconnected app; parts used from vans re-keyed into inventory journals days later, causing negative stock and wrong COGS	No mobile execution or truck-location model	Offline PWA with append-only action log; TRUCK locations with consumption at completion	Parts re-keying 0; inventory accuracy on trucks ≥ 99%; completion-to-invoice 5–10 days → same day
Service billing in AX goes through project invoice proposals (service orders post to ProjEmplTrans/ProjItemTrans), a second module to learn; SAP DP90 again	Service reuses project billing plumbing	Work order lines carry coverage (BILLABLE/WARRANTY/AGREEMENT) and post directly to wave-29 invoicing	Billing steps 6 → 1; warranty-vs-billable misclassification 0 on 30 fixtures

8.45.2 AX / SAP parity map

Capability	AX / D365	SAP ECC / S/4	Passage equivalent	Parity
Installed base / service objects	SMAServiceObjectTable, SMAServiceObjectRelationTable; D365 FS customer assets	IBase (IBIN/IBIB), equipment with customer, serial numbers (SERI)	`ServiceObject` from wave-28 shipment serials	full

Capability	AX / D365	SAP ECC / S/4	Passage equivalent	Parity
Warranties	Limited (SMAServiceObject + agreement) ; D365 FS warranties	Warranty master (BGMK), warranty check on order	`Warranty` (parts/labor/travel, start trigger)	improved
Service agreements	SMAServiceAgreementTable/Line, SMAAgreementLine, service intervals	Service contracts (VA41, VBAK type WV/SC), SLA profiles	`ServiceAgreement`, `Entitlement`, `SlaProfile`	full
Cases/notifications	smmCase (CRM case), D365 FS cases	QMEL service notifications (IW51), CRM cases	`Case`	full
Service orders	SMAServiceOrderTable/Line, stages	Service orders (IW31 type SM01/SM02), CS order	`ServiceWorkOrder`	full
Dispatch board	SMADispatchBoard (Gantt), SMAServiceTechnician	SAP FSM planning board / Resource Scheduling Optimizer; CRM MRS	`dispatch` service (OR-Tools CP-SAT) + board	improved
Technician mobile	D365 FS mobile (Power Apps)	SAP FSM mobile	technician PWA	full
Parts from trucks	D365 FS warehouse per resource	Van stock (storage location per technician)	wave-16 location type TRUCK per technician	full
Billing	SMAServiceOrder → project invoice (service orders post via projects)	DP90 to billing, warranty claim	`ServiceWorkOrder` → wave-29 invoice; WARRANTY lines no-charge	full
KPIs	Service management reports	CS reports	SLA attainment, first-time fix, utilization	improved

8.45.3 Functional requirements

- **FR-45.1 Installed base** — Must. The system shall create service objects automatically from wave-28 shipments of serialized items (and manually), with customer, site address, parent/child structure, install date and current status.
- **FR-45.2 Warranties** — Must. The system shall attach warranty terms (parts/labor/travel coverage flags, duration from SHIP/INSTALL/REGISTRATION, exclusions) to service objects and evaluate coverage per work order line.
- **FR-45.3 Service agreements** — Must. The system shall define agreements per customer with lines covering service objects or categories, entitlements (included visits/hours, discounts, price lists), SLA profile, validity and billing (prepaid, per-visit, subscription via wave 29).
- **FR-45.4 SLA profiles** — Must. The system shall define response and resolution targets by priority with business-hour calendars and pause states per BR-45.2.
- **FR-45.5 Cases** — Must. The system shall capture cases (phone, email, portal, IoT from wave 44/25) and resolve entitlement to the best agreement per BR-45.1, starting the SLA clock.
- **FR-45.6 Service work orders** — Must. The system shall create work orders from cases or agreements (preventive visits) with incident type, required skills, estimated duration, time window, service object, address, and lines (labor, parts, travel, fees) each billable, covered-by-warranty or covered-by-agreement.
- **FR-45.7 Dispatch optimization** — Must. The system shall produce daily schedules per territory with OR-Tools CP-SAT satisfying BR-45.3 constraints, with manual override on the board and incremental re-optimization on changes.
- **FR-45.8 Technician mobile PWA** — Must. The system shall provide an offline-capable PWA: today's route, navigate, start travel/work, checklists, photos, parts from truck stock, signature capture, customer-facing summary.
- **FR-45.9 Truck stock** — Must. The system shall treat each technician's vehicle as a wave-16 location (type TRUCK) with replenishment transfer orders and consumption on work order completion.
- **FR-45.10 Billing** — Must. The system shall post completed work orders to wave-29 invoicing: billable lines at agreement/standard prices, warranty lines at zero with a warranty cost record, agreement-covered lines consuming entitlement.
- **FR-45.11 Service KPIs** — Should. The system shall compute SLA attainment, first-time-fix rate, mean time to respond/resolve, technician utilization and travel ratio.

8.45.4 Data model

ServiceAgreement

Field	Type	Required	Notes
id	uuid	yes	PK; number SA-{seq}
customer_party_id	uuid	yes	
legal_entity_id	uuid	yes	
validity	daterange	yes	
sla_profile_id	uuid	yes	
billing_model	enum	yes	PREPAID, PER_VISIT, SUBSCRIPTION
price_list_id, discount_pct	uuid, numeric(5,2)	no	
priority_rank	smallint	yes	entitlement tie-break
lines[]	AgreementLine	yes	(service_object_id or category, included_visits, included_hours, parts_covered bool, labor_covered bool, travel_covered bool)
status	enum	yes	DRAFT, ACTIVE, SUSPENDED, EXPIRED, CANCELED

Case

Field	Type	Required	Notes
id	uuid	yes	PK; number CS-{seq}
customer_party_id, service_object_id	uuid	yes/no	
channel	enum	yes	PHONE, EMAIL, PORTAL, IOT, INTERNAL
priority	smallint	yes	1–4
agreement_id, entitlement_result	uuid, jsonb	no	resolution trace

Field	Type	Required	Notes
sla_response_due, sla_resolution_due	timestamptz	computed	
sla_state	enum	yes	RUNNING, PAUSED, MET, BREACHED
sla_paused_minutes	int	yes	
status	enum	yes	NEW, IN_PROGRESS, WAITING_CUSTOMER, WAITING_PARTS, RESOLVED, CLOSED, CANCELED
opened_at, responded_at, resolved_at	timestamptz		

ServiceWorkOrder

Field	Type	Required	Notes
id	uuid	yes	PK; number SWO-{seq}
case_id, agreement_id, service_object_id	uuid	no/no/no	
customer_party_id, address_id, geo (lat, lon)		yes	
territory_id	uuid	yes	
incident_type_id, required_skills[]	uuid, uuid[]	yes	
est_duration_minutes	int	yes	
time_window	tstzrange	no	customer-promised
priority, sla_due	smallint, timestamptz		
status	enum	yes	UNSCHEDULED, SCHEDULED, DISPATCHED, TRAVELING, IN_PROGRESS, ON_HOLD, COMPLETED, POSTED, CANCELED
assignment (technician_id, start, end, sequence)		no	from dispatch
lines[]	WorkOrderLine		(kind LABOR/PART/TRAVEL/FEE, qty, unit_price, coverage BILLABLE/WARRANTY/AGREEMENT, amount)
signature_ref, photos[]		no	object store
first_time_fix	bool	no	set at completion

ServiceObject: id, item_id, serial, customer_party_id, site_address_id, parent_id, installed_at, status (ACTIVE, INACTIVE, RETURNED), shipment_line_id, warranty_id. **Warranty**: id, template, starts_on (SHIP/INSTALL/REGISTRATION), months, parts_covered, labor_covered, travel_covered, exclusions[]. **Technician**: worker_id, territory_ids[], skills[] (from wave 42), shift calendar, home geo, truck_location_id, max_daily_minutes, overtime_allowed. **DispatchRun**: id, territory_id, date, status, solver_seconds, objective, assignments[].

Invariants: a case has at most one resolved agreement; SLA due timestamps are recomputed whenever paused minutes change; a work order has at most one active assignment; parts lines must reference the technician's truck location or a wave-16 reservation; invoiced amount = Σ BILLABLE lines. Indexing: `ServiceWorkOrder (territory_id, status, sla_due)`; geo via PostGIS point index; `Case (sla_state, sla_resolution_due)` for the breach monitor.

8.45.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/service/agreements`	Create agreement	key	SvcAgreementMaintain
POST	`/api/v1/service/cases`	Create case (entitlement + SLA start)	key	SvcCaseCreate
POST	`/api/v1/service/cases/{id}/status`	Transition (pauses/resumes SLA)	key + version	SvcCaseMaintain
POST	`/api/v1/service/work-orders`	Create WO	key	SvcWoMaintain
POST	`/api/v1/service/dispatch/runs`	Optimize territory/date	key per (territory, date, version)	SvcDispatch
POST	`/api/v1/service/dispatch/runs/{id}/publish`	Publish schedule	key	SvcDispatch
POST	`/api/v1/service/work-orders/{id}/actions`	Mobile action log (travel start, work start, part, photo, signature, complete)	key per action id	SvcTechnician
POST	`/api/v1/service/work-orders/{id}/post`	Bill / warranty post	key	SvcBillingPost
GET	`/api/v1/service/kpis?territory=&from=&to=`	KPIs	n/a	SvcInquire

Validation: case creation requires a customer; dispatch run requires all WOs to have geo and skills; complete action requires signature unless incident type exempts; post requires COMPLETED and all parts lines consumed.

Event	Trigger	Payload	Consumers
`dev.passage.service.case.opened.v1`	case created	caseId, customerId, priority, agreementId, responseDue, resolutionDue	notifications, analytics
`dev.passage.service.sla.breached.v1`	breach monitor	see 8.0	notifications, analytics
`dev.passage.service.dispatch.published.v1`	publish	see 8.0	PWA, customer notifications
`dev.passage.service.workorder.completed.v1`	complete action	workOrderId, technicianId, lines[], firstTimeFix, durationMinutes	billing, inventory (truck consumption), KPIs
`dev.passage.service.workorder.posted.v1`	billing post	workOrderId, invoiceId, billableTotal, warrantyCost, entitlementConsumed	invoicing, analytics

8.45.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Service workspace	tiles: SLA at risk (> 75% elapsed), unscheduled WOs, today's completion rate		SvcInquire
Case detail	entitle, create WO, pause/resume, resolve	SLA timers, agreement, object, timeline	SvcCaseMaintain
Agreement detail	lines, entitlement usage, SLA profile, billing		SvcAgreementMaintain
Dispatch board (Gantt + map)	optimize, drag assign, lock, publish, re-optimize	technician rows, WO blocks, travel segments, SLA due markers	SvcDispatch
Work order detail	lines, coverage, parts, photos, signature, post		SvcWoMaintain
Technician PWA	route, navigate, start/stop, checklist, parts, photos, signature, complete	offline indicator, sync queue	SvcTechnician
Installed base explorer	search by customer/serial, hierarchy, history		SvcInquire
KPI dashboard	filter by territory/technician	SLA %, FTF %, utilization	SvcInquire

Rules: board locks (pinned assignments) survive re-optimization; dragging a WO outside skill/time window shows a hard-block explanation; PWA photo max 5 per action at 1,600 px long edge; signature stored as PNG with timestamp and geo.

8.45.7 Business rules and edge cases

- **BR-45.1 Entitlement resolution** — candidates = ACTIVE agreements of the customer (or parent party) whose validity covers opened_at and whose lines match the service object (exact object > object category > any) ; rank by match specificity, then priority_rank, then latest start; the chosen agreement sets the SLA profile and coverage; if none, the default SLA profile of the legal entity applies and all lines are BILLABLE unless warranty covers them.
- **BR-45.2 SLA clock rules** — response target measured from opened_at to first of (responded_at, assignment DISPATCHED); resolution target from opened_at to resolved_at. Clocks count only business minutes of the SLA profile calendar (e.g. Mon–Fri 08:00–18:00 in the service site's time zone, holidays excluded), unless the profile is 24×7. Pause states: WAITING_CUSTOMER pauses both clocks; WAITING_PARTS pauses resolution only if profile.pause_on_parts = true. Due timestamps are stored as absolute times and recomputed on each pause/resume: due = add_business_minutes(opened_at, target_minutes + paused_minutes). Warning event at 75% elapsed; breach event when now > due while RUNNING; a breached SLA stays BREACHED even if resolved later (attainment reporting counts it as missed). Example: priority 2 resolution target 480 business minutes, opened Fri 2026-10-23 16:00 (America/Chicago), paused 120 minutes Mon → due Tue 2026-10-27 12:00.
- **BR-45.3 Dispatch optimization constraints** — decision: assign each WO to at most one technician and a start time. Hard constraints: required_skills ⊆ technician.skills (certifications unexpired per BR-42.8); WO.territory ∈ technician.territory_ids; start ≥ time_window.lower and end ≤ time_window.upper if set; start/end within technician shift; sequence feasibility with travel time from an OSRM/Valhalla matrix (self-hosted) plus 5-minute buffer; Σ (work + travel) ≤ max_daily_minutes (+ overtime up to 120 min if overtime_allowed); pinned assignments fixed; parts required must be on the truck or a reservation picked. Objective (lexicographic weights): minimize SLA lateness minutes × 100, unassigned priority-weighted WOs × 1,000, travel minutes × 1, overtime minutes × 5, technician count × 50. Solver: CP-SAT with 30 s time limit, 8 workers; fallback greedy insertion when no feasible solution in time; incremental re-optimization only moves WOs not yet DISPATCHED or with the dispatcher's consent. Scale target: 200 WOs × 25 technicians per territory-day.
- **BR-45.4 Coverage per line** — PART: WARRANTY if warranty.parts_covered and within term and not excluded; else AGREEMENT if line.parts_covered; else BILLABLE. LABOR/TRAVEL analogous. Agreement included_hours consumed FIFO; when exhausted, remaining hours BILLABLE at agreement price list.
- **BR-45.5 Work order status machine** — UNSCHEDULED → SCHEDULED → DISPATCHED → TRAVELING → IN_PROGRESS → COMPLETED → POSTED; IN_PROGRESS ↔ ON_HOLD; any pre-COMPLETED → CANCELED; a COMPLETED WO with follow-up creates a new WO linked by `follow_up_of` and sets first_time_fix = false.
- **BR-45.6 Offline merge** — mobile actions are append-only with client-generated UUIDs and device timestamps; server orders by (device timestamp, receipt order); conflicting status transitions resolve to the furthest valid state; free text last-writer-wins; parts consumption is idempotent per action id.
- **BR-45.7 Truck stock** — consumption posts an inventory issue from the TRUCK location at completion; negative truck stock blocked unless policy `service.allow-negative-truck`; replenishment transfer orders generated when below min per item.
- **BR-45.8 Pricing** — BILLABLE labor at agreement price list → customer price group → standard rate; parts via wave-27 pricing engine with agreement discount; rounding per line half-even.

8.45.8 Acceptance criteria

- **AC-45.1** (FR-45.5, BR-45.1) Given 50 entitlement fixtures (overlapping agreements, categories, parent parties), when cases open, then each resolves to the expected agreement and SLA profile.
- **AC-45.2** (FR-45.4, BR-45.2) Given the BR-45.2 example, when the case is paused 120 minutes on Monday, then resolution due = 2026-10-27T12:00-05:00.
- **AC-45.3** (BR-45.2) Given a RUNNING case at 75% elapsed, when the monitor runs, then a warning notification fires; at 100% a `sla.breached` event fires once.
- **AC-45.4** (FR-45.7, BR-45.3) Given 1,000 generated territory-days (≤ 200 WOs, ≤ 25 technicians), when optimized, then every assignment satisfies skills, windows, shifts and travel feasibility, and solver time p95 < 30 s (property test).
- **AC-45.5** (FR-45.7) Given a pinned assignment, when re-optimization runs, then it is unchanged.
- **AC-45.6** (FR-45.10, BR-45.4) Given a WO with a part covered by warranty and 3 labor hours under an agreement with 1 included hour, when posted, then the invoice contains 2 billable labor hours, the part is zero-priced with warranty cost recorded, and entitlement usage shows 1 hour consumed.
- **AC-45.7** (FR-45.8, BR-45.6) Given a technician offline for 8 h performing start/part/complete actions, when reconnected, then the server state equals the device state and inventory shows one issue per part action.
- **AC-45.8** (FR-45.1) Given a wave-28 shipment of 10 serialized units, when confirmed, then 10 ACTIVE service objects exist with warranty starting on ship date.
- **AC-45.9** (phase exit) Given the demo group, when the recorded walkthrough runs projects → HR → payroll → maintenance → field service, then every step completes on `stage` with reconciliations at zero variance.

8.45.9 Test plan and fixtures

Fixtures: `svc-installed-base` (derived from wave-28 shipped serials, 2,000 objects); `svc-entitlement-50`; `svc-dispatch-gen` (Hypothesis generator for WOs/technicians with seeded travel matrix); `svc-mobile-replay` (offline action logs); `svc-billing-30`.

Test	Type	Target	Pass criterion
Entitlement	unit	resolver	50/50
SLA calendar arithmetic	property	business-minute functions	add/subtract inverse; DST-safe
Dispatch feasibility	property	CP-SAT model	1,000/1,000 feasible; p95 < 30 s
Offline merge	unit/E2E	action log merge	deterministic result
Billing coverage	contract	invoicing API	30/30
PWA	E2E (Playwright, offline)	technician flow	complete with signature offline
Breach monitor	performance	100k open cases	scan < 60 s

Reconciliation: Σ POSTED billable = wave-29 invoices with source SWO; Σ warranty cost = GL warranty expense; truck stock issues = Σ part actions.

8.45.10 Deployment and operations

Charts `passage-service` (TS API, breach monitor, mobile BFF) and `passage-dispatch` (Python OR-Tools worker, OSRM sidecar with regional extracts); Postgres `service` with PostGIS. Flags: `service.allow-negative-truck`. Schedules: `service-sla-monitor` every 60 s; `service-dispatch-auto` 18:00 territory-local for next day; `service-truck-replenish` nightly. SLOs: case create p95 < 300 ms; dispatch run ≤ 30 s; PWA sync p95 < 2 s per 100 actions; availability 99.9%. Alerts: SLA monitor lag > 2 min; solver fallback rate > 10%; OSRM unavailable. Runbook: re-run dispatch with greedy; replay mobile actions; refresh OSRM extract. Rollback by digest; dispatch runs are versioned and republishable.

8.45.11 Migration from AX / SAP

Source	Passage entity	Notes
SMAServiceObjectTable, SMAServiceObjectRelationTable	ServiceObject	link to inventory serials where found
SMAServiceAgreementTable/Line, SMAAgreementLine	ServiceAgreement, lines	interval lines → preventive visit WOs
SMAServiceOrderTable/Line (open)	ServiceWorkOrder	stage map to status
SMAServiceTechnician	Technician	links to wave-42 workers
D365 FS msdyn_workorder, msdyn_agreement, msdyn_customerasset	ServiceWorkOrder, ServiceAgreement, ServiceObject	via Dataverse export
IBase (IBIN/IBIB/IBST), EQUI with customer	ServiceObject	
BGMK warranties	Warranty	
VBAK/VBAP contracts type WV/SC, SLA profiles	ServiceAgreement, SlaProfile	
QMEL service notifications (open), CS orders (open)	Case, ServiceWorkOrder	
SAP FSM activities (open)	ServiceWorkOrder assignments	via FSM API

Validation: installed-base count per customer equal; open WO count and value equal; active agreement count equal. Not migrated: closed service orders (archive), AX subscriptions (recreated as wave-29 recurring invoices), FSM historical GPS traces.

8.45.12 Dependencies, risks, open questions

Depends on: wave 16 locations (TRUCK), wave 27 pricing, wave 28 serial shipments, wave 29 invoicing, wave 35 notification hub, wave 42 workers/skills, wave 44 condition rules, wave 23 scheduler patterns.

Risk	Likelihood	Impact	Mitigation
Travel matrix accuracy/availability	Medium	Medium	Self-hosted OSRM; cached matrix; straight-line fallback × 1.3
Offline conflict edge cases	Medium	Medium	Append-only actions; replay fixtures
Phase-exit demo breadth	High	High	Walkthrough script frozen at wave start; dry run at week 4

Open questions: (1) Customer self-scheduling portal in v1? Default v1.x; decide by wave 46. (2) Should IoT-originated cases bypass the case stage and create WOs directly? Default create case with auto-conversion for priority 1; decide by wave 45 start.

Part 9 — Phase 9: Intelligence, Compliance & Go-Live (waves 46–50)

9.0 Phase overview

Business outcome at phase exit. A manufacturer runs its entire business on Passage in production, off AX/SAP, with the vendor dependencies of the legacy systems removed: analytics no longer needs Entity store, BYOD or BW extractors because every domain streams into one governed lakehouse; users ask questions and delegate work to agents that act only within their own permissions; the tenant runs legally in seven countries with statutory outputs validated against regulator schemas; segregation of duties, SOC 2 evidence and privacy requests are system functions rather than spreadsheets; the customer's open items, balances and history were moved by a migration factory that reconciled every control total; and the platform has demonstrated 10k concurrent users, a region failover and a reconciled production month-end. v1.0 is tagged, signed and published.

Phase-exit sign-off criteria (all measured in the production-candidate environment unless stated)

- Lakehouse freshness: p95 OLTP-commit-to-dashboard-visible lag < 60 s while the wave-50 load profile runs.
- Semantic-layer `revenue` equals GL revenue (universal journal, account category REVENUE) to the cent for 30 consecutive days; `inventory_value` equals the inventory subledger within USD 0.01 / EUR 0.01.
- Agent evals in CI: AP coding ≥ 92% top-1 accuracy, cash application ≥ 90% precision; permission-inheritance contract tests 100% pass; per-tenant token budget enforced with an alert fired in a test.
- Seven country packs (US, UK, DE, FR, IN, MX, CA) each ship a statutory-report fixture that validates against the regulator's XSD/JSON schema in CI.
- 50 planted SoD conflicts detected (0 misses); mitigation workflow records approver, reason and expiry.
- Backup/restore drill: tenant restored to a point in time with RPO ≤ 5 min and RTO ≤ 1 h; external penetration test shows zero open high/critical findings.
- Synthetic AX and SAP datasets (full year, 3 entities each) migrate with 100% match on all control totals; a parallel-run month shows zero unexplained variances; cutover rehearsal < 48 h; rollback rehearsal restores legacy coexistence.
- Load test: 10k concurrent users, 1M journal lines/h, 100k order lines/h with p99 within the SLO table in 9.50.10.
- DR game day: region failover RTO ≤ 1 h, RPO ≤ 5 min, zero committed-transaction loss (ledger hash chain verified).
- First customer closes a full production month-end with all subledgers reconciled to the journal; hypercare exit criteria (9.50.1) met.
- v1.0 tag signed (cosign), SBOMs attached, release notes and documentation published, v2 roadmap opened.

Cross-wave data model summary

Entity	Owning service	Key relationships
`lake_table` / `cdc_pipeline_state`	analytics-ingest	1 per (domain, table); references schema-registry subject
`semantic_metric`, `semantic_model`	semantic (Cube)	metric → model → lake tables; metric → `reconciliation_check`
`dashboard`, `dashboard_tile`, `report_schedule`	analytics-ui	tile → metric; schedule → dashboard or export
`document_template`, `document_render`	docgen	template → source entity type (SalesInvoice, PurchOrder, PackingSlip, Check)
`tool_definition`, `agent`, `agent_run`, `agent_action`, `approval_task`	mcp-gateway, agent-runtime	run → agent version; action → tool_definition; approval → Temporal task
`eval_dataset`, `eval_run`, `prompt_version`, `token_budget`	agent-evals	eval_run → agent version; budget → tenant
`country_pack`, `er_format`, `er_mapping`, `mandate`	localization, ereporting	format → mandate → country; pack → legal entity
`sod_rule`, `sod_conflict`, `mitigation`, `certification_campaign`	sod	rule → (duty_a, duty_b); conflict → user, rule; mitigation → conflict
`control`, `control_evidence`, `subject_request`, `retention_policy`, `legal_hold`	compliance, privacy	evidence → control; hold overrides policy
`migration_pack`, `pack_run`, `control_total`, `legacy_xref`, `parallel_run`, `variance`	migration-factory	run → pack; variance → parallel_run; xref → every migrated entity
`cutover_plan`, `cutover_step`, `go_nogo_item`	cutover	step → plan; item → step, owner
`region`, `failover_event`, `capacity_model`, `tenant_cost`	platform-ops	failover → region pair; cost → tenant, month

Cross-wave event catalog

Event (CloudEvents type)	Producer	Consumers	Payload summary
`dev.passage.analytics.snapshot.committed.v1`	analytics-ingest	semantic, catalog	table, snapshot_id, row_delta, max_source_lsn, watermark
`dev.passage.analytics.metric.reconciled.v1`	semantic	compliance, alerting	metric_id, date, semantic_value, source_value, delta, status
`dev.passage.agent.run.completed.v1`	agent-runtime	audit, copilot-ui, evals	run_id, agent, version, tool_calls[], tokens, outcome
`dev.passage.agent.approval.requested.v1`	agent-runtime	workflow inbox	run_id, action_id, proposed_change, risk_score
`dev.passage.ereporting.document.submitted.v1`	ereporting	AR/AP, compliance	mandate, format_version, document_ref, regulator_ack
`dev.passage.sod.conflict.detected.v1`	sod	compliance, workflow	user_id, rule_id, roles[], severity
`dev.passage.privacy.request.completed.v1`	privacy	audit	request_id, type, subject_ref, affected_domains[]
`dev.passage.migration.pack.reconciled.v1`	migration-factory	cutover	run_id, pack, control_totals[], status
`dev.passage.migration.parallelrun.diffed.v1`	migration-factory	cutover	run_id, period, variances_unexplained, variances_explained
`dev.passage.ops.failover.completed.v1`	platform-ops	all services (fencing)	from_region, to_region, promoted_lsn, rpo_seconds, rto_seconds

Phase-level risks

Risk	Likelihood	Impact	Mitigation	Owner
Lakehouse metrics drift from GL (duplicate CDC, late data)	Medium	High	Daily reconciliation job with hard fail; equality-delete upserts keyed on LSN	Analytics lead
Agent takes an action outside user permission	Low	Critical	Tool calls carry the user's token; OPA evaluated at the API, never in the agent; contract tests per tool	Security lead
Country-pack legal error (wrong tax box, wrong XML)	Medium	High	Regulator schema fixtures in CI; provider adapters (Avalara, Sovos) as fallback; packs behind flags	Localization lead
Migration reconciliation hides netting errors	Medium	High	Control totals at line grain, not just sums (counts, hashes, by-dimension)	Migration lead
Cutover window overrun	Medium	High	Two full rehearsals; runbook timed per step; abort criteria at T+36 h	Cutover manager
Region failover loses committed journal entries	Low	Critical	Synchronous quorum replication for ledger Postgres; hash-chain verification post-failover	SRE lead
Two-person team cannot staff hypercare	High	High	Implementation partner on-call rota; agents in suggest-mode triage support tickets	Product

9.46 Wave 46: Data Platform, Analytics & Embedded BI

9.46.1 Scope and business context

AX customers build analytics on Entity store aggregate measurements refreshed by batch, BYOD exports to Azure SQL and Power BI embedded workspaces; SSRS runs document layouts through `SrsReportDataProvider` classes. SAP customers extract into BW/4HANA InfoProviders via ODP, expose `@Analytics.dataCategory: #CUBE` CDS views and embed SAC stories. Both put analytical load on OLTP or on a copy that is hours stale. This wave gives every Passage domain a CDC path into an Apache Iceberg lakehouse, a Cube semantic layer with governed metrics whose row-level security mirrors OPA, embedded dashboards in each workspace, a self-service explorer, scheduled reports, an operational document designer that replaces SSRS layouts, and OpenMetadata for catalog and lineage. At wave exit a controller sees revenue on a dashboard within 60 s of posting and knows it equals the GL. Out of scope: natural-language querying (wave 47), historical legacy data (wave 49), customer-built ML models.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Entity store refresh (`AggregateMeasurement`, batch job "Refresh entity store") runs hourly at best; Power BI tiles show numbers up to 4 h old and controllers re-query the GL to confirm	Batch copy from OLTP into a separate star schema; no change feed	Debezium CDC → Iceberg → Cube; freshness badge on every tile	Latency 4 h batch → p95 < 60 s
Revenue on a dashboard and revenue on the trial balance disagree; nobody knows which is right (BYOD export gaps, BW extractor delta failures in `RSA7`)	Export jobs drop deltas silently; no reconciliation step	Daily automated metric ↔ GL reconciliation with hard fail and banner	Reconciliation effort 2 person-days/month → 0; delta tolerance 0.005
Changing an SSRS invoice layout needs a developer: edit `.rdl`, extend `SrsReportDataProvider`, build, deploy through LCS; SAP needs `SE71`/`SFP` ABAP skills	Layout bound to code artifacts in a deployable package	Metadata document designer with fixture preview and visual regression	Turnaround 2 weeks → same day; 0 code deploys
Building one BW query or SSAS cube perspective for a new KPI takes an analyst through `RSA1`, transformations, DTPs and authorizations (`RSECADMIN`)	Separate security and modeling stack from the application	Metric registry; RLS compiled from the same OPA policies as OLTP	New governed metric 10 days → 1 h; 0 separate security objects
Exporting to Excel from a data entity (`Open in Office`) caps at 10k rows or times out; users build shadow spreadsheets	OLTP OData serving analytical loads	Explorer exports via Trino as Temporal jobs	1M rows in < 10 min; OLTP load from reporting 0

9.46.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Analytical store	Entity store, `AggregateMeasurement` (e.g. `LedgerActivityMeasure`), batch refresh	BW/4HANA ADSO, CompositeProvider, ODP extractors (`RSA1`)	Debezium CDC → Iceberg `raw.*`, dbt `core.*`/`mart.*`	Improved (streaming, no batch refresh)
Data export for BI	BYOD (`DMFDataEntity` export to Azure SQL), Export to Data Lake	SLT, Datasphere replication flows	Iceberg tables queryable via Trino; CDC is the export	Improved
Embedded dashboards	Power BI embedded workspaces (`PBIWorkspace`, `SysPBIReportController`)	SAC stories in Fiori launchpad, Overview pages	`<DashboardHost>` web component per workspace	Full
Operational documents	SSRS `SalesInvoice.Report`, `PurchPurchaseOrder.Report`, `SrsReportDataProvider`, Print management	SAPscript (`SE71`), Smart Forms, Adobe Forms (`SFP`), `NACE` output	Document designer, `document_template` with print-management rules	Full
OLAP cubes	SSAS cubes (`SalesCube`, `GeneralLedgerCube`), perspectives	BEx queries (`RSRT`), BW queries	Cube semantic models and measures	Improved
Embedded analytical views	Data entities with `IsAggregateDataEntity`	CDS `@Analytics.dataCategory #CUBE`, `@Analytics.query`	Cube `cubes`/`views` generated from entity metadata	Full
Self-service / Excel	Open in Excel (`DataEntity` + OData), Power BI Desktop over BYOD	Analysis for Office, Query Browser (`/UI2/QUERY_BROWSER`)	Explorer; export to XLSX, Power BI (ODBC via Trino), Tableau	Full
Row-level security	XDS policies, Power BI RLS roles	Analysis authorizations (`RSECADMIN`)	OPA policy compiled to Cube `queryRewrite` filters	Full
Financial statements	Management Reporter	FSV (`FSE2`), Fiori financial statement apps	Wave-15 statement designer; marts feed trend tiles	Full
Catalog and lineage	None (AOT only)	BW metadata repository, Datasphere catalog	OpenMetadata with Debezium/dbt/Cube lineage ingestion	Improved

9.46.3 Functional requirements

- **FR-46.1 CDC from every domain** — Must. The system shall run one Debezium Postgres connector per domain database using `pgoutput`, publishing to topics `cdc.<domain>.<table>` keyed on `(tenant_id, primary key)`, Avro-encoded with schema-registry compatibility `BACKWARD_TRANSITIVE`.
  - Notes: every table listed in entity metadata with `analytics: true` is captured; `REPLICA IDENTITY FULL` enforced by the Atlas migration generator; snapshot mode `initial` on first run then `never`; heartbeat every 10 s to advance slots on idle domains; slot lag > 1 GB alerts.
- **FR-46.2 Iceberg landing contract** — Must. The system shall write each CDC topic into Iceberg table `raw.<domain>.<table>` (format v2, merge-on-read, equality deletes on the key) partitioned by `days(_source_ts)` and `bucket(16, tenant_id)`.
  - Notes: every row carries `_op` (c/u/d), `_source_lsn`, `_source_ts`, `_ingested_at`; commits every 30 s or 50 MB, whichever first; snapshot expiry 7 days, orphan-file cleanup daily; compaction rewrites files < 64 MB nightly.
- **FR-46.3 Schema evolution** — Must. The system shall apply additive source changes automatically and quarantine incompatible ones.
  - Notes: add column → add nullable Iceberg column; type widening (`int→bigint`, `decimal(18,4)→decimal(28,6)`) applied; dropped column retained and tagged `deprecated_since`; rename → new column plus `renamed_from` property in catalog; narrowing or NOT NULL additions pause the sink, route to `cdc.dlq.<domain>` and alert within 60 s.
- **FR-46.4 Late-arriving data and exactly-once effect** — Must. The system shall produce exactly one logical state per key regardless of Kafka redelivery.
  - Notes: sink deduplicates on `(tenant_id, pk, _source_lsn)`; Iceberg commit and Kafka offset are written atomically by the sink coordinator (control topic `iceberg.control`), so replay after crash is idempotent; watermark = max(`_source_ts`) – 5 min; rows with `_source_ts` older than the watermark are merged into closed partitions; dbt incremental models use a 3-day lookback; a row arriving more than 30 days late is written and flagged `late=true` for reconciliation.
- **FR-46.5 dbt transformation layers** — Must. The system shall maintain `core.*` (deduplicated current-state and history tables, SCD2 on master data) and `mart.*` (finance, sales, procurement, inventory, production, projects, HR) dbt models with tests on every primary key, every FK and every reconciled measure.
  - Notes: `mart.fin_journal_line` is the only source for monetary metrics; models run every 60 s incremental, nightly full for SCD2 rebuild; dbt test failure blocks downstream models and raises `metric.reconciled` with status `BLOCKED`.
- **FR-46.6 Governed metrics** — Must. The system shall define metrics in the semantic layer with owner, version, grain, allowed dimensions and formula; the initial set is:
  - `revenue = -SUM(amount_reporting) FROM mart.fin_journal_line WHERE account_category='REVENUE' AND status='POSTED'` (credit shown positive).
  - `cogs = SUM(amount_reporting) WHERE account_category='COGS'`; `gross_margin = revenue - cogs`; `gross_margin_pct = gross_margin / NULLIF(revenue,0)`.
  - `dso_days = ar_open_balance_at(end) / NULLIF(revenue_trailing(90d),0) * 90`; countback variant selectable per tenant.
  - `dpo_days = ap_open_balance_at(end) / NULLIF(cogs_trailing(90d),0) * 90`.
  - `inventory_turns = cogs_trailing(365d) / NULLIF(AVG(inventory_value_eom over last 13 month-ends),0)`.
  - `otif_pct = COUNT(sales_line WHERE shipped_ts <= requested_ship_date AND qty_shipped = qty_ordered) / NULLIF(COUNT(sales_line shipped),0)`.
  - `oee_pct = availability * performance * quality` where `availability = run_time / planned_time`, `performance = (ideal_cycle_time * total_count) / run_time`, `quality = good_count / total_count`, sourced from `mart.prod_resource_shift`.
- **FR-46.7 Row-level security mirrored from OPA** — Must. The system shall compile the user's OPA data-scope decision (legal entities, sites, dimension ranges) into Cube `queryRewrite` filters on every query, using the same Keycloak token as the OLTP APIs.
  - Notes: cache keyed by `(user_id, policy_bundle_version)` for 5 min; a query touching a model with no scope mapping is rejected (`403 SCOPE_UNMAPPED`); aggregations never bypass filters (pre-aggregations are built per security context).
- **FR-46.8 Embedded dashboards** — Must. The system shall render a dashboard host in every workspace with tiles bound to semantic metrics, auto-refresh ≤ 60 s, and drill-through to the OLTP record.
- **FR-46.9 Self-service explorer and export** — Should. The system shall let users compose metric × dimension × filter queries, save them, and export to XLSX (≤ 1M rows), CSV, Power BI (Trino ODBC/JDBC connection string) and Tableau (same).
  - Notes: exports > 100k rows run as Temporal jobs and deliver via signed URL valid 24 h; every export is audit-logged with row count and filter.
- **FR-46.10 Scheduled reports** — Should. The system shall schedule dashboards and explorer queries (cron, time-zone aware) with delivery to email, SFTP or in-app inbox, PDF/XLSX, retry 3× with backoff.
- **FR-46.11 Operational document designer** — Must. The system shall provide a layout designer (bands, repeaters, barcodes, conditional sections, locale formats) for SalesInvoice, PurchOrder, PackingSlip, Check, Statement, PickList and Remittance with versioned templates and print-management rules (per entity, per customer/vendor, per destination).
  - Notes: render via headless Chromium to PDF/A-3; data source is the document's REST read model, not the lakehouse; p95 render < 800 ms for a 50-line invoice.
- **FR-46.12 Catalog and lineage** — Should. The system shall register every Postgres table, Iceberg table, dbt model, Cube metric and dashboard in OpenMetadata with column-level lineage and owner, refreshed hourly.
- **FR-46.13 Daily metric reconciliation** — Must. The system shall compare `revenue`, `cogs`, `ar_open_balance`, `ap_open_balance`, `inventory_value` against the universal journal and subledgers at 02:00 tenant local time and fail the dashboard with a banner if Ideltal > 0.005.

9.46.4 Data model

`semantic_metric` (semantic service, Postgres `analytics`)

Field	Type	Required	Notes
metric_id	text	Y	PK, e.g. `revenue`
version	int	Y	monotonically increasing; (metric_id, version) unique
formula	text	Y	Cube measure SQL
grain	text	Y	`journal_line`, `sales_line`, `resource_shift`
allowed_dimensions	text[]	Y	from `semantic_model.dimensions`
owner_user_id	uuid	Y	FK users

Field	Type	Required	Notes
status	enum	Y	DRAFT, PUBLISHED, DEPRECATED
reconciliation_source	text	N	e.g. `ledger.balance_by_account`
tolerance	numeric(18,6)	N	default 0.005

`cdc_pipeline_state` (analytics-ingest)

Field	Type	Required	Notes
domain	text	Y	PK part
table_name	text	Y	PK part
schema_version	int	Y	registry version last applied
last_lsn	bigint	Y	last committed source LSN
last_snapshot_id	bigint	Y	Iceberg snapshot
watermark_ts	timestamptz	Y	
status	enum	Y	RUNNING, PAUSED_SCHEMA, PAUSED_ERROR
dlq_count_24h	int	Y	default 0

`document_template` (docgen)

Field	Type	Required	Notes
template_id	uuid	Y	PK
tenant_id	uuid	Y	RLS
document_type	enum	Y	SalesInvoice, PurchOrder, PackingSlip, Check, Statement, PickList, Remittance
version	int	Y	unique with template_id
locale	text	Y	BCP 47, e.g. `de-DE`
layout_json	jsonb	Y	band tree; max 512 KB
status	enum	Y	DRAFT, PUBLISHED, RETIRED
visual_hash	text	N	perceptual hash of fixture render
published_at	timestamptz	N	

`report_schedule`: schedule_id, tenant_id, target_type (DASHBOARD/QUERY), target_id, cron, timezone, format, destination_json, last_run_at, next_run_at, failure_count.

Invariants: a PUBLISHED metric is immutable (changes create a new version); every mart measure has exactly one reconciliation source or is tagged `non_financial`; a template can be PUBLISHED only if its visual regression delta ≤ 0.1%; `last_lsn` never decreases. Indexing: `cdc_pipeline_state` PK (domain, table_name); Iceberg partition spec as FR-46.2 plus sort order `(tenant_id, _source_lsn)`; `document_template` index (tenant_id, document_type, status).

9.46.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	`/api/v1/analytics/metrics`	list published metrics with formulas	n/a	AnalyticsView
POST	`/api/v1/analytics/query`	Cube query (measures, dimensions, filters, timeDimensions)	n/a (read)	AnalyticsView + data scope
GET	`/api/v1/analytics/dashboards/{id}`	dashboard definition and tiles	n/a	AnalyticsView
PUT	`/api/v1/analytics/dashboards/{id}`	create/update dashboard	ETag	AnalyticsAuthor
POST	`/api/v1/analytics/exports`	start export job	`Idempotency-Key`	AnalyticsExport
POST	`/api/v1/analytics/schedules`	create schedule	`Idempotency-Key`	AnalyticsAuthor
GET	`/api/v1/analytics/reconciliation?date=`	reconciliation results per metric	n/a	FinanceView
POST	`/api/v1/documents/templates/{id}/versions`	publish template version	`Idempotency-Key`	DocumentDesign
POST	`/api/v1/documents/render`	render document (type, record_id, template override)	by (record, version)	per-document duty
GET	`/api/v1/catalog/lineage/{asset_fqn}`	upstream/downstream lineage	n/a	CatalogView

Query validation: ≤ 10 measures, ≤ 8 dimensions, time range ≤ 5 years at day grain, result cap 50k rows interactive; otherwise redirected to export.

Event	Trigger	Payload fields	Consumers
`dev.passage.analytics.snapshot.committed.v1`	Iceberg commit	domain, table, snapshot_id, rows_added, rows_deleted, max_lsn, watermark_ts	semantic (cache invalidation), catalog
`dev.passage.analytics.metric.reconciled.v1`	daily job	metric_id, tenant_id, date, semantic_value, source_value, delta, status	compliance evidence, alerting
`dev.passage.analytics.report.delivered.v1`	schedule run	schedule_id, run_id, format, destination, row_count, bytes	audit
`dev.passage.documents.template.published.v1`	version publish	template_id, version, document_type, visual_hash	print management, catalog

9.46.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Workspace dashboard host	refresh, drill-through, change period, pin tile	metric tiles, sparklines, freshness badge	AnalyticsView

Screen	Primary actions	Key fields/columns	Duty
Explorer	add measure/dimension, filter, pivot, save, export	metric picker with formula tooltip, dimension tree	AnalyticsView
Metric registry	publish version, set owner, view reconciliation history	metric_id, formula, grain, tolerance, last delta	AnalyticsAuthor
Pipeline health	pause/resume, replay from LSN, view DLQ	domain, table, lag_s, status, dlq_count	AnalyticsAdmin
Document designer	drag bands, bind fields, preview with fixture, publish	band tree, locale, print-management rules	DocumentDesign
Schedules	create, run now, view deliveries	cron, timezone, destination, last status	AnalyticsAuthor

Interaction rules: explorer queries run on 500 ms debounce; `Ctrl+E` exports current grid; freshness badge turns amber at lag > 60 s, red at > 300 s; designer preview reruns on each property change with a 300 ms debounce; personalization (pinned tiles, saved explorer views) stored per user and exported with user profile.

9.46.7 Business rules and edge cases

- **BR-46.1** Monetary metrics use `amount_reporting` in the ledger's reporting currency; rounding at display only, half-even, to the currency's minor units.
- **BR-46.2** A deleted OLTP row (`_op='d'`) is tombstoned in `raw.*`, removed from `core.*` current tables and kept in history with `valid_to`.
- **BR-46.3** Reversal journals net to zero in `revenue` on the reversal's posting date, not the original date.
- **BR-46.4** DSO/DPO use calendar days; trailing windows end at the selected period end inclusive; if the trailing window has zero revenue, the metric is NULL, never infinity.
- **BR-46.5** OEE `planned_time` excludes shifts flagged `no_production`; `performance` is capped at 100%.
- **BR-46.6** Pipeline state machine: RUNNING → PAUSED_SCHEMA (incompatible change) → RUNNING after operator approves mapping; RUNNING → PAUSED_ERROR after 5 consecutive sink failures → RUNNING on replay from `last_lsn`.
- **BR-46.7** Export files are deleted 24 h after creation; a second identical request within 1 h returns the existing file.
- **BR-46.8** Template version state: DRAFT → PUBLISHED (requires visual regression pass) → RETIRED; a RETIRED version remains renderable for re-prints of documents posted under it.
- **BR-46.9** Dashboard queries for a user with no legal-entity scope return an empty result with reason `NO_SCOPE`, not an error.
- **BR-46.10** A reconciliation delta inside tolerance for 30 days auto-marks the metric `TRUSTED`; one failure resets the counter.

9.46.8 Acceptance criteria

- **AC-46.1** (FR-46.1, 46.2) Given a journal posted in `ledger` with 1,000 lines, when 60 s elapse, then `mart.fin_journal_line` contains all 1,000 rows with matching `_source_lsn` order.
- **AC-46.2** (FR-46.3) Given a migration adds nullable column `memo2` to `sales_line`, when CDC resumes, then the Iceberg table has the column and no DLQ rows.
- **AC-46.3** (FR-46.3) Given a migration changes `qty` from `decimal(28,6)` to `decimal(18,4)`, when CDC resumes, then the pipeline is PAUSED_SCHEMA within 60 s and an alert fires.
- **AC-46.4** (FR-46.4) Given Kafka redelivers 10,000 already-applied change events, when the sink processes them, then row counts and sums in `core.*` are unchanged.
- **AC-46.5** (FR-46.4) Given a row with `_source_ts` 10 days in the past arrives, when dbt runs, then the row is present in the correct day partition and `late=true`.
- **AC-46.6** (FR-46.6, 46.13) Given a day's postings, when reconciliation runs, then semantic `revenue` equals ledger REVENUE balance movement to USD 0.00 and EUR 0.00.
- **AC-46.7** (FR-46.7) Given user A scoped to legal entity USMF and user B to DEMF, when both open the sales dashboard, then A sees only USMF rows and B only DEMF rows, including in totals.
- **AC-46.8** (FR-46.11) Given the 7 standard document templates and their fixtures, when rendered in CI, then perceptual-hash delta vs baseline ≤ 0.1% for each.
- **AC-46.9** (FR-46.9) Given an explorer query returning 250k rows, when the user exports XLSX, then a Temporal job produces the file and a signed URL within 5 min.
- **AC-46.10** (FR-46.10) Given a schedule `0 6 * * 1` Europe/Berlin, when Monday 06:00 CET arrives, then the PDF is delivered and `report.delivered` is emitted.

9.46.9 Test plan and fixtures

Fixtures: `fx-lake-usmf-year` (one fiscal year, 3 entities, 2.4M journal lines, 180k sales lines, from the wave-30 demo dataset replayed); `fx-schema-evolution` (12 migration scripts: add, widen, drop, rename, narrow, not-null); `fx-late-data` (5k events with `_source_ts` skewed 1 h–45 d); `fx-docs-7` (7 document types × 3 locales × fixture records); `fx-oee-shifts` (90 days, 12 resources).

Test	Type	Target	Pass criterion
Sink idempotency	property	Iceberg sink	any permutation/duplication of events yields identical table state
Schema evolution matrix	contract	connector + sink	12 scripts classified correctly (8 auto, 4 pause)
Metric formula unit tests	unit	dbt/Cube	each metric matches hand-computed value on 20-row fixture
OPA→Cube filter equivalence	contract	semantic	for 200 users, Cube result set ⊆ OLTP API result set
Dashboard lag	performance	end-to-end	p95 < 60 s at 1M journal lines/h
Visual regression	E2E	docgen	21 renders ≤ 0.1% delta
Export at scale	performance	explorer	1M-row XLSX in < 10 min
Lineage completeness	contract	OpenMetadata	100% of dbt models and metrics have upstream edges

Reconciliation: `revenue`, `cogs` ↔ ledger balance movements by account category; `ar_open_balance`, `ap_open_balance` ↔ open-item subledgers; `inventory_value` ↔ inventory close valuation; row counts `raw.*` ↔ `core.*` current ↔ Postgres `COUNT(*)` per tenant.

9.46.10 Deployment and operations

Helm charts: `passage-cdc` (Kafka Connect cluster with Debezium and Iceberg sink, 3 workers), `passage-lakehouse` (Iceberg REST catalog, Nessie-compatible, object storage via S3 API), `passage-trino` (coordinator + 4 workers, autoscaling to 16), `passage-clickhouse` (optional accelerator for `mart.*` pre-aggregations), `passage-cube` (API + refresh workers), `passage-dbt-runner` (Temporal worker), `passage-docgen`, `passage-openmetadata`. Config: `CDC_DOMAINS` list, `ICEBERG_COMMIT_INTERVAL=30s`, `LATE_DATA_WATERMARK=5m`, `RECON_TOLERANCE=0.005`, object-store credentials from Vault. Migrations: `analytics` database (Atlas); Iceberg schemas managed by the sink. Feature flags: `analytics.dashboards`, `analytics.explorer`, `docgen.designer` per tenant. Jobs: dbt incremental every 60 s, SCD2 nightly 01:00, reconciliation 02:00, compaction 03:00, lineage sync hourly. SLOs: dashboard lag p95 < 60 s; query p95 < 2 s for ≤ 50k rows; document render p95 < 800 ms; pipeline availability 99.9%. Alerts: slot lag > 1 GB, DLQ > 0 for 5 min, reconciliation delta > tolerance, Trino queue > 50. Runbook: replay from LSN, rebuild `core.*` from `raw.*`, rotate object-store keys. Rollback: Helm rollback of services; Iceberg `rollback_to_snapshot` per table; connectors restart from stored offsets.

9.46.11 Migration from AX / SAP

Source	Passage	Notes
Entity store `AggregateMeasurement` definitions	`semantic_metric`	mapped by hand; measures become Cube measures, dimensions become joins
Power BI `.pbix` over BYOD	Dashboards	not converted; rebuilt on Cube; Power BI Desktop can connect to Trino
SSRS `.rdl` layouts via `SrsReportDataProvider`	`document_template`	semi-automatic: bands, fields and expressions imported; SSRS expressions in VB are flagged for manual rewrite
Print management settings (`PrintMgmtSettings`)	print-management rules	full
BW InfoProviders, BEx queries	`mart.*` + metrics	metadata import creates draft metrics; data not migrated here (wave 49)
SAC stories	Dashboards	rebuilt
Smart Forms / Adobe Forms	`document_template`	layout imported from XDP where possible; scripting rewritten

Not migrated: SSAS cube partitions, BYOD Azure SQL contents, Power BI datasets — all derivable from the lakehouse.

9.46.12 Dependencies, risks, open questions

Depends on: wave 3 (outbox/CDC infrastructure), wave 5 (audit from CDC), wave 11–15 (universal journal, statements), wave 30 (sales and AR marts), wave 25 (OEE source data), wave 45 (all domains emitting).

Risk	Likelihood	Impact	Mitigation
Iceberg small-file explosion at 30 s commits	High	Medium	nightly compaction; adaptive commit size
Cube pre-aggregations leak across security contexts	Low	Critical	pre-aggregations partitioned by legal entity; contract test
Trino cost at 10k users	Medium	Medium	ClickHouse accelerator for top-20 dashboards

Open questions: (1) Trino-only vs Trino+ClickHouse by default — proposed default: Trino only, ClickHouse behind flag; decide by wave 47. (2) Document rendering engine fallback (Chromium vs Typst) — default Chromium; decide by wave 48.

9.47 Wave 47: AI Copilots, Agents & Automation

9.47.1 Scope and business context

D365 ships Copilot for finance and operations (summaries, draft e-mails, Copilot in workflow history) and Power Automate flows triggered from business events; SAP ships Joule and SAP Build Process Automation. Neither lets a governed agent execute a multi-step ERP process under the invoking user's exact permissions with an eval gate. This wave exposes every Passage API, entity and report as MCP tools through a gateway that forwards the user's token to OPA-protected APIs; adds a copilot in the shell (ask-the-data over the wave-46 semantic layer, explain-this-record, draft documents, summarize workflows); runs agents on Temporal with approval gates; gates deployment on an eval harness; and enforces responsible-AI controls with a provider-agnostic model layer (Claude default; Bedrock, Azure OpenAI, open-weights via vLLM selectable). Out of scope: training custom models, autonomous posting without approval in v1, voice.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Coding a vendor invoice in `VendInvoiceJour`/`MIRO` means opening the PO, the vendor history and the chart of accounts in three forms, then keying account, dimensions and tax code per line	No suggestion engine; copilots cannot read the posting context	AP coding agent proposes account, dimensions, tax code, PO match per line; approver confirms	Clicks per invoice 18 → 2; coding time 4 min → 20 s; top-1 accuracy ≥ 92%
Cash application: `CustPaymMatch` rules and `F-28` match exact amounts only; partials and multi-invoice remittances go to a manual queue	Rule-based matching without reading remittance text	Cash application agent ranks settlements from remittance text and tolerances	Auto-matched share 60% → 90%+; unapplied cash aged > 7 d reduced 10×
Power Automate / SAP Build flows need a connector project, a service account with broad rights and a separate audit trail	Automation runs outside the ERP's authorization model	MCP tools with user-token inheritance; Temporal agent runtime; `agent_action` audit	Integration project 6 weeks → 0 (tools generated); 0 actions beyond user permission
"Why is this number what it is?" requires reading workflow history, `SysDatabaseLog` and voucher transactions across forms	Context scattered across tables with no narrative	Explain-this-record over audit trail and related documents with citations	Investigation 15 min → 30 s
D365 Copilot / Joule answer from a vendor model with no accuracy gate, no cost control and no kill switch	Vendor-managed black box	Eval harness in CI, token budgets, kill switch, provider choice	100% of agent releases gated by evals; kill in ≤ 10 s

9.47.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Conversational assistant	D365 Copilot sidecar (`CopilotFeatureManagement`), Copilot summaries on workflow history	Joule in S/4HANA Cloud, Fiori launchpad	Copilot panel in shell	Improved (acts, not only answers)
Ask the data	Copilot "ask about data" over Dataverse virtual entities	Joule analytics via SAC	`ask_data` tool → Cube query with OPA scope	Full
Process automation	Power Automate flows on business events (`BusinessEventsCatalog`), Flow connectors	SAP Build Process Automation, workflow agents	Agent runtime on Temporal, triggered by CloudEvents	Full
Invoice coding suggestions	Vendor invoice automation, `VendInvoiceAutomation`	Cash Application ML, Invoice assistant (Business AI)	AP coding agent (suggest → approve)	Full
Cash application	Customer payment advanced matching rules (`CustPaymMatch`)	Cash Application ML (`FINS_CASH_ML`)	Cash application agent	Full
Planning exceptions	Planning Optimization action messages	MRP Cockpit (`MD04`, `PPH_MRP_COCKPIT`), Situation Handling	MRP exception triage agent	Improved
Collections outreach	Collections workspace, `CustCollectionLetter`	FSCM Collections worklist (`UDM_SUPERVISOR`)	Collections agent drafts outreach	Full
Anomaly detection	Finance insights (cash-flow forecast, late-payment prediction)	Business AI journal anomaly detection	Journal anomaly agent	Full
Audit of AI actions	None dedicated	None dedicated	`agent_action` audit with inputs, tool calls, outputs, approver	Improved

9.47.3 Functional requirements

- **FR-47.1 MCP gateway** — Must. The system shall expose Passage capabilities as MCP tools from a single gateway (`mcp.passage.<tenant>`), generated from OpenAPI/AsyncAPI and entity metadata.
  - Notes: tool names `passage.<domain>.<entity>.<verb>` (e.g. `passage.ap.invoice.list`, `passage.ap.invoice.propose_coding`, `passage.ledger.journal.get`); `inputSchema` is the endpoint's JSON Schema with tenant_id removed (injected from token); each tool declares `annotations.readOnlyHint`, `destructiveHint`, `idempotentHint`, and a Passage extension `x-passage-duty` (required duty) and `x-passage-approval` (`none`, `suggest`, `approve_before_commit`); read tools return ≤ 200 rows per call with cursor; a tool call carries `traceparent` and `x-agent-run-id`.
- **FR-47.2 Permission inheritance** — Must. The system shall execute every tool call with the invoking user's Keycloak token (on-behalf-of exchange, `act` claim = agent id), so OPA evaluates the user's duties and data scope at the API; agents hold no credentials of their own.
  - Notes: token lifetime for agent runs 15 min, refreshed by Temporal activity; a tool whose `x-passage-duty` the user lacks is not listed to the model; contract test: for each of 400 tools, a user without the duty receives 403 from the gateway and the API.
- **FR-47.3 Copilot UI** — Must. The system shall provide a shell panel with modes ask-the-data, explain-this-record, draft (e-mail, PO text, dunning letter), summarize (workflow history, case) with citations to records and metrics used.
  - Notes: streaming responses; every answer lists tool calls made; "explain" uses the record's audit trail and related documents; drafts are inserted only on user click.
- **FR-47.4 Agent runtime** — Must. The system shall run agents as Temporal workflows (`AgentRunWorkflow`) with steps plan → tool calls → proposal → approval gate → commit, each tool call an activity with retry policy and 60 s timeout, run timeout 24 h including approval wait.
- **FR-47.5 AP invoice coding agent** — Must. The system shall propose account, dimensions, tax code and PO match for an unposted vendor invoice using vendor history, PO lines and text, in suggest mode (approver confirms) or auto mode when confidence ≥ 0.95 and amount < tenant threshold (default USD 5,000 / EUR 4,500).
- **FR-47.6 Cash application agent** — Must. The system shall propose settlements for unapplied customer payments (remittance text, amounts, tolerances), ranked by confidence; applies only after approval in v1.
- **FR-47.7 MRP exception triage, collections outreach, journal anomaly agents** — Should. The system shall ship these three in suggest mode: triage groups action messages by cause and proposes expedite/postpone; collections drafts outreach per dunning policy; anomaly scores journals on amount, account pairing, time of posting and user, flagging score > 0.8.
- **FR-47.8 Approval gates** — Must. The system shall route any `approve_before_commit` action to the wave-4 workflow inbox with proposed change, diff, confidence and rationale; approver decisions are recorded on `agent_action`.
- **FR-47.9 Eval harness** — Must. The system shall maintain golden datasets per agent and run evals in CI on every prompt, model or tool change, blocking deploy below thresholds.
  - Metrics and thresholds: AP coding top-1 accuracy ≥ 0.92, top-3 ≥ 0.98; cash application precision ≥ 0.90, recall ≥ 0.75; MRP triage macro-F1 ≥ 0.80; anomaly AUROC ≥ 0.85, false-positive rate ≤ 0.05; ask-the-data metric-selection accuracy ≥ 0.95 and SQL-equivalence ≥ 0.90; refusal correctness ≥ 0.99 on an adversarial set of 500 prompts; p95 latency per agent run < 30 s excluding approval; cost per run ≤ budgeted USD; regression gate = no metric drops > 1 point vs last release.
- **FR-47.10 Prompt and version registry** — Must. The system shall version system prompts, tool lists, model id and parameters as `agent_version`; production pins a version; rollback is a pointer change.
- **FR-47.11 Token budgets** — Must. The system shall enforce per-tenant monthly token budgets (default 50M input / 10M output), soft alert at 80%, hard stop at 100% with copilot degraded to deterministic features; per-run cap 400k tokens.
- **FR-47.12 Responsible-AI controls** — Must. The system shall: redact PII (names, IBAN, tax ids, emails) from prompts to external providers using the wave-42 PII classification, replacing with stable pseudonyms resolved after response; log every action (prompt hash, tool calls, outputs, approver); provide a tenant and global kill switch that halts all runs within 10 s; prohibit model-initiated posting without `approve_before_commit` or auto-mode policy; label all generated text in UI; keep a model card per agent version; allow provider selection per tenant with data-residency constraint checked against wave-48 tenant residency.
- **FR-47.13 Provider layer** — Must. The system shall abstract chat completion, tool use and embeddings behind one interface with adapters for Anthropic API, Amazon Bedrock, Azure, and OpenAI-compatible open-weights endpoints; feature parity test per adapter.

9.47.4 Data model

`tool_definition` (mcp-gateway)

Field	Type	Required	Notes
tool_name	text	Y	PK, `passage.<domain>.<entity>.<verb>`
version	int	Y	from OpenAPI version
input_schema	jsonb	Y	JSON Schema draft 2020-12
output_schema	jsonb	Y	
required_duty	text	Y	OPA duty id
approval_mode	enum	Y	none, suggest, approve_before_commit
read_only	bool	Y	
source_ref	text	Y	OpenAPI operationId or report id

`agent_run` (agent-runtime)

Field	Type	Required	Notes
run_id	uuid	Y	PK = Temporal workflow id
tenant_id	uuid	Y	RLS
agent_id	text	Y	e.g. `ap_coding`
agent_version	int	Y	FK agent_version
invoked_by_user_id	uuid	Y	whose permissions apply
trigger	enum	Y	USER, EVENT, SCHEDULE
status	enum	Y	PLANNING, ACTING, AWAITING_APPROVAL, COMMITTED, REJECTED, FAILED, KILLED
tokens_in / tokens_out	bigint	Y	
cost_usd	numeric(12,6)	Y	
started_at / ended_at	timestamptz	Y/N	

`agent_action`: action_id, run_id, seq, tool_name, input_json (redacted), output_hash, output_json (redacted, ≤ 64 KB), approval_task_id, approver_user_id, decision (APPROVED/REJECTED/AUTO), decided_at, committed_entity_ref.

`eval_run`: eval_run_id, agent_id, agent_version, dataset_id, dataset_version, metrics_json, passed bool, ci_ref, created_at. **`token_budget`**: tenant_id, month, limit_in, limit_out, used_in, used_out, state (OK/SOFT/HARD).

Invariants: `agent_action.input_json` never contains raw PII (checked by a redaction scanner at write); an action with approval_mode `approve_before_commit` has `committed_entity_ref` only if decision ∈ {APPROVED}; a run's `invoked_by_user_id` is immutable; a production `agent_version` has a passing `eval_run`. Indexing: `agent_run` (tenant_id, status, started_at), partition by month; `agent_action` (run_id, seq).

9.47.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
GET	`/mcp` (SSE/streamable HTTP)	MCP server endpoint; `tools/list`, `tools/call`	per JSON-RPC id	user token
POST	`/api/v1/copilot/chat`	copilot conversation turn (stream)	n/a	CopilotUse
POST	`/api/v1/agents/{agent_id}/runs`	start a run	`Idempotency-Key`	AgentInvoke + agent's duty
GET	`/api/v1/agents/runs/{run_id}`	run status, actions, costs	n/a	AgentView
POST	`/api/v1/agents/runs/{run_id}/actions/{id}/decision`	approve/reject	by action id	approver duty
POST	`/api/v1/agents/kill-switch`	halt tenant or global	by scope	AgentAdmin / PlatformAdmin
GET	`/api/v1/agents/budgets/{tenant}`	budget status	n/a	AgentAdmin
POST	`/api/v1/agents/evals/runs`	trigger eval (CI)	by commit sha	CI service account
PUT	`/api/v1/agents/{agent_id}/versions/{v}/promote`	pin production version	ETag	AgentAdmin

Validation: run start rejects when tenant budget state HARD, kill switch active, or agent version lacks passing eval; decision endpoint rejects if the approver equals `invoked_by_user_id` and the action's amount exceeds the SoD self-approval limit (wave 48).

Event	Trigger	Payload fields	Consumers
`dev.passage.agent.run.completed.v1`	run end	run_id, agent_id, version, status, tool_calls[], tokens, cost_usd	audit, evals, copilot
`dev.passage.agent.approval.requested.v1`	gate reached	run_id, action_id, tool_name, proposal, confidence, risk_score	workflow inbox
`dev.passage.agent.budget.threshold.v1`	80%/100%	tenant_id, month, pct, state	alerting, admin UI
`dev.passage.agent.killswitch.changed.v1`	toggle	scope, active, by_user	all agent workers

9.47.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Copilot panel (shell)	ask, explain record, draft, summarize, view tool calls	conversation, citations, "insert draft"	CopilotUse
Agent inbox (in workflow inbox)	approve, reject, edit proposal, bulk approve ≤ 50	proposal diff, confidence, rationale, amount	approver duty
Agent runs	filter, inspect timeline, replay in sandbox	run_id, agent, status, tokens, cost	AgentView
Agent administration	enable/disable per entity, set mode, thresholds, kill switch	agent, version, mode, auto threshold	AgentAdmin

Screen	Primary actions	Key fields/columns	Duty
Eval dashboard	compare versions, open failing cases	metric per version, trend	AgentAdmin

Rules: `Ctrl+/` opens copilot with current record context; approvals support keyboard `A/R`; generated text is marked with a badge and never auto-saved; copilot remembers mode per workspace per user.

9.47.7 Business rules and edge cases

- **BR-47.1** Run state machine: PLANNING → ACTING → (AWAITING_APPROVAL → COMMITTED | REJECTED) | COMMITTED (auto) | FAILED | KILLED; no transition out of COMMITTED/REJECTED/KILLED.
- **BR-47.2** Auto mode commits only when confidence ≥ threshold AND amount < limit AND the vendor/customer is not flagged `high_risk` AND no SoD conflict exists for the invoking user.
- **BR-47.3** Approval requests expire after 72 h → REJECTED with reason `EXPIRED`; the source document stays unposted.
- **BR-47.4** Token accounting uses provider-reported usage; if absent, estimated at 4 chars/token and flagged `estimated`.
- **BR-47.5** Kill switch active → running activities finish their current API call, no new tool calls; runs move to KILLED within 10 s.
- **BR-47.6** A tool error 4xx is not retried; 5xx/timeouts retry 3× with exponential backoff (1 s, 4 s, 16 s).
- **BR-47.7** Redaction pseudonyms are stable per (tenant, value) for 24 h so the model can refer back; the mapping is stored only in memory of the run worker and in Vault transit-encrypted form on `agent_run`.
- **BR-47.8** Eval datasets are versioned; a dataset change requires re-baselining all agent versions before promotion.
- **BR-47.9** Provider failover: if the default provider returns 5xx for 60 s, runs switch to the tenant's secondary provider only if it satisfies residency; otherwise runs pause.
- **BR-47.10** Cash application proposals never split a payment across more than 20 invoices, and never create write-offs above the wave-30 tolerance.

9.47.8 Acceptance criteria

- **AC-47.1** (FR-47.1) Given an authenticated user, when `tools/list` is called, then only tools whose `required_duty` the user holds are returned.
- **AC-47.2** (FR-47.2) Given user without duty `APInvoicePost`, when the agent calls `passage.ap.invoice.post`, then gateway returns 403 and the action is logged `DENIED`.
- **AC-47.3** (FR-47.5, 47.9) Given the `eval-ap-coding-v3` dataset (2,000 invoices), when evals run, then top-1 accuracy ≥ 0.92 and the CI job passes.
- **AC-47.4** (FR-47.6, 47.9) Given `eval-cash-app-v2` (1,500 payments), when evals run, then precision ≥ 0.90.
- **AC-47.5** (FR-47.8) Given a coding proposal with confidence 0.80, when the run reaches the gate, then an approval task appears in the inbox and nothing is posted.
- **AC-47.6** (FR-47.12) Given a tenant kill switch toggled, when 10 s elapse, then all runs for that tenant are KILLED and no provider calls occur.
- **AC-47.7** (FR-47.11) Given a tenant at 100% budget, when a run is started, then it is rejected with `BUDGET_EXHAUSTED` and an alert was raised at 80%.
- **AC-47.8** (FR-47.12) Given an invoice containing a vendor IBAN, when the prompt is sent, then the provider request contains a pseudonym and the response shows the real IBAN only in the UI.
- **AC-47.9** (FR-47.3) Given "what was USMF revenue last month vs the month before", when asked, then the answer cites metric `revenue`, the two periods, and matches the explorer to the cent.
- **AC-47.10** (FR-47.13) Given provider switched to Bedrock, when the AP coding eval runs, then thresholds are still met.

9.47.9 Test plan and fixtures

Fixtures: `eval-ap-coding-v3` (2,000 labeled invoices, 40 vendors, 3 entities, from wave-19 fixtures plus hand labels); `eval-cash-app-v2` (1,500 payments with remittances, 12% partials, 5% multi-invoice); `eval-mrp-triage-v1` (800 action messages); `eval-anomaly-v1` (50k journals, 400 injected anomalies); `eval-ask-data-v1` (300 questions with expected Cube queries); `adv-prompts-500` (injection, exfiltration, policy bypass).

Test	Type	Target	Pass criterion
Permission inheritance	contract	gateway + 400 tools	0 cases where agent exceeds user
Tool schema conformance	contract	gateway	every tool validates against MCP spec and `x-passage-*`
Agent evals	eval	5 agents	thresholds in FR-47.9
Adversarial refusal	eval	copilot	≥ 0.99 correct refusals
Redaction scanner	unit/property	redaction	0 PII patterns in stored prompts across 100k generated samples
Kill switch latency	E2E	runtime	≤ 10 s
Run throughput	performance	runtime	500 concurrent runs, p95 < 30 s
Provider parity	contract	4 adapters	identical tool-use round trips

Reconciliation: tokens billed by provider ↔ `agent_run.tokens_*` within 1%; `agent_action` COMMITTED count ↔ posted documents tagged `origin=agent`.

9.47.10 Deployment and operations

Helm: `passage-mcp-gateway` (TS, 3 replicas), `passage-agent-runtime` (TS workers on Temporal namespace `agents`), `passage-copilot-bff`, `passage-agent-evals` (Python job runner). Config: `MODEL_PROVIDER_DEFAULT=anthropic`, `MODEL_ID` pinned per agent version, provider keys in Vault, `TOKEN_BUDGET_DEFAULT_IN=50000000`. Flags: `copilot.enabled`, `agent.<id>.mode` (off/suggest/auto) per tenant and legal entity. Jobs: budget rollover monthly, eval nightly on production datasets, model-card export weekly. SLOs: copilot first-token p95 < 2 s; run success rate ≥ 99%; gateway availability 99.9%. Alerts: eval regression, budget 80/100%, provider error rate > 5%, kill switch toggled. Runbook: rotate provider keys, replay a run in sandbox, demote version. Rollback: `promote` previous `agent_version`; Helm rollback.

9.47.11 Migration from AX / SAP

Source	Passage	Notes
Power Automate flows on business events	Agent or Temporal workflow definitions	catalogued; rebuilt by hand; the business-event catalog maps to CloudEvents types
`VendInvoiceAutomation` parameters	AP coding agent thresholds	tolerance and auto-post limits imported
`CustPaymMatch` rules	Cash application agent rules	matching rules become deterministic pre-filters
SAP Build Process Automation projects	Agent definitions	not convertible; documented
Joule / D365 Copilot settings	Copilot flags	n/a

Not migrated: conversation histories, Copilot usage data (no business value).

9.47.12 Dependencies, risks, open questions

Depends on: wave 46 (semantic layer for ask-the-data), wave 4 (Temporal, workflow inbox), wave 2 (OPA duties, token exchange), wave 19/30 (AP invoices, cash application APIs), wave 42 (PII classification).

Risk	Likelihood	Impact	Mitigation
Prompt injection via invoice text	High	High	tool allow-list per agent; adversarial eval; `approve_before_commit`
Eval datasets overfit to demo data	Medium	Medium	customer-labeled holdout from wave 49 parallel run
Provider cost spikes	Medium	Medium	budgets; prompt caching; smaller models for classification

Open questions: (1) Auto mode for AP coding in v1 for amounts below threshold — proposed default on, decide by wave 49 with first customer. (2) Open-weights model as residency fallback for DE tenants — default: vLLM adapter shipped, not certified; decide by wave 50.

9.48 Wave 48: Globalization, Localization & Compliance Controls

9.48.1 Scope and business context

AX delivers country/region localizations, Electronic Reporting (`ERModel`, `ERFormat`, `ERModelMapping` configurations from the Regulatory Configuration Service), segregation of duties (`SysSecSegregationOfDutiesRule`, conflict table `SysSecSegregationOfDutiesConflict`) and `SysDatabaseLog`. SAP delivers country versions, Document and Reporting Compliance (e-documents, statutory reports), GRC Access Control rulesets with risk ids, and the Audit Information System. This wave adds a localization framework with seven country packs, an electronic reporting designer with a mandates registry, an SoD engine with mitigation and access certification, security hardening with SOC 2 control mapping and evidence automation, and GDPR/CCPA subject requests with retention and legal holds. Out of scope: additional countries (v1.x), payroll localizations beyond wave 43, ISO 27001 certification itself (evidence only).

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
A regulator changes an XML schema; the ER configuration must be re-imported from RCS/LCS, tested in a sandbox and deployed, or an SAP note applied via `SNOTE` with a transport	Formats bundled with the application release cycle	Mandates registry with effective-dated `er_format` versions, schema validation in CI	Schema update 3 weeks → 1 day; 0 regulator rejections for schema errors
SoD conflicts in `SysSecSegregationOfDutiesConflict` are found after assignment; GRC AC risk analysis runs as a batch and mitigations live in a separate system	Detection decoupled from role assignment	`sod/simulate` before save; nightly evaluation; mitigation workflow in the same workspace	Detection latency 1 day batch → < 2 s at assignment; unmitigated HIGH conflicts 0
SOC 2 evidence is screenshots and exports gathered by hand each audit (realm settings, backup logs, scan reports)	No system of record for controls	Control library with scheduled collectors and hashed artifacts	Evidence effort 20 person-days/audit → 1; 100% evidence machine-collected
GDPR requests require searching every table by hand (person search report in AX, IRF in SAP) and erasure risks breaking postings	PII not classified; no fan-out workflow	Classification-driven discovery, Temporal fan-out per domain, pseudonymization	Request fulfillment 25 days → 2; 0 broken financial references
`SysDatabaseLog` slows posting and fills the database; `CDHDR/CDPOS` is incomplete for custom tables	Application-side audit writes	CDC-derived audit, configuration only chooses what to surface	Posting overhead from audit 15% → 0%; 100% field coverage

9.48.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Country localization	Country/region-specific features gated by `CountryRegionCodes`, Localization feature keys	Country versions, `T005` country settings, localization BC sets	`country_pack` with tax, reports, layouts, bank formats, calendars	Full (7 countries)
Electronic reporting	ER `ERModel`, `ERFormat`, `ERModelMapping`, RCS, `ERFormatMappingRun`	DRC e-documents (`EDOC_COCKPIT`), statutory reports (`SRF`), Advanced Compliance Reporting	ER designer: `er_model`, `er_mapping`, `er_format` with mandates registry	Full
E-invoicing mandates	Electronic invoicing service (Italy SDI, Mexico CFDI)	DRC eDocument Framework (SDI, CFDI, Peppol, India IRN)	Mandate adapters: MX CFDI 4.0, IN IRP e-invoice, Peppol BIS 3 (wave 35), UK MTD VAT	Full for first set
Statutory tax reports	Sales tax reporting, EU sales list, Intrastat (`Intrastat` form)	`RFUMSV00` VAT return, ESL (`RFASLM00`), Intrastat (`VE01`)	ER formats: US sales tax, UK MTD VAT return, DE USt/VA/ZM, FR CA3/DEB, IN GSTR-1/3B, MX DIOT, CA GST/HST	Full
Segregation of duties	`SysSecSegregationOfDutiesRule`, conflict detection, `SysSecSegregationOfDutiesConflict` resolution	GRC Access Control rulesets, risk ids (e.g. P001, F002), mitigating controls, UAR	`sod_rule`, `sod_conflict`, `mitigation`, `certification_campaign`	Improved (campaigns included)

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Audit log	`SysDatabaseLog`, `DatabaseLog` form	`CDHDR/CDPOS`, AIS (`SECR`), `SM20` security audit log	wave-5 CDC audit + compliance evidence views	Full
Security compliance	Microsoft-managed SOC reports	SAP-managed for RISE; customer for on-prem	SOC 2 control mapping with automated evidence	Improved (customer-visible)
Data privacy	Person search report, GDPR request form (`GDPR request`)	Information Retrieval Framework (IRF), ILM retention (`IRM_CUST`), blocking	`subject_request`, `retention_policy`, `legal_hold`	Full
Data residency	LCS region selection	RISE region	tenant residency policy enforced on storage, lakehouse, AI provider	Full

9.48.3 Functional requirements

- **FR-48.1 Country pack framework** — Must. The system shall package per-country tax rules, statutory report formats, document layouts, bank formats (NACHA, BACS, SEPA CT/DD, NEFT/RTGS, SPEI, EFT), holiday calendars and address/tax-id validators as a versioned `country_pack`, installable per legal entity.
  - Notes: packs are metadata plus ER formats, never code forks; a legal entity has exactly one primary pack; pack upgrade is a versioned migration with preview.
- **FR-48.2 First seven packs** — Must. US (sales/use tax, 1099-NEC/MISC, ACH), UK (MTD VAT, BACS, Companies House XBRL tags), DE (UStVA, ZM, SEPA, GoBD export, DATEV), FR (CA3, DEB, FEC file, SEPA), IN (GST returns GSTR-1/3B, e-invoice IRP, TDS), MX (CFDI 4.0 with PAC, DIOT, SPEI), CA (GST/HST/PST, EFT, T4A).
- **FR-48.3 Electronic reporting designer** — Must. The system shall provide `er_model` (logical data model), `er_mapping` (Passage entities → model), `er_format` (XML/JSON/CSV/fixed-width output with XSD/JSON-schema validation), versioned with effective dates and a mandates registry (`mandate`: regulator, jurisdiction, schema version, effective_from, sunset).
  - Notes: format execution is a Temporal job; every output is stored with hash and regulator acknowledgment; schema validation failure blocks submission.
- **FR-48.4 SoD rules engine** — Must. The system shall evaluate rules of the form `duty_a CONFLICTS duty_b` (and privilege-level) against every user's effective roles on each assignment change and nightly, raising conflicts with severity.
  - Rule schema: `rule_id`, `name`, `duty_a`, `duty_b`, `severity` (HIGH/MEDIUM/LOW), `risk_id` (GRC-style, e.g. `P2P-001`), `description`, `mitigation_allowed` bool, `max_mitigation_days` (default 365), `status`.
  - Seed rules (10 of 50): `P2P-001` MaintainVendors ↔ ProcessVendorPayments (HIGH); `P2P-002` CreatePurchaseOrders ↔ ApprovePurchaseOrders (HIGH); `P2P-003` PostVendorInvoices ↔ ApproveVendorInvoices (HIGH); `P2P-004` ReceiveGoods ↔ PostVendorInvoices (MEDIUM); `O2C-001` MaintainCustomers ↔ PostCustomerPayments (HIGH); `O2C-002` CreateSalesOrders ↔ ApproveCreditLimits (HIGH); `O2C-003` IssueCreditNotes ↔ PostCustomerPayments (HIGH); `GL-001` PostGeneralJournals ↔ ApproveGeneralJournals (HIGH); `GL-002` MaintainChartOfAccounts ↔ PostGeneralJournals (MEDIUM); `SEC-001` MaintainUsersAndRoles ↔ any posting duty (HIGH); `HR-001` MaintainWorkerBankDetails ↔ RunPayroll (HIGH).
- **FR-48.5 Mitigation workflow** — Must. The system shall allow a conflict to be mitigated with a compensating control, approver (not the conflicted user), reason, and expiry ≤ `max_mitigation_days`; unmitigated HIGH conflicts block role assignment when tenant policy `sod.block_high=true`.
- **FR-48.6 Access certification campaigns** — Should. The system shall run campaigns where managers certify or revoke each direct report's roles within a window (default 14 days), auto-revoking uncertified roles at close when policy says so, with evidence export.
- **FR-48.7 SOC 2 control mapping and evidence** — Must. The system shall ship a control library mapped to TSC and automate evidence collection on a schedule.
  - Excerpt: CC6.1 logical access — evidence: Keycloak realm export, OPA bundle hash, RLS policy list (daily); CC6.2 provisioning — SCIM join/leave log, certification results (weekly); CC6.3 role changes — `sod_conflict` report, role-assignment audit (daily); CC6.6 boundary protection — Envoy Gateway TLS config, NetworkPolicy manifests (daily); CC6.7 data in transit/at rest — cert-manager inventory, CNPG encryption settings (weekly); CC6.8 malware/unauthorized software — image signature verification logs, SBOM diff (per deploy); CC7.1 vulnerability detection — Trivy scan results, pen-test tracker (daily/quarterly); CC7.2 anomaly monitoring — alert rule inventory, journal-anomaly agent summary (daily); CC7.3 incident evaluation — incident records with severity and timeline (per incident); CC7.4 incident response — post-incident review, runbook link (per incident); CC7.5 recovery — backup/restore drill results, RPO/RTO measured (monthly).
- **FR-48.8 Security hardening** — Must. The system shall rotate all secrets ≤ 90 days via Vault, enforce signed images, run Trivy and dependency scanning per build, remediate external pen-test findings (high/critical = 0 at exit), and enforce tenant data residency (region for Postgres, object store, lakehouse, AI provider).
- **FR-48.9 Backup and restore** — Must. The system shall take continuous WAL archives and daily base backups per domain cluster (CloudNativePG), support point-in-time restore of one tenant into an isolated namespace, and run a monthly drill measuring RPO ≤ 5 min and RTO ≤ 1 h.
- **FR-48.10 Subject requests** — Must. The system shall process GDPR/CCPA requests of type ACCESS, PORTABILITY, ERASURE, RECTIFICATION, RESTRICTION with the flow: intake (portal or admin) → identity verification (two attributes + e-mail link) → scope discovery (search party, worker, user, contact across all domains via the PII classification) → legal-hold and retention check → domain fan-out (Temporal child workflow per domain) → compile (ACCESS/PORTABILITY: JSON + PDF bundle) or apply (ERASURE: pseudonymize PII fields, keep financial records with `erased=true`) → approval by privacy officer → deliver via signed link → close within 30 days (GDPR) / 45 days (CCPA), with the deadline tracked.
- **FR-48.11 Retention and legal hold** — Must. The system shall apply retention policies per record class (e.g. financial documents 10 years DE, 7 years US; HR 6 years post-termination; logs 1 year) and legal holds that suspend erasure and retention purge for named scopes.
- **FR-48.12 Database log equivalent** — Should. The system shall offer `SysDatabaseLog`-style configuration: pick entities/fields to surface in the compliance audit view, backed by the wave-5 CDC audit, with no extra write path.

9.48.4 Data model

`sod_rule` (sod)

Field	Type	Required	Notes
rule_id	text	Y	PK, e.g. `P2P-001`
duty_a / duty_b	text	Y	OPA duty ids; `*POST*` wildcard allowed
severity	enum	Y	HIGH, MEDIUM, LOW
risk_id	text	Y	GRC-style id
mitigation_allowed	bool	Y	default true
max_mitigation_days	int	Y	default 365
status	enum	Y	ACTIVE, RETIRED
version	int	Y	

`sod_conflict`: conflict_id, tenant_id, user_id, rule_id, roles_json (which roles grant each duty), severity, detected_at, state (OPEN, MITIGATED, RESOLVED, ACCEPTED_RISK), mitigation_id, resolved_at.

`er_format` (ereporting)

Field	Type	Required	Notes
format_id	uuid	Y	PK
mandate_id	uuid	Y	FK mandate
version	int	Y	
output_type	enum	Y	XML, JSON, CSV, FIXED, PDF
schema_ref	text	N	XSD/JSON schema location, hash
template_json	jsonb	Y	element tree with bindings
effective_from / effective_to	date	Y/N	
status	enum	Y	DRAFT, PUBLISHED, SUPERSEDED

`subject_request`: request_id, tenant_id, type, subject_ref (party/worker/user), verified_at, deadline_at, state (RECEIVED, VERIFIED, DISCOVERED, HELD, APPROVED, DELIVERED, CLOSED, REJECTED), domains_json, bundle_ref, privacy_officer_id.

`control_evidence`: evidence_id, control_id (e.g. `CC6.1`), collected_at, collector, artifact_ref (object store), hash, period_start, period_end, status (COLLECTED, REVIEWED, EXCEPTION).

Invariants: a user cannot approve their own mitigation; a PUBLISHED `er_format` with a schema_ref has a validated fixture; an ERASURE cannot complete while a legal hold covers the subject; `control_evidence.hash` matches the stored artifact. Indexing: `sod_conflict` (tenant_id, state, severity); `subject_request` (tenant_id, deadline_at); `control_evidence` (control_id, collected_at).

9.48.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/localization/packs/{code}/install`	install pack on legal entity	by (entity, version)	LocalizationAdmin
POST	`/api/v1/ereporting/formats/{id}/run`	execute format for period	`Idempotency-Key`	per report duty
POST	`/api/v1/ereporting/submissions/{id}/submit`	submit to regulator adapter	by submission id	per report duty
GET	`/api/v1/sod/conflicts?state=OPEN`	list conflicts	n/a	SoDReview
POST	`/api/v1/sod/conflicts/{id}/mitigations`	create mitigation	`Idempotency-Key`	SoDApprove
POST	`/api/v1/sod/simulate`	simulate role assignment	n/a	SecurityAdmin
POST	`/api/v1/sod/campaigns`	start certification campaign	`Idempotency-Key`	SoDAdmin
GET	`/api/v1/compliance/controls/{id}/evidence?period=`	evidence listing	n/a	ComplianceView
POST	`/api/v1/privacy/requests`	intake subject request	`Idempotency-Key`	PrivacyIntake or portal
POST	`/api/v1/privacy/holds`	create legal hold	`Idempotency-Key`	LegalHoldAdmin
POST	`/api/v1/ops/backups/restore-drill`	run PITR drill into namespace	by drill id	PlatformAdmin

Validation: pack install rejects if the entity's fiscal calendar is open-period incompatible; `sod/simulate` returns conflicts before assignment is saved; a `run` of a format outside `effective_from..to` is rejected.

Event	Trigger	Payload fields	Consumers
`dev.passage.ereporting.document.submitted.v1`	regulator ack	mandate_id, format_version, document_ref, ack_id, status	AR/AP (mark invoice sent), compliance
`dev.passage.sod.conflict.detected.v1`	evaluation	user_id, rule_id, severity, roles[]	workflow, compliance
`dev.passage.sod.mitigation.approved.v1`	approval	conflict_id, approver, expires_at	audit
`dev.passage.privacy.request.completed.v1`	close	request_id, type, affected_domains[], delivered_at	audit
`dev.passage.compliance.evidence.collected.v1`	collector	control_id, evidence_id, hash	compliance UI

9.48.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Country packs	install, upgrade preview, view contents	code, version, entity, components	LocalizationAdmin
ER designer	edit model/mapping/format, validate against schema, run with fixture	element tree, binding, schema status	ERDesign
Mandates registry	add mandate, set effective dates, link formats	regulator, jurisdiction, schema version	ERDesign

Screen	Primary actions	Key fields/columns	Duty
SoD conflicts	mitigate, resolve, accept risk, export	user, rule, severity, roles, age	SoDReview/SoDApprove
Role assignment (wave 2)	simulate conflicts before save	conflicts preview	SecurityAdmin
Certification campaign	certify/revoke per user, progress	manager, user, roles, decision	Manager
Compliance workspace	evidence calendar, exceptions, export audit package	control, last evidence, status	ComplianceView
Privacy requests	verify, review scope, approve, deliver	type, subject, deadline, state	PrivacyOfficer

Rules: SoD grid supports bulk mitigation ≤ 25 with one approver; evidence exports as a ZIP with manifest hash; privacy request deadline shows days remaining with red at ≤ 5.

9.48.7 Business rules and edge cases

- **BR-48.1** Conflict state machine: OPEN → MITIGATED (expires → OPEN) | RESOLVED (role removed) | ACCEPTED_RISK (LOW/MEDIUM only, by SoDAdmin).
- **BR-48.2** Conflicts are evaluated on effective duties including inherited roles and temporary delegations; a delegation that creates a HIGH conflict is refused.
- **BR-48.3** ER format outputs are immutable once submitted; corrections create a new submission with `supersedes`.
- **BR-48.4** Tax-period statutory runs lock the source tax settlement; re-running requires reversal of the settlement (wave 13).
- **BR-48.5** Erasure pseudonymizes `name`, `address`, `email`, `phone`, `tax_id`, `bank_account`, `date_of_birth` fields; monetary documents keep amounts, dates and dimension values.
- **BR-48.6** Retention purge runs monthly and skips any record under hold, in an open fiscal period, or referenced by an open document.
- **BR-48.7** Residency: a tenant with `residency=EU` cannot be assigned an AI provider endpoint outside the EU; the API rejects the configuration.
- **BR-48.8** Backup drill failing RPO or RTO opens a Sev-2 incident automatically.
- **BR-48.9** Certification campaign close: uncertified roles are revoked only when `campaign.auto_revoke=true`; otherwise flagged.
- **BR-48.10** Country-pack upgrade with a changed tax rate effective-dates the new rate; it never rewrites posted tax.

9.48.8 Acceptance criteria

- **AC-48.1** (FR-48.2, 48.3) Given each of the 7 packs' statutory fixture, when the format runs, then the output validates against the regulator schema with 0 errors.
- **AC-48.2** (FR-48.4) Given 50 users with planted conflicting role combinations, when evaluation runs, then 50 conflicts are detected with the correct rule ids and 0 false negatives.
- **AC-48.3** (FR-48.5) Given an OPEN HIGH conflict and `sod.block_high=true`, when the role assignment is saved, then it is rejected with the rule id.
- **AC-48.4** (FR-48.5) Given a mitigation, when approved by a non-conflicted approver with reason and 180-day expiry, then the conflict is MITIGATED and `mitigation.approved` is emitted.
- **AC-48.5** (FR-48.9) Given a tenant with 10 GB across domains, when a PITR drill to T–20 min runs, then the restored namespace is queryable within 1 h and the lost-data window ≤ 5 min.
- **AC-48.6** (FR-48.10) Given an ACCESS request for a customer contact, when processed, then the bundle contains data from every domain holding the subject and the request closes within 30 days.
- **AC-48.7** (FR-48.10, 48.11) Given an ERASURE request for a party under legal hold, when processed, then state is HELD and no field is changed.
- **AC-48.8** (FR-48.7) Given control CC7.1, when the daily collector runs, then a Trivy report evidence record exists with a matching hash.
- **AC-48.9** (FR-48.8) Given the external pen-test report, when wave exit is reviewed, then open high/critical findings = 0.
- **AC-48.10** (FR-48.1) Given the DE pack installed on DEMF, when an invoice is printed, then the layout uses `de-DE` formats and the UStVA mapping includes the invoice's tax lines.

9.48.9 Test plan and fixtures

Fixtures: `fx-statutory-7` (one period of postings per country with expected outputs and regulator schemas: UK HMRC MTD VAT JSON, DE ELSTER UStVA XML, FR FEC, IN GSTR JSON, MX CFDI 4.0 XSD, US 1099 IRS FIRE, CA GST34 fields); `fx-sod-50` (50 planted users); `fx-privacy-subjects` (20 subjects spanning 9 domains); `fx-residency` (3 tenants EU/US/IN).

Test	Type	Target	Pass criterion
Schema validation	contract	7 formats	0 schema errors
SoD detection	unit/property	engine	50/50; random role sets never miss a seeded conflict
Mitigation workflow	E2E	sod	approver ≠ user enforced; expiry reopens
Privacy fan-out	E2E	privacy	all domains respond; bundle complete
PITR drill	performance	CNPG	RPO ≤ 5 min, RTO ≤ 1 h
Secrets rotation	contract	Vault	all leases ≤ 90 days
Residency enforcement	contract	config API	cross-region assignment rejected
Evidence collectors	unit	11 controls	artifact + hash per schedule

Reconciliation: statutory VAT return boxes ↔ tax settlement totals (wave 13) to the cent; conflicts count ↔ independent recomputation from role assignments; evidence count ↔ schedule expectations.

9.48.10 Deployment and operations

Helm: `passage-localization`, `passage-ereporting` (TS + Temporal), `passage-sod` (Go), `passage-compliance`, `passage-privacy`. Config: `COUNTRY_PACKS_ENABLED`, regulator adapter credentials (HMRC, PAC, IRP) in Vault, `SOD_BLOCK_HIGH`, `PRIVACY_DEADLINE_DAYS`. Migrations: `compliance` and `localization` databases. Flags per tenant: `pack.<code>`, `sod.enforce`, `privacy.portal`. Jobs: SoD nightly 00:30, evidence collectors per control

schedule, retention purge monthly, backup drill monthly, secrets rotation. SLOs: format run p95 < 5 min for 100k lines; SoD evaluation < 2 s per assignment. Alerts: schema validation failure, regulator rejection, HIGH conflict unmitigated > 7 days, drill failure. Rollback: pack version downgrade with preview; ER format supersede; Helm rollback.

9.48.11 Migration from AX / SAP

Source	Passage	Notes
`SysSecSegregationOfDutiesRule`	`sod_rule`	duties mapped via wave-2 role mapping; severity imported
`SysSecSegregationOfDutiesConflict` with resolution	`sod_conflict` + `mitigation`	open conflicts re-evaluated; mitigations imported with original approver and expiry
GRC AC rulesets (risk ids, functions, actions)	`sod_rule`	function→duty mapping table; mitigating controls → `mitigation`
ER configurations (`ERModel`, `ERFormat` XML exports)	`er_model`, `er_format`	importer parses ER XML; formulas in ER expression language are translated where possible, else flagged
`SysDatabaseLog` configuration	FR-48.12 configuration	table/field list
`CDHDR/CDPOS` history	wave-49 lakehouse history	read-only
GDPR request logs	`subject_request` history	import closed requests for evidence

Not migrated: Microsoft/SAP SOC reports (vendor-owned), regulator credentials (re-issued).

9.48.12 Dependencies, risks, open questions

Depends on: wave 2 (roles/duties, OPA), wave 5 (CDC audit), wave 13 (tax), wave 35 (Peppol, e-invoicing transport), wave 42 (PII classification), wave 46 (evidence from reconciliation).

Risk	Likelihood	Impact	Mitigation
Regulator schema changes mid-wave	High	Medium	mandates registry with effective dates; schema fixtures pinned
SoD rule set incomplete for customer	Medium	High	50-rule seed plus GRC import
Erasure breaks referential integrity	Medium	High	pseudonymize, never delete rows

Open questions: (1) PAC provider for MX (default: Finkok sandbox); decide by wave 49. (2) Whether to pursue SOC 2 Type I before go-live — proposed yes, readiness assessment by wave 50.

9.49 Wave 49: Migration Factory: Full AX/SAP Transactional Cutover Tooling

9.49.1 Scope and business context

AX 2012 → D365 uses the data upgrade scripts (`ReleaseUpdateDB` classes, LCS data-upgrade deployable package) and DIXF; SAP uses SUM with DMO for conversion, the Migration Cockpit (`LTMC`/`LTMOM`, staging tables) and Selective Data Transition. None of them moves a customer to a different ERP. Building on the wave-10 framework (connectors, declarative mapping, idempotent runs), this wave adds transactional packs with control totals, historical loads into the lakehouse with legacy cross-references, a parallel-run harness, a cutover runbook generator and coexistence adapters. Out of scope: code migration (X++/ABAP), custom-table migration beyond the mapping DSL, SAP archive (ADK) files.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
Open items loaded through DIXF (`DMFDefinitionGroup`) or Migration Cockpit staging tables (`/1LT/DS*`) are reconciled in Excel against `VendTransOpen`/`BSIK` totals; mismatches found days later	No built-in control totals at sub-total grain	`control_total` per pack at (entity, account, currency) grain with hashes, computed in-run	Reconciliation 5 person-days per entity → 10 min automated; mismatches found in-run
Re-running a failed load creates duplicates (`InventSum` doubled, duplicate `BKPF` opening documents)	Loaders without natural-key idempotency	Idempotent packs keyed on `legacy_xref`	Duplicate rate after re-run 0
Parallel runs compare printed trial balances by eye; variances are explained in e-mail threads	No diff tooling	Normalized tuple diff with EXPECTED/EXPLAINED/UNEXPLAINED classification and sign-off	Variance review 3 weeks → 2 days; 0 unexplained at sign-off
Cutover plans are spreadsheets; step durations are guesses; rollback is untested	No runbook tooling tied to the data loads	Generated `cutover_plan` with timed steps from rehearsals, go/no-go gates, rehearsed rollback	Cutover < 48 h; 2 rehearsals; rollback rehearsed
Users lose the legacy voucher/document id after migration and keep the old system online for lookups	No cross-reference model	`legacy_xref` on every record and in the lakehouse history	Legacy lookups needing the old system 100% → 0

9.49.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Open AP/AR items	`VendTrans`/`VendTransOpen`, `CustTrans`/`CustTransOpen`, `VendSettlement`	`BSIK`/`BSID` (ECC), `ACDOCA` open items (S/4), `BSEG`	pack `open_items` → AP/AR open items with `legacy_xref`	Full
GL balances	`GeneralJournalAccountEntry`, `DimensionAttributeValueCombination`, `LedgerBalance*`	`GLT0`/`FAGLFLEXT` (ECC), `ACDOCA` (S/4)	pack `gl_balances` → opening journal per dimension	Full
On-hand inventory	`InventSum`, `InventTrans`, `InventDim`, `InventTransOrigin`	`MARD`, `MCHB` (batch), `MSKA`, `MBEW` valuation	pack `onhand` → inventory opening transactions by dimension	Full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Open POs / SOs	`PurchTable` / `PurchLine`, `SalesTable` / `SalesLine` with remaining qty	`EKKO` / `EKPO` / `EKET`, `VBAK` / `VBAP` / `VBEP`	pack `open_orders` → orders with delivered/invoiced quantities	Full
In-process production	`ProdTable`, `ProdBOM`, `ProdRoute`, `ProdJournalProd`	`AFKO` / `AFPO` / `AFVC`, `AUFK`	pack `wip_prod` → orders at status with consumed/reported quantities and WIP value	Full
Fixed assets	`AssetTable`, `AssetBook`, `AssetTrans`	`ANLA` / `ANLB` / `ANLC` / `ANEP`	pack `fixed_assets` → NBV per book with depreciation profile	Full
Project WIP	`ProjTable`, `ProjTrans*`, `ProjWIP*`	`PRPS`, `COSP` / `COSS`, RA results	pack `project_wip`	Full
Employee/payroll YTD	`HcmWorker`, `PayrollEarningStatement*`	`PA0001` / `PA0008`, payroll results cluster (`PCL2`)	pack `payroll_ytd`	Partial (US + DE only)
Upgrade tooling	AX 2012→D365 data upgrade (LCS), DIXF	SUM/DMO, Migration Cockpit, SDT	migration factory packs + runbook generator	Improved (cross-vendor)
History	Legacy database kept online	Archive / SDT history	lakehouse `history.*` with `legacy_xref`	Full (read-only)

9.49.3 Functional requirements

- **FR-49.1 Transactional packs** — Must. The system shall provide packs `open_items`, `gl_balances`, `onhand`, `open_orders`, `wip_prod`, `fixed_assets`, `project_wip`, `payroll_ytd`, `bank_balances`, each with AX and SAP extractors, mapping, validation (Great Expectations suites), load and reconciliation steps.
- **FR-49.2 Control totals** — Must. The system shall compute and compare per pack: `open_items`: count, sum amount in transaction currency per currency, sum in accounting currency, per (entity, account, currency) sub-totals, hash of sorted (legacy_id, amount); `gl_balances`: per (entity, account, dimension combination, currency) balance, trial balance debit = credit, count of non-zero combinations; `onhand`: per (item, site, warehouse, location, batch, serial) quantity and per (item, site) value, total value per entity; `open_orders`: order count, line count, remaining quantity, remaining amount; `wip_prod`: order count, WIP value per order; `fixed_assets`: count, acquisition cost, accumulated depreciation, NBV per book; `project_wip`: WIP value per project; `payroll_ytd`: gross, tax, net per worker; `bank_balances`: per account.
  - Notes: every total must match exactly (0.00) unless a tolerance rule is explicitly configured with reason; totals are stored on `control_total` with source and target values and timestamp.
- **FR-49.3 Legacy cross-reference** — Must. The system shall store `legacy_xref` (system, company, table, legacy_id → passage entity, id) for every migrated record and expose it on every record detail and in the lakehouse.
- **FR-49.4 Historical load** — Must. The system shall load closed legacy transactions (journals, invoices, inventory transactions, orders) for N years (default 7) into Iceberg `history.<system>.*` with the legacy schema preserved, mapped views `history.unified.*` on the Passage model, queryable in explorer with `source=legacy` filter.
- **FR-49.5 Parallel-run harness** — Must. The system shall replay a legacy period's source documents (vendor invoices, customer invoices, inventory movements, journals, production postings) into a Passage rehearsal tenant through the public posting APIs in original sequence and compare results.
  - Diff algorithm: (1) normalize both sides to canonical tuples `(entity, account, dimension set, currency, period, amount)` for trial balance, `(item, dims, qty, value)` for inventory, `(subledger, open item id, amount)` for subledgers; (2) outer-join on key; (3) classify each difference: EXPECTED (rule-based: known rounding ≤ 0.01, policy differences registered in `variance_rule`, e.g. different FX rounding step), EXPLAINED (operator attaches a reason), UNEXPLAINED; (4) aggregate by rule; (5) report unexplained count and amount; (6) exit criterion unexplained = 0 and explained amount ≤ 0.1% of period revenue.
- **FR-49.6 Cutover runbook generator** — Must. The system shall generate `cutover_plan` with timed steps from T−14d to T+30d from the selected packs, entities and legacy systems, with owners, durations from the last rehearsal, go/no-go items and rollback steps.
  - Timeline baseline: T−14d freeze master-data changes in legacy, final pack dry run; T−7d second rehearsal, sign-off on variance report; T−3d freeze configuration in Passage; T−1d 18:00 legacy transaction freeze, final extracts of master data and static reference data; T−0 00:00 extract transactional packs; T+4h load and reconcile master and balances; T+12h load open items, orders, on-hand, WIP; T+24h reconciliation complete, go/no-go meeting; T+30h user access opened; T+36h abort deadline if go/no-go fails; T+1d first postings, hypercare starts; T+7d first weekly reconciliation; T+30d first month-end close, hypercare review.
- **FR-49.7 Coexistence adapters** — Should. The system shall support entity-by-entity cutover with intercompany documents and consolidation data exchanged with the legacy system through wave-35 integration hub adapters (AX OData/`DMF` entities; SAP IDoc/BAPI) for `INVOIC`, `ORDERS`, GL postings, and master-data sync one way (legacy → Passage) until the last entity cuts over.
- **FR-49.8 Rehearsal environment** — Must. The system shall provision a rehearsal tenant from pipeline (Terraform + Helm + seed) in ≤ 2 h, with masked extracts, and tear it down.
- **FR-49.9 Rollback plan** — Must. The system shall generate a rollback plan per cutover: stop Passage access, restore legacy write access, export any Passage postings made after go-live as journal/document files for re-keying, with a decision point at T+36h.
- **FR-49.10 Idempotent re-runs** — Must. The system shall re-run any pack without duplicates (wave-10 natural keys plus `legacy_xref` uniqueness) and record every run with inputs' hashes.
- **FR-49.11 Validation suites** — Must. The system shall run Great Expectations suites per pack before load: referential (every vendor on open items exists), domain (currency codes ISO 4217), balance (debit = credit), uniqueness; failures above 0 block load unless the record is quarantined to `pack_quarantine` with reason.

9.49.4 Data model

`pack_run` (migration-factory)

Field	Type	Required	Notes
run_id	uuid	Y	PK = Temporal workflow id
tenant_id	uuid	Y	
pack	enum	Y	the 9 packs

Field	Type	Required	Notes
source_system	enum	Y	AX2012, D365FO, SAPECC, S4HANA
source_company	text	Y	`DataAreaId` / `BUKRS`
cutoff_ts	timestamptz	Y	extraction cutoff
input_hash	text	Y	sha256 of extract
status	enum	Y	EXTRACTED, VALIDATED, LOADED, RECONCILED, FAILED, ROLLED_BACK
records_in / records_loaded / records_quarantined	int	Y	

`control_total`: total_id, run_id, name, key_json (e.g. {entity, account, currency}), source_value numeric(28,6), target_value numeric(28,6), delta, tolerance, status (MATCH, WITHIN_TOLERANCE, MISMATCH), computed_at.

`legacy_xref`: xref_id, tenant_id, source_system, source_company, source_table, source_id, passage_entity, passage_id, run_id; unique (source_system, source_company, source_table, source_id).

`parallel_run`: parallel_run_id, tenant_id, period, source_system, documents_replayed, state (REPLAYING, DIFFING, REVIEW, SIGNED_OFF), unexplained_count, unexplained_amount, explained_amount, signed_off_by. **`variance`**: variance_id, parallel_run_id, area (TB/INV/AP/AR/FA), key_json, legacy_value, passage_value, delta, class (EXPECTED/EXPLAINED/UNEXPLAINED), rule_id, reason.

`cutover_step`: step_id, plan_id, seq, offset_minutes (from T–0), name, owner, planned_duration_min, actual_start, actual_end, status, rollback_step_id, go_nogo bool.

Invariants: a pack run reaches RECONCILED only if every `control_total.status != MISMATCH`; `legacy_xref` is append-only; a `parallel_run` can be SIGNED_OFF only when unexplained_count = 0; cutover steps with `go_nogo=true` must all be complete before the "open user access" step can start. Indexing: `legacy_xref` unique index as above plus (passage_entity, passage_id); `control_total` (run_id, status); `variance` (parallel_run_id, class).

9.49.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/migration/packs/{pack}/runs`	start extract/validate/load	by input_hash	MigrationRun
GET	`/api/v1/migration/runs/{id}/control-totals`	totals and status	n/a	MigrationView
POST	`/api/v1/migration/runs/{id}/quarantine/{rec}/release`	fix and release quarantined record	by record	MigrationRun
POST	`/api/v1/migration/history/loads`	start historical load	by (system, years)	MigrationRun
GET	`/api/v1/migration/xref?system=&table=&id=`	resolve legacy id	n/a	any authenticated
POST	`/api/v1/migration/parallel-runs`	start replay for period	by (period, system)	MigrationRun
POST	`/api/v1/migration/parallel-runs/{id}/variances/{v}/explain`	attach reason	by variance	MigrationReview
POST	`/api/v1/cutover/plans/generate`	generate plan from packs	by inputs hash	CutoverManager
POST	`/api/v1/cutover/plans/{id}/steps/{s}/complete`	mark step done	by step	step owner
POST	`/api/v1/cutover/plans/{id}/rollback`	execute rollback plan	by plan	CutoverManager + second approver

Validation: pack run rejects extracts with `cutoff_ts` later than the legacy freeze; `complete` rejects out-of-sequence go/no-go steps; `rollback` requires two distinct approvers.

Event	Trigger	Payload fields	Consumers
`dev.passage.migration.pack.reconciled.v1`	reconciliation done	run_id, pack, totals_match, totals_mismatch, quarantined	cutover, notifications
`dev.passage.migration.parallelrun.diffed.v1`	diff done	parallel_run_id, period, unexplained_count, unexplained_amount	cutover
`dev.passage.cutover.step.completed.v1`	step done	plan_id, step_id, actual_duration_min, status	cutover dashboard
`dev.passage.cutover.decision.recorded.v1`	go/no-go	plan_id, decision, approvers[], timestamp	audit, platform-ops

9.49.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Migration factory workspace	run pack, view totals, open quarantine, re-run	pack, source, status, records, mismatches	MigrationRun
Control-total board	drill mismatch to records, export	name, key, source, target, delta	MigrationView
Quarantine	edit field, release, reject	record, rule failed, reason	MigrationRun
Parallel-run review	classify variances, attach reason, sign off	area, key, legacy, passage, delta, class	MigrationReview
Cutover runbook	start/complete step, timer, go/no-go checklist, rollback	seq, T-offset, owner, planned vs actual, status	CutoverManager
Legacy reference (on any record)	open xref	system, company, table, id	all

Rules: control-total board auto-refreshes every 10 s during a run; variance grid supports bulk classify by rule; runbook shows elapsed vs planned with amber at 110%, red at 125%.

9.49.7 Business rules and edge cases

- **BR-49.1** Pack run state machine: EXTRACTED → VALIDATED → LOADED → RECONCILED; any step → FAILED; LOADED/RECONCILED → ROLLED_BACK (compensating journals and deletion of unposted documents).
- **BR-49.2** Open items are loaded against an opening-balance clearing account; the sum of open items per control account must equal the GL balance migrated for that account to 0.00 before RECONCILED.

- **BR-49.3** On-hand is loaded at legacy cost per dimension; the inventory GL account opening balance is migrated as a single line and must equal the loaded value per entity.
- **BR-49.4** Open orders are loaded with `qty_delivered` and `qty_invoiced` history so remaining quantities match; prior deliveries are not re-posted.
- **BR-49.5** WIP production orders carry consumed material and reported hours as a WIP opening entry; remaining operations are rebuilt from the Passage route.
- **BR-49.6** Fixed assets load acquisition date and accumulated depreciation per book; the next depreciation run must equal legacy's projected amount within 0.01 or the asset is flagged.
- **BR-49.7** Amount rounding during replay follows Passage rules; a difference ≤ 0.01 per document is classified EXPECTED by rule `RND-001`.
- **BR-49.8** Historical loads are read-only; a legacy document with no mapped Passage type lands in `history.<system>.unmapped`.
- **BR-49.9** Cutover abort at T+36h: if any go/no-go item is open, the rollback plan executes automatically after CutoverManager confirmation.
- **BR-49.10** Coexistence: a document received from legacy that already exists by `legacy_xref` is acknowledged and ignored.

9.49.8 Acceptance criteria

- **AC-49.1** (FR-49.1, 49.2) Given the synthetic AX dataset (3 entities, full year), when all packs run, then every `control_total.status = MATCH`.
- **AC-49.2** (FR-49.1, 49.2) Given the synthetic SAP dataset, when all packs run, then every `control_total.status = MATCH`.
- **AC-49.3** (FR-49.11) Given an open item for a vendor missing in master data, when validation runs, then the record is quarantined and load proceeds for the rest.
- **AC-49.4** (FR-49.10) Given a completed `open_items` run, when the same extract is run again, then records_loaded = 0 and totals are unchanged.
- **AC-49.5** (FR-49.5) Given one month of legacy documents replayed, when the diff runs, then unexplained_count = 0 and explained amount ≤ 0.1% of revenue.
- **AC-49.6** (FR-49.6) Given the generated runbook, when the rehearsal is executed, then all steps from T–0 to "open user access" complete in < 48 h.
- **AC-49.7** (FR-49.9) Given a rehearsal cutover, when rollback executes, then the legacy coexistence adapters resume and Passage postings since T–0 are exported as re-keying files.
- **AC-49.8** (FR-49.3) Given migrated customer invoice legacy id `CIV-00123` (AX `CustInvoiceJour.InvoiceId`), when a user opens the Passage invoice, then the legacy reference panel shows system, company and id.
- **AC-49.9** (FR-49.4) Given 7 years of legacy journals loaded, when the explorer queries `revenue` with `source=legacy` for a closed year, then it equals the legacy trial balance for that year.
- **AC-49.10** (FR-49.8) Given a pipeline trigger, when a rehearsal tenant is requested, then it is ready in ≤ 2 h with masked data.

9.49.9 Test plan and fixtures

Fixtures: `fx-ax-synth-year` (AX 2012 R3 schema: `CustTrans`, `VendTrans`, `InventSum`, `GeneralJournalAccountEntry`, `PurchLine`, `SalesLine`, `ProdTable`, `AssetBook`, 3 `DataAreaId`, 1.1M rows); `fx-sap-synth-year` (ECC 6.0 and S/4 2023 variants: `BSID/BSIK/ACDOCA`, `MARD/MBEW`, `EKPO`, `VBAP`, `AFKO`, `ANLC`, 3 `BUKRS`, 1.3M rows); `fx-replay-month` (one month, 42k documents); `fx-cutover-plan` (plan with 84 steps).

Test	Type	Target	Pass criterion
Pack reconciliation	E2E	9 packs × 2 systems	100% MATCH
Idempotent re-run	property	loader	re-run of any subset loads 0 new rows
GE suites	unit	validation	all expectations defined per pack with ≥ 95% coverage of mapped fields
Parallel-run diff	E2E	harness	0 unexplained
Diff classifier	unit	classifier	100 seeded variances classified per rule
Cutover timing	performance	runbook	< 48 h; each step within 125% of plan
Rollback	E2E	rollback	coexistence resumes; exported files complete
History load throughput	performance	Iceberg	≥ 200k rows/min

Reconciliation: trial balance (Passage opening) ↔ legacy trial balance per entity/currency; open items ↔ control accounts; on-hand value ↔ inventory account; NBV ↔ asset accounts; parallel-run TB ↔ legacy period TB.

9.49.10 Deployment and operations

Helm: `passage-migration-factory` (Go extractors, TS orchestration on Temporal namespace `migration`), `passage-legacy-adapters` (AX OData/SQL, SAP RFC/ODP via wave-35 hub), `passage-rehearsal-provisioner` (pipeline job). Config: legacy connection strings in Vault, `HISTORY_YEARS=7`, `MASK_PII=true` for rehearsal. Flags: `migration.coexistence` per entity. Jobs: nightly delta extract during coexistence; weekly reconciliation post go-live for 30 days. SLOs: pack load ≥ 50k records/min; control totals computed within 10 min of load. Alerts: MISMATCH, quarantine > 1%, step overrun > 125%. Runbook entries: resume a failed pack from VALIDATED; rebuild xref; execute rollback. Rollback: per BR-49.1 and FR-49.9.

9.49.11 Migration from AX / SAP

This wave is the migration. Summary mapping:

Source	Passage entity	Transformation
`CustTransOpen/VendTransOpen`, `BSID/BSIK`	AR/AP open items	amount in transaction and accounting currency; due date; settlement history collapsed to remaining amount
`GeneralJournalAccountEntry` + `DimensionAttributeValueCombination`, `ACDOCA/FAGLFLEXT`	opening journal	balances per account × dimension × currency at cutoff; P&L closed to retained earnings if cutoff is year-end
`InventSum/InventDim/MBEW/MARD/MCHB`	inventory opening transactions	qty and cost per dimension; batch/serial attributes
`PurchLine/SalesLine`, `EKPO/VBAP`	open order lines	remaining = ordered − delivered; price and tax re-derived and compared

Source	Passage entity	Transformation
`ProdTable/AFKO`	production orders	status mapping (Started → RELEASED+WIP); WIP value
`AssetBook/ANLC`	asset books	NBV, accumulated depreciation, remaining life
`ProjTrans*/COSP`	project WIP	WIP per project, by category

Not migrated: legacy number sequences (new sequences, old ids kept in xref); AX `SysDatabaseLog` rows and SAP `CDHDR` (loaded to history only); settled items and closed orders (history only); X++/ABAP customizations.

9.49.12 Dependencies, risks, open questions

Depends on: wave 10 (framework, connectors), wave 35 (integration hub for coexistence), wave 46 (lakehouse for history), wave 48 (PII masking rules), all transactional waves for posting APIs.

Risk	Likelihood	Impact	Mitigation
Legacy dimension model does not map 1:1	High	High	mapping DSL with default dimension and exception report; sign-off before load
Extract window exceeds freeze	Medium	High	delta extracts; parallelized extractors; rehearsal timing
Coexistence creates duplicate intercompany documents	Medium	Medium	xref dedupe (BR-49.10)

Open questions: (1) History years default 7 vs 10 for DE GoBD — proposed 10 when DE pack installed; decide by wave 50. (2) Payroll YTD for countries without a Passage payroll engine — proposed: load as balances only; decide by wave 50.

9.50 Wave 50: Scale Hardening, Multi-Region & Production Go-Live

9.50.1 Scope and business context

AX customers rely on LCS-managed production environments and Microsoft SLAs; SAP customers on Basis operations, HA/DR setups (HANA system replication, cluster failover) or RISE. Passage must prove the same in a self-operated, cloud-agnostic stack: a performance campaign to 10k concurrent users, 1M journal lines/h and 100k order lines/h; active/passive multi-region with cross-region Postgres replication, Kafka MirrorMaker 2 and DNS failover; SRE runbooks, capacity and cost models, zero-downtime upgrades, tenant provisioning automation; complete documentation; the first customer's production go-live and the public v1.0 release. Out of scope: active/active multi-region, per-tenant dedicated clusters, marketplace listings.

Pain points solved (10x lens)

Pain point in AX/SAP today	Root cause	Passage answer in this wave	Measurable 10x target
D365 One Version updates and SAP SUM upgrades require a downtime window (hours) and a sandbox cycle; customers defer and accumulate risk	Monolithic application server; schema changes applied in place	Rolling deploys with expand/contract migrations and N/N–1 API contract tests	Upgrade downtime 4–8 h → 0; release cadence quarterly → weekly
DR is a service request to LCS or a Basis project with HANA system replication; failover is rarely rehearsed and RPO/RTO are unknown	DR owned by vendor or built per customer	One-command failover, quarterly game day with measured RTO/RPO	RTO ≤ 1 h, RPO ≤ 5 min, measured every quarter
Performance regressions appear in production (`SysOperation` batch queues, `ST05` traces after the fact)	No standing load test tied to releases	k6 scenarios S1–S7 and SLO burn-rate gates in the release pipeline	Regressions caught pre-release 100%; p99 SLOs published
Environment sizing is a vendor estimator (LCS subscription estimator, Quick Sizer) with opaque cost	No per-tenant metering in the product	Capacity model and `tenant_cost` meter per month	Cost estimate error ±50% → ±5%
Provisioning a new company environment takes days of service requests	Manual environment lifecycle	One-call tenant provisioning with residency and license	Provisioning 3 days → ≤ 15 min

Cutover go/no-go checklist (phase exit, executed at T+24h of the production cutover): (1) all wave-49 packs RECONCILED, 0 MISMATCH; (2) parallel run for the prior month SIGNED_OFF with 0 unexplained; (3) opening trial balance signed by the customer controller; (4) SoD evaluation shows 0 unmitigated HIGH conflicts for go-live users; (5) country packs installed and one statutory fixture executed per entity; (6) load test on the production cluster at 50% of target passed within SLOs; (7) DR game day passed within 30 days before cutover; (8) backup verified: PITR drill on production data within 7 days; (9) security: pen-test high/critical = 0, secrets rotated, images signed; (10) integration endpoints (bank, EDI, e-invoicing, carriers) smoke-tested in production; (11) user acceptance sign-off for 12 core processes; (12) support rota, escalation and rollback decision owner named; (13) monitoring dashboards and alerts green for 24 h; (14) legacy freeze confirmed and rollback plan rehearsed. Any open item = NO-GO; the decision is recorded via `cutover.decision.recorded`.

Hypercare (T+1d to T+30d): 24×5 on-call with 15-minute response for Sev-1, 1 h for Sev-2; daily 09:00 customer stand-up; daily reconciliation of all subledgers to the journal and of lakehouse metrics; weekly reconciliation against legacy coexistence feeds; agents in suggest mode only; change freeze except Sev-1 fixes; exit criteria: first month-end closed and reconciled, Sev-1 count = 0 for 14 consecutive days, Sev-2 backlog ≤ 5, SLOs met for 30 days, customer sign-off.

v1.0 release criteria: all 50 waves' proofs green in CI on the release commit; phase-exit criteria of 9.0 met; CVE scan 0 critical, 0 high older than 30 days; API reference generated from OpenAPI/AsyncAPI with 0 undocumented operations; administrator, implementer and per-module functional guides published; upgrade path from the last stage build tested; license and pricing published; tag `v1.0.0` signed with cosign, SBOM (SPDX) attached, Helm charts and images published by digest.

9.50.2 AX / SAP parity map

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
Production environment management	LCS production, sandbox tiers (Tier-2–5), service requests	Basis: `SM51` servers, `RZ10` profiles, RISE managed	Tenant provisioning automation, environment tiers via Helm values	Full

Capability	AX 2012 / D365 F&SCM	SAP ECC / S/4	Passage equivalent	Parity
HA/DR	Microsoft-managed geo-secondary, LCS disaster recovery	HANA system replication, Pacemaker clusters, DR site	CNPG replica clusters cross-region, MirrorMaker 2, DNS failover	Full
Capacity and sizing	LCS subscription estimator, usage profile	Quick Sizer, SAPS	Capacity model per tenant (users, lines/h → nodes, storage)	Full
Performance tooling	Performance SDK, Trace parser, `SysOperation` batch tuning	`ST05`, `ST12`, `SAT`, `SQLM`, HANA PlanViz	k6 scenarios, OpenTelemetry traces, pg_stat_statements dashboards	Full
Monitoring	LCS Environment monitoring, Application Insights	`RZ20` CCMS, Solution Manager, Focused Run	Prometheus/Grafana SLO dashboards, Loki, Tempo	Full
Upgrades	One Version monthly updates, LCS deployable packages, downtime	SUM, maintenance planner, zero-downtime option (ZDO)	Expand/contract Atlas migrations, rolling deploys, no downtime	Improved
Batch and job scheduling	Batch framework (`SysOperation`), batch groups	`SM36`/`SM37`, Redwood	Temporal schedules (from wave 4) with multi-region continuity	Full
Licensing	Dynamics 365 per-user licenses	SAP named user licenses, FUE	Passage license file per tenant, user-type metering	Full
Documentation	Microsoft Learn	SAP Help Portal	Docs site: admin, implementer, API, functional guides, training	Full

9.50.3 Functional requirements

- **FR-50.1 Load-test scenarios** — Must. The system shall meet the SLOs below under the k6 scenario set run for 2 h sustained with 10k virtual users on the production-sized cluster:
  - S1 journal posting: 1M lines/h via `/api/v1/ledger/journals` (batches of 500); S2 order intake: 100k order lines/h via sales order API plus pricing and ATP; S3 interactive mix: 10k users with 60% list/detail reads, 25% form saves, 10% dashboard loads, 5% explorer queries (think time 10 s); S4 warehouse mobile: 2k scanners at 1 scan/5 s; S5 MRP: 50k-item regeneration concurrent with S3; S6 month-end: inventory close, FX revaluation, allocations concurrent with S1 at 25%; S7 CDC lag: lakehouse freshness measured during S1+S2.

Operation	SLO (p99 unless noted)	Error budget
Entity read (generated)	p95 < 50 ms, p99 < 150 ms	0.1%
Form save	p99 < 400 ms	0.1%
Order intake (line)	p99 < 500 ms	0.1%
Pricing	p95 < 20 ms	0.1%
Journal post (500-line batch)	p99 < 2 s; sustained 5k lines/s	0.05%
Mobile scan round trip	p99 < 300 ms	0.1%
Dashboard load	p95 < 2 s	0.5%
Lakehouse lag	p95 < 60 s	n/a
MRP 50k items	< 10 min	n/a
Availability per tenant	99.9% (99.95% with multi-region)	monthly

- **FR-50.2 Postgres tuning** — Must. The system shall partition the journal line table by month (range) and hash by tenant (16), inventory transactions by month, audit by week; CNPG clusters with 1 primary + 2 synchronous-quorum replicas (`synchronous_standby_names ANY 1`); PgBouncer transaction pooling; `pg_stat_statements` top-50 reviewed with indexes added before exit.
- **FR-50.3 Kafka and cache tuning** — Must. The system shall run Strimzi with 6 brokers, `min.insync.replicas=2`, `acks=all`, topic partitions sized at ≥ 2× consumer parallelism, and Redis-compatible caches for pricing and policy bundles with ≤ 5 s invalidation.
- **FR-50.4 Multi-region active/passive** — Must. The system shall run a passive region with CNPG replica clusters streaming from the primary (async, lag SLO < 30 s; ledger cluster quorum-synchronous across regions when the customer accepts the latency), MirrorMaker 2 for all topics with offset translation, object-store replication for lakehouse and documents, Temporal multi-cluster replication, and DNS failover (health-checked, TTL 60 s).
- **FR-50.5 Failover procedure** — Must. The system shall promote the passive region by one command (`passage-ops failover --to eu-west`), which fences the old primary (revoke writes, scale to zero), promotes CNPG replicas, switches MirrorMaker direction, activates Temporal, flips DNS, and verifies the ledger hash chain; RTO ≤ 1 h, RPO ≤ 5 min.
- **FR-50.6 DR game day** — Must. The system shall run a quarterly game day: (1) announce window, freeze deploys; (2) take baseline hashes (ledger chain head per tenant, row counts per domain, Kafka end offsets); (3) inject failure (primary region network partition via NetworkPolicy, then node pool scale-down); (4) on-call executes failover runbook with timer; (5) verify: hash chain head matches or exceeds baseline, no gap in gap-free sequences, in-flight Temporal workflows resumed, reconciliation jobs green; (6) run smoke suite of 12 processes; (7) measure RTO/RPO; (8) fail back during a planned window; (9) post-game review with actions. Pass: RTO ≤ 1 h, RPO ≤ 5 min, zero committed-transaction loss.
- **FR-50.7 Zero-downtime upgrades** — Must. The system shall deploy services with rolling updates and expand/contract schema migrations (add → backfill → switch → drop in separate releases), with API compatibility tests between N and N−1 and a 30-minute automatic rollback window on SLO breach.
- **FR-50.8 Tenant provisioning** — Must. The system shall create a tenant (Keycloak realm, databases/schemas with RLS, Kafka ACLs, lakehouse namespace, object-store prefixes, residency region, license) from one API call in ≤ 15 min, and decommission with retention export.
- **FR-50.9 Capacity and cost model** — Must. The system shall publish a model mapping (concurrent users, journal lines/h, order lines/h, storage GB) to node counts, Postgres sizes, Kafka brokers and monthly cost per provider, and meter per-tenant resource usage monthly into `tenant_cost`.
- **FR-50.10 SRE runbooks and on-call** — Must. The system shall ship runbooks for the 30 alert rules shipped by the platform, incident severity definitions, escalation, and a status page; every alert links to its runbook.
- **FR-50.11 Documentation** — Must. The system shall publish the administrator guide, implementer guide (migration, localization, cutover), API reference (OpenAPI/AsyncAPI generated), functional guides per module (one per phase 1–8 module), and a training curriculum (role-based, 6 tracks).
- **FR-50.12 Licensing and release** — Must. The system shall enforce a signed license file per tenant (user types: full, operational, self-service; expiry; grace 30 days), and the release pipeline shall tag, sign, attach SBOMs and publish charts and images by digest.

9.50.4 Data model

`region` (platform-ops)

Field	Type	Required	Notes
region_id	text	Y	PK, e.g. `us-east`, `eu-west`
role	enum	Y	PRIMARY, PASSIVE, FENCED
provider	text	Y	eks/gke/aks/bare
cnpg_lag_seconds	int	Y	observed
mm2_lag_messages	bigint	Y	observed
last_failover_event_id	uuid	N	
dns_weight	int	Y	0 or 100

`failover_event`: event_id, from_region, to_region, initiated_by, started_at, dns_flipped_at, promoted_at, completed_at, rpo_seconds, rto_seconds, ledger_hash_ok bool, gapfree_ok bool, notes.

`tenant_license`: tenant_id, license_id, issued_at, expires_at, full_users, operational_users, selfservice_users, signature, state (VALID, GRACE, EXPIRED).

`tenant_cost`: tenant_id, month, cpu_core_hours, memory_gb_hours, storage_gb, kafka_gb, lakehouse_gb, ai_tokens, cost_usd, cost_eur, model_version.

`capacity_model`: model_version, inputs_json (users, lines/h, storage), outputs_json (nodes, pg size, brokers), provider, cost_usd_month.

Invariants: exactly one region is PRIMARY; a FENCED region accepts no writes (enforced by revoking the application role); license `full_users` ≥ active full users else state GRACE; `tenant_cost` is computed from Prometheus data with a stored query version. Indexing: `failover_event` (started_at); `tenant_cost` PK (tenant_id, month).

9.50.5 APIs and events

Method	Path	Purpose	Idempotency	Auth duty
POST	`/api/v1/ops/tenants`	provision tenant	`Idempotency-Key`	PlatformAdmin
DELETE	`/api/v1/ops/tenants/{id}`	decommission with export	by id	PlatformAdmin + second approver
POST	`/api/v1/ops/failover`	initiate failover	by event id	PlatformAdmin (break-glass)
GET	`/api/v1/ops/regions`	region status and lag	n/a	PlatformView
POST	`/api/v1/ops/gameday`	start game day workflow	by id	PlatformAdmin
GET	`/api/v1/ops/capacity/estimate`	capacity estimate for inputs	n/a	PlatformView
GET	`/api/v1/ops/tenants/{id}/cost?month=`	metered cost	n/a	TenantAdmin
PUT	`/api/v1/ops/tenants/{id}/license`	install license	ETag	PlatformAdmin
GET	`/api/v1/ops/slo`	SLO status and error budgets	n/a	PlatformView
GET	`/healthz/region`	region health for DNS checks	n/a	public

Validation: failover rejects if target region replication lag > 300 s unless `--accept-data-loss` with second approver; provisioning rejects a residency region not deployed.

Event	Trigger	Payload fields	Consumers
`dev.passage.ops.failover.completed.v1`	promotion done	from_region, to_region, promoted_lsn, rpo_seconds, rto_seconds	all services (reconnect), audit
`dev.passage.ops.tenant.provisioned.v1`	provisioning done	tenant_id, region, realm, databases[]	billing, catalog
`dev.passage.ops.slo.breached.v1`	SLO burn rate	slo_id, window, burn_rate, tenant_id	alerting, release gate
`dev.passage.ops.license.state_changed.v1`	license check	tenant_id, state, expires_at	shell banner, admin

9.50.6 User experience

Screen	Primary actions	Key fields/columns	Duty
Platform operations console	view regions, initiate failover (two-person), view game-day timeline	region, role, lag, last failover	PlatformAdmin
Tenant management	provision, license, residency, decommission	tenant, region, license state, users	PlatformAdmin
SLO dashboard (Grafana embedded)	filter by tenant/service, view burn rate	SLO, target, current, budget remaining	PlatformView
Cost and capacity	estimate, view monthly meter, export	inputs, outputs, cost USD/EUR	TenantAdmin
Status page (public)	subscribe	component, status, incidents	public
Release notes / docs site	search, version switch	version, module	public

Rules: failover button requires two distinct PlatformAdmin confirmations within 5 min; SLO dashboard default window 30 days; license banner appears in the shell at GRACE.

9.50.7 Business rules and edge cases

- **BR-50.1** Region state machine: PASSIVE → PRIMARY (promotion) with the former PRIMARY → FENCED; FENCED → PASSIVE only after re-sync verified; never two PRIMARY.

- **BR-50.2** During failover, any request to the fenced region returns 503 with `Retry-After: 30`; clients retry idempotent requests with the same key, so no double posting.
- **BR-50.3** Ledger hash chain: each journal carries `prev_hash`; after promotion the chain head per tenant must be verifiable; a break is a Sev-1 and blocks user access.
- **BR-50.4** Gap-free sequences (fiscal invoice numbers) are allocated in the ledger Postgres (synchronous replication), never in caches, so failover cannot duplicate numbers.
- **BR-50.5** Load tests run against a copy of the first customer's masked data, never production.
- **BR-50.6** An SLO burn rate > 14.4× over 1 h (fast burn) pages; > 6× over 6 h (slow burn) tickets; a release with any fast-burn alert in the 30-minute window auto-rolls back.
- **BR-50.7** Expand/contract: a migration that drops a column cannot ship in the same release as the code that stops using it.
- **BR-50.8** License GRACE allows logins for 30 days with a banner; EXPIRED blocks new logins but never data export or admin access.
- **BR-50.9** Tenant decommission retains an encrypted export for the retention period defined by the tenant's country packs before deletion.
- **BR-50.10** Cost model rounds to USD 0.01 / EUR 0.01; FX for EUR at the month-average ECB rate.

9.50.8 Acceptance criteria

- **AC-50.1** (FR-50.1) Given scenarios S1–S7 run concurrently for 2 h, when results are published, then every SLO row is met and error rates are within budget.
- **AC-50.2** (FR-50.2) Given 1M journal lines/h, when posting runs, then ledger Postgres replica lag p95 < 5 s and no query exceeds 1 s in pg_stat_statements.
- **AC-50.3** (FR-50.5, 50.6) Given the game day partition, when on-call runs the failover command, then RTO ≤ 1 h, RPO ≤ 5 min, ledger hash chain verified, gap-free sequences continuous.
- **AC-50.4** (FR-50.6) Given failover completed, when the 12-process smoke suite runs in the new primary, then all pass and CDC lag recovers to < 60 s within 15 min.
- **AC-50.5** (FR-50.7) Given release N+1 with an expand migration, when rolled out, then zero failed requests and N clients continue to work.
- **AC-50.6** (FR-50.8) Given a provisioning request with residency EU, when it runs, then the tenant exists in `eu-west` with RLS, realm, ACLs and license in ≤ 15 min.
- **AC-50.7** (FR-50.9) Given one month of production, when the cost meter runs, then `tenant_cost` is within 5% of provider billing for the cluster.
- **AC-50.8** (FR-50.12) Given a tenant whose license expired 31 days ago, when a user logs in, then login is blocked and admin export works.
- **AC-50.9** (9.50.1 checklist) Given the production cutover, when T+24h arrives, then all 14 go/no-go items are complete and the decision is recorded.
- **AC-50.10** (9.50.1 hypercare) Given the first production month, when month-end closes, then all subledgers reconcile to the journal, lakehouse revenue equals GL, and the customer signs off.
- **AC-50.11** (FR-50.11, 50.12) Given the release commit, when the pipeline runs, then `v1.0.0` is tagged and signed, SBOMs attached, docs published with 0 broken links and 0 undocumented API operations.

9.50.9 Test plan and fixtures

Fixtures: `fx-load-10k` (masked copy of the first customer plus synthetic growth to 10k users, 3 years of transactions, 40M journal lines); `k6-scenarios` (S1–S7 scripts with data pools); `fx-gameday` (baseline hash manifest generator); `fx-smoke-12` (12 core processes: PO→receipt→invoice→payment, SO→ship→invoice→cash, production order→RAF, journal post, bank reconciliation, asset depreciation, project invoice, payroll run, statutory report, dashboard, copilot query, mobile pick).

Test	Type	Target	Pass criterion
Load campaign	performance	whole platform	SLO table met, 2 h
Soak	performance	whole platform	24 h at 50% load, no memory growth > 10%
Failover	E2E	multi-region	RTO/RPO met, hash chain ok
Fail-back	E2E	multi-region	re-sync and promotion without loss
Upgrade compatibility	contract	APIs N vs N–1	100% of contract tests pass
Expand/contract linter	unit	migrations	no drop in same release as last use
Provisioning	E2E	tenant API	≤ 15 min, all components
Chaos	E2E	Kafka broker loss, Postgres primary kill	no failed postings, recovery < 5 min
Docs	contract	docs site	0 broken links, OpenAPI coverage 100%
Smoke 12	E2E	production	all pass post-cutover and post-failover

Reconciliation: post-failover row counts and ledger hash heads ↔ baseline; production daily subledger ↔ journal; lakehouse metrics ↔ GL; `tenant_cost` ↔ provider billing; license active users ↔ Keycloak sessions.

9.50.10 Deployment and operations

Helm: `passage-platform` umbrella with per-region values; `passage-mm2` (MirrorMaker 2), `passage-ops` (Go control plane), `passage-status-page`, `passage-docs`. CNPG `Cluster` with `replica.enabled` in the passive region; Temporal multi-cluster. Config: `REGION_ROLE`, `DNS_PROVIDER` (thin Terraform module per provider), `FAILOVER_MAX_LAG_S=300`, `LICENSE_PUBLIC_KEY`. Migrations: `ops` database. Flags: `ops.failover.enabled`, `license.enforce`. Jobs: cost meter monthly, capacity model nightly, game day quarterly (scheduled Temporal workflow with human task), backup drill monthly, SLO report weekly. SLOs: as FR-50.1; control-plane API p99 < 500 ms. Alerts: replication lag > 60 s, MM2 lag > 100k messages, SLO fast/slow burn, license expiry 30/7/1 days, failover in progress. Runbooks: failover, fail-back, broker replacement, Postgres primary loss, slot bloat, cache stampede, Temporal namespace replication lag, certificate expiry, object-store replication lag, release rollback. Rollback: Argo CD rollback by digest; database contract step deferred to the next release; DNS weight restore.

9.50.11 Migration from AX / SAP

Source	Passage	Notes
LCS environment inventory (production, sandboxes)	Tenant and environment tiers	documented; sandboxes re-created from masked production via provisioning
LCS deployable packages / Basis transports (`STMS`)	GitOps releases	not migrated; release notes map customer changes to metadata extensions
Batch groups and job schedules (`BatchJob`, `SM37`)	Temporal schedules	imported in wave 4 and verified for production cadence here
Monitoring thresholds (CCMS `RZ20`, LCS alerts)	Prometheus alert rules	mapped per alert category
D365 / SAP user licenses	`tenant_license` user types	full ↔ Dynamics full user / SAP professional; operational ↔ activity / functional; self-service ↔ team member / productivity

Not migrated: Microsoft/SAP SLAs and support contracts (replaced by Passage support terms), proprietary performance traces.

9.50.12 Dependencies, risks, open questions

Depends on: wave 1 (pipeline, Argo CD), wave 3 (Kafka, outbox), wave 4 (Temporal), wave 11 (ledger hash chain, gap-free sequences from wave 5), wave 46 (lakehouse lag measurement), wave 48 (backup drills, residency, pen test), wave 49 (customer migration and cutover runbook).

Risk	Likelihood	Impact	Mitigation
Cross-region synchronous ledger replication adds latency beyond SLO	Medium	High	quorum within region plus async cross-region by default; customer opt-in to cross-region sync
First customer month-end reveals unmapped edge cases	High	High	hypercare change process; parallel-run month before cutover; agents in suggest mode
Load test environment differs from production	Medium	Medium	same Helm values and node classes; 50% load re-run on production before cutover
DNS TTL caching delays failover for some clients	Medium	Medium	TTL 60 s; client retry with alternate endpoint list

Open questions: (1) Active/active for read-only analytics in the passive region — proposed yes for Trino only; decide before v1.1. (2) Public pricing unit (per full user per month vs per tenant tier) — proposed per user type; decide at v1.0 release. (3) Licensing model (open-core vs source-available) carried from PRD §12 — must be decided before the v1.0 tag.

Part 10 — Appendices

These appendices are normative for every phase file (`docs/prd/phase-N-*.md`). Where a phase file and an appendix disagree, the appendix wins for conventions (naming, envelopes, error codes, security, testing) and the phase file wins for business behavior. The stack is fixed by ADR-0001; nothing here proposes infrastructure outside it.

A. Glossary

Terms are drawn from AX 2012 / D365 F&SCM, SAP ECC / S/4HANA, the standards Passage implements, and Passage itself. The last column is the name used in Passage metadata, APIs and UI.

Term	Origin	Definition	Passage term
AOT	AX	Application Object Tree: the AX metadata repository for tables, classes, forms, menus.	Metadata registry
DDIC	SAP	ABAP Data Dictionary: domains, data elements, tables, append structures.	Metadata registry
EDT	AX	Extended Data Type: a typed, reusable field definition (e.g. `CustAccount`).	Field type
Base enum	AX	Fixed enumeration stored as an integer; extensible via enum extensions.	Enum (metadata)
Extension (vs overlayering)	AX	D365 model that adds to standard objects instead of editing them.	Entity extension / hook
DataAreaId	AX	The company (legal entity) column present on every company-specific table.	`legal_entity_id`
MANDT	SAP	Client key present on every client-dependent table.	`tenant_id`
Tenant	Passage	Top-level isolation boundary: one customer organization, one RLS scope, one Keycloak realm.	Tenant
Legal entity	AX / Passage	A company that produces its own financial statements. SAP: company code (BUKRS).	LegalEntity
Operating unit	AX	Department, cost center, business unit, value stream: non-legal organization units.	OperatingUnit
Company code (BUKRS)	SAP	Smallest organizational unit with a complete self-contained set of accounts.	LegalEntity
Plant (WERKS)	SAP	Physical production/logistics location. AX: site.	Site
Storage location (LGORT)	SAP	Sub-location of a plant where stock is held. AX: warehouse.	Warehouse
Role / duty / privilege	AX	Three-layer security model: privileges grant entry points, duties group privileges, roles group duties.	Role / Duty / Privilege
Authorization object	SAP	Named set of fields (e.g. `F_BKPF_BUK`) checked at runtime.	Privilege
SoD	AX / SAP GRC	Segregation of duties: rules preventing one user from holding conflicting duties.	SoD rule
Number sequence	AX	Generator of document identifiers per scope; continuous (gap-free) or not. SAP: number range object (SNRO).	NumberSequence
Database log	AX	Table/field change log. SAP: change documents (CDHDR/CDPOS).	Audit log (CDC)
Workflow	AX / SAP	Approval and process orchestration (AX workflow framework, SAP Business Workflow).	Temporal workflow
DIXF / DMF	AX	Data Import Export Framework / Data Management Framework. SAP: LSMW, Migration Cockpit.	Migration pipeline
LCS	AX	Lifecycle Services: Microsoft's hosted deployment/upgrade portal. SAP: STMS/CTS+.	Argo CD GitOps pipeline
DirParty / Global Address Book	AX	Shared party model with customer/vendor/contact roles.	Party
Business Partner (BP)	SAP	S/4 single party record with roles; CVI links to KNA1/LFA1.	Party
CustTable / KNA1	AX / SAP	Customer master table.	Customer (party role)
VendTable / LFA1	AX / SAP	Vendor master table.	Vendor (party role)
Posting profile	AX	Mapping from a customer/vendor/item group to GL control accounts. SAP: reconciliation account.	PostingProfile
Product master / variant	AX	A product with dimensions (size, color, style, configuration) that explode into distinct products. SAP: configurable material.	ProductMaster / Product
Released product	AX	A product made available to one legal entity with entity-specific defaults. SAP: MARC plant view.	ReleasedProduct
MARA / MARC / MARD	SAP	Material master: general, plant and storage-location views.	Product / ReleasedProduct / on-hand
Dimension group	AX	Product, storage, and tracking dimension groups controlling which dimensions apply to an item.	DimensionGroup
Item model group	AX	Costing method, negative inventory policy, physical/financial posting flags. SAP: valuation class + MRP/accounting views.	ItemModelGroup
GTIN / SSCC	GS1	Global Trade Item Number; Serial Shipping Container Code (18-digit pallet/case id).	GTIN / SSCC
BOM	AX / SAP	Bill of materials: component list by version with effective dates. SAP: STKO/STPO.	BOM
Formula	AX	Process-industry BOM with potency, co-/by-products, scalable quantities. SAP: master recipe.	Formula
Route / routing	AX / SAP	Operation sequence with resources and times. SAP: PLKO/PLPO.	Routing
ECM / ECO	AX / SAP	Engineering change management; engineering change order.	ChangeOrder
Universal journal	SAP	ACDOCA: single line-item table for FI, CO, AA and ML. AX: GeneralJournalAccountEntry.	JournalEntry / JournalLine
Financial dimension	AX	Analytical attribute on a ledger line (department, cost center, project). SAP: CO object / profit center / segment.	Dimension
Account structure	AX	Rules for which dimension combinations are valid per main account.	AccountStructure
Posting layer	AX	Parallel accounting layer (Current, Operations, Tax). SAP: ledger group / non-leading ledger.	Ledger (parallel)
Currency types	SAP	10 company code, 30 group, 40 hard currency amounts per line. AX: accounting/reporting currency.	Transaction / accounting / reporting currency

Term	Origin	Definition	Passage term
OB52	SAP	Posting period control per account type and ledger. AX: ledger calendar period status per module.	PeriodControl
Allocation rule	AX	Basis/fixed distribution of balances across dimensions. SAP: assessment/distribution (KSU5/KSV5).	AllocationRule
FSV	SAP	Financial statement version: hierarchical statement definition (OB58). AX: Management Reporter row/column definitions.	StatementDefinition
InventTrans	AX	Immutable inventory transaction with issue/receipt status. SAP: MSEG material document line.	InventoryTransaction
InventSum	AX	On-hand aggregate per dimension combination. SAP: MARD/MCHB stock tables.	OnHand (projection)
Physical / financial status	AX	Transaction lifecycle: ordered → reserved → picked → deducted / received → registered → posted (financially).	InventoryTransaction status
Reservation	AX / SAP	Allocation of on-hand or expected receipts to a demand.	Reservation
Inventory close	AX	Periodic settlement of issues against receipts to finalize cost. SAP: material ledger closing (CKMLCP).	Cost close
Standard cost / costing version	AX	Frozen unit cost by version and site, activated per date. SAP: standard price, costing run (CK40N).	CostingVersion
Costing sheet	AX / SAP	Indirect cost (overhead) rule tree applied during cost roll-up.	CostingSheet
Product receipt	AX	Posting that records goods received against a PO. SAP: goods receipt (MIGO, movement type 101).	ProductReceipt
GR/IR	SAP	Goods-receipt / invoice-receipt clearing account; accrues liability between receipt and invoice. AX: accrued purchases.	GR/IR accrual account
Three-way match	AX / SAP	Invoice compared to PO and receipt within price/quantity tolerances (MIRO, tolerance keys).	Matching
ERS	SAP	Evaluated receipt settlement: self-billing from receipts without a vendor invoice.	ERS
Open item	SAP	Unsettled AP/AR line (FBL1N/FBL5N). AX: customer/vendor transaction.	OpenItem
Settlement	AX	Application of payment or credit against open items. SAP: clearing.	Settlement
F110	SAP	Automatic payment program. AX: payment proposal.	PaymentProposal
pain.001 / camt.053	ISO 20022	Credit-transfer initiation message / end-of-day bank statement message.	Payment file / BankStatement
NACHA	Standard	US ACH file format.	Payment file (US)
Production order	AX / SAP	Order to make a quantity of an item via BOM and route (CO01).	ProductionOrder
RAF	AX	Report as finished: posts finished goods receipt from a production order. SAP: order confirmation (CO11N) + GR 101.	ReportAsFinished
Coverage group	AX	Planning parameters per item: requirement/period/min-max coverage, time fences, safety stock. SAP: MRP type + lot-sizing procedure.	CoverageGroup
MRP / MPS	Standard	Material requirements planning; master production schedule.	Planning run
Planned order	AX / SAP	MRP proposal for purchase, production or transfer, to be firmed.	PlannedOrder
Action / delay message	AX	Planner advice: advance, postpone, cancel; or dates that cannot be met. SAP: exception messages.	PlanningMessage
Finite / infinite capacity	AX / SAP	Scheduling that respects or ignores resource load (CM01, PP/DS).	Scheduling mode
Catch weight	AX	Dual-UoM inventory (nominal units plus actual weight).	CatchWeight
Potency	AX	Active-ingredient concentration used to scale formula consumption.	Potency
OEE	ISO 22400	Overall equipment effectiveness = availability × performance × quality.	OEE
Trade agreement	AX	Price/discount list by account/group/all × item/group/all with effective dates. SAP: condition record.	PriceCondition
Condition technique	SAP	Pricing procedure of condition types resolved via access sequences (VK11).	Pricing procedure / access sequence
ATP / CTP	AX / SAP	Available-to-promise from on-hand and supply; capable-to-promise considers capacity.	ATP
Packing slip	AX	Delivery document posting inventory deduction and COGS. SAP: outbound delivery + post goods issue.	PackingSlip (Shipment posting)
Handling unit / license plate	SAP / AX	Physical container (pallet, case) with its own id holding nested items.	LicensePlate
Location directive	AX	WMS rule choosing put-away/pick locations. SAP: storage type search strategy.	LocationDirective
Quality order	AX	Inspection with tests and results leading to a disposition. SAP: inspection lot + usage decision.	QualityOrder
NCR / CAPA	Quality	Nonconformance report; corrective and preventive action. SAP: quality notification.	Nonconformance / CAPA
AQL	ANSI/ASQ Z1.4	Acceptable quality level sampling plan.	SamplingPlan
Landed cost	Standard	Freight, duty and handling allocated to inventory cost.	LandedCost
EDI 850 / 856 / 810	X12	Purchase order / advance ship notice / invoice transaction sets. EDIFACT: ORDERS / DESADV / INVOIC.	EDI message
Peppol / UBL	Standard	Pan-European e-invoicing network; Universal Business Language XML.	E-invoice
Dunning	SAP	Reminder letters by escalation level (F150). AX: collection letters.	CollectionLetter
RMA	Standard	Return merchandise authorization. AX: return order; SAP: RE order.	ReturnOrder
SSP	ASC 606	Standalone selling price used to allocate transaction price across performance obligations.	SSP
Performance obligation	ASC 606 / IFRS 15	Distinct promise in a contract to which revenue is allocated.	PerformanceObligation
ROU asset	IFRS 16 / ASC 842	Right-of-use asset recognized for a lease.	Lease
Budget control / encumbrance	AX / SAP	Reservation of budget by requisitions and POs (commitments).	Encumbrance
Elimination	Consolidation	Removal of intercompany balances and unrealized profit in group statements.	EliminationRule
POC	Accounting	Percentage-of-completion revenue estimate. SAP: results analysis.	ProjectEstimate
Functional location	SAP PM	Hierarchical place where equipment is installed. D365: functional location.	FunctionalLocation

Term	Origin	Definition	Passage term
Installed base	SAP CS	Serialized equipment at customer sites covered by service.	InstalledBase
Entity store	AX	D365 aggregate measurement store for Power BI. SAP: BW/CDS views.	Lakehouse (Iceberg)
MCP	Standard	Model Context Protocol: tool interface for AI agents.	MCP server
Copilot / agent	Passage	Copilot answers and drafts; an agent executes multi-step work under approval gates.	Copilot / Agent
Outbox	Pattern	Table written in the same transaction as the state change and relayed to Kafka by Debezium.	Outbox
CloudEvent	CNCF	Standard event envelope (specversion, id, source, type, subject, time, data).	Event
RLS	Postgres	Row-level security policy evaluated on every statement.	Tenant isolation
Wave (delivery)	Passage	A 4–8 week increment with deliverable, proof, deploy and references. Not to be confused with a WMS pick wave.	Wave

B. Cross-module data model

B.1 Bounded contexts

Passage has 12 core bounded contexts. Each is a deployable set of services with exactly one Postgres database (CloudNativePG cluster) whose schema is generated from entity metadata. Three supporting contexts (`integration`, `analytics`, `agents`) own only operational tables (partner profiles, message logs, prompt registry, eval runs) and never master or transactional business data.

Context	Database	Owns	Language	Waves
`platform`	`passage_platform`	Tenants, users, roles/duties/privileges, metadata registry, number sequences, audit log, workflow/batch definitions, legal entities, operating units, sites, warehouses, hierarchies, fiscal calendars, ISO reference data, UoM	Go (sequences, audit), TS (admin, workflow)	1–6
`party`	`passage_party`	Party, roles (customer, vendor, contact, employee link), addresses, tax registrations, groups, payment and delivery terms, credit limits	TS	7
`product`	`passage_product`	Product, product master, released product, dimension groups, attributes, BOM, formula, routing, resources, calendars, change orders	Go (explosion), TS	8–9
`ledger`	`passage_ledger`	Accounts, account structures, dimensions, journal entries/lines, balances, ledgers, exchange rates, tax codes and transactions, period control, allocations, accruals, statements, budgets, cost accounting ledger, consolidation groups	Go	11–15, 38, 40
`inventory`	`passage_inventory`	Inventory transactions, on-hand projection, reservations, inventory journals, counting, cost layers, costing versions, costing sheets, quality orders, nonconformances	Go	16, 20, 33, 39
`procurement`	`passage_procurement`	Requisitions, RFQs, purchase agreements, purchase orders, product receipts, vendor invoices, match results, AP open items, payment proposals, payments	Go (matching, settlement), TS	17–20
`manufacturing`	`passage_manufacturing`	Production/batch orders, picking lists, route cards, planning runs, planned orders, planning messages, schedules, kanbans, time registrations, machine events	Go; Python (forecasting)	21–25
`sales`	`passage_sales`	Price conditions, quotations, sales orders, holds, shipments/packing slips, customer invoices, revenue schedules, AR open items, cash applications, collection cases, return orders, rebate agreements	Go (pricing, ATP, settlement), TS	26–30
`warehouse`	`passage_warehouse`	Locations, license plates, location directives, work templates, work, waves, replenishment, loads, routes, carrier rates, freight invoices, mobile flow definitions	Go (directives), TS	28, 31–34
`treasury`	`passage_treasury`	Bank accounts, statements, reconciliations, cash forecasts, checks, letters of credit, fixed assets, asset books, asset transactions, leases	Go	36–37
`workforce`	`passage_workforce`	Workers, positions, jobs, employment, absence, benefits, payroll runs, pay rules, expense reports	TS; Go (payroll calc)	25 (time), 42–43
`projects`	`passage_projects`	Projects, WBS, project transactions, estimates, invoice proposals, functional locations, maintenance assets, maintenance plans, maintenance work orders, installed base, service agreements, cases, service work orders	TS	41, 44–45

B.2 Cross-context reference rule

A row in one database may reference an entity in another context only by its identifier (UUIDv7 or, for user-facing documents, the number-sequence id) in a column named `<entity>_id` plus, when needed for display, a denormalized `<entity>_number` and `<entity>_name` snapshot. There are no foreign keys, views or `dblink` connections across databases. The owning context publishes `created`/`changed`/`deleted` events; consumers maintain a local read model (`<context>_ref.<entity>` tables) refreshed from those events and never query another context's database. Referential existence is validated at write time by calling the owning context's API (`HEAD /v1/<context>/<collection>/{id}`) or against the local read model when staleness of up to 5 s is acceptable (documents, not postings). The universal journal is the only synchronous cross-context write: subledgers call `ledger.Post` and store the returned `journal_entry_id`.

Identity rules: every table has `id uuid` (UUIDv7, time-ordered), `tenant_id uuid NOT NULL`, `legal_entity_id uuid` where the entity is company-specific, `version int` for optimistic concurrency, `created_at/created_by/updated_at/updated_by`. Fiscal documents additionally have `document_number text` (from a number sequence) and `posted_at timestamptz`.

B.3 Core entities

Entity	Context	Database	Key identifiers	Main relationships
Tenant	platform	passage_platform	`id`, `slug`	has many LegalEntity, User, Role
User	platform	passage_platform	`id`, `keycloak_sub`, `email`	many-to-many Role scoped by LegalEntity with `valid_from/to`

Entity	Context	Database	Key identifiers	Main relationships
Role / Duty / Privilege	platform	passage_platform	`code` (e.g. `AP.Clerk`, `MaintainVendors`, `vendor.write`)	Role ⊃ Duty ⊃ Privilege; compiled to OPA data
EntityDefinition	platform	passage_platform	`name`, `version`	has Fields, Enums, Relations, Extensions
NumberSequence	platform	passage_platform	`code`, `legal_entity_id`, `scope`	referenced by every document type (Appendix G)
AuditEntry	platform	passage_platform	`id`, `entity`, `record_id`, `trace_id`, `hash`	hash-chained to previous entry; immutable
LegalEntity	platform	passage_platform	`id`, `code` (e.g. `USMF`), `country`	has Sites, FiscalCalendar, ledger set
OperatingUnit	platform	passage_platform	`id`, `type`, `code`	effective-dated Hierarchy nodes by purpose
Site / Warehouse	platform	passage_platform	`id`, `code`, `legal_entity_id`	Warehouse belongs to Site; referenced by inventory and warehouse contexts
FiscalCalendar / Period	platform	passage_platform	`id`, `fiscal_year`, `period_no`	PeriodControl rows per Ledger × module
Party	party	passage_party	`id`, `party_number`	has Roles, Addresses, TaxRegistrations, Contacts
Customer	party	passage_party	`id`, `party_id`, `customer_account`	belongs to CustomerGroup, PostingProfile; credit limit
Vendor	party	passage_party	`id`, `party_id`, `vendor_account`	belongs to VendorGroup, PostingProfile, payment terms
Product / ProductMaster	product	passage_product	`id`, `product_number`, `gtin`	Master has Variants by dimension values; DimensionGroups
ReleasedProduct	product	passage_product	`product_id`, `legal_entity_id`	ItemModelGroup, defaults per Site/Warehouse
BOM / BOMLine	product	passage_product	`id`, `bom_number`, `version`	lines reference Product ids; effective dates; phantom, scrap
Routing / Operation	product	passage_product	`id`, `routing_number`, `version`	Operations require Resource capabilities
Account / AccountStructure	ledger	passage_ledger	`main_account`, `structure_id`	Dimensions allowed per structure
JournalEntry / JournalLine	ledger	passage_ledger	`id`, `document_number`, `idempotency_key`	lines carry 3 currency amounts, dimension set, `source_document_id`; `reverses_id`
Balance	ledger	passage_ledger	(`ledger_id`, `account`, `dimension_hash`, `period`)	projection rebuilt from JournalLine
Ledger / RateType / ExchangeRate	ledger	passage_ledger	`ledger_code`, `rate_type`, `from/to`, `valid_from`	LegalEntity has leading + parallel Ledgers
TaxCode / TaxTransaction	ledger	passage_ledger	`tax_code`, `jurisdiction_id`	TaxTransaction references source line id and JournalLine
InventoryTransaction	inventory	passage_inventory	`id`, `lot_id`, `reference_type/id`	dimensions (site, warehouse, location, batch, serial, LP); status pair
OnHand	inventory	passage_inventory	(`product_id`, dimension hash)	projection = Σ InventoryTransaction
CostLayer / CostingVersion	inventory	passage_inventory	`id`, `receipt_transaction_id`, `version_code`	settlements link issue to receipt layers
QualityOrder	inventory	passage_inventory	`id`, `qo_number`	triggered by receipt/production/shipment reference; blocks OnHand
PurchaseRequisition / RFQ / PurchaseAgreement	procurement	passage_procurement	`document_number`	Requisition lines → PO lines; Agreement releases
PurchaseOrder / Line	procurement	passage_procurement	`document_number`, `version`	`vendor_id`, `product_id`, `site/warehouse_id`; delivery schedules
ProductReceipt	procurement	passage_procurement	`document_number`	references PO lines, creates InventoryTransaction ids and JournalEntry id
VendorInvoice / MatchResult	procurement	passage_procurement	`document_number`, `vendor_invoice_number`	matches PO + Receipt lines; posts JournalEntry; creates APOpenItem
APOpenItem / Payment	procurement	passage_procurement	`id`, `payment_number`	Settlement links Payment to OpenItems; payment file id
ProductionOrder	manufacturing	passage_manufacturing	`document_number`, `status`	BOM/Routing snapshot; PickingList, RouteCard, JournalEntry ids for WIP
PlannedOrder / PlanningRun	manufacturing	passage_manufacturing	`id`, `run_id`	pegged to demand ids (SalesOrderLine, ProductionOrder, Forecast)
TimeRegistration / MachineEvent	manufacturing	passage_manufacturing	`id`, `worker_id`, `resource_id`	pay types exported to PayrollRun; OEE aggregates
PriceCondition	sales	passage_sales	`id`, `condition_type`, access key	resolved by access sequence; effective dates
SalesOrder / Line	sales	passage_sales	`document_number`, `status`	`customer_id`, `product_id`; Holds; ATP results; linked ProductionOrder/PlannedOrder ids
Shipment / PackingSlip	sales	passage_sales	`document_number`, `sscc`	references SalesOrder lines, Load id, InventoryTransaction ids, COGS JournalEntry id
CustomerInvoice / RevenueSchedule	sales	passage_sales	`document_number`	invoice lines → PerformanceObligations → schedule lines → JournalEntry ids
AROpenItem / CashApplication	sales	passage_sales	`id`, `receipt_number`	settlement to invoices; BankStatement line id
CollectionCase / ReturnOrder	sales	passage_sales	`case_number`, `rma_number`	customer, disputed invoice ids; RMA disposition → InventoryTransaction
Location / LicensePlate	warehouse	passage_warehouse	`location_code`, `lp_number`	Zone ⊃ Aisle ⊃ Rack ⊃ Bin; LP nests LPs
Work / PickWave	warehouse	passage_warehouse	`work_id`, `wave_id`	work lines reference order lines; directives; assigned user
BankAccount / BankStatement	treasury	passage_treasury	`id`, `iban/account_number` (P3)	statement lines reconcile to JournalLines, Payments, CashApplications
FixedAsset / AssetBook / AssetTransaction	treasury	passage_treasury	`asset_number`, `book_code`	acquisition from VendorInvoice/Project ids; depreciation JournalEntry ids
Worker / Position / Job	workforce	passage_workforce	`personnel_number`, `position_id`	effective-dated Employment; Position reports to Position; `party_id`

Entity	Context	Database	Key identifiers	Main relationships
PayrollRun / ExpenseReport	workforce	passage_workforce	`run_id`, `report_number`	pay statements; JournalEntry ids; reimbursement → VendorInvoice id
Project / WBS / ProjectTransaction	projects	passage_projects	`project_number`, `wbs_code`	transactions reference TimeRegistration, ExpenseReport, PO, inventory ids; JournalEntry ids
MaintenanceAsset / MaintenanceWorkOrder	projects	passage_projects	`asset_number`, `work_order_number`	FunctionalLocation tree; parts → InventoryTransaction; labor → TimeRegistration
ServiceAgreement / Case / ServiceWorkOrder	projects	passage_projects	`agreement_number`, `case_number`	InstalledBase serials from Shipment; billing → CustomerInvoice id

C. Event catalog conventions

C.1 Envelope

Every event is a CloudEvents 1.0 message in structured JSON mode on Kafka with the following attributes. Payload schemas are JSON Schema 2020-12 registered in the schema registry and described in AsyncAPI 3.0 documents generated from entity metadata.

Attribute	Rule
`specversion`	`1.0`
`id`	UUIDv7; equals the outbox row id, so redelivery carries the same id
`source`	`/passage/<context>/<service>` e.g. `/passage/ledger/posting`
`type`	`dev.passage.<context>.<entity>.<verb>.v<n>` — lowercase, no hyphens; verb is past tense (`posted`, `confirmed`, `changed`)
`subject`	the aggregate id (UUIDv7)
`time`	commit time of the producing transaction, RFC 3339 UTC
`datacontenttype`	`application/json`
`dataschema`	registry URL `https://schemas.passage.dev/<type>.json`
`tenantid` (ext)	tenant UUID; always present
`legalentityid` (ext)	legal entity UUID when company-scoped
`correlationid` (ext)	id of the originating request or workflow; propagated unchanged
`causationid` (ext)	id of the event or command that directly caused this event
`traceparent` (ext)	W3C trace context from the producing span
`actor` (ext)	`user:<sub>`, `agent:<agent-id>@user:<sub>`, or `system:<job>`
`sequence` (ext)	monotonically increasing per aggregate (`version` after the change)

C.2 Topics and partitions

- Topic: `passage.<context>.<entity>` (one topic per entity, all verbs and versions share it); 12 partitions default, 48 for `ledger.journalentry`, `inventory.transaction`, `manufacturing.machineevent`.
- Partition key: `tenantid + ":" + subject` so events for one aggregate are ordered; `machineevent` uses `tenantid:resource_id`.
- Retention: 30 days for transactional topics, 7 days for `machineevent`, compacted `passage.<context>.<entity>.snapshot` topics for reference-data read models.
- Dead letter: `passage.<context>.<entity>.dlq.<consumer>` with the original event, error and attempt count in headers.
- Access: ACLs per consumer service principal (SCRAM-SHA-512); cross-tenant consumers (analytics) run under a dedicated principal and are the only ones allowed to read all tenants.

C.3 Schema evolution

1. Changes within a version are additive only: new optional fields, new enum values flagged `x-passage-added-in`. Removing, renaming or retyping a field, or making an optional field required, requires a new `v<n+1>` type.
2. Schema registry compatibility mode is `BACKWARD_TRANSITIVE`; CI rejects a non-additive change (wave 4 proof).
3. When `v<n+1>` ships, the producer emits both `v<n>` and `v<n+1>` for a deprecation window of two minor releases of the producing service, then drops `v<n>`. Consumers must migrate inside the window; a consumer still subscribed to a dropped type fails its startup health check.
4. Enum consumers must treat unknown values as `other` and log at warn, never fail.

C.4 Outbox table DDL

Every service database contains exactly this table; Debezium's outbox event router relays rows to Kafka and deletes them after acknowledgment.

```
CREATE TABLE outbox (
  id             uuid        PRIMARY KEY,                    -- UUIDv7, becomes CloudEvents id
  tenant_id      uuid        NOT NULL,
  aggregate_type text        NOT NULL,                       -- e.g. 'journalentry'
  aggregate_id   uuid        NOT NULL,                       -- CloudEvents subject
  event_type     text        NOT NULL,                       -- full CloudEvents type
  partition_key  text        NOT NULL,                       -- tenant_id:aggregate_id
  sequence       bigint      NOT NULL,
  headers        jsonb       NOT NULL DEFAULT '{}'::jsonb,   -- correlationid, causationid, traceparent, actor
  payload        jsonb       NOT NULL,
  created_at     timestamptz NOT NULL DEFAULT clock_timestamp()
```

```
);
CREATE INDEX outbox_created_at_idx ON outbox (created_at);
ALTER TABLE outbox REPLICA IDENTITY FULL;
```

Rules: the outbox insert happens in the same transaction as the state change; the payload is the event `data` only (envelope attributes come from columns); payload size limit 256 KiB (larger content is stored and referenced by URL); the `outbox` table is excluded from RLS because only the service role writes it.

C.5 Consumer idempotency

Every consumer keeps an inbox table and processes each event inside one database transaction:

```
CREATE TABLE consumer_inbox (
  consumer   text        NOT NULL,
  event_id   uuid        NOT NULL,
  processed_at timestamptz NOT NULL DEFAULT now(),
  PRIMARY KEY (consumer, event_id)
);
```

1. `INSERT INTO consumer_inbox ... ON CONFLICT DO NOTHING`; if zero rows were inserted, skip the event (duplicate delivery).
2. Apply the effect and write any outgoing outbox rows in the same transaction; commit; then acknowledge the Kafka offset.
3. Out-of-order protection: consumers compare `sequence` with the stored aggregate version and park events that arrive early for up to 60 s before sending to the DLQ.
4. Inbox rows are pruned after 30 days (the topic retention) by the batch framework.
5. Retries: 5 attempts with exponential backoff (1 s → 60 s), then DLQ; DLQ depth per consumer is an alert at 100 and a page at 1,000.

C.6 Replay procedure

1. Open a change ticket naming consumer, topic, time window and reason; replay of a fiscal consumer (ledger, AP, AR) requires controller approval via the platform workflow.
2. Pause the consumer group (`passage-ctl consumer pause <group>`), snapshot its inbox table.
3. Reset the offset to the timestamp (`kafka-consumer-groups --reset-offsets --to-datetime`) or to a DLQ range; for a rebuild, truncate the read model and reset to earliest on the compacted snapshot topic.
4. Resume; the inbox guarantees already-processed events are skipped, so replay is safe by construction. A replay that must reprocess (bug fix) deletes inbox rows for the window first, inside the ticket.
5. Verify with the reconciliation checks in Appendix I.6 and close the ticket with the before/after totals.

C.7 Principal event types

Event type	Producer	Wave	Key consumers
`dev.passage.platform.entitydefinition.changed.v1`	platform/metadata	3	generators, shell, analytics
`dev.passage.platform.user.provisioned.v1`	platform/identity	2	workforce, audit
`dev.passage.platform.workflowtask.completed.v1`	platform/workflow	4	originating context, notifications
`dev.passage.platform.legalentity.created.v1`	platform/org	6	ledger, party, product, number sequences
`dev.passage.platform.periodcontrol.changed.v1`	platform/org	14	all posting services (cache invalidation)
`dev.passage.party.party.created.v1`	party	7	sales, procurement, workforce read models
`dev.passage.party.party.merged.v1`	party	7	every context holding `party_id` (rewrite references)
`dev.passage.party.customer.creditchanged.v1`	party	7	sales (credit checks)
`dev.passage.product.product.released.v1`	product	8	inventory, sales, procurement, manufacturing
`dev.passage.product.bom.versionactivated.v1`	product	9	manufacturing, inventory (cost roll)
`dev.passage.product.changeorder.released.v1`	product	9	manufacturing, warehouse, projects
`dev.passage.ledger.journalentry.posted.v1`	ledger	11	balances, analytics, audit, budgets
`dev.passage.ledger.journalentry.reversed.v1`	ledger	11	subledgers, analytics
`dev.passage.ledger.exchangerate.published.v1`	ledger	12	sales, procurement, treasury
`dev.passage.ledger.period.closed.v1`	ledger	14	all posting services, analytics
`dev.passage.ledger.taxsettlement.posted.v1`	ledger	13	statutory reporting, analytics
`dev.passage.inventory.transaction.posted.v1`	inventory	16	on-hand, sales (ATP), planning, analytics
`dev.passage.inventory.reservation.changed.v1`	inventory	16	sales, warehouse, manufacturing
`dev.passage.inventory.costclose.completed.v1`	inventory	20	ledger reconciliation, analytics
`dev.passage.inventory.qualityorder.decided.v1`	inventory	33	warehouse, procurement (scorecards), sales
`dev.passage.procurement.purchaseorder.confirmed.v1`	procurement	17	planning, budgets (encumbrance), integration (EDI 850)
`dev.passage.procurement.productreceipt.posted.v1`	procurement	18	inventory, ledger (GR/IR), quality, warehouse
`dev.passage.procurement.vendorinvoice.posted.v1`	procurement	18	ledger, treasury (forecast), budgets, analytics
`dev.passage.procurement.payment.posted.v1`	procurement	19	ledger, treasury, integration (payment file)
`dev.passage.manufacturing.productionorder.released.v1`	manufacturing	21	inventory (reservations), warehouse, shop floor
`dev.passage.manufacturing.productionorder.reportedasfinished.v1`	manufacturing	21	inventory, quality, sales (ATP)
`dev.passage.manufacturing.productionorder.ended.v1`	manufacturing	21	ledger (variances), inventory (cost), analytics

Event type	Producer	Wave	Key consumers
`dev.passage.manufacturing.plannedorder.firmed.v1`	manufacturing	22	procurement (PO creation), manufacturing orders
`dev.passage.manufacturing.schedule.changed.v1`	manufacturing	23	production orders, planning, shop floor
`dev.passage.manufacturing.machineevent.received.v1`	manufacturing/iot	25	OEE, projects (condition monitoring)
`dev.passage.sales.salesorder.confirmed.v1`	sales	27	inventory (reservation), planning, warehouse, integration (EDI 855)
`dev.passage.sales.shipment.posted.v1`	sales	28	inventory, ledger (COGS), billing, integration (ASN 856)
`dev.passage.sales.invoice.posted.v1`	sales	29	ledger, AR, revenue schedules, integration (810/Peppol), analytics
`dev.passage.sales.payment.applied.v1`	sales	30	ledger, collections, treasury
`dev.passage.sales.returnorder.received.v1`	sales	30	inventory, quality, billing (credit note)
`dev.passage.warehouse.work.completed.v1`	warehouse	31	inventory (status change), sales, procurement
`dev.passage.warehouse.load.shipped.v1`	warehouse	34	sales (shipment posting), integration (EDI 214), treasury (freight accrual)
`dev.passage.treasury.bankstatement.reconciled.v1`	treasury	36	ledger, AR, AP
`dev.passage.treasury.asset.depreciated.v1`	treasury	37	ledger, analytics
`dev.passage.workforce.payrollrun.posted.v1`	workforce	43	ledger, treasury (payments), projects
`dev.passage.projects.transaction.posted.v1`	projects	41	ledger (WIP), billing, analytics
`dev.passage.agents.action.executed.v1`	agents	47	audit, cost guard, analytics

D. API conventions

D.1 Resources and naming

- Base path `https://api.<env>.passage.dev/v1/<context>/<collection>`; collections are plural kebab-case (`/v1/ledger/journal-entries`, `/v1/sales/sales-orders`, `/v1/product/released-products`). Ids in paths are UUIDv7; document numbers are reachable through `?filter=documentNumber eq 'INV-USMF-2026-000123'`.
- Sub-resources only for composition (`/sales-orders/{id}/lines`). Lifecycle actions are custom methods with a colon: `POST /v1/ledger/journal-entries/{id}:post`, `:reverse`, `:confirm`, `:release`, `:cancel`. Actions are the only way to change status; `PATCH` on a `status` field returns 422 `state_transition_invalid`.
- JSON field names are camelCase in the wire format and snake_case in Postgres; the generator maintains the mapping. Money is `{ "amount": "1234.56", "currency": "USD" }` with a decimal string; dates are ISO 8601 (`2026-10-21`, `2026-10-21T14:03:00Z`); quantities are decimal strings with `unit`.
- Every entity declared in metadata gets `GET` list, `GET` by id, `POST`, `PUT`, `PATCH` (JSON Merge Patch), `DELETE` (soft, unless metadata says hard), plus `:batch` and `:export`, generated. Hand-written endpoints are limited to engines (post, price, plan, schedule, match, settle).

D.2 Versioning

`/v1` is the only version in v1.0. Within a major version changes are additive only (new optional fields, new endpoints, new enum values). Breaking changes ship as `/v2` for the affected context with both versions served for at least 12 months; the `Sunset` and `Deprecation` headers (RFC 8594) announce retirement. The `X-Passage-Api-Build` response header carries the service build for support.

D.3 Pagination, filtering, sorting, shape

- Cursor pagination: `?limit=50&cursor=<opaque>`; `limit` max 500. Response: `{ "items": [...], "nextCursor": "...", "estimatedTotal": 1234 }`. Cursors encode the sort key and are valid for 24 h; a cursor from a different filter returns 400 `invalid_cursor`.
- Filter grammar (subset of OData): `field op value` with `eq ne gt ge lt le in like isnull`, combined with `and or not` and parentheses; nested fields by dot (`customer.group eq 'WHOLESALE'`); dimension filters via `dimensions.costCenter eq 'CC-100'`. Max 20 predicates; filters on non-indexed fields over 100k rows are rejected with 422 `filter_not_indexed`.
- `?sort=-postedAt,documentNumber`; `?fields=id,documentNumber,total` for projection; `?expand=lines,customer` for one level of embedded reads (never across contexts; cross-context fields come from the local read model and are marked `x-passage-snapshot` in OpenAPI).
- `?asOf=2026-01-01` on effective-dated entities (hierarchies, BOMs, positions, prices).

D.4 Error envelope

Errors are RFC 9457 problem details (`Content-Type: application/problem+json`):

```
{
  "type": "https://passage.dev/problems/period_closed",
  "title": "Period is closed",
  "status": 422,
  "detail": "Ledger USMF-IFRS period 2026-09 is closed for module AP since 2026-10-05T18:00:00Z.",
  "instance": "/v1/procurement/vendor-invoices/0192f3a1-...:post",
  "code": "period_closed",
  "traceId": "4bf92f3577b34da6a3ce929d0e0e4736",
  "errors": [ { "field": "postingDate", "code": "period_closed", "message": "..." } ]
}
```

Code	HTTP	Meaning
`validation_failed`	400	Payload fails JSON Schema; `errors[]` lists fields
`required_field_missing`	400	A metadata-required field is absent

Code	HTTP	Meaning
`invalid_enum_value`	400	Value not in the enum (including extensions)
`invalid_cursor`	400	Cursor expired or from another query
`filter_not_indexed`	422	Filter would scan an unindexed column on a large table
`unauthenticated`	401	Missing or invalid token
`forbidden_duty`	403	OPA denied: principal lacks the duty/privilege
`forbidden_tenant`	403	Resource belongs to another tenant or legal entity outside scope
`sod_conflict`	403	Action would violate a segregation-of-duties rule
`approval_required`	403	Action requires an approval workflow; a task was created
`not_found`	404	No such resource in the caller's scope
`conflict_version`	412	`If-Match` does not equal the current ETag
`precondition_required`	428	Mutating request without `If-Match`
`idempotency_key_reuse`	422	Same `Idempotency-Key` with a different payload
`idempotency_in_progress`	409	Original request with this key is still executing
`unique_violation`	409	Unique constraint (e.g. vendor invoice number per vendor)
`state_transition_invalid`	422	Action not allowed from the current status
`period_closed`	422	Posting date in a closed/on-hold period for this module
`unbalanced_entry`	422	Debits ≠ credits in any currency
`dimension_invalid`	422	Dimension combination violates the account structure
`inventory_insufficient`	422	On-hand would go negative and the item model group forbids it
`credit_hold`	422	Customer on hold or limit exceeded; hold created
`number_sequence_exhausted`	422	Sequence reached its upper bound or is stopped
`rate_limited`	429	Tenant tier quota exceeded; `Retry-After` set
`payload_too_large`	413	Body over 10 MiB (bulk) or 1 MiB (single)
`upstream_unavailable`	503	Tax/rate/carrier/bank provider failed and no fallback applied
`internal_error`	500	Unexpected; `traceId` for support

D.5 Idempotency

`Idempotency-Key` (UUID or ≤ 128-char string) is required on every `POST` that creates or posts a document and on every custom method; it is optional on `PUT/PATCH` (which rely on ETags). The key is scoped to tenant + principal + method + path. The server stores the request body hash and the final response for 24 h: a repeat with the same hash returns the stored response with `Idempotent-Replayed: true`; a repeat with a different hash returns 422 `idempotency_key_reuse`; a repeat while the first is in flight returns 409 `idempotency_in_progress`. Posting engines additionally persist the key on the document (`idempotency_key` column, unique per tenant) so the guarantee survives the 24 h cache.

D.6 Concurrency

Every readable resource returns a strong `ETag` derived from `version` (e.g. `"v17"`). `PUT`, `PATCH`, `DELETE` and custom methods require `If-Match`; a missing header is 428, a mismatch is 412 `conflict_version` with the current representation in `detail`. List responses include per-item `etag`.

D.7 Rate limits

Tier	Sustained (requests/min per tenant)	Burst	Bulk rows/day	Webhook deliveries/min	Concurrent exports
Starter	600	100	200k	60	2
Professional	3,000	500	2M	300	5
Enterprise	12,000 (negotiable)	2,000	20M	1,200	20

Enforced at Envoy Gateway per tenant and per API key; headers `RateLimit-Limit`, `RateLimit-Remaining`, `RateLimit-Reset` (IETF draft) on every response. Agents share the tenant budget and have a separate per-agent cap (Appendix E). Shop-floor and mobile apps use a dedicated limit class (`device`) of 120 requests/min per device.

D.8 Bulk endpoints

`POST /v1/<context>/<collection>:batch` accepts up to 1,000 items with `"atomic": true|false`; atomic batches roll back on the first error, non-atomic batches return per-item results with the same problem-details shape. Larger loads use NDJSON upload to `POST /v1/platform/jobs` (type `import`), which runs in the batch framework with checkpoints and returns a job id, progress and an error file. `:export` streams NDJSON or CSV with the same filter grammar. Postings via bulk are still one journal entry per document; the batch framework never bypasses posting rules.

D.9 OpenAPI and SDKs

The metadata generator emits one OpenAPI 3.1 document per context (`openapi/<context>.v1.yaml`) with schemas, examples, error codes and `x-passage-duty` on every operation; hand-written engine endpoints are merged from annotated Go/TS handlers. CI runs Spectral (style) and oasdiff (breaking-change detection) against the last released spec; the developer portal (wave 35) publishes specs and the AsyncAPI documents. SDKs generated per release: TypeScript (first-party, used by the shell), Go, Python in v1.0; C# and Java in v1.x for AX and SAP shops respectively. SDKs expose idempotency keys, ETags and pagination as first-class helpers and never hide problem details.

E. Security and authorization model

E.1 Tenancy

A tenant is one customer organization: one Keycloak realm, one `tenant_id`, one row-level security scope across all 12 databases, one Kafka ACL principal set, one Vault namespace, one cost budget. Legal entities are scopes inside the tenant; users are assigned roles per legal entity (or `*`). Data residency is per tenant (wave 48): a tenant is pinned to a region's cluster set and its Iceberg warehouse. Cross-tenant access does not exist in the application; only the analytics and platform-operations principals can read across tenants and only through audited service accounts.

The request path sets the scope: Envoy validates the JWT, the service opens a Postgres connection from a pool bound to the non-superuser application role and executes `SET LOCAL passage.tenant_id = '<uuid>'; SET LOCAL passage.legal_entity_ids = '<uuid>,<uuid>'; SET LOCAL passage.actor = '<sub>'` inside the transaction. The application role has no `BYPASSRLS`.

E.2 RLS policy template

Generated for every tenant-scoped table; the generator adds the `legal_entity_scope` policy only when the entity declares `legal_entity_id`.

```
ALTER TABLE ledger.journal_entry ENABLE ROW LEVEL SECURITY;
ALTER TABLE ledger.journal_entry FORCE ROW LEVEL SECURITY;

CREATE POLICY tenant_isolation ON ledger.journal_entry
  AS RESTRICTIVE FOR ALL TO passage_app
  USING      (tenant_id = current_setting('passage.tenant_id', true)::uuid)
  WITH CHECK (tenant_id = current_setting('passage.tenant_id', true)::uuid);

CREATE POLICY legal_entity_scope ON ledger.journal_entry
  AS RESTRICTIVE FOR ALL TO passage_app
  USING (
    current_setting('passage.legal_entity_ids', true) = '*'
    OR legal_entity_id = ANY (string_to_array(current_setting('passage.legal_entity_ids', true), ',')::uuid[])
  )
  WITH CHECK (
    legal_entity_id = ANY (string_to_array(current_setting('passage.legal_entity_ids', true), ',')::uuid[])
  );

-- Batch and CDC roles: separate role with the same policies; Debezium reads the WAL and is outside RLS by design,
-- so the outbox payload carries tenant_id and the relay sets the CloudEvents tenantid from it.
```

A missing `passage.tenant_id` makes `current_setting(..., true)` return NULL and the policy returns no rows: the fail-safe is empty, never cross-tenant.

E.3 Role / duty / privilege schema

```
CREATE TABLE platform.privilege (
  code text PRIMARY KEY,                  -- 'vendor.write', 'journalentry.post', 'payment.approve'
  resource text NOT NULL,                 -- entity or endpoint family
  action text NOT NULL,                   -- read | write | post | approve | delete | export | admin
  description text NOT NULL
);
CREATE TABLE platform.duty (
  code text PRIMARY KEY,                  -- 'MaintainVendors', 'PostVendorInvoices', 'ApprovePayments'
  tenant_id uuid,                         -- NULL = shipped by Passage, else customer-defined
  description text NOT NULL
);
CREATE TABLE platform.duty_privilege (duty_code text REFERENCES platform.duty, privilege_code text REFERENCES platform.privilege, PRIMARY
KEY (duty_code, privilege_code));
CREATE TABLE platform.role (code text PRIMARY KEY, tenant_id uuid, description text NOT NULL);
CREATE TABLE platform.role_duty (role_code text REFERENCES platform.role, duty_code text REFERENCES platform.duty, PRIMARY KEY (role_code,
duty_code));
CREATE TABLE platform.user_role (
  user_id uuid NOT NULL, role_code text NOT NULL, tenant_id uuid NOT NULL,
  legal_entity_id uuid,                   -- NULL = all legal entities in the tenant
  valid_from date NOT NULL DEFAULT current_date, valid_to date,
  granted_by uuid NOT NULL, PRIMARY KEY (user_id, role_code, legal_entity_id, valid_from)
);
CREATE TABLE platform.sod_rule (
  code text PRIMARY KEY, duty_a text NOT NULL, duty_b text NOT NULL,
  severity text NOT NULL CHECK (severity IN ('high','medium','low')), description text NOT NULL
);
```

Passage ships ~60 roles, ~400 duties and ~2,500 privileges, named after AX (e.g. role `AccountsPayableClerk`, duty `MaintainVendorInvoices`). Customers add duties and roles; they cannot add privileges, which are generated from metadata (`<entity>.<action>`) and engine endpoints.

E.4 OPA policy structure

The compiled policy bundle contains `data.passage.grants[tenant][user] = {duty: [legal_entity_ids]}` and `data.passage.duty_privileges[duty] = [privileges]`, rebuilt by the platform service on every role change (push to the OPA sidecars via bundle server, ≤ 10 s propagation). The input is `{ "principal": {sub, tenant, agent?, on_behalf_of?}, "action": "<privilege>", "resource": {context, entity, id?, legal_entity_id?, amount?, owner?} }`. Every service calls `POST /v1/data/passage/authz/allow` on its sidecar; the gateway performs a coarse check (token, tenant, route duty) and the service performs the fine-grained one (record scope).

Example 1 — duty-based allow with legal-entity scope:

```
package passage.authz

default allow := false

allow if {
  some duty
  priv_set := data.passage.duty_privileges[duty]
  priv_set[_] == input.action
  scopes := data.passage.grants[input.principal.tenant][input.principal.sub][duty]
  scope_ok(scopes, input.resource.legal_entity_id)
}

scope_ok(scopes, le) if scopes[_] == "*"
scope_ok(scopes, le) if scopes[_] == le
```

Example 2 — agents never exceed the user they act for, and are additionally limited by their own allow-list:

```
allow if {
  input.principal.agent != null
  user_input := object.union(input, {"principal": {"sub": input.principal.on_behalf_of, "tenant": input.principal.tenant}})
  data.passage.authz.allow with input as user_input
  data.passage.agents[input.principal.agent].allowed_actions[_] == input.action
  not data.passage.agents[input.principal.agent].kill_switch
}
```

Example 3 — amount-banded approval (payment release above a threshold needs `ApprovePaymentsHighValue`, and the approver may not be the preparer):

```
deny contains msg if {
  input.action == "payment.approve"
  input.resource.amount_usd > 50000
  not has_duty("ApprovePaymentsHighValue")
  msg := "payments over USD 50,000 require ApprovePaymentsHighValue"
}

deny contains msg if {
  input.action == "payment.approve"
  input.resource.prepared_by == input.principal.sub
  msg := "sod: preparer cannot approve own payment"
}

has_duty(d) if data.passage.grants[input.principal.tenant][input.principal.sub][d]
```

E.5 Segregation-of-duties enforcement points

Point	Behavior
Role design (wave 48 SoD engine)	Creating or editing a role whose duties conflict is blocked (high) or warned (medium/low)
User assignment	Assigning a role that creates a conflict with the user's existing roles opens a mitigation workflow; the assignment is inactive until a mitigating control (compensating review, time-boxed access) is approved
Runtime (OPA `deny`)	Same-person checks on document pairs: prepare/approve journals, create vendor/approve vendor invoice, create PO/post receipt over threshold, enter payment/release payment, maintain bank account/run payments
Agents	An agent's action set is checked against SoD rules at registration; an agent cannot hold both sides of any high-severity rule
Certification	Quarterly access-certification campaigns; uncertified roles expire after 30 days
Evidence	Every conflict, mitigation and certification decision is an audit entry exported to the SOC 2 evidence store

E.6 PII classification and handling

Level	Examples	Storage	Logging / tracing	Display	Export / analytics	Retention
P0 Public	Product catalog, public prices, legal entity name	Plain	Allowed	Full	Allowed	Business need
P1 Internal	Order numbers, amounts, dimension values, internal notes	Plain, RLS	Allowed (ids only, no free text)	Full with duty	Tenant-scoped	Business need
P2 Personal	Person names, business emails, phone, postal address, personnel number	Plain, RLS, `pii: true` in metadata	Never in logs; hashed in traces	Full with duty; masked in list exports by default	Lakehouse with column tags; erasure propagates by `party_id`	Per GDPR/CCPA policy; erasure request honored within 30 days except fiscal documents (pseudonymized)
P3 Sensitive	National IDs, tax IDs of persons, bank account/IBAN, card tokens, salary, health/absence reasons, biometrics, dates of birth	Column-level encryption (pgcrypto, key per tenant in Vault transit), separate `*_secure` tables, no full-text index	Never; access itself is audited	Masked (`****1234`) unless a `ViewSensitive*` duty is held; reveal is an audited action	Excluded from lakehouse except aggregated; excluded from exports except statutory formats	Minimum legal; vaulted in `workforce` with per-field retention

Metadata flags (`pii: P2|P3`) drive generation: masking in generated APIs and forms, exclusion from the Debezium CDC allow-list for P3, and the subject-request tooling (wave 48) that enumerates every P2/P3 field per `party_id`.

E.7 Secrets management

HashiCorp Vault is the only secret store: Kubernetes auth per service account, one namespace per tenant for customer-specific credentials (bank, tax provider, carrier). External Secrets Operator materializes only the minimum into Kubernetes secrets with 1 h TTL. Database credentials are dynamic (Vault database engine, 24 h leases). Bank signing keys and EBICS keys live in Vault transit and never leave it; files are signed by calling transit. Rotation: 90 days for static secrets, automatic on lease expiry for dynamic ones, immediate on offboarding. Git is scanned by gitleaks in CI; a leak fails the build. Keycloak client secrets rotate per release.

E.8 Audit and retention

Audit is derived from Debezium CDC into an append-only `platform.audit_entry` table with a per-tenant SHA-256 hash chain (`hash = sha256(prev_hash || canonical_json(entry))`) verified nightly; the chain head is written to the lakehouse and to object storage with object lock. Retention: audit entries for fiscal documents (journal entries, invoices, payments, receipts, packing slips, production orders, asset transactions, payroll) 7 years from fiscal-year end, extended by country pack (DE and FR 10 years, IN 8 years); other entities 3 years; agent actions 7 years. Audit entries cannot be deleted; GDPR erasure replaces P2/P3 values with pseudonyms and records the erasure itself.

E.9 Encryption

In transit: TLS 1.3 at the gateway (cert-manager, ACME or customer CA), mTLS between services and to Postgres and Kafka with cert-manager-issued certificates rotated every 30 days; Kafka SCRAM-SHA-512 plus TLS; Debezium and Temporal over mTLS. At rest: encrypted storage class on every persistent volume (provider KMS or LUKS on bare metal), CloudNativePG backups and WAL archives encrypted with a per-tenant key via Vault transit (envelope), Kafka tiered storage and Iceberg objects encrypted server-side; P3 columns encrypted additionally at the application level. Key rotation re-wraps envelopes without rewriting data; a tenant's crypto-shred (key deletion) is the final step of tenant offboarding.

F. Non-functional requirements detail

ID	Area	Metric	Target	Measurement	First enforced
NFR-A1	Availability	Monthly availability per tenant (API + shell)	≥ 99.9% (v1), ≥ 99.95% with multi-region	Synthetic probes every 30 s from two regions; SLO in Grafana	Wave 5 (measured), wave 50 (99.95%)
NFR-A2	Availability	RPO	≤ 5 min	Backup/restore drill; WAL archive lag metric	Wave 48
NFR-A3	Availability	RTO	≤ 1 h	DR game day, timed	Wave 48, region failover wave 50
NFR-A4	Availability	Zero-downtime deploys and migrations	0 failed requests during rollout	Argo rollout with k6 traffic; expand/contract migrations only	Wave 3
NFR-A5	Availability	Rollback time	≤ 15 min to prior digest	Timed in Argo CD history	Wave 1
NFR-P1	Performance	Journal posting throughput	≥ 5,000 lines/s sustained, p99 < 200 ms	k6 against ledger on 3-node stage	Wave 11
NFR-P2	Performance	Generated entity simple read	p95 < 50 ms at 1,000 rps	k6	Wave 3
NFR-P3	Performance	Price resolution (10-line order)	p95 < 20 ms	Go benchmark + k6	Wave 26
NFR-P4	Performance	Order intake API	500 orders/min, p99 < 500 ms	k6	Wave 27
NFR-P5	Performance	MRP run	50k items × 12 periods × 5 levels < 10 min	Scheduled benchmark in CI nightly	Wave 22
NFR-P6	Performance	Scheduling	5k operations < 60 s	Benchmark fixtures	Wave 23
NFR-P7	Performance	BOM explosion (7 levels, 2,000 components)	< 500 ms	Go benchmark	Wave 9
NFR-P8	Performance	Inventory close (1M transactions)	< 20 min	Batch job timing	Wave 20
NFR-P9	Performance	Cost roll (50k items)	< 15 min	Batch job timing	Wave 39
NFR-P10	Performance	Dashboard freshness (OLTP → lakehouse)	p95 < 60 s	Debezium/Iceberg lag metric	Wave 46
NFR-P11	Performance	Shell first contentful paint (warm)	< 1.5 s on a mid-range laptop	Lighthouse in CI	Wave 5
NFR-P12	Performance	Warehouse task assignment	p95 < 1 s	Simulation	Wave 31
NFR-S1	Scale	Concurrent users	10,000	k6 load campaign	Wave 50
NFR-S2	Scale	Journal lines per hour	1M	Load campaign	Wave 50
NFR-S3	Scale	Order lines per hour	100k	Load campaign	Wave 50
NFR-S4	Scale	IoT ingest	10k events/min per plant, no loss on broker restart	Simulator + chaos	Wave 25
NFR-S5	Scale	Migration load	50k customers, 20k vendors, 100k items, 10k BOMs < 30 min	Migration run ledger	Wave 10
NFR-S6	Scale	Tenants per cluster set	100 with noisy-neighbor limits	Capacity model, rate limits	Wave 50
NFR-SEC1	Security	Authorization on every request	100% of routes carry `x-passage-duty`; 0 unprotected	OpenAPI lint + gateway audit	Wave 2
NFR-SEC2	Security	Tenant isolation	0 cross-tenant rows in RLS test suite	Automated RLS test per table	Wave 2

ID	Area	Metric	Target	Measurement	First enforced
NFR-SEC3	Security	Supply chain	100% images signed, SBOM attached, 0 critical CVEs at deploy	cosign verify, Grype gate	Wave 1
NFR-SEC4	Security	Penetration test	0 high/critical open at wave 48 exit	External test report	Wave 48
NFR-SEC5	Security	Secrets in source	0	gitleaks	Wave 1
NFR-SEC6	Security	Agent permission ceiling	0 actions exceeding the user's OPA decision	Contract tests	Wave 47
NFR-C1	Compliance	SOC 2 TSC mapping	100% of in-scope controls with automated evidence	Compliance workspace	Wave 48
NFR-C2	Compliance	Subject access/erasure request	Fulfilled ≤ 30 days; export ≤ 24 h	Workflow SLA timer	Wave 48
NFR-C3	Compliance	Statutory outputs validate	SAF-T, Peppol, pain.001, X12 pass schema/schematron	CI fixture validation	Waves 13, 19, 35
NFR-C4	Compliance	Fiscal document immutability	0 updates to posted documents (DB trigger)	Trigger + CDC audit test	Wave 11
NFR-ACC1	Accessibility	WCAG 2.1 AA on generated forms	100% pass axe, 0 serious/critical	Playwright + axe in CI	Wave 5
NFR-ACC2	Accessibility	Keyboard operability	100% of workspaces navigable without a mouse	Keyboard test suite	Wave 5
NFR-ACC3	Accessibility	Mobile/kiosk apps	WCAG 2.1 AA, touch targets ≥ 44 px	axe on emulated devices	Waves 25, 32
NFR-O1	Observability	Trace coverage	100% of services emit OTel traces; one trace id per request end-to-end	Tempo query in CI smoke	Wave 5
NFR-O2	Observability	SLO dashboards and alerts	Every service has latency/error/saturation SLOs with burn-rate alerts	Grafana provisioning review	Wave 5
NFR-O3	Observability	Audit latency	Field change visible in audit viewer ≤ 2 s	E2E test	Wave 5
NFR-PT1	Portability	Provider-managed services in the critical path	0	Architecture review per wave	Wave 1
NFR-PT2	Portability	Local parity	`make deploy-local` on k3d < 10 min	Timed in CI	Wave 1
NFR-PT3	Portability	Provider modules	EKS, GKE, AKS, bare-metal Terraform modules with identical interface	Module tests	Wave 1 (k3d + one cloud), wave 50 (all)
NFR-D1	Data integrity	Subledger-to-GL reconciliation	0 unexplained variance, daily	CI reconciliation job (Appendix I.6)	Wave 16
NFR-D2	Data integrity	Balance projection = journal re-aggregation	Exact	Nightly rebuild compare	Wave 11
NFR-D3	Data integrity	Gap-free sequences	0 gaps, 0 duplicates under 10k concurrent requests	Jepsen-style test	Wave 5
NFR-D4	Data integrity	Negative inventory	0 unless item model group allows	Concurrency test	Wave 16
NFR-D5	Data integrity	Exactly-once event effect	0 duplicate effects after chaos kill	Outbox/inbox chaos test	Wave 4
NFR-D6	Data integrity	Migration reconciliation	100% control totals match	Migration run ledger	Waves 10, 49

G. Numbering and document conventions

G.1 Number sequence service

Sequences are defined per tenant with a scope of `tenant`, `legal_entity`, `legal_entity+fiscal_year` or `site`. Format strings use `{LE}` (legal entity code), `{FY}` (fiscal year), `{YY}`/`{YYYY}`, `{SITE}`, and `{######}` (zero-padded counter). Two modes:

- **Continuous (gap-free)**: the number is allocated inside the posting transaction from a per-sequence row locked with `SELECT ... FOR UPDATE`; if the transaction aborts the number is never consumed. Throughput is bounded (≈ 2,000/s per sequence on stage); used only where the law or audit requires it.
- **Non-continuous**: numbers are pre-allocated in blocks of 100 per service instance (cached); gaps are possible on restart. Used for everything that is not a fiscal document.

A sequence can be `stopped` (posting returns `number_sequence_exhausted`), has `lowest/highest` bounds, an optional yearly reset (only with `{FY}` in the format) and an alert at 90% consumption.

G.2 Formats per document type

Document	Sequence code	Format (default)	Example	Mode	Scope / reset
General journal entry (posted)	`GJ`	`{LE}-GJ-{FY}-{#######}`	`USMF-GJ-2026-0001234`	Gap-free	Legal entity × fiscal year
Journal batch (unposted)	`JB`	`JB-{########}`	`JB-00004521`	Non-continuous	Tenant

Document	Sequence code	Format (default)	Example	Mode	Scope / reset
Customer invoice / credit note	`INV` / `CN`	`{LE}-INV-{FY}-{######}`	`DEMF-INV-2026-000087`	Gap-free	Legal entity × fiscal year (legal requirement in DE, FR, IN, MX)
Vendor invoice (internal number)	`VINV`	`{LE}-VINV-{FY}-{######}`	`USMF-VINV-2026-003310`	Gap-free	Legal entity × fiscal year
Purchase requisition	`REQ`	`REQ-{######}`	`REQ-001204`	Non-continuous	Tenant
RFQ	`RFQ`	`RFQ-{######}`	`RFQ-000051`	Non-continuous	Tenant
Purchase order	`PO`	`{LE}-PO-{######}`	`USMF-PO-000912`	Non-continuous	Legal entity
Product receipt	`PR`	`{LE}-PR-{FY}-{######}`	`USMF-PR-2026-000455`	Gap-free	Legal entity × fiscal year
Quotation	`QUO`	`QUO-{######}`	`QUO-000210`	Non-continuous	Tenant
Sales order	`SO`	`{LE}-SO-{######}`	`USMF-SO-004417`	Non-continuous	Legal entity
Packing slip / delivery note	`PS`	`{LE}-PS-{FY}-{######}`	`USMF-PS-2026-002031`	Gap-free	Legal entity × fiscal year
Return order (RMA)	`RMA`	`RMA-{######}`	`RMA-000078`	Non-continuous	Tenant
Vendor payment	`PAY`	`{LE}-PAY-{FY}-{######}`	`USMF-PAY-2026-000660`	Gap-free	Legal entity × fiscal year
Customer receipt / cash application	`RCPT`	`{LE}-RCPT-{FY}-{######}`	`USMF-RCPT-2026-001102`	Gap-free	Legal entity × fiscal year
Check number	`CHK-<bank>`	`{######}`	`100245`	Gap-free	Bank account (void keeps the number)
Production / batch order	`PRD`	`{SITE}-PRD-{######}`	`S1-PRD-000318`	Non-continuous	Site
Planned order	`PLN`	`PLN-{########}`	`PLN-00019902`	Non-continuous	Tenant (reset per planning run)
Warehouse work	`WRK`	`WRK-{########}`	`WRK-00031177`	Non-continuous	Tenant
Load	`LD`	`LD-{######}`	`LD-000504`	Non-continuous	Tenant
Quality order	`QO`	`QO-{######}`	`QO-000912`	Non-continuous	Tenant
Fixed asset	`FA`	`{LE}-FA-{#####}`	`USMF-FA-00317`	Non-continuous	Legal entity
Asset transaction	`FAT`	`{LE}-FAT-{FY}-{######}`	`USMF-FAT-2026-000045`	Gap-free	Legal entity × fiscal year
Maintenance work order	`MWO`	`MWO-{######}`	`MWO-001560`	Non-continuous	Tenant
Service case / service work order	`CASE` / `SWO`	`CASE-{######}`	`CASE-000731`	Non-continuous	Tenant
Project	`PRJ`	`PRJ-{######}`	`PRJ-000142`	Non-continuous	Tenant
Payroll run	`PAYR`	`{LE}-PAYR-{FY}-{###}`	`USMF-PAYR-2026-021`	Gap-free	Legal entity × fiscal year
Expense report	`EXP`	`EXP-{######}`	`EXP-004012`	Non-continuous	Tenant
Bank statement	`BST`	`{bank}-{YYYY}-{####}`	`CHASE01-2026-0210`	Non-continuous	Bank account

Country packs may override formats and modes (Appendix H), for example a per-series gap-free invoice number for MX CFDI (`serie` + `folio`) or the FR requirement that credit notes use a separate chronological sequence.

G.3 Fiscal document immutability

A document is fiscal when it posts to the universal journal or is legally required to be retained (invoices, credit notes, receipts, packing slips, payments, journal entries, asset transactions, payroll statements, production order settlements). Rules:

1. After `posted_at` is set, the header and lines are immutable: a database trigger rejects `UPDATE`/`DELETE` on posted rows except the status columns `settled_amount`, `settlement_status` and `printed_at`, and the metadata generator marks all other fields read-only in the API.
2. Corrections are new documents: reversal (an exact negating journal entry linked by `reverses_id`), credit note, correcting invoice (country packs that require it), or an adjustment journal. The original stays and shows its correction links.
3. The rendered PDF/XML of a fiscal document is stored at posting time in object storage with object lock for the retention period and its SHA-256 is written to the document; re-rendering produces a copy marked `Duplicate`.
4. The posting date determines the fiscal period; the document date may differ. Both are immutable after posting.
5. Document numbers are unique per tenant forever; a sequence reset per fiscal year includes the year in the number.

G.4 Attachments

Attachments are stored in S3-compatible object storage (bucket per tenant, server-side encrypted) and referenced by `platform.attachment` rows (`entity`, `record_id`, `kind`, `content_sha256`, `size`, `retention_class`). Limits: 25 MiB per file, 100 files per record, file types allow-listed (PDF, images, Office, XML, CSV, TXT, ZIP ≤ 25 MiB); antivirus scan (ClamAV) before the attachment becomes visible. Attachments on fiscal documents inherit the document's retention and object lock; attachments on other records follow the entity's retention. Deleting a record soft-deletes attachments; hard purge runs after the retention window. Inbound documents (OCR invoices, EDI, e-invoices) keep the original payload as a `source` attachment.

H. Localization scope

Country packs are metadata and code bundles installed per legal entity (wave 48). Every pack ships tax determination data, statutory report formats for the electronic reporting designer, document layouts, bank formats, calendar/number defaults and a regulator-schema fixture set. The matrix below is the v1.0 scope; items marked v1.x ship after go-live.

Capability	US	UK	DE	FR	IN	MX	CA
Tax model	Sales & use tax by jurisdiction (state/county/city/district), nexus, exemption certificates, use-tax accrual; Avalara/Vertex adapter	VAT 20/5/0%, reverse charge, postponed import VAT, CIS withholding, Northern Ireland protocol	USt 19/7%, §13b reverse charge, intra-EU supplies, withholding for construction (Bauabzugsteuer)	TVA 20/10/5.5/2.1%, autoliquidation, intra-EU, DAS2 withholding hooks	GST (CGST/SGST/IGST/UTGST), HSN/SAC codes, reverse charge, TDS/TCS withholding sections	IVA 16/8/0%, ISR and IVA retentions, IEPS, SAT product/service catalogs	GST/HST 5–15%, provincial PST/QST, input tax credits, non-resident rules
Statutory reports (v1.0)	1099-MISC/NEC/INT, sales tax returns by state (data extract), W-2 structure	MTD VAT return (HMRC API), VAT100 boxes, EC sales list for NI	UStVA (ELSTER XML), ZM (EC sales list), Intrastat, GoBD-compliant export (GDPdU), E-Bilanz v1.x	CA3 VAT return, DEB/Intrastat, FEC (fichier des écritures comptables), liasse fiscale v1.x	GSTR-1, GSTR-3B, e-way bill payload, TDS Form 26Q structure, 26AS reconciliation hook	DIOT, Contabilidad Electrónica (catálogo, balanza, pólizas XML), ISR provisional data	GST/HST return (GST34 boxes), T4/T4A/T5018 structures, QST return
E-invoicing mandate	None; Peppol/DBNAlliance optional	None in v1.0 (B2B mandate consultation); Peppol BIS optional	B2G XRechnung mandatory; B2B receive (2025) and issue (2027) via XRechnung/ZUGFeRD 2.x	B2B mandate via PDP: receive 2026, issue phased 2026–2027; Factur-X profiles	IRP e-invoice (IRN + signed QR) mandatory above turnover threshold; GSP adapter	CFDI 4.0 via PAC mandatory for every invoice, payment complement, carta porte v1.x; cancellation workflow	None; Peppol optional
Bank formats	NACHA ACH (PPD/CCD/CTX), check printing (MICR), positive pay, BAI2 statements, lockbox BAI2, ISO 20022 for wires	BACS Standard 18, Faster Payments via ISO 20022 pain.001, CHAPS, MT940/camt.053 statements	SEPA SCT/SDD (pain.001/pain.008), camt.052/053/054, EBICS 3.0	SEPA SCT/SDD, camt.053, EBICS TS (signed)	NEFT/RTGS/IMPS bank-specific layouts via adapter, ISO 20022 pain.001 for H2H banks, MT940	SPEI bank layouts via adapter, ISO 20022 for multinational banks, BAI2/camt.053	CPA Standard 005 EFT, Lynx wires via ISO 20022, cheque printing, camt.053
Payroll	Full engine: federal/state/local withholding (Pub. 15-T), FICA, FUTA/SUTA, garnishments, W-2/941 structures	PAYE RTI (FPS/EPS), NI, auto-enrolment — engine only if chosen as EU/UK reference (open question 3), else provider adapter	Lohnsteuer, Sozialversicherung (DEÜV), ELStAM — reference EU engine candidate (open question 3), else provider adapter	DSN monthly declaration — v1.x, provider adapter in v1.0	PF, ESI, professional tax, Form 16 — v1.x, provider adapter in v1.0	CFDI nómina, IMSS, ISR tables — v1.x, provider adapter in v1.0	CRA/RQ source deductions, CPP/EI, ROE — v1.x, provider adapter in v1.0
Calendar and formats	Fiscal year any; MM/DD/YYYY display; USD 2 decimals; Letter paper	Tax year 6 April (payroll), company FY any; DD/MM/YYYY; GBP; A4	Calendar FY default; DD.MM.YYYY; EUR with `1.234,56`; A4; DIN 5008 letters	Calendar FY default; DD/MM/YYYY; EUR with `1 234,56`; A4	FY 1 April–31 March; DD/MM/YYYY; INR lakh/crore grouping (`12,34,567.00`)	Calendar FY; DD/MM/YYYY; MXN; Letter	FY any; YYYY-MM-DD; CAD; Letter; bilingual EN/FR documents (Quebec)
Document layouts	Invoice, PO, packing slip, check, 1099	Invoice with VAT breakdown, credit note, remittance	Rechnung with §14 UStG mandatory fields, Gutschrift, Mahnung levels	Facture with mentions obligatoires, avoir, relance	Tax invoice with GSTIN/HSN/IRN QR, debit/credit note, e-way bill	CFDI PDF representation with UUID/QR, complemento de pago	Invoice with GST/HST registration, bilingual variants
Fiscal retention	7 years	6 years	10 years (GoBD)	10 years	8 years	5 years	6 years

Common to every pack: ISO 3166/4217/639 reference data, address formats and postal validation rules, tax-registration validators (EIN, UTR/VAT, USt-IdNr, SIREN/SIRET/TVA, GSTIN/PAN, RFC, BN/GST), number-sequence overrides, holiday calendars, and a regulator-schema fixture that must validate in CI before the pack can be released.

I. Testing standards

I.1 Test pyramid per service type

Service type	Unit	Property	Contract	Integration	E2E	Performance
Go engine (ledger, inventory, pricing, MRP, scheduling, costing, matching, settlement)	`go test`, table-driven, ≥ 85% line coverage on domain packages	`rapid` on invariants (balance, conservation, determinism, state machines)	OpenAPI contract via `oapi-codegen` validators; Pact for consumer contracts between contexts	Testcontainers Postgres + Kafka; outbox/inbox chaos	Through the shell for the wave demo path	Go benchmarks in CI (`benchstat` regression gate ±10%); k6 for throughput targets

Service type	Unit	Property	Contract	Integration	E2E	Performance
TypeScript service (workflow, integration, BFF, admin)	Vitest, ≥ 80% coverage	`fast-check` for mappers, parsers (EDI, GS1, bank files), policy routing	OpenAPI + AsyncAPI schema validation; Pact	Testcontainers + Temporal test server	Playwright	k6
Python service (forecasting, ML, agent evals)	pytest, ≥ 80%	`hypothesis` for numeric routines and parsers	JSON Schema on inputs/outputs	Docker compose fixtures	n/a	Benchmark notebooks exported to CI metrics
React shell and module-federated workspaces	Vitest + Testing Library	`fast-check` for form state reducers	Storybook interaction tests; generated-form snapshot per entity	MSW-backed integration	Playwright + axe across every generated form and workspace	Lighthouse CI
PWA (shop floor, warehouse, field)	Vitest	`fast-check` for the offline queue (commutativity, idempotence)	Same as shell	Device emulation	Playwright on emulated Android and tablet, offline toggling	Scan-to-ack latency budget 300 ms

The ratio target per service is roughly 70% unit/property, 20% contract/integration, 10% E2E/performance by count; every wave's proof items map to at least one row.

I.2 Fixture naming

```
fixtures/<context>/w<NN>-<slug>/<nnn>-<scenario>.<json|csv|ndjson>
fixtures/<context>/w<NN>-<slug>/<nnn>-<scenario>.expected.<json|csv>
fixtures/<context>/w<NN>-<slug>/README.md               -- what the set covers, source, size
fixtures/<context>/w<NN>-<slug>/REVIEW.yaml              -- accountant review record (I.5)
```

Example: `fixtures/ledger/w12-currency-parallel-ledgers/017-revaluation-eur-usd-unrealized.json`. Scenario numbers are stable once published (a retired scenario keeps its number with `retired: true`). Shared master-data seeds live in `fixtures/seed/<demo-company>/` (`USMF` manufacturer, `DEMF` process plant, `GBSI` services entity, `MXDS` distributor) and are loaded by the migration framework, so fixtures and demos use the same data.

I.3 Property-based testing

Libraries are fixed: Go `pgregory.net/rapid`, TypeScript `fast-check`, Python `hypothesis`. Every engine declares its invariants in a `PROPERTIES.md` next to the code, and each invariant has one property test with a minimum of 1,000 generated cases in CI (10,000 nightly). Required invariant families: conservation (Σ debits = Σ credits; Σ inventory transactions = on-hand; Σ allocations = source), determinism (same input → same output across runs and goroutines), idempotence (apply twice = apply once), monotonicity (sequences, versions), state machines (every illegal transition rejected), and round-trips (serialize/parse, UoM conversions, currency rounding). Shrunken counterexamples are committed as regression fixtures with the originating seed.

I.4 Golden files

Golden outputs (generated migrations, OpenAPI specs, rendered documents, statement packs, bank files, EDI messages, Rego bundles) live in `testdata/golden/` next to the test. A golden test fails on any byte difference. Updates happen only through `make update-golden PKG=<path>`, which rewrites files and records the reason in the commit; a PR that touches `testdata/golden/` must carry the label `golden-update` and a reviewer other than the author must approve (enforced by CODEOWNERS). Rendered PDFs are compared as rasterized pages with a 0.1% pixel tolerance; XML and JSON are canonicalized before comparison; timestamps and ids are normalized by the test harness.

I.5 Accountant-reviewed fixtures

Finance waves (11–15, 18–20, 29, 36–41, 43) require fixture sets reviewed by a practicing accountant before the wave can close. Process: (1) the engineer drafts scenarios and expected results in a workbook (`.xlsx`) exported to `.expected.csv`; (2) the reviewer works the scenarios independently and signs `REVIEW.yaml` (`reviewer`, `credential`, `reviewed_on`, `standard` such as ASC 606, IAS 21, IRS Pub. 946, `scope`, `exceptions[]`); (3) CI fails if a finance fixture set lacks a `REVIEW.yaml` dated after its last change; (4) any later change to the set re-opens review. The reviewer owner is tracked in the PRD validation checklist.

I.6 Reconciliation checks in CI

A scheduled job (`reconcile-daily`) runs at 02:00 UTC on stage against the demo companies and on every merge to `main` against the fixture databases; any non-zero variance fails the job and pages the wave owner.

Check	Ties	Tolerance	From wave
Trial balance	Σ debits − Σ credits per ledger per period	0.00	11
Balance projection	`Balance` table vs re-aggregated `JournalLine`	0.00	11
Parallel ledgers	Each ledger's trial balance sums to zero	0.00	12
Tax	Σ `TaxTransaction` per tax code vs GL tax accounts	0.00	13
Statements	Balance sheet balances; P&L net = retained earnings movement; cash flow = Δ cash	0.00	15
Inventory quantity	Σ `InventoryTransaction` per dimension vs `OnHand`	0	16
Inventory value	Inventory value report vs GL inventory accounts by item group	0.00	20
GR/IR	Σ receipts − Σ invoices per PO line vs GR/IR account	0.00	18
AP	Σ open vendor items vs AP control accounts by posting profile	0.00	19
WIP	Open production order cost vs GL WIP accounts	0.00	21

Check	Ties	Tolerance	From wave
AR and deferred revenue	Σ open customer items vs AR control; revenue schedule balance vs deferred revenue	0.00	29–30
Bank	Reconciled statement balance vs GL bank account	0.00	36
Fixed assets	Σ NBV per book vs GL asset and accumulated depreciation	0.00	37
Cost accounting	Primary costs in cost ledger vs GL	0.00	38
Intercompany	Due-to/due-from pairs net to zero per pair	0.00	40
Projects	Project WIP/accrued revenue vs GL	0.00	41
Payroll	Payroll register vs GL payroll accounts	0.00	43
Semantic layer	Revenue/COGS metrics vs GL	0.00	46
Audit chain	Hash chain verifies end-to-end	0 breaks	5

I.7 Performance and E2E harness

k6 scripts live in `perf/<context>/<scenario>.js` with thresholds encoded in the script (`http_req_duration{scenario:post}: p(99)<200`); CI runs smoke profiles on every merge and full profiles nightly on stage, publishing results to Grafana with a trend alert at +10% over 7 days. Playwright suites live in `e2e/<workspace>/` with page objects per generated form type and run against the stage URL after Argo reports healthy; every suite calls `axe` on each page it visits and fails on serious/critical violations. Flaky tests are quarantined by label for at most one wave before deletion or repair.

J. Traceability matrix

Each master-PRD functional requirement maps to its primary wave (FR-p.n → wave 5p + n), the later waves that extend it, the phase-exit criterion that proves it, and the test types that are primary evidence. Phase-exit criteria are the proof items of waves 5, 10, 15, 20, 25, 30, 35, 40, 45 and 50.

FR	Requirement (short)	Primary wave	Extended by	Phase-exit criterion	Primary test types
FR-0.1	GitOps deploy, signed images, SBOM, digest rollback	1	48, 50	Wave 5 platform demo: deploy → login → extend → approve → trace	CI negative tests, timed rollback, cosign/Grype gates
FR-0.2	Federated identity; roles/duties/privileges → OPA; RLS	2	47, 48	Wave 5 demo (login + permission viewer)	Contract (20 duties), RLS per table, federation in CI
FR-0.3	Metadata-declared entities generate schema/API/UI; no-code extensions	3	5, 46	Wave 5 demo (extend entity live)	Golden generation, backward-compat, load 1,000 rps
FR-0.4	CloudEvents via outbox; Temporal workflows and batch	4	47	Wave 5 demo (approval + trace)	Chaos outbox, Temporal timer tests, schema-registry gate
FR-0.5	Shell (WCAG AA), gap-free sequences, immutable audit, tracing	5	46	Wave 5 demo signed off	Playwright + axe, Jepsen-style sequence test, audit E2E
FR-1.1	Enterprise structure, hierarchies, calendars, ISO data, UoM	6	14, 40	Wave 10: demo master data fully migrated	Property (as-of hierarchy), UoM round-trip, calendar fixtures
FR-1.2	Single party model, dedup/merge, tax-id validation, governance	7	30, 42	Wave 10	Dedup precision/recall, merge integrity audit, Temporal approval
FR-1.3	Products, dimension groups, released products, GS1, configurator	8	24, 27	Wave 10	Variant explosion perf, configurator fixtures, GTIN vectors
FR-1.4	BOMs, formulas, routings, resources, ECM	9	21, 24, 39	Wave 10	Explosion golden + perf, effective-date tests, ECO workflow
FR-1.5	Migration framework with AX/SAP connectors and reconciliation	10	49	Wave 10: synthetic AX extract loads at 100% reconciliation	Load timing, idempotent re-run, mapping DSL units
FR-2.1	Append-only universal journal, dimensions, idempotent posting, balances	11	14, 15, 40	Wave 15: closed period with statements tying to TB	Property (1M entries), negative posting tests, 5k lines/s k6
FR-2.2	Three currencies, rate providers, parallel ledgers, revaluation	12	40	Wave 15	Rounding property, parallel-ledger fixtures, revaluation workbook
FR-2.3	Tax determination/calculation/posting, providers, SAF-T	13	35, 48	Wave 15	500-scenario matrix, settlement reconciliation, XSD validation
FR-2.4	Period control, close checklist, allocations, accruals, year-end	14	38	Wave 15: recorded close walkthrough	Contract (closed period on every path), allocation fixtures
FR-2.5	Statement designer, standard statements, drill-down, audit report	15	40, 46	Wave 15 sign-off by finance stakeholders	Golden statements, drill-down property, designer round-trip
FR-3.1	Immutable inventory transactions, journals, reservations, counting	16	31, 33	Wave 20: buy → receive → pay → value walkthrough	Concurrency (no negative), rebuild compare, FEFO test
FR-3.2	Requisition → RFQ → agreement → PO with policies and versions	17	35, 38	Wave 20	Approval matrix, versioning tests, price-lookup property
FR-3.3	Receiving with GR/IR, invoice capture, 2/3-way match, ERS	18	35	Wave 20	200-scenario matching fixtures, GR/IR nets to zero
FR-3.4	AP settlement, payment proposals, ISO 20022/NACHA/check, positive pay	19	36	Wave 20	300 settlement fixtures, pain.001 XSD, idempotent run
FR-3.5	FIFO/LIFO/average/standard costing, close, GL reconciliation, portal	20	39	Wave 20	Costing workbooks, close timing, value report ties to GL
FR-4.1	Production order lifecycle, WIP, variances, subcontracting	21	24, 25, 39	Wave 25: planned order → finished goods with full costing	State-machine property, costing fixtures, backflush test
FR-4.2	MRP/MPS, coverage groups, messages, pegging, forecasting	22	23, 27	Wave 25	40 APICS scenarios, scale 50k items, pegging completeness

FR	Requirement (short)	Primary wave	Extended by	Phase-exit criterion	Primary test types
FR-4.3	Finite/infinite scheduling, Gantt, CP-SAT	23	44, 45	Wave 25	OR-Library fixtures, no-double-booking property, 5k ops perf
FR-4.4	Process manufacturing and lean	24	33	Wave 25	Potency/co-product fixtures, kanban simulation, catch-weight tests
FR-4.5	Shop-floor kiosk, time registration, IoT, OEE, offline	25	43, 44	Wave 25	Kiosk E2E, 10k events/min ingest, 30-min offline test
FR-5.1	Deterministic pricing engine with explain	26	30, 40	Wave 30: one simulated month buy → make → sell → collect, all subledgers reconcile	500 pricing fixtures, determinism property, p95 < 20 ms
FR-5.2	Quotations, orders, ATP/CTP, credit, holds, intake API	27	35	Wave 30	ATP fixtures + over-promise property, credit workflow, 500 orders/min
FR-5.3	Release, pick/pack/ship, packing slips with COGS, carriers, drop-ship	28	31, 32, 34	Wave 30	Posting contract tests, 1,000-order E2E, SSCC validation
FR-5.4	Invoicing variants and ASC 606/IFRS 15	29	40, 41	Wave 30	Totals property, 40 ASC 606 fixtures, roll-forward reconciliation
FR-5.5	Cash application, collections, RMA, rebates, customer portal	30	36, 47	Wave 30 GO-LIVE CANDIDATE sign-off	Cash-application fixtures, dunning tests, RMA E2E
FR-6.1	Location directives, work, waves, license plates, replenishment, loads	31	34	Wave 35: EDI PO → ASN → invoice between demo partners	100 directive fixtures, 10k-line simulation, capacity property
FR-6.2	Mobile scanner app, flow designer, GS1, offline, labels	32	45	Wave 35	Device E2E, GS1 parser suite, 200-transaction offline sync
FR-6.3	Quality orders, sampling, usage decisions, NCR/CAPA, CoA	33	44	Wave 35	Trigger fixtures, blocking property, Z1.4 tables
FR-6.4	Load/route planning, rating, tendering, freight reconciliation, landed cost	34	40	Wave 35	Solomon VRP fixtures, 200 rating fixtures, tender timeouts
FR-6.5	EDI X12/EDIFACT, Peppol, integration hub, developer portal	35	48	Wave 35	Partner-simulator round trips, schematron, webhook at-least-once
FR-7.1	Bank accounts, statements, reconciliation, cash forecast, connectivity	36	48	Wave 40: group consolidated statements, finance parity sign-off	1,000-line reconciliation, forecast property, EBICS contract tests
FR-7.2	Fixed assets, multi-book depreciation, CIP, IFRS 16	37	41	Wave 40	100 asset fixtures vs MACRS tables, lease workbook, subledger tie
FR-7.3	Budgeting, budget control, cost accounting, ABC	38	41	Wave 40	Budget-check fixtures, allocation convergence, cost ledger tie
FR-7.4	Costing sheets, roll-up, standard cost activation, variances	39	—	Wave 40	50-product workbook, activation revaluation, variance sums
FR-7.5	Intercompany trade/accounting, consolidation with eliminations	40	—	Wave 40 sign-off by a practicing controller	IC balance property, 3-entity eliminations, mismatch detection
FR-8.1	WBS, resources, journals, billing modes, WIP/POC, EVM	41	45	Wave 45: projects, HR, payroll, maintenance, field service walkthrough	POC workbook, invoice-proposal property, WBS scheduling
FR-8.2	Workers/positions/jobs, absence, benefits, self-service, PII	42	43	Wave 45	Effective-date property, accrual fixtures, PII contract tests
FR-8.3	Country-pluggable payroll, expense management	43	48	Wave 45	200 Pub. 15-T scenarios, retro fixtures, payroll GL tie
FR-8.4	Functional locations, maintenance plans, work orders, IoT triggers	44	—	Wave 45	Plan fixtures, cost posting tie, 5 s IoT trigger
FR-8.5	Installed base, agreements/SLAs, cases, dispatch, technician app	45	—	Wave 45	Entitlement fixtures, dispatch property, warranty billing
FR-9.1	Lakehouse, semantic layer, dashboards, document designer, catalog	46	—	Wave 50: first customer month-end in production; load and DR targets met	Lag < 60 s, metric = GL check, RLS in dashboards, visual regression
FR-9.2	MCP APIs, copilot, governed agents, evals, cost budgets	47	—	Wave 50	Eval thresholds in CI, permission-ceiling contracts, audit of actions
FR-9.3	Localization framework, country packs, ER designer, SoD, SOC 2, privacy	48	—	Wave 50	Regulator-schema fixtures, 50 SoD conflicts, restore drill, pen test
FR-9.4	Transactional migration packs, parallel-run diff, cutover, coexistence	49	—	Wave 50	100% control totals, zero unexplained variance, 48 h rehearsal
FR-9.5	Scale targets, multi-region DR, operations, first customer, v1.0	50	—	Wave 50	k6 campaign, DR game day, production close, signed release

K. Reference index

Consolidated from the `references` arrays in `roadmap/waves.json` (183 distinct URLs, deduplicated by URL). Wave numbers in parentheses. SAP Help URLs are the identifiers recorded in `waves.json`; verify against the current SAP Help portal before citing in ADRs.

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L. Competitive benchmark — top-tier manufacturing ERPs

L.1 Capability matrix

The matrix compares the six systems a mid-to-large manufacturer shortlists against the Passage v1.0 target. Cells name the actual feature or module and give typical numbers where they are public or widely reported by customers and analysts; "n/a" means the capability is absent or only available through a

partner product. Figures marked "typical" are customer-reported ranges, not vendor commitments, and should be re-validated per deal.

Capability	SAP S/4HANA	D365 F&SCM	Infor LN / CloudSuite Industrial	Epicor Kinetic	IFS Cloud	Plex	Passage target
MRP full regeneration, 50k items	MRP Live on HANA: minutes (typically < 10 min; MD01N)	Planning Optimization (Azure service): typically 5–30 min; legacy master planning: hours	LN: batch overnight typical; CSI (SyteLine) APS: minutes to hours	MRP regeneration batch, typically overnight; net-change intraday	MRP/MS batch, typically hours; incremental supported	Net-change MRP continuous, minutes	< 10 min for 50k items × 12 periods × 5 BOM levels; deterministic; net-change continuous
Live pegging / explain plan	MD04 stock/requirements list; PP/DS pegging	Planning Optimization pegging view	LN order pegging (project pegging); CSI APS pegging	Time Phase inquiry pegging	Supply/demand pegging in Supply Chain Planning	Limited (order-level links)	Pegging tree API explaining 100% of any planned order's quantity
Finite capacity scheduling	Embedded PP/DS with optimizer (extra license)	Finite capacity in master planning + Gantt; no optimizer	Infor Scheduling / CSI APS finite scheduling	APS module (add-on), finite with rules	Constraint-Based Scheduling (CBS) module	Production Scheduling (finite, drag-drop)	Finite/infinite + CP-SAT optimizer built in; 5k operations < 60 s
Shop-floor execution UX	SAP Digital Manufacturing (separate cloud, extra license) or MII	Production floor execution (touch, job cards)	Infor Factory Track (separate)	Kinetic MES (Advanced MES, ex-Mattec)	Shop Floor Workbench; IFS MES	Native MES, machine-integrated; strongest of the group	Touch PWA kiosk, 30-min offline, time → costing and payroll in one flow
Machine / IoT connectivity	SAP DM Edge, SAP IoT	Sensor Data Intelligence (Azure IoT Hub)	Infor OS IoT + Factory Track	Advanced MES direct PLC monitoring	IoT Business Connector	Native PLC data collection	OPC-UA/MQTT edge connector → Kafka; 10k events/min per plant, no loss on broker restart
OEE out of the box	In SAP DM	Via Sensor Data Intelligence / Power BI; limited native	Factory Track	Advanced MES real-time OEE	Via MES; limited native	Native real-time OEE	Native, ISO 22400 definitions, 5 s refresh
Product configurator (CTO)	Variant Configuration (LO-VC / AVC)	Product configuration models (constraint-based)	Infor CPQ (separate); LN configurator	Kinetic Configurator (rules-based)	Sales Configurator	Basic option selection	Constraint-based (OR-Tools CP-SAT), 50-constraint fixtures, drives BOM/route
Process manufacturing: formula, potency, co/by-products, catch weight	PP-PI recipes, batch management, active ingredient	Full: formulas, potency, co/by-products, catch weight, batch attributes	LN: limited; CloudSuite Food & Beverage (M3) strong	Limited in Kinetic	Recipe/formula, batch; limited potency	Lot, catch weight; formulas for food/bev	Wave 24: formulas, potency, co/by allocation, catch weight, shelf life, batch disposition
Lean / kanban	KANBAN (PK*), REM	Lean manufacturing: production flows, kanban rules/boards	CSI kanban basic	Kanban basic	Kanban/repetitive basic	Kanban native	Production flows, event/fixed/withdrawal kanban, backflush costing
Quality management	QM: inspection lots, usage decisions, notifications; CAPA via QIM	Quality orders, nonconformances, CoA, test groups	Infor Quality / LN quality basic	Advanced Quality (AQM) with NCR/CAPA	Quality Management: control plans, nonconformance	Native: control plans, SPC, gage R&R, PPAP	Rule-triggered quality orders, ANSI Z1.4 AQL, 8D CAPA, SPC, CoA, supplier scorecards, calibration
Advanced WMS	EWM (basic embedded or advanced, extra license)	Advanced warehouse: location directives, waves, LPs	Infor WMS (separate); LN warehouse basic	Kinetic warehouse basic + Advanced Mobile Warehouse	Warehouse data collection (WADACO) basic	Native basic/mid WMS	Location directives, work templates, waves, LPs, replenishment, slotting, dock scheduling
WMS mobile	EWM RF framework / Fiori	Warehouse Management mobile app (Android, configurable flows)	Factory Track mobile	Mobile Warehouse	Aurena mobile scanning	Native mobile	PWA + Android shell, flow designer, offline queue, GS1 AI parsing, ZPL printing
Transportation management	SAP TM (embedded)	TMS native (rating, routing, load building)	Infor Nexus (separate)	n/a (partner)	Limited	n/a	VRP load/route planning, rate shopping, tendering, freight reconciliation, landed cost
Inventory costing methods	Standard, moving average, actual (material ledger)	FIFO, LIFO, weighted average, standard, moving average	Standard, actual, FIFO/LIFO (lot)	Standard, average, last, FIFO, lot FIFO	Standard, weighted average, FIFO (lot), actual	Standard, actual (average)	FIFO, LIFO, weighted average (running/periodic), standard, moving average; close ties to GL
Universal journal (single line-item truth)	ACDOCA: one table for FI/CO/AA/ML	GeneralJournalAccountEntry + separate subledger tables	GL with integration transactions	GL with posting (booking) rules	GL vouchers	GL	Append-only journal; balances are rebuilt projections; every subledger reconciles daily by automated check
Parallel ledgers / multi-GAAP	Leading + non-leading ledgers, currency types 10/30/40	Posting layers (current/operations/tax + custom)	Limited (parallel companies / dimensions)	Multiple GL books	Parallel accounting (IFRS/local)	n/a	Ledger groups per legal entity; 3 currency amounts per line
Month-end close, typical	3–5 days with Advanced Financial Closing	5–8 days	5–10 days	5–10 days	5–10 days	5–10 days	≤ 3 days; continuous close via daily reconciliation (Appendix I.6)

Capability	SAP S/4HANA	D365 F&SCM	Infor LN / CloudSuite Industrial	Epicor Kinetic	IFS Cloud	Plex	Passage target
ASC 606 / IFRS 15	Revenue Accounting and Reporting (RAR)	Revenue recognition module (being replaced by subscription billing)	n/a (partner)	n/a	Project-based rev rec	n/a	Native: performance obligations, SSP allocation, ratable/milestone/usage schedules
Intercompany & consolidation	Cross-company postings; Group Reporting	Intercompany trade; consolidation	CloudSuite consolidation	Multi-company, consolidation	Group consolidation	Multi-plant; limited consolidation	Mirror documents, due-to/due-from matching, eliminations, IAS 21 translation
EAM and field service	PM + Field Service Management	Asset Management + Field Service (separate app)	Infor EAM (now HxGN EAM, separate)	Maintenance basic	Strongest EAM/FSM of the group	Basic maintenance	Native EAM (plans, work orders, IoT triggers) + field service (dispatch optimization, technician app)
Localization coverage	60+ country versions	40+ countries	40+ (LN)	~20 country packs	50+ countries	US-centric, ~10	7 packs in v1.0 (US, UK, DE, FR, IN, MX, CA); framework for v1.x
Extensibility model	ABAP (key-user, developer, BTP side-by-side)	X++ extensions (no overlayering), Power Platform	ION, Mongoose (CSI), LN Extensions	Application Studio (low-code), BPM, C#	Aurena low-code configuration + PL/SQL	Configuration; limited custom code	Metadata extensions (entities, enums, fields, workflows, forms, reports) + sanctioned hooks; no proprietary language
API openness	OData/SOAP on SAP Business Accelerator Hub (thousands, uneven coverage)	OData data entities (2,000+), custom services	ION APIs (REST), BODs	REST API v2 (OData) across the product	REST projections (OData) across the product	REST/SOAP via developer portal	OpenAPI 3.1 generated for every entity and engine; GraphQL read layer; additive-only within /v1
Event streaming	Event Mesh / Advanced Event Mesh (CloudEvents)	Business events → Event Grid / Service Bus / Dataverse	ION BOD publishing (XML)	Limited (Epicor Functions, webhooks)	IFS Connect events	Limited webhooks	Every state change is a replayable CloudEvent on Kafka via outbox; AsyncAPI published
Analytics / data access	Embedded CDS analytics; Datasphere	Entity store; Synapse/Fabric link	Birst; Data Lake	Epicor Data Analytics (Grow)	Analysis models; IFS Analytics	Plex Intelligence	Debezium CDC → Iceberg lakehouse, Cube semantic layer, p95 lag < 60 s, open to any BI tool
AI agents	Joule agents; Business AI	Copilot; autonomous agents (reconciliation, supplier comms)	Infor GenAI / Industry AI agents	Prism agents	IFS.ai agents	Copilot features	MCP-exposed APIs; agents inherit user OPA permissions; evals gate deploys; model-agnostic
Audit trail	Change documents (per table, configurable); AIS	Database log (opt-in per table, performance cost)	Configurable audit trail	Change log per table (opt-in)	History logging (opt-in)	Change logs	Field-level from CDC on every table, hash-chained, 7-year fiscal retention
Segregation of duties	GRC Access Control (separate)	Native SoD rules	Infor GRC (separate)	n/a	n/a	n/a	Native SoD engine with mitigation workflow and certification campaigns
Deployment model	Public cloud, private cloud (RISE), on-premise	Microsoft-managed Azure only (on-premise LBD retired)	Multi-tenant AWS (CloudSuite) or on-premise LN	Epicor-managed Azure or on-premise	IFS-managed Azure or customer-managed Kubernetes (remote deployment)	SaaS only, multi-tenant	Any conformant Kubernetes (EKS/GKE/AKS/bare-metal/k3d); self-hosted or managed; no provider-managed service in the critical path
Upgrade cadence and effort	Annual (on-prem) / biannual (public cloud) mandatory; on-prem upgrades are multi-month projects	4 mandatory service updates per year (One Version); regression testing each	Monthly cloud updates; LN on-prem by project	Biannual releases; cloud updates mandatory	Biannual releases + monthly service updates (evergreen)	Continuous (weekly)	Continuous; zero-downtime expand/contract migrations; digest rollback < 15 min; customer-controlled cadence
Licensing	Named users (Professional/Functional), FUE under RISE; list prices not public	Per user per month (Finance / SCM list ≈ USD 180–210 per full user; Activity/Team tiers lower)	Subscription per user; not public	Subscription per user; not public	Subscription per user; not public	Subscription per plant and user; not public	Open question 1 (decide by wave 35): open-core or source-available; no per-seat charge for shop-floor, portal or agent users
Implementation, mid-size manufacturer (typical)	12–24 months	9–18 months	6–12 months (CSI)	6–12 months	9–18 months	6–12 months	3–6 months using the migration factory (wave 49) and parallel-run diff
Accessibility	Fiori: WCAG 2.1 AA for most apps	WCAG 2.1 AA	Partial	Partial	Partial	Partial	WCAG 2.1 AA on every generated form and workspace, axe in CI

Reading the matrix: the incumbents win today on breadth of localization, installed-base references and depth in a few modules (SAP finance, IFS EAM, Plex MES). Passage does not try to out-feature them row by row in v1.0; it competes on the rows where every incumbent is structurally weak — one journal with automated reconciliation, metadata extensibility without a proprietary language, event streaming and open APIs as the default integration surface, cloud-agnostic deployment, continuous zero-downtime upgrades, and governed AI agents that act under the same authorization as users.

L.2 Interoperability contract

Principle: any system on the customer's network — MES, PLM, SCADA/OPC-UA, a third-party WMS, EDI trading partners, BI tools, AI agents — consumes Passage through the same APIs, events and MCP tools that Passage's own modules use. There is no separate "integration layer" product, no licensed middleware, and no integration project to stand up before the first interface works. What SAP sells as Integration Suite, Microsoft as dual-write plus Logic Apps, and Infor as ION is, in Passage, the integration hub shipped in wave 35: a catalog, keys and quotas, a message monitor and replay over surfaces that already exist from wave 3 (REST), wave 4 (Kafka) and wave 47 (MCP).

The contract every surface honors:

1. **One authorization model.** Every call, event subscription, webhook and tool invocation is authorized by the same OPA bundle against the same roles/duties/privileges (Appendix E). Partner credentials are users or service accounts in the tenant's Keycloak realm; there is no integration back door.
2. **One schema source.** OpenAPI, AsyncAPI, GraphQL SDL, MCP tool definitions and EDI mappings are generated from the same entity metadata, so a field added by a customer extension appears on every surface in the same release.
3. **Additive evolution.** `/v1` and `v<n>` event types change additively only; breaking changes ship as a new version with a 12-month (API) or two-minor-release (event) overlap, announced with `Deprecation`/`Sunset` headers and registry flags.
4. **Correlation and idempotency everywhere.** Every inbound request carries or receives a `correlationid`; every outbound event and webhook carries it; every mutating call is idempotent by key (Appendix D.5); every consumer is expected to be idempotent by event id (Appendix C.5).
5. **Replayability.** Events are retained 30 days and replayable per consumer; webhooks can be re-driven from the message monitor; file and EDI inbound payloads are kept as source attachments.
6. **Sandbox and conformance.** Every tenant gets a sandbox tenant with the demo companies; a conformance kit (Postman/k6/Pact suites) lets a partner prove their integration before production keys are issued.
7. **Observability shared with the customer.** Per-key traces, delivery logs, DLQ contents and SLA dashboards are visible in the integration hub, not only to Passage operators.
8. **No data hostage.** Full export (NDJSON per entity, Iceberg tables, event archive) is a standard API, not a service engagement.

Surface	Typical consumers	Protocol / format	Authentication and authorization	SLA (per tenant, measured)	Notes
REST	MES, PLM, custom apps, iPaaS, mobile	HTTPS/JSON, OpenAPI 3.1, RFC 9457 errors	OAuth 2.0 (Keycloak): authorization code for users, client credentials for services; JWT validated at Envoy, OPA per request; tenant API keys for partner quotas	99.9% availability; p95 < 50 ms simple reads, p99 < 500 ms document posts; rate limits per tier (D.7)	Idempotency keys, ETags, cursor pagination, bulk `:batch`/`:export`
GraphQL	BI tools, portals, composite screens, low-code builders	HTTPS, GraphQL over the generated SDL (read-only in v1.0; mutations via REST)	Same JWT and OPA; field-level masking for P2/P3 per duty	99.9%; p95 < 200 ms for queries ≤ 1,000 cost points; depth ≤ 8, cost-limited	Persisted queries for portals; mirrors REST filter grammar
Kafka topics	MES, third-party WMS, data platform, CRM sync, customer event consumers	Kafka protocol, CloudEvents JSON, AsyncAPI 3.0, schema registry	mTLS + SCRAM-SHA-512 per principal; topic ACLs per tenant; cross-tenant principals only for the customer's own analytics	99.9% broker availability; p95 commit-to-consumer < 5 s; 30-day retention; replay on request	External consumers connect through the tenant's dedicated consumer group; DLQ visible in hub
Webhooks	SaaS partners, iPaaS (Workato, Make), Salesforce/Shopify connectors	HTTPS POST of the CloudEvent, JSON	HMAC-SHA256 signature in `X-Passage-Signature` with per-endpoint secret (rotation supported), optional mTLS, IP allow-list	At-least-once; p95 first delivery < 10 s; retries with exponential backoff for 24 h, then DLQ; per-tier delivery quotas	Subscriptions filtered by event type and legal entity; re-drive from monitor
MCP	AI agents (Passage copilot, Claude, customer-built agents, partner agents)	MCP over HTTPS (streamable), tool/resource definitions generated from OpenAPI and the semantic layer	OAuth 2.1 with user-delegated token; agent identity registered in the tenant; OPA evaluates `agent@user` so the agent never exceeds the user (E.4 example 2); kill switch	99.9%; p95 tool call < 1 s excluding downstream posting time; per-tenant token and cost budgets	Every tool call audited with inputs/outputs; approval gates for posting tools
EDI	Trading partners, 3PLs, carriers, retailers	X12 (850/855/856/810/820/214) and EDIFACT (ORDERS/DESADV/INVOIC) over AS2 or SFTP; Peppol UBL 2.1 via access point	AS2 certificates per partner, SFTP key pairs, Peppol access-point credentials; partner profile binds to a service account with scoped duties	997/CONTRL functional acknowledgment within 15 min; inbound processing p95 < 5 min; partner SLA monitor with alerting	Mappings versioned per partner; replay from message monitor; partner simulator in sandbox
OPC-UA / MQTT	SCADA, PLCs, machines, sensors, historians, Andon displays	OPC-UA client (Sign & Encrypt) and MQTT 3.1.1/5 over TLS via the edge connector; events normalized to CloudEvents on Kafka	X.509 device certificates issued per plant; per-device MQTT credentials; connector runs under a plant service account	10k events/min per plant; store-and-forward on edge for 24 h; end-to-end p95 < 5 s to OEE/condition monitoring; no loss on broker restart	Tag/topic mapping is metadata; thresholds raise maintenance requests (wave 44)
File / SFTP	Banks, payroll providers, tax providers, legacy systems, lockbox	SFTP (ISO 20022 pain/camt, NACHA, BAI2, MT940, CSV/NDJSON), optional PGP	SFTP key authentication per partner, PGP keys and bank signing keys in Vault (never exported), checksum manifests	Polling every 5 min; inbound processing p95 < 15 min; outbound files signed and audited (E.7); bank acknowledgments (pain.002) ingested	Idempotent by file hash; duplicates rejected and logged; retention per fiscal rules

A customer therefore adds an MES by subscribing it to `passage.manufacturing.productionorder` and `passage.inventory.transaction`, pointing it at `/v1/manufacturing/production-orders/{id}:report` for confirmations, and issuing it a service account with the `ReportProduction` duty — the same three artifacts Passage's own shop-floor kiosk uses. Replacing that MES later means revoking one principal and one subscription, not unwinding a middleware project.

Part 12 — Coverage Inventory: Dynamics AX 2012 R3 / D365 F&SCM → Passage

Status	Draft v0.1 — completeness oracle for the 50-wave roadmap
Date	2026-10-21
Source of truth for waves	roadmap/waves.json (wave numbers 1–50); roadmap/phases.json
Scope rules	docs/PRD.md §2 goals and non-goals, §5 scope map
Purpose	The team's fear is "missing something". This part walks the AX 2012 R3 navigation pane module by module (plus the D365 F&SCM additions that matter to a manufacturer) and assigns every form, inquiry, report, periodic job and setup form to a Passage wave — or says, honestly, that nothing covers it yet.

12.0 How to read this inventory

Each module section mirrors the AX 2012 R3 menu structure and has four tables:

1. **Common / Journals** — every list page, details form and journal the user opens from *Common* or *Journals*. Columns: Form name | AOT form or menu item (best effort from memory of the AX 2012 R3 AOT; treat as a search key, not gospel) | Purpose | Passage status | Wave | Notes.
2. **Inquiries & Reports** — every *Inquiries* entry and every SSRS report under *Reports*. Columns: Name | Purpose | Passage status | Wave | Notes.
3. **Periodic** — every periodic job or process. Same columns as 2.
4. **Setup** — every setup form, with the module's *parameters* form listed first and its tabs spelled out as separate rows where the tab changes behavior that a wave must implement. Columns: Name | Key parameters / options | Passage status | Wave | Notes.

Status legend (the *Passage status* column always starts with one of these tokens so the document can be counted mechanically):

- `COVERED-<wave>` — a deliverable in waves.json for that wave clearly implements the AX capability, at parity or better.
- `PARTIAL-<wave> (what's missing)` — the wave covers the core but a named piece of the AX feature is not in any wave deliverable. The missing piece is listed in the gap register (§12.99) with a proposed wave.
- `GAP` — no wave in waves.json clearly covers the item. The Notes column always begins with `Propose wave NN:` so the gap register can be generated from the rows. Wave column shows `-`.
- `OUT-OF-SCOPE (reason)` — excluded by a PRD non-goal (CRM/marketing automation, retail POS/e-commerce, country packs beyond US/UK/DE/FR/IN/MX/CA, X++/ABAP code migration) or because AX itself deprecated it before R3. Wave column shows `-`.
- `SUPERSEDED (how)` — Passage deliberately does it a different way; the replacing mechanism is named. Wave column shows the wave that ships the replacement.

Conventions

- Wave numbers refer to `n` in waves.json. Phase boundaries: 0 = 1–5, 1 = 6–10, 2 = 11–15, 3 = 16–20, 4 = 21–25, 5 = 26–30, 6 = 31–35, 7 = 36–40, 8 = 41–45, 9 = 46–50.
- "SSRS report" rows are assessed against two things: the *data* (does a Passage wave produce it?) and the *rendering* (wave 46 replaces every SSRS layout with the operational report designer and embedded dashboards). A report is `COVERED-<n>` when the data wave exists and wave 46 renders it; it is `PARTIAL` only when the data itself is missing.
- Forms that are shared across modules (Terms of payment, Methods of payment, Charges codes, Print management, Form setup, Number sequences) are listed in every module where AX shows them, so each module section is self-contained for its reviewer. They map to the same wave each time.
- Where AX 2012 R3 and D365 F&SCM differ (e.g. Collections, Credit management, Cost accounting ledger, Landed cost), the D365 name is given in Notes and the row is assessed against the D365 behavior because that is what a manufacturer migrating today expects.
- Country-specific forms are listed only where they belong to one of the seven v1 country packs; everything else is `OUT-OF-SCOPE (country pack beyond v1 set)`.

12.1 General ledger

The general ledger is the heart of AX. In Passage it is the universal journal (wave 11) plus currencies/parallel ledgers (12), tax (13), period close (14) and financial reporting (15). Consolidation moves to wave 40; budgeting to wave 38.

12.1.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Main accounts (list page)	MainAccountListPage	Browse chart of accounts with balances FactBox	COVERED-11	11	Metadata-driven list page with balance projection FactBox
Main account details	MainAccount	Account type, category, DB/CR default, suspended, validation (currency/user/posting type), allocation terms, default dimensions, translations	COVERED-11	11	Account structure validation rules at posting API
Chart of accounts	LedgerChartOfAccounts	Shared COA header; main accounts belong to a COA shared by ledgers	COVERED-11	11	Shared COA across legal entities (tenant-level)

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Ledger	Ledger	Per-legal-entity ledger: COA, fiscal calendar, accounting/reporting currency, exchange rate types, account structures, balance control	COVERED-11	11	Ledger service entity; parallel ledgers in 12
Financial dimensions	DimensionAttribute	Define dimensions backed by entities (dept, cost center, customer…) or custom lists	COVERED-11	11	Up to 20 dims per ledger
Financial dimension values	DimensionAttributeValue (DimensionValueDetails)	Values with active dates, suspended, owner, group dimension	COVERED-11	11	Effective-dated values
Financial dimension sets	DimensionFocus	Balance views by dimension set, rebuild balances	COVERED-11	11	Balance projection by account × dimension × period (event-sourced)
General journal (list)	LedgerJournalTable (menu LedgerJournalTable)	List of journal batches by name with status/approval	COVERED-11	11	Journal workspace
General journal lines	LedgerJournalTransDaily	Voucher lines, account/offset, dimensions, tax, invoice, due date, payment info	COVERED-11	11	
Journal voucher (lines) — Fixed assets journal	LedgerJournalTransAsset	FA transactions from GL journal	COVERED-37	37	FA transaction types posted through journal
Periodic journals	LedgerJournalPeriodic	Recurring journal templates with period unit, last date, end date	COVERED-14	14	Recurring journal scheduling
Allocation journal	LedgerJournalTable (type Allocation)	Journal created by allocation rule processing, review before post	COVERED-14	14	Allocation engine output
Elimination journal	LedgerJournalTable (type Elimination)	Journal created by elimination proposals in consolidation company	COVERED-40	40	
Intercompany journal	LedgerJournalTable (intercompany lines)	Lines posting due-to/due-from across legal entities	COVERED-40	40	Due-to/due-from entries wave 40; posting API wave 11
Global general journal (D365)	LedgerJournalTable (global)	Enter journals for any legal entity without company switch	COVERED-11	11	Shell re-scopes per legal entity; cross-entity journal entry in workspace
Journal approval (workflow)	LedgerJournalTable + workflow	Submit journal for approval	COVERED-11	11	Temporal approval template (wave 4) applied to journals in 11
Posted journals (list)	LedgerJournalTable (posted)	Review posted batches, reverse	COVERED-11	11	Reversal creates linked negating entry
Voucher attachments / document handling	DocuView	Attach documents to vouchers	PARTIAL-11 (no document types, archiving policy or external store yet)	11	Attachments named in wave 11; general document management is a gap under Org admin §12.29
Advanced ledger entries (public sector)	LedgerAdvancedEntry (PSA)	Multi-line entries with posting definitions	OUT-OF-SCOPE (public sector feature set is not in the PRD scope map)	—	Public sector is not a target vertical
Journal names list	LedgerJournalName	Also reachable from Common in some role centers	COVERED-11	11	See Setup
Reverse transaction (from voucher)	LedgerTransReverse	Reverse one voucher / trace reversal	COVERED-11	11	
Reverse entire journal	LedgerJournalTableReverse	Reverse all vouchers of a batch	COVERED-11	11	
Fiscal calendars (Common link in R3)	FiscalCalendars	Shared calendars, years, periods	COVERED-6	6	Owned by Org admin in Passage

12.1.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Trial balance (inquiry, dimension set)	Opening/debit/credit/closing by account and dimension set, date range	COVERED-15	15	Drill-down to lines
Ledger transactions (Voucher transactions)	Lines by voucher with origin trace	COVERED-11	11	Journal lines inquiry with source-document link
Audit trail	Who posted which voucher when, journal/source	COVERED-15	15	Audit trail report (wave 15) over CDC audit (wave 5)
Transaction origin	Drill from a GL line to originating subledger document	COVERED-15	15	Statement → balance → lines → source document
Posted sales tax (Sales tax transactions inquiry)	Tax amounts by code/period	COVERED-13	13	
Settled sales tax (sales tax payments inquiry)	Settlement history	COVERED-13	13	
Ledger budget transactions inquiry	Budget register entries by account	COVERED-38	38	
Currency exchange rate inquiry	Rates by type/pair/date	COVERED-12	12	Rate audit
Dimension statement (Dimension balance inquiry)	Balances for a single dimension value	COVERED-11	11	Balance projection query
Closing sheet (periodic, country-dependent)	Closing adjustments worksheet (used in ES/IT/EE)	OUT-OF-SCOPE (country pack beyond v1 set; closing adjustments in Passage are allocation/accrual journals)	—	
Ledger reconciliation inquiry (Inventory/ledger, AR/AP)	Compare subledger to GL	COVERED-20	20	Daily automated subledger-to-GL reconciliation from wave 16; inventory value report wave 20
SSRS: Trial balance	Printed trial balance	COVERED-15	15	Rendered via wave 46 designer

Name	Purpose	Passage status	Wave	Notes
SSRS: Trial balance summary	Summary by account	COVERED-15	15	
SSRS: Ledger transaction list	Lines by date range	COVERED-11	11	
SSRS: Ledger account balance	Balance per account/period	COVERED-15	15	
SSRS: Dimension statement	Dimension-based P&L/balance	COVERED-15	15	
SSRS: Financial statement (old LedgerBalanceSheet row/column definitions)	Legacy AX statement engine	SUPERSEDED (report definitions in wave 15 designer, Management Reporter model)	15	AX 2012 had both; Passage keeps only the MR-style model
SSRS: Chart of accounts	List of accounts with attributes	COVERED-11	11	
SSRS: Journal (LedgerJournal report)	Unposted journal printout	COVERED-11	11	
SSRS: Posted journals	Posted batch printout	COVERED-11	11	
SSRS: Voucher (LedgerTransVoucher)	Single voucher with all lines	COVERED-11	11	
SSRS: Audit trail	Printed audit trail	COVERED-15	15	
SSRS: Ledger reconciliation (Ledger/Inventory, Ledger/AR, Ledger/AP, Ledger/Project, Ledger/Bank)	Reconciliation listings	COVERED-20	20	Reconciliation checks per subledger, wave 20 report; bank in 36; project in 41
SSRS: Sales tax payments	Settlement report	COVERED-13	13	
SSRS: Sales tax specification by ledger transaction	Tax by voucher	COVERED-13	13	
SSRS: Sales tax transactions — details	Tax detail listing	COVERED-13	13	
SSRS: Sales tax list / Sales tax codes	Setup listing	COVERED-13	13	
SSRS: EU sales list	Statutory EU sales listing	COVERED-13	13	
SSRS: Exchange rate (Currency exchange rates)	Rate table printout	COVERED-12	12	
SSRS: Fiscal year close report / Opening balances	Closing results	COVERED-14	14	
SSRS: Consolidation (Consolidated balances)	Consolidated trial balance	COVERED-40	40	
SSRS: Eliminations	Elimination detail	COVERED-40	40	
SSRS: Allocation rule report	Allocation definitions	COVERED-14	14	
SSRS: Journal names, Posting setup listings	Setup listings	COVERED-11	11	Any entity can be exported/printed from the shell (wave 5)
SSRS: Cash flow forecast (GL cash flow)	Cash flow forecast from open items	COVERED-36	36	
SSRS: Balance control (balance sheet with balance control accounts)	Control accounts	COVERED-15	15	
SSRS: Fiscal LIFO reports (Italy)	Italian fiscal LIFO valuation	OUT-OF-SCOPE (country pack beyond v1 set)	—	
SSRS: Withholding tax reports	Withholding by vendor/period	COVERED-13	13	1099 in AP
Management Reporter / Financial reporting (D365)	Row/column/tree report designer and viewer	COVERED-15	15	

12.1.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Journals → Periodic journals (retrieve)	Generate journal lines from periodic journal template	COVERED-14	14	
Journals → Allocation journal (Process allocation request)	Run allocation rules, create allocation journal	COVERED-14	14	Fixed, basis, iterative
Journals → Accrual schemes (Ledger accruals)	Spread an expense/revenue across periods	COVERED-14	14	
Journals → Ledger accrual (from journal line)	Apply scheme on a journal line	COVERED-14	14	
Fiscal year close → Fiscal year close (year-end close)	Transfer P&L to retained earnings, opening balances	COVERED-14	14	Per-dimension carry-forward
Fiscal year close → Opening transactions	Create opening transactions per account/dimension	COVERED-14	14	
Fiscal year close → Transfer opening balances (D365 year-end close templates)	Templates for retained earnings by dimension	COVERED-14	14	
Consolidate → Consolidate [Online]	Pull subsidiary balances into consolidation company	COVERED-40	40	
Consolidate → Consolidate [Export] / [Import]	File-based consolidation	COVERED-40	40	Import path for non-Passage subsidiaries
Consolidate → Elimination proposal	Generate elimination journal	COVERED-40	40	
Currency revaluation (Foreign currency revaluation, GL)	Unrealized gain/loss on monetary accounts	COVERED-12	12	Reversal next period
Currency revaluation → Reporting currency adjustment (D365)	Restate reporting currency balances	COVERED-12	12	Currency translation
Transaction reversal (Reverse transaction / Trace reversal)	Reverse posted vouchers	COVERED-11	11	
Sales tax payment (Settle and post sales tax)	Settle tax period to authority	COVERED-13	13	

Name	Purpose	Passage status	Wave	Notes
Sales tax → Sales tax report / EU sales list / VAT return	Statutory extracts	COVERED-13	13	VAT return, EU sales list, Intrastat, SAF-T
Conditional sales tax settlement	Settle tax when payment made (cash basis)	COVERED-13	13	Rules engine supports cash-basis trigger; verify fixture
Ledger balances rebuild (Financial dimension sets balance update)	Rebuild balance table	COVERED-11	11	Balance projection rebuild from journal
Close period (Ledger calendar period status change)	Open/on hold/closed per module	COVERED-14	14	Period control per ledger × module × user group
Financial period close workspace (D365)	Close checklist, tasks, dependencies	COVERED-14	14	
Delete / Compress transactions (Ledger transaction compression)	Compress old GL transactions	SUPERSEDED (append-only journal; history archived to lakehouse wave 46, never compressed in place)	46	
Reason code assignment (reversal reasons)	Require reasons on reversal	PARTIAL-11 (reason codes as a shared framework are not named in any wave)	11	See Org admin §12.29 reason codes
Intercompany journal post (periodic)	Post pending intercompany lines	COVERED-40	40	
Advanced ledger entries (public sector posting definitions)		OUT-OF-SCOPE (public sector)	—	
Fiscal LIFO calculation (IT)		OUT-OF-SCOPE (country pack beyond v1 set)	—	
Ledger budget: transfer budget to ledger (old budget model)		SUPERSEDED (budget register entries, wave 38)	38	
Financial reports generation (MR schedule)	Scheduled report packs	COVERED-15	15	Scheduled distribution PDF/XLSX/email
Export ledger (SAF-T, audit file)	Statutory audit file	COVERED-13	13	SAF-T extract; other country audit files in wave 48

12.1.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
General ledger parameters — Ledger tab	Check for voucher used, correction (storno), journal-based accrual, transaction reversal, batch transfer rules, use document date for voucher, balance control	COVERED-11	11	Storno (negative debit) handled as posting option
General ledger parameters — Sales tax tab	Apply sales tax taxation rules, cash discount on invoice incl. tax, tax on prepayment, rounding, conditional sales tax, sales tax settlement account, reverse charge	COVERED-13	13	
General ledger parameters — Inventory dimensions tab	Dimensions shown on ledger lines	COVERED-16	16	
General ledger parameters — Number sequences tab	Voucher sequences for journals, tax settlement	COVERED-5	5	Number sequence service
General ledger parameters — Batch transfer rules (D365 summarize subledger)	Asynchronous subledger transfer, summarize by account	SUPERSEDED (subledgers post synchronously via the universal journal API; summarization is a projection)	11	
Ledger	COA, calendar, currencies, account structures, balance control	COVERED-11	11	
Chart of accounts → Chart of accounts	Shared COA	COVERED-11	11	
Chart of accounts → Main accounts	Accounts	COVERED-11	11	
Chart of accounts → Main account categories	Categories for reporting/cubes	COVERED-15	15	Used by statement designer and wave 46 semantic layer
Chart of accounts → Account structures (Configure account structures)	Segments, allowed values, constraints, activation	COVERED-11	11	
Chart of accounts → Advanced rule structures / Advanced rules	Conditional additional segments	PARTIAL-11 (advanced rules (conditional extra segments) are not explicitly listed; account structures are)	11	Validation rules may need conditional branch
Chart of accounts → Account structure activation (periodic)	Activate draft structures	COVERED-11	11	
Chart of accounts → Financial dimensions		COVERED-11	11	
Chart of accounts → Financial dimension sets		COVERED-11	11	
Chart of accounts → Derived financial hierarchies (R3)	Category hierarchies used to derive dimension groupings for reporting	PARTIAL-15 (derived hierarchies for reporting are not named)	15	Report tree designer likely covers; confirm
Chart of accounts → Default dimension templates (D365)	Templates for default dimensions on master data	COVERED-11	11	Default dimensions on master data
Ledger calendar (Fiscal calendars + Ledger calendar)	Period statuses per module (open/on hold/closed), per user group access	COVERED-14	14	Calendar definition in 6; period control in 14
Fiscal calendars (Org admin shared)	Years, periods, closing periods	COVERED-6	6	

Name	Key parameters / options	Passage status	Wave	Notes
Journals → Journal names	Type, voucher series, approval workflow, posting restrictions, lines limit, fixed offset account, default currency/exchange rate type, private for user group	COVERED-11	11	
Journals → Journal texts	Reusable transaction text codes	COVERED-11	11	Journal templates
Journals → Journal controls (posting restrictions by account type/segment)	Restrict accounts per journal name	COVERED-11	11	
Journals → Default descriptions (Transaction text)	Auto text by transaction type with %1 placeholders	PARTIAL-11 (default description templates per posting type not named)	11	Small; put in wave 11 journal workspace
Journals → Accrual schemes	Ledger accrual schemes with debit/credit accounts, period key	COVERED-14	14	
Journals → Periodic journal names		COVERED-14	14	
Posting → Accounts for automatic transactions (System accounts)	Penny difference, exchange rate gain/loss, year-end result, error account, cash discount etc.	COVERED-11	11	
Posting → Posting definitions (public sector)		OUT-OF-SCOPE (public sector)	—	
Posting → Transaction posting definitions (PSA)		OUT-OF-SCOPE (public sector)	—	
Posting → Intercompany accounting	Due-to/due-from account pairs between legal entities, journal names	COVERED-40	40	
Posting → Reason codes (financial reasons)	Reasons for reversals, FA, customer/vendor transactions	PARTIAL-11 (shared reason code framework not named)	11	Gap register
Posting → Ledger posting groups (sales tax)	Tax payable/receivable/settlement accounts	COVERED-13	13	
Posting → Inventory posting (link)	Inventory posting profiles	COVERED-16	16	
Allocation → Ledger allocation rules	Fixed/basis/spread, source filters, destination, offset, reversal	COVERED-14	14	
Allocation → Allocation terms (on main account)	Fixed percentage allocation on account	COVERED-14	14	
Currency → Currencies	ISO code, rounding (price/unit/sales/purchase/general rounding rules), symbol	COVERED-6	6	ISO 4217 reference data
Currency → Currency exchange rates	Rates by type/pair/date, triangulation	COVERED-12	12	
Currency → Exchange rate types	Rate types (default, budget, closing)	COVERED-12	12	
Currency → Currency revaluation accounts	Realized/unrealized gain/loss by currency	COVERED-12	12	
Currency → Import currency exchange rates (provider)	Configure providers (ECB, OANDA) and schedule	COVERED-12	12	ECB/OpenExchangeRates adapters
Currency → Euro triangulation / Denomination currency	Legacy euro conversion	OUT-OF-SCOPE (obsolete; triangulation itself is in wave 12)	—	
Sales tax → Sales tax codes	Rate, limits, unit, calculation method (whole amount/interval), period, ledger posting group, use tax, exempt	COVERED-13	13	
Sales tax → Sales tax groups	Grouping by customer/vendor with exempt/reverse charge flags	COVERED-13	13	
Sales tax → Item sales tax groups	Grouping by item	COVERED-13	13	
Sales tax → Sales tax authorities	Authority vendor account, rounding	COVERED-13	13	
Sales tax → Sales tax settlement periods	Period intervals per authority, terms	COVERED-13	13	
Sales tax → Ledger posting groups		COVERED-13	13	
Sales tax → Sales tax exempt codes	Reason codes for exemption	COVERED-13	13	Exemption certificates
Sales tax → Sales tax jurisdictions (US)	Jurisdiction hierarchy, codes	COVERED-13	13	
Sales tax → Sales tax reporting codes	Codes for statutory return boxes	COVERED-13	13	VAT return box mapping
Sales tax → Sales tax (country-specific tab e.g. VAT registration numbers by country)		COVERED-7	7	Tax registration numbers on party
Sales tax → Withholding tax codes / groups	Withholding rates, item withholding groups (D365)	COVERED-13	13	
Sales tax → Tax branch (IN/other)		OUT-OF-SCOPE (country pack beyond v1 set)	—	India is in v1 pack set but tax branch is GST-era D365; handled in wave 48 India pack as needed
Sales tax → External (Sales tax, Avalara-type)	External provider connection	COVERED-13	13	
Sales tax → Conditional sales tax	Tax on payment	COVERED-13	13	

Name	Key parameters / options	Passage status	Wave	Notes
Sales tax → Sales tax on prepayment		COVERED-13	13	
Foreign trade → Intrastat parameters, commodity codes, transaction codes, transport methods, ports, statistics procedure	Intrastat master data	PARTIAL-13 (Intrastat extract is named; the setup forms (commodity codes, transaction codes, transport methods, ports, country properties) are not)	13	Gap register proposes wave 13
Foreign trade → EU sales list parameters / reporting codes		COVERED-13	13	
Consolidation → Consolidation accounts, Elimination rules, Consolidation groups	Account mapping, elimination rules, legal entity groups	COVERED-40	40	
Fiscal LIFO (IT)		OUT-OF-SCOPE (country pack beyond v1 set)	—	
Financial reporting → Report definitions (MR)	Rows, columns, trees, report	COVERED-15	15	
Financial reporting → Financial statements (legacy)	Row/column definitions	SUPERSEDED (wave 15 designer)	15	
Financial reporting → Financial reporting parameters (D365)	Integration settings	COVERED-15	15	
Budgeting (links)		COVERED-38	38	See §12.4
Ledger → Balance control accounts	Accounts for balance control on journal name	COVERED-11	11	
Ledger → Posting layers	Current/operations/tax layers	COVERED-12	12	Posting layers = ledger groups
Cash flow forecasts (GL setup: main accounts cash flow, dependent accounts)	Forecast settings per account	COVERED-36	36	
Organization → Legal entities (link)		COVERED-6	6	
Number sequences (GL)	Voucher series	COVERED-5	5	
Print management (GL)		COVERED-46	46	Document designer
Form setup (GL letters)		COVERED-46	46	
Workflow → General ledger workflows (journal approval, advanced ledger entry)	Workflow configuration per journal	COVERED-4	4	Workflow designer
Ledger → Closing sheets setup (country)		OUT-OF-SCOPE (country pack beyond v1 set)	—	
Tax (IN) → India-specific setup (tax components, TDS/TCS)		PARTIAL-48 (India pack listed for wave 48 but TDS/TCS/GST depth unspecified)	48	

12.2 Accounts payable

Accounts payable in AX spans vendor master (Passage wave 7), purchase orders (17), receiving and invoice matching (18), payments and settlement (19) and 1099/positive pay (19, 36). Vendor collaboration goes to wave 20.

12.2.1 Common / Journals

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Vendors (list page)	VendTableListPage	All vendors with FactBoxes (balance, contacts, recent POs)	COVERED-7	7	Party model with Vendor role
Vendor details	VendTable	General, addresses, contacts, misc details, vendor profile, purchasing demographics, invoice and delivery, purchase order defaults, payment, tax 1099, financial dimensions	COVERED-7	7	Role-specific vendor data entity
Vendors on hold	VendTable (Blocked = Invoice/All/Payment/Requisition/Never)	Hold statuses	COVERED-7	7	Hold types in vendor role
Vendor requests (R3)	VendRequest (VendRequestListPage)	New vendor / change request workflow (employee-initiated, unsolicited via EP)	PARTIAL-20 (vendor request workflow and prospective-vendor self-registration are not named; supplier portal is)	20	Gap register
Vendor bank accounts	VendBankAccount	Bank details per vendor, SEPA, prenote status	COVERED-19	19	
Purchase orders (list page)	PurchTableListPage	All POs	COVERED-17	17	
Purchase order details	PurchTable	Header/lines, charges, confirmation, change management	COVERED-17	17	

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Purchase orders → Received but not invoiced	PurchTableListPage (filter)	POs awaiting invoice	COVERED-18	18	GR/IR reconciliation
Purchase orders → Open purchase order lines	PurchLine list	Open line view	COVERED-17	17	
Purchase orders → Backordered purchase lines	PurchLineBackOrder	Overdue deliveries	COVERED-17	17	
Purchase orders → Purchase order confirmations (list of journals)	PurchConfirmJournal (VendPurchOrderJour)	Confirmation history	COVERED-17	17	
Purchase orders → Product receipts (Packing slip journal)	VendPackingSlipJour	Posted receipts	COVERED-18	18	
Purchase orders → Prepayment invoices	PurchPrepayTable	Prepayment on PO	COVERED-19	19	Prepayments
Vendor invoices → Open vendor invoices	VendInvoiceJournal (CustVendInvoiceJour)	Posted invoices not yet paid	COVERED-19	19	Open-item AP
Vendor invoices → Pending vendor invoices	VendInvoiceInfoListPage (VendInvoiceInfoTable)	Unposted header/lines invoices with matching	COVERED-18	18	
Vendor invoices → Vendor invoice register (journal)	LedgerJournalTable (VendInvoiceRegister)	Register invoice to arrival account before approval	COVERED-18	18	Invoice capture (two-step register/approve)
Vendor invoices → Invoice approval journal	LedgerJournalTable (VendInvoiceApprove)	Approve registered invoices, assign ledger accounts	COVERED-18	18	
Vendor invoices → Invoice journal	LedgerJournalTable (VendInvoiceRecord)	Direct expense invoices (no PO)	COVERED-18	18	
Vendor invoices → Invoice pool excl. posting	VendInvoicePool (LedgerJournalPool)	Pending pool of invoices	COVERED-18	18	
Vendor invoices → Vendor invoice workflow (D365)	VendInvoiceInfoTable + workflow	Header/line approval	COVERED-18	18	Exception workflow
Vendor invoices → Invoice matching details	VendInvoiceMatchingDetails	Price/qty/total match status per line	COVERED-18	18	
Vendor invoices → Vendor invoice automation (D365 OCR / "Invoice capture")	VendInvoiceCapture	Capture, classify, auto-apply prepayment, submit	COVERED-18	18	OCR/AP-automation adapter; AI coding agent wave 47
Payments → Payment journal	LedgerJournalTable (VendPaymentJournal)	Payment lines, proposals, generate payments, settle	COVERED-19	19	
Payments → Promissory note journals (Draw / Redraw / Remittance / Settle)	LedgerJournalTable (VendPromissoryNote*)	Vendor bills of exchange	GAP	—	Propose wave 36: promissory notes / bills of exchange (vendor side) with remittance and settlement journals; not in any wave
Payments → Payment proposal	VendPaymProposal	Select by due date/discount date	COVERED-19	19	
Payments → Vendor payment advice (print)	VendPaymAdvice	Advice document	COVERED-19	19	
Payments → Positive pay (link)	BankPositivePayTable	Export issued checks	COVERED-19	19	
Payments → Prenotes (ACH)	VendPaymPrenote	Prenote generation and confirmation	PARTIAL-19 (ACH prenote (zero-dollar validation) flow is not named)	19	Gap register
Payments → Centralized payments (cross-company)	VendPaymentJournal (centralized)	Pay on behalf of other legal entities	PARTIAL-40 (centralized payments across entities are not named in 19 or 40)	40	Gap register
Vendor transactions	VendTrans	Posted vendor ledger	COVERED-19	19	
Vendor open transactions (Settle open transactions)	VendOpenTrans	Settlement, mark, cross-currency, cash discount	COVERED-19	19	
Closed transaction editing	VendTransClosed? (VendSettledTrans)	Reverse settlements	COVERED-19	19	Reversal handling
Vendor invoice (from PO) — Invoice posting form	PurchEditLines (VendInvoiceInfoTable)	Post PO invoice	COVERED-18	18	
Purchase order → Receipts list (Arrival overview)	WMSArrivalOverview	Inbound arrival planning	COVERED-18	18	Arrival/registration

Form	AOT form / menu item	Purpose	Passage status	Wave	Notes
Purchase order → Registration	InventTransRegister	Register receipt by dimension/serial	COVERED-18	18	
Vendor → Consignment (D365 consignment replenishment orders)	VendConsignmentReplenishmentOrder	Vendor-owned stock, ownership change	GAP	—	Propose wave 16: consignment inventory (vendor-owned on-hand, ownership-change journal, consignment replenishment orders)
1099 → Vendor 1099 transactions	Vend1099Trans (Tax1099Summary)	Tax 1099 amounts per vendor	COVERED-13	13	1099 reporting hooks
Vendor → Vendor collaboration invoices (D365)	VendCollabInvoice	Vendor-submitted invoices	COVERED-20	20	Supplier portal invoice submission
Vendor → Vendor evaluation (R3 procurement)	VendEvaluation	Rate vendors on criteria	COVERED-33	33	Supplier quality scorecards
Purchase agreements (link)	PurchAgreement	Blanket agreements	COVERED-17	17	
Vendor → Charges transactions	MarkupTrans	Posted charges	COVERED-17	17	
Vendor → Deliveries (Vendor delivery schedule)	PurchLineDeliverySchedule	Delivery schedules on PO lines	COVERED-17	17	

12.2.2 Inquiries & Reports

Name	Purpose	Passage status	Wave	Notes
Vendor transactions inquiry	Posted transactions by vendor	COVERED-19	19	
Vendor balance (Vendor balance list)	Balances by vendor	COVERED-19	19	
Vendor aging (Aged balances inquiry)	Aging by period definition	COVERED-19	19	Vendor aging
Invoice history and matching details	Match status per invoice	COVERED-18	18	
Invoice journal inquiry / Invoice (Vendor invoice journal)	Posted invoice headers	COVERED-18	18	
Payment history / Payment inquiry	Payments by vendor	COVERED-19	19	
Settlement inquiry (Settled transactions)	Settlement pairs, cash discounts	COVERED-19	19	
Purchase order lines / Purchase order history	PO status history	COVERED-17	17	Versioned change management
Product receipt journal inquiry (Packing slip journal)	Receipts	COVERED-18	18	
Cash discount inquiry	Discounts taken/lost	COVERED-19	19	
Vendor invoice transactions (line-level)	Invoice lines	COVERED-18	18	
Journals → Confirmations / Receipts lists / Product receipts / Invoices (journal inquiries)	Posted document journals	COVERED-17	17	
Accrued purchases (Received not invoiced)	GR/IR accrual balance	COVERED-18	18	
Charges inquiry	Charges by PO	COVERED-17	17	
Withholding tax inquiry	Withholding by vendor	COVERED-13	13	
1099 → Tax 1099 summary / detail / state	1099 inquiries	COVERED-13	13	US country pack wave 48 produces IRS file formats
Vendor hold history		COVERED-7	7	Audit log
Positive pay inquiry	Exported checks	COVERED-19	19	
SSRS: Vendor aging (Vendor aging report)	Aged AP	COVERED-19	19	
SSRS: Vendor balance list with credit limit		COVERED-19	19	
SSRS: Vendor invoice (printed invoice copy)		COVERED-18	18	
SSRS: Vendor transactions report		COVERED-19	19	
SSRS: Vendor payment advice / Payment summary		COVERED-19	19	
SSRS: Check (BankChequePrint) / Check register	Check layout printing	COVERED-19	19	Check printing wave 19; check management wave 36
SSRS: Vendor 1099 (1099-MISC/INT/DIV/S/G forms and electronic file)	IRS forms and e-file	COVERED-48	48	US country pack; hooks in 13
SSRS: Vendor invoice specification / Invoice register journal		COVERED-18	18	
SSRS: Purchase order (document), Purchase order confirmation, Purchase inquiry (price inquiry)	PO documents	COVERED-17	17	Rendered wave 46
SSRS: Product receipt (packing slip) document		COVERED-18	18	
SSRS: Receipts list		COVERED-18	18	
SSRS: Vendor statement (Vendor account statement)		COVERED-19	19	
SSRS: Vendor base data / Vendor list		COVERED-7	7	
SSRS: Purchase order status / Purchase order overview / Backorder lines		COVERED-17	17	
SSRS: Purchase analysis (Purchase by vendor/item/period)	Statistics	COVERED-46	46	Semantic layer
SSRS: Vendor payment journal / Payment history		COVERED-19	19	

Name	Purpose	Passage status	Wave	Notes
SSRS: Cash discount list / Terms of payment list	Setup listings	COVERED-7	7	
SSRS: Expected cash flow (vendor)		COVERED-36	36	
SSRS: Vendor posting profile / Vendor groups listing		COVERED-7	7	
SSRS: Withholding tax (US/other)		COVERED-13	13	
SSRS: Purchase order lines — Vendor delivery performance (On-time delivery)	Vendor OTD	COVERED-33	33	Supplier scorecards
SSRS: Promissory note (vendor) documents		GAP	—	Propose wave 36: with promissory notes
SSRS: Vendor due date list (Open invoices by due date)		COVERED-19	19	
SSRS: Prepayment statement		COVERED-19	19	
SSRS: Vendor invoice totals (country)		OUT-OF-SCOPE (country pack beyond v1 set)	—	

12.2.3 Periodic

Name	Purpose	Passage status	Wave	Notes
Payment proposal (batch)	Generate payment lines	COVERED-19	19	
Generate payments (payment media)	Produce ACH/SEPA/check/wire files	COVERED-19	19	pain.001, NACHA, check print
Payment file return import (pain.002)	Bank status ingestion	COVERED-19	19	
Positive pay export		COVERED-19	19	
Foreign currency revaluation (AP)	Revalue open vendor items	COVERED-12	12	
Settlement → Settle open transactions (batch) / Automatic settlement	Auto-settle by rules	COVERED-19	19	
Reverse settlement		COVERED-19	19	
Cash discount → Recalculate cash discount dates		COVERED-19	19	
Invoice → Post pending vendor invoices (batch)		COVERED-18	18	
Invoice → Vendor invoice policy violations (R3 audit)	Policy check batch	PARTIAL-48 (audit policies are not named; SoD engine is)	48	Gap register
Purchase order → Confirm / Receive / Invoice (batch posting)	Mass posting	COVERED-17	17	Batch job framework wave 4
Purchase order → Update → Delivery date / Finalize PO (close)		COVERED-17	17	
Purchase order → Purchase order confirmation requests (vendor collaboration)		COVERED-20	20	
Purchase order → Evaluated receipt settlement (ERS)	Self-billing invoices from receipts	COVERED-18	18	
Purchase order → Delete purchase orders / lines (clean up)	Housekeeping	COVERED-17	17	
Purchase order → Vendor delivery schedule update		COVERED-17	17	
1099 → Update 1099 (year-end) / Print 1099 / electronic filing		COVERED-48	48	US pack
Vendor → Vendor hold release (mass)		COVERED-7	7	
Vendor → Update vendor bank accounts status (prenote results)		PARTIAL-19 (prenote)	19	
Promissory notes → Draw / Redraw / Remit / Settle (periodic batches)		GAP	—	Propose wave 36: promissory notes
Clean up → Delete settled transactions / Compress vendor transactions		SUPERSEDED (append-only ledger; archive to lakehouse wave 46)	46	
Interest/collection letters (vendor side)	Not in AX AP	—	—	n/a (listed for completeness; no AX feature)
Consignment → Consignment replenishment order processing		GAP	—	Propose wave 16: consignment
Invoice capture → Process OCR queue (D365)		COVERED-18	18	
Vendor collaboration → Provision users / invite (D365)		COVERED-20	20	
Purchase requisition → Release approved requisitions / Consolidate (procurement, link)		COVERED-17	17	

12.2.4 Setup

Name	Key parameters / options	Passage status	Wave	Notes
Accounts payable parameters — General tab	Posting profile default, settlement priority, cash discount administration, credit note as correction, delivery address, intercompany	COVERED-19	19	
Accounts payable parameters — Updates tab	PO defaults (confirmation/receipt/invoice posting rules), safety levels on PO, enable change management, cancel/reverse rules, product receipt posting	COVERED-17	17	

Name	Key parameters / options	Passage status	Wave	Notes
Accounts payable parameters — Invoice tab	Invoice matching (2-way/3-way/none), post product receipt in ledger, post invoice automatically, approval workflow, invoice totals matching, charges matching, allow price tolerance, use invoice register, posting of invoice: tax, prepayment	COVERED-18	18	
Accounts payable parameters — Settlement tab	Automatic settlement, cash discount mark, exchange rate at settlement, reverse settlement options	COVERED-19	19	
Accounts payable parameters — Ledger and sales tax tab	Sales tax, posting type, financial dimension defaults	COVERED-13	13	
Accounts payable parameters — Credit rating / Blocking	Block types	COVERED-7	7	
Accounts payable parameters — Number sequences	Invoice, payment, PO sequences	COVERED-5	5	
Accounts payable parameters — Summary update	Consolidation of POs at invoicing by invoice account/order/automatic summary	COVERED-18	18	Summary invoice posting across POs
Accounts payable parameters — Prices (Price tolerance)	Default price tolerance percentage, charges tolerance	COVERED-18	18	
Accounts payable parameters — Vendor invoice workflow (D365 Invoice capture)		COVERED-18	18	
Vendors → Vendor groups	Group, default terms, default tax group	COVERED-7	7	
Vendors → Vendor posting profiles	Summary account, settle account, sales tax prepayment, arrival account per group/vendor/table	COVERED-7	7	Posting profile hooks in 7; accounts resolved in 11
Vendors → Vendor holds reasons (D365)		COVERED-7	7	
Vendors → Vendor request types (R3)	Request workflow types	PARTIAL-20 (vendor request workflow)	20	
Vendors → Vendor evaluation criteria / criterion groups (R3)	Scoring criteria	COVERED-33	33	
Vendors → Vendor profiles / Vendor user requests (EP)		SUPERSEDED (supplier portal wave 20 with Keycloak identities)	20	
Payment → Terms of payment	Net days, cash on delivery, current month, COD, payment schedule, grace	COVERED-7	7	
Payment → Cash discounts	Discount %, days, next discount code, ledger account	COVERED-7	7	Also in 19 settlement
Payment → Methods of payment (vendor)	Period, account type, file format (export), bridging posting, payment specification, remittance, bank transaction type, check/EFT/SEPA, positive pay format, validation	COVERED-19	19	
Payment → Payment specifications	Per method of payment code (e.g. SEPA purpose)	COVERED-19	19	
Payment → Payment fees	Fee per bank/method with ledger account	COVERED-19	19	
Payment → Payment days	Fixed days of week/month for payment date	COVERED-19	19	
Payment → Payment schedules	Installments (fixed amount/quantity/percentage)	COVERED-7	7	
Payment → Payment calendar (D365)	Bank holidays by country for due dates	PARTIAL-19 (payment calendar / bank holiday due-date shifting is not named)	19	Gap register
Payment → Promissory note setup (journal names, posting profiles, protest)		GAP	—	Propose wave 36: promissory notes
Payment → Positive pay formats	XSLT-defined bank formats	COVERED-19	19	Positive pay export
Payment → Payment purpose codes (country)		COVERED-19	19	SEPA purpose codes
Payment → Direct debit (vendor)		OUT-OF-SCOPE (vendor-side direct debit is country-specific; SEPA creditor side covered via customer direct debit in AR)	—	
Charges → Charges codes (purchase)	Debit/credit type (item/ledger/customer/vendor), tax group, posting	COVERED-17	17	
Charges → Auto charges	Automatic charges by account/item code, mode of delivery	COVERED-17	17	
Charges → Vendor charges groups / Item charges groups / Delivery charges groups		COVERED-17	17	
Charges → Charges allocation (to lines)	Allocate header charges to lines by net amount/quantity	COVERED-17	17	Landed cost in 34
Invoice matching → Price tolerances (item/vendor/group)	Tolerance % per item/vendor combination	COVERED-18	18	
Invoice matching → Charges tolerances		COVERED-18	18	
Invoice matching → Invoice totals tolerances (D365: Invoice totals matching)		COVERED-18	18	
Invoice matching → Matching policy by item/vendor	2-way/3-way per item/vendor	COVERED-18	18	

Name	Key parameters / options	Passage status	Wave	Notes
Invoice → Vendor invoice policies (R3 audit policy)	Policy rules for invoice audit (duplicates, missing receipts)	PARTIAL-48 (audit policies)	48	
Invoice → Invoice register setup (arrival accounts)	Via posting profile	COVERED-18	18	
Invoice → Vendor invoice automation parameters (D365)		COVERED-18	18	
Forms → Form setup (PO, receipt, invoice print options)	Document notes, print settings, number of copies	COVERED-46	46	
Forms → Form notes	Notes per document type	COVERED-46	46	
Forms → Print management	Conditional print destinations per document/vendor	PARTIAL-46 (print management conditional routing per account is not named; document designer is)	46	Gap register
Tax 1099 → 1099 fields / 1099 boxes / State 1099 / 1099 software vendor info	IRS box mapping	COVERED-48	48	US pack; hooks in 13
Distribution → Financial dimension defaults on vendor		COVERED-11	11	
Vendor bank → Bank transaction types, Bank parameters (link)		COVERED-36	36	
Trade agreements → Price/discount agreement journal names		COVERED-26	26	Pricing engine applies to purchase prices too (wave 17 uses it)
Trade agreements → Activate price/discount (Trade agreement activation)		COVERED-26	26	
Trade agreements → Price/discount groups (vendor)		COVERED-26	26	
Trade agreements → Purchase price tolerance		COVERED-18	18	
Purchase → Purchase order defaults (order type, charges)		COVERED-17	17	
Purchase → Purchase order change management (activation per vendor)		COVERED-17	17	
Purchase → Prepayment setup (prepayment posting profile)		COVERED-19	19	
External codes (vendor)	Partner codes for EDI	COVERED-35	35	Partner profiles
Vendor → Vendor item catalog (Vendor external item numbers)	Vendor item numbers, descriptions	COVERED-8	8	External item numbers on released product
Withholding tax (vendor group assignment)		COVERED-13	13	
Vendor → Purchase order approval policies (link to Procurement)		COVERED-17	17	
Vendor → Vendor consignment setup (D365 consignment parameters, ownership change journal names)		GAP	—	Propose wave 16: consignment
Workflow → Accounts payable workflows (vendor invoice, PO, vendor change)		COVERED-4	4	
Number sequences (AP)		COVERED-5	5	
Vendor → Vendor collaboration parameters (D365)		COVERED-20	20	
Payment → Electronic payment file formats (AIF-based/ER)		COVERED-19	19	ISO 20022 generator; ER designer in 48 for exotic formats
Payment → Check layout (bank account)	Check printing layout per bank account	COVERED-19	19	Check layout in document designer; bank check management in 36

Part 13 — Coverage Inventory: SAP ECC 6.0 / S/4HANA → Passage

This part is the completeness oracle for the SAP side of the parity goal (PRD §2 goal 1, §5 scope map). It inventories SAP ECC 6.0 and S/4HANA functionality module by module — by transaction code and by IMG (SPRO) node — and maps every row to one of the 50 Passage waves in roadmap/waves.json. Its purpose is to make "we missed something" a question that can be answered by grep rather than by fear. The companion inventory for Dynamics AX / D365 F&SCM is Part 12.

13.0 How to read this inventory

Each SAP application component gets one section with four tables:

1. **Master data & documents** — the create/change/display transactions and the documents they produce (`T-code | Name | Purpose | Passage status | Wave | Notes`).
2. **Reports & inquiries** — standard list, line-item and analysis reports (`T-code | Name | Passage status | Wave | Notes`).
3. **Periodic processing** — closing, batch and mass-processing transactions (`T-code | Name | Passage status | Wave | Notes`).
4. **Customizing (SPRO)** — the IMG nodes, their customizing tables and the key options Passage must offer an equivalent for (`IMG node / T-code | Table(s) | Key options | Passage status | Wave | Notes`).

Status legend (exactly one token per row, always in the "Passage status" column):

- `COVERED-<wave>` — the wave's `deliverable/proof` in waves.json clearly includes an equivalent. The wave column repeats the number.
- `PARTIAL-<wave>` — the wave covers the core but the Notes column names what is missing. Every PARTIAL row is repeated in the gap register (§13.99).
- `GAP` — no wave clearly covers the item. The wave column holds a *proposed* assignment (`propose <n>`); the gap register collects them for a roadmap decision.
- `OUT-OF-SCOPE` — excluded by a PRD non-goal (§2) or by the manufacturer focus; the reason is in Notes.
- `SUPERSEDED-<wave>` — Passage deliberately does it a different way; the Notes column names the Passage mechanism. A SUPERSEDED row is **not** a gap: the SAP transaction has no one-to-one twin because the underlying mechanism (e.g. the condition technique's separate rebate module, SAP GUI, ABAP enhancement spots) is replaced by something structurally different.

Conventions:

- T-codes are given in their ECC 6.0 form; where S/4HANA replaced a transaction with a Fiori app or a new transaction, both are named (e.g. `FD32 → UKM_BP`, `VBO1 → WCOCO`). Fiori apps are cited by name, not by app number.
- "Wave" refers to the `n` in waves.json. Phase-exit waves (5, 10, 15, 20, 25, 30, 35, 40, 45, 50) are the sign-off points at which a module's rows must be demonstrably closed.
- IMG rows name the customizing table where it is well known (T001, T001B, T030, T156, T161, T685, TVAK…). Passage stores all customizing as metadata-registry configuration (wave 3) scoped per tenant / legal entity (wave 6); the "Key options" column lists the switches Passage must expose, not every SAP field.
- Rows are intentionally granular (a create/change/display triple is one row; a report family such as `S_ALR_87012*` is one row per distinct report, not one row per variant).
- Where SAP offers both an ECC transaction and a New GL / S/4 successor, the successor is inventoried and the predecessor noted, so classic GL (`GLT0`-based) transactions appear once under the FI-GL section.

Status totals across all four tables of every section (computed from the rows, excluding §13.99):

{{STATUS_TOTALS}}

13.1 FI-GL — General Ledger (classic GL, New GL, ledgers, ACDOCA)

SAP context: FI-GL in ECC 6.0 runs either classic GL (`GLT0` totals, special-purpose ledgers) or New GL (`FAGLFLEXT`, document splitting, parallel ledgers). S/4HANA collapses FI and CO totals into the universal journal `ACDOCA` (one line item table; CO objects and ledgers as columns). Passage's wave 11 universal journal is built on the S/4 model: one append-only line-item store, balances as projections, dimensions as columns, parallel ledgers via ledger groups (wave 12).

13.1.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
FS00	G/L account master (centrally)	Create/change/display chart-of-accounts and company-code segment of a G/L account	COVERED-11	11	Account entity with per-legal-entity attributes (currency, open-item flag, posting block, tax category, field status)
FSP0	G/L account in chart of accounts	Chart segment only	COVERED-11	11	Same entity; chart-level vs entity-level attributes split in metadata
FSS0	G/L account in company code	Company-code segment only	COVERED-11	11	Per-legal-entity release of an account
FS15 / FS16	Copy G/L accounts (send/receive)	Copy company-code segments between company codes	COVERED-11	11	"Release account to legal entity" bulk action

T-code	Name	Purpose	Passage status	Wave	Notes
FSM1 / FSM2 / FSM3	Sample accounts	Templates for company-code segments	SUPERSEDED-11	11	Account templates in the journal workspace; no separate sample-account object
KDH1 / KDH2 / KDH3	Account groups (G/L account groups hierarchy)	Hierarchies for FSV/reporting	COVERED-15	15	Account structures and report trees
FB50	Enter G/L account document	Single-screen G/L posting (Enjoy)	COVERED-11	11	Manual journal; idempotent Post API
FB50L	Enter G/L account document for ledger group	Ledger-specific manual posting	COVERED-12	12	Journal with ledger-group scope
F-02	Enter G/L account posting (classic)	Posting-key based general posting	SUPERSEDED-11	11	Posting keys replaced by debit/credit amount sign on lines
FB01	Post document (general)	Any document type, any account type	COVERED-11	11	Single Post API for all subledger-aware lines
FV50 / FBV0 / FBV2 / FBV3	Park / post / change / display parked G/L document	Draft documents for later posting or approval	COVERED-11	11	Draft status + approval workflow (Temporal)
FBV4 / FBV5 / FBV6	Change parked header / display changes / reject	Parked document administration	COVERED-11	11	Draft audit from CDC (wave 5)
FB02 / FB03 / FB04	Change / display / display changes to document	Document maintenance (header text, reference, assignment only)	COVERED-11	11	Post-posting edit limited to non-financial fields; audit log
FB08 / F.80	Reverse document / mass reversal	Negating entry linked to the original	COVERED-11	11	Reversal API creates linked negating entry; mass reversal via batch job (wave 4)
FBRA	Reset cleared items	Undo clearing before reversal	COVERED-19	19	Settlement reversal in AP/AR; GL open-item accounts see §13.1 note on F-03
FBD1 / FBD2 / FBD3 / FBD4	Recurring entry document create/change/display/changes	Recurring journals with run schedule	COVERED-11	11	Recurring journals with schedule; execution via wave 4 batch
F.14 / F.15	Execute recurring entries / list recurring entries	Batch creation from recurring templates	COVERED-11	11	Scheduled recurring-journal job
FBS1	Enter accrual/deferral document	Reversing document with reversal date	COVERED-14	14	Accrual schemes with automatic reversal
F.81	Reverse accrual/deferral documents	Mass reversal of FBS1 documents	COVERED-14	14	Automatic reversal on schedule
FBM1 / FBM2 / FBM3	Sample documents	Non-posting template documents	SUPERSEDED-11	11	Journal templates
F-03 / F-04	Clear G/L account / post with clearing	Open-item clearing on OI-managed G/L accounts	PARTIAL-11	11	Open-item management specified only for AP/AR (waves 19/30) and GR/IR (wave 18); generic OI-managed G/L accounts (clearing accounts, bank sub-accounts) need the same clearing engine exposed on GL
F-05	Post foreign currency valuation (manual)	Manual FX adjustment posting	COVERED-12	12	Manual journal in reporting currency
F-65	Preliminary posting (park)	Classic parking	COVERED-11	11	Draft journals
FB41	Post tax payable	Transfer of tax balances	COVERED-13	13	Tax settlement job posts to tax payable
OB08	Enter exchange rates (also master data)	Rates by rate type, date, pair	COVERED-12	12	Exchange rate service with provider adapters and audit
FBCJ	Cash journal	Petty cash with receipts/expenses and GL posting	GAP	propose 36	No wave mentions a petty-cash/cash-journal document; small but common on plant floors — add to wave 36 (cash & bank) as "cash journal with receipts, business transactions and daily closing"
FBCJC0 / FBCJC1 / FBCJC2 / FBCJC3	Cash journal setup, number ranges, business transactions, print	Cash journal customizing	GAP	propose 36	Follows FBCJ
FAGLL03 (document)	Ledger-specific line item entry context	See reports	COVERED-12	12	—
GB01 / GB02	Enter/post G/L documents with document splitting characteristics	Classic FI-SL documents	SUPERSEDED-12	12	Special-purpose ledger replaced by parallel ledgers + dimensions
OB_GLACC11 / OB_GLACC12 / OB_GLACC13	Mass maintenance of G/L accounts (chart / company code / names)	Mass change	COVERED-5	5	Grid mass edit + Excel import/export on any entity
FAGL_ACTIVATE_OP	Activate open-item management on G/L accounts	Conversion utility	PARTIAL-11	11	Depends on the GL OI gap above
FAGL_SWITCH_TO_OPEN_ITEM	Switch to OI management (S/4)	Same	PARTIAL-11	11	Same
GCAC	Ledger comparison	Compare two ledgers	COVERED-12	12	Per-ledger trial balances with diff
FAGLB03 (document view)	Ledger balance display → line items	See reports	COVERED-15	15	—
FINS_ACDOC_CUST	Universal journal customizing overview (S/4)	ACDOCA extensibility	SUPERSEDED-3	3	Journal-line extensions via the metadata kernel (customer fields on JournalLine)
FAGL_MIG_*	New GL migration cockpit	Classic → New GL migration	SUPERSEDED-49	49	Migration factory loads GL balances by dimension directly
ACC_DOCUMENT (BAPI_ACC_DOCUMENT_POST)	Accounting document BAPI	Programmatic posting	COVERED-11	11	Post API (REST/gRPC) with idempotency keys

13.1.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FBL3N	G/L account line items (classic)	COVERED-11	11	Journal line inquiry with saved views, grid, export
FAGLL03	G/L account line items (New GL, ledger-aware)	COVERED-12	12	Ledger filter on the same inquiry
FAGLL03H	G/L line item browser (HANA)	COVERED-46	46	Lakehouse-backed line-item browser for history; OLTP inquiry for recent
FS10N / FAGLB03	G/L account balance display (classic / New GL)	COVERED-11	11	Balance projection by account × dimension × period with drill to lines
FS10NA	Balance display with drill-down	COVERED-15	15	Statement → balance → lines drill
F.01	Financial statements (RFBILA00)	COVERED-15	15	Balance sheet / P&L from financial statement version equivalent
S_ALR_87012284	Financial statements (FSV, with comparison)	COVERED-15	15	Comparative columns in report designer
S_ALR_87012277	G/L account balances	COVERED-15	15	Trial balance
S_ALR_87012301	Totals and balances	COVERED-15	15	Trial balance by period
S_ALR_87012282	G/L line items, list for printing	COVERED-11	11	Printable line-item list
S_ALR_87012287	Document journal	COVERED-15	15	Audit trail / document journal
S_ALR_87012289	Compact document journal	COVERED-15	15	Same, compact layout
S_ALR_87012291	Line item journal	COVERED-15	15	Same
S_ALR_87012293	Display of changed documents	COVERED-5	5	Audit log viewer
S_ALR_87012308	Display changes to G/L accounts	COVERED-5	5	Audit log on Account entity
S_ALR_87012326	Chart of accounts list	COVERED-11	11	Account list export
S_ALR_87012328	G/L account list	COVERED-11	11	Same
S_ALR_87012332	G/L account statements	COVERED-15	15	Account statement report
S_ALR_87012341 / 87012342	Invoice numbers allocated twice / gaps in document number assignment	COVERED-5	5	Number-sequence gap report (gap-free mode audit)
S_ALR_87012357 / 87012356	Advance return for tax on sales/purchases / sales tax report	COVERED-13	13	VAT return extracts
S_ALR_87012382	Cash flow (direct method, from FSV)	COVERED-15	15	Indirect cash flow standard; direct via report designer
S_PL0_86000028 / S_PL0_86000029	Financial statement: actual/actual comparison, plan/actual	COVERED-15	15	Comparative statements; plan column after wave 38
S_PL0_86000030	G/L account balances (New GL)	COVERED-12	12	Ledger-aware trial balance
S_PL0_86000031	Transaction figures: account balance	COVERED-11	11	—
S_E38_98000088	Profit center group: plan/actual/variance	COVERED-38	38	Dimension = profit center; plan from budgeting
S_AC0_52000887 / 52000888	Receivables/payables by profit center (New GL)	COVERED-15	15	Dimension-level subledger report
FAGLF03	Reconciliation ledger vs. documents (New GL)	COVERED-16	16	Daily automated subledger-to-journal reconciliation check
F.03	Financial accounting comparative analysis (SAPF190)	COVERED-16	16	Same reconciliation job
GR55 / GRR1 / GRR2 / GRR3	Report Painter / Report Writer report groups and reports	SUPERSEDED-15	15	Row/column report designer (Management-Reporter model) replaces Report Painter; wave 46 semantic layer for ad-hoc
FGI0 / FGI1 / FGI3	Drill-down reporting (New GL): execute / create report / display	SUPERSEDED-15	15	Report designer + drill-down
FGI4 / FGI5 / FGI6	Drill-down report forms	SUPERSEDED-15	15	Same
FAGL_SCHEDULE_LOG? (FAGLCOFITRACEADMIN)	Trace for CO-FI real-time integration	SUPERSEDED-11	11	No CO→FI reconciliation needed: single journal
S_ALR_87012309	Cash book / cash journal report	GAP	propose 36	Follows FBCJ gap
FAGLGVTR (log)	Balance carry-forward log	COVERED-14	14	Year-end job log
Fiori: Trial Balance	Trial balance (ledger, dimensions, comparison)	COVERED-15	15	—
Fiori: Display Line Items in General Ledger	S/4 line-item app	COVERED-11	11	—
Fiori: Display Financial Statement	S/4 statements app	COVERED-15	15	—
Fiori: Balance Sheet/Income Statement — Multidimensional	Dimension-pivot statements	COVERED-46	46	Semantic layer pivot
Fiori: Manage Journal Entries	Search/display/reverse journal entries	COVERED-11	11	—
Fiori: Verify General Journal Entries (approval)	Workflow-based journal approval	COVERED-11	11	Approval workflow on journals
Fiori: Audit Journal	Chronological audit journal	COVERED-15	15	Audit trail report
Fiori: Journal Entry Analyzer	Analyze journal entries by dimension	COVERED-46	46	—

13.1.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
OB52 (operational use)	Open/close posting periods	COVERED-14	14	Period control per ledger × module × user group

T-code	Name	Passage status	Wave	Notes
FAGL_FC_VAL / F.05 (classic)	Foreign currency valuation (GL, AP, AR open items)	COVERED-12	12	Revaluation job with next-period reversal
FAGL_FC_TRANS	Currency translation (balances into group currency)	COVERED-12	12	Reporting-currency translation
FAGLGVTR / F.16 (classic)	Balance carry-forward (New GL / classic)	COVERED-14	14	Year-end close: P&L to retained earnings, BS carry-forward per dimension
F.07	Balance carry-forward AP/AR	COVERED-14	14	Subledger carry-forward in the same job
F.13 / F13E	Automatic clearing (with/without specification of clearing currency)	PARTIAL-11	11	Automatic clearing specified for GR/IR (wave 18) and AP/AR settlement; generic rule-based GL auto-clearing is part of the GL OI gap
FAGLF101 / F101	Regroup receivables/payables (sorted list, reclassification by remaining term)	GAP	propose 14	Balance-sheet reclassification (short/long-term, debit-balance vendors, credit-balance customers) is a standard close step not named in wave 14's checklist/allocations — add as a close task type
F.19	GR/IR analysis / regrouping	COVERED-18	18	GR/IR reconciliation report and reclass
F.50 / F.52	G/L interest calculation (balance / account interest)	GAP	propose 30	Interest on GL balances (e.g. intercompany loans) is not named; wave 30 covers AR interest notes only — extend the interest engine to GL/intercompany accounts
ACACACT / ACACTREE01 / ACACPSITEMS	Accrual engine: periodic run / accrual objects / display postings	COVERED-14	14	Accrual schemes; recurring accrual objects
ACACADJ	Accrual engine: adjust accrual	COVERED-14	14	—
FAGL_CL_REGROUP / FAGL_CL_MIGRATION	Closing: regrouping (S/4)	GAP	propose 14	Same as FAGLF101
F.97 / F.98 / F.99	Balance confirmations for G/L / AP / AR	GAP	propose 19	Balance confirmation letters (vendor/customer/GL) not named in waves 19 or 30 — add to wave 30 customer statements + wave 19 vendor confirmations
FB1S / FB1D / FB1K	Clear G/L / customer / vendor (classic clearing)	COVERED-19	19	Settlement; GL part in GL OI gap
GCAC	Ledger comparison (closing check)	COVERED-12	12	—
FINS_CO_RECON? (FINS_RECON_RC0..RC3)	S/4 reconciliation reports (ACDOCA vs. BSEG/COEP)	SUPERSEDED-11	11	No dual tables to reconcile; balance projection vs journal check is the equivalent
FINSC_LEDGER (admin run)	Ledger settings activation	COVERED-12	12	Ledger group configuration
CLOCO / CLOCOC / CLOCOS	Financial Closing Cockpit: task list / templates / monitor	COVERED-14	14	Close checklist with templates, dependencies, assignees, status board
SCMA	Schedule Manager (classic)	SUPERSEDED-14	14	Close checklist + wave 4 batch scheduler
Fiori: Manage Closing Tasks / Advanced Financial Closing	S/4 close orchestration	COVERED-14	14	Close checklist across entities
SAFT / Fiori: Run Statutory Reports (ACR)	Advanced Compliance Reporting (SAF-T, VAT returns)	COVERED-13	13	SAF-T extract validated to OECD XSD; VAT returns; statutory packs in wave 48
FAGL_SCHEDULE? (FAGL_PLAN_*)	Plan data entry in New GL (GP12N etc.)	COVERED-38	38	Budget register against ledger dimensions
GP12N / GP12NA	Plan data entry / display (New GL)	COVERED-38	38	—
FAGL_CO_PLAN	Transfer CO plan data to New GL	SUPERSEDED-38	38	One planning store
S_ALR_87099918? (RFUMSV00 run)	Advance tax return run	COVERED-13	13	Tax settlement job

13.1.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OX02 — Company code	T001	Country, currency, language, fiscal year variant, chart of accounts	COVERED-6	6	LegalEntity entity
OBY6 — Company code global parameters	T001	FYV, posting period variant, field status variant, negative postings, business area FS, tax base net/gross, cash management active	COVERED-11	11	LegalEntity accounting settings
OB13 / OB62 — Chart of accounts / assign to company code	T004, T001	Operational / group / country chart; length; integration with CO	COVERED-11	11	Account structure per ledger; alternative account numbers via dimension mapping
OBD4 — Account groups	T077S	Number ranges, field status per group	COVERED-11	11	Account groups with field visibility rules (metadata field attributes)
OB53 — Retained earnings account	T030 (BIL)	P&L statement account type	COVERED-14	14	Year-end close settings
OB29 — Fiscal year variants	T009, T009B	Calendar/non-calendar, special periods, shortened FY	COVERED-6	6	Fiscal calendars: 4-4-5, 13-period, calendar; special periods needed — covered by period types
OB37 — Assign fiscal year variant to company code	T001	—	COVERED-6	6	Calendar per legal entity / ledger
OB52 — Posting period variants: open/close periods	T001B	Account type ranges, from/to period, authorization group (2nd interval)	COVERED-14	14	Period control per ledger × module × user group
OBBO / OBBP — Posting period variant define / assign	T010O, T001	—	COVERED-14	14	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OBA7 — Document types	T003	Number range, account types allowed, reversal doc type, net posting, authorization group, reference/doc header text required	COVERED-11	11	Journal/document types with number sequence and allowed line types
FBN1 / OBH2 — Number ranges for accounting documents / copy	NRIV	Year-dependent, external, interval	COVERED-5	5	Number sequence service, gap-free mode for fiscal docs
OB41 — Posting keys	TBSL	D/C, account type, field status, sales-related, special G/L	SUPERSEDED-11	11	No posting keys: line = account + signed amount + line type
OBC4 — Field status variants / groups	T004F, T004V	Required/optional/suppressed per field group	COVERED-3	3	Metadata field attributes per account group / line type
OBC5 — Assign company code to field status variant	T001	—	COVERED-11	11	—
OB28 / GGB0 — Validations (FI)	GB90*, T001Q	Prerequisite/check/message at header, line, document level	PARTIAL-11	11	Wave 11 has dimension validation rules; a general user-defined validation rule language (Boolean rules over header/line fields with message classes) is only implied by wave 3 "validation" — scope it explicitly
OBBH / GGB1 — Substitutions (FI)	GB91*, T001R	Field substitution with exits at line/document	PARTIAL-11	11	Default dimensions on master data exist; rule-based substitution (derive dimension from another field) needs an explicit rule engine on the Post API
OBBZ / OBBW — Substitution/validation for cost of sales accounting	—	Functional area derivation	PARTIAL-11	11	Functional area as a dimension with derivation rules (see above)
OKKP (FI side) — Controlling area assignment	TKA01, TKA02	Company code ↔ controlling area, currency type	SUPERSEDED-11	11	No separate controlling area: dimensions on the journal; wave 38 management reporting
OB22 / FINSC_LEDGER — Parallel currencies per company code / ledger settings (S/4)	T001A, FINSC_LEDGER	Currency types 10/30/40, valuation view, ledger type (standard/extension), accounting principle, FYV per ledger	COVERED-12	12	Three-currency lines; ledger groups; extension ledgers (adjustment-only) need confirming — treat as covered by ledger groups
FAGL_ACTIVATION / FAGL_ACTIVEC — Activate New GL	FAGL_ACTIVEC	—	SUPERSEDED-11	11	Single journal always on
Define ledgers for GL accounting (FAGL_LEDGER)	T881, T882G	Leading ledger, representative ledger	COVERED-12	12	—
Define ledger groups (FAGL_LEDGER_GRP)	T8G*, FAGL_TLDGRP	Group of ledgers for postings	COVERED-12	12	—
Define accounting principles / assign to ledger groups	TACC_PRINCIPLE, TACC_TRGT_LDGR	IFRS/local GAAP/tax	COVERED-12	12	—
Document splitting: classify G/L accounts, document types, splitting method, zero-balance, constants (FAGL_SPLIT*)	T8G01…T8G40	Active/passive split, characteristics (PC, segment, business area), zero-balance clearing account, inheritance	PARTIAL-11	11	Wave 11 requires every line to carry dimensions, so dimension-complete documents are native; but *inheritance to balance-sheet lines* (AP/AR/tax lines split by the expense lines' profit centers) and zero-balance clearing per dimension are not named — needed for balance sheet by profit center/segment
Define segment / derive segment from profit center	FAGL_SEGM, CEPC	—	COVERED-11	11	Segment as a dimension; derivation default from profit center dimension
OB08 / OBBS / OBBT — Exchange rate types, ratios, rate maintenance	TCURR, TCURV, TCURF, TCURX	Rate types (M, EURX, P), inversion, triangulation via reference currency, decimals per currency	COVERED-12	12	Rate types, triangulation, decimals from ISO 4217
OB07 — Exchange rate types	TCURV	—	COVERED-12	12	—
OBA1 — Accounts for exchange rate differences / FX valuation	T030H	Realized/unrealized gain-loss, valuation and translation accounts per G/L account	COVERED-12	12	Posting profile for realized/unrealized
OB59 / OBBD? — Valuation methods and valuation areas (FC valuation)	T044A, T044G, FAGL_VAL_AREA?	Lowest value principle, always valuate, reset, exchange rate type per method; valuation area ↔ accounting principle	COVERED-12	12	Revaluation method per ledger / accounting principle
OBXR / OBYR — Special G/L indicators (customers / vendors)	T074	Down payments, bills of exchange, guarantees, noted items	PARTIAL-19	19	Prepayments/down payments covered (waves 19/30); bills of exchange and noted items not named — see FI-AP/AR sections
OBY7 / OBY8 / OBY9 — Copy chart of accounts / company code / delete	—	—	COVERED-6	6	Entity templates ("copy legal entity") in enterprise structure
OBYA — Clearing accounts for cross-company-code transactions	T001U	Due-to/due-from per company-code pair	COVERED-40	40	Intercompany accounting due-to/due-from
OBYC (FI-side view) — Automatic postings (MM/SD integration)	T030	Transaction keys BSX, GBB, WRX, PRD, KDM, UMB, FRL, BSV, AUM, VAX/VAY	COVERED-16	16	Inventory posting profiles (item model group × site × account)
OB40 — Tax accounts	T030K	Per tax transaction key (MWS, VST, NAV…)	COVERED-13	13	Tax posting profile
OBB8 / OBB9 — Terms of payment / installment terms	T052, T052R	Day limits, baseline date, discount days/percent, installment split	PARTIAL-7	7	Payment terms covered; installment terms (split one invoice into several due dates) not named — add to wave 7 terms model
OBA3 / OBA4 — Tolerance groups (customers/vendors) / (employees, posting amount limits)	T043G, T043T	Max amount per document, per line, cash discount, payment difference tolerances	COVERED-19	19	Amount limits via OPA policy + settlement tolerances (wave 19)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OBBE / OBXL — Reason codes for payment differences / accounts	T053R, T030	Charge off, residual, disputes	COVERED-30	30	Reason codes on cash application
OBD2 — Customer account groups / OBD3 — Vendor account groups	T077D, T077K	Number range, field status, one-time account	COVERED-7	7	Party role groups, field attributes; one-time vendors/customers — see §13.2
OBXU / OBXI — Cash discount received/granted accounts	T030	Net/gross method	COVERED-19	19	—
OBXK / OBXB — Bank charges / bank sub-accounts	T030	—	COVERED-36	36	Bank posting profiles
OBYP — GR/IR clearing accounts	T030	WRX	COVERED-18	18	—
OBXM / OBXO / OBXW — Retained earnings-related: acquisition/retirement, rounding, over/under payments	T030	—	COVERED-19	19	Rounding and payment difference accounts
OBA0 — Tolerance groups for G/L accounts (clearing)	T043S	—	PARTIAL-11	11	GL OI gap
OB32 — Document change rules	TBAER	Which fields editable after posting	COVERED-11	11	Field-level edit rules post-posting
OB46 — Interest indicators / OB82 / OBAA	T056*	Balance interest, item interest, calendar type	GAP	propose 30	Interest engine (see F.52)
OBD5? / Define sort keys (allocation field)	TZUN	Assignment field fill rule	SUPERSEDED-11	11	Sort/assignment via configurable reference fields and saved views
OBR1 / OBR2 / OBR3 — Reset transaction data / delete master data / delete customer-vendor	—	Test-client tooling	SUPERSEDED-1	1	Environment reset through pipeline/tenant provisioning
FINSC_FIELDSTATUS? / Define field status for ledger (S/4)	—	—	COVERED-3	3	Metadata
OB74 — Additional rules for automatic clearing	TF123	Criteria fields per account type/range	PARTIAL-11	11	GL OI gap (AP/AR criteria covered by settlement rules)
OBB1 / OBB2? — Define accounts for bank interest / tax on cash discount	T030	—	COVERED-13	13	—
SNUM / SNRO (FI number range objects)	NRIV	RF_BELEG, etc.	COVERED-5	5	Number sequences
OBBG — Assign country to tax calculation procedure	T005	TAXUSJ, TAXUSX, TAXEUR…	COVERED-13	13	Jurisdiction hierarchy per country
OBH1 — Copy number range intervals to fiscal year	NRIV	—	COVERED-5	5	Year-dependent sequences
Define Reasons for Reversal (OB_REVERSE? / T041C)	T041C	Negative posting allowed, alternative posting date	COVERED-11	11	Reversal reason codes
FAGL_CL_RGRP? / Define sort method & adjustment accounts for regrouping	T043?/ T030	Remaining term intervals	GAP	propose 14	FAGLF101 gap
Define Posting Keys for Clearing (OBXH etc.)	T030	—	SUPERSEDED-11	11	No posting keys
OBD4 → "Define Screen Layout per Company Code"	T001	—	COVERED-3	3	—
Cost of sales accounting: functional areas (OKBD), activate (FAGL_FUNC_AREA?)	TFKB, T001	—	PARTIAL-11	11	Functional area dimension + derivation rules (substitution gap)
Define Business Areas (OX03) / Consolidation business areas	TGSB, TGSBK	—	COVERED-6	6	Dimension values
Profit center: standard hierarchy, dummy PC (KCH1/KE59 — see CO-PA)	CEPC, SETHEADER	—	COVERED-6	6	Profit center as dimension with hierarchy purposes

13.2 FI-AP — Accounts Payable

SAP context: vendor master (LFA1/LFB1/LFM1, replaced by Business Partner with FLVN00/FLVN01 roles in S/4), invoices and credit memos (FB60/FB65 or MIRO for PO-based), open items (BSIK/BSAK), payment program F110, dunning (rare for AP), down payments, withholding tax. Passage maps vendor master to wave 7 (party model), non-PO invoices to wave 18 (capture) and open items/payments to wave 19.

13.2.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
XK01 / XK02 / XK03 (→ BP)	Vendor master create/change/display centrally	General, company-code and purchasing data	COVERED-7	7	Party + Vendor role + per-legal-entity and per-purchasing-org data
FK01 / FK02 / FK03	Vendor master (accounting)	Company-code segment only	COVERED-7	7	Vendor role, legal-entity scope
MK01 / MK02 / MK03	Vendor master (purchasing)	Purchasing-org segment	COVERED-7	7	Vendor role, purchasing scope; see MM-PUR

T-code	Name	Purpose	Passage status	Wave	Notes
XK05 / XK06 / FK05 / FK06	Block / mark for deletion	Posting/payment/purchasing blocks	COVERED-7	7	Role status and block flags with approval
XK07	Change account group	Reassign vendor group	COVERED-7	7	Group change under governance workflow
XK99 / MASS (LFA1)	Mass maintenance vendor	Mass change	COVERED-5	5	Grid mass edit
BP (role FLVN00/FLVN01)	Business partner — vendor roles (S/4)	Unified BP	COVERED-7	7	Native model
CVI_FS_CHECK_CUSTOMIZING / MDS_LOAD_COCKPIT	Customer-vendor integration synchronization	ECC → S/4 BP sync	SUPERSEDED-7	7	One party model, no sync
One-time vendor (account group CPD)	One-time accounts with address on document	Ad-hoc payee	PARTIAL-7	7	Not named in wave 7; add "one-time party" role with document-level address capture
FK08 / FK09	Confirm vendor master changes (dual control)	Sensitive-field dual control	COVERED-7	7	Change approval workflow on master data (bank details, payment terms)
FB60	Enter vendor invoice (non-PO)	Single-screen AP invoice	COVERED-18	18	Manual invoice capture without PO
FB65	Enter vendor credit memo	—	COVERED-19	19	Vendor credit notes
F-43 / F-41	Enter vendor invoice / credit memo (classic)	Posting-key based	SUPERSEDED-18	18	Single invoice form
FV60 / FV65	Park vendor invoice / credit memo	Draft for approval	COVERED-18	18	Invoice draft + approval workflow
FBV0	Post parked document	—	COVERED-18	18	—
F-48	Vendor down payment	Special G/L down payment posting	COVERED-19	19	Prepayments
F-47	Down payment request	Noted item for F110	COVERED-19	19	Prepayment request on PO/invoice workflow
F-54	Clear vendor down payment	Transfer down payment against invoice	COVERED-19	19	Prepayment application on settlement
F-53 / F-58	Post outgoing payment / with printed form	Manual payment with clearing	COVERED-19	19	Manual payment + settlement; check print
F-44	Clear vendor	Account clearing without payment	COVERED-19	19	Settlement without payment (credit vs invoice)
F-51	Post with clearing (transfer)	Transfer with clearing	COVERED-19	19	—
F-40 / F-36? (AP side: FBW6?)	Bill of exchange payment (AP)	Bills of exchange payable	GAP	propose 36	Bills of exchange (AP and AR) are not in any wave; relevant in IT/ES/FR/JP/BR; propose as a wave 36 (bank) option or defer to a localization v1.x pack under wave 48 — needs a decision
FB10	Invoice/credit memo fast entry	Multi-document fast entry	COVERED-18	18	Grid entry / import
FBL1N (as document finder)	—	—	COVERED-19	19	—
FCHN / FCH5 / FCH6	Check register / create manual check / change check info	Check management	COVERED-36	36	Check management (void/reprint), check register
FCH7 / FCH8 / FCH9	Reprint / cancel payment + check / void check	—	COVERED-36	36	—
FCHR / FCHT	Online cashed checks / assign checks to payments	—	COVERED-36	36	Bank statement cashed-check matching
FCHU / FCHX	Create reference for check / check extract (positive pay)	Positive pay file	COVERED-19	19	Positive pay export
F110 (document side: payment documents)	—	—	COVERED-19	19	—
FBZ5	Print check for payment document	—	COVERED-19	19	Check printing
FBZ8? / FBZP	—	See customizing	—	—	—
FB03 (vendor doc)	Display document	—	COVERED-19	19	—
FBD1 (vendor recurring)	Recurring vendor invoice	—	COVERED-11	11	Recurring journal with vendor lines
J1INCHLN / J1INBANK / J1INCERT (IN WHT)	Withholding tax challan / bank challan / certificate (India)	Country-specific WHT remittance	PARTIAL-13	13	Withholding engine + 26AS hooks named; Indian TDS challan/certificate flows belong to the IN country pack (wave 48) — partial until pack ships
S_P00_07000134 / RFIDYYWT	Generic withholding tax reporting (1099, DME)	Statutory WHT reporting	COVERED-13	13	1099 reporting hooks (year-end forms in wave 43 share structures)
FIBLFFP / F111	Payment request / payment program for payment requests	Treasury payment requests outside AP	PARTIAL-36	36	Payment requests from treasury (not vendor items) not named; add to wave 36 in-house banking
VIM (OpenText) / Fiori: Manage Supplier Invoices / Upload Supplier Invoices	AP automation	OCR capture, approval	COVERED-18	18	OCR/AP automation adapter
Ariba Network invoice	Supplier-submitted invoice	—	COVERED-20	20	Supplier portal invoice submission

13.2.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FBL1N / FBL1H	Vendor line items / line item browser	COVERED-19	19	Vendor open/cleared item inquiry

T-code	Name	Passage status	Wave	Notes
FK10N / FK10NA	Vendor balance display	COVERED-19	19	Vendor balance with drill
S_ALR_87012077	Vendor information system	COVERED-46	46	Vendor analytics
S_ALR_87012078	Due date analysis for open items	COVERED-19	19	Aging
S_ALR_87012082	Vendor balances in local currency	COVERED-19	19	—
S_ALR_87012083	List of vendor open items for printing	COVERED-19	19	—
S_ALR_87012084	Open items — vendor due date forecast	COVERED-36	36	Cash forecast input
S_ALR_87012085	Vendor payment history	COVERED-19	19	—
S_ALR_87012086	Vendor list	COVERED-7	7	—
S_ALR_87012087	Address list	COVERED-7	7	—
S_ALR_87012089	Display changes to vendors	COVERED-5	5	Audit viewer
S_ALR_87012090	Display/confirm critical vendor changes	COVERED-7	7	Dual-control queue
S_ALR_87012093	Vendor business (purchasing volume)	COVERED-46	46	—
S_ALR_87012103	List of vendor line items	COVERED-19	19	—
S_ALR_87012104	Vendor open items (OI list)	COVERED-19	19	—
S_ALR_87012105	List of down payments open at key date	COVERED-19	19	Prepayment report
S_P99_41000099	Payment list (F110)	COVERED-19	19	Payment run output
S_ALR_87012119	Cashed checks per bank account	COVERED-36	36	—
S_P99_41000101	Check register	COVERED-36	36	—
S_ALR_87012176? (AP aging via RFKOPR00)	Vendor open item aging	COVERED-19	19	—
S_ALR_87012116?	Bill of exchange list	GAP	propose 36	Follows bills gap
FBL1N variant: withholding tax	Withholding tax line items	COVERED-13	13	—
S_ALR_87012111? / S_P00_07000134	Withholding tax report / 1099-MISC	COVERED-13	13	—
Fiori: Supplier Balances / Display Supplier List / Manage Supplier Line Items	S/4 apps	COVERED-19	19	—
Fiori: Accounts Payable Overview (analytical)	AP KPI page	COVERED-46	46	Embedded dashboards
Fiori: Cash Discount Utilization / Days Payable Outstanding	AP KPIs	COVERED-46	46	Governed metrics

13.2.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
F110	Automatic payment program (proposal, edit, schedule, print)	COVERED-19	19	Payment proposal → approval → run → media; idempotent
F110S	Payment program scheduled (batch variant)	COVERED-19	19	Scheduled via wave 4 jobs
F111	Payment program for payment requests	PARTIAL-36	36	See payment requests
FBPM / FBPM1	Payment medium workbench (PMW) run / DME engine	COVERED-19	19	ISO 20022 pain.001, NACHA, check files
DMEE / DMEE1	DME engine format tree maintenance	SUPERSEDED-19	19	Format adapters (ISO 20022, NACHA) + wave 48 electronic reporting designer for new formats
FDTA	DME administration (download files)	COVERED-19	19	Payment file vault with audit (keys never leave Vault — wave 36)
SAPF110V? (F110 proposal log)	Payment proposal log	COVERED-19	19	—
FBZ0	Edit payment proposal (display/change)	COVERED-19	19	—
F150 (AP dunning)	Dunning vendors (debit balances)	PARTIAL-30	30	Dunning covers customers; dunning vendors with debit balances is an edge case — reuse engine
FAGL_FC_VAL (AP)	FX valuation of vendor open items	COVERED-12	12	—
FAGLF101 (AP)	Regroup vendors with debit balances	GAP	propose 14	Reclassification gap
F.98	Vendor balance confirmation	GAP	propose 19	Balance confirmations gap
MRRL (ERS — see MM-IV)	Evaluated receipt settlement	COVERED-18	18	—
J1INCHLN run / withholding tax remittance	WHT payment to authority	COVERED-13	13	Tax settlement for withholding
S_P00_07000134 (year-end 1099 run)	1099 generation	COVERED-13	13	—
F.07 (AP)	Carry forward vendor balances	COVERED-14	14	—
FBWE? (AP bill of exchange presentation)	Bill of exchange presentation	GAP	propose 36	Bills gap
Fiori: Schedule Accounts Payable Jobs	Job templates (payment proposal, correspondence)	COVERED-4	4	Job framework
Fiori: Manage Automatic Payments	S/4 payment run app	COVERED-19	19	—

T-code	Name	Passage status	Wave	Notes
FPAYM? / FBPM payment advice	Payment advice generation (REMADV)	COVERED-19	19	Vendor payment advice email/portal
F.64 (AP correspondence)	Correspondence: maintain / print	PARTIAL-19	19	Correspondence types (account statement, payment notice) only partially named (payment advice) — define correspondence framework (templates + triggers) on wave 46 document designer

13.2.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OBD3 — Vendor account groups / field status	T077K	Number range, one-time, field status	COVERED-7	7	—
XKN1 / OBAS — Number ranges for vendor accounts / assign	NRIV, T077K	Internal/external	COVERED-5	5	—
OBB8 — Terms of payment (vendors)	T052	—	COVERED-7	7	—
FBZP — Payment program configuration: all company codes, paying company codes, payment methods per country, per company code, bank determination (ranking, bank accounts, available amounts, value date), house banks	T042, T042A, T042B, T042D, T042E, T042I, T042Z, T042V	Payment method classification (check/transfer/bill), min/max amounts, foreign payments allowed, form/DME program, optimization by bank group/postal code, bank selection ranking	COVERED-19	19	Payment methods, bank ranking and limits per legal entity; house banks in wave 36
FI12 / FI13 — House banks and bank accounts	T012, T012K	Bank key, account id, G/L account, currency	COVERED-36	36	Bank account master (BAM)
OBPM1 / OBPM2 / OBPM3 / OBPM4 — Payment medium formats / note-to-payee / payment method / selection variants	TFPM*, T042Z*	—	COVERED-19	19	Format adapters
OBYR — Special G/L indicators vendors (down payments, guarantees, BoE)	T074	Reconciliation accounts per SGL	PARTIAL-19	19	Down payments yes; BoE no
OBXT / OBA3 — Tolerances for vendors / payment differences	T043G	Residual, charge-off limits	COVERED-19	19	Settlement tolerances
OBA4 — Tolerance groups for employees (posting limits)	T043T	Max amount per doc/line/discount	COVERED-2	2	OPA policy on amounts
Define withholding tax types / codes / assign to company code (extended WHT)	T059P, T059Z, T001WT	Invoice/payment time, base, rates, exemptions, certificates, accumulation	COVERED-13	13	Withholding engine
Classic withholding tax (OBCS? / T059Q)	T059Q	—	SUPERSEDED-13	13	Extended model only
OBZT — Tax codes allowed in FB60/FB70	T007V?	Per transaction	COVERED-13	13	Allowed tax codes per document type
OBCL — Default tax code for non-taxable transactions	T001	—	COVERED-13	13	—
OB71 — Define reason codes for payment block / OB27	T008	Block for payment, manual/automatic	COVERED-19	19	Payment holds
OBB9 — Installment terms	T052R	—	PARTIAL-7	7	Installment gap
Define alternative payee / permitted payee (BP role)	LFA1-LNRZA, LFZA	—	COVERED-7	7	Alternate payee on vendor role
Define correspondence types / forms (OB77, OB78, OB79)	T048, T048B	—	PARTIAL-19	19	Correspondence framework gap
Define payment advice forms (SAPscript/PDF)	—	—	COVERED-19	19	Payment advice templates; wave 46 designer
Check lots (FCHI) / check number ranges	PCEC	Lot per house bank/account	COVERED-36	36	Check lots
OBYP? / OBA7 (vendor doc types KR, KG, KZ, KA)	T003	—	COVERED-11	11	Document types
Define sensitive fields for dual control (S_ALR_87003378 / T055F)	T055F	Fields requiring confirmation	COVERED-7	7	Change approval on sensitive fields
OBXS? — Down payment request accounts	T030	—	COVERED-19	19	—
Define vendor activity groups? (no) — Define payment block reasons	T008	—	COVERED-19	19	—
Evaluated receipt settlement defaults (vendor master ERS flag, OMRR?)	LFM1	—	COVERED-18	18	—
OBXF? — Posting keys for AP (OBXK group)	TBSL	—	SUPERSEDED-11	11	—
OBD3 "one-time account"	T077K	—	PARTIAL-7	7	One-time party gap
Define intercompany vendor/customer clearing (OBYA)	T001U	—	COVERED-40	40	—
Bank-to-bank transfers (payment request types, FIBL)	—	—	PARTIAL-36	36	Payment requests gap

13.3 FI-AR — Accounts Receivable

SAP context: customer master (KNA1/KNB1/KNVV → BP roles FLCU00/FLCU01), customer invoices via SD billing (VF01 → RV document) or FB70, incoming payments (F-28), lockbox (FLB2), dunning (F150), correspondence, interest calculation, credit management (classic FD32 or FSCM-CR). Passage: customer master wave 7, billing wave 29, AR/cash application/collections wave 30, credit wave 27.

13.3.1 Master data & documents

T-code	Name	Purpose	Passage status	Wave	Notes
XD01 / XD02 / XD03 (→ BP)	Customer master centrally	General, company-code, sales-area data	COVERED-7	7	Party + Customer role + sales-scope data
FD01 / FD02 / FD03	Customer master (accounting)	Company-code segment	COVERED-7	7	—
VD01 / VD02 / VD03	Customer master (sales)	Sales-area segment	COVERED-7	7	Sales-scope role data (see SD)
XD05 / XD06 / FD05 / FD06 / VD05 / VD06	Block / flag for deletion	—	COVERED-7	7	—
XD07	Change customer account group	—	COVERED-7	7	—
XD99 / MASS (KNA1)	Mass maintenance customers	—	COVERED-5	5	—
FD08 / FD09	Confirm critical customer changes	Dual control	COVERED-7	7	—
FD32 → UKM_BP / UKM_MY_DCDS	Customer credit management master	Credit limit, risk category, blocks	COVERED-27	27	Credit limits, aging-based rules, risk scoring (wave 30)
FD24 / FD33	Credit management changes / display	—	COVERED-27	27	—
FB70	Enter customer invoice (FI, non-SD)	Direct AR invoice	COVERED-29	29	Free-text invoice
FB75	Enter customer credit memo	—	COVERED-29	29	Credit notes
F-22 / F-27	Customer invoice / credit memo (classic)	—	SUPERSEDED-29	29	Single invoice form
FV70 / FV75	Park customer invoice / credit memo	—	COVERED-29	29	Draft + approval
F-28	Post incoming payment	Manual cash application with clearing	COVERED-30	30	Manual cash application
F-26	Incoming payment fast entry	—	COVERED-30	30	—
F-29	Customer down payment	—	COVERED-30	30	Customer prepayments (deposit on order)
F-37	Customer down payment request	—	COVERED-27	27	Deposit request on order
F-39	Clear customer down payment	—	COVERED-30	30	—
F-30 / F-32	Post with clearing / clear customer	—	COVERED-30	30	Settlement without payment
F-34? / F-36 / F-33	Bill of exchange payment / discount (AR)	—	GAP	propose 36	Bills of exchange gap
FBE1 / FBE2 / FBE3	Payment advice create/change/display	Remittance advices for matching	COVERED-30	30	Remittance advice ingestion and matching
FBE6?	Delete payment advices	—	COVERED-30	30	—
FLB1 / FLB2	Postprocess lockbox / import lockbox file (BAI2)	Lockbox	COVERED-30	30	Lockbox BAI2
FEBA_LOCKBOX	Postprocess lockbox (New)	—	COVERED-30	30	—
F-21 / F-30	Transfer posting (customer)	—	COVERED-30	30	—
FDM1? / UDM_DISPUTE	Create dispute case (FSCM-DM)	—	COVERED-30	30	Disputes workflow
F.27 (document side)	Account statement (periodic correspondence)	—	COVERED-30	30	Customer statements
FBL5N (document finder)	—	—	COVERED-30	30	—
FB12 / F.61 / F.64	Request / print correspondence	Correspondence requests	PARTIAL-30	30	Statements and dunning named; general correspondence framework (payment notices, individual letters, open item lists) only partial
F.2A / F.2B	Interest calculation (customer items) run	—	COVERED-30	30	Interest notes
VKM1 / VKM4 / VKM3	Release blocked SD documents (credit)	Credit release	COVERED-27	27	Credit hold release workflow
BP (FLCU00/FLCU01)	Customer roles in BP (S/4)	—	COVERED-7	7	—
Fiori: Manage Customer Line Items / Post Incoming Payments / Reprocess Bank Statement Items	S/4 AR apps	COVERED-30	30	—	
Biller Direct (FSCM-BD)	Customer self-service for invoices/payments	—	SUPERSEDED-30	30	Customer portal with payment adapter

13.3.2 Reports & inquiries

T-code	Name	Passage status	Wave	Notes
FBL5N / FBL5H	Customer line items / browser	COVERED-30	30	—
FD10N / FD10NA	Customer balance display	COVERED-30	30	—
FD11	Customer account analysis	COVERED-30	30	—
S_ALR_87012167	Accounts receivable information system	COVERED-46	46	AR analytics

T-code	Name	Passage status	Wave	Notes
S_ALR_87012168	Due date analysis for open items	COVERED-30	30	Aging
S_ALR_87012169 / 87012170	Transaction figures: account balance / special sales	COVERED-30	30	—
S_ALR_87012172	Customer balances in local currency	COVERED-30	30	—
S_ALR_87012173	List of customer open items for printing	COVERED-30	30	—
S_ALR_87012174	List of customer open items	COVERED-30	30	—
S_ALR_87012175	Open items — customer due date forecast	COVERED-36	36	Cash forecast
S_ALR_87012176	Customer evaluation with OI sorted list (aging)	COVERED-30	30	—
S_ALR_87012177	Customer payment history	COVERED-30	30	Payment behaviour feeds risk scoring
S_ALR_87012178	Customer open item analysis by balance of overdue items	COVERED-30	30	—
S_ALR_87012179	Customer list	COVERED-7	7	—
S_ALR_87012180	Address list	COVERED-7	7	—
S_ALR_87012182	Display changes to customers	COVERED-5	5	—
S_ALR_87012183	Display/confirm critical customer changes	COVERED-7	7	—
S_ALR_87012186	Customer sales	COVERED-46	46	—
S_ALR_87012197	List of customer line items	COVERED-30	30	—
S_ALR_87012198	List of down payments open on key date	COVERED-30	30	—
S_ALR_87012199	Customer: bill of exchange list	GAP	propose 36	Bills gap
S_ALR_87012215	Customer sales per period? (RFDUML00)	COVERED-46	46	—
F.21? / S_ALR_87012218?	Interest scale / interest calculation list	COVERED-30	30	—
F.31	Credit management overview	COVERED-27	27	Credit exposure
F.32? / FDK43	Credit management: missing data / master data list	COVERED-27	27	—
FCV3? / S_ALR_87012215 (credit)	Credit overview (SD)	COVERED-27	27	—
UKM_COMMITMENTS / UKM_LOGS_DISPLAY	FSCM credit exposure / logs	COVERED-27	27	—
UDM_SUPERVISOR / UDM_SPECIALIST	Collections worklists (FSCM-COL)	COVERED-30	30	Collections workspace
UDM_DISPUTE (list)	Dispute case list	COVERED-30	30	—
S_ALR_87012085? (customer) / F.23?	Dunning history	COVERED-30	30	—
FBMP (display)	Dunning procedure list	COVERED-30	30	—
Fiori: Days Sales Outstanding / Overdue Receivables / Future Receivables / Cash Collection Tracker	AR KPIs	COVERED-46	46	—
Fiori: Process Receivables	Collections work center	COVERED-30	30	—

13.3.3 Periodic processing

T-code	Name	Passage status	Wave	Notes
F150	Dunning run (parameters, schedule, print)	COVERED-30	30	Dunning levels, grace days, blocks; letters only for expected customers
F.24 / F.26 / F.2A / F.2B	Interest calculation (items / balances)	COVERED-30	30	Interest notes
F.27	Periodic account statements	COVERED-30	30	Customer statements
FLB2 / FEBA_LOCKBOX	Lockbox import and postprocessing	COVERED-30	30	—
FF_5 / FEBAN (AR side)	Bank statement → customer clearing	COVERED-30	30	camt.053 auto-match for AR (wave 36 for the bank side)
F.13 (AR)	Automatic clearing customers	COVERED-30	30	Rule-based auto-match
FAGL_FC_VAL (AR)	FX valuation of customer open items	COVERED-12	12	—
FAGLF101 (AR)	Regroup customers with credit balances	GAP	propose 14	Reclassification gap
F.99	Customer balance confirmation	GAP	propose 30	Balance confirmation gap
F.07 (AR)	Carry forward	COVERED-14	14	—
UKM_MALUS_DSP / UKM_MASS? / UKM_CREDIT_LIMIT_MASS?	Mass credit limit recalculation / scoring run	COVERED-30	30	Risk scoring batch
F.28? / FD24 mass	Reset credit limit data / recalculate	COVERED-27	27	—

T-code	Name	Passage status	Wave	Notes
UDM_BP_PROF? / UDM_GENWL	Generate collections worklist (FSCM-COL)	COVERED-30	30	Worklist generation
F110 (customer direct debits)	Direct debit run (SEPA SDD)	COVERED-30	30	Customer direct debit via payment adapter / pain.008 — confirm pain.008 is in wave 19 formats (wave 19 lists pain.001 only)
FDTA (DD)	DME admin for direct debits	COVERED-19	19	—
SEPA mandate management (FSEPA_M1..M4)	Mandates	PARTIAL-30	30	SEPA direct-debit mandates not named; required for EU customers paying by direct debit — add mandate entity to wave 30
VF44 / VF45? (see SD)	Revenue recognition run (classic)	COVERED-29	29	—
FBWD? / FBWE (AR BoE)	Bill of exchange presentation	GAP	propose 36	Bills gap
Fiori: Schedule Accounts Receivable Jobs	Job templates (dunning, statements, interest)	COVERED-4	4	—

13.3.4 Customizing (SPRO)

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
OBD2 — Customer account groups / field status	T077D	—	COVERED-7	7	—
XDN1 / OBAR — Number ranges customers / assign	NRIV, T077D	—	COVERED-5	5	—
OBB8 — Terms of payment (customers)	T052	—	COVERED-7	7	—
OBXR — Special G/L customers (down payments, security deposits, BoE)	T074	—	PARTIAL-30	30	BoE gap
OBA3 — Customer tolerances	T043G	—	COVERED-30	30	—
FBMP — Dunning procedures (levels, intervals, charges, min amounts, grace days, forms)	T047*, T047A..T047M	Dunning levels 1–9, dunning charges by amount, interest indicator, print parameters, dunning by dunning area	COVERED-30	30	Dunning levels + letters
OBL6? / OB61 — Dunning areas / block reasons	T047M, T040S	—	COVERED-30	30	—
OB46 / OB82 / OB83 — Interest indicators, calculation types, reference rates	T056*, T056P, T056R	Item/balance interest, calendar, rounding	COVERED-30	30	AR interest; GL interest is a gap
OB45 — Credit control areas (classic)	T014	Credit limit currency, update group, risk categories	SUPERSEDED-27	27	FSCM-style credit segments modeled as credit profiles per legal entity/group
OVA8 — Automatic credit control (SD)	T691F	Static/dynamic check, horizon, open orders/deliveries, reactions	COVERED-27	27	Credit rules (limit, aging, dynamic horizon)
OB01 — Risk categories / OVFL credit groups	T691A, T691B	—	COVERED-27	27	—
UKM_* FSCM credit customizing (credit segments, rules, scoring formulas, BRF+ rules)	UKMCRED_SGM*, UKM_*	Score formulas, limit rules, check rules per segment	COVERED-27	27	Credit profiles + scoring (wave 30)
UDM_* collections customizing (collection strategies, segments, profiles, priorities, worklist rules)	UDM_*	Strategy rules (BRF), priorities, groups	COVERED-30	30	Collection strategies
UDM_* dispute case customizing (case types, attributes, status, reasons, RMS)	SCMG_*	—	COVERED-30	30	Disputes
OBB9 — Installment terms (customers)	T052R	—	PARTIAL-7	7	Installment gap
OB10? / Lockbox control (OBAY, OBAX)	T049A, T049B, T049L	Lockbox destination, BAI format, posting rules	COVERED-30	30	—
OBXL / OBBE — Payment difference reason codes and accounts	T053R, T030	—	COVERED-30	30	—
Define payment advice types / notes to payee (OBBE?)	T053?	—	COVERED-30	30	—
Define correspondence types (OB77) / periodic account statements	T048	—	PARTIAL-30	30	Correspondence framework gap
SEPA mandate customizing	SEPA_*	—	PARTIAL-30	30	Mandate gap
Define alternate payer / head office–branch	KNA1-KNRZE, KNB1	Head office/branch accounting	PARTIAL-7	7	Head-office/branch (invoice to branch, pay from HQ) not named in wave 7 party roles — add "bill-to/pay-from parent" relationship
OBYR? (customer side) sales/FI account determination — see VKOA	T030, C001..	—	COVERED-29	29	Revenue account determination
Define reason codes for credit holds (OVA8/ OVAK)	T691*	—	COVERED-27	27	—
OB38? — Assign credit control area to company code	T001	—	SUPERSEDED-27	27	—
Define cash discount base, tax on discount (OBY6)	T001	—	COVERED-13	13	—

IMG node / T-code	Table(s)	Key options	Passage status	Wave	Notes
Define dunning forms (SAPscript F150_DUNN_01)	—	—	COVERED-30	30	Letter templates; wave 46 designer